SEMICONDUCTOR MATERIALS

LECTURE NOTES
FOR
MSE 223

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TABLE OF CONTENTS

Page

1. Review of Semiconductor Basics

1.1. Intrinsic Semiconductors 2
1.1.1. Electrical Properties 2
1.1.2. Optical Properties (near the bandgap) and Bandstructure 8
1.1.3. Vibrational Modes (phonons) 19

1.2. Extrinsic Semiconductors 31
1.2.1. Shallow Level Impurities 31
1.2.2. Multivalent Impurities 35
1.2.3. Deep Level Impurities and Defects 38
1.2.4. Compensation 41
1.2.5. Free Carrier Statistics 42
1.2.6. Electronic Transport 51

1.3. Carrier Generation and Recombination 59
1.3.1. Generation 59
1.3.2. Recombination 62
1.3.2.1. Radiative Recombination 62
1.3.2.2. Non-Radiative Recombination 64

2. Semiconductor Purification

2.1. Chemical Purification - Physical Purification 67

2.2. Normal Freezing 69

2.3. Zone Purification or "Zone Melting" 70

2.4. Equilibrium versus Effective Segregation Coefficient 74

3. Crystal Growth

3.1. Introduction 78

3.2. Crystallization 78

3.3. Bulk Crystal Growth Techniques 81
3.3.1. Bulk Growth of Si and Ge from a Melt 81
3.3.2. Silicon Floating Zone (FZ) Crystal Growth 84
3.3.3. Bulk Growth of III-V Compound Semiconductors from the Melt 90
3.3.4. Melt Convection and Impurity Distribution 91
3.3.5. Semiconductor Bulk Crystal Growth Literature 95

3.4. Epitaxial Growth Techniques 98
3.4.1. Introduction 98
3.4.2. Liquid Phase Epitaxy (LPE) 99
3.4.3. Vapor Phase Epitaxy (VPE) 102
3.4.4. Molecular Beam Epitaxy (MBE) 123

4. Point and Line Defects

4.1. Introduction 134

4.2. Point Defects 134

4.3. Line and Two-Dimensional Defects 137

5. Doping of Semiconductors

5.1. Introduction 142

5.2. Dopant Introduction During Crystal Growth 145

5.3. Neutron Transmutation Doping 151

5.4. Alloyed (or fused) Contacts 156

5.5. Diffusion 158
5.5.1. Phenomenological, Macroscopic Aspects 158
5.5.2. Practical Aspects 166
5.5.3. Microscopic Diffusion Mechanisms 173
5.5.3.1. Self-diffusion 174
5.5.3.2. Dopant Diffusion 182

5.6. Ion Implantation 191
5.6.1. Ion-Solid Interactions and Ion Range 194
5.6.1.1. Nuclear and Electronic Interactions 194
5.6.1.2. Ion Range and Range Straggling 199
5.6.1.3. Channeling and Dose Rate Effects 204
5.6.2. Annealing of the Radiation Damage and Impurity Activation 210
5.6.3. Solid Phase Epitaxy 220
5.6.4. Ion Implantation Technology 223
5.6.5. Recent Developments 230
5.6.5.1. Ion Implantation in Compound Semiconductors 230
5.6.5.2. Plasma Immersion Ion Implantation 233

6. Metal-Semiconductor and Metal-Oxide-Semiconductor (MOS) Junctions

6.1. Introduction 235

6.2. The Ideal Metal-Semiconductor Junction 236

6.3. Real Metal-Semiconductor Junctions (including Silicides) 241

6.4. The Metal-Oxide-Semiconductor (MOS) Junction 249

7. Electronic Materials Analysis

7.1. Introduction 252

7.2. Electrical Measurements 253
7.2.1. Type of Conduction 253
7.2.2. Electrical Conductivity and Resistivity 254

7.2.3. Hall Effect Measurements....263
7.2.4. Space Charge and Diode Capacitance....269
7.2.5. Deep Level Transient Spectroscopy (DLTS)....274
7.2.6. Minority Carrier Lifetime....289

7.3. Optical Characterization....292
7.3.1. Introduction....293
7.3.2.1. Bandgap (or Fundamental) Absorption Spectroscopy....293
7.3.2.2. Far Infrared Spectroscopy of Shallow Acceptors and Donors....296
7.3.3. Vibronic Transitions: Local Vibrational Mode Spectroscopy....305

7.4. Rutherford Backscattering (RBS) and Ion Channeling Spectrometry....315

7.5. Electron Beam Induced Current Microscopy (EBIC)....331

7.6. Chemical Methods....335

DEPARTMENT OF MATERIALS SCIENCE
AND MINERAL ENGINEERING
UNIVERSITY OF CALIFORNIA AT BERKELEY

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SEMICONDUCTOR MATERIALS

(or "The Materials Basis of Semiconductor Device Technology")

This graduate course was offered by the Department of Materials Science and Mineral Engineering, University of California at Berkeley, for the first time in the Spring Quarter, 1981. Semiconductor science and technology have dramatically changed in these past nineteen years and will undoubtedly continue to do so in the future.

This, the seventh (2002) edition of the lecture notes, reflects many of these changes. A great effort has been taken to familiarize the students not only with the standard, well-established semiconductor processing and characterization techniques but also with recent advances and developments. I am asking the readers of these notes to communicate to me any suggestions, comments or personal points of view which could improve these notes (please send your suggestions to: EEHALLER@LBL.GOV)

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1. REVIEW OF SEMICONDUCTOR BASICS

1.1. Intrinsic Semiconductors

1.1.1. Electrical Properties

Semiconductors have energy bandgaps of between 0 and 3.5 eV. At T = 0 K, electrons fill all the states of the valence band, and there are no free carriers in the conduction or valence bands. Semiconductor properties can be visualized in a number of ways. Two are shown in Fig. 1(a) and 1(b).

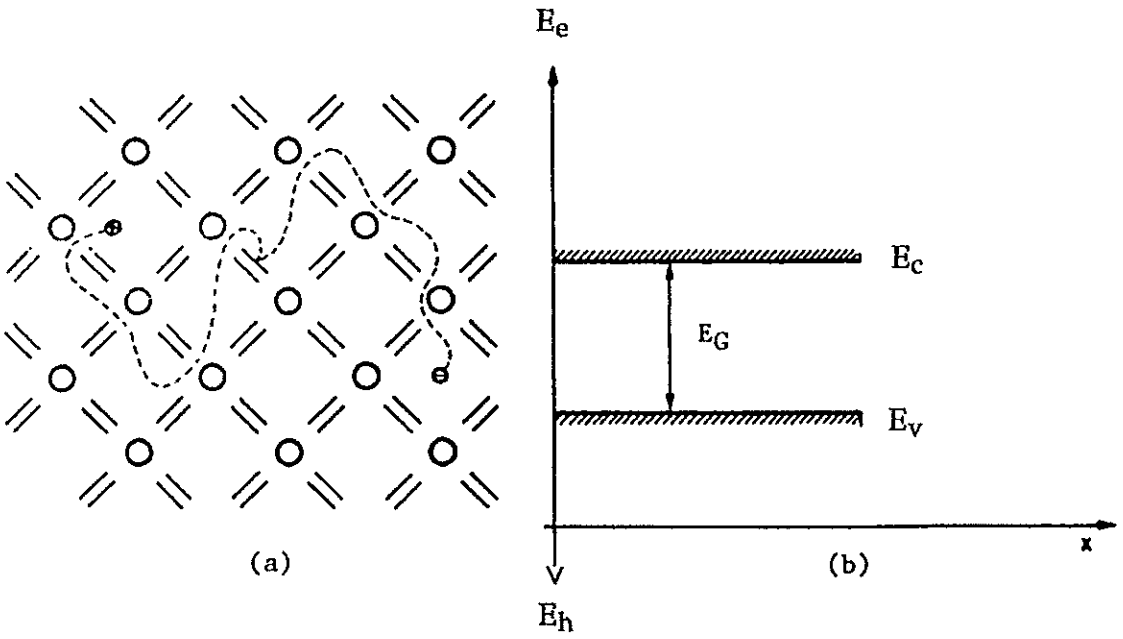


Figure 1

(a) Projection of a Si or Ge crystal lattice (diamond structure) in the <100> direction. Electrons and holes can be generated either by optical (photons) or thermal (phonons) excitations. Electrons and holes can move freely through the crystal until they recombine.

(b) One-dimensional band diagram; E_e, E_h are the energies of electrons and holes, respectively; E_g is the energy of the forbidden gap.

One of the fundamental properties of intrinsic semiconductors is the strong dependence of the concentration of electrons (n) and holes (p) on absolute temperature (T). The rapid increase of these concentrations leads to a sharp decrease of the intrinsic resistivity with increasing temperature, opposite to the behavior of the resistivity one observes for metals.

Assuming that the reader is familiar with basic band theory, we briefly derive the dependence of n and p on T. We know that electrons (and holes) obey Fermi statistics, i.e., the probability for an electron to occupy a state at energy E is:

$$F(E) = \frac{1}{1 + \exp\frac{E - E_F}{k_B T}}$$

with E_F = Fermi energy and k_B = Boltzmann constant. Electrons near the minimum of a conduction band can be treated as if they were "free" electrons. In a three-dimenstional solid the density of available electron states Z(E) in a parabolic band ($E \propto k^2$) is:

$$Z(E) = 4\pi\left(\frac{2m_e}{h^2}\right)^{3/2}(E - E_G)^{1/2}$$

with h = Planck's constant, E_G = energy of the forbidden gap, and m_e = effective electron mass (we arbitrarily set E = 0 at the top of the valence band; this sets the conduction band minimum E_C = E_G). In order to obtain the electron concentration in the conduction band we need to integrate from E_C (=E_G) to infinity:

$$n = \int_{E_G}^{\infty} Z(E)\,F(E)\,dE\,.$$

This is most easily achieved by approximating F(E) with the Boltzmann distribution. For $E–E_F > k_BT$, which applies to intrinsic semiconductors up to high temperatures, we find $\exp[(E–E_F)/k_BT] \gg 1$ and

$$F(E) \cong \exp\left(-\frac{E - E_F}{k_B T}\right).$$

This leads to

$$n = \int_{E_G}^{\infty} Z(E) \exp\left(-\frac{E - E_F}{k_B T}\right) dE$$

Let us expand the exponential term:

$$\exp\left(-\frac{E - E_F}{k_B T}\right) = \exp\left(-\frac{E - E_G + E_G - E_F}{k_B T}\right)$$

and set

$$x = \frac{E - E_G}{k_B T} \quad , \quad (dE = k_B T\, dx) .$$

This leads to

$$n = 4\pi \left(\frac{2m_e}{h^2}\right)^{3/2} (k_B T)^{3/2} \exp\left(-\frac{E_G - E_F}{k_B T}\right) \int_0^{\infty} x^{1/2} \exp(-x)\, dx$$

The value of the integral is $\frac{\sqrt{\pi}}{2}$ and

$$n = 2(2\pi\, m_e\, k_B T/h^2)^{3/2} \exp\left(-\frac{E_G - E_F}{k_B T}\right) .$$

The pre-exponential term is called "the effective density of states of the conduction band, N_C" and it is weakly temperature-dependent ($\propto T^{3/2}$):

$$n = N_C \exp\left(-\frac{E_G - E_F}{k_B T}\right) .$$

In perfect analogy we find for holes in the valence band:

$$p = N_V \exp\left(-\frac{E_F}{k_B T}\right) .$$

(Just remember that the probability of finding a hole at energy E is equal to the probability of *not* finding an electron, i.e., 1 – F(E)!)

The effective density of states for holes in the valence band $N_V = 2(2\pi m_h k_B T/h^2)^{3/2}$. The product of the free electron and hole concentrations is independent of E_F:

$$n\,p = N_C N_V \exp\left(-\frac{E_G}{k_B T}\right).$$

The Fermi level can be found in cases where m_e and m_h are known.

We apply the neutrality condition $n = p$:

$$E_F = \frac{E_G}{2} + \frac{3}{4} k_B T \ln \frac{m_h}{m_e}.$$

Figure 2 displays the free carrier concentration (also called intrinsic carrier concentration n_i=n=p) for three common semiconductors. Can you determine the bandgap energy E_G?

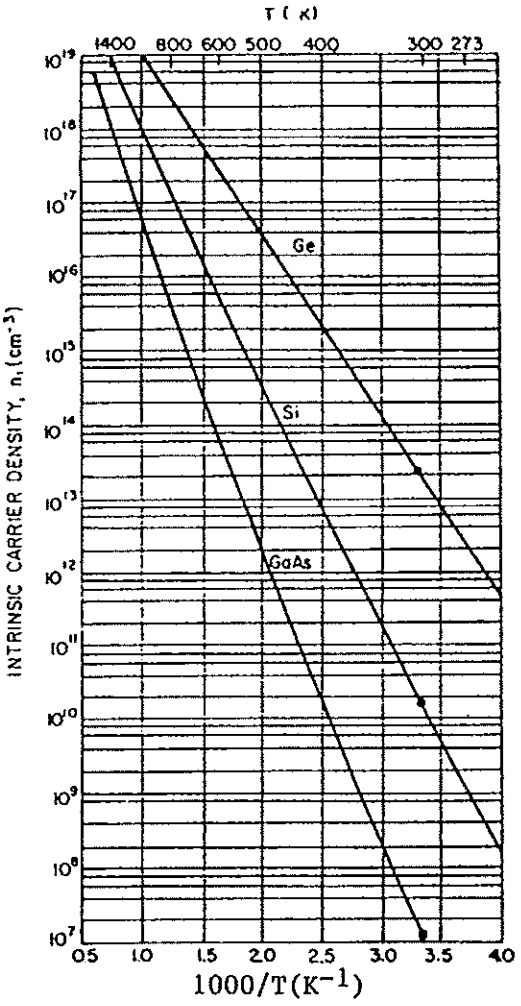


Fig. 2. Arrhenius plot (ln versus 1/T) of the intrinsic carrier densities of Ge, Si and GaAs as a function of inverse absolute temperature. [After C. D. Thurmond, J. Electrochem. Soc. **122**, 1133 (1975).]

Table 1 contains a collection of important properties of a large number of semiconductors. Beyond the well known digital and analog device applications of Si, GaAs and its alloys, there exists a very large number of sensor and display applications, e.g., infrared detectors (Ge, Si, InSb, etc.), radiation detectors (x-rays: Si; γ-rays: Ge), light-sensitive resistors (CdS), TV and oscilloscope screens (ZnS, ZnSe, etc.).

Table 1

PROPERTIES OF IMPORTANT SEMICONDUCTORS

	Element or Compound	Name	Crystal Structure	Lattice Constant at 300 K (Å)	Bandgap (eV) 300 K	0 K
Element	C	Carbon (diamond)	D	3.56683	5.47(I)	5.48
	Ge	Germanium	D	5.64613	0.66(I)	0.74
	Si	Silicon	D	5.43095	1.12(I)	1.17
	Sn	Grey tin	D	6.48920		0.082
IV-IV	SiC	Silicon Carbide	W	3.086 (a) 15.117 (c)	2.996(I)	3.03
III-V	AlAs	Aluminum arsenide	Z	5.6605	2.17(I)	
	AlP	Aluminum phosphide	Z	5.4635	2.45(I)	
	AlSb	Aluminum antimonide	Z	6.1355	1.58(I)	1.68
	BN	Boron nitride	Z	3.6150	≈7.5	
	BP	Boron phosphide	Z	4.5380	2.0	
	GaAs	Gallium arsenide	Z	5.6533	1.42(D)	1.52
	GaN	Gallium nitride	W	3.189 (a) 5.185 (c)	3.36(D)	3.50
	GaP	Gallium phosphide	Z	5.4512	2.26(I)	2.34
	GaSb	Gallium antimonide	Z	6.0959	0.72(D)	0.86
	InAs	Indium arsenide	Z	6.0584	0.36(D)	0.42
	InP	Indium phosphide	Z	5.8686	1.35(D)	1.42
	InSb	Indium antimonide	Z	6.4794	0.17(D)	0.23
II-VI	CdS	Cadmium sulfide	Z	5.8320	2.42(D)	2.56
	CdS	Cadmium sulfide	W	4.16 (a) 6.756 (c)	2.49(D)	2.59
	CdSe	Cadmium selenide	Z	6.050	1.70(D)	1.85
	CdTe	Cadmium telluride	Z	6.482	1.43(D)	1.60
	ZnO	Zinc oxide	R	4.580	3.35(D)	3.42
	ZnS	Zinc sulfide	Z	5.430	3.68(D)	3.84
	ZnS	Zinc sulfide	W	3.82 (a) 6.26 (c)	3.80(D)	3.91
	ZnSe	Zinc selenide	Z	5.669	2.71(D)	2.82
	ZnTe			6.1		
IV-VI	PbS	Lead sulfide	R	5.9362	0.41(D)	0.286
	PbTe	Lead telluride	R	6.4620	0.31(D)	0.19

D = Diamond, W = Wurtzite, Z = Zincblende, R = Rock Salt

Bandgap: (D) direct; (I) indirect

The intrinsic carrier concentration as a function of temperature is rather insensitive to the detailed bandstructure of a semiconductor. This is because the conduction band minimum and the valence band maximum are to first approximation parabolic and the intrinsic carrier distributions are highly peaked very near the band extrema ($k_BT \ll E_G$ for practically all cases).

1.1.2. Optical Properties (near the Bandgap) and Bandstructure

The situation changes drastically when we use *photons* to probe the optical properties of a semiconductor. Photons with energies ranging from μeV to many eV and beyond are conveniently handled with modern spectrometers. Absorption of photons can be used to explore the bandstructure in detail over a wide energy range. The linear absorption coefficient α is defined by:

$$I = I_0 \exp(-\alpha x)$$

with I_0 = photon flux impinging on a wafer of semiconductor with parallel faces, I = photon flux exiting the semiconductor sample, and x = thickness of the semiconductor wafer. Reflection effects are assumed to be negligible, a condition which can be closely approximated through efficient antireflection coatings. Solving the above equation for α, we find:

$$\alpha = \frac{1}{x} \ln (I_0/I) \quad .$$

The linear absorption coefficient is a strong function of photon energy. This is shown in Fig. 3, where log α is plotted as a function of photon energy for three important semiconductors for two temperatures each. GaAs shows one very sharp rise in α near 1.4 eV for 300 K and 1.5 eV for 77 K, while Si and Ge display more complicated dependencies. Analyzing the energy E versus crystal momentum $\vec{k}$ plots shown in Fig. 4, we can understand the dependence of α on photon energy in Fig. 3.

At first sight the calculated energy bands shown in Fig. 4 look rich and complicated. Upon closer examination, however, we see that in all four examples three valence band maxima are located at the Γ point of the Brillouin zone, two of which are degenerate in energy (in the case of Si this is not visible on the energy scale chosen but it is well established experimentally!). Similarly conduction band minima appear at L, Γ and X. These common features are the consequence of the physical crystal structure of these four semiconductors. Silicon and germanium crystallize in the diamond lattice structure while gallium arsenide and indium antimonide assume the zinc blende lattice. In all four cases every atom is located at the center of an equal-sided tetrahedron formed by

four neighbors located at the four corners of the tetrahedron. This highly symmetric crystal structure leads to the well-known first Brillouin zone (a truncated octahedron) shown in Fig. 5 and to the basic bandstructure shown in Fig. 6. The magnitude of the three energy gaps E(L), E(Γ) and E(X) depends on the chemical composition of the semiconductor under discussion but the existence of three conduction band minima and the valence band maxima is tied directly to the crystal structure.

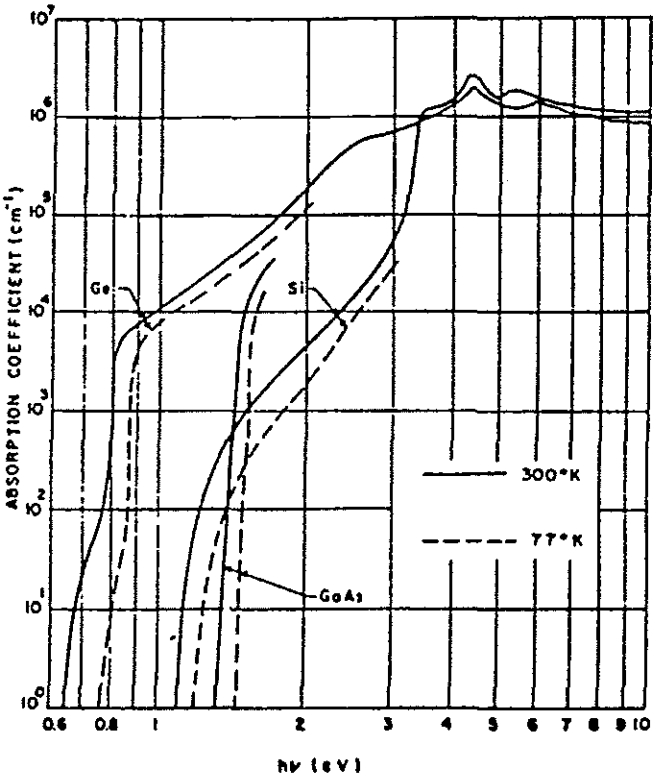


Fig. 3. Measured absorption coefficients near and above the fundamental absorption edge for pure Ge, Si and GaAs. (After S. M. Sze, *Physics of Semiconductor Devices*, 2nd. ed., p. 42, John Wiley & Sons, 1981.)

GaAs has the conduction band minimum (Γ_6) and the valence band maximum (Γ_8) both located at Γ, the center of the Brillouin zone. Photons can excite electrons across the minimum bandgap without a change in momentum (no change in $\vec{k}$). This is an efficient process involving only two particles, which leads to the sharp rise of the absorption coefficient α for energies larger than E_G. For Si and Ge the conduction band minima do not occur at Γ but at finite values of $\vec{k}$ (at L for Ge and near X for Si). In this case a photon and a phonon are required to satisfy the energy and momentum changes necessary for the excitation of electrons across the minimum bandgap.

Photons have virtually no momentum (often referred to as "photons cause vertical transitions") and phonons carry very little energy (horizontal transition) (Fig. 7). A three-particle process is less likely than the two-particle process discussed above. This is evident from the rise of α at the minimum bandgap energy of 0.66 eV at 300 K to a value of only 10 cm^{-1} for Ge. A second sharp rise up to $10^4 cm^{-1}$ occurs at higher energies where direct (phonon-less) transitions become possible. In Si the indirect gap is 1.1 eV at room temperature, and photon energies larger than 3 eV are required for direct transitions.

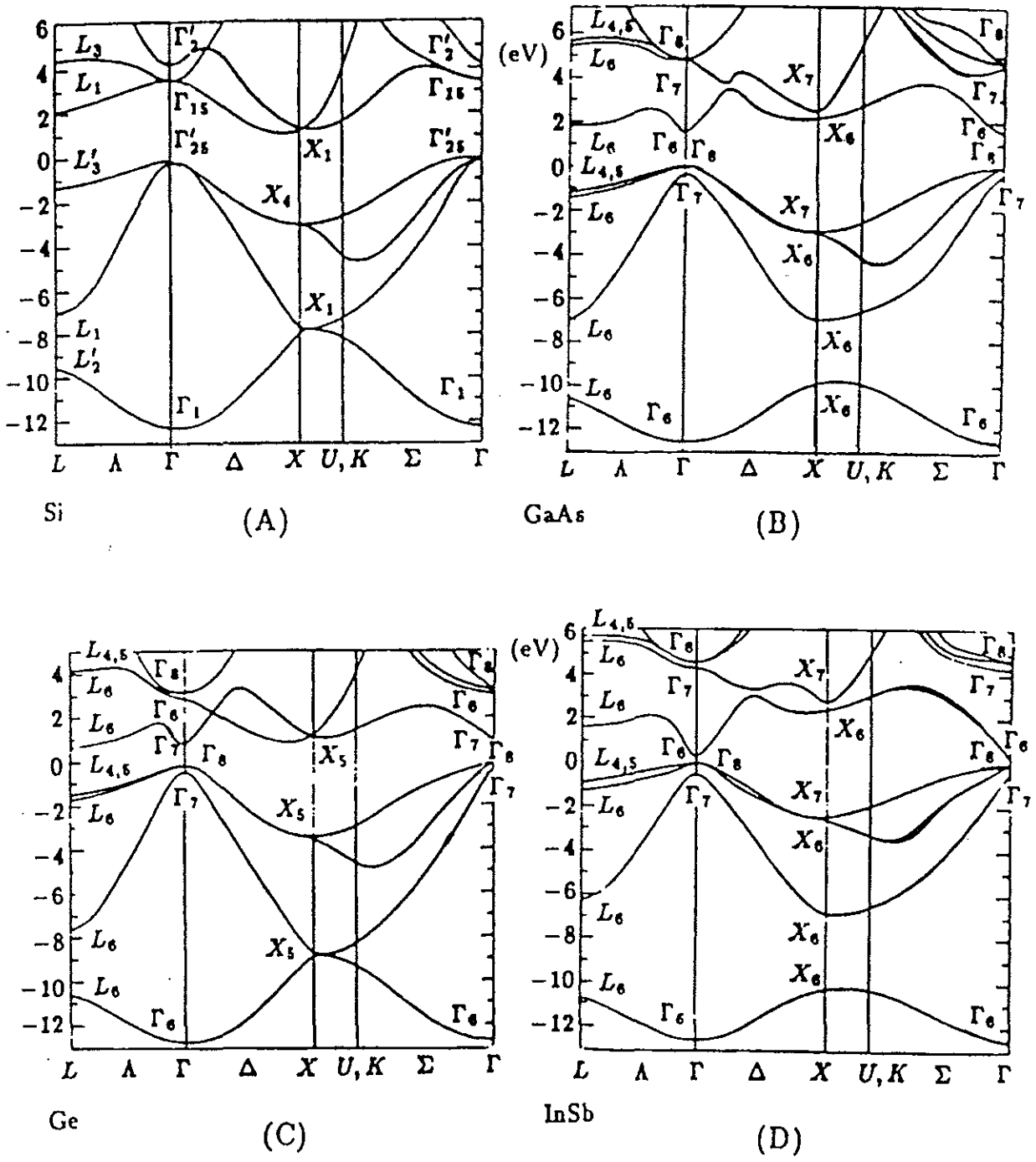


Fig. 4. Energy bands of Si, Ge, GaAs, and InSb in subfigures (A)-(D) respectively, as calculated with empirical nonlocal pseudopotentials. Energy scale in eV. [After Chelikowsky and Cohen, Phys. Rev. B 14, 100, 139, 556 (1976).]

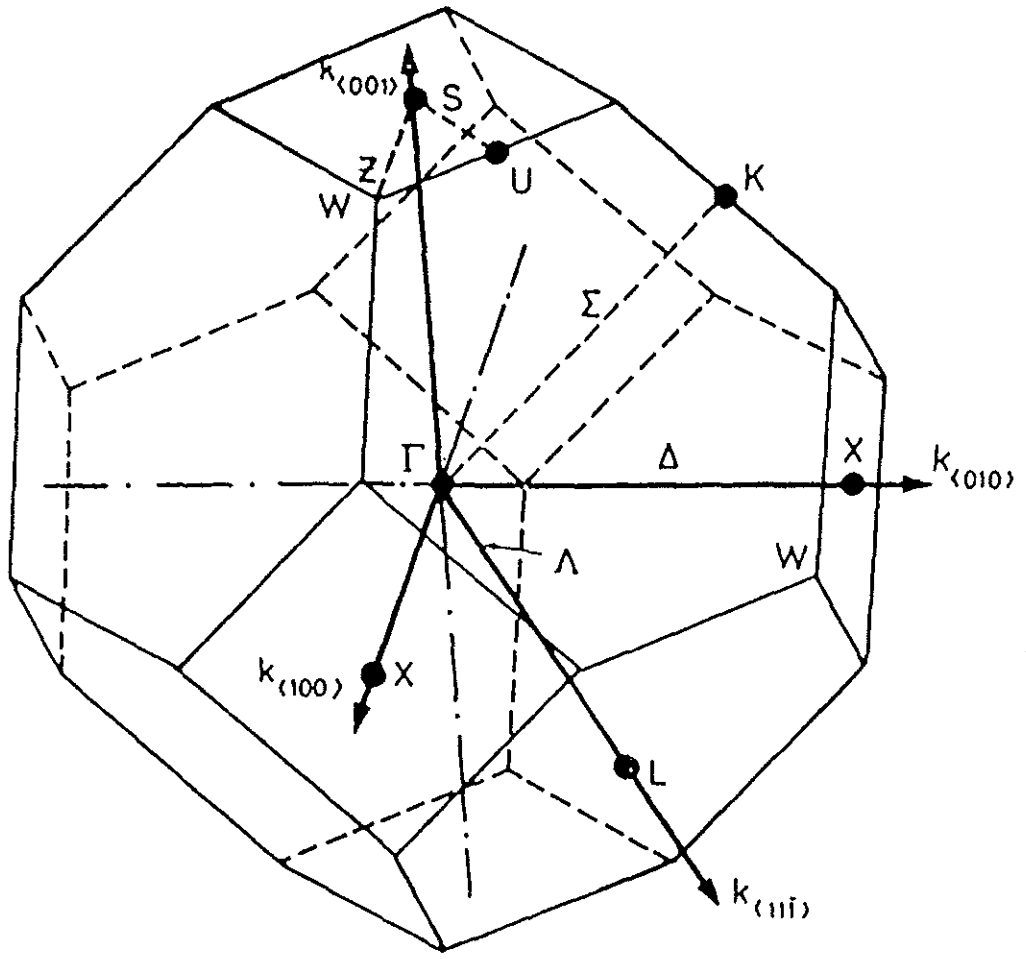


Fig. 5. First Brillouin zone for the diamond and zinc blende lattice, including important symmetry points and lines. [After D. Long, "Energy Bands in Semiconductors" (J. Wiley and Sons, 1968).]

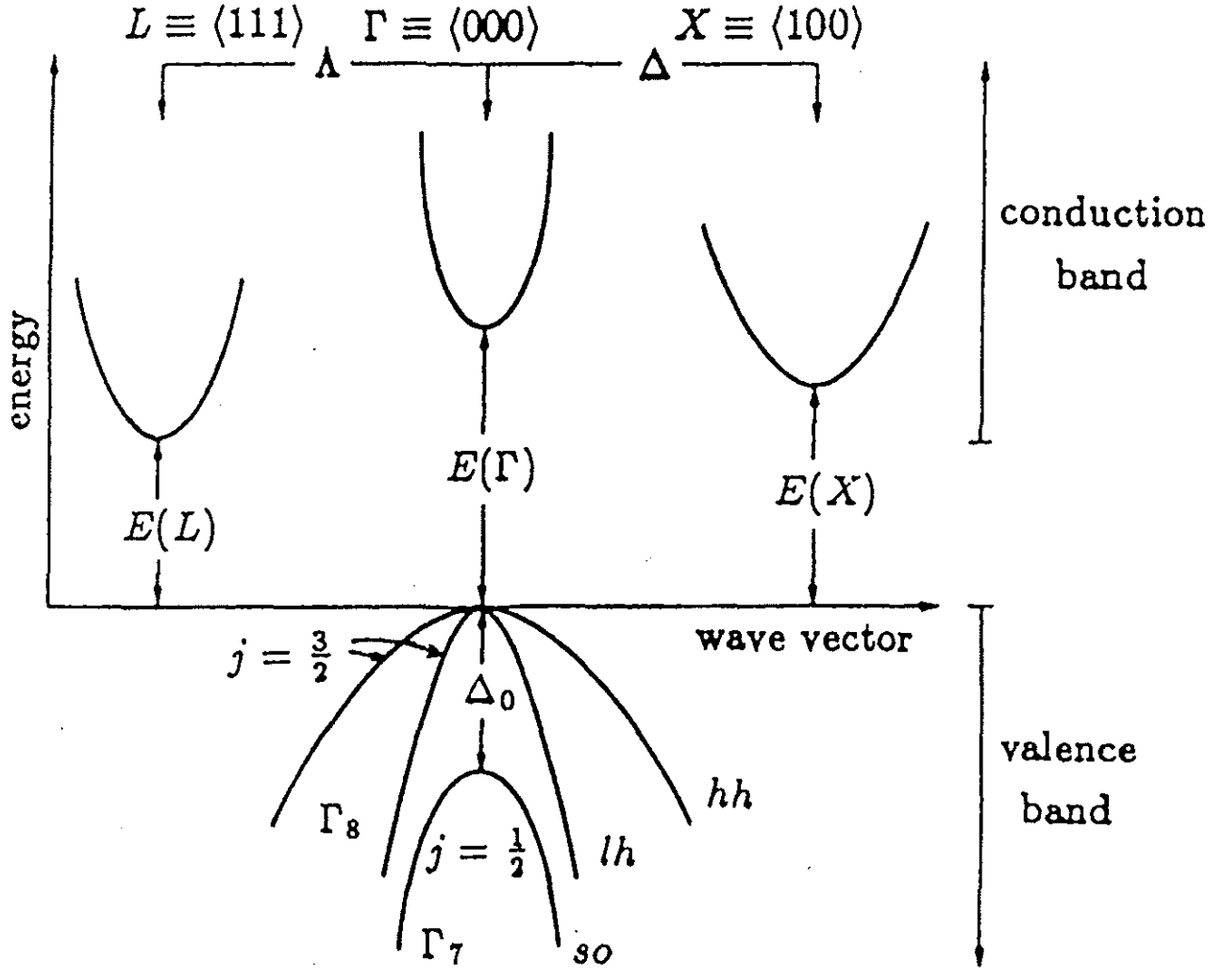


Fig.6. All semiconductors crystallizing in the diamond or zinc blende crystal structure share the bandstructure features shown in this figure (hh = heavy holes, lh = light holes, so = split off valence band). [After Karl W. Böer, "Survey of Semiconductor Physics," (Van Nostrand Reinhold, NY, 1990, p. 214).]

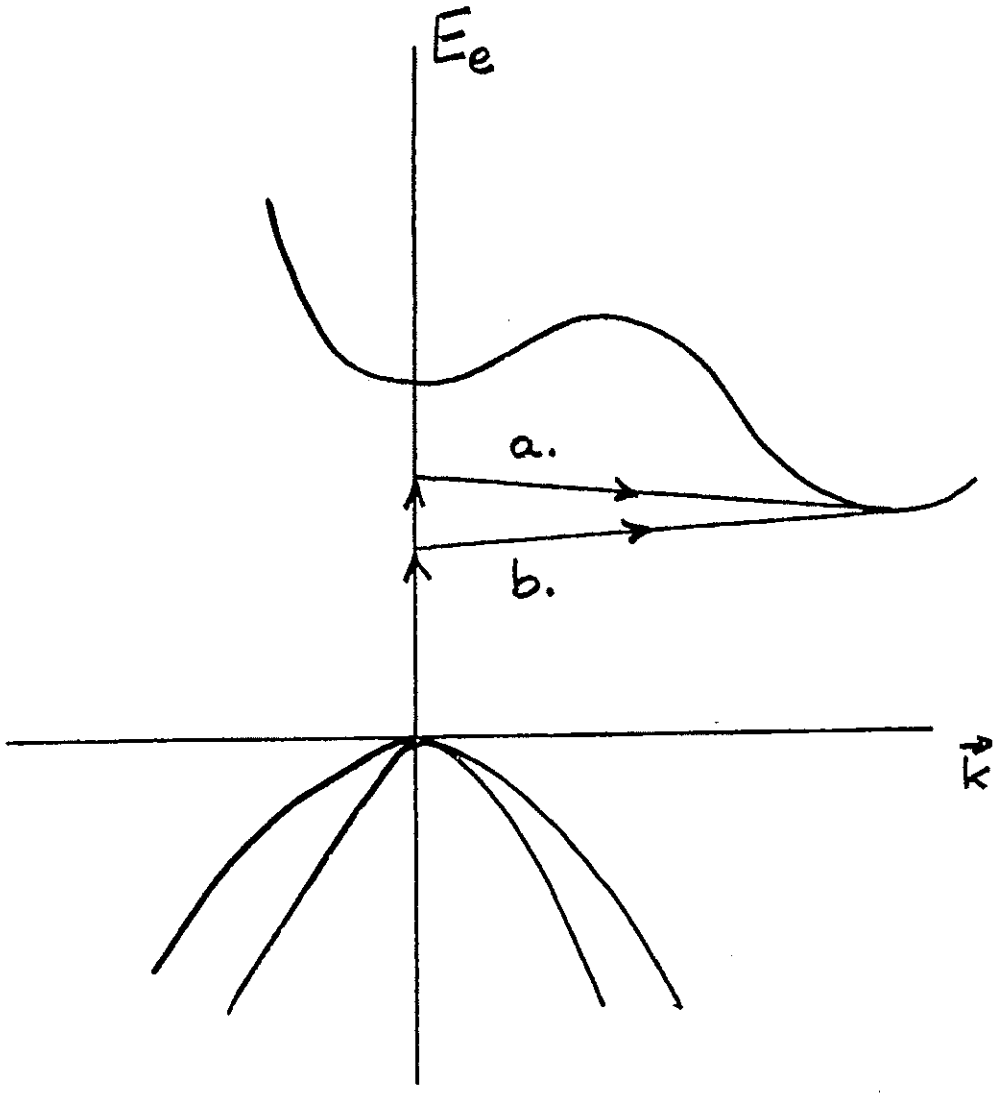


Fig. 7. An electron can be excited across an indirect bandgap by the simultaneous absorption of a photon and a phonon (b.), or by the absorption of a photon followed by emission of a phonon (a.).

These examples illustrate why one classifies semiconductors as *direct* or *indirect* bandgap materials. We will understand the striking consequences of this classification when we discuss optical and electrical devices made from semiconductors.

The conduction band minima of many semiconductors have been analyzed in great detail. Table 2 lists some energy gap values for several elemental and compound semiconductors. The energy shifts of the various minima with pressure have been extensively explored, and it has been found that direct bandgap semiconductors can become indirect bandgap materials above well-defined values of pressure.

TABLE 2

Direct (*d*) and indirect (*i*) band gaps for various semiconductors and their temperature and pressure coefficients.

Crystal	E_g (0 K)	E_g (300 K)	$\frac{dE_g}{dT}\left(\frac{\text{meV}}{\text{K}}\right)$	$\frac{dE_g}{dp}\left(\frac{\text{meV}}{\text{kbar}}\right)$
Si	1.1695 (*i*)	1.110 (*i*)	– 0.28	– 1.41
Ge	0.744 (*i*)	0.664 (*i*)	– 0.37	5.1
α-Sn	0		– 0.5 (2nd)	
GaP	2.350 (*i*)	2.272 (*i*)	– 0.37	10.5
GaAs	1.519 (*d*)	1.411 (*d*)	– 0.39	11.3
GaSb	0.812 (*d*)	0.70 (*d*)	– 0.37	14.5
InP	1.4236 (*d*)	1.34 (*d*)	– 0.29	9.1
InAs	0.418 (*d*)	0.356 (*d*)	– 0.34	10.0
InSb	0.2352 (*d*)	0.180 (*d*)	– 0.28	15.7
ZnS_{cub}	3.78 (*d*)	3.68 (*d*)	– 0.47	– 5.8
ZnS_{hex}	3.91 (*d*)	3.8 (*d*)	– 0.30	
ZnSe	2.820 (*d*)	2.713 (*d*)	– 0.45	0.7
ZnTe	2.391 (*d*)	2.26 (*d*)	– 0.52	8.3
CdS	2.585 (*d*)	2.485 (*d*)	– 0.41	4.5
CdSe	1.841 (*d*)	1.751 (*d*)	0.36	5
CdTe	1.606 (*d*)	1.43 (*d*)	– 0.54	8

[After Karl W. Böer, "Survey of Semiconductor Physics," (Van Nostrand Reinhold, NY, 1990, p. 235).]

We stated earlier that the dispersion in the vicinity of the band minima and maxima can be approximated by isotropic parabolas. The proportionality constant between E and $|\vec{k}|^2$ is the product of $\hbar^2$ and the effective mass ($\hbar=h/2\pi$, m_e) for conduction band minima, m_h for valence band maxima). The effective mass is proportional to the inverse of the second derivative of the $E/\vec{k}$ parabola:

$$m_{effective} = \hbar^2 \left(\frac{\partial^2 E}{\partial |\vec{k}|^2} \right)^{-1} .$$

Electrons do not travel through a crystal with their "free" mass but with a modified or effective mass. This is an important consequence of the wave nature of the electron and the regular spacing between the atoms in the semiconductor crystal. Table 3 lists the effective masses of electrons at various conduction band minima for several semiconductors. For the minimum at Γ we obtain one value for $m_{effective}$ because the mass is isotropic. For band minima along a certain direction of $\vec{k}$ (<111>, <100>, etc.), a parallel $m_{\|}$ and a perpendicular mass $m_{\perp}$ are listed. It is left to the reader's imagination to figure out what happens to the effective mass at the values of $\vec{k}$ where the second derivative becomes zero!

The valence band maximum of all elemental and III-V compound semiconductors is degenerate and is described by two curved surfaces which touch at Γ_8. Again, the parabolic approximation is taken. The "heavy" holes belong to the band with the smaller curvature. The "light" holes define the larger curvature band. As in the case of electrons the effective hole masses differ from the free electron mass. Table 4 contains a collection of effective light and heavy hole masses for a number of semiconductors. T. M. Luttinger [Phys. Rev. **102**, 1030 (1956)] showed that the $E/\vec{k}$ dependence can be described accurately by:

$$E_{l,h} = \frac{\hbar^2}{2m_0} \left[\gamma_1 k^2 \pm \left(4\gamma_2^2 k^4 + 12(\gamma_3^2 - \gamma_2^2)(k_x^2 k_y^2 + k_y^2 k_z^2 + k_z^2 k_x^2) \right)^{1/2} \right]$$

with γ_1, γ_2, and γ_3 called Luttinger parameters. In addition we find at some energy Δ_0 below the valence band top, the so-called split-off band. Depending upon how large Δ_0 is, this split-off band plays a major or minor role. The effective mass of the split-off band is given in the last column under m_{so}.

TABLE 3

Electron effectivemasses in units of m0, conduction bands in eV (at 0 K) with characteristic point and band numbers identified, and camel's back parameters in eV (Δ^c and Δ_0^c).*

Crystal	1. Indir. Gap	$m_{n\|\|}$	$m_{n\perp}$	2. Direct Gap	m_n	3. Gap	$m_{n\|\|}$	$m_{n\perp}$	Δ^c	Δ_0^c or ΔE
Si	(X_1) 1.16	0.191	0.916	(Γ_2) 4.19		(K_3) 1.70				
Ge	(L_6) 0.76	1.57	0.081	(Γ_7) 0.898	0.038					
GaP	(Δ_1) 2.35	0.91	0.25	(Γ_1) 2.895	0.093	(L_1) 2.64			(Δ) 0.355	(Δ) 0.433 (Δ_0^c)
GaAs	(L_6) 1.82	1.9	0.075	(Γ_6) 1.519	0.067	(X_6) 2.03	1.8	0.257	(X) 0.304	(X) 0.009 (ΔE)
GaSb	(L_6) 1.22	0.95	0.11	(Γ_6) 0.86	0.041	(X_6) 1.72	1.2	0.250	(X) 0.178	(X) 0.025 (ΔE)
InP	(L_6) 2.19			(Γ_6) 1.50	0.077	(X_6) 2.44				
InAs	(L_6) 1.53			(Γ_6) 0.37	0.024	(X_6) 2.28				
InSb	(L_6) 1.03	0.09		(Γ_6) 0.25	0.0136	(X_6) 1.71				
ZnS_{cub}				(Γ_6) 3.78	0.34					
ZnS_{hex}				(Γ_1) 3.91	0.28					
ZnSe	(L_6) 3.96			(Γ_6) 2.82	0.16	(X_6) 4.57				
ZnTe				(Γ_6) 2.394	0.122					
CdS				(Γ_6) 2.583	0.21					
CdSe				(Γ_7) 1.829	0.112					
CdTe	(L_6) 2.82			(Γ_6) 1.475	0.096	(X_6) 3.48				

* [After Karl W. Böer, "Survey of Semiconductor Physics," (Van Nostrand Reinhold, NY, 1990, p. 216).]

TABLE 4

Hole effective masses, valence band splitting,* and Luttinger valence band parameters.**

Crystal	γ_1	γ_2	γ_3	Δ_0 (T(K))	Δ_1	m_h (100)	m_h (110)	m_h (111)	m_l	m_{so}
Si	4.285	0.339	1.446	0.045 (10)		0.537			0.153	0.234
Ge	13.38	4.28	5.69	0.297 (10)		0.284	0.352	0.376	0.044	0.095
GaP	4.05	0.49	1.25	0.08 (100)		0.419		0.997	0.16	0.465
GaAs	6.95	2.25	2.86	0.34 (10)		0.51			0.082	0.154
GaSb	13.3	4.4	5.7	0.76		0.28			0.05	
InP	5.15	0.94	1.62			0.56		0.60	0.12	0.121
InAs	20.4	8.3	9.1	0.38 (1.5)		0.35		0.43	0.026	
InSb	3.25	–0.20	0.90	0.85 (100)		0.34	0.42	0.45	0.016	
ZnS_{cub}				0.67 (80)		1.76			0.169	
ZnS_{hex}				0.86 (77)	0.055	$1.4_{\parallel}$	$0.49_{\perp}$			
ZnSe	4.3	1.14	1.84	0.403 (80)						
ZnTe	3.9	0.83	1.30	0.97 (80)						
CdS				0.062 (77)	0.027	0.64			0.64	
CdSe				0.416 (77)	0.039	≥ 1	0.45		0.9	
CdTe	5.3	1.7	2.0	0.811 (80)		0.72	0.81	0.84	0.13	

* The two top valence bands are split at $k = 0$ by Δ_1 due to drystal field coupling; $\Delta_1 \neq 0$ in anisotropic crystals, such as hexagonal structures. Δ_0 is the spin-orbit splitting. Δ_0 and Δ_1 are given in eV.

** [After Karl W. Böer, "Survey of Semiconductor Physics," (Van Nostrand Reinhold, NY, 1990, p. 219).]

From this brief description of some "real" bandstructures, we see that this is a very complicated topic. Fortunately, we can restrict ourselves in most cases to the bandedges which create the minimum gap.

1.1.3. Vibrational Modes (Phonons)

(For details consult the texts by Kittel and/or Blakemore.)

Our experience with the macroscopic world shows that most solids display, within limits, elastic properties. Hooke's law tells us that the relative dimensional changes are fully reversible and proportional to the applied forces. A familiar and simple case is the reversible stretching Δl of a steel wire of length l by a force F such as exerted by a weight suspended from the wire. The strain $\varepsilon = \Delta l/l$ is proportional to the stress σ = F/cross sectional area

$$\varepsilon = \sigma \cdot \frac{1}{E} .$$

E is a materials constant called Young's modulus.

We can relate the macroscopic law to changes at the atomic level. Figure 8 shows the total energy per molecule of KCl in a KCl crystal. The attractive (Coulomb) and repulsive (mostly due to the Pauli exclusion principle) contributions add up to give a minimum energy at an interionic distance $r_0 = 3.15$ Å. The first derivative of total energy E_{tot} curve shown in Fig. 8 corresponds to the force between the ions:

$$F = -\left(\frac{\partial E_{tot}}{\partial r}\right)_{r_0} .$$

We clearly see that the linear approximation is only valid near r_o.

We are now prepared to explore dynamic properties of crystals, such as elastic vibrations. Quantized vibrations are called **phonons**. They play important roles for a wide range of properties, including electrical and optical properties. To begin, we will focus on a simple cubic lattice with one atom per primitive cell. Looking at a set of neighboring (100) planes (Fig. 9), we write down Hooke's law for plane s. We further assume that only the nearest neighbor planes, s–1 and s+1, exert a force on plane s when they are displaced from their equilibrium position:

$$F_s = C(u_{s+1} - u_s) + C(u_{s-1} - u_s) .$$

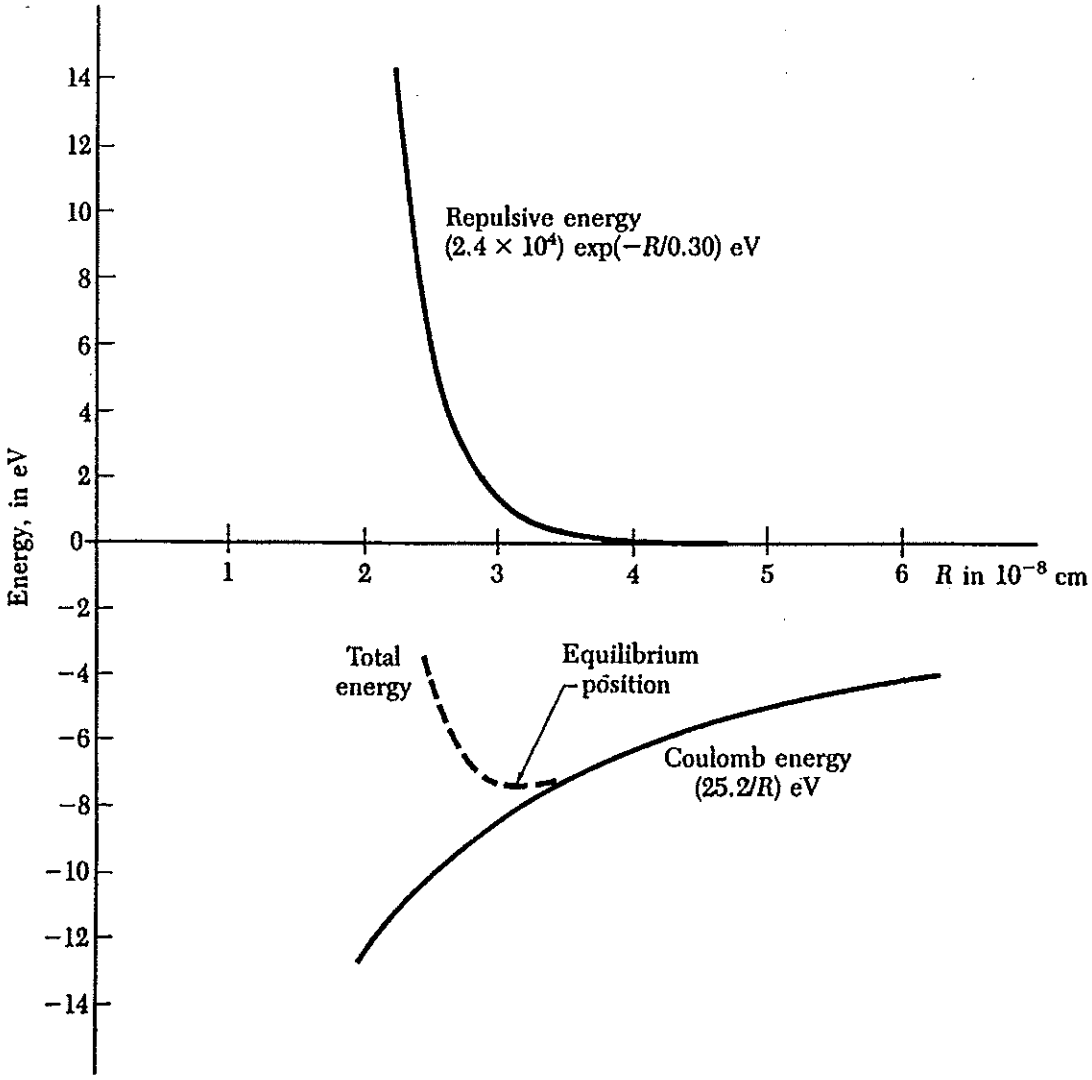


Fig. 8. Energy per molecule of KCl crystal, showing Madelung and repulsive contributions. (After Charles Kittel, "Introduction to Solid State Physics," 6th edition, John Wiley & Sons, Inc. New York, 1986, p. 70.)

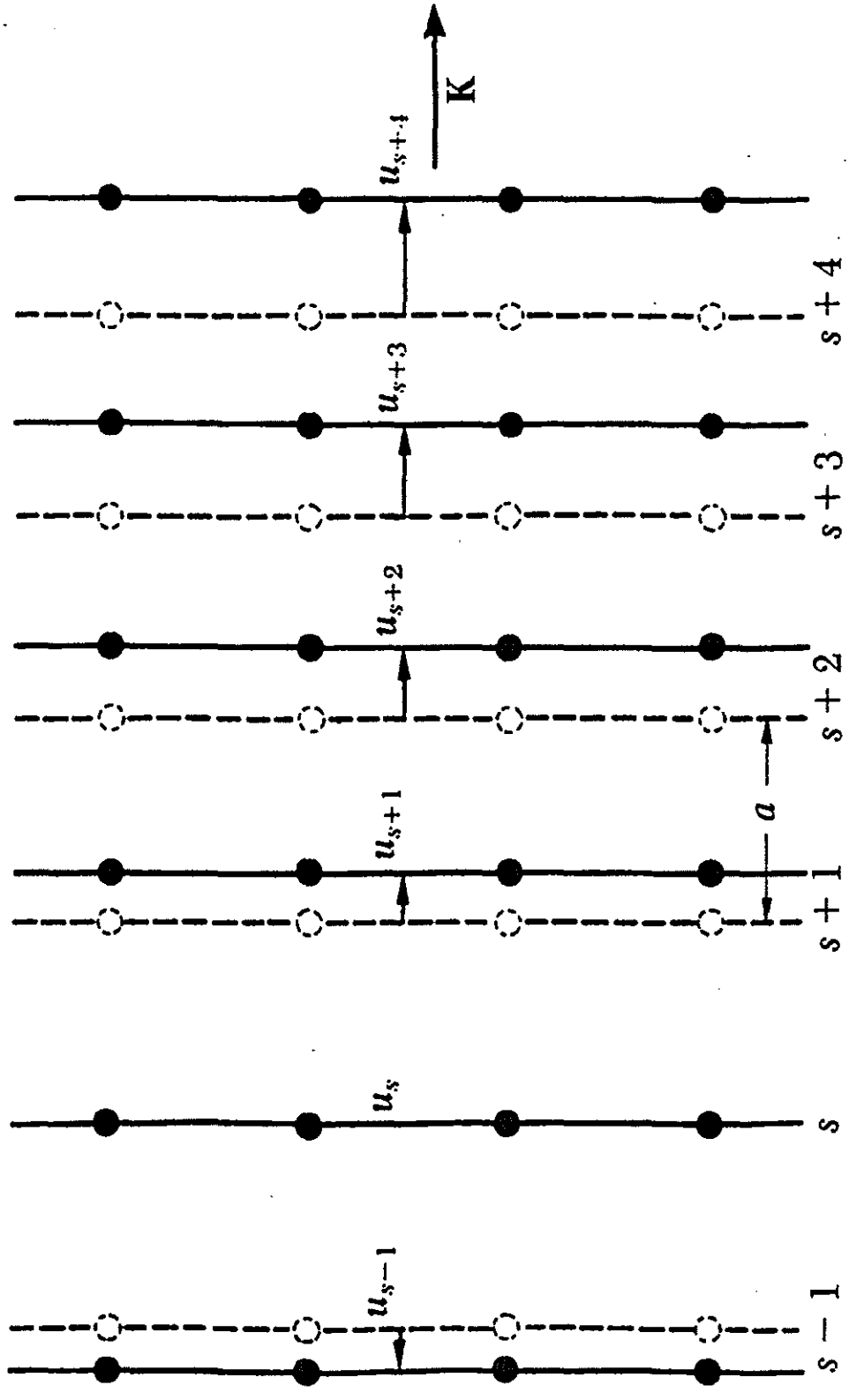


Fig. 9. (Dashed lines) Planes of atoms when in equilibrium. (Solid lines) Planes of atoms when displaced as for a longitudinal wave. The coordinate u measures the displacement of the planes. (After Charles Kittel, "Introduction to Solid State Physics," 6th edition, John Wiley & Sons, Inc. New York, 1986, p. 82.)

This force leads to an acceleration d^2u_s/dt^2 of the atoms with mass M:

$$F_s = M\frac{d^2u_s}{dt^2} = C(u_{s+1} + u_{s-1} - 2u_s) .$$

Whenever the second derivative of a variable is proportional to the variable itself, we find solutions of the following kind:

$$u_s \propto \exp(-i\omega t)$$

since

$$\frac{d^2u_s}{dt^2} = -\omega^2 u_s .$$

Inserting this in the above equation we find:

$$-M\omega^2 u_s = C(u_{s+1} + u_{s-1} - 2u_s) .$$

The solution is a traveling wave:

$$u_{s\pm1} = u \exp(i(s\pm1)ka)\exp(-i\omega t) . \quad \text{(three equations for } u_{s-1}, u_s \text{ and } u_{s+1})$$

Remember, s is the coordinate of the plane and a is the distance between planes. The wavevector k $= 2\pi/\lambda$ with λ = wavelength. For the special case we have chosen, namely (100) planes, a coincides with the lattice constant. For other planes we would have found different interplane distances. Inserting the expression for $u_{s\pm1}$ into the preceding equation and dividing both sides by exp(–iωt) we find:

$$-\omega^2 Mu \exp(iska) = uC\{\exp[i(s+1)ka] + \exp[i(s-1)ka] - 2\exp[iska]\} .$$

Division by – Mu exp[iska].... leads to:

$$\omega^2 = (2C/M)(1 - \cos ka) .$$

This is the dispersion relation between ω and k. The first derivative of this expression

$$d(\omega^2)/dk = (2Ca/M) \sin ka$$

becomes zero at the Brillouin zone boundary $k = \pm \pi/a$.

Using the trigonometric identity $1 - \cos\alpha = 2 \sin^2(\alpha/2)$, we can rewrite the expression for ω^2.

$$\omega^2 = (4C/M) \sin^2(\tfrac{1}{2} ka)$$
$$\omega = (4C/M)^{1/2} \left|\sin(\tfrac{1}{2} ka)\right|$$

Fig. 10 shows a plot of $\omega/(4C/M)^{1/2}$ versus k. The first Brillouin zone extends from $-\pi/a$ to $+\pi/a$.

It can be shown rigorously that only values for k in the first Brillouin zone have a physical meaning. We refer to the arguments of Kittel or Blakemore for a detailed proof. Fig. 11 illustrates the problem. As k increases, the wavelength becomes shorter. Once the wavelength falls below two lattice spacings, there are simply not enough lattice points left to define the wave. The critical wavelength is reached when neighboring atoms are moving against each other with exactly opposite phase ($k=\pi/a$). We clearly see the difference between waves propagating in a continuous medium and waves on a lattice with discrete mass points. In a continuous medium, $k \rightarrow \pm \infty$ because $a = 0$.

Another way of looking at wavevectors k larger (smaller) than π/a ($-\pi/a$) can be found by rewriting the displacement ratio:

$$u_{s+1}/u_s = \exp(ika) .$$

We generate a new wavevector k' which lies in the first Brillouin zone by subtracting a sufficient integer number n of $2\pi/a$:

$$k' \equiv k - 2\pi n/a .$$

Inserting k' into the displacement ratio equation we find:

$$u_{s+1}/u_s = \exp(ika) = \exp(i2\pi n) \exp[i(ka - 2\pi n)] = \exp(ik'a) .$$

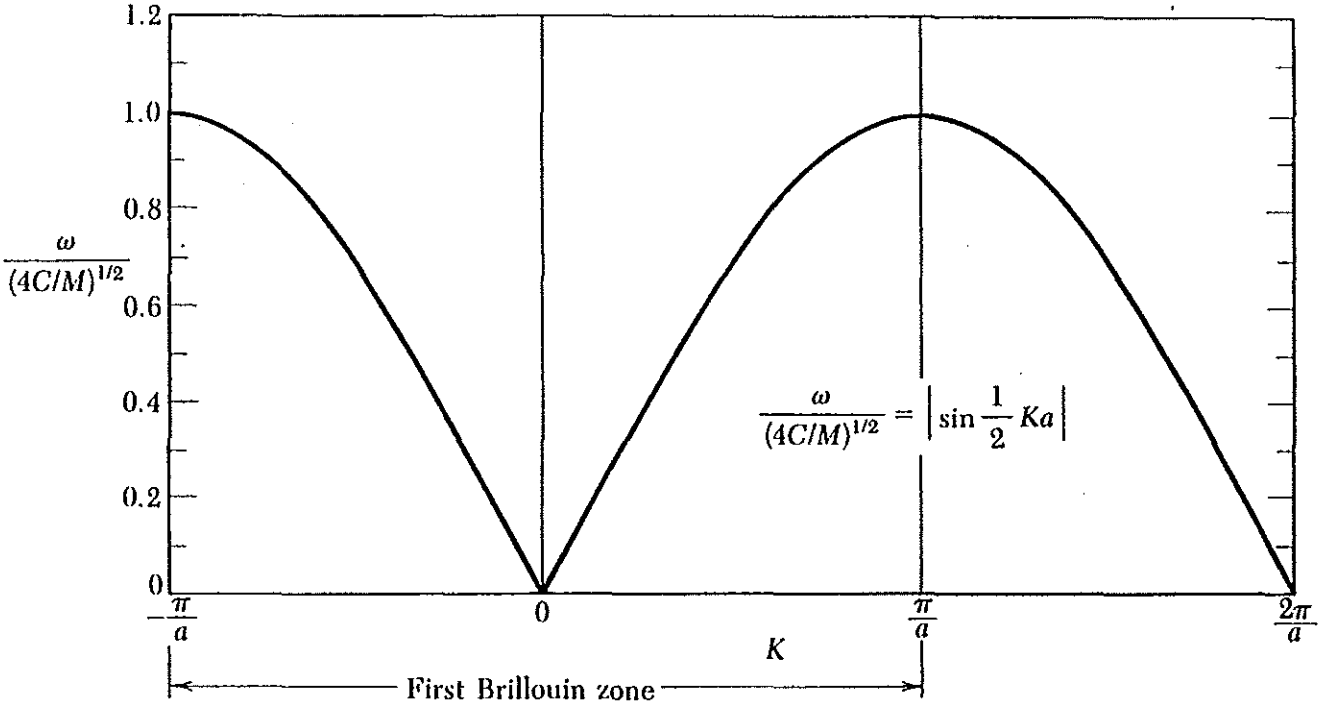


Fig. 10. Plot of ω versus K. The region of $K \ll 1/a$ or $\lambda \gg a$ corresponds to the continuum approximation; here ω is directly proportional to K. (After Charles Kittel, "Introduction to Solid State Physics," 6th edition, John Wiley & Sons, Inc. New York, 1986, p. 85.)

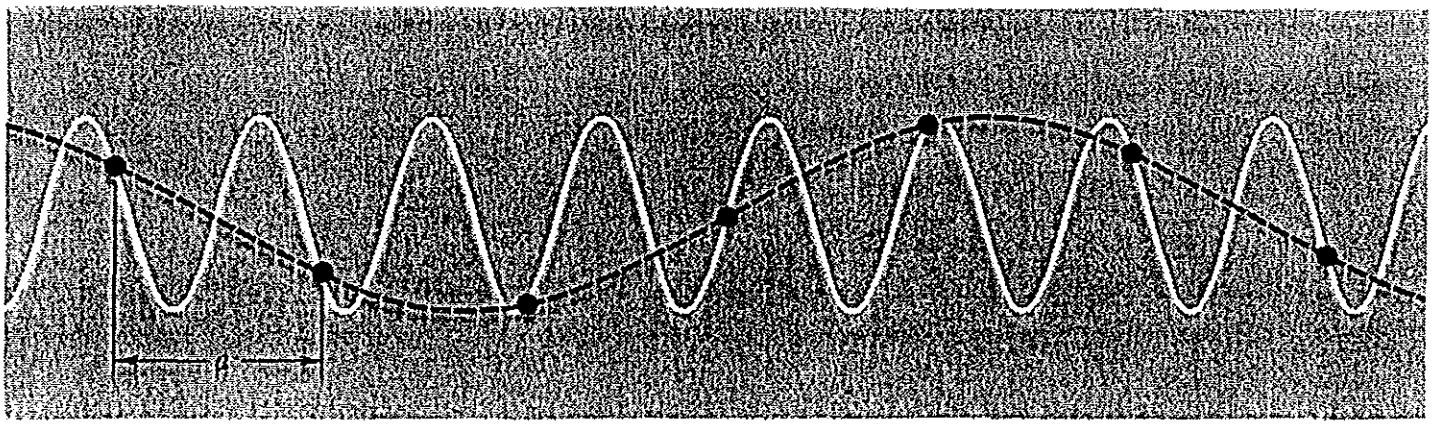


Fig. 11. The wave represented by the solid curve conveys no information not given by the dashed curve. Only wavelengths longer than $2a$ are needed to represent the motion. (After Charles Kittel, "Introduction to Solid State Physics," 6th edition, John Wiley & Sons, Inc. New York, 1986, p. 85.)

The difference between k' and k is merely a phase without any physical significance!

Let us now inspect what happens for the two limits of k, $k \rightarrow 0$ and $k \rightarrow \pi/a$. For $ka \ll 1$ we can rewrite the dispersion relation using $\cos ka \cong 1 - \frac{1}{2}k^2a^2$:

$$\omega^2 = (C/M)\, k^2a^2 \quad ,$$

i.e., the frequency is proportional to the wavevector which means the velocity is constant ($v = d\omega/dk$)

$$v = a(C/M)^{1/2} \quad .$$

For $k = \pi/a$, $d\omega/dk = 0$, i.e., we observe a standing wave with alternating planes of atoms moving towards each other or away from each other.

So far we have discussed waves with the displacements of the individual planes pointing along the propagation direction. Such waves are called longitudinal. We could have derived the same equations with different force constants for waves with displacements perpendicular to the propagation direction, i.e., transversal waves. We want to expand our discussion of lattice vibrations to the case where we find two atoms in the primitive basis. Let us assume that the two types of atoms have different masses M_1 and M_2. Atoms of mass M_1 shall be described by the displacement coordinates u, those with M_2 with v. We obtain two equations for motion:

$$M_1\, d^2u_s/dt^2 = C(v_s + v_{s-1} - 2u_s)$$
$$M_2\, d^2v_s/dt^2 = C(u_{s+1} + u_s - 2v_s)$$

As in the monatomic case discussed above we choose traveling wave solutions:

$$u_s = u \exp(iska) \exp(-i\omega t)$$
$$v_s = v \exp(iska) \exp(-i\omega t)$$

After inserting these solutions into the equations of motion we find:

$$-\omega^2 M_1 u = Cv[1 + \exp(-ika)] - 2Cu$$
$$-\omega^2 M_2 v = Cu[\exp(ika) + 1] - 2Cv$$

This is a system of **homogeneous linear equations.** Solutions exist only if the determinant of the coefficients of u and v is zero:

$$\begin{vmatrix} 2C - M_1\omega^2 & -C[1+\exp(-ika)] \\ -C[1+\exp(ika)] & 2C - M_2\omega^2 \end{vmatrix} = 0$$

$$M_1 M_2 \omega^4 - 2C(M_1 + M_2)\omega^2 + 2C^2(1 - \cos ka) = 0$$

Let us examine the solutions of this equation for $ka \ll 1$ (near the center of the Brillouin zone) and for $ka = \pm\pi$.

For small ka we use the approximation:

$$\cos ka = 1 - \tfrac{1}{2}k^2a^2 + \ldots$$

and we find:

$$\omega_1^2 \cong 2C\left(\frac{1}{M_1} + \frac{1}{M_2}\right)$$

and

$$\omega_2^2 \cong \frac{C}{2(M_1 + M_2)} k^2 a^2 \quad .$$

ω_1 is a solution for the so-called optical branch while ω_2 belongs to the acoustical branch. The significance of these names becomes clear shortly!

For $k = \pi/a$ we obtain:

$$\omega_1^2 = 2C/M_2 \quad \text{and} \quad \omega_2^2 = 2C/M_1$$

The complete dependence of ω on k is shown in Fig. 12 for $M_1 > M_2$.

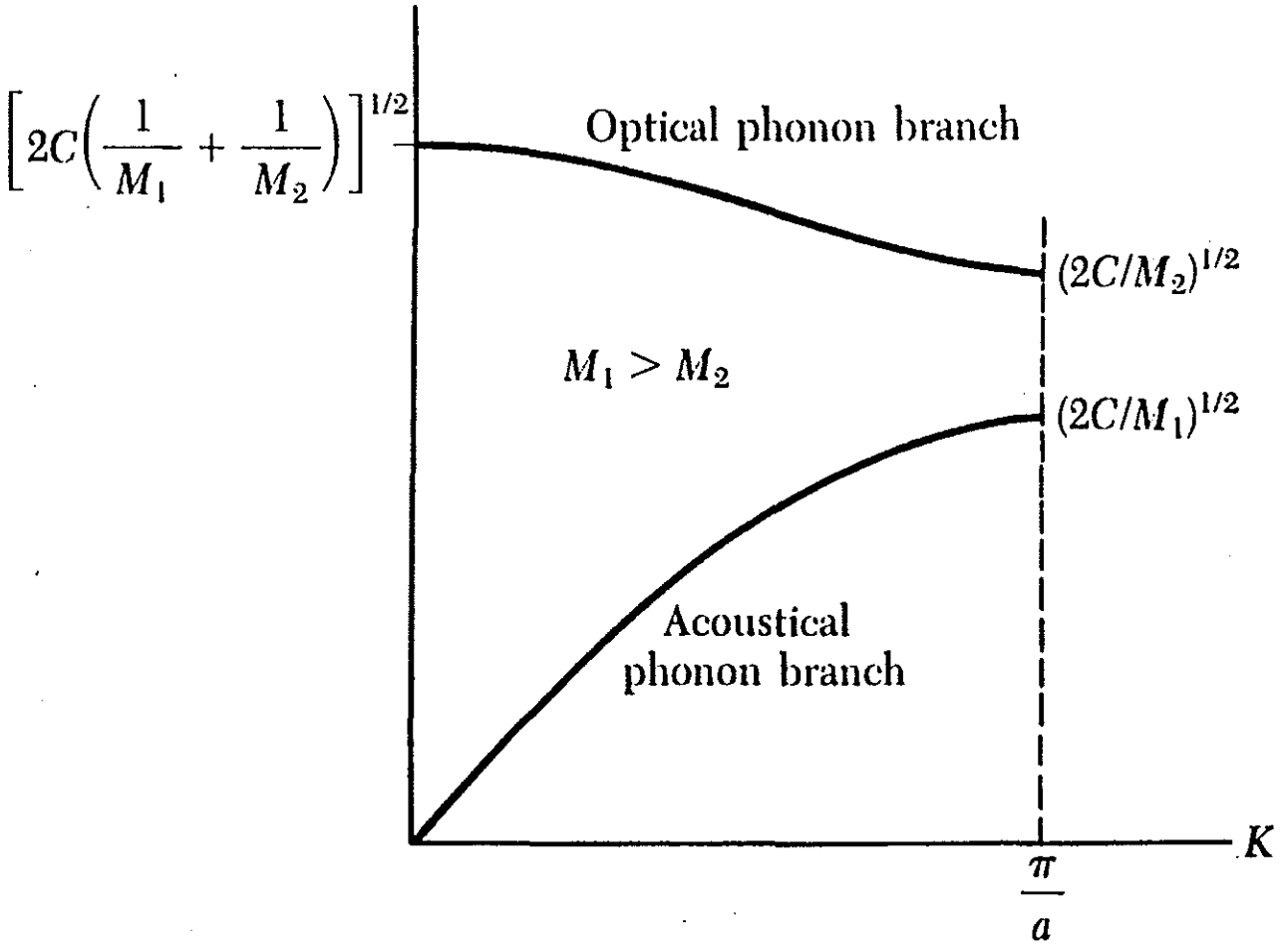


Fig. 12. Optical and acoustical branches of the dispersion relation for a diatomic linear lattice, showing the limiting frequncies at $K = 0$ and $K = K_{max} = \pi/a$. The lattice constant is a. (After Charles Kittel, "Introduction to Solid State Physics," 6th edition, John Wiley & Sons, Inc. New York, 1986, p. 89.)

The names optical and acoustical branch come from the study of ionic crystals (e.g., KCl, NaCl, etc.). In Fig. 13, it is shown how the oppositely charged ions can vibrate in two different ways, both generating a wave with the same wavelength. In the optical mode the plus and minus ions move in opposite directions, creating strong electric dipoles. These rapidly changing dipole fields can emit electromagnetic radiation (photons) or vice versa--electromagnetic radiation can excite optical mode vibrations. The acoustical mode does not generate dipole fields of large magnitude and is not excited by photons.

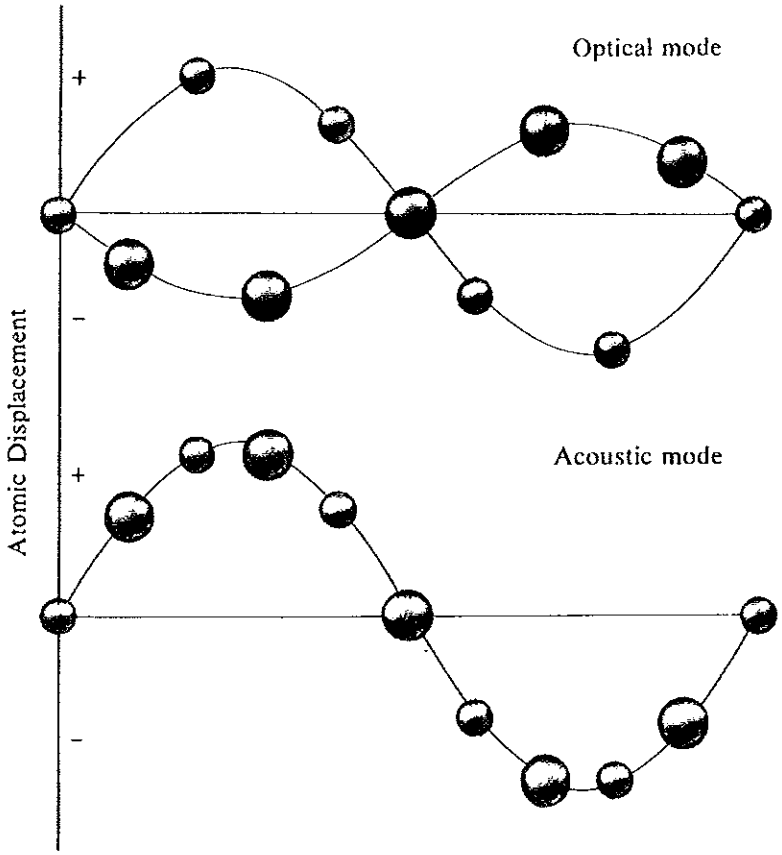


Fig. 13. The sequence of atomic displacements involved in the transmission of a transverse wave along a diatomic chain. The difference between the acoustic and optical modes can bee seen more clearly when transverse displacements are sketched rather than a longitudinal wave. (After J.S. Blakemore, "Solid State Physics," 2nd edition, Cambridge Univ. Press, Cambridge, 1985, p. 109.)

At the Brillouin zone boundary we find an energy gap for which no values for ω exist for the full range of k values between $-\pi/a$ and $+\pi/a$. If one tries to force a crystal to vibrate with such a "forbidden" frequency one generates a strongly damped wave which does not propagate.

Now let us take into account that there are both longitudinal and transversal modes for both optical and acoustical phonons. We find four types of vibrational modes: LO, LA, TO, TA. (Rule: for p-atoms in the primitive cell there are 3p branches to the dispersion relation: three acoustical and 3p – 3 optical. E.g., KBr: 2 atoms: one LA, one LO, two TA and two TO.) In Fig. 14, the phonon dispersion in the [111] direction is shown for Ge. In this case we do not have two different masses but two atoms in the primitive cell which lead to both acoustical and optical phonon branches.

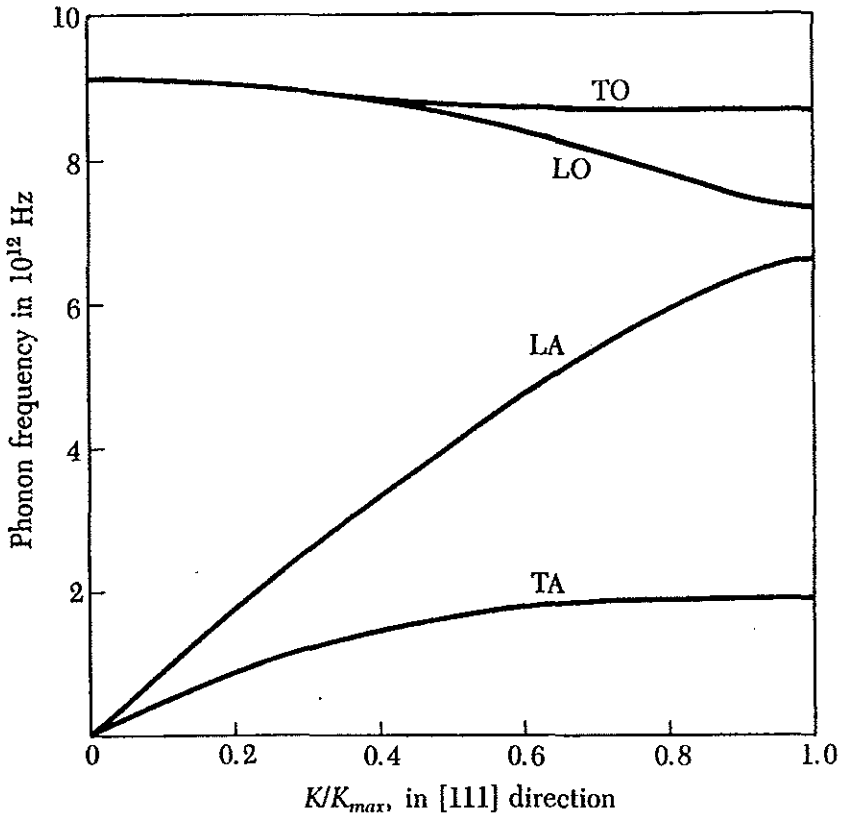


Fig. 14. Phonon dispersion relations in the [111] direction in germanium at 80 K. The two degenerate TA phonon branches are horizontal at the zone boundary position, $K_{max}=(2\pi/a)(\frac{1}{2}\ \frac{1}{2}\ \frac{1}{2})$. The LO and TO branches coincide at K = 0; this also is a consequence of the crystal symmetry of Ge. The results were obtained with neutron inelastic scattering by G. Nilsson and G. Nelin, Phys. Rev. B **3**, 364 (1971).

Quantization of Lattice Vibrations

Lattice vibrations are quantized into phonons. A thermal vibration is a number of thermally excited phonons. The energy of an elastic mode of frequency ω is:

$$E = (n + \tfrac{1}{2})\,\hbar\omega \quad .$$

$\frac{1}{2}\ \hbar\omega$ is the zero point energy, a quantum mechanical phenomenon. Phonons carry **crystal momentum, but no real mechanical momentum**. Think about a vibrating H_2 molecule: Only the relative coordinates are involved in the vibration. The center of gravity may be at rest p = mv = 0! The crystal momentum is $\hbar k$. It is effective in scattering processes with photons and particles (e.g., neutrons).

Properties of Solids Related to Phonons

- Reststrahl bands: ionic solids show strong reflection for photons with frequencies between ω_{TO} and ω_{LO}.
- Covalent solids have phonon modes but these cannot be excited by photons (no dipole moments!). Combination modes ("two phonon overtone bands") can be excited if $\Delta K = 0$. (Anharmonicity in the elastic potential is the source for the creation of combination modes!)
- Local(ized) phonon modes exist for low mass impurities (e.g., H in diamond, Si in GaAs, O in Si, etc.). Such modes are localized in space at the impurity and have frequencies well above the phonon modes of the host lattice. Localized vibrational modes (LVMs) are often used for impurity characterization.
- The specific heat of an insulating solid is due to phonons.
- Velocity of sound.
- Free carrier mobility (phonons modulate the band edges via the "deformation constant.")
- Thermal conductivity (low energy phonons near $\vec{k} = 0$)

1.2. Extrinsic Semiconductors

1.2.1. Shallow Level Impurities

The replacement of atoms in a semiconductor crystal with impurities (so-called substitutional impurities), interstitial impurities, vacancies, interstitial host atoms, complexes of some or all of these and dislocations, can lead to electron states inside the forbidden energy gap. The existence of such energy states, also called energy levels, is the major reason why semiconductors have gained enormous scientific and technical importance. Among the hundreds of known energy levels (Fig. 15) there are only a few which are used to fabricate electronic devices.

Energy levels which lie close to the conduction band minimum or the valence band maximum are called shallow. In the group IV semiconductors Si and Ge, the elements of the third group (B, Al, Ga, In, Th) require an extra electron to form a complete Lewis octet when they substitute a host atom. We call these impurities acceptors. Elements of the fifth group (P, As, Sb, Bi) have one valence electron more than required to form a Lewis octet. This extra electron is given up easily to the nearest conduction band minimum, and we call these impurities donors.

Shallow donors and acceptors are called "hydrogenic" or hydrogen-like because they can be described with a very simple and elegant model. The shallow impurity is compared to a hydrogen atom embedded in a homogeneous medium with a relative dielectric constant ε and an electron or a hole mass m* which corresponds to the effective mass of the particular band extremum. In order to estimate the binding energy of the electron to the shallow donor core or the hole to the shallow acceptor core we write down the equation for the binding energy (E_H) for the free hydrogen atom modified by ε and m*:

$$E = E_H \frac{m^*}{m_e} \frac{1}{\varepsilon^2} = \frac{e^4 m_e}{2(4\pi \varepsilon_0 h)^2} \frac{m^*}{m_e} \frac{1}{\varepsilon^2} = 13.6\,\text{eV} \frac{m^*}{m_e} \frac{1}{\varepsilon^2}$$

For Ge (ε=16, $m^* \cong 0.2\ m_e$) we find a binding energy for electrons and holes of the order of 10 meV, over 1000 times smaller than for hydrogen. The large differences in the electron and hole masses in Si and GaAs lead to different values for the electron and hole binding energies:

$$\text{Si}(\varepsilon=11.7):\ E_{donor} \cong 33\ \text{meV},\ E_{acceptor} \cong 75\ \text{meV}$$
$$\text{GaAs}(\varepsilon=13.2):\ E_{donor} \cong 6\ \text{meV},\ E_{acceptor} \cong 28\ \text{meV}$$

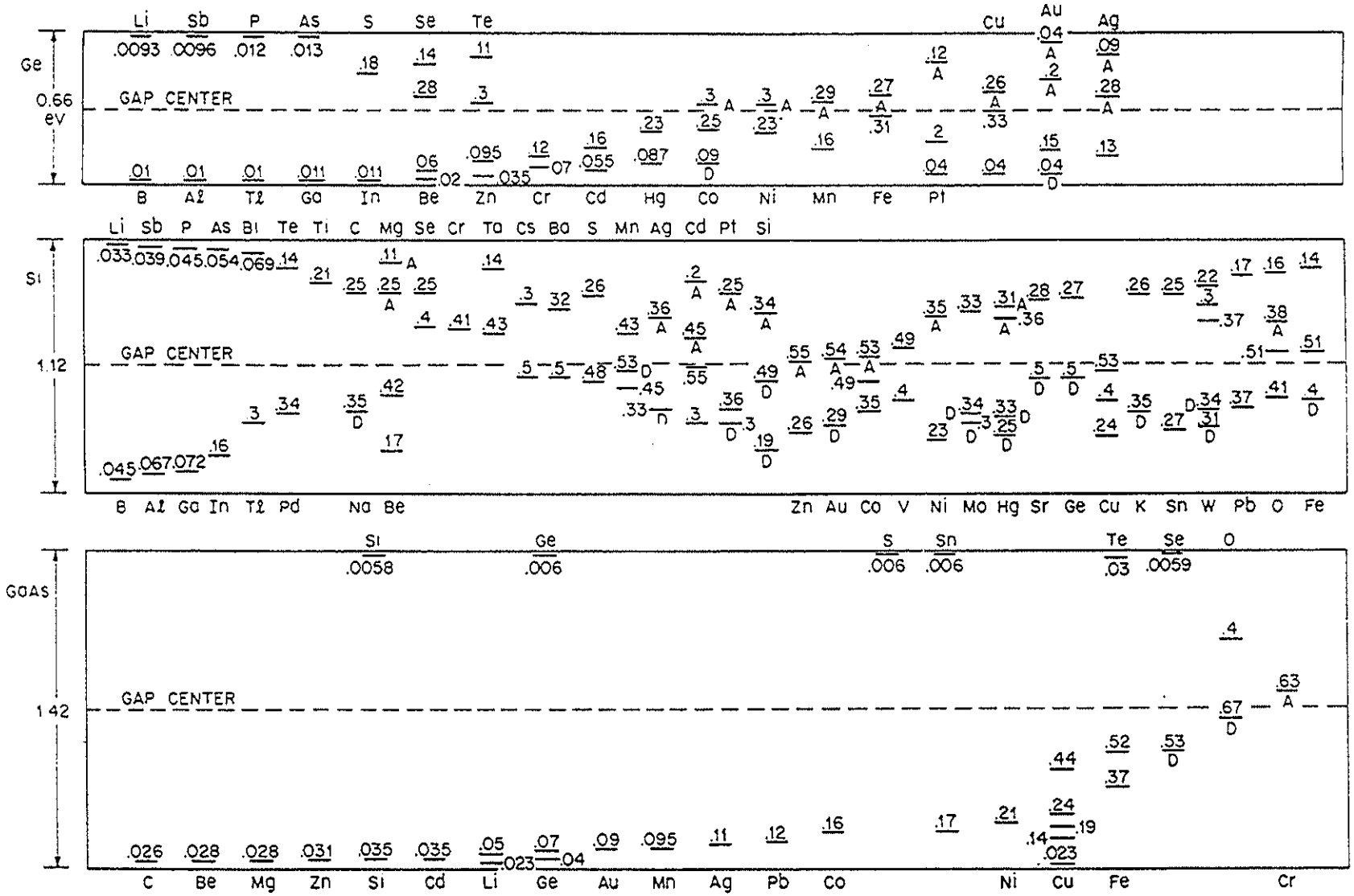


Fig. 15. Measured ionization energies for various impurities in Ge, Si, and GaAs. The levels below the gap centers are measured from the top of the valence band and are acceptor levels unless indicated by D for donor level. The levels above the gap centers are measured from the bottom of the conduction-band level and are donor levels unless indicated by A for acceptor level. The bandgaps at 300 K are 0.66, 1.12, and 1.42 eV for Ge, Si, and GaAs, respectively. (After S.M. Sze, "Physics of Semiconductor Devices," 2nd edition, John Wiley & Sons, Inc., NY, 1981, p. 21.)

The fact that the Effective Mass Theory (EMT) works as well as it does is strongly related to the very large Bohr radii for electrons and holes bound to shallow level impurities. We again use the equation for hydrogen and modify it with ε and m*:

$$r_{Bohr} = \frac{4\pi\ \varepsilon_0\ h^2}{m_e\ e^2} \cdot \frac{m_e}{m^*} \cdot \varepsilon \quad = 0.5\,\text{Å} \cdot \frac{m_e}{m^*} \cdot \varepsilon$$

For the case of donors in GaAs we find a Bohr radius of ≅86 Å. There are over 10^5 lattice points within the sphere with its radius = r_{Bohr}! This very large Bohr radius justifies the use of the simple hydrogenic model. The electrons (holes) of shallow local impurities are strongly **delocalized**, i.e., their wavefunctions are spread over a very large volume compared to the space taken up by each host. This averaging over so many lattice points allows us to use a continuous medium with an average dielectric constant ε. The result in turn justifies the use of such a simple model.

A further consequence of the large Bohr radii of the ground and bound excited states (or: the strong delocalization) is the small extent of the impurity wavefunctions in k-space. This inverse relationship of real space and k-space originates in the Heisenberg uncertainty relationship: $\Delta x\ \Delta k \geq \frac{1}{2}$.

In addition to the ground state, one finds bound excited states just as in the case of the hydrogen atom. The energy spacings between the various states do not follow precisely the simple $1/n^2$ law (n=principle quantum number) valid for hydrogen, because hydrogenic impurities are embedded in a crystal lattice.

We have seen earlier that the effective masses are proportional to the inverse of the curvature of the various band minima. The conduction band minima for Si and Ge are found along the [100] and the [111] directions of k-space respectively. The orientation dependence of the electron mass leads to a partial lifting of the degeneracy of the 2p state into a $2p_0$ state for the orbitals along the k-directions, along which the conduction band minima occur; and a $2p_\pm$ for the two orbitals perpendicular to these axes. Similar degeneracy lifting is found for higher-lying bound excited states (3p, 4p, 5p, etc.).

Fig. 16 shows the linear absorption coefficient of phosphorus-doped Si. Photons with energies corresponding to the energy differences between the ground state (ls) and a bound excited state (np) will be strongly absorbed because these photons cause transitions of electrons (holes)

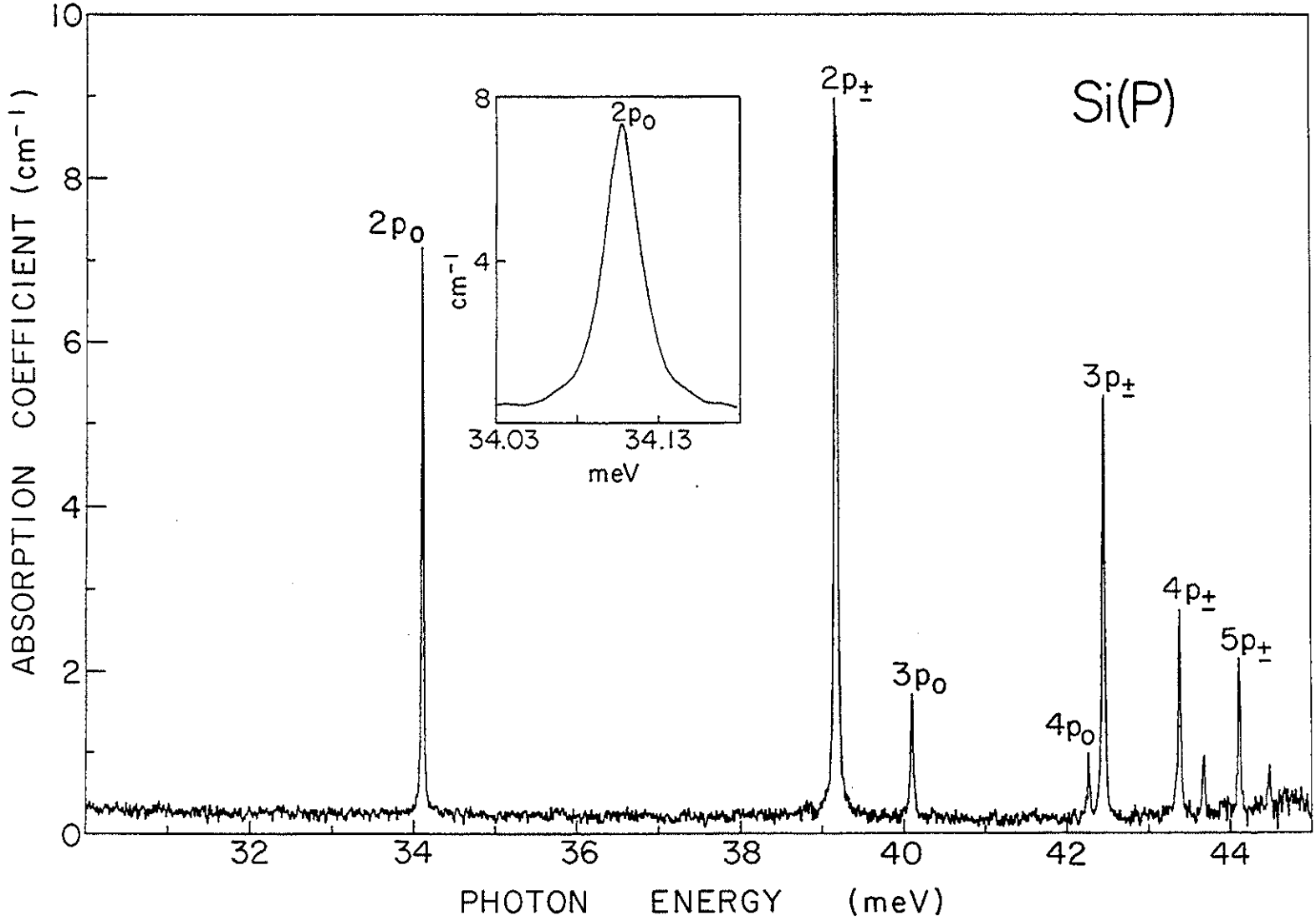


Fig. 16. Linear absorption coefficient for a phosphorus-doped Si single crystal. The sharp peaks correspond to electron transitions from the 1s ground state to p-like bound excited states. [After C. Jagamuth, Z.W. Grawolski, A.K. Ramdas, Phys. Rev. B 23, 2082 (1981).]

between such states. In lightly doped crystals for which the average inter-impurity distance is much larger than the Bohr orbit diameter, we observe a series of very sharp absorption lines corresponding to transitions between well defined energy levels. For more heavily doped crystals the absorption lines broaden because of orbit overlap.

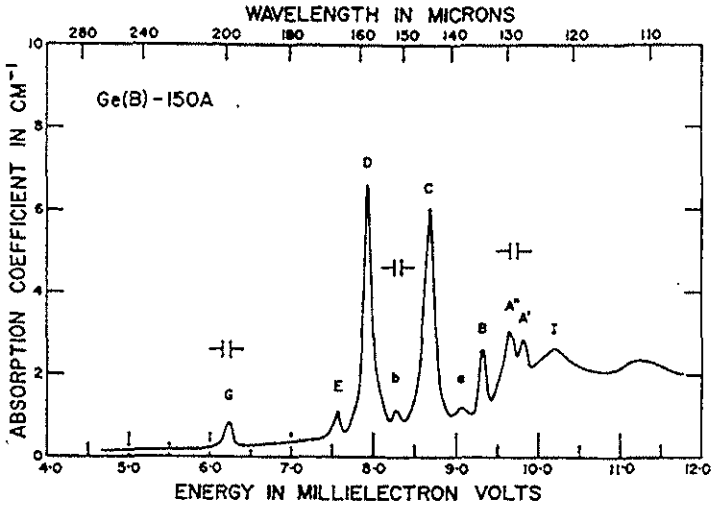
Fig. 17 shows absorption spectra of the first four group III acceptors in Ge. The strong similarities between these few spectra are quite apparent. This is not too surprising when we recall the hydrogenic model. The model only required a charge at the impurity lattice location which generated the Coulomb field binding the electron or hole. No mention was made of the particular impurity species. For bound excited states with orbitals which vanish at the impurity core (e.g., 2p), we expect no dependence of the energy of the state on the impurity species. An electron (hole) in the 1s ground state, on the other hand, experiences the potential in the immediate vicinity of the dopant impurity. This core potential affects the ground state energy of the impurity species. The resulting spectra have series of lines with identical spacings between the lines but with the individual lines at different photon energies.

Infrared absorption spectroscopy has evolved into one of the finest semiconductor characterization tools used in research, development and device fabrication laboratories. The large redundance in the spectrum of a particular impurity given by the up to 10 or more lines at very precise photon energies, makes for the unambiguous identification of donors and acceptors in semiconductors.

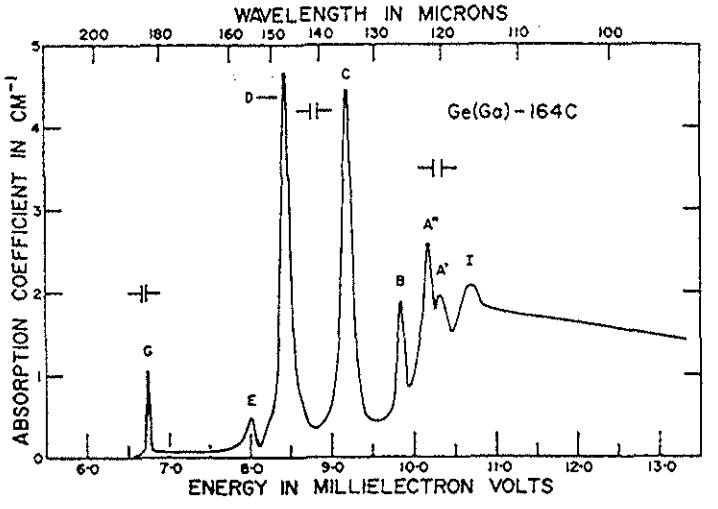
In order to illustrate the extreme sensitivity that can be achieved with infrared spectroscopy, the residual acceptor line spectra of B, Al and Ga in ultra-pure Ge are shown in Fig. 18. The major lines belong to the Al acceptors which are present at a concentration of $<10^{10}$ cm^{-3}. The sample volume was approximately 0.1 cm^3. The extreme sensitivity was obtained with a photo-conductivity technique described in detail in the reference given in the figure caption.

1.2.2. Multivalent Impurities

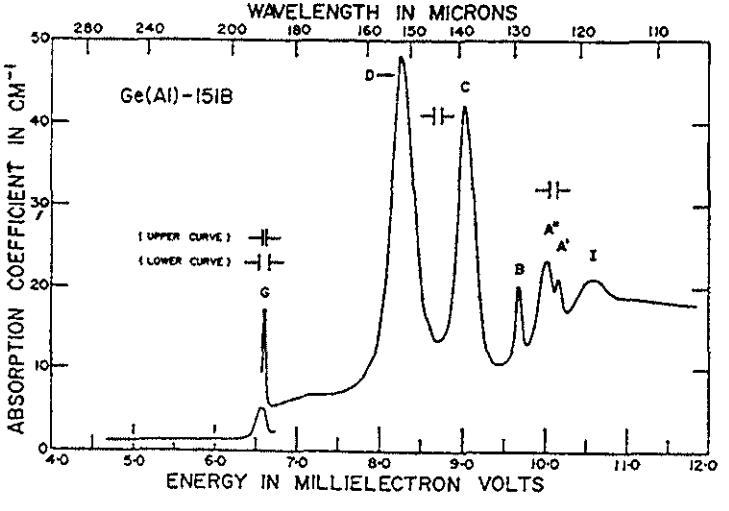
A natural extension of hydrogenic impurities are "helium-like" centers. Substitution of a Si atom in a Si crystal with a S or Se atom leads to a so-called double donor. There are two electrons in excess of what is required to obtain a Lewis octet. Will both electrons be bound with the same energy? The answer is No, since removing the first electron requires sufficient energy to overcome the binding energy in the Coulomb field of one positive charge. The electron in the excited state experiences the Coulomb field of only one charge because the second electron left in the ground state screens the two impurity core charges, reducing it to one effective charge.



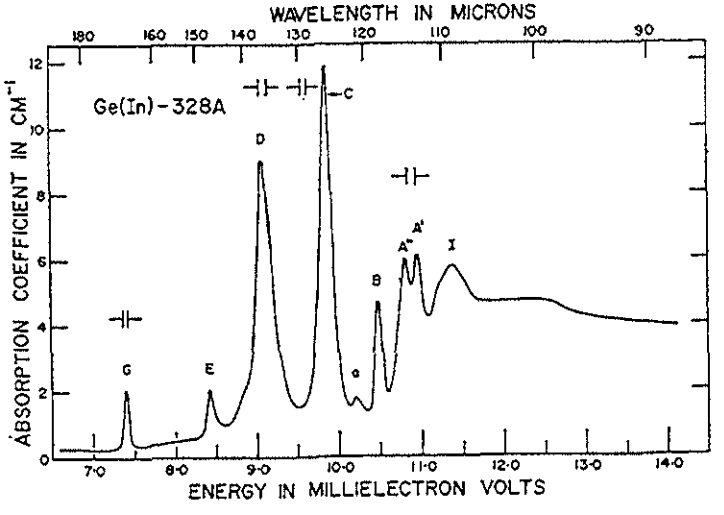
The excitation spectrum of boron impurity in germanium. Temp. $\simeq 10°K$; $N_A \simeq 2\times10^{14}$ cm^{-3}.



The excitation spectrum of gallium impurity in germanium. Temp. $\simeq 9°K$; $N_A \simeq 2\times10^{14}$ cm^{-3}.



The excitation spectrum of aluminum impurity in germanium. Temp. $\simeq 9°K$; $N_A \simeq 10^{15}$ cm^{-3}.



The excitation spectrum of indium impurity in germanium. Temp. $\simeq 8°K$; $N_A \simeq 4\times10^{14}$ cm^{-3}.

Fig. 17. After R. L. Jones and P. Fisher, *J. Phys. Chem. Solids* **26**, 1125 (1965).

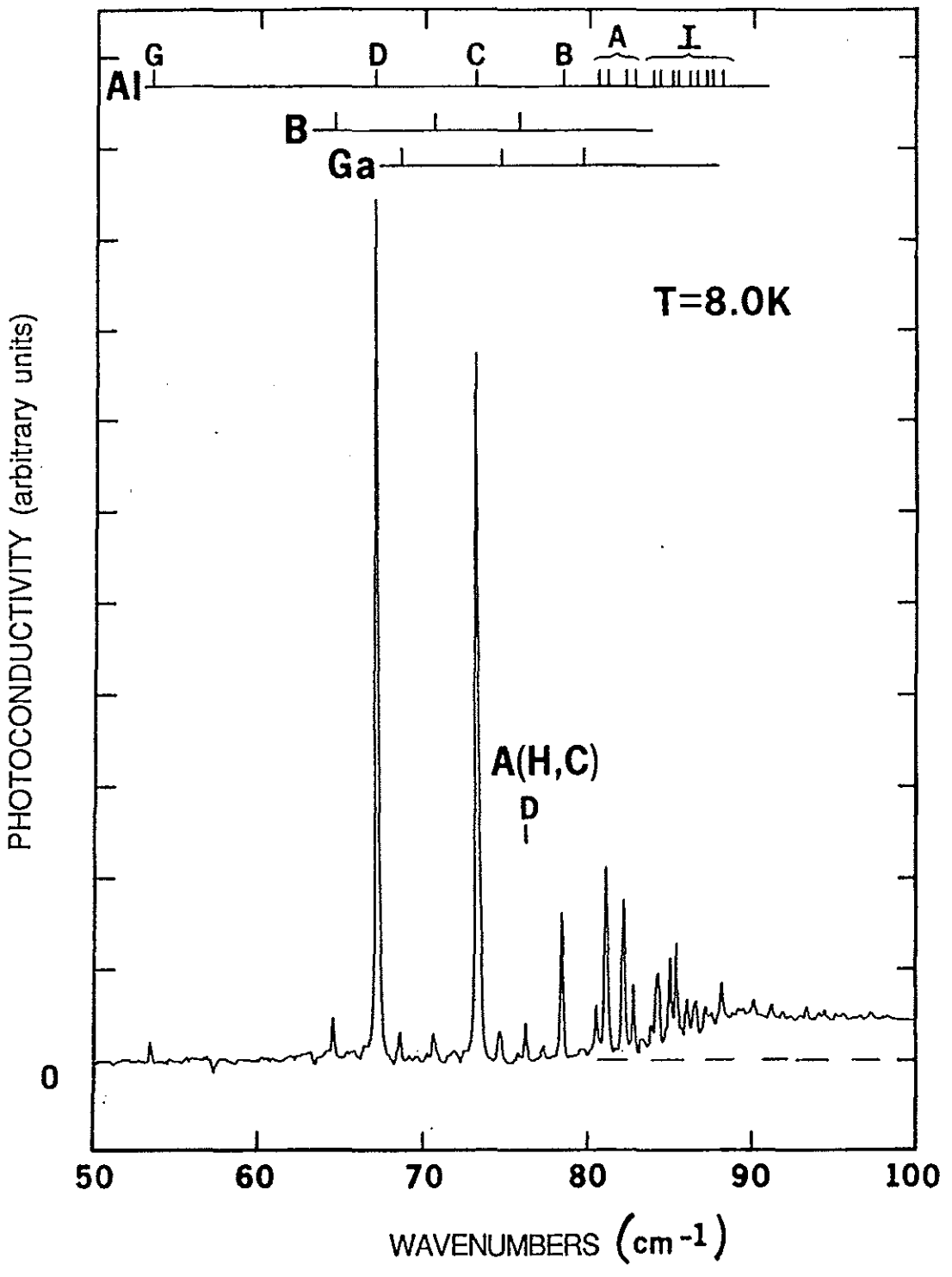


Fig. 18. Bound excited states spectrum of Al, B and Ga acceptors in ultra-pure Ge. The ground state lies at E = 0 (off to the left of the figure) and each sharp peak corresponds to a bound excited state. Transitions from the ground state into the valence band give rise to the increase seen just below 90cm^{-1} (1 meV = 8.065 cm^{-1}). [After E. E. Haller *et al.*, Adv. in Physics **30**, 93 (1981).]

Removing the second electron requires more energy because the Coulomb field is now twice as strong with two positive charges. We indicate the two ionization energies with two energy levels in the bandgap, one underneath the other (Fig. 15). In addition, one often indicates the charge state of the impurity. With the Fermi level above the upper energy level, the double donor is neutral with both electrons bound. When the Fermi level is pulled down (e.g., by adding acceptors), the first electron will be removed and the center becomes singly positively charged. A further drop in the Fermi level leads to the full ionization of the double donor, i.e., it turns into a doubly ionized center.

Figure 19 shows the infrared absorption spectra of the double acceptors Be, Zn and the triple acceptor Cu in Ge. These spectra are obtained when one hole is excited by photons from the ground state to one of the excited states. The other hole(s) remains in the ground state, closely bound to the impurity core. This situation is quite similar to a hydrogenic impurity, i.e., an electron (hole) on a large Bohr orbit is bound in the Coulomb field generated by one net charge at the impurity core. The strong similarity between neutral helium and neutral hydrogen-like spectra leads to the very similar energy spacings between bound excited states. For comparison, the spectrum of the single acceptor Al is shown. The energy axes of the four spectra are shifted such that equivalent lines coincide.

In Fig. 20 the infrared absorption spectrum of the neutral double donor Se in Ge is shown. The doping of the Ge crystal was achieved by the formation of the Se through the double beta decay of the isotope ^{77}Ge. Chapter 5.3. Neutron Transmutation Doping gives details of this unique semiconductor doping process. Just as in the previous figure showing the absorption spectra of four acceptors in Ge, the spectrum of neutral Se shows a series of sharp absorption lines which occur at photon energies which correspond to 1s -> np transitions. The spacings between the lines are the same as the spacings found in shallow donor spectra. As in the case of the multivalent acceptors, this is due to the fact that the electron which is lifted from the ground state to a bound excited orbit experiences the Coulomb field of only one positive charge at the impurity core. The second charge is perfectly screened by the electron which remains in the 1s ground state.

1.2.3. Deep Level Impurities and Defects

As their name indicates, deep impurities create energy levels far from the bandgap edges. Such impurities do not generate a hydrogenic ground state because the electron (hole) is bound strongly at a short distance from the impurity core. The averaging of the ground state wave-function over a large volume does not occur. For these reasons deep level impurity ground state

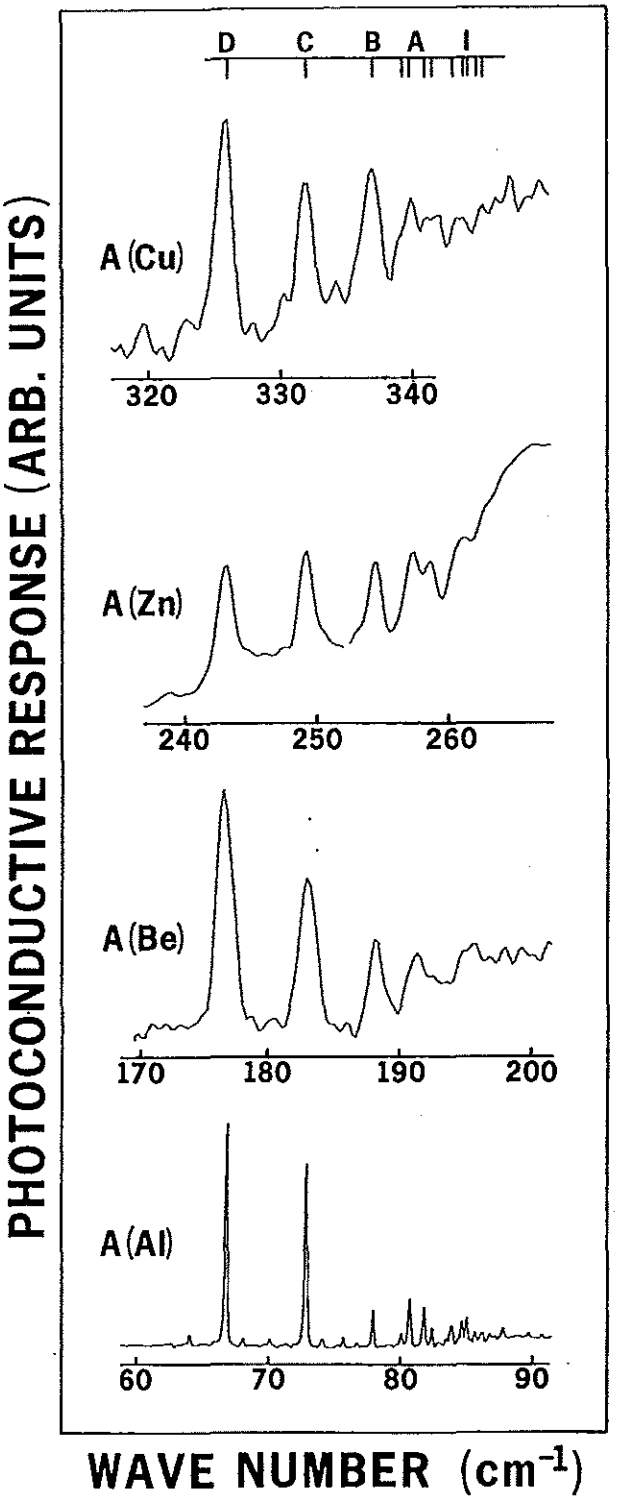


Fig. 19. Excited state spectra of Al, Be, Zn and Cu (all acceptors) in pure Ge. The scale of the x-axes is identical with the axes' origins shifted so that the corresponding lines line up. This shows that the excited state wavefunctions are not influenced by the specific impurity. [After E. E. Haller *et al.*, Adv. in Physics **30**, 93 (1981).]

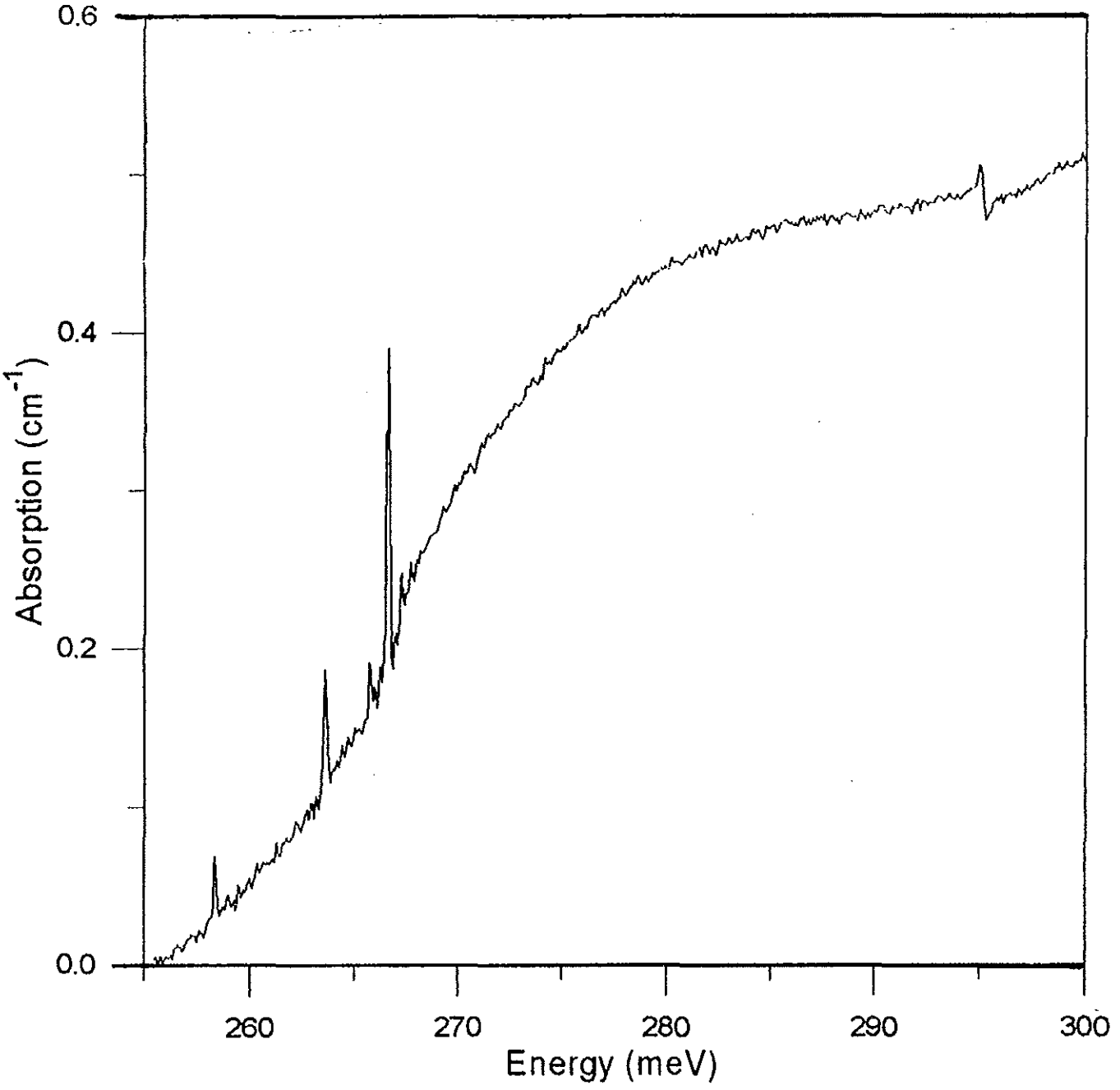


Fig. 20. Infrared absorption spectrum of double donor Se doped Ge. The sharp absorption peaks in the 255 to 270 meV range correspond to 1s -> np transitions of one of the two electrons bound to the Se double donor. The Se donors were formed by a nuclear double beta decay of ^{77}Ge which in turn was created through the capture of thermal neutrons by the stable isotope ^{76}Ge. [see chapter 5.3. Neutron Transmutation Doping; after C. S. Olsen *et al.*, Solid State Commun.**108**, 895-8 (1998).]

energies strongly depend on the particular details of the potential near the impurity core, and there exists no simple model which accurately describes deep levels.

Impurities which generate deep levels in the bandgap typically do not fit well into the crystal. Fe and Au are much-studied deep level impurities in Si. These two impurities are diffusing rapidly into Si crystals, and they dominate certain crystal properties even at very low concentrations. The minority carrier lifetime was controlled in the early days of Si technology by the addition of Au. The first fast switching diodes were fabricated from such crystals.

An important deep level defect in GaAs is the arsenic antisite As_{Ga}, an arsenic atom on a gallium lattice site. This deep double donor controls the position of the Fermi level of so-called semi-insulating GaAs used for integrated circuits. The As_{Ga} center has a number of unusual properties which are related, in part, to the change of the relative positions of As_{Ga} and its four As neighbors with a change in charge state. This so-called lattice relaxation is now found to occur for an increasing number of deep level centers in various semiconductors. In extreme cases the lattice relaxation can be strong enough to allow a center to bind two electrons and assume a lower total energy (Coulomb energy and lattice distortion energy) than binding one electron, despite the fact that two equally charged carriers repel each other.

We will return to some of these interesting deep level impurities when we discuss semiconductor crystal growth, devices, and characterization techniques.

1.2.4. Compensation

In a "real" semiconductor we find several donor and acceptor species, each at its specific concentration. Careful purification followed by doping with an ultra-pure dopant leads to crystals which have perhaps one or two types of impurities present at significant concentrations. Let us assume we have a Si crystal with P as the majority donor dopant and some residual B as a minority acceptor dopant (it is indeed very difficult to remove B from Si, as will be discussed later).

The electron energy levels of the P donors lie ~45 meV beneath the conduction band minimum. The B acceptor levels are located ~45 meV above the valence band top. According to the **minimum energy principle** the donor electrons will fill all available energy states from the lowest energy upwards. Because the valence band is full, no donor electrons can be accommodated by valence band states. B acceptors, however, require additional electrons to form Lewis octets. Instead of raising electrons from the valence band into the acceptor energy levels, the B acceptor can bind donor electrons. After all acceptors have an extra electron (and have become negatively

charged), there are only N_P–N_B donors left which can give their electrons to the conduction band. N_P–N_B is often called net-donor concentration. This Si crystal is partially compensated, i.e., some of the P donors give their electrons to low-lying acceptor states. One defines the compensation ratio k as

$$k = \frac{\text{N minority}}{\text{N majority}} \quad .$$

With this definition k can lie between 0 (uncompensated) and 1 (fully compensated).

For certain applications low compensation is advantageous; for others we need high compensation. The important point is that we need to control the compensation in semiconductor crystals for many applications. The reader is encouraged to use the minimum energy principle to consider compensation for the case where [Acceptor] > [Donor], i.e., for a p-type crystal.

1.2.5. Free Carrier Statistics

Having discussed the electronic properties of various types of impurities, we are now prepared to analyze quantitatively the drastic changes in the free carrier concentration effected by the addition of such impurities to the semiconductor. Specifically, we want to find out how many free electrons and holes exist near their respective band edges as a function of impurity concentration and temperature. This topic is often called "free carrier statistics" and has been thoroughly discussed by a number of authors (J. Blakemore's *Semiconductor Statistics*, Dover Publ., Inc., 1987 is an excellent and inexpensive account of this and related topics).

In our discussion of intrinsic semiconductor properties we derived the intrinsic carrier concentration

$$n_i^2 = N_C N_V \exp(-E_G/kT) \quad .$$

For a doped semiconductor we must take into account all the donors and acceptors with their respective energy levels and concentrations. We begin our discussion with a semiconductor which contains one kind of donor with energy level E_D and concentration N_D, and one kind of acceptor with energy level E_A and concentration N_A. We arbitrarily choose the conduction band edge (≡ conduction band minimum) as the zero energy point (Fig. 21). The position of the Fermi level E_F will be determined self-consistently by the temperature T, the impurity concentrations N_D, N_A, the impurity binding energies E_D, E_A and the bandgap energy E_G. In addition we have to introduce

the degeneracy g of a dopant level which is typically 2 for donors and 4 for acceptors (spin 1/2, spin 3/2 respectively). The probability that a donor level at E_D with degeneracy g is occupied is:

$$F(E_D) = \frac{1}{1 + g^{-1}\exp\left(\frac{E_D - E_F}{kT}\right)} = \frac{1}{1 + \frac{1}{2}\exp\left(\frac{E_D - E_F}{kT}\right)} .$$

The total number of donors N* which are occupied by electrons is:

$$N^* = N_D F(E_D) = \frac{N_D}{1 + \frac{1}{2}\exp\left(\frac{E_D - E_F}{kT}\right)} .$$

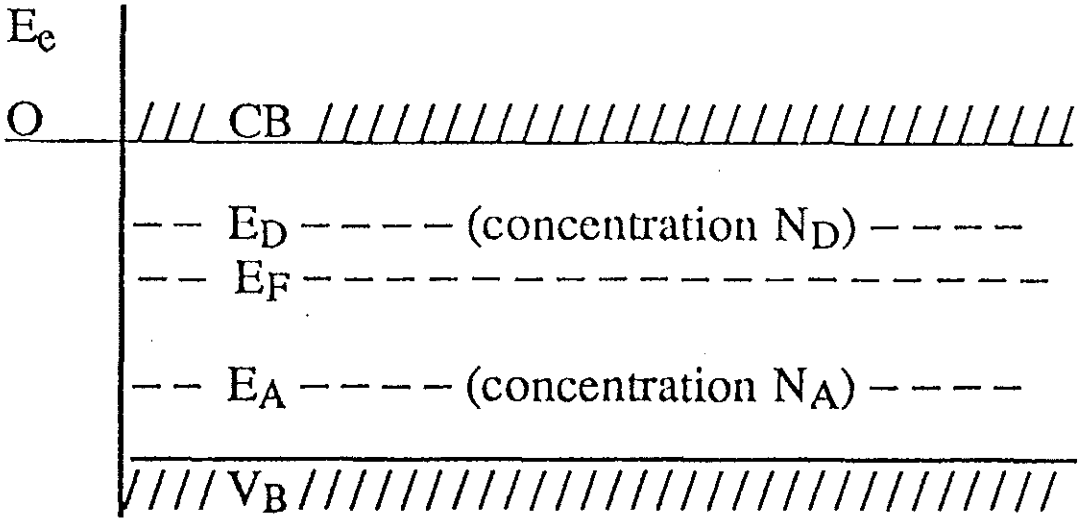


Fig. 21.

Let us choose the specific case for which $N_D > N_A$, i.e., an n-type semiconductor. The maximum number of electrons which can be promoted to the conduction band is $N = N_D - N_A$. N_A electrons are "lost" to acceptor states which lie far below the donor levels, located close to the valence band top. These electrons require almost as much energy to reach the conduction band edge as the valence band electrons. Therefore they can be totally neglected at low temperature. Setting n = number of electrons in the conduction band we can write:

$$N = N_D - N_A = N^* + n$$

or

$$N = \frac{N_D}{1 + \frac{1}{2}\exp\left(\frac{E_D - E_F}{k_B T}\right)} + n$$

We can remove the Fermi energy from this expression because we know that in the Boltzmann approximation

$$n = N_C \exp\left(\frac{E_F}{k_B T}\right) .$$

N_C is the same effective density of states we derived in Chapter 1, page 4. The zero point of the energy axis which can be arbitrarily chosen is set at E_C, in contrast to Chapter 1 where we set $E_V = 0$!

Inserting $E_F = k_B T \ln (n/N_C)$ yields:

$$\frac{n(N_D - N + n)}{N - n} = \frac{1}{2} N_C \exp\left(\frac{E_D}{k_B T}\right) .$$

We define

$$\phi = \frac{1}{2} N_C \exp\left(\frac{E_D}{k_B T}\right)$$

and write

$$n = \frac{1}{2}(\phi + N_D - N)\left(\left[1 + \frac{4N\phi}{(\phi + N_D - N)^2}\right]^{\frac{1}{2}} - 1\right) .$$

This expression is accurate to within the Boltzmann approximation of the Fermi distribution for $E \gg E_F$. The number of free electrons in the conduction band n can be computed for a wide range of T. The expression for n can be simplified for important cases. We will now discuss some of these.

Setting $N_A = 0$ leads to $N = N_D$. At very low temperature only a very small number of electrons can gain sufficient energy to reach the conduction band, $n \ll N_D$. We approximate

$$\frac{n(N_D - N + n)}{N - n} \cong \frac{n^2}{N_D} = \frac{1}{2} N_C \exp\left(\frac{E_D}{k_B T}\right)$$

and find

$$n = \left(\frac{1}{2} N_D N_C\right)^{1/2} \exp\left(\frac{E_D}{2k_B T}\right) .$$

In an Arrhenius plot of the free carrier concentration and the absolute temperature this equation generates a straight line with a slope $E_D/2k_B$. This dependence can be used to determine E_D.

We introduce acceptors with a concentration N_A and determine what happens in the very low temperature range where $n \ll N_A, N_D$ (and $N_A < N_D$):

$$\frac{n(N_D - N + n)}{N - n} = \frac{n(N_A + n)}{N_D - N_A - n} \cong \frac{n N_A}{N_D - N_A} \cong \frac{1}{2} N_C \exp\left(\frac{E_D}{k_B T}\right)$$

and

$$n = \frac{1}{2} N_C \frac{N_D - N_A}{N_A} \exp\left(\frac{E_D}{k_B T}\right) .$$

The slope in an Arrhenius plot, E_D/k_B, is now twice as big as in the case without acceptors.

Increasing the temperature we can reach a regime in which $n > N_A$ and $n < N_D$. In this intermediate temperature regime we find the following approximation:

$$\frac{n(N_A + n)}{N_D - N_A - n} = \frac{n^2}{N_D} \cong \frac{1}{2} N_C \exp\left(\frac{E_D}{k_B T}\right)$$

and

$$n = \left(\frac{1}{2} N_D N_C\right)^{1/2} \exp\left(\frac{E_D}{2k_B T}\right) .$$

As in the case with $N_A = 0$ we find an Arrhenius plot slope of $E_D/2k_B$. Upon further increase in temperature we reach the so-called "exhaustion" or "saturation" regime in which the free electron concentration stays constant:

$$n = N_D - N_A \quad .$$

Figure 22 displays so-called "freeze-out curves" for the cases discussed above.

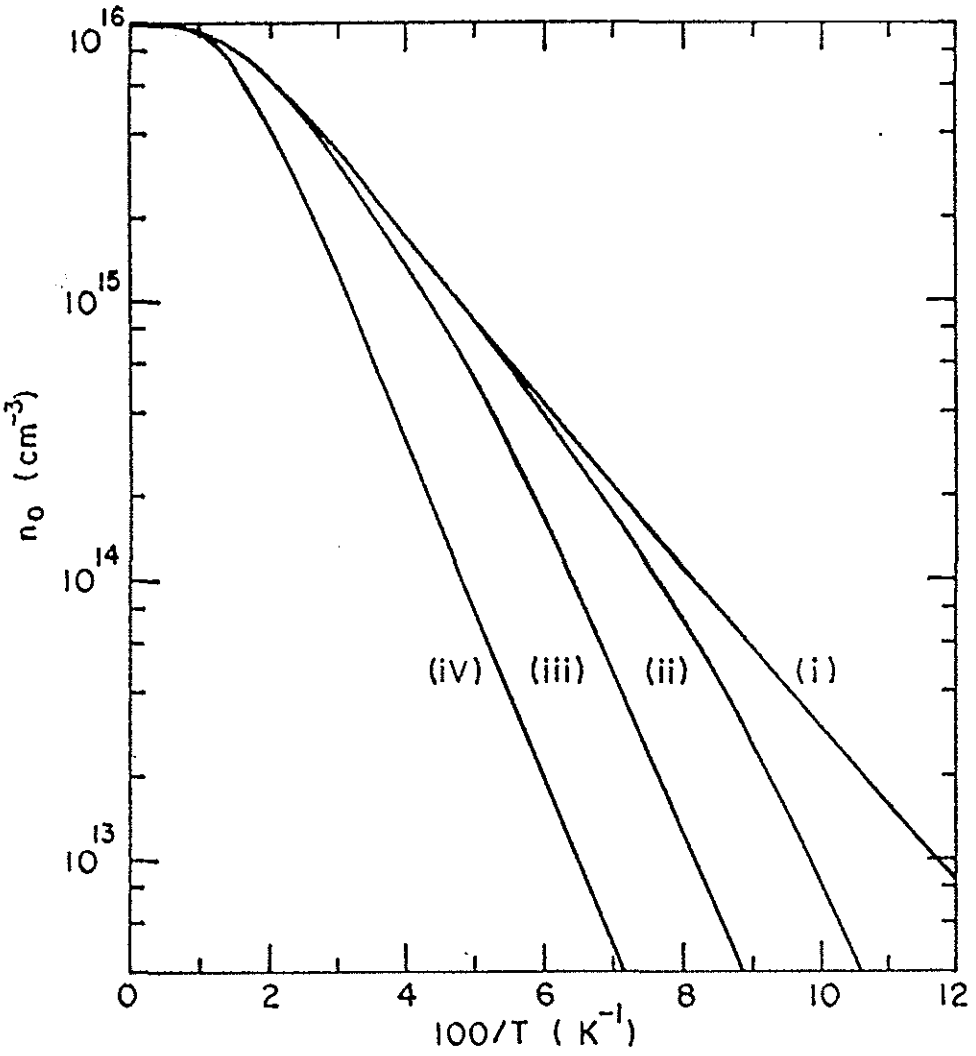


Fig. 22. Arrhenius plot of the logarithm of the free electron concentration as a function of inverse absolute temperature. N_D–N_A = 10^{16} cm^{-3} is kept constant for all cases. E_D = 10 meV, m^* = 0.25 m and g = 2. (i) N_A = 0, (ii) N_A = 10^{14} cm^{-3}, (iii) N_A = 10^{15} cm^{-3}, and (iv) N_A = 10^{16} cm^{-3}. The half slope ($E_D/2k_B$) and full slope (E_D/k_B) regimes are clearly visible.

It is worthwhile to assess what can be learned from the free electron concentration displayed in an Arrhenius plot ($\ln(n)/T^{-1}$). The electron saturation concentration corresponds to $N_D - N_A$. The concentration at which the half-slope regime turns into the full-slope regime corresponds to N_A

(remember that the half-slope regime occurs for $N_D > n > N_A$ while for the full slope regime we assumed $N_D > N_A > n$). The slopes of the freeze-out curve yield E_D. We cannot determine E_A. We simply assumed that it is lying at least several k_BT below E_D.

What happens to the freeze-out curve for higher temperatures? Eventually the intrinsic carrier concentration will first become comparable to, then larger than $N_D - N_A$. The semiconductor assumes intrinsic properties from this temperature on upwards (Fig. 23). Let us

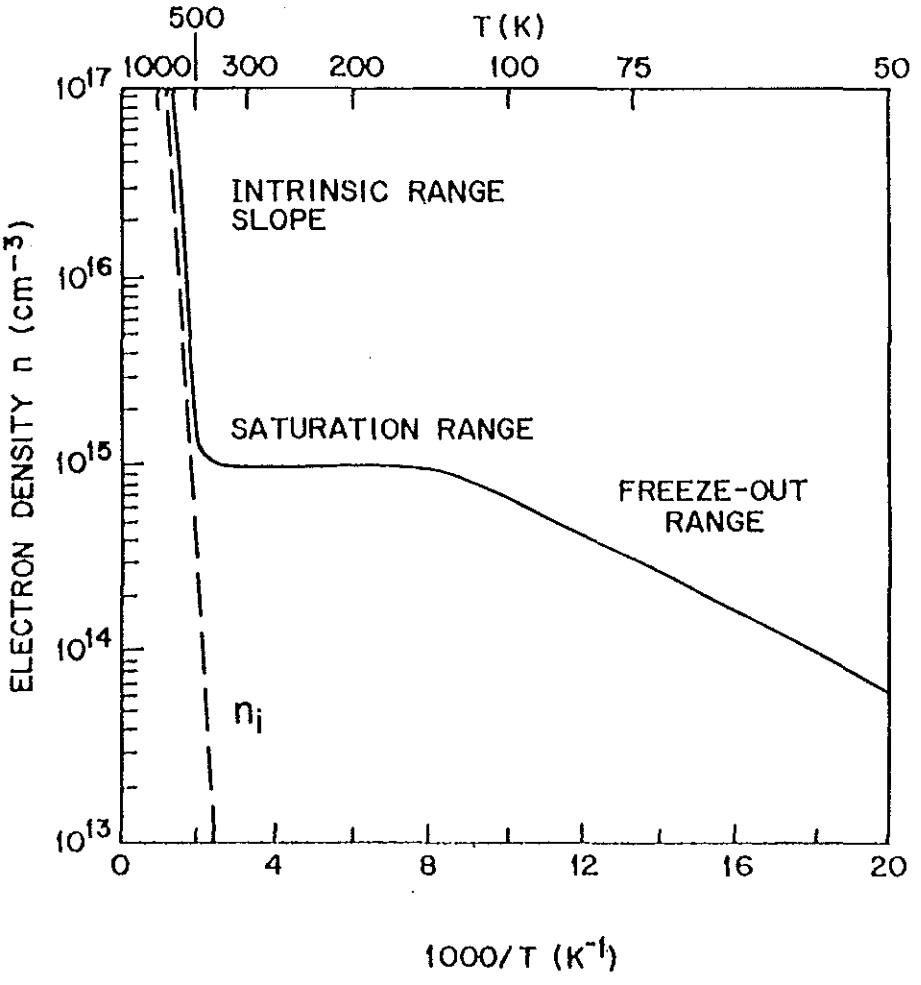


Fig. 23. Arrhenius plot of the electron concentration as a function of the inverse absolute temperature of an uncompensated silicon sample doped with 10^{15} cm^{-3} donors.

discuss a slightly more complicated case in which a deep level is present in addition to the shallow donors and acceptors. Such a situation is found for dislocation-free, ultra-pure, p-type Ge. Fig. 24 shows the Arrhenius plots for a dislocation-free and a dislocated piece of Ge single crystal

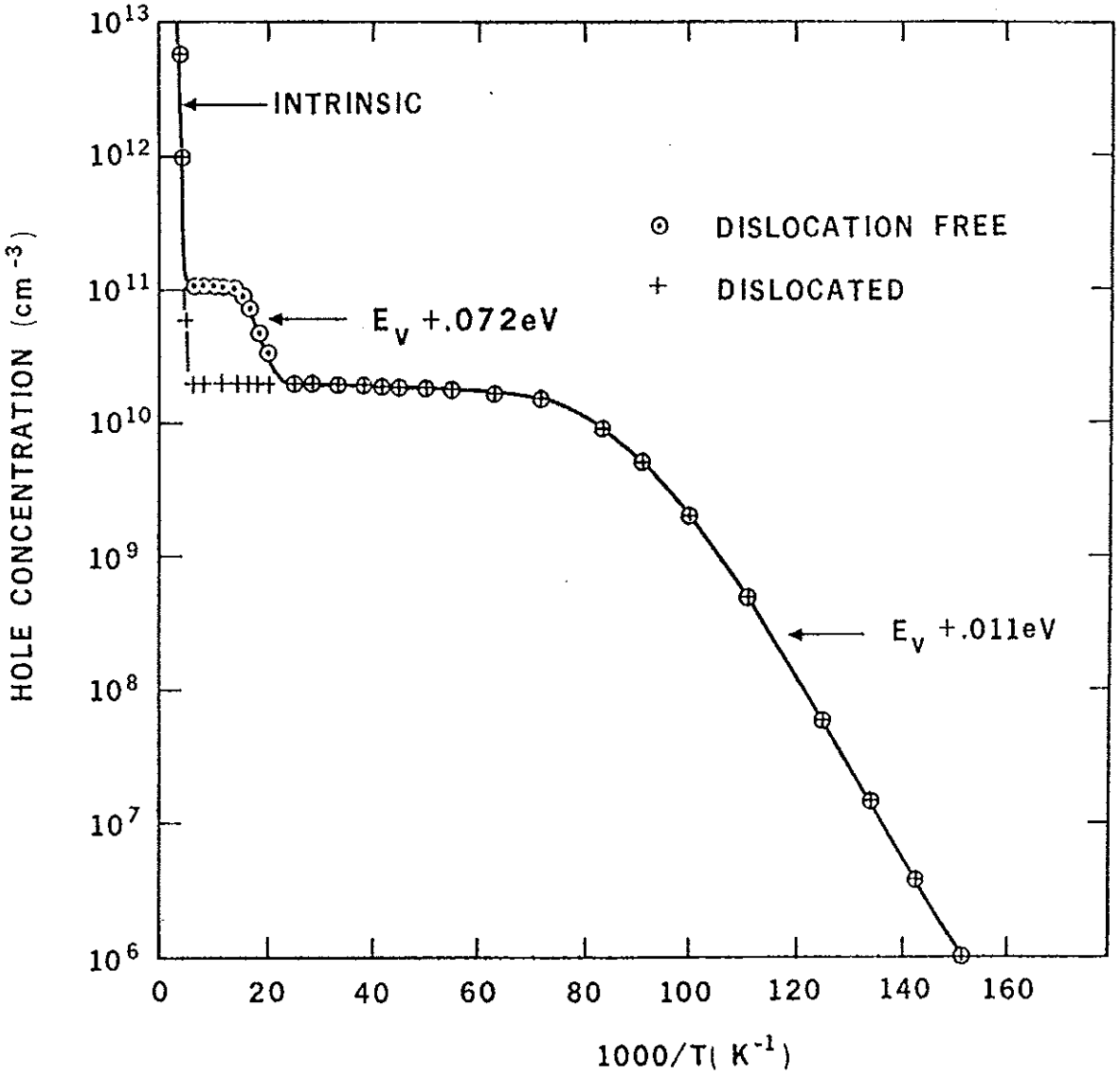


Fig. 24. Arrhenius plots of the free hole concentrations of a dislocation-free and a dislocated piece of single crystal high-purity Ge grown in a H_2 atmosphere [After E. E. Haller *et al.*, Adv. in Physics **30**, 93 (1981)].

cut from a slice of a partially dislocated crystal. We find $N_A - N_D = 2\times10^{10}$ cm^{-3}. The low temperature freeze-out slope is 11 meV, which is the full slope for shallow acceptors. No half-slope regime can be seen, which tells us that N_D is only little smaller than N_A, i.e., we have a "highly" compensated crystal (compensation ratio = minority impurity concentration/majority impurity concentration; in this case N_D/N_A). Above 50 K the hole concentration rises again for the dislocation-free piece. This indicates that holes from an acceptor level which lies deeper in the forbidden band than the shallow acceptors are promoted to the valence band. Careful analysis yields E_{Adeep} = 72 meV (energy above the valence band top). We have proposed that this acceptor level is due to divacancy-hydrogen (V_2H) complexes which form during the slow cooling of the crystal after growth in a H_2 atmosphere. The excess vacancies cannot find sinks in the dislocation-free crystal and form V_2H centers. These centers are not found in crystals which are not grown in a H_2 atmosphere. In the dislocated crystal the excess vacancies condense at dislocations causing dislocation climb.

Free carrier statistics offers a powerful semiconductor characterization tool. When precisely doped and compensated crystals are developed for a specific application, free carrier freeze-out curves provide quantitative analysis. When an unknown impurity is introduced into a crystal one can determine the electron or hole binding energy. Many of the energy levels shown in Fig. 15 were determined in this way.

Before leaving the topic of free carrier statistics, we should briefly discuss the motion of the Fermi level in a doped semiconductor as a function of temperature. Scientists and engineers with an interest in electronic devices often discuss, for the sake of convenience, various device characteristics in terms of the Fermi level. We have defined the Fermi level under equilibrium conditions. When electric fields or currents are present, one and possibly two "quasi" Fermi levels can be introduced.

At temperatures sufficiently high for the semiconductor to be intrinsic, the Fermi level lies close to the middle of the band gap. As the temperature drops, the Fermi level moves either towards the conduction band edge for n-type semiconductors or towards the valence band top for p-type semiconductors. At the temperature for which 50% of the majority dopants are frozen out (neutral), the Fermi level coincides with the majority dopant energy level. In the ideal case of a totally uncompensated crystal the Fermi level would approach a position close to the middle between the dopant energy level and the conduction band minimum at very low temperatures. In compensated crystals a fraction of the majority dopants remains ionized at the lowest temperatures. With the Fermi distribution becoming more abrupt as we go to lower temperatures, the Fermi level will be

locked very closely to the dopant energy level. Figure 25 displays the position of E_F for a number of p- and n-type dopant concentrations as a function of temperature for Si. The reduction of bandgap energy with increasing temperature is also shown.

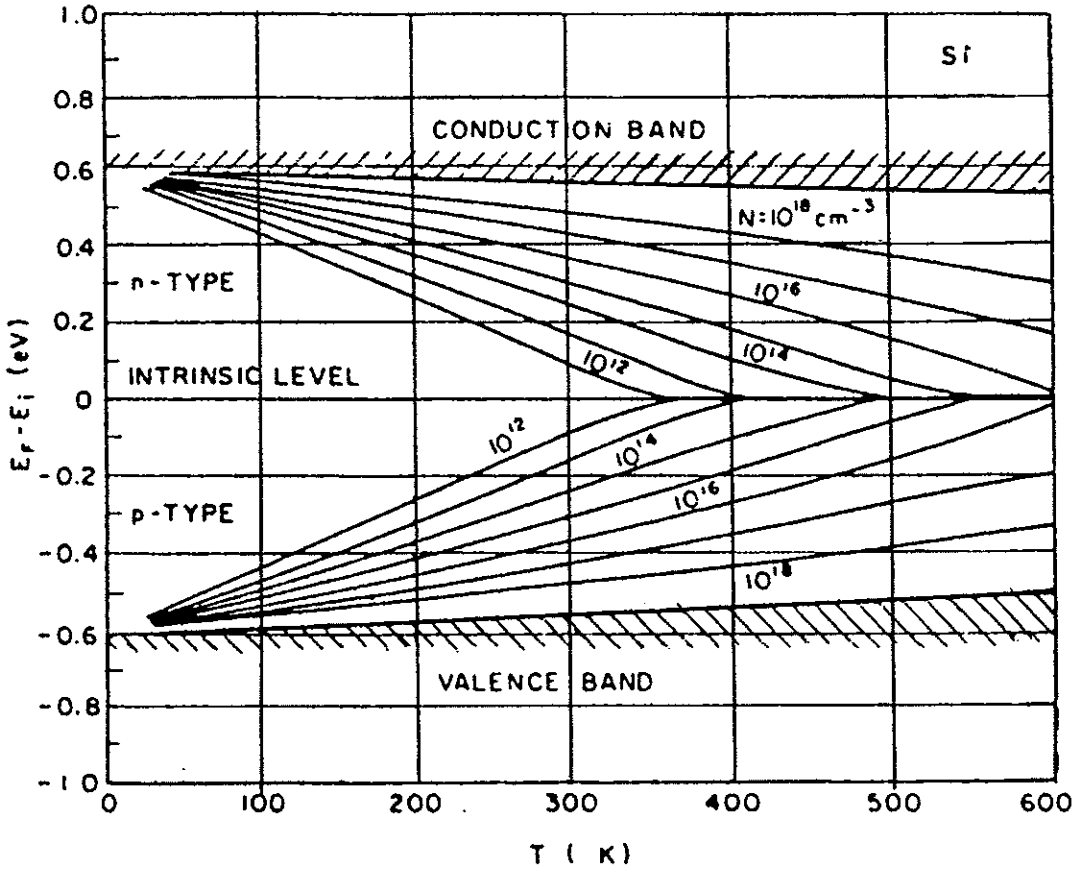


Fig. 25. Fermi level for Si as a function of temperature and impurity concentration. The dependence of the bandgap on temperature is incorporated in the figure. (After A.S. Grove, "Physics and Technology of Semiconductor Devices," John Wiley & Sons, Inc., NY, 1967.)

1.2.6. Electronic Transport

Under the broad term electronic transport we bring together a large number of phenomena associated with the motion of free carriers, electrons in the conduction band and holes in the valence band. Motion in low and high electric fields, in combined electric and magnetic fields, elastic and inelastic scattering of free carriers from neutral and ionized impurities and from phonons belong, among other effects, to this topic. We will restrict ourselves to the most basic aspects of carrier transport and refer to excellent texts† for the reader with an in-depth interest.

In low electric fields (E) the motion of a free carrier of charge e can be described with the effective mass (m*) and Newtonian mechanics:

$$F = eE = am^* \quad .$$

The average velocity increase produced by the acceleration a is a small effect superimposed on the thermal motion of the carrier. The energy gain of the carrier in the electric field is lost in inelastic collisions. Such collisions are infrequent compared to the number of elastic collision events, which is equivalent to our assumption that the average gain in kinetic energy is small compared to the thermal energy.

A careful and detailed analysis is required to arrive at the finding that the electric field-induced drift velocity V_d is proportional to the applied electric field:

$$V_d = \frac{e\tau}{m^*}E = \mu E$$

where τ = scattering time (average time between collisions), μ = mobility. Because the electrical conductivity σ can be expressed with the mobility μ:

$$\sigma = \mu e \text{ conc.} \ ,$$

it is most practical to express the various carrier scattering effects in terms of mobilities.

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†See for example K. Böer, "Survey of Semiconductor Physics," Van Nostrand Reinhold (1990).

In an intrinsic semiconductor with no defects or dopants, the free carriers only encounter phonon scattering at moderate temperatures ($n_i < 10^{16}$ cm^{-3}). We can describe phonons in this context as a kind of particle. The actual interaction of an electron and a phonon can be depicted easily when we remember that phonons are lattice vibrations involving motion of lattice atoms. We further remember that the bandgap depends strongly on externally applied pressure. The motion of the lattice atoms during passage of a phonon in turn leads to local, oscillatory bandgap fluctuations. The electrons interact with these bandgap gradients, i.e., they get scattered by phonons (Fig. 26).

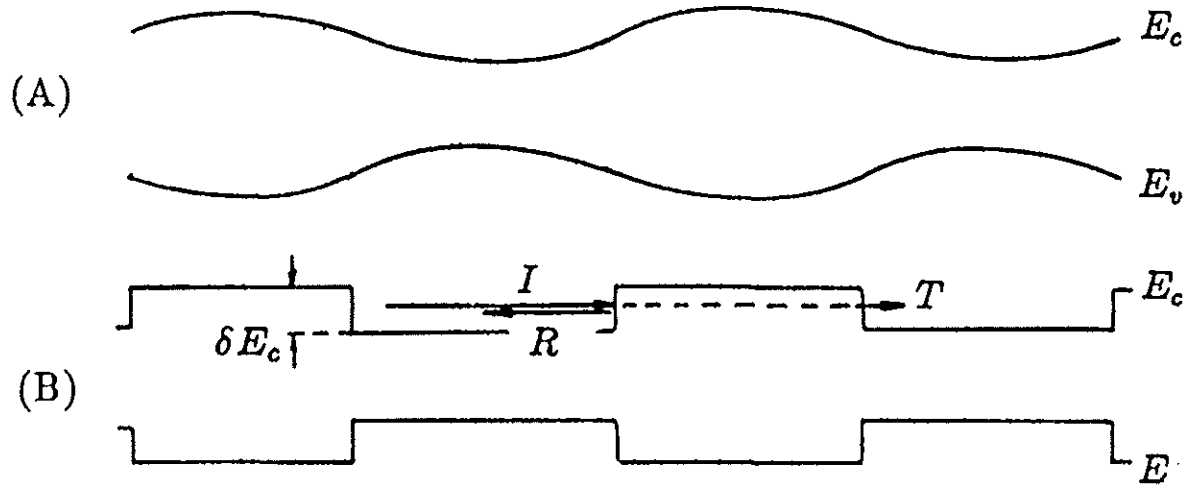


Fig. 26. (A) Undulation of the band edges due to "pressure waves," e.g., in the long wave range of longitudinal acoustic phonons. (B) Step-like approximation of these undulations. [After Karl W. Böer, "Survey of Semiconductor Physics," (Van Nostrand Reinhold, NY, 1990, p. 783).]

Phonon scattering leads to the following expression for the mobility:

$$\mu_{\text{phonon}} = A T^{-3/2} .$$

The constant A contains a number of materials constants. The mobilities of direct bandgap semiconductors follow the above expression rather accurately at high temperatures. Deviations in the temperature exponent are found for indirect semiconductors such as Ge and Si. For pure Ge one finds a $T^{-5/3}$ dependence for electrons above 70K. Intervalley scattering is the major cause for these deviations.

At low temperature Coulomb scattering by ionized impurities becomes dominant. The lower the thermal energy is, the larger the relative Coulomb field forces are. Early in this century, Rutherford solved this fundamental problem when he analyzed the scattering of α particles by charged nuclei. A lengthy derivation yields:

$$\mu_{\text{ion. imp.}} = B\frac{T^{3/2}}{Z^2 N} \quad .$$

Z = number of electron charges on the impurity, N = ionized impurity concentration, B = materials constant.

The third scattering mechanism which we consider is neutral impurity scattering. This is a rather weak interaction and can only be observed in doped semiconductors with very low compensation. At low temperatures the dopant impurities are neutral (the free carriers are frozen out) and few ionized impurities remain because of the low compensation. Neutral impurity scattering is temperature independent:

$$\mu_{\text{neutral. imp.}} = \frac{C}{N} \quad .$$

C = materials constant.

In order to obtain the total mobility in a semiconductor we have to add the inverse of the various mobilities (just like adding parallel resistors! also called Matthiessen's rule):

$$\frac{1}{\mu_{\text{total}}} = \frac{1}{\mu_{\text{phonon}}} + \frac{1}{\mu_{\text{ion. imp.}}} + \frac{1}{\mu_{\text{neutral imp.}}} \quad .$$

Fig. 27 shows the mobility as a function of temperature for two Ga-doped Ge crystals. Above 70 K a $T^{-3/2}$ law is obeyed. Below 8 K the mobility becomes constant for crystal #112–17.5, which can be interpreted as neutral impurity scattering. In the temperature range from 10 to 70 K several scattering mechanisms are effective. Because both crystals are of low compensation, the mobility increases slowly in the temperature range where the Ga acceptors are ionized but rises sharply as soon as the holes freeze out, yielding the Ga acceptors neutral (<20 K). In older semiconductor samples, purity and compensation were often coupled. This is nicely illustrated in Fig. 28 which shows the free hole mobility as a function of temperature for a series of B doped Si

samples. The extreme reduction of the mobility with temperature in sample #141 is a clear sign for a high degree of compensation by donors leading to a large number of ionized donors and acceptors causing ionized impurity scattering.

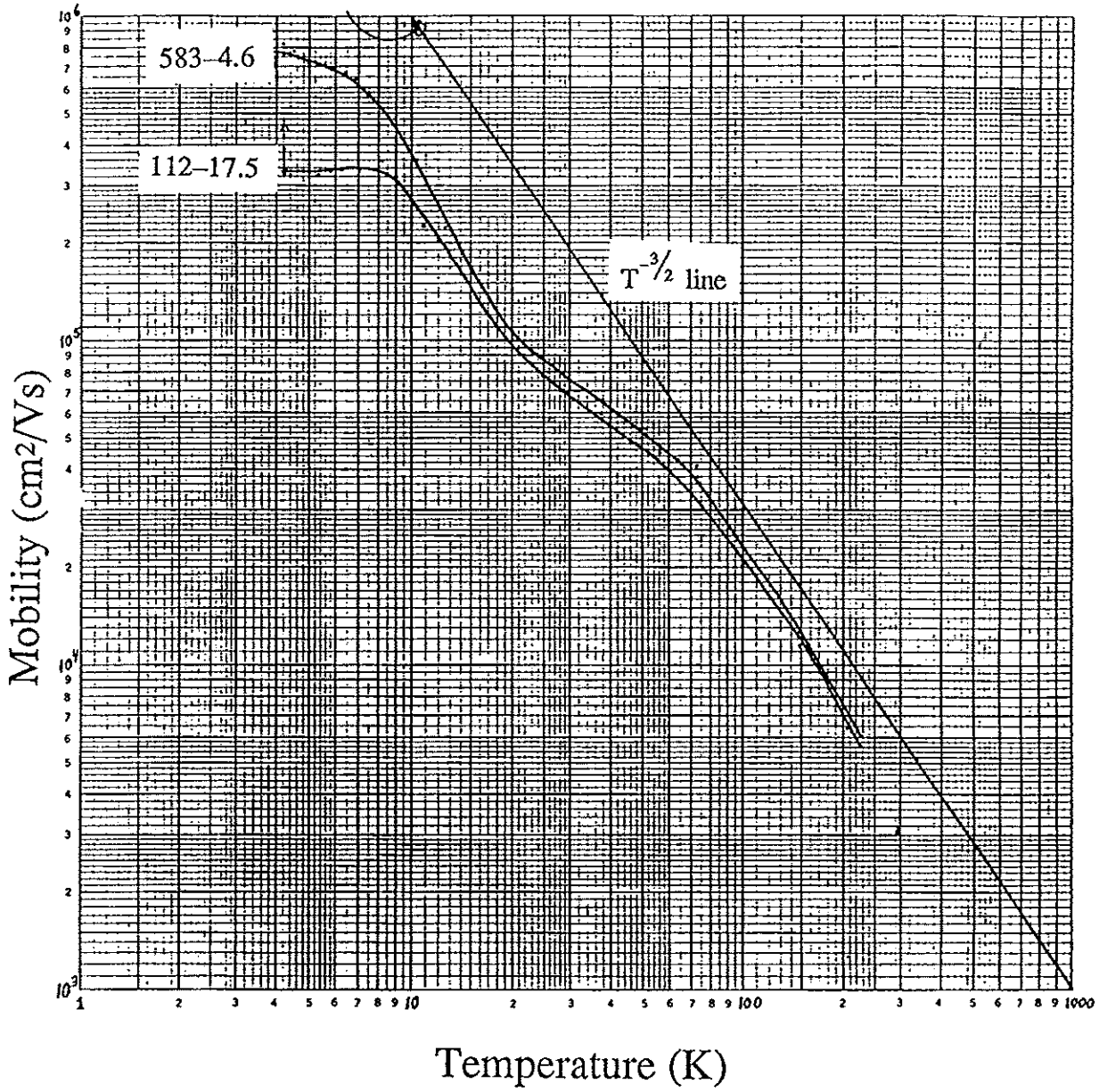


Fig. 27. Hole mobility as a function of temperature for two Ga-doped Ge crystals. [Ga] ≅ 10^{14} cm^{-3}. Above ~70 K the mobility is phonon-dominated ($T^{-3/2}$). The rapid increase in mobility below ~20 K is a result of Ga acceptor freeze-out, i.e., a rapid reduction of ionized scattering centers. Only low compensation crystals show this dependence of μ at low temperatures. Such crystals are used for sensitive far-infrared photoconductive detectors.

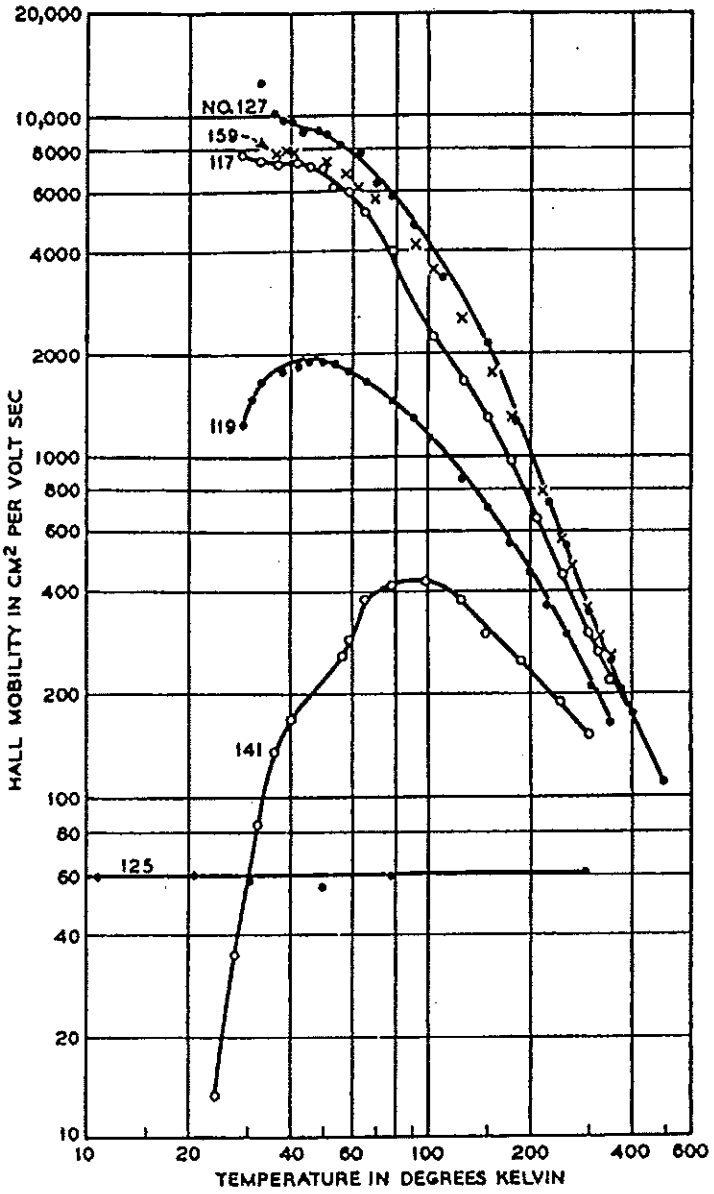


Fig. 28. Hall mobility vs. temperature for a series of B-doped Si samples whose purity increases from sample 125 to 127 in order of mobility at the higher temperatures. [After F.J. Morita and J.P. Maita, Phys. Rev. **96**, 28 (1954).]

The discussion so far has focused on low electric fields. At high fields carriers can become "hot," i.e., their energy exceeds kT significantly. A host of new phenomena can occur in this regime. The frequent emission of optical phonons typically limits the high field drift velocity.

Figure 29 shows the drift velocities for holes and electrons in Si and Ge. In GaAs an interesting effect occurs for electrons at electric fields exceeding ~ 4kVcm^{-1}. The drift velocity goes through a maximum of $\cong 2\times10^{7}$cms^{-1} and then drops (Fig. 30). The main reason for this drop in velocity with increasing field is the transfer of electrons from the central Γ valley with very low mass to the local band minimum along the L direction with a much higher mass (Fig. 31). Maximum drift velocities are obtained in GaAs circuits at electric fields near 4kV/cm. GaAs offers two high frequency advantages over Si circuits: the maximum electron mobility in GaAs is higher than in Si and this maximum mobility is reached at low electric fields. Si circuits must be driven by very high electric fields to reach electron saturation velocity. This leads to much higher power dissipation and heating. There are of course several disadvantages regarding compound semiconductors. We will encounter some of these in the following chapters.

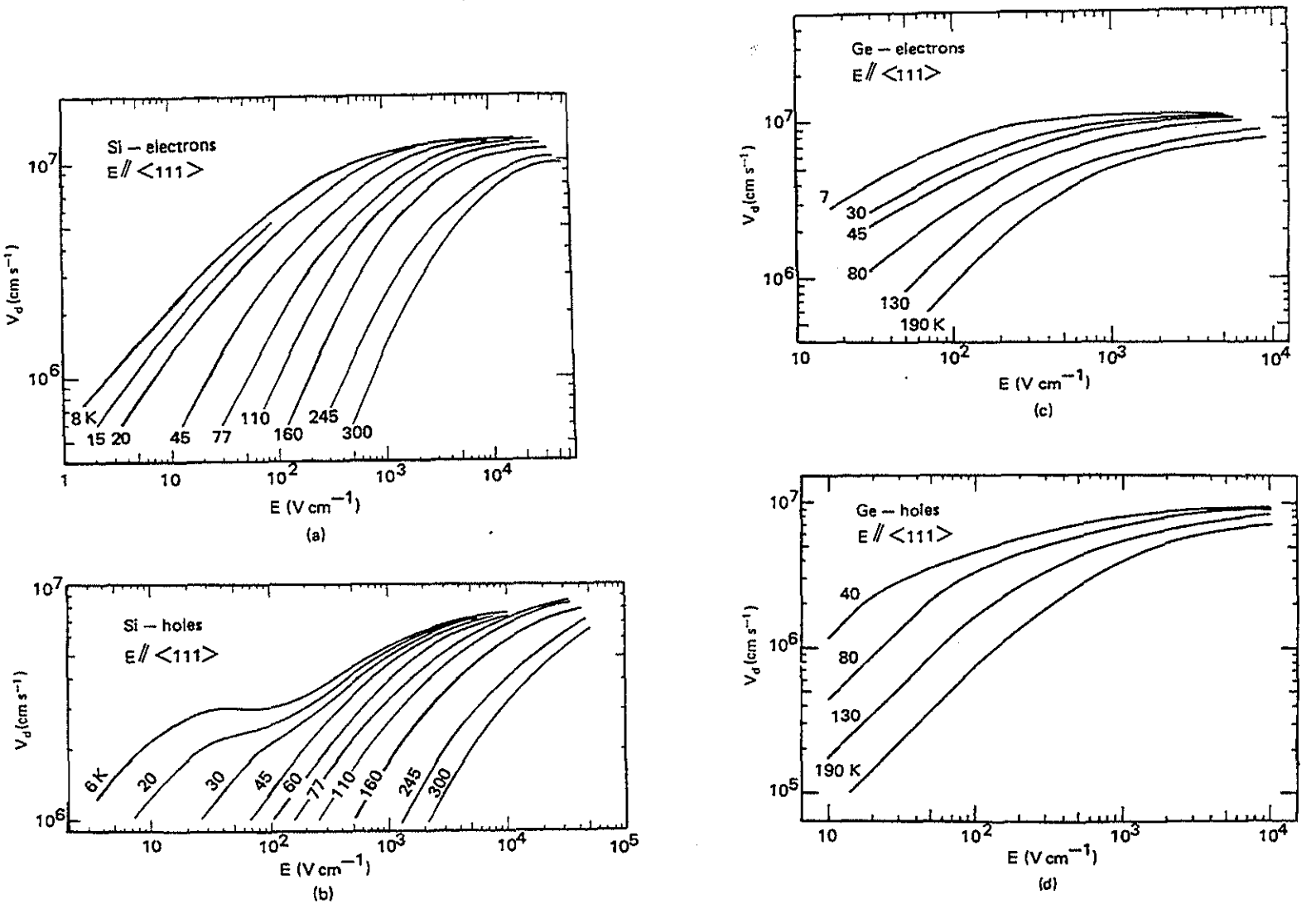


Fig. 29. Drift velocity as a function of electric field applied parallel to the <111> crystallographic direction. Absolute temperature is the parameter for the different curves. (a) Electrons in silicon; (b) holes in silicon; (c) electrons in germanium; (d) holes in germanium. [After G. Ottaviani *et al.*, IEEE Trans. Nucl. Sci. **NS-22**(1), 192 (1975).]

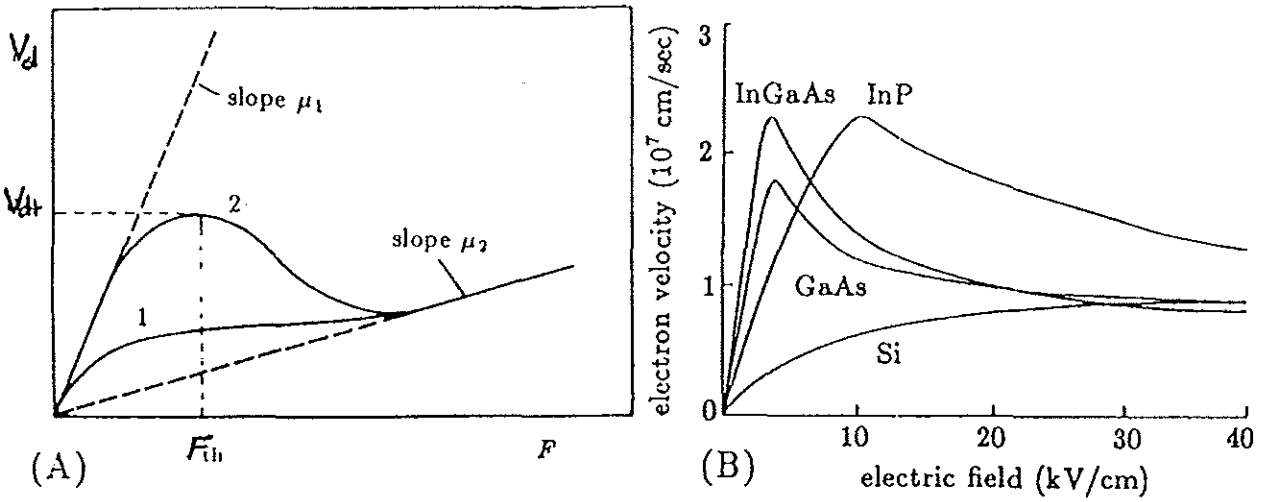


Fig. 30. (A) Drift velocity as a function of the field when major repopulation of the higher satellite valley with substantially lower effective mass becomes effective at the threshold field F_{th} (schematic). (B) Measured field dependence of the drift velocity for several semiconductors. [After A.G.R. Evans and P.N. Robson, Solid State Electronics **17**, 805 (1974).]

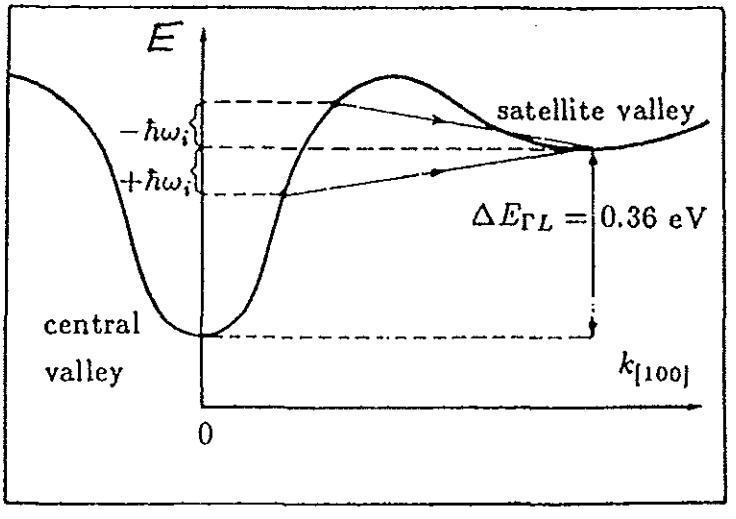


Fig. 31. Electron scattering into a higher satellite valley with higher effective mass (GaAs). [After Karl W. Böer, "Survey of Semiconductor Physics," (Van Nostrand Reinhold, NY, 1990, p. 829).]

1.3. Carrier Generation and Recombination

1.3.1. Generation

Our discussions of various semiconductor properties have never addressed time-dependent effects. We have only considered equilibrium and steady state situations for which the time derivatives are zero. In most semiconductor device applications we encounter time-dependent phenomena. A bipolar transistor in a digital circuit may be turned "on" and "off" billions of times per second. The fastest resonant tunneling diodes can switch "on" and "off" close to a trillion times per second (10^{12} s^{-1})! It is quite clear that we have to understand how rapidly the electron or hole population in the respective bands can be changed and what processes limit the speed with which such changes can be brought about.

An increase in the free carrier concentration is caused by carrier **generation**. The return to equilibrium or steady state is accomplished by **recombination**. There are four major processes by which free carriers can be generated:

- thermal generation (phonons)
- electromagnetic generation (photons, x-rays, γ-rays)
- electrical field generation (injection, field ionization, etc.)
- energetic particle induced generation (electrons, protons, etc.)

Generation occurs between initial states and final states. These states can be localized in the bandgap (impurity and defect states) or delocalized in the band.

Let us discuss *thermal* generation and recombination for a specific case. We consider an n-type semiconductor with a small concentration of deep donor-like levels (N_t) near the middle of the bandgap at an energy E_t. Transitions of electrons from the conduction band (N_c) into the empty deep levels (N_t^+) and from the full deep levels (N_t^0) to the conduction band must lead to a steady state situation if the semiconductor is not disturbed by some external effects.

$$e_t N_t^0 N_c = c_t n N_t^+ = c_t n (N_t - N_t^0)$$

where e = emission coefficient, c = capture coefficient. In the case where $E_t = E_F$ we know that $F(E) = \frac{1}{2}$, i.e., half of the deep traps are full and the other half are empty:

$$N_t^+ = N_t^0$$

and

$$e_t N_c \cong c_t n = c_t N_c \exp\left(-\frac{E_c - E_t}{kT}\right) .$$

The ratio of emission to capture coefficient is

$$\frac{e_t}{c_t} = \exp\left(-\frac{E_c - E_t}{kT}\right) .$$

The capture coefficient can be expressed as a product of capture cross-section s and the average thermal velocity of the electrons v_{th}:

$$c_t = s\, v_{th} .$$

Typical values for capture cross sections range from 10^{-13} cm^2 for Coulomb attractive centers to 10^{-21} cm^2 for Coulomb repulsive deep levels.

In a widely used deep level characterization technique (Deep Level Transient Spectroscopy, DLTS), one measures directly the thermal emission rate of electrons from deep traps by monitoring the capacitance of a reverse-biased diode. The electron emission leads to an increase in positively charged deep levels which in turn reduces the width of the depletion layer, increasing the capacitance. The deep levels are refilled at constant time intervals with a reverse bias reduction pulse. For a detailed discussion of DLTS, consult the semiconductor characterization sections.

Electromagnetic (optical) generation of carriers can occur through several processes. Five possibilities are shown in Fig. 32. Process 4, excitation of an electron from a deep level to the conduction band, is the optical equivalent of the thermal ionization process discussed above. Optical generation occurs in photoconductors, photodiodes, solar cells (which are a special kind of photodiodes), charged coupled devices (two-dimensional CCD's are used in picture phones, video recorders, telescopes for astronomy, etc.), and many other sensors and detectors. Photoluminescence requires carrier excitation by photons, which is followed by the emission of lower energy photons generated by specific carrier recombination transitions. X-ray and γ-ray detector diodes measure the energy of the individual rays by collecting the electrons and holes produced in the depletion layer of the detector. The total free charge produced by the x-ray or γ-ray is proportional to its energy.

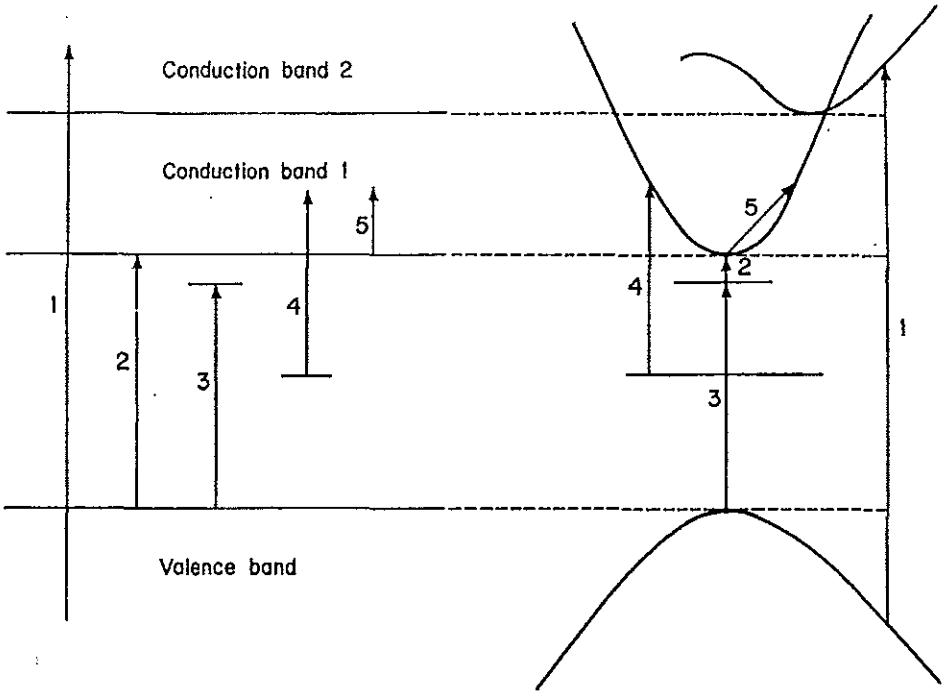


Fig. 32. Characteristic types of optical transitions shown both for the flat-band model and the for E vs. k plot. (1) Excitation from the valence band to higher-lying conduction bands, (2) excitation across the bandgap, (3) exciton formation, (4) excitation from imperfections, (5) free-carrier excitation. (After Richard H. Bube, "Electrons in Solids," Academic Press, NY, 1981, p. 112.)

Electric field induced excess carriers occur under various conditions. Injection of carriers from a metal into a semiconductor can be electric field induced. Impact ionization of neutral impurities at low temperature generates free carriers. At sufficiently high fields electrons can be ionized across the bandgap. Such a situation is most often uncontrolled and called breakdown.

Thermal ionization of shallow or deep level centers can be electric field assisted. The lowering of the binding energy carries the name Poole-Frenkel effect. It is the result of the linear superposition of the Coulomb field at an impurity and the externally applied electric field.

This brief survey of some processes which move the semiconductor free carrier concentration away from equilibrium or steady state conditions shall suffice to illustrate what is meant by the general expression "generation." A key question which needs to be answered is how much time will it take for the semiconductor to return to its pre-excitation condition. In many, though not all the cases, recombination is the most effective return path.

1.3.2. Recombination

Free carrier recombination is the inverse process of free carrier generation. Just as there exist several generation processes, we distinguish between a number of recombination processes. In most cases a specific carrier generation path is not retraced inversely by the subsequent recombination process. For example, a photon-generated electron-hole pair in Si recombines without emission of a photon ~10^6 times more frequently than by photon emission. This is of course due, in part, to the indirect bandgap of Si. As we will see, there always are some deep levels in Si which act as very effective recombination centers already at very low concentrations. Even in the purest, structurally most perfect Si single crystals grown today, the recombination of excess electrons or holes is completely dominated by deep level imperfections.

It is convenient to differentiate between radiative and non-radiative recombination. In radiative recombination, photons are generated. The recombination energy which is freed when an electron and hole recombine across the bandgap, E_G, leaves the crystal in the form of electromagnetic radiation. In non-radiative processes the recombination energy is transformed through one or more steps into heat.

1.3.2.1. Radiative Recombination

Radiative recombination is a process of great technical and scientific importance. All solid state lasers and light-emitting diodes work on the basis of this process. A direct bandgap is most favorable for radiative recombination because no phonons are required for the conservation of momentum. Several III-V compound and alloy semiconductors (e.g., GaAs, $Al_xGa_{1-x}As$, etc.) have direct bandgaps and are widely used for the fabrication of efficient light-emitting sources. A high quality laser is 10-20% efficient at converting electric energy into light.

Indirect bandgap semiconductors such as Si and Ge also exhibit radiative recombination although the probability for such processes is much reduced. Table 5 lists the recombination coefficients c_{cvr} (conduction band - valence band recombination) and the cross-sections s_{nr} (n: electron, r: recombination) for two direct and two indirect bandgap semiconductors.

Radiative recombination is most likely to occur between electrons and holes near the band extrema of a direct bandgap semiconductor. There are, however, other intrinsic radiative processes. An electron and a hole can bind together to form an exciton with a binding energy E_{exc}. The electron and the hole have opposite charges and attract each other just like a positive proton binding a negative electron. Because the electron mass and the hole mass are rather similar, we find the center of mass lying between the two. The center of mass of a hydrogen atom or a

Table 5

RECOMBINATION COEFFICIENTS AND CROSS SECTIONS FOR RADIATIVE BAND-TO-BAND RECOMBINATION
[after Y.P. Varshni, Phys. Stat. Sol. **19**, 459 and **20**, 9 (1967).]

Semiconductor	T	c_{cvr}	s_{nr}
	(K)	$cm^3 s^{-1}$	cm^2
Si	90	$1.3 \cdot 10^{-15}$	$2.4 \cdot 10^{-22}$
	290	$1.8 \cdot 10^{-15}$	$1.9 \cdot 10^{-22}$
Ge	77	$4.1 \cdot 10^{-13}$	$8.5 \cdot 10^{-20}$
	300	$5.3 \cdot 10^{-14}$	$5.5 \cdot 10^{-21}$
GaAs	90	$1.8 \cdot 10^{-8}$	$3.3 \cdot 10^{-15}$
	294	$7.2 \cdot 10^{-9}$	$7.6 \cdot 10^{-17}$
GaSb	80	$2.8 \cdot 10^{-8}$	$5.6 \cdot 10^{-15}$
	300	$2.4 \cdot 10^{-10}$	$2.5 \cdot 10^{-17}$

hydrogenic donor, on the other hand, lies very close to the center of the proton or the donor impurity, respectively. Excitons are very mobile and can move through the crystal. They can get trapped at many different kinds of impurities and defects. Such excitons have lifetimes of up to several ms in high-purity indirect bandgap semiconductors at low temperatures. When an exciton decays radiatively, a photon of energy

$$h\nu = E_G - E_{exc}$$

is emitted. Excitons, multiexcitons and a condensed phase of excitons called electron-hole liquid have been studied in several semiconductors (see for example J.P. Wolfe and C. D. Jeffries, in *Electron-Hole Droplets in Semiconductors*, C. D. Jeffries and L. V. Keldysh, eds., North- Holland, Amsterdam, 1983).

Extrinsic radiative processes occur between free carriers near bandedges and deep levels, between bound states of impurity and defect levels, at acceptor-donor pairs, and in impurity-bound excitons. Several of these processes produce valuable signatures for characterization purposes. When the excess carriers are generated by photons, one describes radiative recombination processes with the name **photoluminescence**. Electron bombardment-produced light is called **cathodoluminescence**. Carrier injection through contacts can lead to **electroluminescence**.

1.3.2.2. Non-Radiative Recombination

There exists a large number of possibilities for an electron and/or a hole to recombine without the emission of light. Such non-radiative recombination processes include:

- Capture of a free carrier by a shallow dopant impurity under the emission of a phonon.
- Capture of a free carrier by deep trap accompanied by a multiphonon emission cascade.
- Capture of an electron (hole) by a deep trap followed by the capture of a hole (electron) under emission of phonons (the deep trap impurity or defect facilitates the phonon generation).
- Transfer of the recombination energy to a free electron which is raised into a high energy state in its band (this process is called Auger effect, leading to an Auger electron; the Auger effect is used, perhaps too often, to explain why radiative recombination does not occur under circumstances where light emission is expected!).

By far the most effective traps and recombination centers are formed by deep level impurities in indirect bandgap semiconductors. Table 6 contains values for capture cross-sections s_n of some deep level impurities in Si and CdS. Values reach from 10^{-10} to 10^{-21} cm^2 but more often lie between 10^{-15} and 10^{-16} cm^2.

The time constant for an exponential return of a free carrier population back to steady state or equilibrium is given by

$$\tau_r = (v_{th} N_t^* s_r)^{-1}$$

with N_t^* = concentration of empty traps. Figure 33 summarizes the various recombination and trapping processes which we have discussed.

Table 6

ELECTRON AND HOLE CAPTURE CROSS SECTIONS OF DEEP LEVEL IMPURITIES AT ROOM TEMPERATURE IN CM^2 AND LEVEL DEPTH IN eV

(collected from Landoldt Börnstein, 1982)

Dopant	Host Lattice	
	Si	CdS
Cu		$s_n(E_c - 0.09) = 4 \cdot 10^{-18}$ $s_n(E_c - 0.2) = 3 \cdot 10^{-17}$ $s_n(E_v + 1.1) = 3 \cdot 10^{-21}$ $s_p(E_v + 0.34) \geq 10^{-15}$
Ag	$s_n(77\,\mathrm{K}) = 10^{-12}$	$s_n(E_c - 0.23) = 10^{-13}$
Au	$s_p(E_c - 0.55) = 10^{-15}$ $s_n(E_v + 0.35) = 10^{-14}$	$s_n(E_c - 0.065) = 10^{-19}$ $s_n(E_c - 0.15) = 10^{-17}$
Zn	$s_p(E_v + 0.26) = 10^{-10}$	
S	$s_n(E_c - 0.59) = 2 \cdot 10^{-15}$	
W	$s_n(E_c - 0.22) = 10^{-16}$ $s_n(E_c - 0.3) = 6 \cdot 10^{-17}$ $s_n(E_c - 0.37) = 3 \cdot 10^{-18}$	
Co	$s_p(E_v + 0.29) = 2 \cdot 10^{-16}$ $s_p(E_v + 0.4) = 10^{-16}$	
Ni	$s_n(E_c - 0.35) = 10^{-15}$ $s_n(E_v + 0.23) = 10^{-15}$	$s_n = 10^{-17}$ $s_p(E_v + 1.44) = 10^{-15}$
Rh	$s_n(E_c - 0.31) = 7 \cdot 10^{-16}$ $s_p(E_v + 0.52) = 6 \cdot 10^{-16}$	
Pd	$s_n(E_c - 0.22) = 3 \cdot 10^{-15}$ $s_p(E_c - 0.19) = 10^{-16}$	
Pt	$s_n(E_c - 0.23) = 5 \cdot 10^{-15}$ $s_p(E_v + 0.32) = 10^{-15}$	

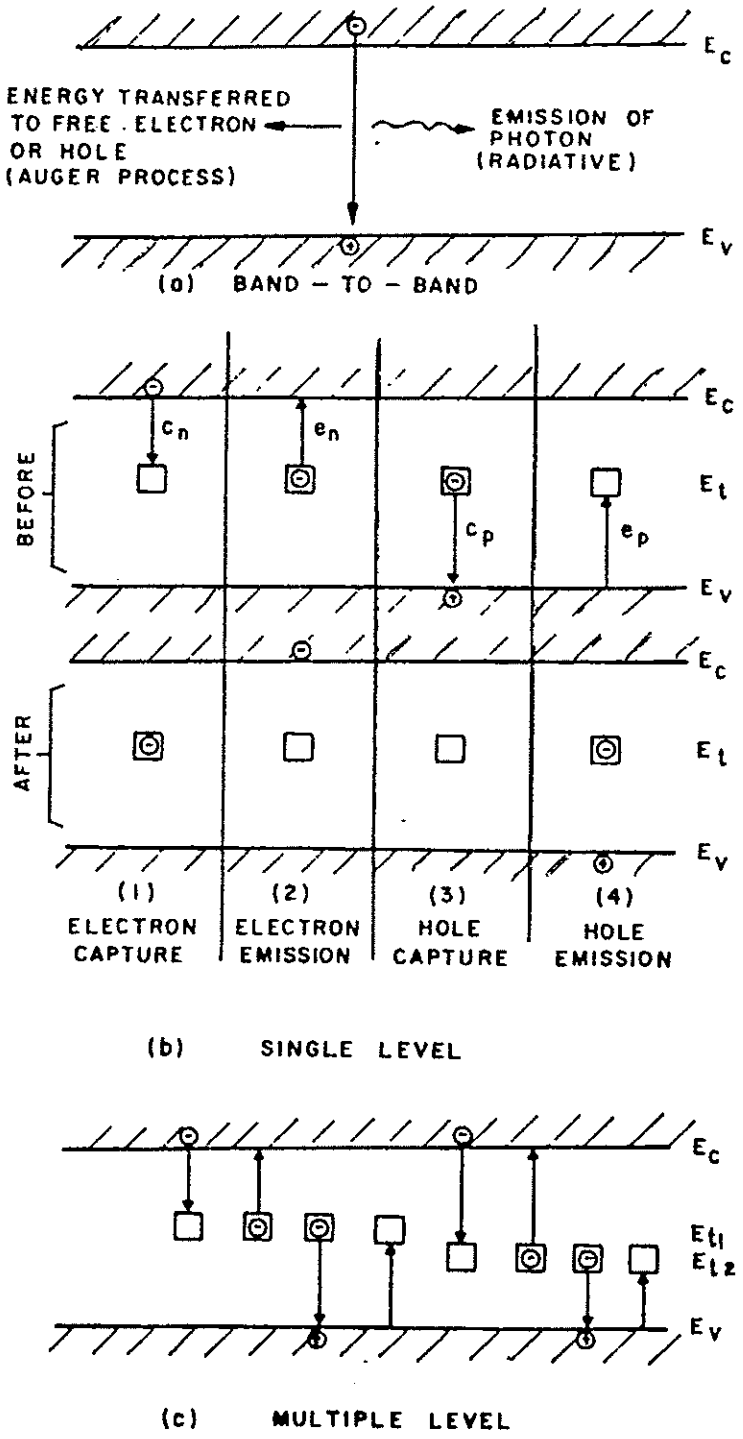


Fig. 33. Recombination processes: (a) band-to-band recombination (radiative or Auger process), (b) single-level recombination, (c) multiple-level recombination. (After S.M. Sze, "Physics of Semiconductor Devices," 2nd edition, John Wiley & Sons, Inc., NY, 1981, p. 36.)

2. SEMICONDUCTOR PURIFICATION

2.1. Chemical Purification - Physical Purification

Extremely small amounts of certain impurities change the type of semiconductor crystals and their conductivity by orders of magnitude. Other impurities do not influence the electrical properties of semiconductors, at least not in first order.

We shall focus on the electrically active impurities which lead to shallow hydrogenic donors and acceptors, as well as deep levels. The dopant concentrations in typical electronic devices range from 10^{20} cm^{-3} down to 10^{13} cm^{-3}. Therefore, the starting material should contain residual impurity concentrations below these levels. The measurement of such small concentrations in crystalline semiconductors is possible because the electrically active impurities affect the conductivity so strongly.

The first and by far the largest purification step is achieved by chemical purification of a substance containing the semiconductor element(s). Chemical purification can be done at *low temperatures* at which contamination from containers and ambients can be reduced to almost any level. The processing can be done *continuously* and with *very large quantities*--factors helping purification. The foremost purification method is distillation in multistage columns of $SiCl_4$, $GeCl_4$, $GaCl_3$, etc. (see Fig. 34).

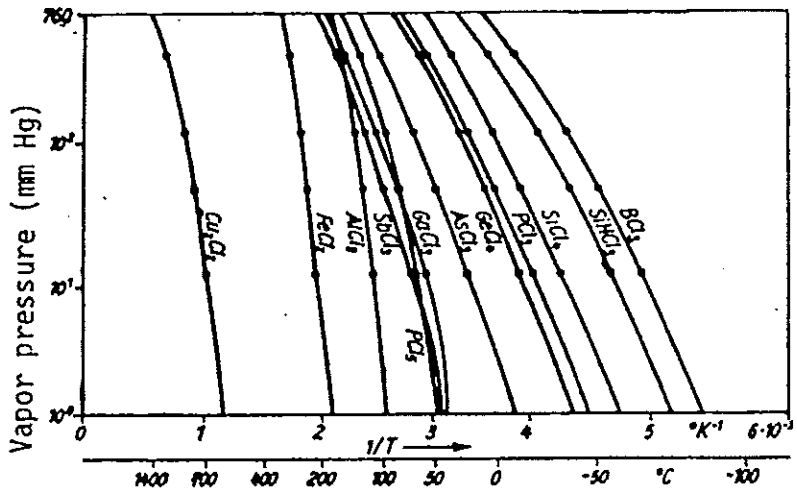


Fig. 34. Vapor pressure of some chlorides. (After H.-F. Hadamovski, "Halbleiterwerkstoffe," 2nd edition, VEB Deutscher Verlag für Grundstoffindustrie, Leipzig, 1972, p 174.)

One problem associated with chemical purification is the lack of analytical techniques with sufficient sensitivity. We have to transform the chemical component into a single crystal before we can determine the concentration and species of relevant impurities. In order to develop chemical processes, one has to go back and forth between chemical purification and physical analysis. A second problem with chemical purification occurs when the boiling or sublimation temperature of an impurity compound and the semiconductor compound lie close together. If there are no chemical ways around such a problem, one has to resort to physical purification.

Impurity segregation between a liquid and a solid phase in contact is the principle used for what we call physical purification. We differentiate between "normal freezing" and "zone refining." The melting point of most semiconductors lies close to or above 1000° C, which makes the above processes inherently "less pure" than chemical purification at low temperatures. The segregation coefficient in equilibrium k_o is defined by:

$$k_o = c_s/c_l \quad .$$

$c_{s,l}$ = concentration of an impurity in the solid (s), in the liquid (l). The segregation coefficient can be taken from the binary phase diagram, as can be seen in Fig. 35.

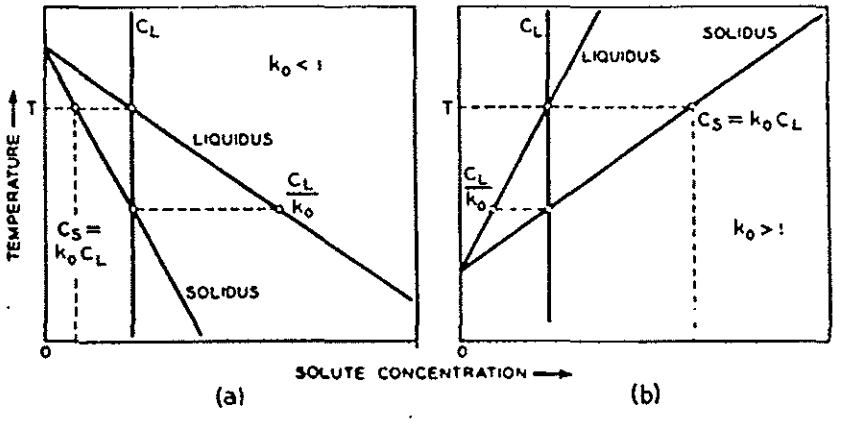


Fig. 35. Portions of constitutional diagrams in which the freezing point of the solvent is (a) lowered, (b) raised, by the solute.

2.2. "Normal Freezing"

The one-dimensional case of normal freezing is very simple. A long, constant cross-sectional area liquid begins to solidify at one end. Disregarding diffusion of impurities in the solid and assuming perfect mixing of the unfrozen liquid, we obtain a very simple expression for the impurity concentration along the frozen material:

$$c_s(x) = c_o k_o (1-x)^{k_o - 1} \quad .$$

$c_s(x)$ = impurity concentration in the solid, x = fraction of melt frozen, c_o = concentration in the original melt. This expression is derived as follows:

$$c_s = k_o c_l \quad .$$

The total number of impurities in the material is $I_o = V_o c_o$ with V_o = total volume. The fraction which is frozen shall be

$$x = V(x)/V_o \quad .$$

We freeze an infinitesimal quantity dV. The impurities dI which are removed from the melt are incorporated in the solid:

$$dI = -k_o c_l(x)\, dV = -k_o \frac{I(x)}{V_o - V(x)} dV \quad .$$

I(x) is the total number of impurities remaining in the melt. Rearrangement of the equation leads to:

$$\frac{dI}{I(x)} = -k_o \frac{dV}{V_o - V(x)} \quad .$$

We integrate

$$\int_{I_o}^{I(x)} \frac{dI}{I(x)} = \int_{o}^{V(x)} -k_o \frac{dV}{V_o - V(x)}$$

and obtain:

$$\ln \frac{I(x)}{I_o} = \ln\left(1 - \frac{V(x)}{V_o}\right)^{k_o} = \ln(1-x)^{k_o}$$

and

$$I(x) = I_o(1-x)^{k_o} .$$

We know that:

$$c_s(x) = k_o c_l(x) = k_o \frac{I(x)}{V_o - V(x)} .$$

Substituting I(x) leads to:

$$\begin{aligned} c_s(x) &= k_o I_o(1-x)^{k_o} \frac{1}{V_o - V(x)} \\ &= k_o \frac{I_o}{V_o} \frac{V_o}{V_o - V(x)} (1-x)^{k_o} \\ &= k_o c_o \frac{1}{1-x} (1-x)^{k_o} . \end{aligned}$$

And finally:

$$c_s(x) = k_o c_o (1-x)^{k_o - 1} .$$

Normal freezing is a nonrepetitive process and is not well suited for an automatic purification process. Crystal growing from a depletable melt is a kind of normal freezing (Czochralski method).

Figure 36 shows the solute (impurity) concentration as a function of fraction solidified for several values of k ranging from 10^{-2} to 5.

2.3. Zone Purification or "Zone Melting"

Instead of the whole quantity of material, only a short section is melted by a resistive or an RF heater coil (Fig. 37).

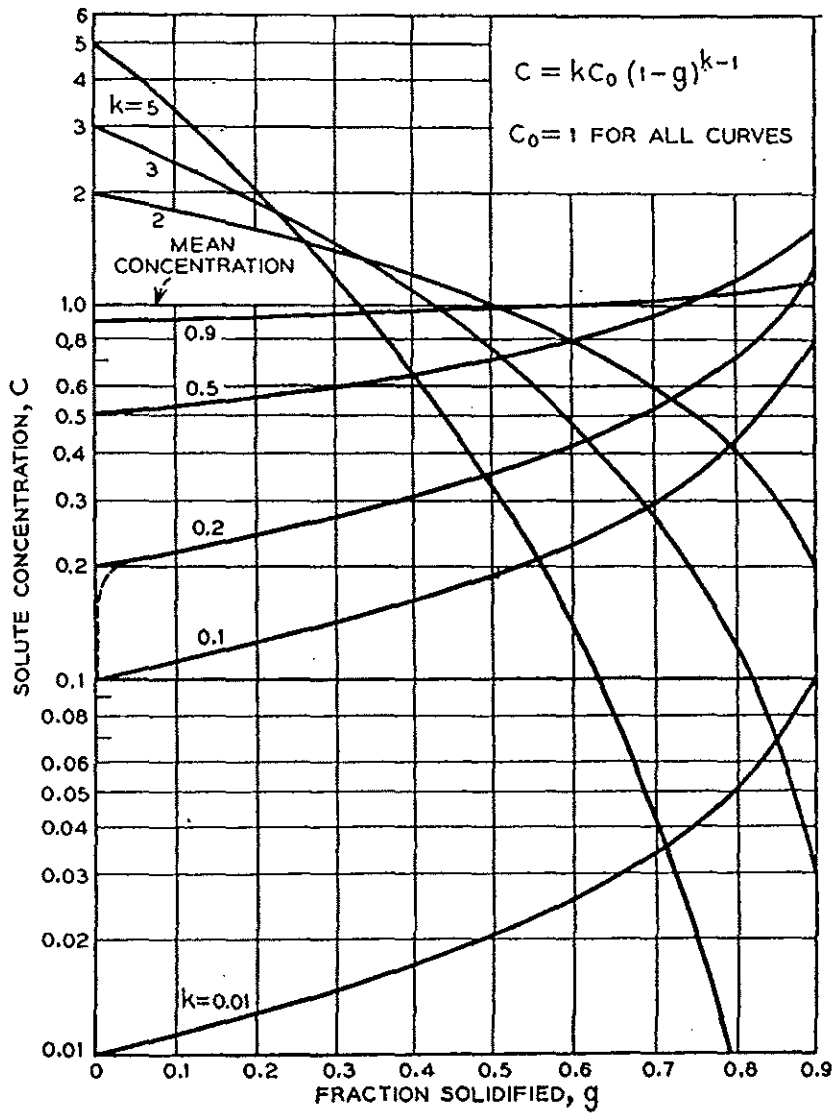


Fig. 36. Curves for normal freezing, showing solute concentration C in the solid versus fraction solidified g. (After W. G. Pfann, "Zone Melting," 2nd edition, John Wiley & Sons, Inc., NY, 1966, p. 12.)

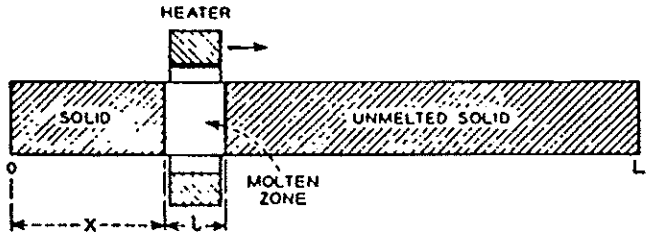


Fig. 37. Molten zone of length ' l ' traversing a cylindrical ingot of length ' L' (schematic). (After W. G. Pfann, "Zone Melting," 2nd edition, John Wiley & Sons, Inc., NY, 1966, p. 28.)

The concentration of impurities in the solid c_s changes as indicated in Fig. 38.

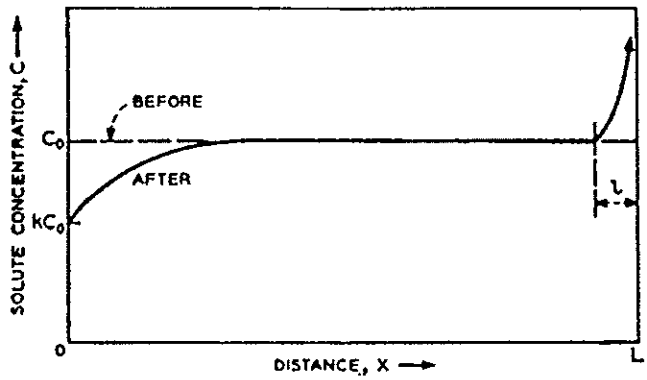


Fig. 38. Approximate concentration of solute in the solid after passage of one molten zone through a charge of uniform mean concentration c_0. (After W. G. Pfann, "Zone Melting," 2nd edition, John Wiley & Sons, Inc., NY, 1966, p. 29.)

At $x = L - l$, a distance l from the end of the bar (l = length of the molten zone) we get into the normal freeze regime. The following expression describes the impurity concentration after a single zone pass.

$$c(x) = c_o[1 - (1 - k)\exp(-kx/l)]$$

x ranges from 0 to $L - l$. Figure 39 shows a number of curves obtained from the above equation for various values of k_o.

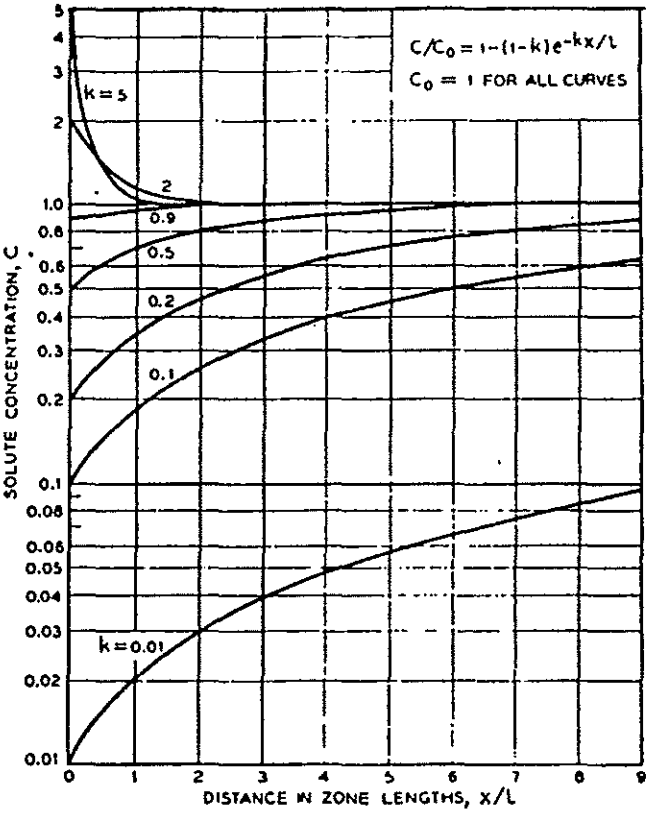


Fig. 39. Curves for single-pass zone melting, showing solute concentration in the solid versus distance in zone lengths from beginning of charge, for various values of the distribution coefficient k. (After W. G. Pfann, "Zone Melting," 2nd edition, John Wiley & Sons, Inc., NY, 1966, p. 30.)

Zone purification lends itself to a repetitive, automatic purification process. The same substance can be "zoned" many times. For a given ratio of *l*/L, we obtain an ultimate impurity distribution after an infinite number of passes. The case shown in Fig. 40 is technically an extremely difficult one (the value of k is very close to unity), atypical for semiconductors.

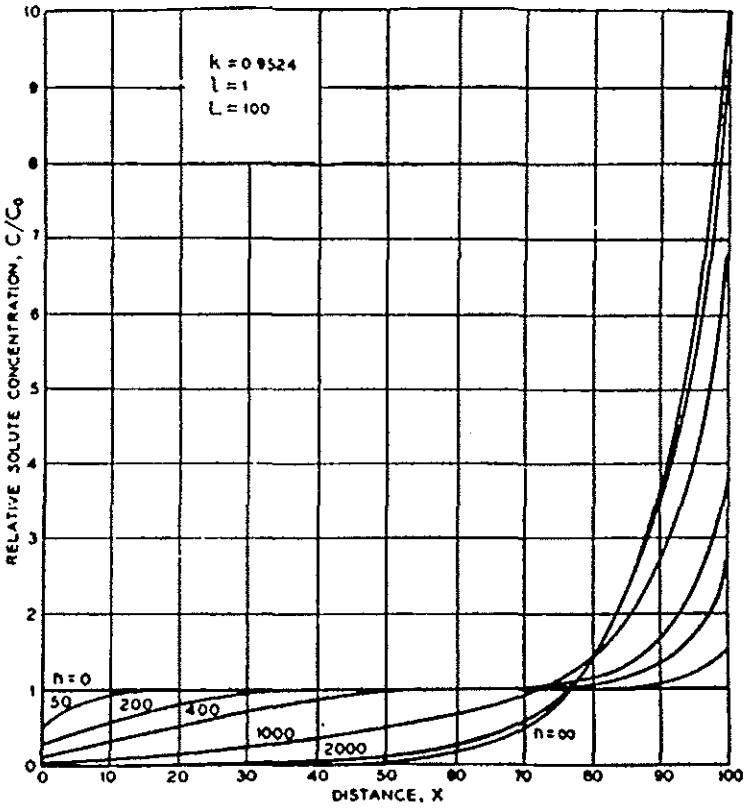


Fig. 40. Relative solute concentration c/c_o versus distance x with number of passes n as a parameter for k = 0.9524, l = 1, L = 100. (After W. G. Pfann, "Zone Melting," 2nd edition, John Wiley & Sons, Inc., NY, 1966, p. 37.)

2.4. Equilibrium versus Effective Segregation Coefficient

The equilibrium segregation coefficient k_o is defined as the ratio of impurity concentration in the solid to the one in the melt at zero solidification rate. As soon as we encounter a finite solidification (crystallization) rate, this equilibrium will be disturbed because impurities have to diffuse away ($k < 1$) or diffuse to ($k > 1$) the interface through a diffusion layer. Figure 41 shows the concentration profile for the equilibrium and the finite growth rate cases.

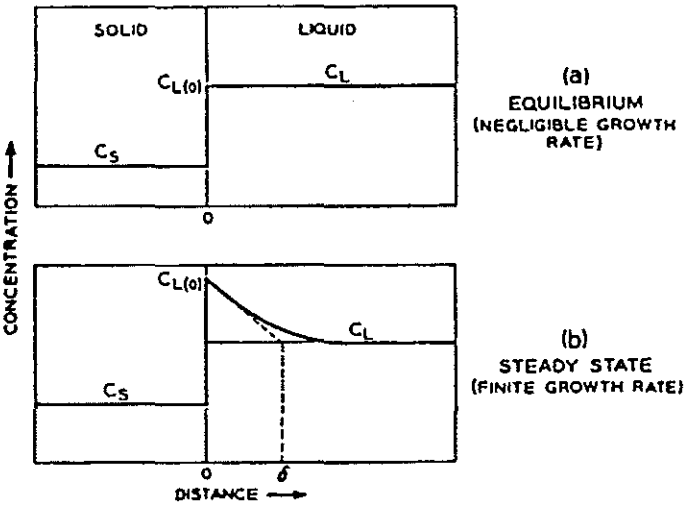


Fig. 41. Solute concentration near an advancing solid-liquid interface for different conditions of crystal growth. [After J.A. Burton, R.C. Prim and W.P. Slichter, J. Chem. Phys. **21**, 1987 (1953).]

The Burton-Prim-Slichter theory finds, with many simplifying assumptions:

$$k = \frac{1}{1 + \left(\frac{1}{k_o} - 1\right) \exp(-f\delta/D)}$$

where D = diffusivity of impurities in the liquid semiconductor ($cm^2\ s^{-1}$);
f = growth rate ($cm\ s^{-1}$); and
δ = width of the diffusion layer (cm).

Figures 42 and 43 show plots of the effective segregation coefficients as a function of the "normalized" growth rate for $k_o < 1$ and $k_o > 1$. Table 7 lists the equilibrium segregation coefficients for several impurities in germanium and silicon.

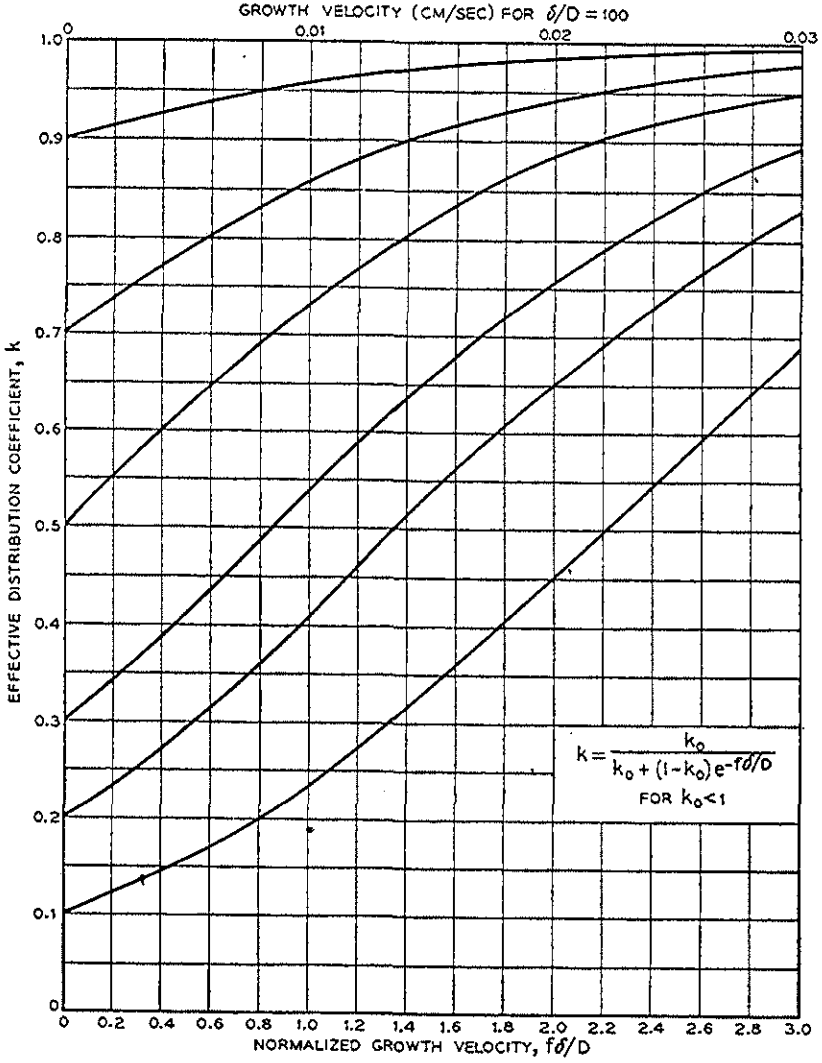


Fig. 42. Dependence of the effective distribution coefficient k, on normalized growth velocity (fδ/D), for values of equilibrium distribution coefficient ($k_o < 1$). (After W. G. Pfann, "Zone Melting," 2nd edition, John Wiley & Sons, Inc. 1966, p. 16.)

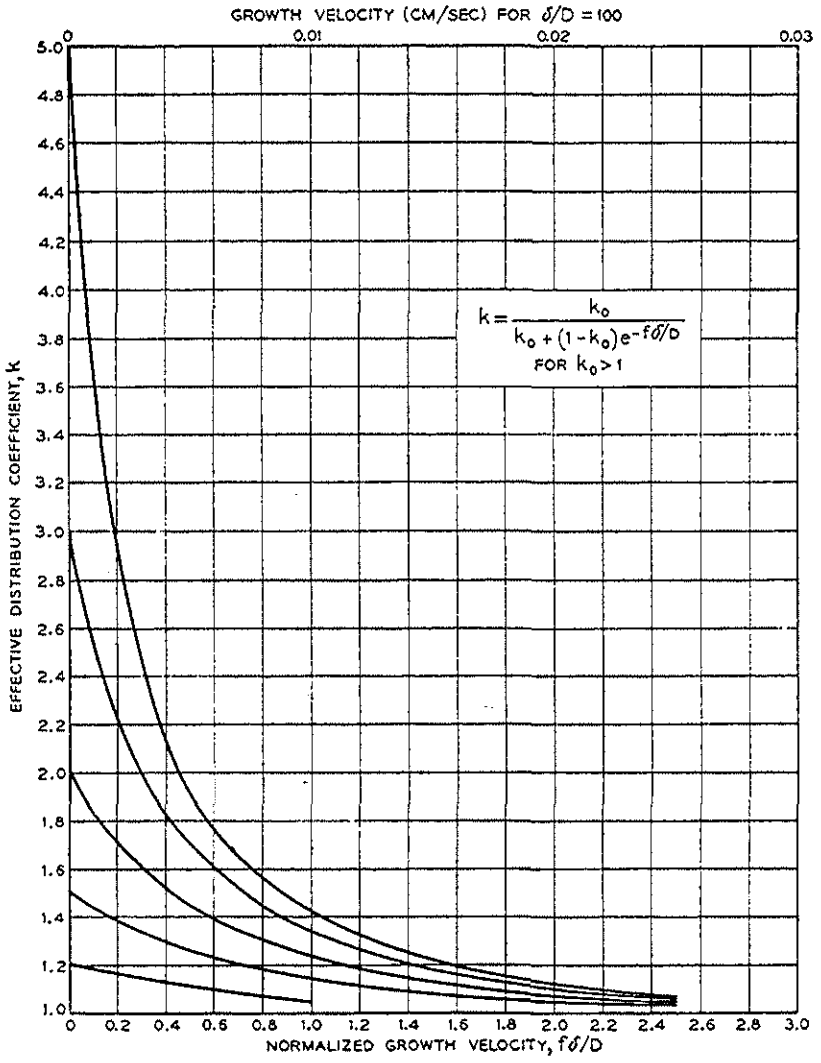


Fig. 43. Dependence of the effective distribution coefficient k, on normalized growth velocity (fδ/D), for values of equilibrium distribution coefficient ($k_o > 1$). (After W. G. Pfann, "Zone Melting," 2nd edition, John Wiley & Sons, Inc. 1966, p. 17.)

Table 7

DISTRIBUTION COEFFICIENTS AT THE MELTING POINTS OF GERMANIUM AND SILICON

[After F. A. Trumbore, Bell Syst. Tech. J. **39**, 205 (1960).]

Element	Germanium	Silicon
Lithium	0.002	0.01
Copper	1.5×10^{-5}	4×10^{-4}
Silver	4×10^{-7}	—
Gold	1.3×10^{-5}	2.5×10^{-5}
Zinc	4×10^{-4}	$\sim 1 \times 10^{-5}$
Cadmium	$> 1 \times 10^{-5}$	—
Boron	17	0.80
Aluminium	0.073	0.0020
Gallium	0.087	0.0080
Indium	0.001	4×10^{-4}
Thallium	4×10^{-5}	—
Silicon	5.5	1
Germanium	1	0.33
Tin	0.020	0.016
Lead	1.7×10^{-4}	—
Nitrogen	—	$< 10^{-7}$ (?)
Phosphorus	0.80	0.35
Arsenic	0.02	0.3
Antimony	0.0030	0.023
Bismuth	4.5×10^{-5}	7×10^{-4}
Oxygen	—	0.5
Sulfur	—	10^{-5}
Tellurium	$\sim 10^{-6}$	—
Vanadium	$< 3 \times 10^{-7}$	—
Manganese	$\sim 10^{-6}$	$\sim 10^{-6}$
Iron	$\sim 3 \times 10^{-5}$	8×10^{-6}
Cobalt	$\sim 10^{-6}$	8×10^{-6}
Nickel	3×10^{-6}	—
Tantalum	—	10^{-7}
Platinum	$\sim 5 \times 10^{-6}$	—

3. CRYSTAL GROWTH

3.1. Introduction

The band theory of solids is based, at least in its most simple form, on ideal crystals. Real crystals can be close to ideal (actually most semiconductor crystals are very close to "ideal") or may be far from ideal (polycrystalline, amorphous). The degree of perfection of a single crystal depends on the purity of the starting materials and on the particular technique used to grow the crystal. A general rule is that the lower the crystal growth temperature, the lower are the native defect and impurity concentrations.

A large number of crystal growth techniques have been invented and developed. Each one can be optimally applied to a range of materials and the final structure which is desired (e.g., bulk crystals, layered structures, etc.). Typically, the crystal growth techniques used for semiconductors are designed for high- to ultra-high purity and for outstanding control of the thermal gradients.

In the following sections we will review briefly some crystallization basics. This will be followed by brief reviews of the major bulk and thin film growth techniques which are used today for elemental (Si, Ge) and certain compound semiconductors (GaAs, GaP, AlAs, etc.).

3.2. Crystallization

The atoms in a liquid (gas) have a higher kinetic energy than in the solid. The heat of crystallization shows directly that energy is released in the phase transition from liquid (gas) to solid. When a melt is slowly cooled from a temperature above the melting temperature $T > T_m$ down to T_m we can, in general, distinguish between two cases:

(a) The melt contains at least one seed crystal. The seed crystal will begin to grow at $T = T_m$. The crystal growth depends strongly on the temperature gradients in the solid and in the liquid both near the solid/liquid interface.

(b) The melt does not contain any seed and is very pure (the second condition is typical for semiconductor crystals). In this case no crystallization occurs upon decreasing the melt temperature below T_m.

We get into the metastable supercooled range. Upon further cooling, supercooling becomes so large that spontaneous nucleation leads to many seeds simultaneously and we obtain a poly-crystalline solid. The existence of a metastable supercooling temperature range (also called Ostwald-Miers range) is of great help for single crystal growth techniques using a melt. The physical causes for the existence of this range lie in the dependence of the amount of energy released per condensed atom on the various types of nucleation sites. Figure 44 shows three different sites on a liquid/solid interface. Atoms condensing on top of a plane (1) release the least amount of energy; the ones lodging in the kink (3) release the most energy. The largest energy release leads to the largest growth velocity. The crystal planes with the smallest energy release per condensed atom grow more slowly than all the other planes and lag behind. This leads to the facets which give crystals their typical appearance.

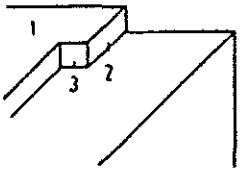


Fig. 44. Three nucleation sites for single-crystal growth.

Using these introductory remarks, we can sketch several general sequences of steps required for crystal growth–Fig. 45.

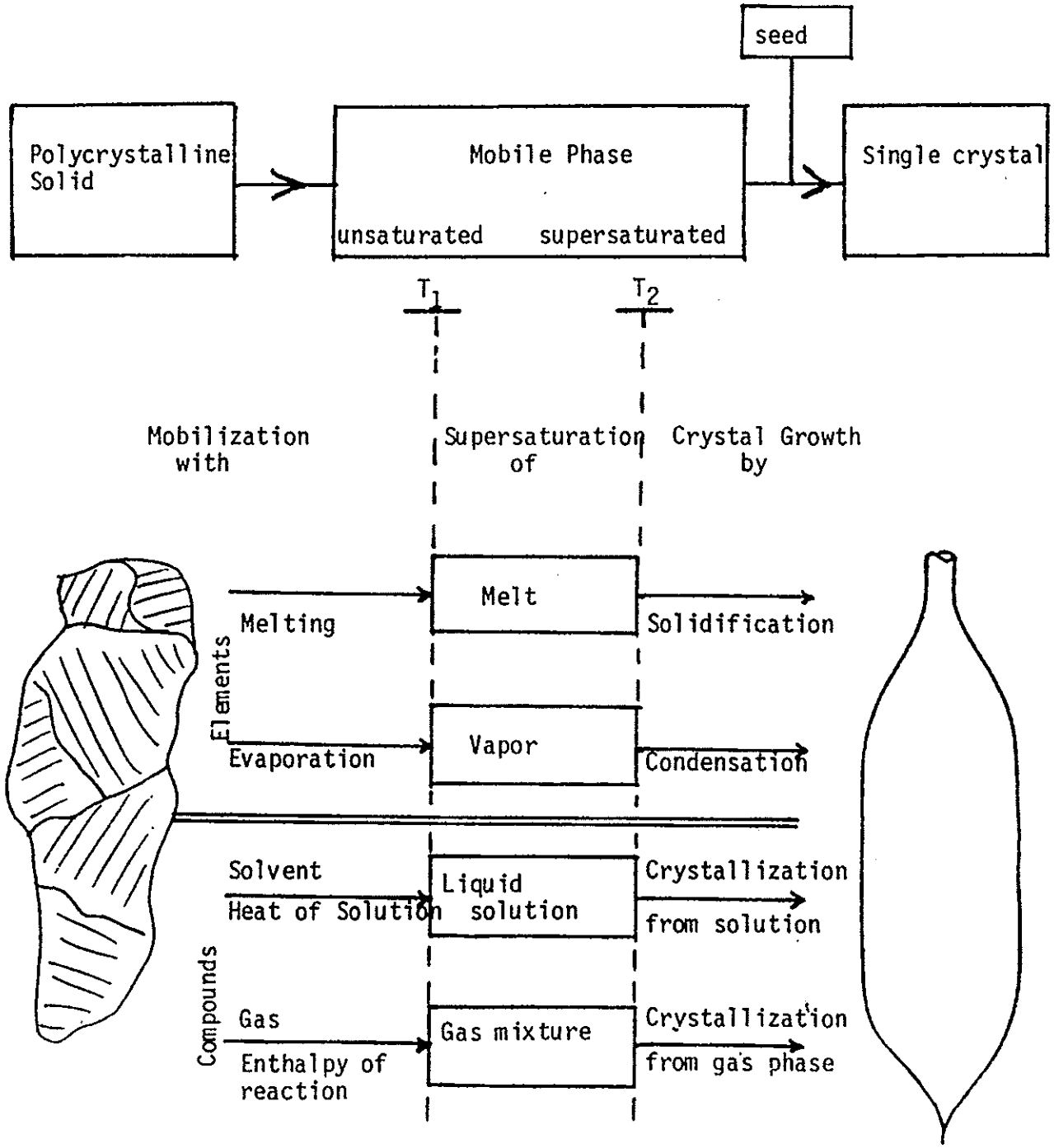

seed
Polycrystalline Solid
Mobile Phase
unsaturated
supersaturated
Single crystal
T1
T2
Mobilization with
Supersaturation of
Crystal Growth by
Melting
Melt
Solidification
Elements
Evaporation
Vapor
Condensation
Solvent
Heat of Solution
Liquid solution
Crystallization from solution
Compounds
Gas
Enthalpy of reaction
Gas mixture
Crystallization from gas phase

Figure 45

3.3. Bulk Crystal Growth Techniques

3.3.1. Bulk Growth of Si and Ge from a Melt

The oldest and still most commonly used technique for the growth of bulk semiconductor crystals was described for the first time by Czochralski.* A melt contained in a crucible (quartz, graphite or other materials) is heated by radio frequency power (RF) or by resistors. A crystalline "seed" of well-defined crystal orientation is dipped into the melt ([111], [100] are most common but many other orientations can be and are used, e.g., [113], etc.). The balance of all heat flows determines the crystal growth rate and crystal diameter. There exists a large amount of literature dealing with the modeling of the particular heat flows in crystal pullers.† We consider only the most basic situation (Fig. 46).

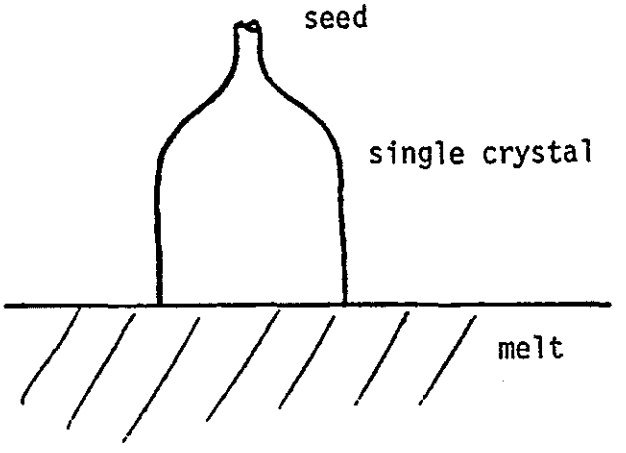


Fig. 46

Latent heat + heat transferred from melt to crystal = heat conducted away from interface by the crystal.

$$L\frac{dm}{dt} + k_l\left(\frac{dT}{dx}\right)_l A = k_s\left(\frac{dT}{dx}\right)_s A$$

L = latent heat of fusion (= heat of crystallization), $\frac{dm}{dt}$ = amount of melt freezing per unit time, $k_{l,s}$ = thermal conductivity of liquid/solid, and A = area of the liquid-solid interface.

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*J. Czochralski, Z. Phys. Chemie 92, 219 (1918).
†W. E. Langlois, Physico Chemical Hydrodynamics 2, 245 (1981).

$\left(\frac{dT}{dx}\right)_{s,l}$ = temperature gradient in the solid (s), in the liquid (l) near the interface.

With $\frac{dm}{dt} = vA\delta$,

v = growth velocity (= sum of pull velocity V_p and drop velocity of liquid surface) and δ = density of the material.

Assuming v ≈ pull velocity V_p, we obtain

$$LV_p \delta A + k_l \left(\frac{dT}{dx}\right)_l A = k_s \left(\frac{dT}{dx}\right)_s A \quad .$$

This equation is independent of A, which means that the crystal diameter could not be controlled if A and $\left(\frac{dT}{dx}\right)$ were independent of each other. The crystal growth apparatus design strongly governs the dependence of the above variables on each other. In a grower where the heat is leaving the crystal mainly by conduction to the ambient gas (Q_c) and by radiation (Q_r) we find:

$$k_s \left(\frac{dT}{dx}\right)_s A = Q_c + Q_r \quad .$$

Q_c and Q_r are proportional to the outer crystal surface S which is in turn proportional to the crystal radius R:

$$S = 2\pi Rx \text{ with } x = \text{length of crystal;}$$

$$Q_c + Q_r = Q'Rx \text{ with } Q' = \text{constant} \quad .$$

We write:

$$LV_p \delta \pi R^2 + k_l \left(\frac{dT}{dx}\right)_l \pi R^2 = Q'R \quad .$$

Division by $L\delta\pi R^2$ leads to:

$$V_p + \alpha_1 = \frac{\alpha_2}{R}$$

with $\alpha_2 = \text{constant}$ and $\alpha_1 = \frac{k_l \left(\frac{dT}{dx}\right)_l}{L\delta}$.

In the above approximation, we find the pull rate V_p to be inversely proportional to the crystal radius R. Other dependencies, such as maximum growth rate or crystal radius on temperature variation of the melt, can be found easily when the boundary conditions are defined. (See W. R. Runyan, *Silicon Semiconductor Technology*, McGraw-Hill, 1965.) Cylinder symmetry is maintained by rotation of the seed. Stirring of the melt is achieved by natural convection and by forced convection caused by the seed rotation and the counter- or co-rotation of the crucible. A typical Czochralski crystal puller is sketched in Fig. 47.

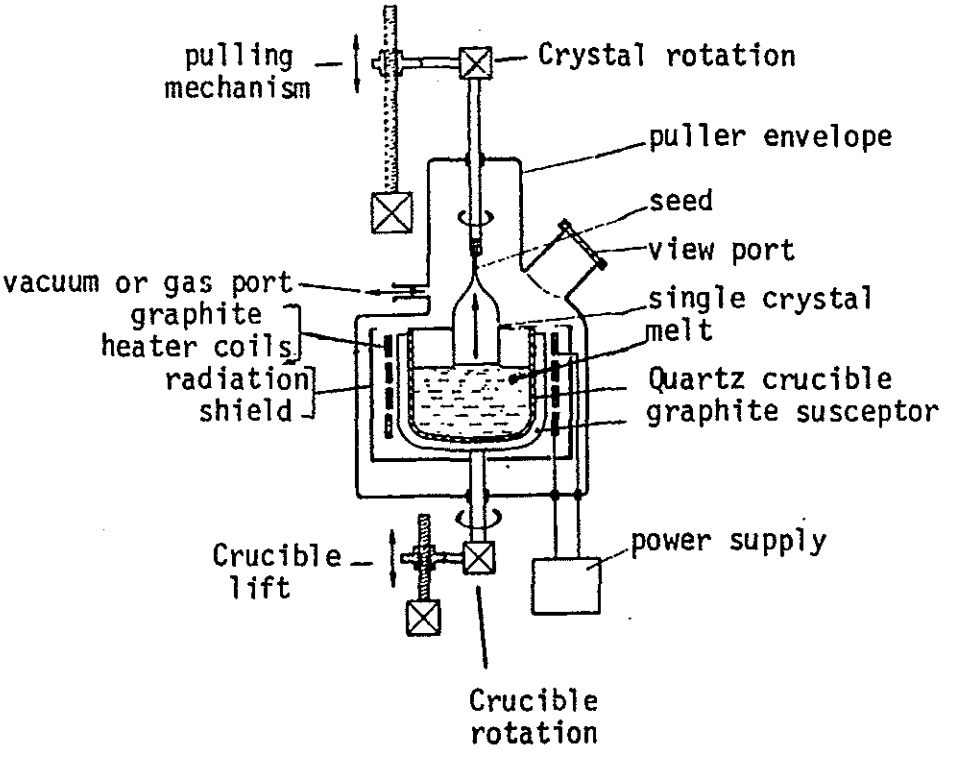


Fig. 47. Czochralski crystal puller. (After I. Ruge, "Halbleiter Technologie," Springer-Verlag, Berlin, 1975, p. 52.)

Over 80% of all single-crystal silicon (~4000 metric tons in 1980) are grown using this technique. Typical crystal diameters range from 6" to 12" diameter crystal wafers are used in device production, with 8" being the most common diameter. Up to more than 16" diameter crystals have been grown. Typical lengths reach over six feet. The advantages of the Czochralski technique lie in the simplicity of the apparatus, the large size of the crystals and ease of doping with

impurities (this is a "normal freeze" case!). Also, there are no special demands on the shape of the starting material. Polycrystalline pieces or powder can be melted down.

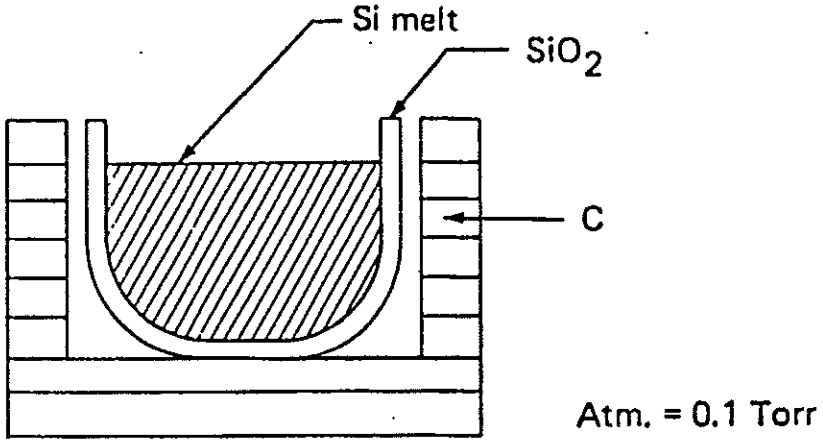
A disadvantage in the case of silicon is the reduction of the quartz (SiO_2) crucible by the melt ($T \gtrsim 1412°$ C). The free oxygen produced by the reduction dopes the crystals with 10^{17} to 10^{18} [O] cm^{-3}. Oxygen further attacks the graphite susceptor which has the function of supporting the soft quartz crucible. CO and CO_2 dope the Si melt and lead to high carbon concentrations ($4 \times 10^{16} - 5 \times 10^{17}$ cm^{-3}). To illustrate the complexity of all the possible reactions, a list of such reactions, studied by Schmidt et al., is included (Fig. 48).

Boron, phosphorus and other impurities in the quartz are released by the reduction of the crucible and they dope the melt and the crystal. The highest resistivity obtainable for Si grown from the melt contained in quartz is typically a few tens Ω cm ($|N_A - N_D| \approx 10^{14} - 10^{15}$ cm^{-3}). It was recognized early on that a reduction of the melt flow velocities could lead to reduction in impurity concentrations because the impurities released in the reduction of crucible material would no longer be transported to the growing crystal. Use of either axial or transverse magnetic fields of a few k Gauss strength can bring all melt convections to a halt. Crystals grown in such machines exhibit, as expected, more uniform and better controlled oxygen concentrations. The residual boron and phosphorus concentrations are lower. The increase in equipment cost has limited "magnetic Czochralski" to the growth of special crystals (see R. N. Thomas et al., *Solid State Technology*, March 1990, p. 83 ff). Figure 49 shows three stages of a CZ silicon pull sequence.

Germanium, with its melting point at 937° C, is much less affected by the quartz reduction. This is the major reason why it has been possible to grow ultra-pure Ge crystals with $|N_A - N_D| \approx 10^{10}$ cm^{-3} and low compensation. Doped Ge crystals can also be grown from a melt contained directly in the graphite susceptor because Ge and C do not react, and the small concentrations of boron and phosphorus introduced by the graphite susceptor are negligible compared to typical dopant concentrations.

3.3.2. Silicon Floating Zone (FZ) Crystal Growth

For silicon it is clearly desirable to find a crystal growth technique which does not rely on any crucible. The floating zone technique has been developed as an answer to this demand. Figure 50 shows a typical floating zone apparatus. Usually the RF coil is located inside the crystal grower. The density of liquid silicon is small (2.42 g cm^{-3}) and its surface tension is high (720 dyn cm^{-1}). This allows one to go to rather large-diameter crystals. In addition, one can



A

1) $2C + SiO_2 \longrightarrow SiC + CO_2$
2) $3C + 3SiO_2 \longrightarrow SiC + 2CO_2 + 2SiO$
3) $4C + 2SiO_2 \longrightarrow SiC + 3CO + SiO$
4) $3C + SiO_2 \longrightarrow SiC + 2CO$
5) $C + SiO_2 \longrightarrow SiO + CO$
6) $C + 2SiO_2 \longrightarrow 2SiO + CO_2$
7) $Si + SiO_2 \longrightarrow 2SiO$

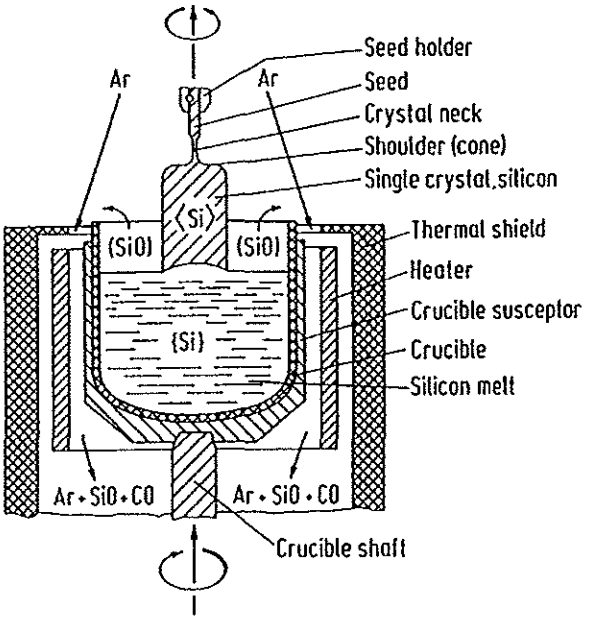
B

8) $C + CO_2 \longrightarrow 2CO$
9) $CO + 3SiO \longrightarrow SiC + 2SiO_2$
10) $2C + SiO \longrightarrow SiC + CO$
11) $3C + 2SiO \longrightarrow 2SiC + CO_2$
12) $C + SiO \longrightarrow Si + CO$

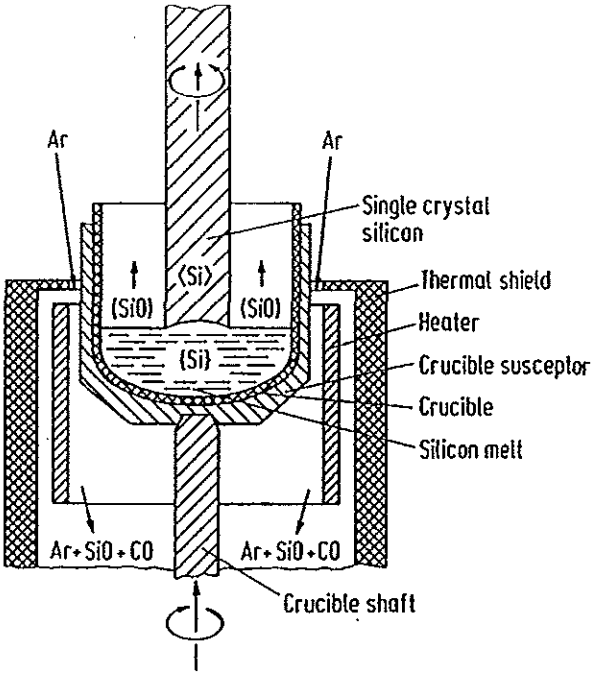
C

13) $Si + CO \longrightarrow SiC + \frac{1}{2}O_2$
14) $3Si + 2CO \longrightarrow 2SiC + SiO_2$
15) $2Si + CO \longrightarrow SiC + SiO$

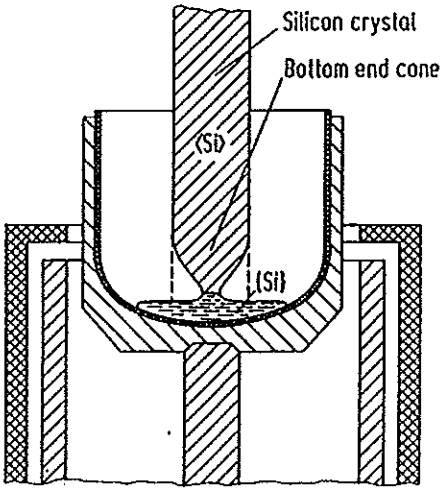
Fig. 48. Example: SiC in Si. [From: Schmidt *et al.*, J. Electrochem. 126, 935 (1979).]



(a) Si. Czochralski single crystal pulling. Initial stage of the process. ⟨Si⟩ indicates solid, and {Si} indicates liquid silicon. (SiO) is a gaseous reactant.



(b) Si. Advanced stage of Czochralski single crystal pulling. Arrows indicate the gas flow. ⟨Si⟩ is solid, and {Si} is liquid silicon.



(c) Si. Final stage of Czochralski single crystal pulling. If the crystal has the shape of the dashed lines, then the crystal cannot be finished dislocation-free. ⟨Si⟩ is solid silicon. {Si} is liquid silicon.

Figure 49

choose appropriate RF coil designs which help support the melt through Eddy-current forces. The melt and coil for large diameters (3-5") are shown in Fig. 51.

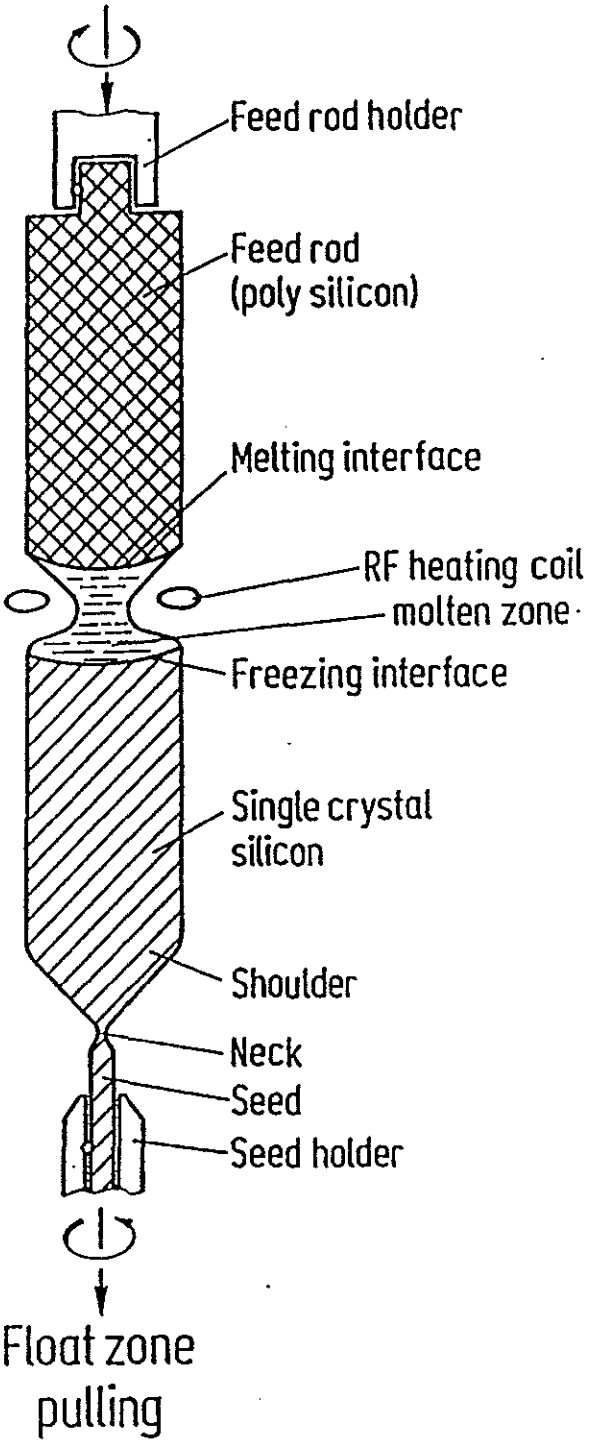


Figure 50

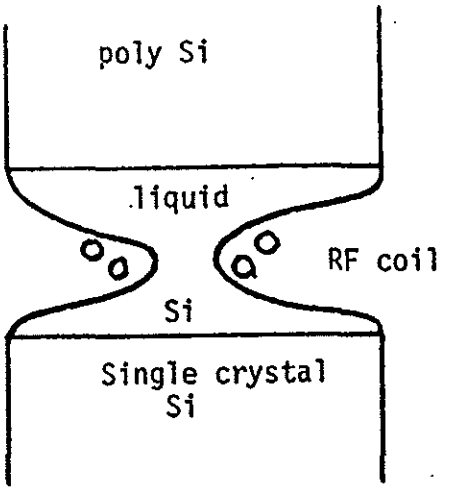


Figure 51

To insure good mixing and uniform growth across the melt/solid interface, the "feed" polycrystal and the single crystal are counter-rotated. The RF coil is usually stationary while the single crystal and polycrystal cylinders are slowly moved. The *mechanical design* of a modern floating zone machine (crystals ~6' long, 5" diameter) is extremely demanding (see Fig. 52). Near the end of the crystal pulling run, the whole crystal is balanced on the seed, with a few support clamps near the seed end.

Typical growth rates for both the Czochralski and the floating zone techniques are up to a few mm per minute. Inert gases (Ar, He for Si; Ar, N_2, H_2 for Ge) or vacuum are used as ambient.

Floating zone combined with vacuum increases considerably the effective purification for impurities which have high vapor pressures. This observation is used to remove phosphorus, usually the dominant impurity in polycrystalline Si produced by the pyrolysis of $SiHCl_3$. After one or two vacuum zone passes, the crystal turns p-type because the boron remains in the crystal (boron has a very low vapor pressure *and* the segregation coefficient close to unity).

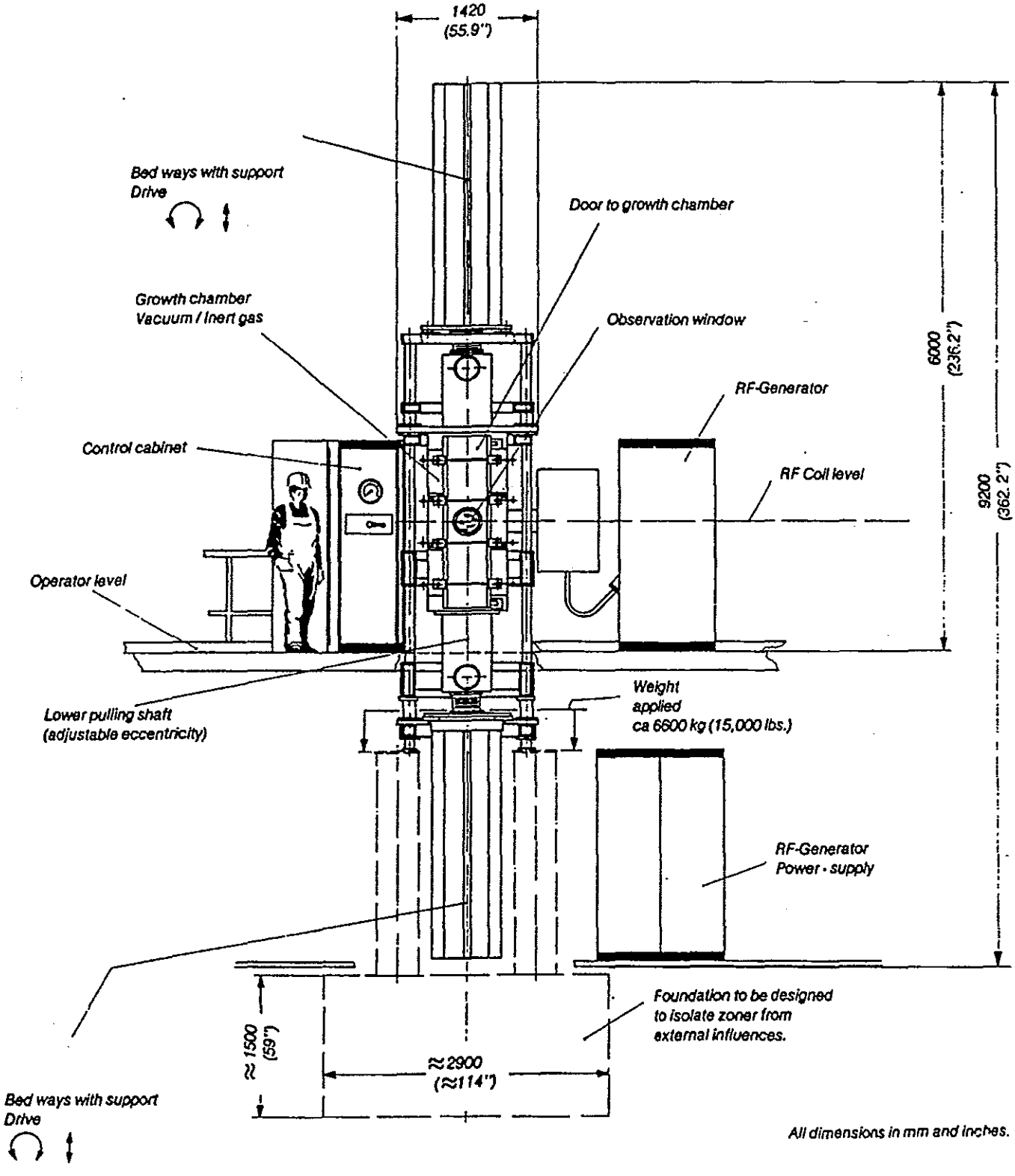


Fig. 52. Siemens Floating Zone Puller, Model VZA9.

Already in the early 1960s it was possible to produce Si single crystals with resistivities close to 100 kΩ cm using the FZ technique. This resistivity is close to intrinsic at room temperature. Oxygen and carbon concentrations are also much lower than in CZ-grown crystals ([O] < 10^{15} cm^{-3}, [C] < 10^{15} cm^{-3}).

3.3.3. Bulk Growth of III-V Compound Semiconductors from the Melt

With the exception of InSb and GaSb, the III-V compound semiconductors exhibit high dissociation pressures and component vapor pressure at the melting point. Both the CZ and FZ crystal growth techniques have been used to grow III-V compound semiconductors; however, special precautions have to be taken in regard to the high vapor pressure component. Figure 53 shows the vapor pressures of some important elements. In the particular case of GaAs (T_{melt} = 1238° C), the As pressure is so high that a stoichiometric melt would rapidly become non-stoichiometric if one did not take countermeasures. One possibility is to keep an As vapor pressure established above the melt which corresponds precisely to the dissociation pressure (~0.9 atm). In the case of phosphides (GaP, InP), one has to go to very high pressures (60 atm for InP). In this case one often chooses the so-called liquid encapsulation technique (LEC). Liquid B_2O_3 swims on top of the melt and forms a transparent seal which suppresses the escape of the volatile element.

Typically III-V bulk crystals are much less pure than Si or Ge. Today III-V crystals are sliced into wafers which are used like Si wafers for device fabrication or as substrate materials. GaAs bulk crystal growth has evolved rapidly in recent years because of an increased interest in both the optical and the high-speed properties of this material. Doped and semi-insulating GaAs crystals are grown with the liquid encapsulated Czochralski and the Bridgman techniques. The latter technique usually uses a horizontal semicircular boat with a seed at one end. Vertical Bridgman growth producing crystals fully enclosed by the crucible are also grown.

Most GaAs crystals are either doped n-type or are semi-insulating (SI). For n-type crystals silica crucibles can be used because Si contamination leads to donor doping (Si occupies preferably Ga sites where it acts as a donor). SI GaAs crystals can be grown by using excess As which leads to Ga vacancies and arsenic antisites As_{Ga} (an arsenic atom occupying a Ga site). The antisite or a defect complex containing an antisite form deep double donors. If a crystal is lightly p-type, without these deep donors, the antisite double donors can overcompensate the net-acceptors and the Fermi level moves towards the center of the bandgap. In this state the crystal has a free carrier concentration which approaches the intrinsic carrier concentration. SI GaAs crystals are highly

resistive, 10^7 to 10^9 Ωcm, an advantage when they are used as substrates for very high speed electronic circuits.

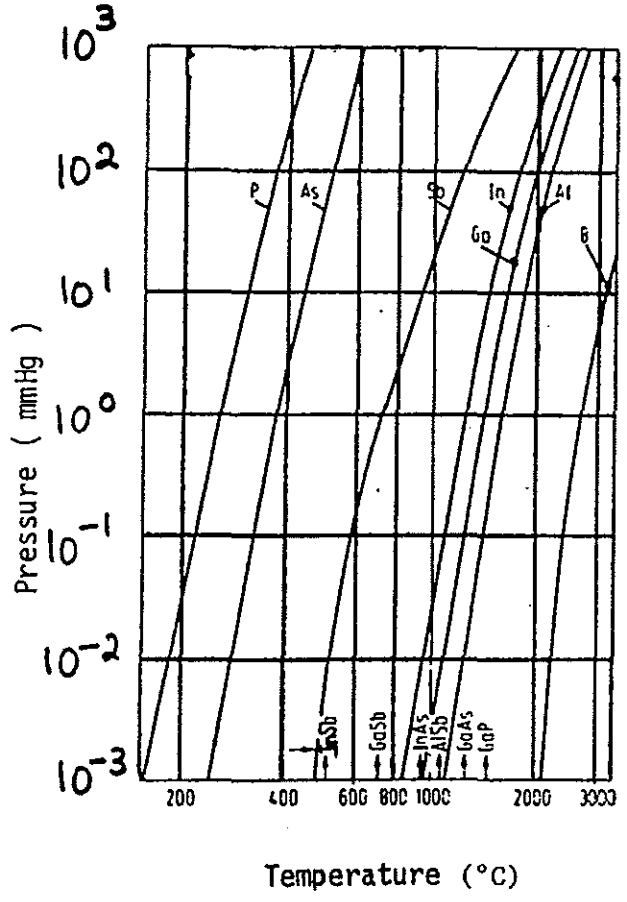


Fig. 53. Vapor pressure curves of various group III and group V elements.

3.3.4. Melt Convection and Impurity Distribution

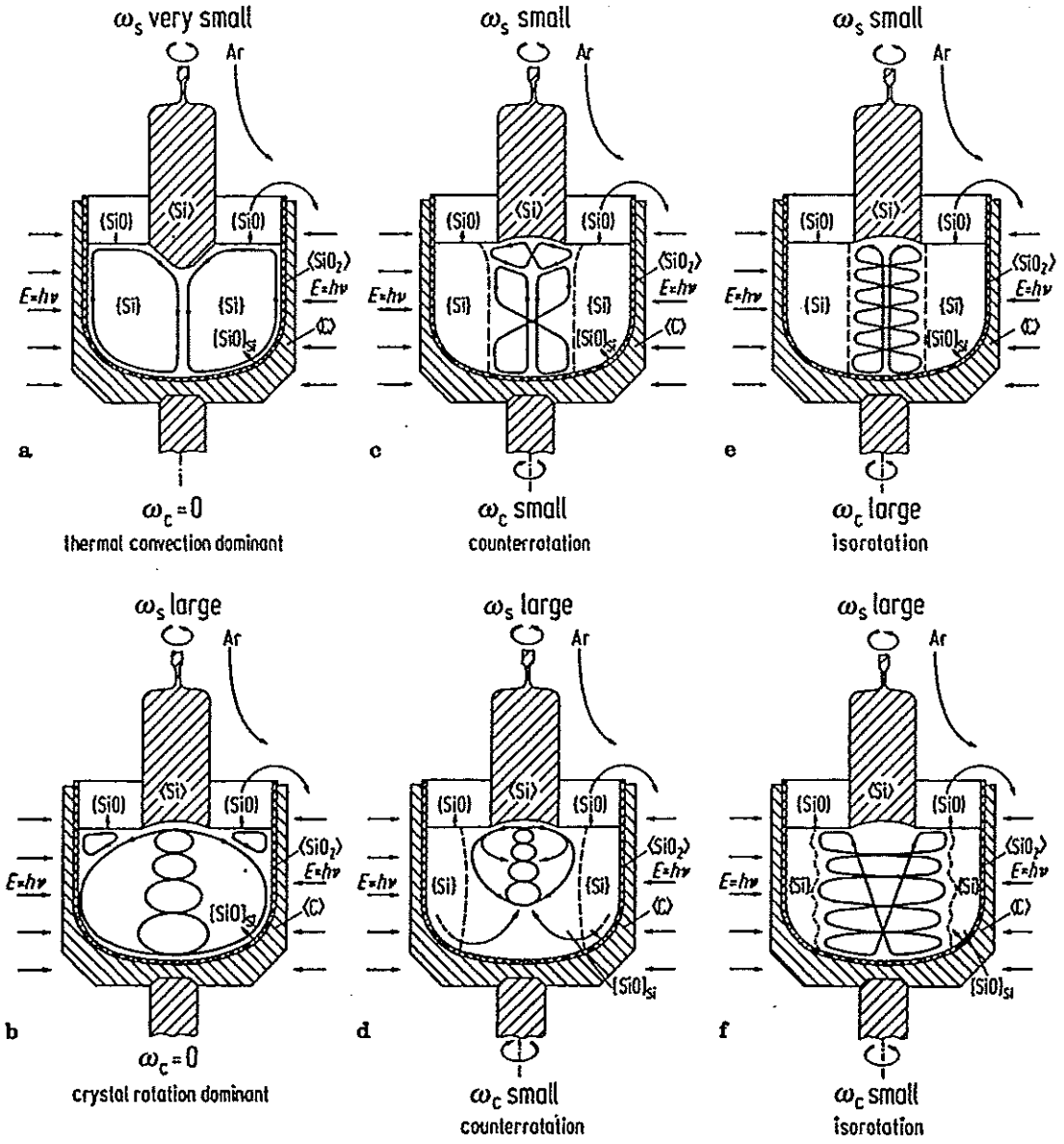
A semiconductor melt in a crucible experiences forces which are due to density variations caused by temperature differences or forces produced by mechanical motion. The former forces lead to natural convection and the latter to forced or driven convection.

Our interest in melt flow patterns and velocities stems from the domination of the thermal gradients near the crystal growth interface by melt flow and from the effect of liquid flow on the effective impurity segregation coefficients. The thermal gradients affect the liquid-solid interface morphology as well as point and line defect generation, while the variations in the effective

segregation coefficients generate dopant fluctuations in the crystal. It is extremely difficult to determine experimentally convective flow patterns in a Si melt at temperatures above 1400° C. Few materials can withstand such an environment. In addition, the melt is highly conductive and absorbs infrared or visible light readily. This in turn makes remote imaging of melt flows difficult if not impossible. Transparent model liquids containing reflective particles have been used for imaging simulation. It is, however, practically impossible to reproduce a representative model situation with liquids which have different densities and viscosities and with different thermal gradients. Computational methods based on finite element analysis techniques have been developed in recent years. These methods produce impressive results.

Figures 54a-f illustrate schematically melt flow patterns for typical Czochralski growth situations. The interaction of the natural and driven convective forces leads to a very rich set of flow patterns.

As mentioned earlier, it has been demonstrated that axial or transverse magnetic fields of a few k Gauss can stop the melt flow. Because mixing in the melt is suppressed, one obtains lower concentrations of oxygen, carbon and electrically active impurities such as P, Al and B. Furthermore, the concentrations of these impurities vary very slowly axially and radially.



a. Si. Flow characteristics in a CZ-crucible dominated by thermal convection [67C, 68C, 76C1, 77C].

b. Si. Flow characteristics in a CZ-crucible dominated by crystal rotation [67C, 68C, 76C1, 77C].

c. Si. Flow characteristics in a CZ-crucible dominated by counterrotation of the crystal and the crucible at small frequencies [67C, 68C, 76C1, 77C].

d. Si. Flow characteristics in a CZ-crucible dominated by counterrotation with a fast rotation of the crystal [67C, 68C, 76C1, 77C].

e. Si. Flow characteristics in a CZ-crucible dominated by fast rotation of the crucible and slow rotation in the same direction of the crystal [67C, 68C, 76C1, 77C].

f. Si. Flow characteristics in a CZ-crucible dominated by fast rotation of the crystal and slow rotation in the same direction of the crucible [67C, 68C, 76C1, 77C].

Fig. 54a-f. Schematic illustration of melt flow patterns for various crucible and seed rotation conditions in a typical Czochralski puller.

Compared to melt flow during CZ growth, the flow behavior in the liquid zone during floating zone crystal growth must be called violent (Fig. 55). Large forces are produced by the thermal gradients and electromagnetic induction from the RF heating field.

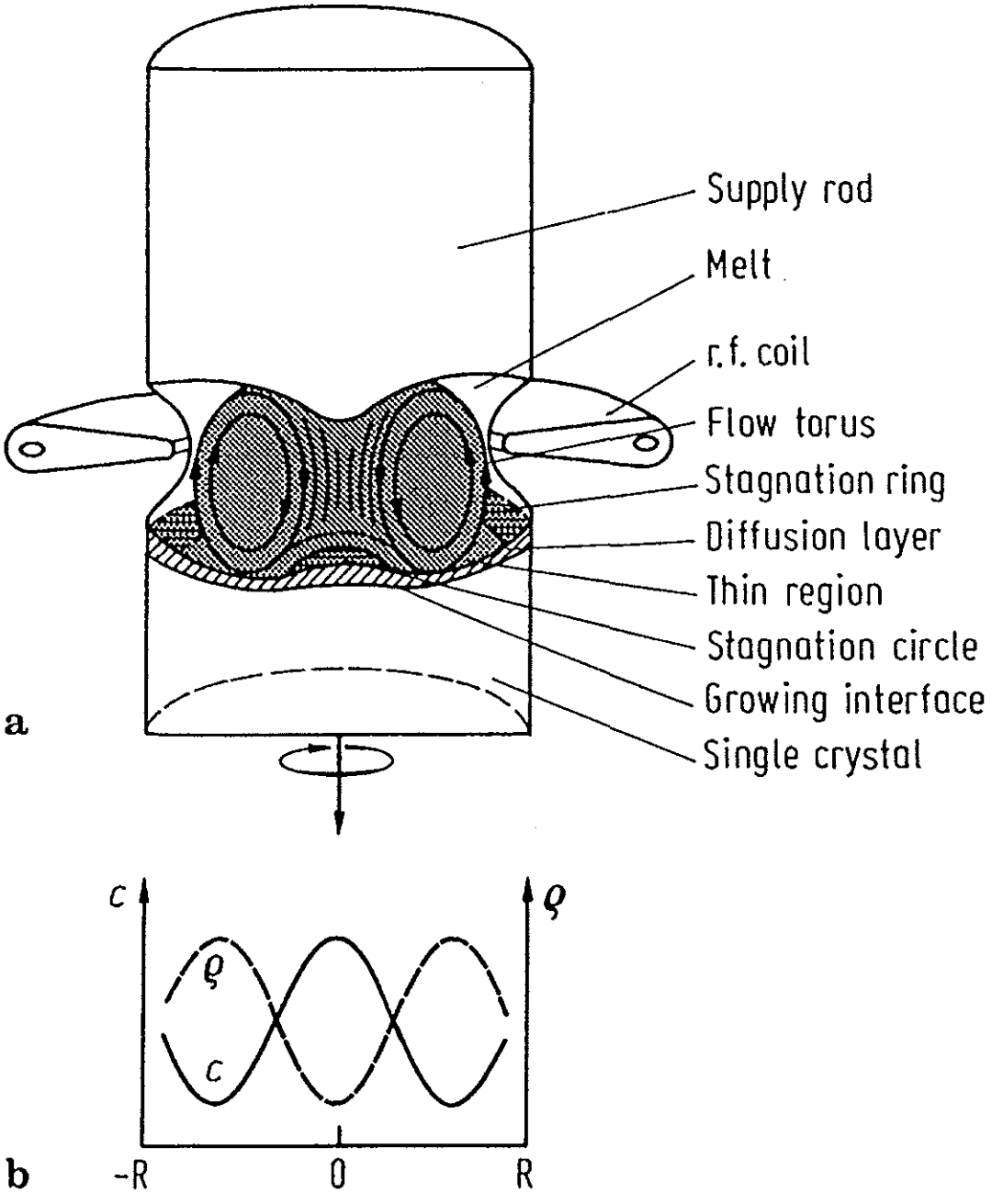


Fig. 55. Si. Model of flow pattern during float-zone growth, and dopant distribution. (a) Flow torus and diffusion layer thickness; (b) radial dopant concentration c and resistivity profile ρ.

In order to estimate the effect of fluid flow on the local effective segregation coefficient one needs to know the stagnant layer thickness δ, the local growth rate f, and the impurity diffusion coefficient in the liquid D (see Fig. 41, page 75). Each of these variables is very difficult to determine and we typically infer them *after* crystal growth by analyzing the impurity distribution patterns in the crystal. The radial microresistivity patterns of 3" diameter crystals grown along the <100> and <111> orientations by both the CZ and the FZ techniques are shown in Fig. 56. The so-called dopant striations lead in some cases to resistivity fluctuations of over 20%. In FZ growth it can happen that some sections of the solid-liquid interface grow and then remelt some distance during each crystal rotation. This "melt-back" causes the large dopant fluctuations seen for doped FZ crystals. We will come back to dopant striations when we discuss semiconductor doping techniques.

3.3.5. Semiconductor Bulk Crystal Growth Literature

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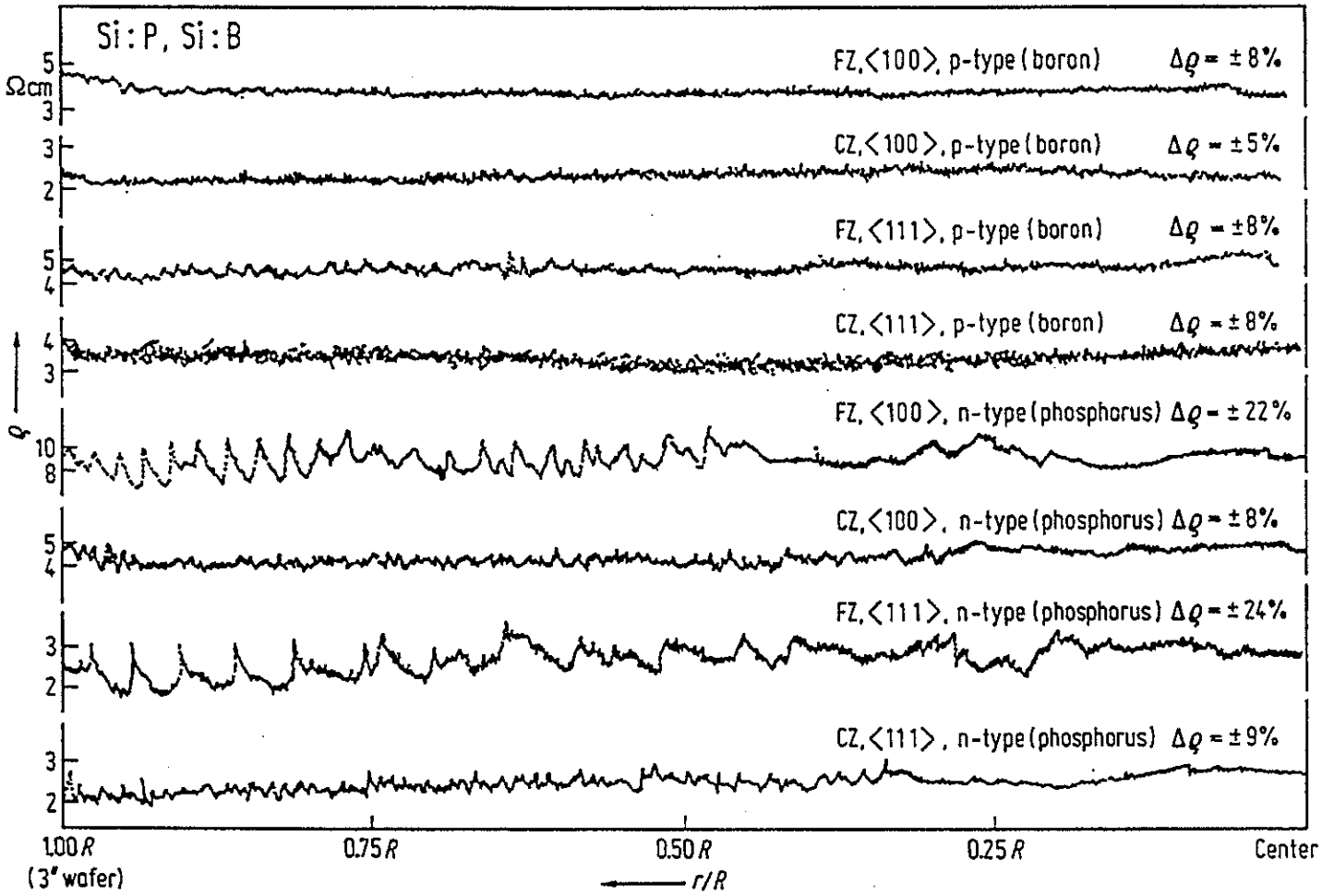


Fig. 56. SiP, Si:B. Spreading resistivity ρ vs. radial position r/R. Comparison of Czochralski grown (CZ) with float zone (FZ) silicon crystals of the same specification by spreading resistance measurements. The dopant striations in float zone crystals are greater due to the pronounced remelt phenomena at every crystal rotation. Parameters are the growth type, the conductivity type and dopant, and the measured resistivity variation $\Delta\rho$.

3.4. Epitaxial Growth Techniques

3.4.1. Introduction (See D. W. Pashley, "The Basics of Epitaxy," in GCS.)

Epitaxy, or the growth of a crystalline film on a crystalline substrate with defined orientational relationship, was first practiced and studied with alkali halides. A serious interest in Si epitaxy arose when lightly doped layers on heavily doped substrates were required for the fabrication of junction field effect transistors. Growth of pure Si on doped Si is a form of **homoepitaxy**. Today there exists a great interest in multilayer growth of different semiconductors on top of each other. Hundreds of $Al_xGa_{1-x}As$ layers with $0 < x < 1$ can routinely be grown today with excellent crystallinity and good control of purity and doping concentration. Thin film growth with layers of different compositions is a form of **heteroepitaxy**.

Heteroepitaxy is used to form **quantum wells**, thin layers of one type of semiconductor sandwiched between another type of semiconductor with bandedges at higher energy. A GaAs layer of a few tens of Å thickness located between two $Al_xGa_{1-x}As$ "barriers" forms a quantum well (QW). A number of new and exciting properties have been discovered with quantum well structures. The integer and fractional quantum Hall effects (QHE) occur in such layered structures. The integer QHE won R. von Klitzing his Nobel prize in 1985 and the fractional QHE followed suit in 1998 for H. Störmer, D. Tsui and R. Laughlin. The resonant tunneling diode uses one quantum well for ultra-fast switching purposes. Such diodes can turn on and off exceeding 6×10^{11} times per second (> 600 GHz).

Superlattices consist of a series of quantum wells grown on top of each other. Structures with several hundred layers have been grown successfully. Besides exhibiting interesting solid state physics, such superlattices are beginning to be used in electronic devices.

As illustrated in Fig. 45 (page 80), every kind of crystal growth requires a means of rendering the atoms of the material to be grown mobile. Epitaxy makes use of essentially three basic techniques: **Liquid phase epitaxy (LPE)** uses a solvent to generate mobility of the growth substance (the solute). **Vapor phase epitaxy (VPE)**, a form of chemical vapor deposition (CVD), is based on the formation of a vapor of molecules containing the growth atoms. Metal organic chemical vapor deposition (MOCVD), also called organometallic vapor phase epitaxy (OMVPE), belongs to the VPE techniques and is used mostly for growth of III-V and II-VI compound semiconductor films. **Molecular beam epitaxy (MBE)** and its recent offspring chemical beam epitaxy (CBE), also called metal organic MBE (MOMBE), is a sophisticated form of ultra-high

vacuum evaporation in which the individual molecules fly along straight trajectories from a source to the substrate without interacting on the way.

Each of the epitaxial growth techniques has a number of advantages and disadvantages. These will be discussed in the following sections.

3.4.2. Liquid Phase Epitaxy (LPE)

Liquid phase epitaxy is a low-temperature process and can produce very pure epitaxial layers. The investment in equipment is very small compared to other epitaxial techniques. The liquid solvent for semiconductor LPE is typically a metal. Ga, Bi, In, Sn and other metals have been used for Si and III-V compound semiconductor epitaxy. In recent years the importance of LPE has diminished because the interfaces between adjacent layers are not atomically sharp and the composition or[?] the dopant concentrations cannot be conveniently varied in a continuous fashion. For the production of light-emitting diodes LPE is still the most efficient epitaxial growth technique.

The basic growth sequence starts with the saturation of the solvent with the materials to be deposited. A clean, flat substrate is brought in contact with the solvent. Upon slow cooling of the solvent, solute material will grow epitaxially onto the substrate. H. Nelson [RCA Review **24**, 603 (1963)] was the first to use this basic idea. Figure 57 shows his simple growth apparatus. A little piece of GaAs floats on Sn, the solvent. The liquid Sn becomes saturated with GaAs at a given

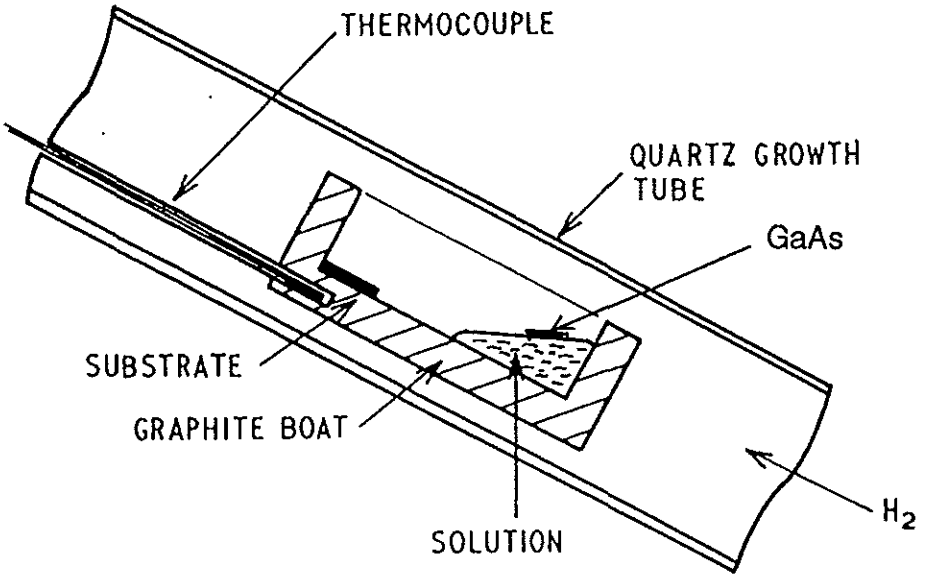


Fig. 57. Schematic diagram of tipping LPE system.

temperature T. Tilting the apparatus so that the Sn flows over a GaAs single crystal substrate followed by slowly lowering the temperature starts the epitaxial growth. Layers of up to 20 μm thickness were grown in this simple apparatus.

Figure 58 shows a modern, multi-solvent epitaxy system. A substrate wafer is held in a graphite slider. With a simple push rod the wafer can be positioned under four different solvent containers, each of which can hold a different combination of semiconductor alloy and dopant. P-n homojunctions and a variety of p-n, n-n and p-p heterojunctions have been grown for LED and laser applications. The performance summary of LEDs given in Table 8 is by now somewhat outdated, but it contains a few basic and valuable pieces of information.

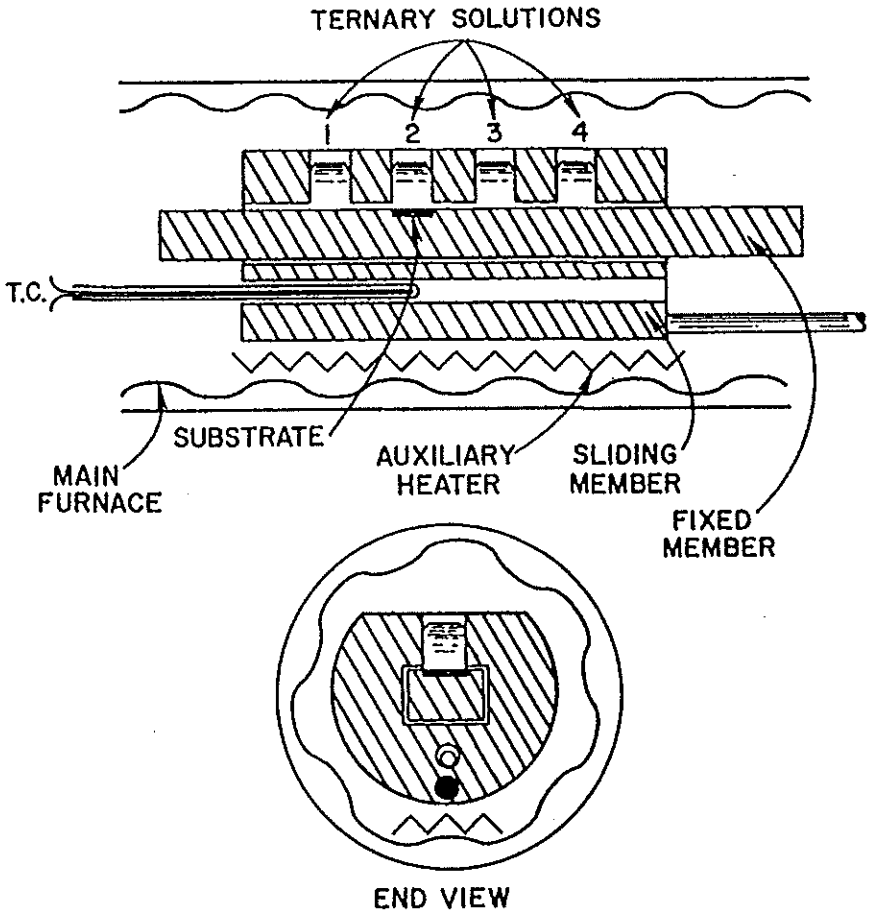


Fig. 58. Multicompartment slider apparatus for growing multiple-layered semiconductor structures by liquid phase epitaxy.

TABLE 8

Reported performance of the commonly used LED materials as affected by composition, substrate material, and junction formation techniques

Semiconductor material		Junction formation		Dominant wavelength (nm)	Colour	External quantum efficiency (%)	Luminosity $\ell m\ W^{-1}$	$\ell m\ A^{-1}$	Reference
Substrate	Junction	n-Side	p-Side						
GaAs	GaAs:Si/Zn	substrate	Zn diff.	~ 910	IR	3·5	–	–	38
GaAs	GaAs:Si	LPE	LPE	~ 970	IR	32	–	–	39
GaAs	$Ga_{1-x}Al_xAs$	LPE	Zn diff.	> 700	IR	13·5	–	–	41
GaAs	$GaAs_{0.7}P_{0.3}$	CVD	Zn diff.	–	Red	0·7	~ 12	0·08	40
GaAs	$GaAs_{0.6}P_{0.4}$	CVD	Zn diff.	~ 680	Red	0·2	~ 100	0·17	42*n*
GaP	$GaAs_{0.6}P_{0.4}$	CVD	Zn diff.	~ 680	Red	00·9	~ 100	0·4	42*n*
GaP	$GaAs_{0.35}P_{0.65}N$	CVD	Zn diff.	~ 630	Red–orange	0·65	~ 150	1·9	42*n*
GaP	$GaAs_{0.1}P_{0.9}$:N	CVD	Zn diff.	~ 580	Yellow	0·08	~ 540	0·85	42*n*
GaP	GaP:Zn–O	LPE	Zn diff.	637	Red	1·5	~ 20	0·57	42, 46
GaP	GaP:Zn–O	LPE	LPE	637	Red	12*	~ 20	4·6	44
GaP	GaP:N	LPE	Zn diff.	–	Yellow	0·1	–	–	42*j*
GaP	GaP:N	CVD	Zn diff.	574	Yellow–Green	0·1	~ 560	1·1	42*m*
GaP	GaP:N	LPE	Zn diff.	570	Yellow–Green	0·1	~ 600	1·2	42*l*
GaP	GaP:N	LPE	LPE	568	Yellow–Green	0·33	~ 630	4·05	42*a*
GaP	GaP:N	LPE	LPE	570	Yellow–Green	0·7	618	8·44	47, 47*a*

* The originally reported value of 15 per cent has been revised to 12 per cent (R. Solomon, private communication).

(after A.A. Bergh and P.J. Dean, "Light Emitting Diodes," Clarendon Press, Oxford, 1976, p. 392.)

Besides the technological applications of LPE, this crystal growth technique has been very useful in the study of fundamental melt growth mechanisms. E. Bauser ["Advances in Solid State Physics" XXII, 141 (1983) and *Crystal Growth of Electronic Materials*, E. Kaldis, ed., Elsevier Science Publ., 1985, p. 44 ff] has demonstrated the strong influence of crystal orientation on crystal growth mechanisms and epi layer morphology. LPE of Si on a spherically ground and polished single crystal substrate results in at least four kinds of growth mechanisms (Fig. 59). A central atomically flat facet of up to 1 cm diameter develops perpendicular to the [111] orientation. This area is surrounded by a terraced surface. Once the surface normal deviates more than 2° from the [111] direction, the surface morphology becomes again very smooth (Fig. 60a, b). Bauser was able to evaluate directly the effect of dislocations on layer morphology and the incorporation of dopants for the various growth mechanisms. Figure 61 illustrates the effects of obstacles in the facet growth and the terraced areas.

In order to grow multilayer systems, Bauser has developed a rotating LPE growth system in which four wafers can be placed (Fig. 62). Four liquids with different compositions and dopants can be transported from one wafer to the next simply by spinning the growth reactor at various speeds. The liquids are forced outwards and rise through thin tubes into intermediate holding containers. When the rotation is slowed down, the metal solvents flow into the next substrate chamber and a new layer can be grown on all four substrates. This advanced LPE technique appears to be quite successful insofar as several hundred layers of n- and p-Si have been grown sequentially. An interesting observation is that the crystalline perfection is improving as the number of layers grown increases.

Advantages of LPE are: high purity, low equipment and operating costs. Disadvantages include: Composition and doping of the films cannot be changed continuously; selective growth through windows in masks is difficult especially for small dimensions; and small throughput.

3.4.3. Vapor Phase Epitaxy (VPE)

Chemical Vapor Deposition (CVD) is most extensively used by the semiconductor industry to grow thin films of crystalline and polycrystalline layers of Si and silicides, of a wide variety of compound semiconductors, of metals and of insulators. A modern Very- (or Ultra-) Large Scale Integrated (VLSI or ULSI) circuit consists of a thick substrate (300 to 500 μm) and numerous thin layers each fulfilling one or more device, interconnect, insulation or other functions. In the following, we will focus on Si and compound semiconductor VPE.

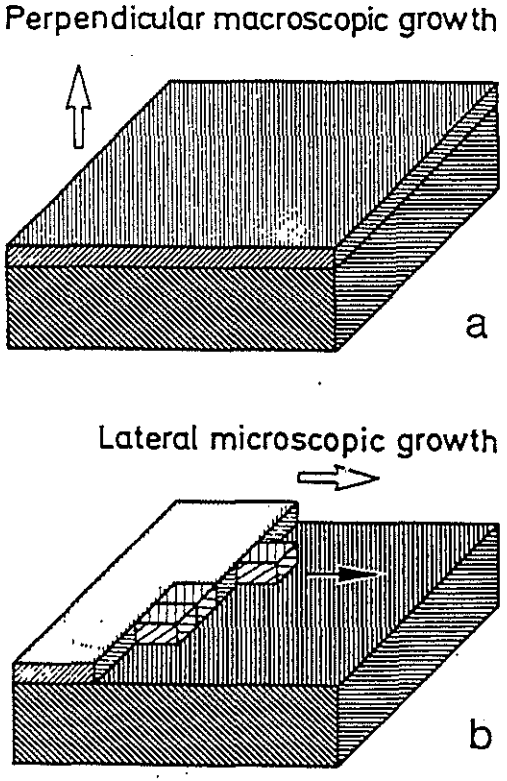
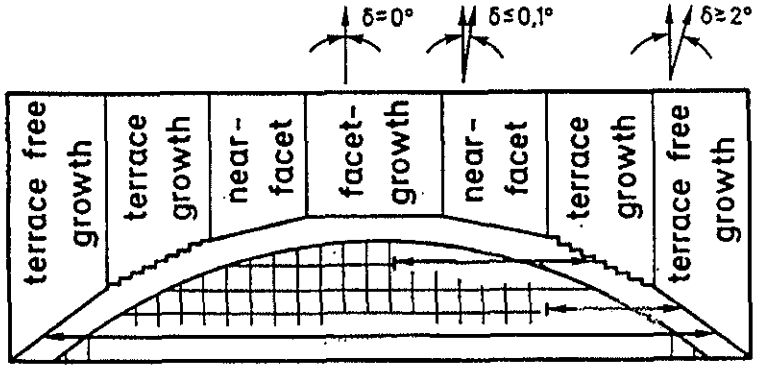


Fig. 59. (a) Macroscopic crystal growth occurs perpendicular to the growth interface. Macroscopic growth arises from the superposition of lateral microscopic growth. (b) Lateral microscopic growth occurs by attachment of atoms at the edges of steps, and hence, a lateral motion of the growth steps.



(c) Spherical substrate with LPE layer, schematic cross section. The misorientation angle δ is exaggerated in the schematic. Dislocations act as growth step sources where misorientation steps are absent, i.e. for $\delta = 0$. In the other areas, misorientation steps are available as growth steps, and dislocations do not influence growth kinetics.

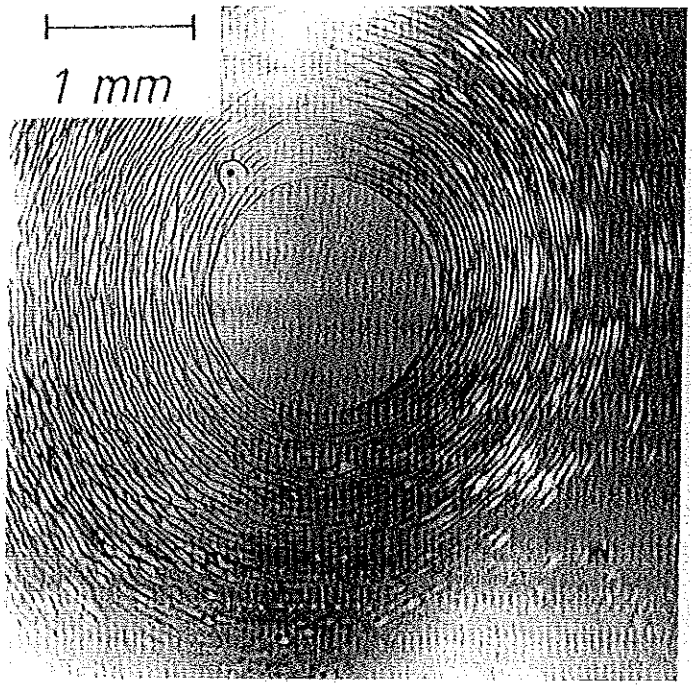
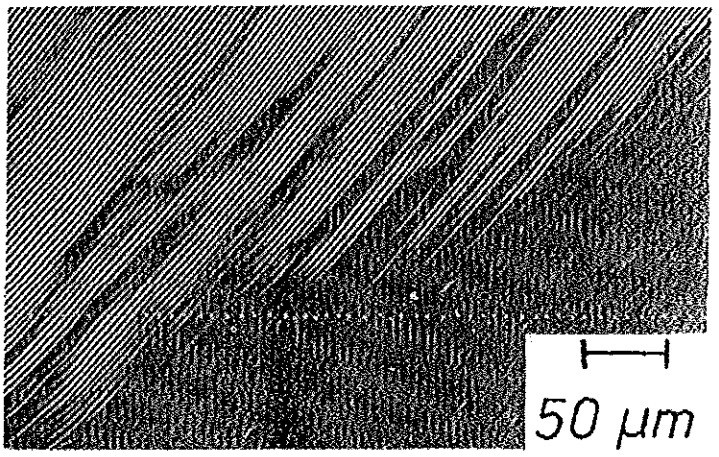


Fig. 60. (a) Surface of GaAs layer grown on spherical substrate (radius of curvature 0.3 m). The central facet is surrounded by a terraced area; and the latter by a terrace-free area near the wafer edge.



(b) Terrace-free surface (right side of figure) near the edge of a GaAs LPE layer on spherical substrate.

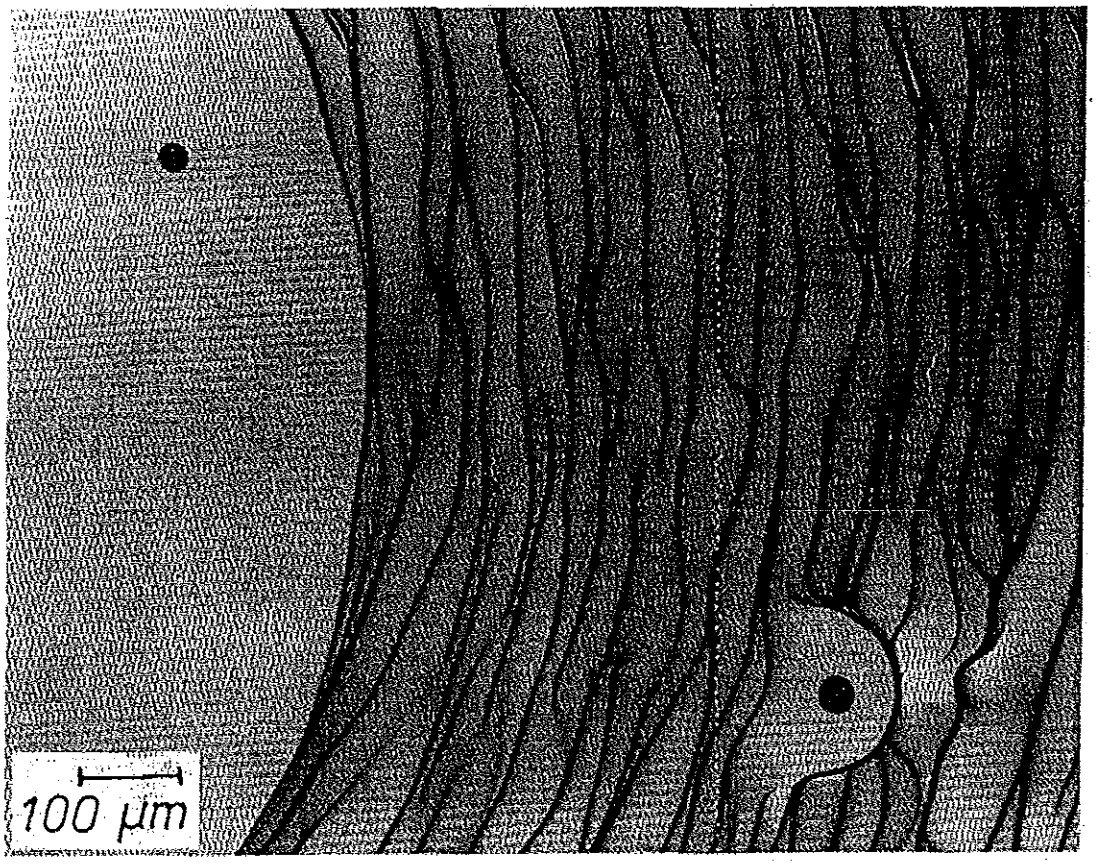


Fig. 61. Obstacles in a facet-area (left) and in a terraced area (right) demonstrate the different morphological stability of these interfaces.

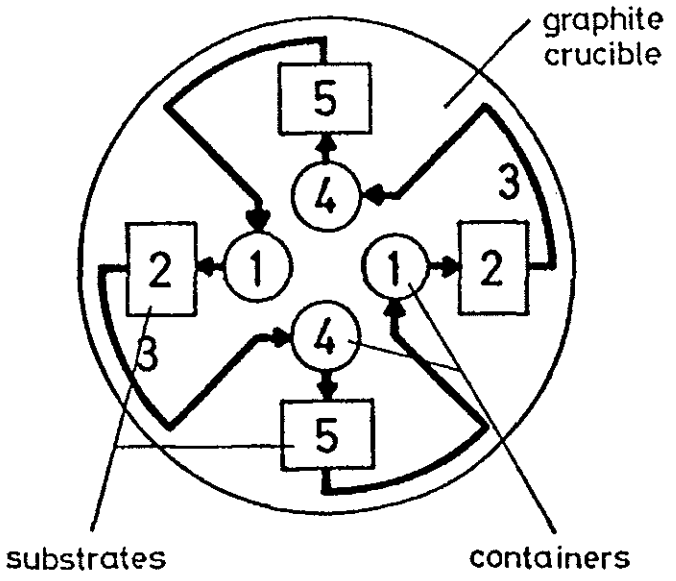
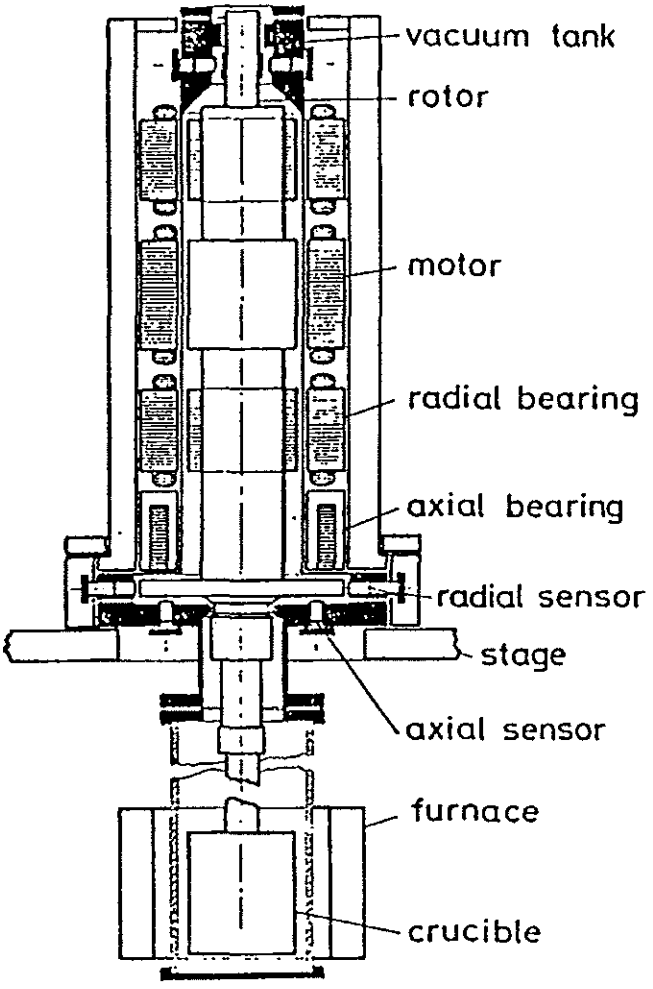


Fig. 62. (a) Crucible of the LPE centrifuge, schematic cross section. The crucible itself contains no moving parts, solutions are transported by centrifugal forces.



(b) LPE centrifuge with contactless, electromagnetic suspension of the rotor, longitudinal section.

Si VPE is performed with $SiCl_4$, SiH_4 or one of the chlorosilanes (SiH_xCl_{4-x}). A small concentration of these chemicals is transported in a H_2 gas stream to heated substrate wafers. The wafers are located inside a quartz tube in a furnace. A surface reaction takes place leaving Si behind on the substrate while the other chemical constituents return to the gas stream. We differentiate between atmospheric and low pressure VPE. The surface reactions can be promoted by photons (typically, one uses lasers). Plasma-enhanced deposition has also been explored for a number of thin films.

$SiCl_4$ VPE is the least expensive method to grow thick films which do not have to be too pure. Figure 63 shows schematically a VPE apparatus. The overall chemical reaction for $SiCl_4$ proceeds as follows in the reactor chamber (Fig. 64):

$$SiCl_4 + 2H_2 \rightarrow Si + 4HCl$$

Substrates are heated to a temperature between 1150 and 1250° C. Deposition rates range from 0.5 to 1.5 μm/min (Fig. 65). It is interesting that $SiCl_4$ epigrowth can be turned into Si etching. At 1270° C and a molar fraction of $SiCl_4$ exceeding approximately 0.28, etching caused by HCl begins to exceed the deposition rate. Typical growth conditions are chosen in the linear region near the origin of Fig. 65 at a few percent $SiCl_4$ in H_2.

Because HCl is generated as a by-product of $SiCl_4$ epitaxy, there exists always a danger of local etching of Si wafers. Etching of a heavily doped substrate leads to volatile dopant containing molecules contaminating the reactant gases and to inadvertent doping of epilayers downstream in the reactor. This effect is called autodoping.

A large number of studies have explored the thermodynamics and the kinetics of VPE. These studies have led to differentiation of the overall process into the following steps:

a. Reactant gas molecules are transported in H_2 to the deposition region;
b. Reactants diffuse through the near-surface gas stagnant layer to the surface;
c. Reactant adsorption;
d. Surface reactions (chemical) and surface diffusion to proper sites;
e. Desorption of reaction products;
f. Out-diffusion of reaction products to the gas stream; and finally
g. Removal of products.

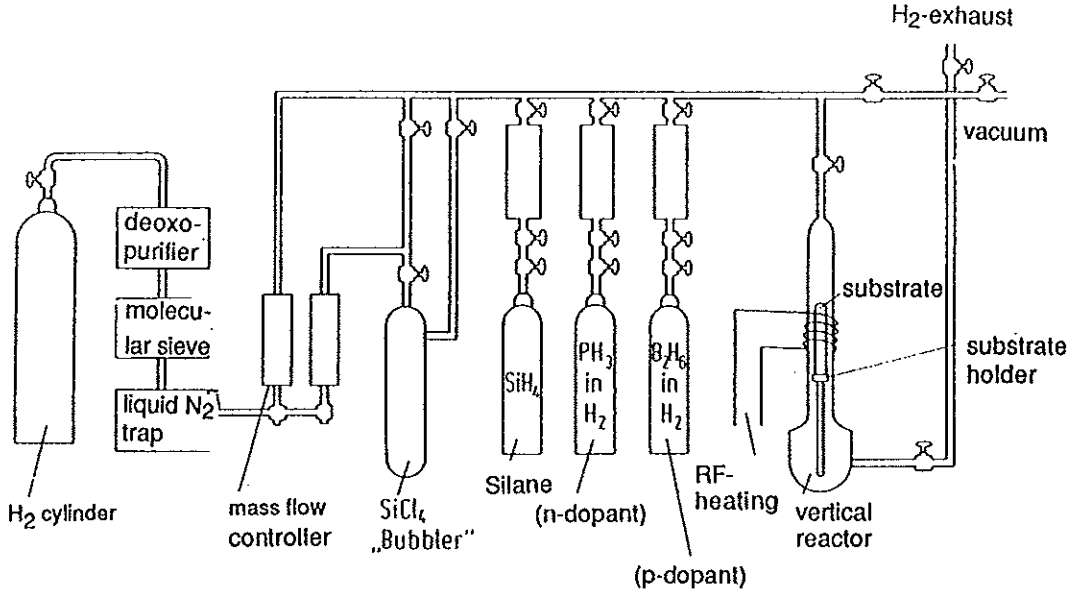


Fig. 63. Schematic of a CVD reactor. (After I. Ruge, "Halbleiter Technologie," Springer-Verlag, Berlin, 1975, p. 121.)

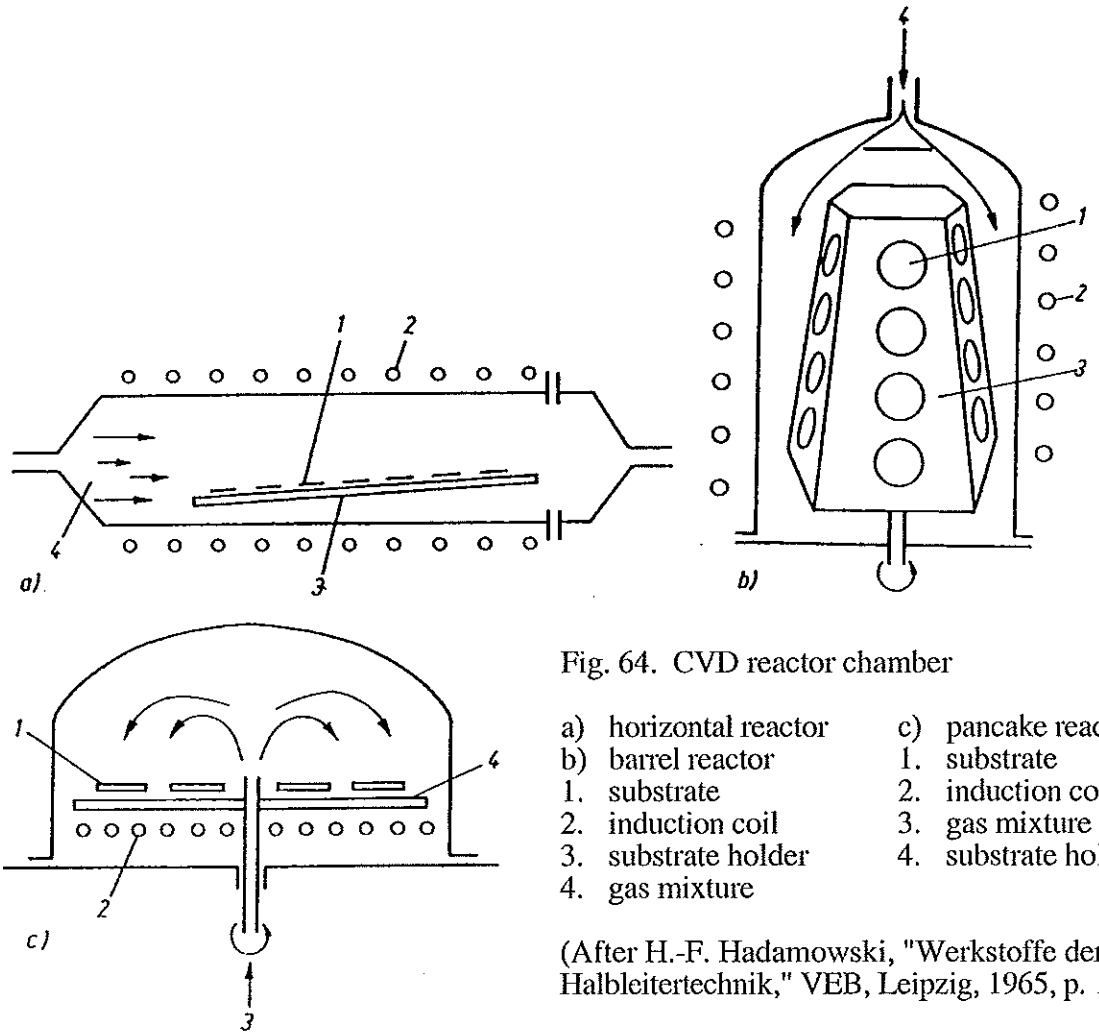


Fig. 64. CVD reactor chamber

a) horizontal reactor
b) barrel reactor
1. substrate
2. induction coil
3. substrate holder
4. gas mixture
c) pancake reactor
1. substrate
2. induction coil
3. gas mixture
4. substrate holder

(After H.-F. Hadamowski, "Werkstoffe der Halbleitertechnik," VEB, Leipzig, 1965, p. 121.)

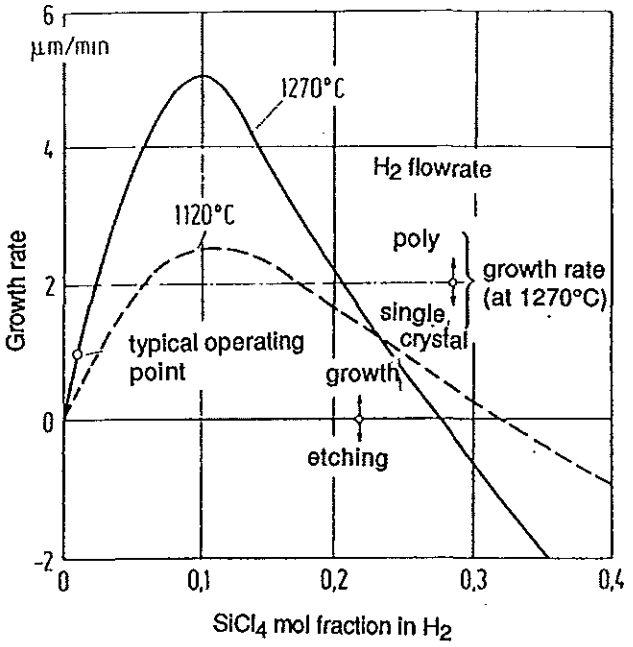


Fig. 65. Growth rate of Si as a function of the $SiCl_4$ mol fraction in H_2. [After H.C. Theurer, J. Electrochem. Soc. **108**, 649 (1961).]

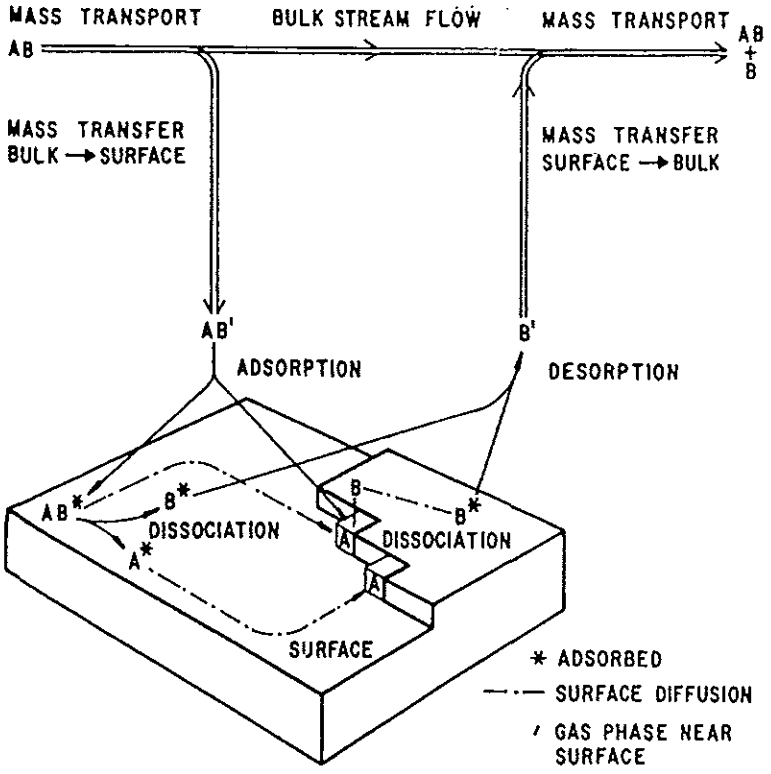


Fig. 66. Sequence of Events in Vapor Phase Epitaxial Growth. [After D.W. Shaw, in "Crystal Growth and Characterization," R. Ueda and J.B. Mullin, eds., North Holland Publ. Co., Amsterdam, 1975, p. 208.)

The slowest step dictates the type of growth regime and the growth rate. Typically, we distinguish between mass transport limited processes (a, b, f and g) and surface reaction limited processes (c-e). Figure 66 is a pictorial account of the reaction steps a-g.

VPE of compound semiconductors faces a problem with the low vapor pressures of metal chlorides at room temperature. In order to obtain acceptably large pressures, a small metal reservoir is held in a preheated zone at the gas inlet of a tube epitaxy reactor. HCl gas transported into the reactor in a H_2 stream reacts with the group III or II metal and forms a volatile chloride. In the case of GaAs VPE we can write down the following overall reactions.

Vapor synthesis:

$$2AsH_3 \rightarrow \tfrac{1}{2}As_4 + 3H_2$$
$$2HCl + 2Ga \xrightarrow[775°C]{} 2GaCl + H_2$$

Epigrowth:

$$2GaCl + \tfrac{1}{2}As_4 \xrightarrow[750°C]{} 2GaAs + 2HCl$$

With the exception of Ga which is a liquid, and GaAs which is a solid, all the reactants are gases. Figs. 67a and b show the schematic of hydride-chloride reactors for the growth of $GaAs_{1-x}P_x$ and of $In_{1-x}Ga_xP$. It is rather difficult to control the generation rate of GaCl in such a system.

A strong development effort resulted in an all-gas technique, the metal-organic-chemical vapor deposition (MOCVD) growth method. Typical metalorganic epi growth molecules are trimethyl gallium (TMGa) $(CH_3)_3Ga$, trimethyl aluminum (TMAl) $(CH_3)_3Al$, triethyl gallium (TEGa) $(C_2H_5)_3Ga$, dopants are dimethyl zinc (DMZn) $(CH_3)_2Zn$, etc. Until the early 1970s (and in Europe still today!) motorists poured large quantities of tetraethyl lead (TEPb) $(C_2H_5)_4Pb$ with gasoline into their cars' tanks. This additive effectively prevented knocking of the engine but at the same time led to significant lead pollution. It is then not too surprising that the companies which produced TEPb are now applying their experience to developing high purity metalorganic sources for semiconductor epitaxy. Compared to TEPb the metalorganics used for semiconductors are very high value-added products!

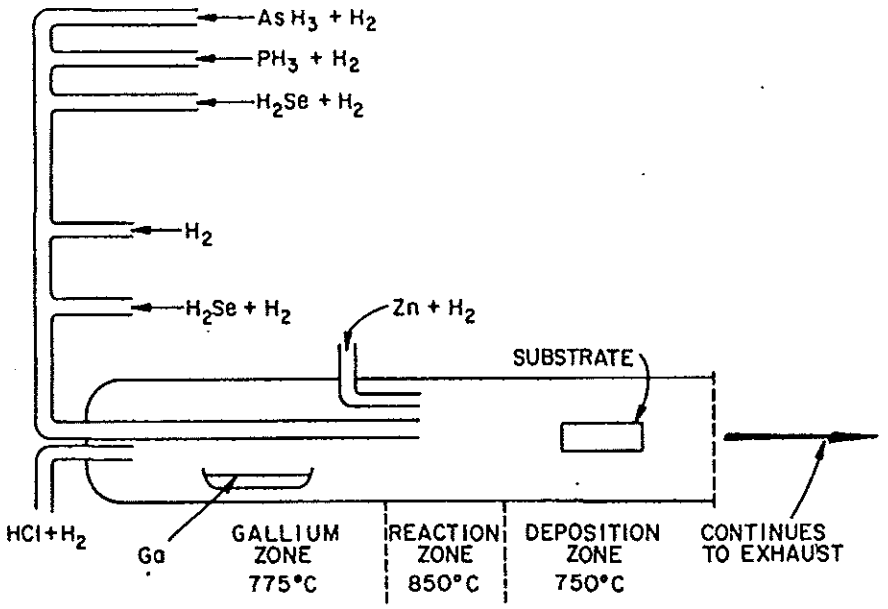


Fig. 67a. Schematic diagram of a hydride-HCl synthesis-transport CVD system. (After A.A. Bergh and P.J. Dean, "Light Emitting Diodes," Clarendon Press, Oxford, 1976, p. 410.)

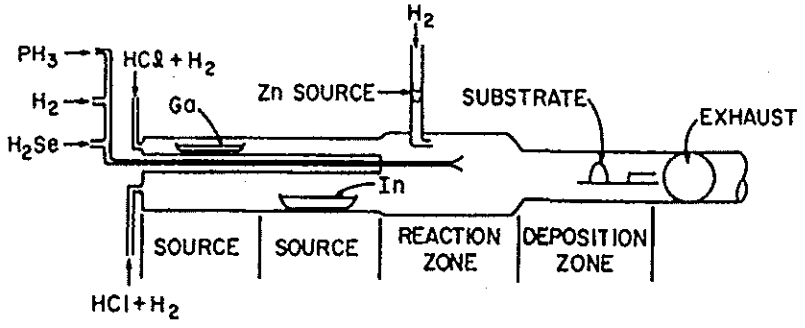


Fig. 67b. Schematic diagram of growth system used for vapor phase deposition of $In_{1-x}Ga_xP$. [After C.J. Nuese *et al.*, Met. Trans. **2**, 789 (1971).]

MOCVD has become the workhorse for all multilayer III-V structures that do not need the ultimate atomic flatness provided only by molecular beam epitaxy. As an all-gas technique, MOCVD offers superb control of composition and doping. Close to atomically sharp interfaces between layers of different composition and doping are achieved by using constant laminar gas flows in the reactor. When the flow of one gas component is abruptly increased or decreased, the ballast H_2 gas flow is accordingly changed to maintain constant flow. The schematic of a typical MOCVD reactor is shown in Fig. 68. A typical reaction describing the formation of GaAs layers can be written down as follows:

$$(CH_3)_3Ga + AsH_3 \rightarrow GaAs + 3CH_4$$

The arsine and the TMGa can be precisely metered with mass flow controllers. The substrate temperature is held constant to within a few degrees. Constant laminar gas flow insures extremely uniform growth and sharp composition and dopant concentration interfaces.

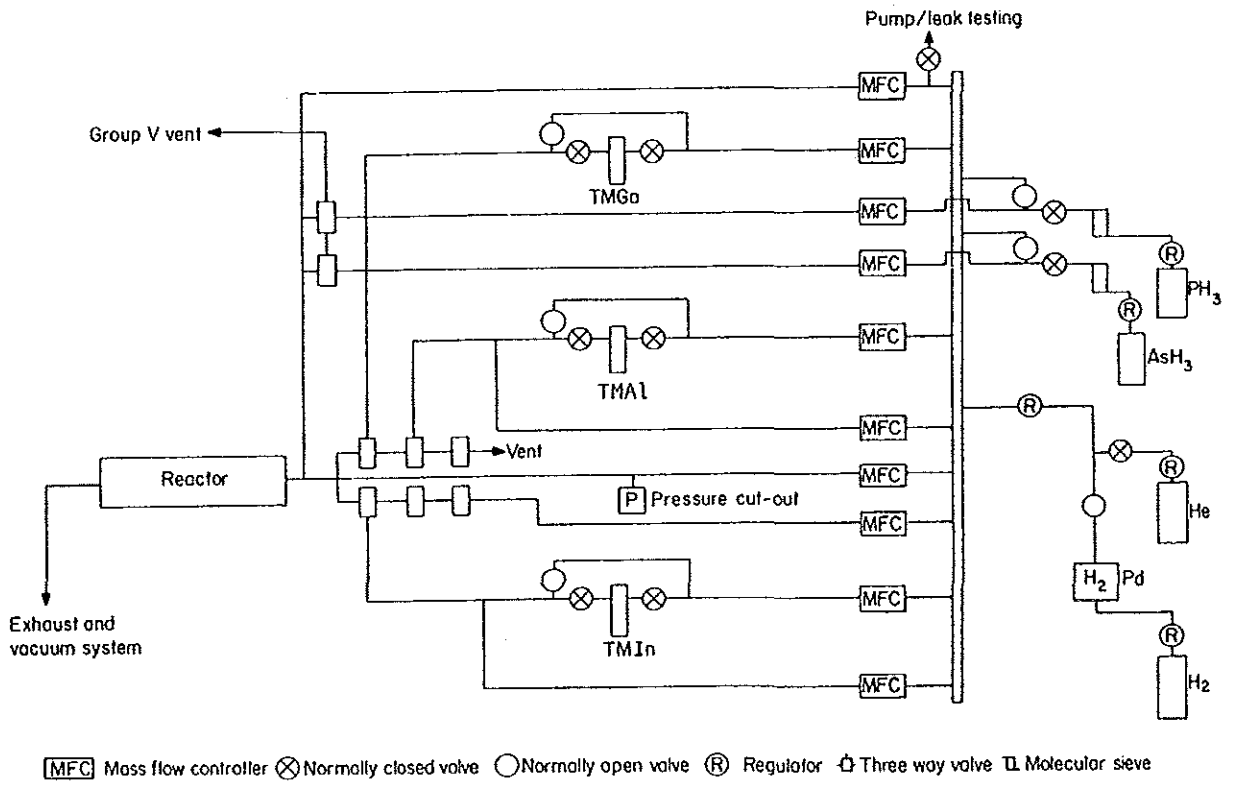


Fig. 68. Schematic representation of the MOVPE reactor. (After J.O. Williams, in "Growth and Characterization of Semiconductors," R.A. Stradling and P.C. Klipstein, eds., Adam Hilger, Bristol & New York, 1990, p. 18.)

In general, MOCVD makes use of reactions between group II or III organometallics (R_nX) and group VI or V hydrides:

$$R_n M + XH_n \rightarrow MX + nRH$$

As an example we use the epi growth of ZnSe and $Al_{1-x}Ga_xAs$:

$$(C_2H_5)_2Zn + H_2Se \rightarrow ZnSe + 2C_2H_6$$

and

$$x(C_2H_5)_3Ga + (1-x)(CH_3)_3Al + AsH_3 \rightarrow Al_{1-x}Ga_xAs + 3(1-x)CH_4 + 3xC_2H_6$$

Pyrolization near or on the substrate surface usually drives the reaction. In some cases organometallics form polymers with hydrides at room temperature. In such cases the gases are kept isolated until they enter the reaction chamber. In severe cases one reduces the pressure to prevent room temperature reactions.

All-organometallic reactions have been studied as well:

$$R_nM + R_nX \rightarrow MX + n(R-R)$$

e.g.,

$$(CH_3)_3Ga + (CH_3)_3Sb \rightarrow GaSb + 3C_2H_6 \quad .$$

In recent years InP has rapidly gained interest because of optoelectronic devices for transmission at the wavelength of $\lambda = 1.55\mu m$, the optimal wavelength for low loss and low dispersion transmission. Organometallics exhibit a strong tendency to what is called "adduct formation." Adduct formation can be described by the following reaction:

$$R_3M + R_3'X \rightarrow \underset{\text{adduct}}{R_3MX\,R_3'}$$

Adduct formation occurs because group III metal alloys are strong Lewis acids while group V hydrides are Lewis bases. Again, separation of the reactants and reduced vapor pressure help in suppressing adduct formation.

The typical MOCVD reactor works in the diffusion-limited mass flow regime. The growth rate g is given by the following reaction:

$$g = K_0(T)\, K_1(v)\, p_m \quad .$$

With $K_0(T)$ = reaction efficiency, $K_1(v)$ = constant depending on geometry and total gas velocity, p_m = inlet partial pressure of the organometallic compound. Figure 69 shows growth rates for four III-V compounds as a function of p_m. The linear relationship over two decades supports the above equation. Figure 70 demonstrates the independence of the growth rate over a wide substrate temperature range in the mass flow regime for GaAs and GaInAs. Only at temperatures exceeding 750° C the growth rate starts to become smaller because of the high dissociation pressure of the growing epilayer.

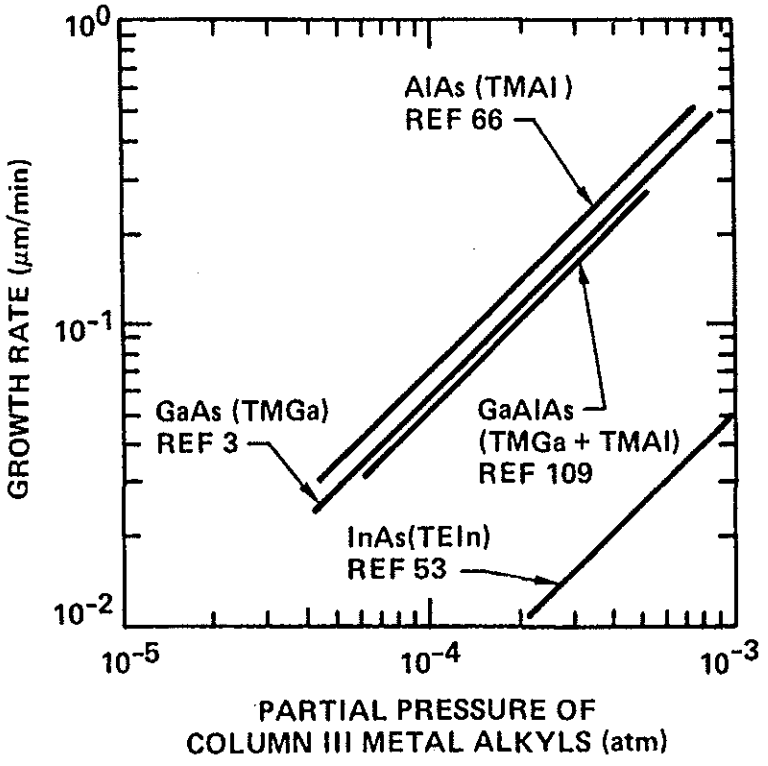


Fig. 69. Dependence of growth rate upon the partial pressure of column III metal alkyls for growth by MOCVD of several III-V semiconductors. [After P.D. Dapkus, Ann. Rev. Mater. Sci. **12**, 243 (1982).]

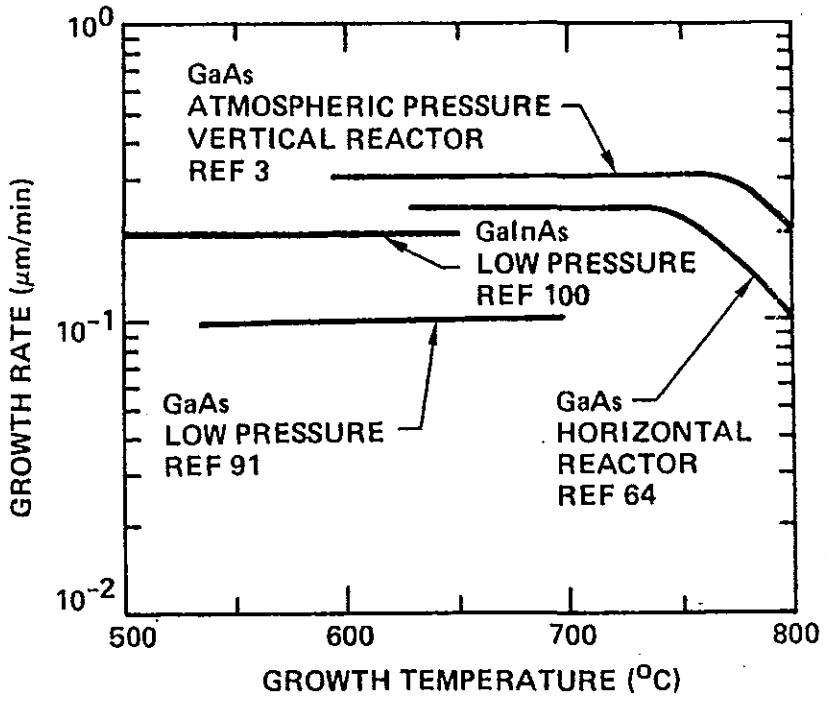


Fig. 70. Dependence of the growth rate upon growth temperature for GaAs and GaInAs by MOCVD. [After P.D. Dapkus, Ann. Rev. Mater. Sci. **12**, 243 (1982).]

The quality of MOCVD-grown layers is often measured by the electron or hole mobility, which can be experimentally determined. A temperature of 77 K is convenient to reach with liquid N_2 and it is a temperature at which ionized impurity scattering dominates transport. Figures 71a-d show the combined free carrier concentration and mobility results of two layers grown with the same arsine source but with two different TMGa sources. The chemical analyses of the TMGa sources is given in the small table above the figure. It is quite obvious that the TMGa 2 source produces layers of much higher purity. The mobility is significantly higher in the low substrate temperature regime and the residual dopant concentration is lower. Table 9 lists a number of III-V alloy semiconductors which have been grown successfully by MOCVD. II-VI and IV-VI compound semiconductors have been grown as well (Table 10). Table 11 contains some basic properties of a number of organometallic compounds.

ICP analysis for impurities in TMGa1 and TMGa2*

Elements	TMGa1	TMGa2
Si	0.2 ppm	0.05 ppm ND
Zn	0.6 ppm	0.3 ppm ND
Sn	< 0.5 ppm ND	0.5 ppm ND
Mg	0.05 ppm	0.05 ppm ND
Mn	< 0.05 ppm ND	0.05 ppm ND
Fe	< 0.3 ppm ND	0.3 ppm ND

*ND = not detected

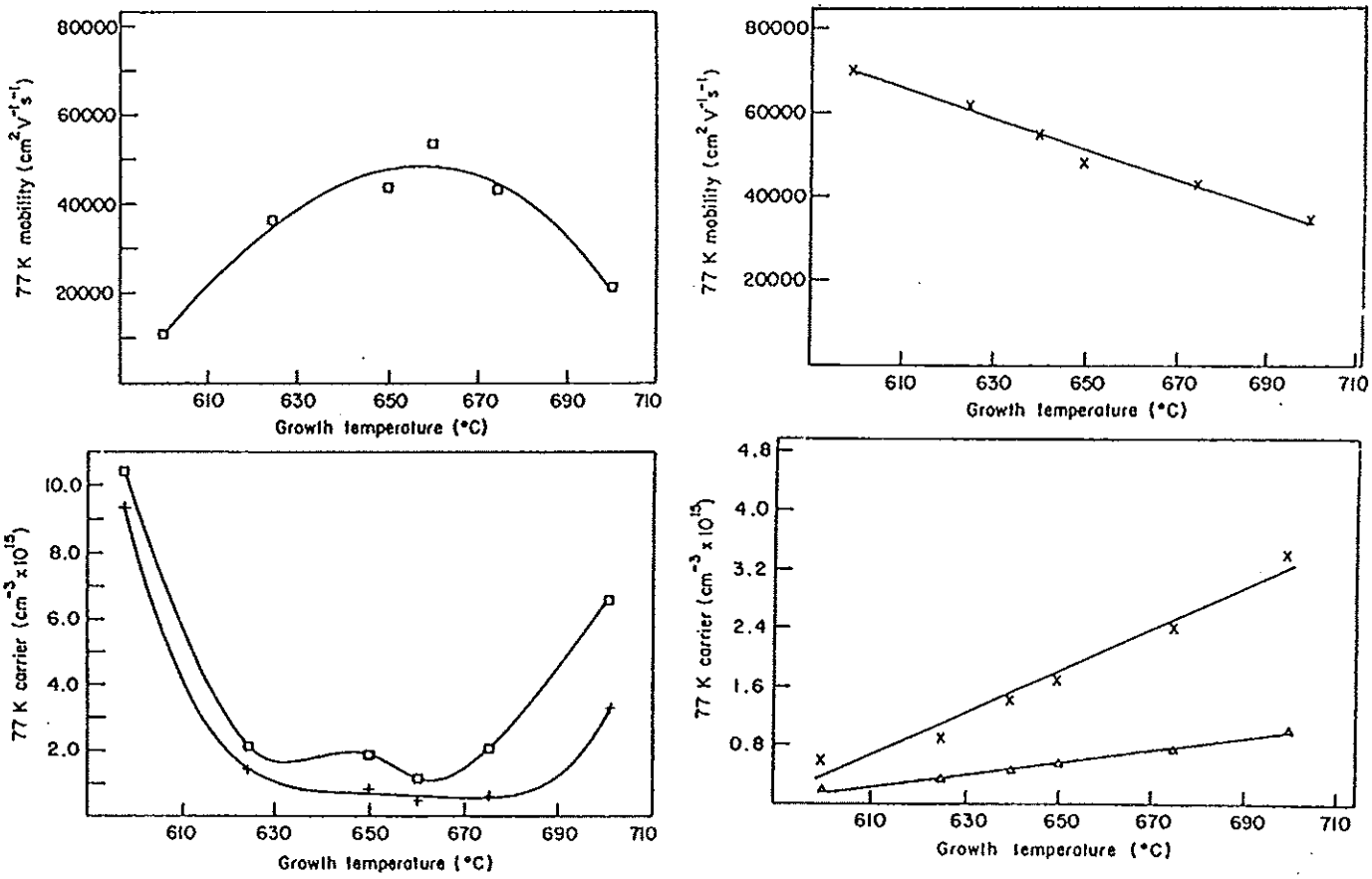


Figure 71

(a) 77 Hall mobility variation with growth temperature for GaAs grown from TMGa1 at constant V/III ratio of 75:1.

(b) 77 K Hall carrier concentration yielding values of (❑) N_D and (+) N_A plotted against growth temperature for GaAs grown from TMGa1 at constant V/III ratio of 75:1.

(c) 77 K Hall mobility variation with growth temperature for GaAs grown from TMGa2 at constant V/III ratio of 75:1.

(d) 77 K Hall carrier concentration yielding values of (X) N_D and (Δ) N_A plotted against growth temperature for GaAs grown from TMGa2 at constant V/III ratio of 75:1.

[After N. Hunt and J.O. Williams, Chemtronics 2, 145 (1987).]

Table 9

III-V SEMICONDUCTORS GROWN BY MOCVD

[adapted from P.D. Dapkus, Ann. Rev. of Mater. Sci. **12**, 243 (1982).]

(see pages 119-22 for reference list)

Material	Reactants	References
AlAs	TMAl-AsH_3	4, 9–11
AlN	TMAl-NH_3	5, 12–15
GaAs	TMGa-AsH_3	2, 3, 10, 11, 16–31
	TEGa-AsH_3	3, 32
	DEGaCl-AsH_3	33, 34
GaN	TMGa-NH_3	5, 12, 13, 35, 36
GaP	TMGa-PH_3	5, 37–40
	TEGa-PH_3	41
GaSb	TEGa-TMSb	42
InAs	TEIn-AsH_3	6
InP	TEIn-PH_3	6, 43–45
	TMIn-PH_3	46
InSb	TMIn-SBH_3	47
$GaAs_{1-x}P_x$	TMGa-AsH_3-PH_3	3, 48–50
$GaAs_{1-x}Sb_x$	TMGa-AsH_3-SbH_3	3
	TMGa-TMSb-AsH_3	3
$Ga_{1-x}Al_xAs$	TMGa-TMAl-AsH_3	3, 10, 11, 29, 31
	TEGa-TMAl-AsH_3	51, 52
$Ga_{1-x}In_xAs$	TMGa-TEIn-AsH_3	53, 54
$InAs_{1-x}P_x$	TEIn-AsH_3-PH_3	8
$Ga_{1-x}In_xAs_{1-y}P_y$	TEIn-TMGa-PH_3-AsH_3	102

Table 10

II-VI AND IV-VI SEMICONDUCTORS GROWN BY MOCVD

Compound	Reactants	References
ZnS	DEZn-H_2S	55
ZnSe	DEZn-H_2Se	55, 56
ZnTe	DEZn-DMTe	55
CdS	DMCd-H_2S	55, 57
CdSe	DMCd-H_2Se	55
CdTe	DMCd-DMTe	55
HgTe	Hg-DMTe	58
HgCdTe	Hg-DMCd-DMTe	58
SnTe	TESn-DMTe	57
PbTe	TMPb-DMTe	59
	TEPb-DMTe	59
$Pb_{1-x}Sn_xTe$	TMPb-TESn-DMTe	59
PbS	TMPb-H_2S	59
PbSe	TMPb-H_2Se	59

Table 11

PROPERTIES OF METALORGANICS USED FOR SEMICONDUCTOR GROWTH

Compound	Symbol	Melting point (°C)	Boiling point at 760 mm (°C)	Vapor pressure (mm)
Dimethylzinc	DMZn	−42 (−29)	46	124 at 0°C
Diethylzinc	DEZn	−28	118	15 at 20°C
Dimethylcadmium	DMCd	−4.2	105.5	350 at 80°C
Trimethylaluminum	TMAl	15.4	126	8.4 at 20°C
Trimethylgallium	TMGa	−15.8	55.7	64.5 at 0°C
Triethylgallium	TEGa	−82.3	143	18 at 48°C
Diethylgalliumchloride	DEGaCl	—	—	—
Trimethylindium	TMIn	88.4	135.8	7.2 at 30°C
Triethylindium	TEIn	−32	184	3 at 53°C
Tetramethyltin	TMSn	−53	78	10 at −20°C
Tetraethyltin	TESn	−112	179.5–181.5	10 at 73°C
Tetramethyllead	TMPb	−27.5	110	10 at 4.4°C
Tetraethyllead	TEPb	−135	198–202	10 at 78°C
Triethylphosphine	TEP	−88 (−85)	127	—
Trimethylantimony	TMSb	−87.6 (−62.0)	80.6	—
Dimethyltelluride	DMTe	−10 (−150)	82 (93.5)	—
Diethyltelluride	DETe	—	137–138	—

[from P.D. Dapkus, Ann. Rev. of Mater. Sci. **12**, 243 (1982).]

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3.4.4. Molecular Beam Epitaxy (MBE)*

Molecular Beam Epitaxy (MBE) is conceptually the simplest but, paradoxically, the most expensive epitaxial growth technique. MBE is the only epi growth method which offers atomic layer growth control. An MBE system consists of one or more large stainless steel, ultra-high-vacuum chambers. (They represent a major part of the total costs of ~ 1M$!) A schematic of an MBE growth chamber is shown in Fig. 72. A number of effusion ovens (Knudsen cells) provide beams of molecules or atoms by heating the various source substances to temperatures at which the required vapor pressures are reached. The molecular and atomic beams travel collisionless through the ultra-high-vacuum chamber to the substrate. Surface reactions and migration, which are necessary for epitaxial growth are promoted through substrate heating. Each of the effusion ovens, for simplicity called sources, is shrouded with liquid-nitrogen-cooled baffles in order to keep as much of the inside surface of the MBE system as cold as possible, in order to achieve the required vacuum. In front of each source sits a shutter which is used to turn the source "on" or "off," either manually or by computer control.

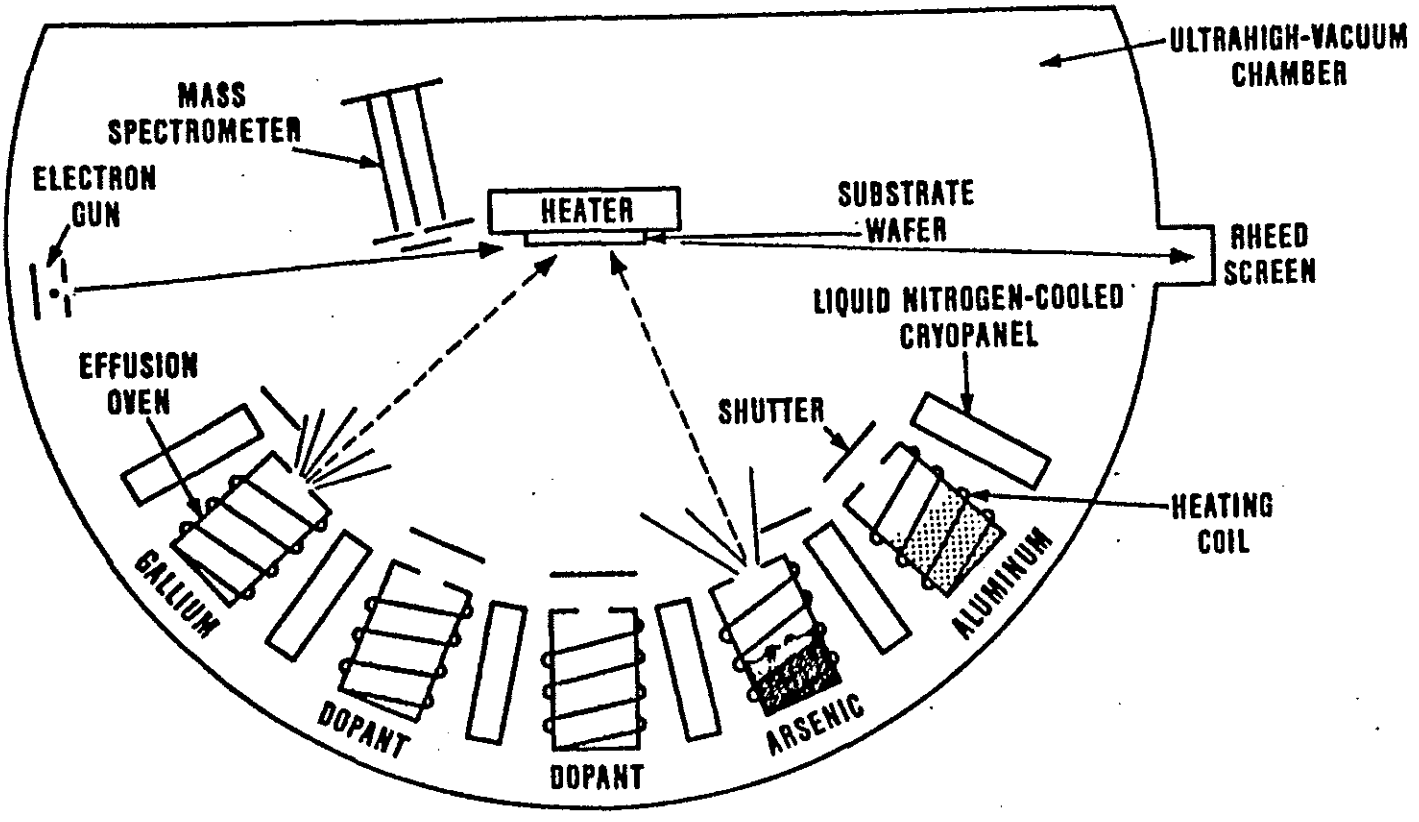


Fig. 72. Typical MBE chamber for the growth of pure and doped GaAs and AlGaAs layers.

* several figures are taken from the article by C.T. Foxon and B.A. Joyce on "Growth and Characterization of Semiconductors," R.A. Stradling and P.C. Klipstein, eds., Adam Hilger, Bristol and New York, 1990, p.35ff.

Two or more analytical techniques are used *in situ* to monitor and control MBE growth. A monoenergetic electron beam is directed at the growing surface under a very shallow angle . The diffracted beam generates a visible light pattern on a ZnS screen. This technique is called Reflection High Energy Electron Diffraction (RHEED). It provides information which makes possible the growth of one atomic layer at a time. A mass spectrometer continuously monitors the quality of the vacuum and the chemical composition of the various beams.

The MBE growth chamber is separated from a second chamber by an ultra-high-vacuum valve. In research MBE machines, this second chamber is often used for detailed studies of the epitaxial film. A number of analytical tools for thin film and surface studies are available in this chamber. A third chamber may be used as a substrate loading lock. The substrate can be moved between the various chambers with a system of magnetically coupled sliding rods.

A large variety of semiconductor epi layers have been grown by MBE. We choose to discuss growth of GaAs, AlAs and their alloys because of the nearly perfect lattice match of the two compounds and because of their scientific and technical importance. Growth of these ternary alloys illustrates most of the critical aspects of MBE epi growth which are also encountered with other materials systems. Naturally, each combination of substrate and epi layer materials has its own set of intricacies and complexities.

MBE of GaAs has been developed for the past 30 years. It has reached a high level of maturity, and a number of devices which require atomic layer control make use of MBE-grown layers. We know that stoichiometry plays a very important role for all compound semiconductors and the first question one may ask is: How can one control stoichiometric epi growth by individually controlling the vapor pressures of a liquid Ga and a solid As sublimation source? Worse yet, the As source emits a molecular beam consisting mostly of As_4 and some As_2 molecules.

Were it not for rather special circumstances which can be characterized by the so-called "sticking coefficient," GaAs and many other compound semiconductors and their alloys could be grown stoichiometrically only with very great difficulties. Figure 73 depicts schematically what happens when GaAs is grown with a Ga and an As_2 beam. As_2 can be formed by breaking up As_4 molecules thermally (cracking) or with an energetic electron beam moving across the As_4 beam. When the As_2 molecules arrive at a Ga covered GaAs surface (Ga stabilized), practically all As_2 molecules stick to this surface, dissociate and form GaAs. The sticking coefficient is unity (sticking coefficient = number of As_2 molecules adsorbed on the surface/number of As_2 molecules

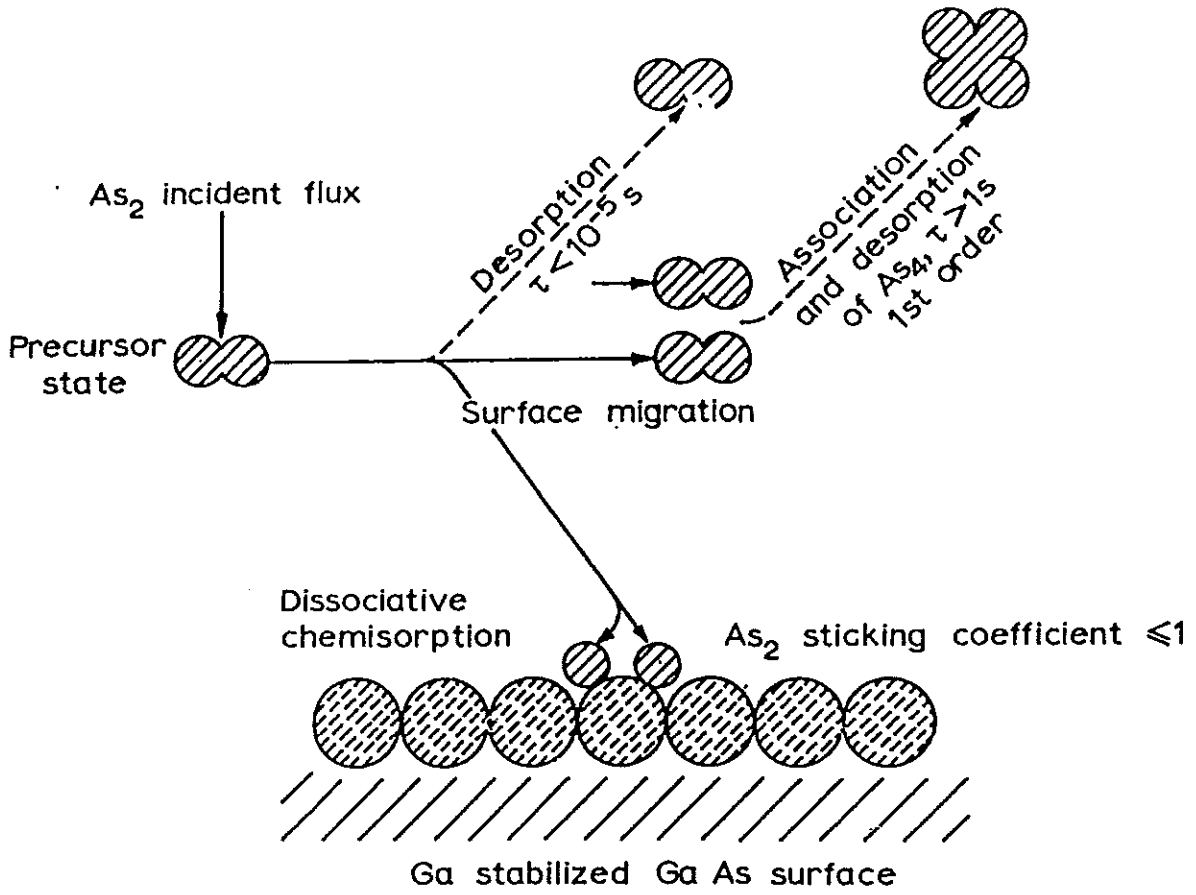


Fig. 73. Model of the growth chemistry of GaAs from molecular beams of Ga and As_2.

arriving at the surface). As the surface gradually becomes As terminated, the sticking coefficient goes to zero.

In a somewhat simplistic fashion the difference in sticking coefficient for As arriving at a Ga or at an As terminated surface can be explained with vapor pressure arguments. A typical substrate temperature of 550° to 650° C is high enough for the additional As to re-evaporate but it is far too low to lead to GaAs dissociation. We see that in this situation the Ga flux is dominating the growth process as long as the arsenic flux is kept larger or much larger than the Ga flux. In practical GaAs epi growth the As_2 flux is kept at least two times, often many more times, the intensity of the Ga flux. This automatic stoichiometry stabilization has been of enormous help in the growth of stoichiometric GaAs epi layers.

In the case of growth with uncracked As_4, one observes a maximum sticking coefficient of 0.5. A model of epi growth for this case is illustrated in Fig. 74. As_4 molecules adsorbed on a Ga stabilized surface appear to dissociate in a first step into two As_2 molecules, one of which desorbs and combines with another As_2 molecule to form As_4.

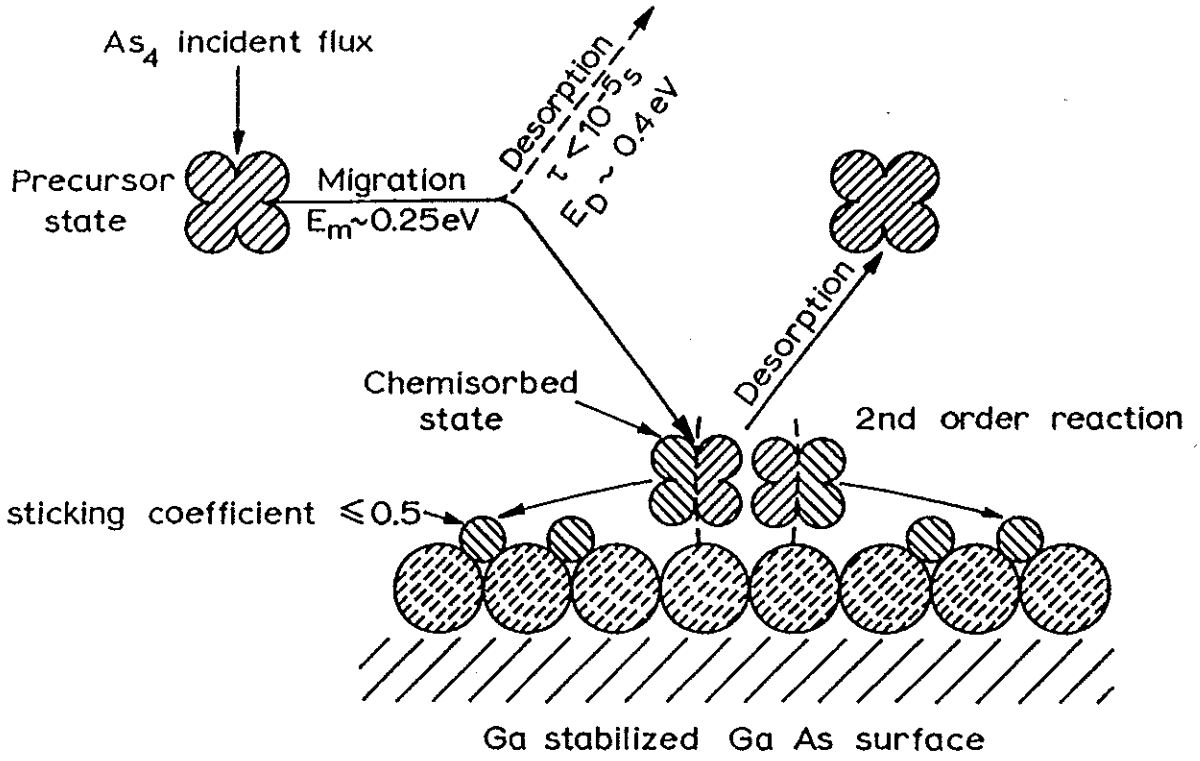


Fig. 74. Model of the growth chemistry of GaAs from molecular beams of Ga and As_4.

Very similar stoichiometry stabilization is observed for many III-V and II-VI compound semiconductors. In their elemental form group V and VI species typically have much higher vapor pressures than when they are forming a compound. They re-evaporate when the surfaces are terminated by their own species (sticking coefficient = 0), but they form compound semiconductor material when they arrive at a group III or group II terminated surface.

Incorporation of dopants during MBE growth is generally straightforward. For each dopant a separate source must be installed. The preferred n-type dopant for $Al_xGa_{1-x}As$ is Si. P-type doping is often achieved with Be. Both of these impurities oxidize readily and special care must be taken in the choice of materials for the construction of the sources. Other acceptor dopants

such as Zn and Cd with very high vapor pressures and/or large diffusion coefficients obviously pose a number of special problems which will not be discussed here.

The combination of source shutter control and vapor pressure for each dopant offers, in principle, great flexibility in dopant concentration profiles in the epi layers. Because the temperature of a large dopant source cannot be changed rapidly, one typically cannot use vapor pressure changes in controlling the dopant profile at the atomic layer level. If such rapid changes are required, growth of the host material is stopped at every layer and the desired dopant atom quantity is added to each layer. An extreme case of this method is the so-called δ-doping in which only one heavily doped atomic layer is formed. Such heavily doped layers exhibit some interesting properties which recently have generated a lot of interest. We will come back to this topic when we discuss some unique MBE-grown structures.

We stated earlier that MBE is the only technique which allows layer-by-layer control of the epitaxial growth. In order to make use of this capability, one requires a tool to monitor growth at the atomic level. RHEED, an electron diffraction technique based on the experiments by Davisson and Germer [Physical Review **30**, 705 (1927)], who confirmed experimentally the wave nature of electrons, is used to obtain layer-by-layer control. The formation of visible RHEED patterns on a ZnS screen depends strongly on the surface structure of the growing crystal. For an atomically flat surface a raster of bright points is observed. The pattern represents the particular surface reconstruction which in turn depends on the surface composition.

The discovery of damped oscillations in the intensity of the RHEED pattern immediately after start of MBE growth was correctly interpreted with a two-dimensional layer-by-layer growth [J. H. Neave et al., Appl. Phys. **A31**, 1 (1983)]. Figure 75 shows two growing, stepped surfaces. In case (a), where the surface diffusion length λ of the arriving atoms is much larger than the width of each step, the epi growth proceeds by lateral one-dimensional growth at the individual steps. The RHEED pattern does not change with time and no information is gained regarding layer-by-layer growth. As the step widths become larger or the adatom diffusion smaller, a new regime of two-dimensional epi growth is observed. Initially, the steps are atomically flat and lead to strong specular electron diffraction spots. As little islands nucleate and grow, the diffraction intensity drops because the steps are no longer flat. The RHEED pattern intensity increases again when each step has been overgrown by one complete layer and conditions become equivalent to the starting surface topography. Figure 76 shows two further cases of strong RHEED oscillations from (001) AlAs and GaAs surfaces. The five rectangles shown in perspective illustrate the two-dimensional growth of one layer. In modern MBE machines one can follow hundreds of RHEED oscillations for carefully aligned and cleaved substrates.

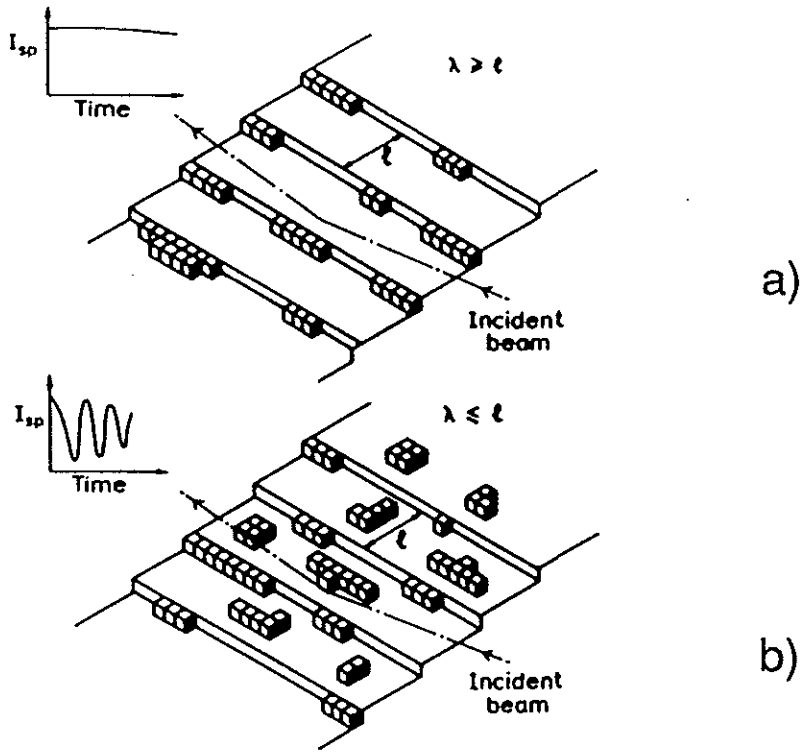


Fig. 75. Schematic illustration of one-dimensional growth a) and two-dimensional growth b), and the effect on the RHEED pattern intensity.

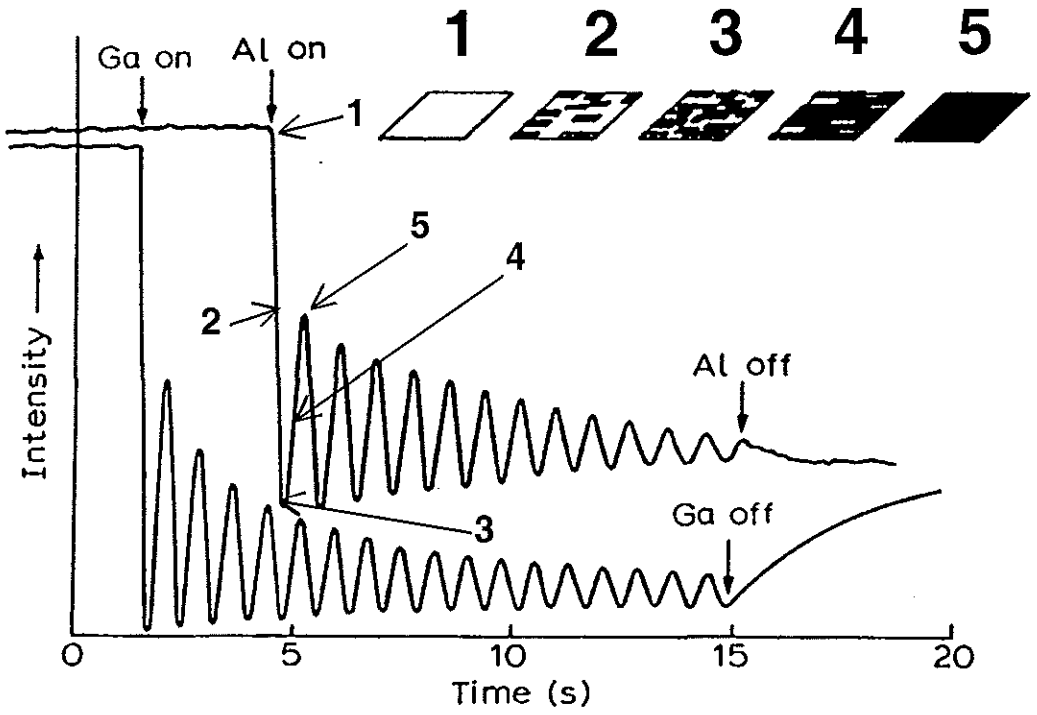


Fig. 76. Intensity oscillations of the specular beam in the RHEED pattern from a AlAs and GaAs (001) - 2 × 4 surface during film growth.

In order to illustrate the unique capabilities of MBE, we will briefly discuss some heteroepitaxial structures. The first heterostructure is n-type $Al_xGa_{1-x}As$ (x ~ 0.3) grown on lightly p-type doped GaAs shown in Fig. 77.

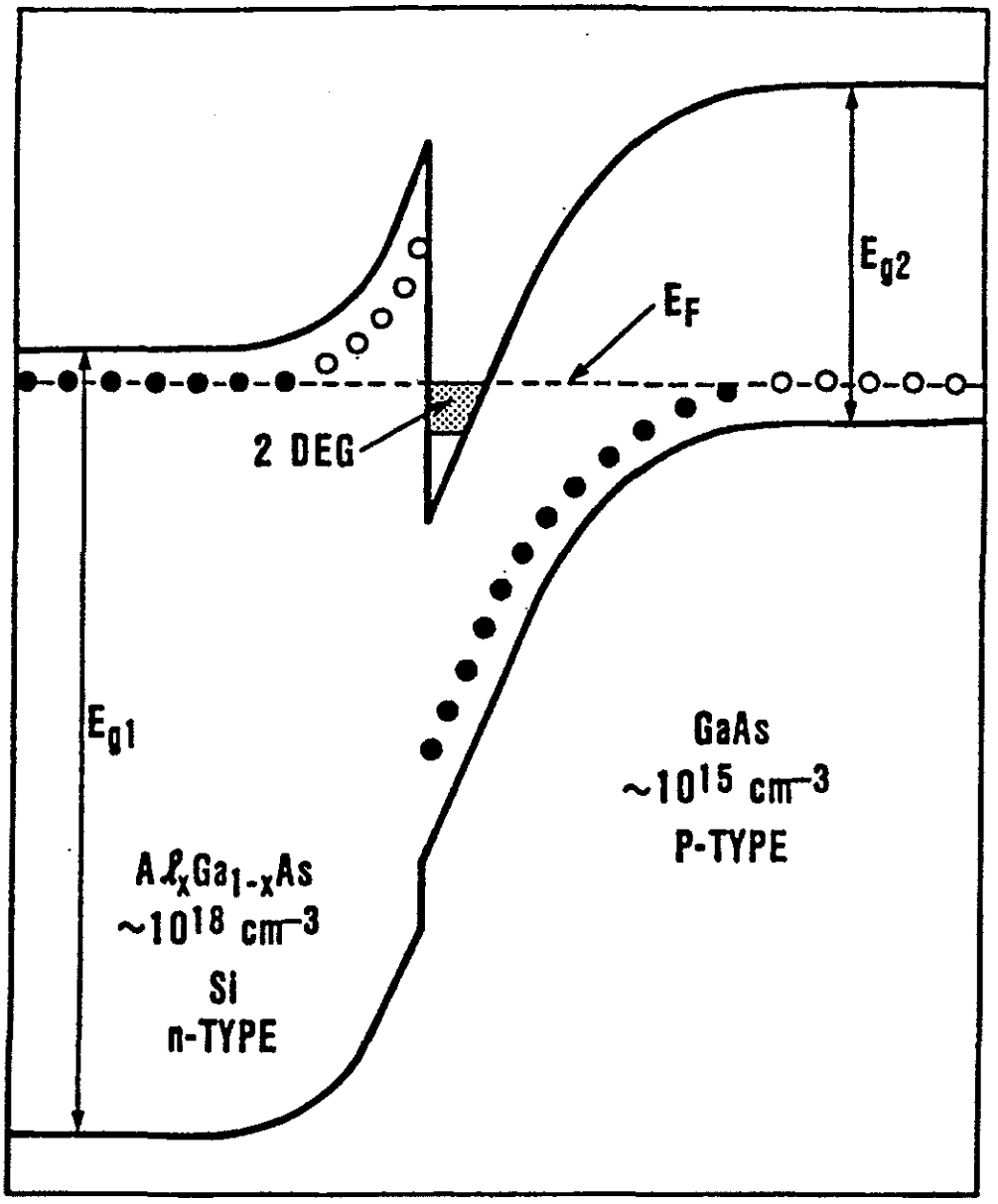


Fig. 77. Two-dimensional electron gas (2DEG) trapped in a heterojunction quantum well structure.

In constructing the band-diagram of this structure we have to know the conduction and the valence band offsets and have to remember that in the absence of any external bias the Fermi levels of the two materials must match. We see that the conduction band of GaAs drops below the Fermi level near the heterointerface. Electrons from the donors in the AlGaAs layer will flow into this region forming what is called a two-dimensional electron gas (2DEG). Because of the small width of the this 2DEG in one dimension ($\leq$ 100 Å), the electron states become quantized. Mobility measurements show that 2DEGs reach low temperature mobility values which far exceed electron mobilities in any other GaAs structures. There are mainly two reasons for this increase. Most importantly, the electrons in the 2DEG are spatially separated from the donors which are the source of these electrons. The mobility limiting ionized impurity scattering is completely eliminated in this 2DEG structure. Secondly, the heterointerface can be made atomically flat with MBE growth, eliminating a second source of scattering for the electrons moving along this interface. Figure 78 shows the evolution of electron mobilities in 2DEG structures over the years. Today's maximum values have reached 10^7 cm^2/Vs. For comparison, the mobility in a VPE-grown, undoped GaAs layer is shown. This comparison is no longer completely fair, insofar as LPE GaAs layers with impurity concentrations in the low 10^{12} cm^{-3} have been grown which show low temperature mobilities of several hundred thousand cm^2/Vs.

It is interesting to note that mobilities as high as 10^7 cm^2/Vs have been reported in bulk ultra-pure Ge in the 1970s; however, only for very low free carrier concentrations.

The second structure which illustrates the capabilities of modern MBE growth is shown in the TEM micrograph of a superlattice consisting of a sequence of three monolayers each of GaAs and AlAs (Fig. 79). Growth started out at the bottom where the layers are not yet perfectly flat. As growth proceeds over more than 100 individual layers, the interfaces become flatter and the quality of the superlattice improves. The highest resolution TEM micrographs clearly show that atomically abrupt interfaces can be grown routinely.

Superlattices and individual quantum wells (e.g., a thin layer of GaAs between AlGaAs) exhibit a host of novel optical and electronic properties which are studied extensively worldwide. A number of new electronic devices have been developed already. The double heterojunction quantum well laser shown in Fig. 80 is used in every compact disc player to read the information stored on a 2000-Å-wide track. It should be noted that commercial production of such lasers is performed with the MOCVD technique, which is discussed in Section 3.4.3.

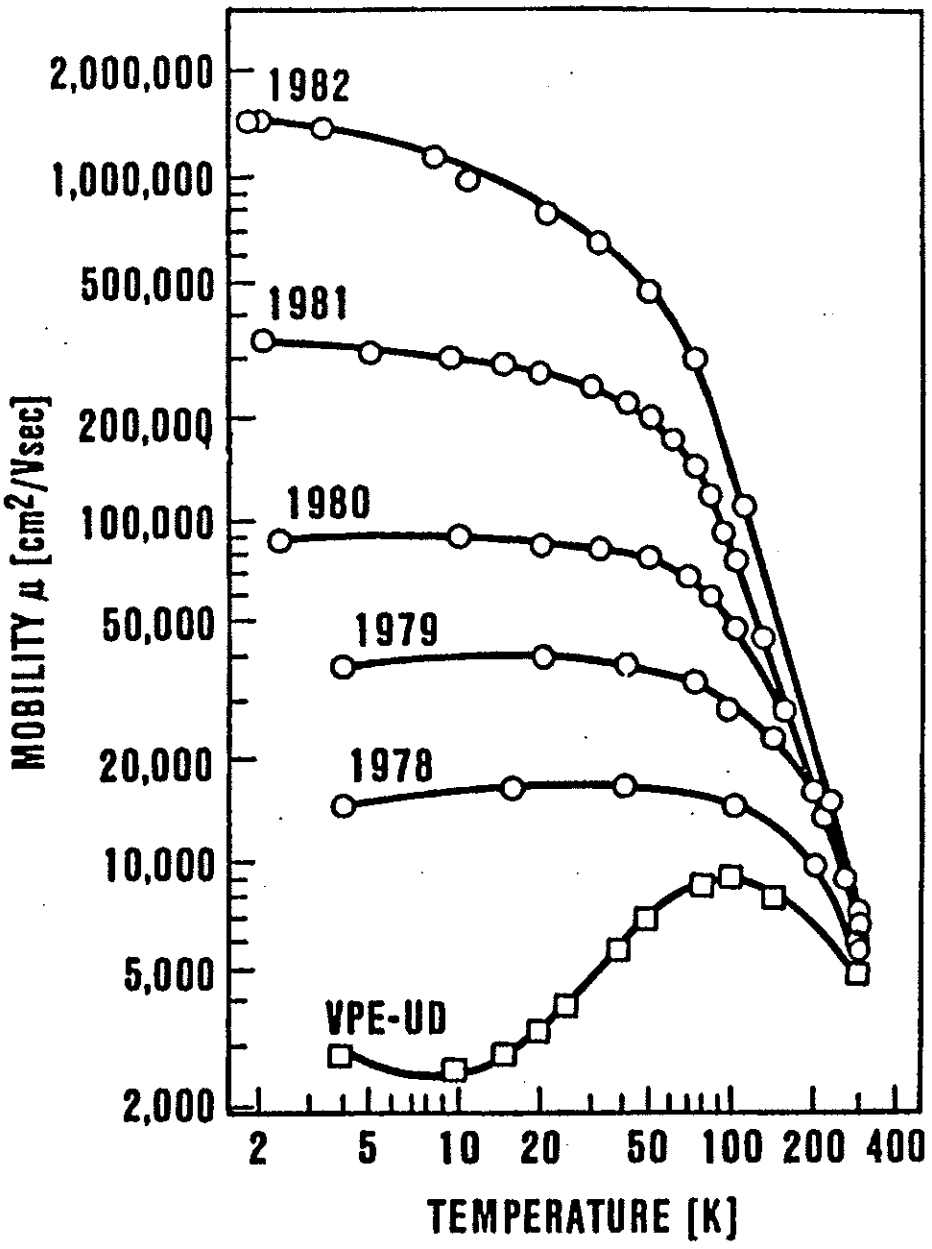


Fig. 78. Electron mobilities as a function of temperature for two-dimensional electron gas heterostructures. For comparison, the mobility in an undoped VPE layer is shown. Today's highest values have reached 10^7 cm^2/Vs at low temperatures.

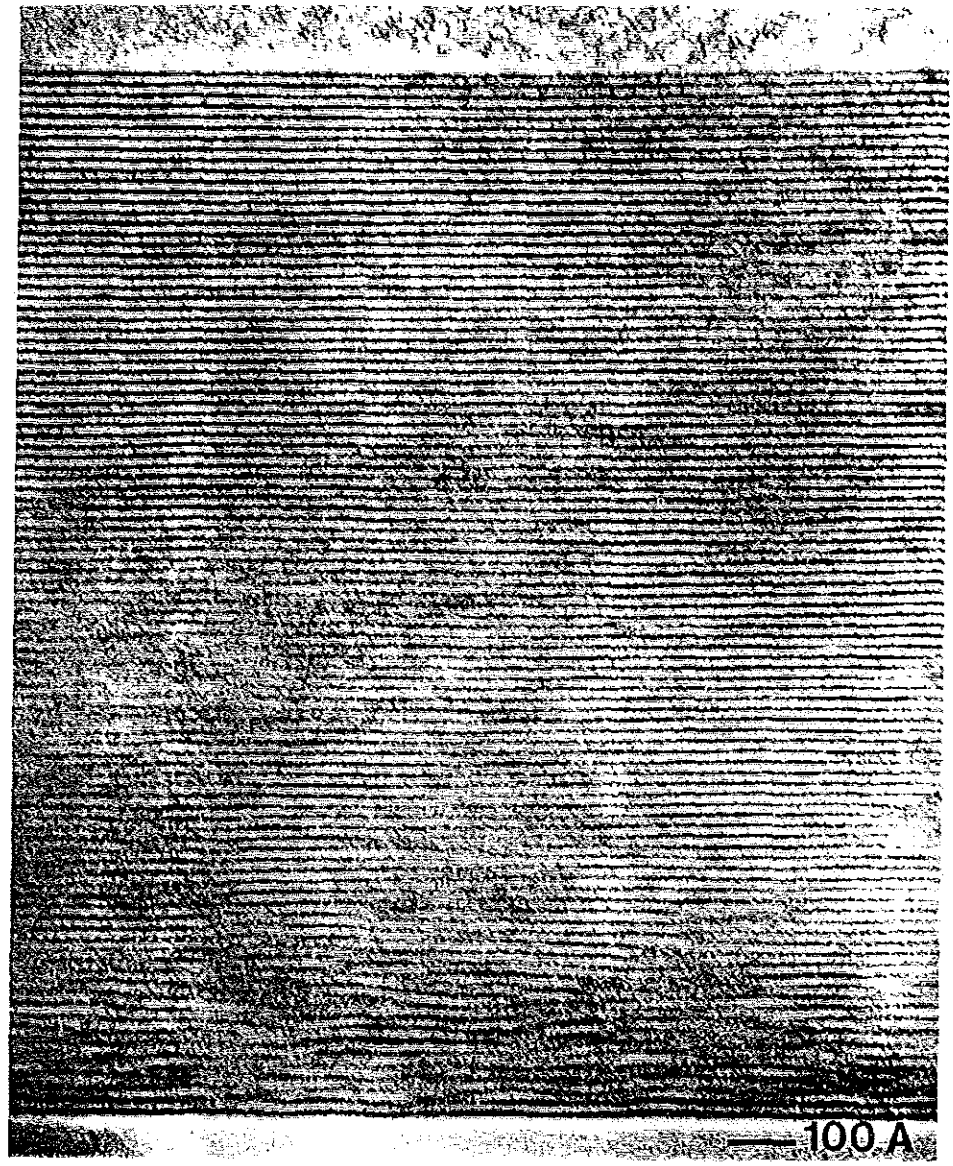


Fig. 79. TEM micrograph of a GaAs (3ML)/AlAs(3ML) superlattice (grown by D. Hilton, TEM micrograph by J. P. Gowers).

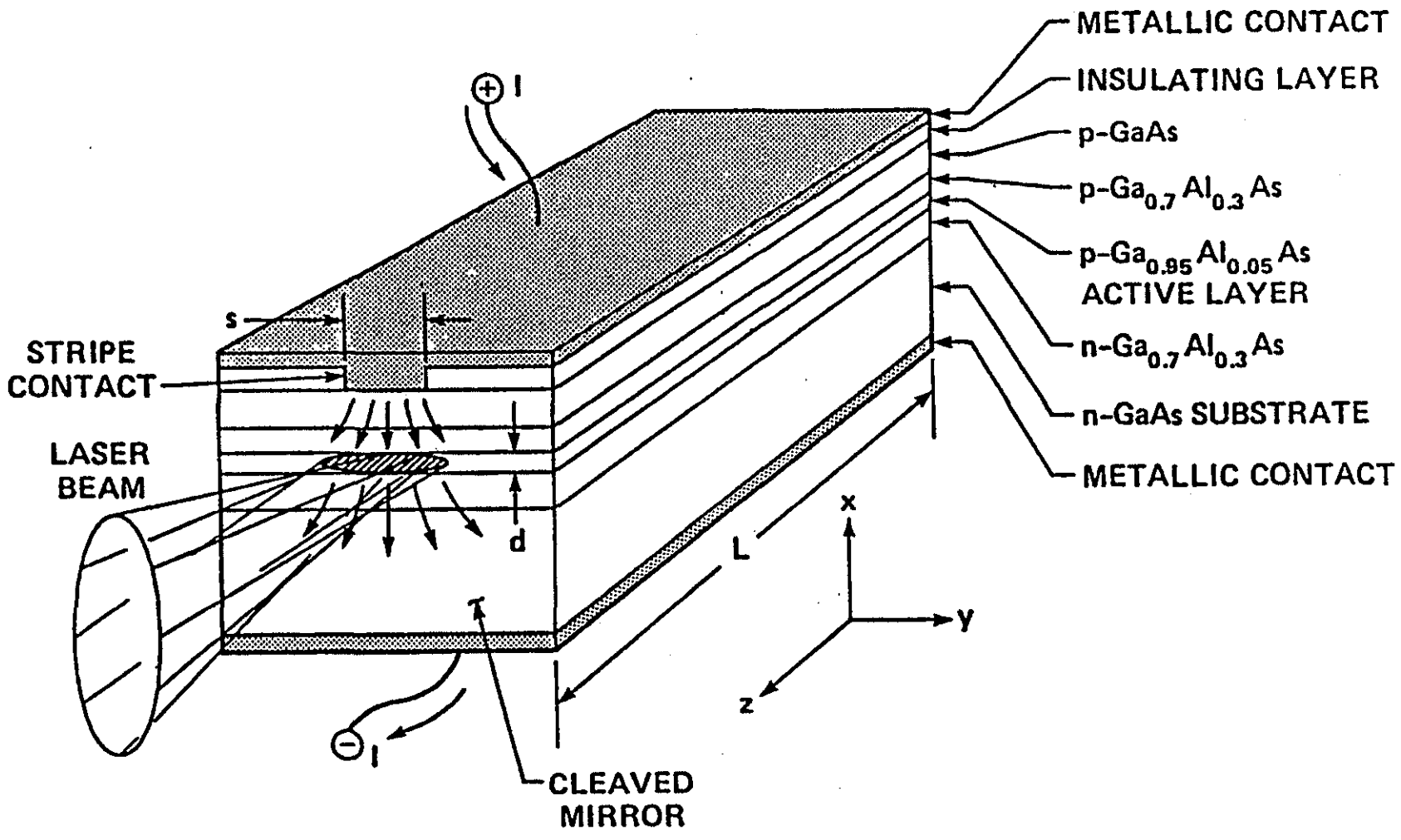


Fig. 75. Double heterojunction quantum well laser.

4. POINT AND LINE DEFECTS

4.1. Introduction

Only theory can deal with perfect crystals! We will show that real crystals must have imperfections. Some of these imperfections make semiconductors highly interesting from an electrical and optical device point of view. These are the electrically active dopant impurities which either replace a crystal host atom (substitutional) or occupy an interstitial site. Other imperfections are highly undesirable because they can produce energy levels in the bandgap which influence device performance in a detrimental way. This chapter deals mainly with the latter type of defects. We distinguish between point, line and two-dimensional defects.

4.2. Point Defects

Four different kinds of intrinsic defects are of importance in elemental semiconductors. they are:

- The lattice vacancy (V).
- The interstitial host atom (A_I).
- The Schottky defect: A host atom leaves its position and moves to the surface.
- The Frenkel defect: A host atom moves into an interstitial position forming a vacancy-interstitial pair.

In compound semiconductors we find two kinds of each of the above defects and a new kind of defect, the so-called anti-sites, e.g., a Ga atom residing on an As site (Ga_{As}) or an As atom sitting on a Ga site (As_{Ga}). Deviation from exact stoichiometry leads to very large concentrations of interstitials and antisites of the excess constituent. A good example of this is the As_{Ga}, a double donor in GaAs which is strongly favored as the deep donor which is responsible for the semi-insulating properties of much of today's GaAs.

A simple and elegant derivation of the dependence of the vacancy concentration on the temperature has been given by C. Kittel (*Introduction to Solid State Physics*, 3rd edition, J. Wiley and Sons, NY, 1966).

A monoatomic crystal with vacancies is considered. The aim is to minimize the free energy F by varying the concentration of vacant lattice sites. The energy required to remove one atom from the interior of the crystal is E_v. In order to create n vacancies, the energy nE_v has to be invested. It is assumed that the vacancies do not interact, i.e., the concentration is sufficiently small.

We now want to determine the entropy of the crystal with vacancies. The number of possibilities to form n vacancies in an N atom crystal is

$$\frac{N(N-1)\ldots(N-n+1)}{n!} = \frac{N!}{n!(N-n)!}$$

We divide by n! because it does not matter in which sequence we remove the n identical vacancies. Boltzmann has shown that the entropy of a system is proportional to the natural logarithm ln of the number of possible arrangements. Formation of n vacancies increases the entropy by

$$S = k_B \ln \frac{N!}{n!(N-n)!} \quad .$$

The change in free energy is

$$F = E - TS = n E_v - k_B T \ln \frac{N!}{n!(N-n)!} \quad .$$

Assuming that the crystal is sufficiently large that the surface does not play any role (not true for today's nanocrystals!) and that the total volume of the crystal stays constant, we know that the free energy will reach a minimum for a specific n, i.e., $\partial F/\partial n = 0$. Using Stirling's approximation $\ln(x!) = x \ln x - x$ for large x, we find

$$\ln \frac{N!}{n!(N-n)!} \cong N \ln N - (N-n) \ln(N-n) - n \ln n \quad .$$

In equilibrium we arrive at

$$\partial F/\partial n = E_v - k_B T \ln \frac{N-n}{n} = 0$$

and

$$\ln \frac{n}{N-n} = -E_v/k_B T$$

and for $n \ll N$

$$n = N \exp(-E_v/k_B T) \quad .$$

Assuming that it costs 2 eV to form a vacancy thermodynamically (in contrast to formation of defects with energetic electrons or ion beams), we find at the melting point of Si (T_m = 1683 K) per cm^{-3} (5×10^{22} atoms):

$$n = 5 \times 10^{22} \exp(-2\text{eV}/k_B\ 1683\ \text{K}) = 5.15 \times 10^{16}\ \text{cm}^{-3} \quad .$$

This computed vacancy concentration is very sensitive to E_v. Changing the value of E_v to 3 eV, n drops to 1.08×10^{15} cm^{-3}.

During cooling the vacancies become supersaturated and recombine with interstitials (which can be evaluated in a fashion similar to vacancies), with dislocations, or they migrate to the crystal surface. They also can condense into an intrinsic stacking fault, a part of a crystal plane which is missing. A stacking fault is surrounded by a dislocation loop.

Although we have performed this exercise for vacancies in silicon, it should be made clear that the Si interstitials (Si_I) have been shown to be the dominant native defect species at high temperatures in this semiconductor. The experimental proof for this finding has come mainly from detailed electron microscopy studies which have shown extrinsic stacking faults in dislocation-free Si crystals. Extrinsic stacking faults are small sections of extra planes which form when interstitial Si atoms condense during cooling. The fact that interstitial Si is the dominant native point defect at elevated temperatures affects all phenomena associated with diffusion, annealing of radiation-induced defects (e.g., during ion implantation), gettering, dislocation formation and motion, etc.

Contrary to silicon, germanium is believed to be dominated by vacancies up to the melting point T_m = 936°C. Because of difficulties with electron microscopy and other characterization techniques, the situation for germanium and most other semiconductors is not nearly as well understood as for silicon.

A direct consequence of the exponential dependence of the native defect concentration on temperature is the advantage gained from growing crystals with increased perfection at low temperatures. This can be achieved through liquid phase epitaxy and to a lesser degree with other low temperature epitaxial techniques. The additional advantage of liquid phase epitaxy over other epitaxial techniques is the gettering of impurities by the liquid metal which is typically used as the solvent.

Besides intrinsic point defects there exist several families of extrinsic defects. In most cases these defects are impurities occupying either a substitutional lattice site or one of several high symmetry interstitial sites. In this section we do not discuss impurities which give rise to energy levels in the forbidden gap of a semiconductor. The electronic properties of these so-called shallow and deep impurities have been introduced in Chapter 1 and the removal of unwanted impurities, i.e., purification, has been treated in Chapter 2.

The two intrinsic point defects which do not form energy levels in Si but still play major roles are oxygen and carbon. Oxygen rests between two neighboring Si atoms, forming bonds with both neighbors. The Si-Si bond length is too short to accommodate the oxygen impurity, which in turn moves perpendicularly to the bond direction, relieving the local strain. In this so-called bond-centered position, the oxygen atom can vibrate at distinct frequencies. In the infrared part of the spectrum these vibrations lead to absorption peaks which act as quantitative fingerprints for the presence of this impurity. Oxygen concentration measurements using these "local vibrational modes" (LVMs) will be discussed in more detail in the chapter on characterization.

The effects of oxygen in silicon on impurity gettering have been studied for many years and play a crucial role in silicon device processing. Oxygen has also been shown to combine with other defects, e.g., the vacancy with which it forms the A-center, a deep level defect complex. Control of the oxygen concentration in modern Si crystals is of paramount importance for device processing.

Carbon is a group IV element and assumes a substitutional position in Si. Because of its small size it produces considerable local strain. One way to relax this strain is to have a pair of carbon atoms occupy one Si site. Although carbon by itself does not produce any energy levels in the gap, it participates in many defect reactions which occur at high temperatures. Just as in the case of oxygen, the carbon concentration in high quality Si crystals is carefully controlled. A Proceedings Volume of the Materials Research Society (Mat. Res. Soc. Symp. Proc. Vol. **59**, J. C. Mikkelson, Jr. et al., eds.) is entitled *Oxygen, Carbon, Hydrogen and Nitrogen in Silicon.*

4.3. Line and Two-Dimensional Defects

Edge and screw dislocations are the two textbook examples of line defects. In real semiconductors one rarely encounters these two "pure" types of dislocations. Figure 81 shows the structure of such dislocations. The Burgers vector is used to characterize a dislocation. It is the closure vector of the Burgers loop around the dislocation. Figure 82 shows that the Burgers vector of an edge dislocation is normal to the line of the dislocation and Figure 83 shows that the Burgers vector of a screw dislocation is parallel to the line of the dislocation.

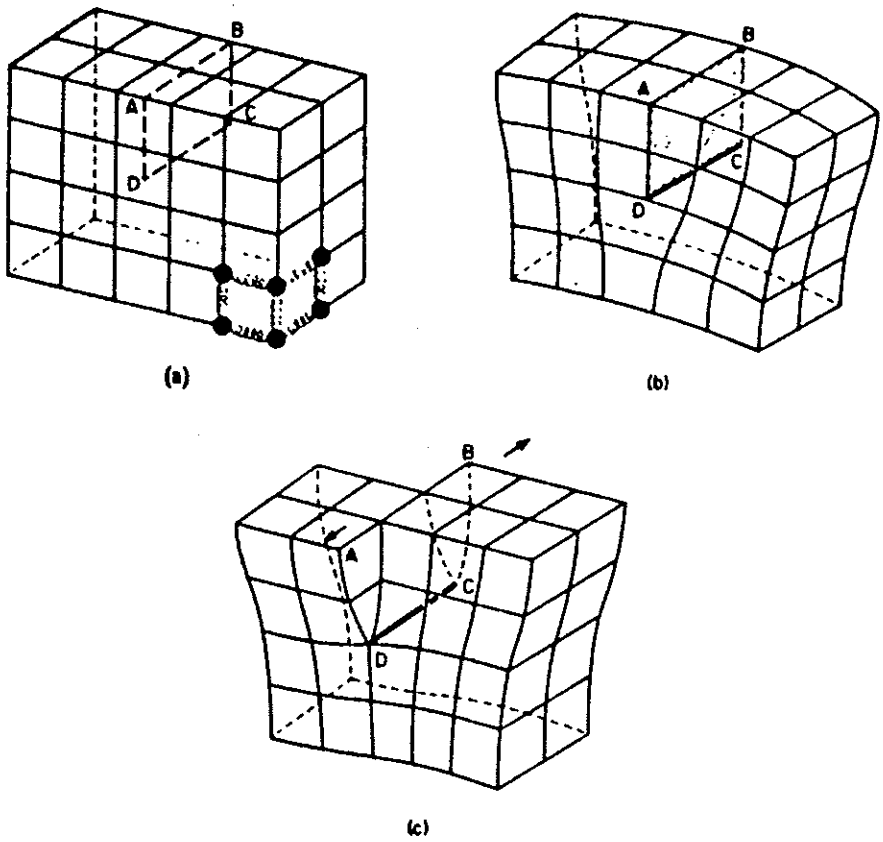


Fig. 81. (a) Model of a simple cubic lattice; the atoms are represented by hard spheres, and the bonds between atoms by springs, only a few of which are shown; (b) positive edge dislocation DC formed by inserting an extra half-plane of atoms in ABCD; (c) left-handed screw dislocation DC formed by displacing the faces ABCD relative to each other in direction AB.

Dislocations can *never* end inside a crystal without interacting with other defects. They can end at the crystal surface, form a closed loop, or branch into more dislocations. A low concentration of dislocations can be advantageous because they can absorb excess vacancies or interstitials present in the crystal after solidification from the melt. In this case, an edge dislocation "climbs," e.g., for the case of vacancies, the atoms of the extra half-plane at the dislocation line are removed when they fill arriving vacancies.

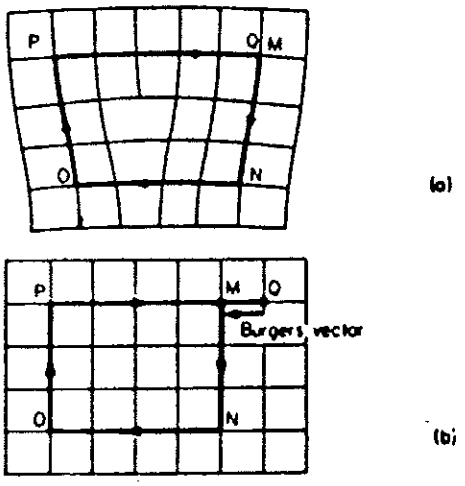


Fig. 82. (a) Burgers circuit round an edge dislocation; (b) the same circuit in a perfect crystal--the closure failure is the Burgers vector.

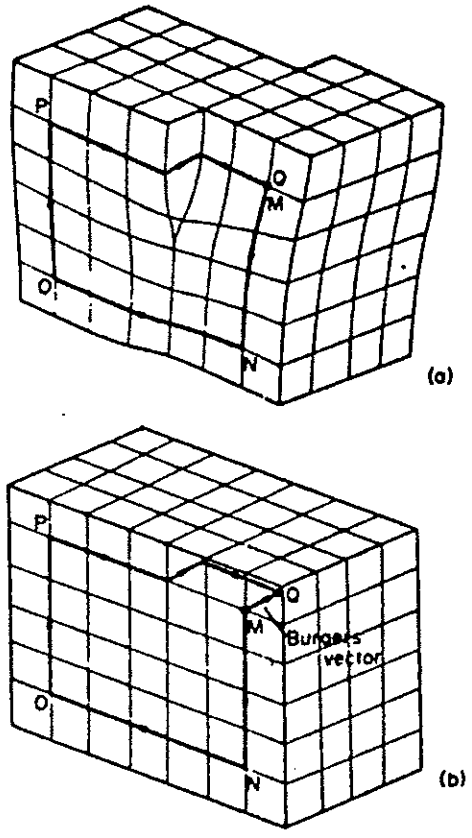


Fig. 83. (a) Burgers circuit round a screw dislocation; (b) the same circuit in a perfect crystal--the closure failure is the Burgers vector.

The model of the simple screw and edge dislocation has been derived from a simple cubic lattice. In the diamond lattice there are several possible dislocation structures, some of which have been characterized in detail using electron microscopy. The screw dislocation is usually dissociated into two 30° partial dislocations. The 60° dislocation, the most common line defect in Si, Ge and III-V compounds, is almost always split into a 30° and a 90° partial dislocation. The splitting of a dislocation into two Shockley partials is described by the following equation for the Burgers vector:

$$\frac{a}{2}[011] \rightarrow \frac{a}{6}[121] + \frac{a}{6}[112] \quad .$$

Figure 84 shows a ball and wire model of a split (aB) 60° dislocation. The stacking fault ribbon is located between the (111) planes. To the right lies the 90° partial dislocation, to the left we see the 30° partial dislocation. A ball and stick model only gives a rather crude picture of a dislocation. The details of the dislocation core are not revealed. It is quite obvious that in the simple ball and stick model a large number of broken or dangling bonds exist.

To understand theoretically and experimentally the core structure is of utmost interest because the core is the location of the possible electrical activity of a dislocation. Recent studies indicate that many of the neighboring dangling bonds in the core form covalent bonds. This is called dislocation core reconstruction or minimization of the dislocation energy. A reconstructed dislocation shows very little electrical activity, confirming many experimental observations on dislocations in a large variety of semiconductor crystals.

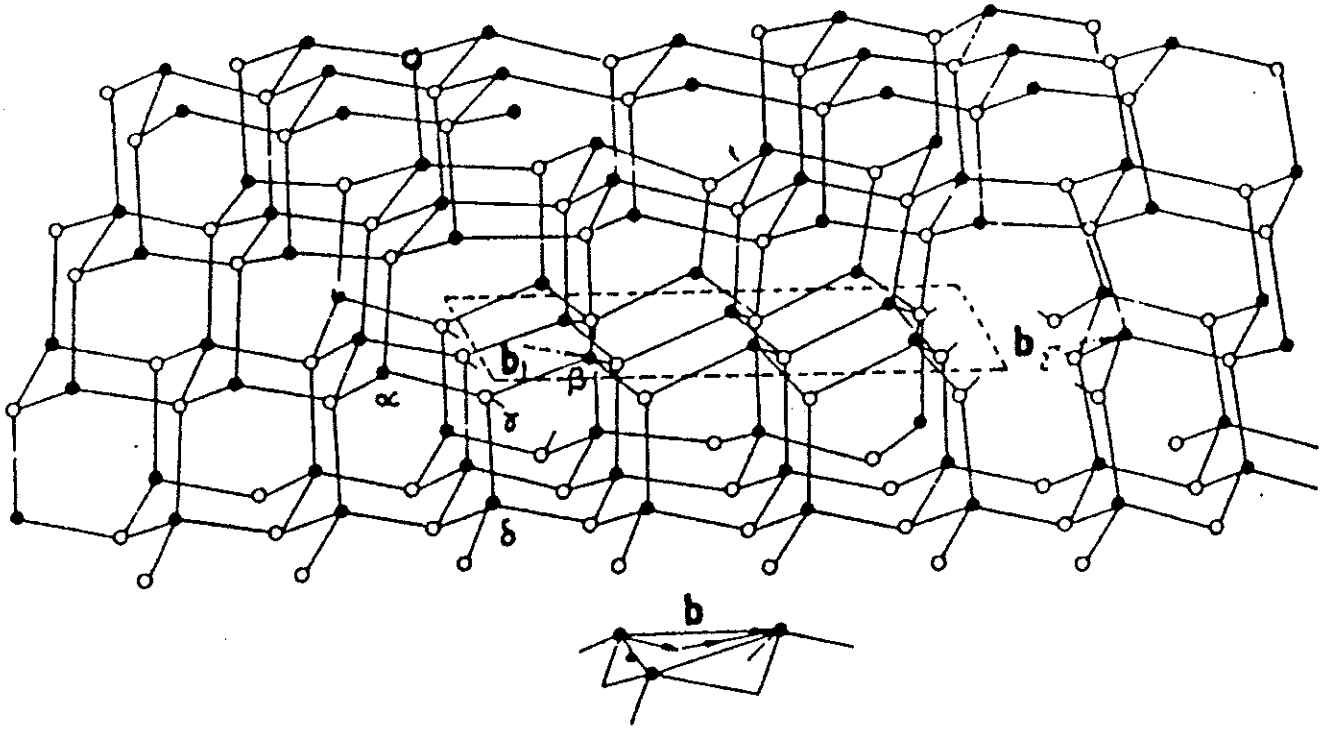


Fig. 84. Ball and wire model of a split (aB) 60° dislocation.

5. DOPING OF SEMICONDUCTORS

5.1. Introduction

Doping of semiconductors means controlled introduction of dopant impurities into well-defined areas of a semiconductor. Because of its extreme importance, doping has been and continues to be studied extensively. Every electronic device from the simple diode to the most sophisticated Ultra Large Scale Integrated (ULSI) circuit depends crucially on at least one, but much more commonly, on multiple doping processes. Doping of elemental semiconductor single crystals (Si, Ge) with the group III elements B, Al, Ga, In and Tl produces shallow*, "hydrogenic" acceptor levels in the bandgap near the valence band, leading to p-type conduction (mobile holes in the valence band). The group V elements P, As, Sb and Bi produce shallow, "hydrogenic" donor levels in the bandgap near the conduction band leading to n-type conduction (mobile electrons in the conduction band). Other elements produce a wide variety of levels in the bandgap. Usually these are not shallow (an exception is lithium, a shallow donor) and quite often more than one carrier can be emitted or absorbed from one impurity center (multi-valent acceptors/donors). In compound semiconductors and their alloys (e.g. GaN, ZnSe, GaAs, GaAsP, $Al_xGa_{1-x}As$, etc.) there exist two sublattices and the dopant impurities must be incorporated on the appropriate one in order to act as desired. In III-V semiconductors one finds that group II elements occupy the group III sublattice and act as acceptors while group VI elements prefer the group V sublattice forming donors. Group IV elements are in some cases amphoteric, i.e., they may occupy group III sublattice sites acting as donors or group V sublattice sited forming acceptors. Much of this has been discussed in chapter 1.2. "Extrinsic Semiconductors" and on page 32 one finds Fig. 15 which displays the energy levels for numerous impurities in Ge, Si and GaAs.

In certain cases one needs to dope to the highest possible concentrations in order to obtain low electrical impendances for high speed or high current density (often synonymous!) applications. The solid solubility (Fig. 85a,b) is a key parameter for these cases. It corresponds to the maximum equilibrium dopant atom concentration which can be introduced at a certain temperature. Introduction of impurities at a high temperature and subsequent cooling leads to dopant supersaturation. If the impurities are immobile at low temperatures (the case for most impurities at room temperature), the supersaturation can be "frozen" into the crystal. If the impurities are mobile, precipitation will occur—usually an undesirable process. Rapid Thermal Annealing (RTA), a recently developed technique based on rapid heating and cooling of

*Shallow means the energy difference between a dopant level and the corresponding band edge is small when compared with the bandgap.

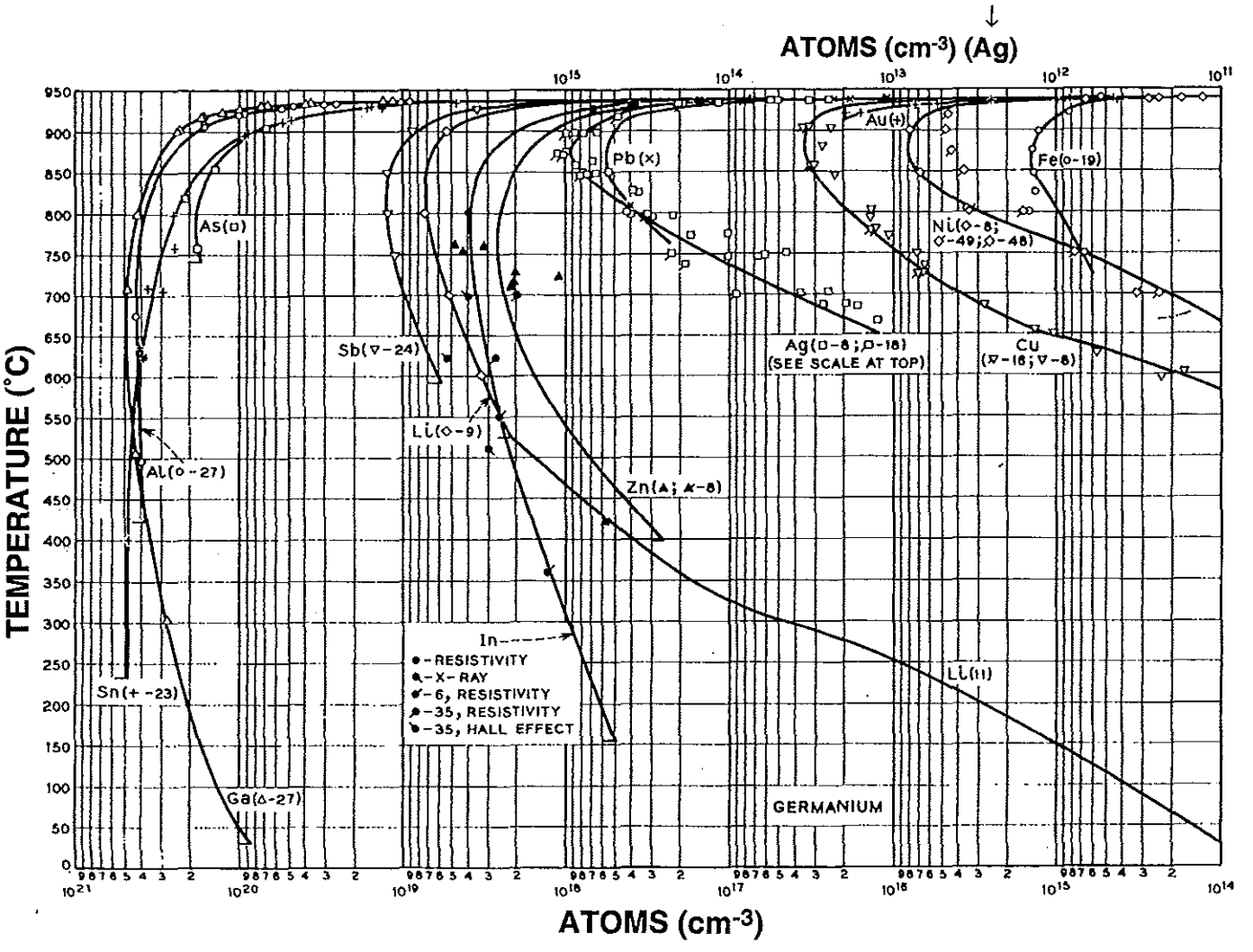


Figure 85a. Equilibrium Solubility of Impurities in Germanium
[after F.A. Trumbore, Bell Syst. Tech. J. **39**, 205 (1960).]

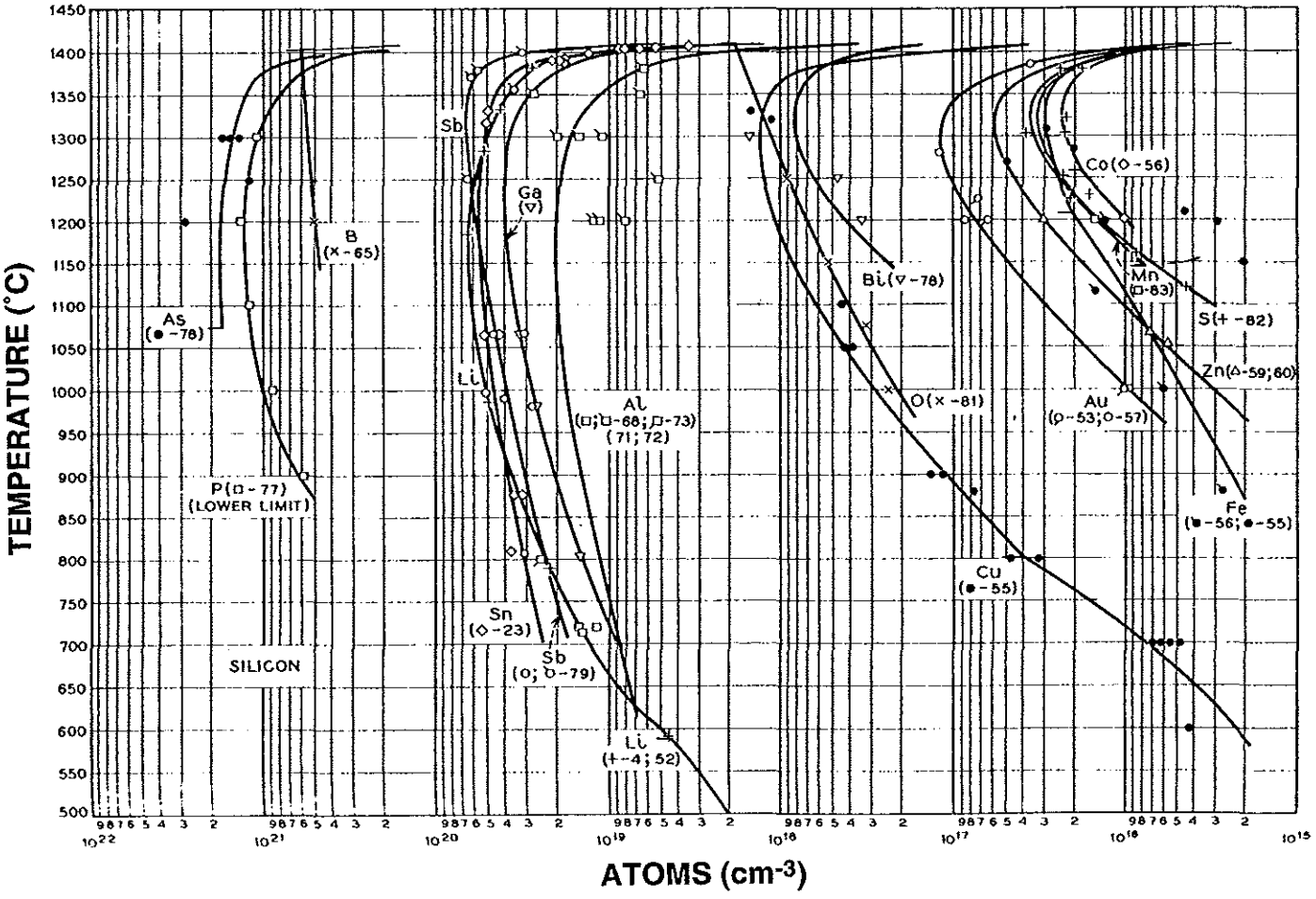


Figure 85b. Equilibrium Solubility of Impurities in Silicon
[after F.A. Trumbore, Bell Syst. Tech. J. **39**, 205 (1960).]

semiconductor wafers by a high power pulsed light source (in one to hundreds of seconds) can be used to obtain active dopant concentrations exceeding equilibrium impurity solubilities.

One can classify doping into several categories depending on the method used to introduce impurities:

- incorporation during crystal growth (bulk and thin films)
- neutron transmutation (bulk and thin films)
- diffusion (shallow layers)
- ion implantation (shallow layers)

5.2. Dopant Incorporation During Crystal Growth

The Czochralski (CZ) and the floating zone (FZ) crystal growth techniques allow more or less uniform doping of bulk single crystals. The upper limit of doping levels is usually given by the solid-solubility of a particular impurity. Most impurities exhibit a "retrograde" solubility near the melting point, i.e., the solubility decreases sharply with increasing temperature near the melting point of the semiconductor. At the phase boundary between solid (s) and liquid (l) we observe an impurity concentration step described by the segregation (or distribution) coefficient k (see chapter 2. "Semiconductor Purification"):

$$k = \frac{c_s}{c_l}$$

The lower limit for dopant impurity concentrations in a crystal is dominated by residual impurities in the polycrystalline starting material, from the melt container walls and from the ambient. The purity record lies around a few times 10^9 cm^{-3} in germanium. In these crystals one finds one dopant impurity for every 10^{13} Ge atoms (in chemist's terms: thirteen nines pure material)! Such crystals are used for the fabrication of detectors (p-i-n diodes) for energetic radiation (gamma rays, particles) (see Haller and Goulding, chapter 11 in the 2nd edition of the *Handbook on Semiconductors*). An important reason for this achievement is related to the fact that the melting point of Ge is 936°C, roughly 500°C lower than the melting point of Si. Lower temperature processing reduces difficulties in obtaining high purity.

The high melting point of Si affects doping and purification especially in regard to the group V donors P, As and to a lesser degree Sb. These high vapor pressure elements can leave the Si melt via the gas phase. Most poly Si produced from pyrolysis of SiH_4 and $SiHCl_3$ is initially n-

type (phosphorus) but turns p-type after one or two floating zone passes in vacuum. Much of the phosphorus evaporates while the boron remains in the crystal and, because the segregation coefficient of B is close to unity, this acceptor is barely transported. The purest Si crystals produced by the floating zone technique exhibit resistivities of > 100k Ω cm or < a few times $10^{10}cm^{-3}$ net impurities. The individual residual dopant concentrations lie between 10^{11} and $10^{12}cm^{-3}$, i.e., these materials are highly compensated.

Intentional doping of CZ crystals is achieved by either adding the dopant element in pure form to the melt (unusual because it is difficult to control the small quantities involved!), or by adding a piece of heavily-doped semiconductor (sometimes called "master alloy"). The impurity concentration changes along the crystal axis are caused by impurity segregation (see chapter 2.). This can be counteracted to some degree by continuously feeding fresh, pure poly material to the melt.

Impurity striations, or variations of the net-impurity concentration along the axis and as a function of the crystal radius, are an undesirable effect observed in almost all crystals grown from a melt. The impurity concentration fluctuations show up to three periodicities. The first one is due to the crystal rotation in a not absolutely perfectly cylindrical group of isotherms. Once every revolution, each point at the solid/liquid interface passes through the coldest and the hottest spots. The crystal growth rate and consequently the effective segregation coefficient are modulated by this regular change in temperature and temperature gradients. This leads to regular impurity concentration fluctuations (Fig. 86). A second effect is due to the fact that the incorporation of impurities changes the melting point (see Fig. 35, p. 68). This can lead to oscillations of the growth rate, to a variation of the stagnation layer width at the solid/liquid interface and to oscillations in the effective segregation coefficient. A third cause for impurity striations is the formation of convection cells in the melt (similar to magma-convection in the earth). The melt flow in these cells stirs the melt in distinct patterns which get "imaged" via the effective segregation coefficient directly into the local impurity concentration. (see Figs. 54a-f, page 93)

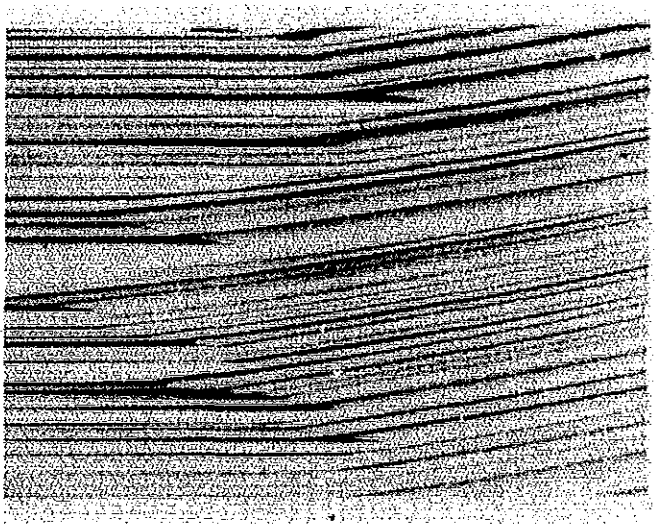


Fig. 86. Nonrotational impurity striations in Te doped InSb as revealed by chemical etching of a surface cut along the longitudinal growth axis. Observe the random nature of the striations which are continuous across the growth interface—curved in the off-facet region (right hand side) and flat in the facet region (left hand side). Magnification ~ 100. [Courtesy A.F. Witt and H.C. Gatos, J. Electrochem. Soc. **113**, 808 (1966).]

The understanding of the effect of melt convection on the local effective segregation coefficient has been significantly advanced in recent years through experimental and theoretical studies. In heavily Ga doped Ge and Te doped InSb crystals the dopant striations are made visible by cutting CZ boules along the growth axis all the way from the seed to the tail, lapping and polish etching the cut surface to remove saw damage, and then chemically attacking this surface with an etchant sensitive to the local doping concentration. The result of such a procedure is shown as black and white striations in Fig. 87 for Ga doped Ge and in Fig. 88 for Te doped InSb. An elegant way to produce a temporal record of the growth interface (liquid-solid interface) makes use of Peltier cooling (or heating) which occurs when an electric current is passed between two dissimilar conductors (in this case a solid semiconductor in contact with its liquid). High current pulses at regular intervals are sent through the seed and the crystal into the melt. At the interface momentary cooling (or heating) increases (reduces) the thermal gradient abruptly, changing the local crystal growth velocity. This change in turn affects the effective segregation coefficient and the dopant concentration. In this fashion a precisely timed record of the interface is stored and it allows one to obtain accurate microscopic growth velocities as well as records of melt convection and melt-back phenomena (Fig. 89). An excellent account of such studies has been given by J.R. Carruthers and A.F. Witt in *Crystal Growth and Characterization* edited by R. Ueda and J.B. Mullin (North Holland 1975).

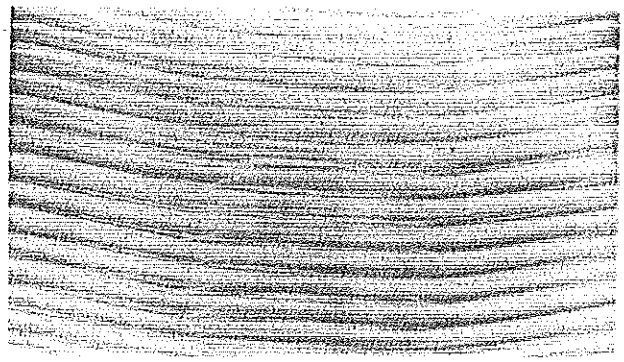


Fig. 87. Rotational and nonrotational impurity striations in Ga doped Germanium ($3 \times 10^{19}cm^3$) as revealed on a longitudinal cut along the 111 growth axis by chemical etching (6 seconds in 1 part H_2O, 1 part HF, 1 part CH_3COOH, at room temperature). Magnification ~ 10. [Courtesy J.R. Carruthers and A.F. Witt in *Crystal Growth and Characterization* edited by R. Ueda and J.B. Mullin (North Holland 1975).]

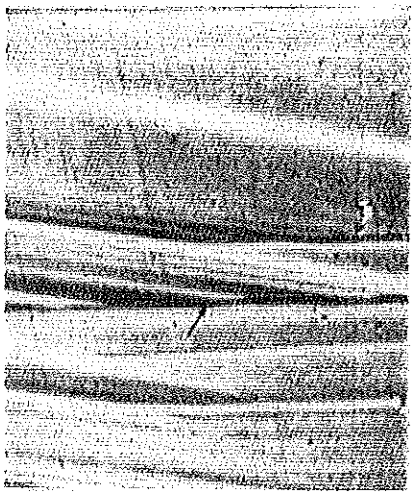
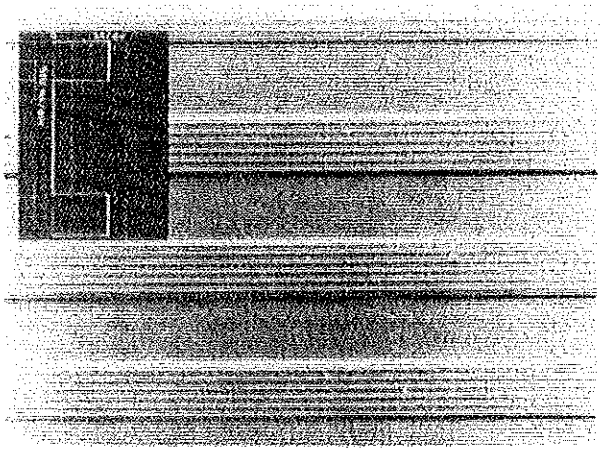


Fig. 88. Rotational and nonrotational impurity striations in Te doped indium antimonide ($6 \times 10^{17}cm^{-3}$) revealed on a longitudinal cut along the <111> growth axis by chemical etching (40 seconds in 1 part $KMNO_4$ 0.1M, 1 part HF, 1 part CH_3COOH). The abrupt termination of nonrotational striations at the pronounced rotational striations as indicated by the arrow is a clear indication of partial remelting. Magnification 160. [Courtesy K. Morizane, A.F. Witt and H.C. Gatos, J. Electrochem. Soc. **114**, 738 (1967).]

Fig. 89. Etched off-facet section of a Te doped indium antimonide single crystal grown without seed rotation and subjected to controlled current pulsing (see insert). The effects of extended current pulses from the crystal to the melt are investigated through superimposed current pulses of 15 msec duration which result in effective interface demarcation. From the spacing of the demarcation lines it can be seen that the extended current pulses lead to abrupt, constant increases in the microscopic growth-rate. Note also that with termination of current flow the microscopic growth-rate reverts without any noticeable transient to its original level. Magnification ~ 75. [Courtesy M. Lichtensteiger, A.F. Witt, and H.C. Gatos, J. Electrochem. Soc. **118**, 1013 (1971).]



Progress in the numerical simulation of heat and mass transfer in a CZ crystal puller has been impressive in recent years, and it is based mostly on the much improved power of desktop computers. K. Kakimoto, [Applied Physics Reviews, J. Appl. Phys. **77**, 1827-42 (1995)] describes experimentally determined flow patterns in a CZ Si melt. Small tungsten spheres 0.5 mm in diameter coated with SiO_2 and graphite were used as traces in the melt. X-ray imaging produced the particle paths shown in Fig. 90. The temperature distribution and the finite-element mesh used for the numerical simulations are displayed in Fig. 91. The author discusses the computation of the temperature distribution in the whole CZ puller and the flow patterns in the melt. One of the main findings is the coexistence of natural convection flow and surface-tension driven flow originating from the relative rotation of the crystal and the crucible. The computed particle paths look remarkably realistic and reflect the enormous complexity of the interior of a CZ crystal puller. (Note: in Kakimoto's article the figure captions got mixed up: the caption of Fig. 16 belongs to Fig. 17, the one of Fig. 17 belongs to Fig. 18 and the one with Fig. 18 belongs to Fig. 16!)

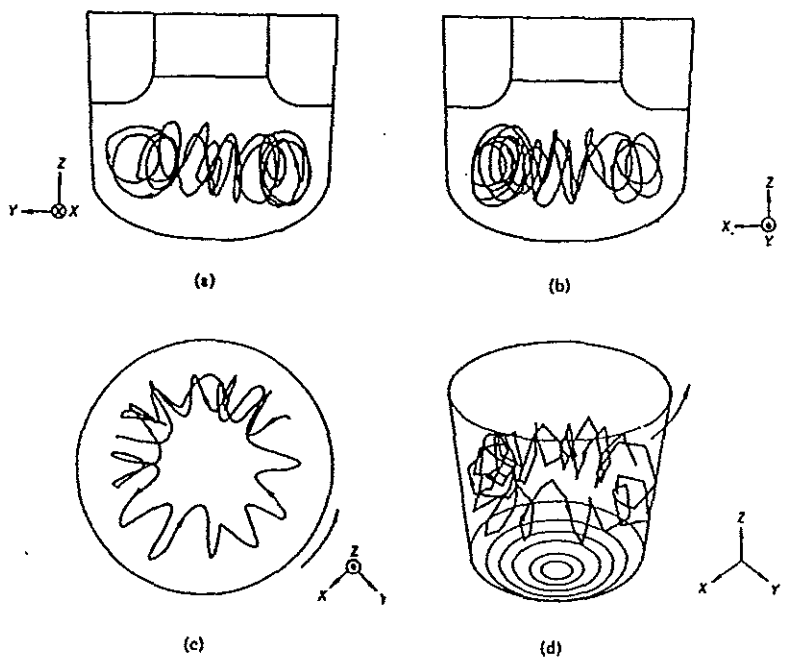


Fig. 90. Particle paths observed from (a) X, (b) Y, (c) Z directions and of bird's-eye view, (d) for axisymmetric flow [Courtesy K. Kakimoto, Applied Physics Review, J. Appl. Phys. **77**, 1829 (1995).]

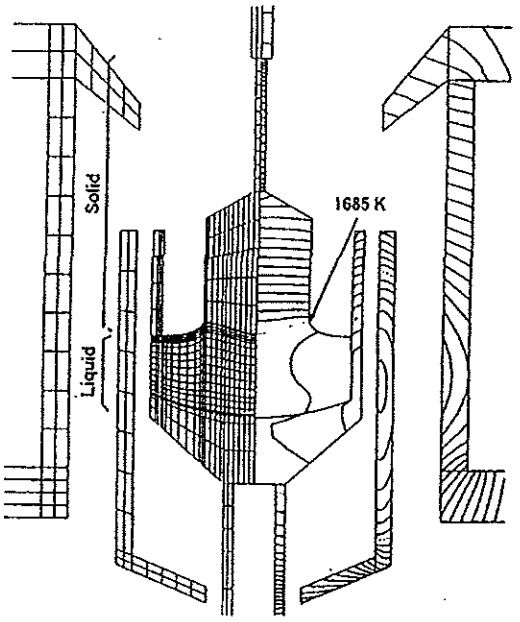


Fig. 91. Temperature distribution (right-hand side) and finite-element mesh (left-hand side) used in the calculation. [Courtesy K. Kakimoto, Applied Physics Review, J. Appl. Phys. **77**, 1832 (1995).]

Because a liquid semiconductor is a good electrical conductor, one can use Eddy current breaking of melt convection. This old principle has for the first time been applied by the Sony Corp. to a large Czochralski Si crystal puller. A very large magnet produces a strong magnetic field (a few kGauss) which causes Eddy currents in the flowing melt. The energy dissipated by the Eddy currents results in an increase of the effective viscosity. The company claims to have substantially reduced impurity striations. Because mixing of the Si melt is much reduced, the oxygen which is freed in the reduction of the SiO_2 crucible walls by the molten Si can not move effectively into the whole melt and the resulting CZ crystals exhibit much lower oxygen concentrations than standard CZ Si crystals.

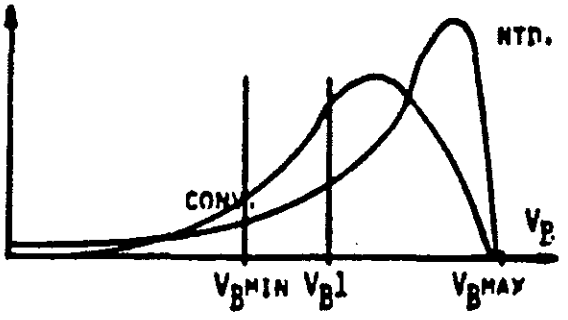
The difficulties encountered in controlling dopant-striation formation in melt grown crystals are massively reduced in thin film growth techniques using liquids and gases and are absent in MBE grown layers. In VPE systems laminar flow can be achieved through various means (e.g. reactor design, susceptor rotation) and in LPE systems the volumes of the liquids are typically too small to develop convection patterns.

5.3. Neutron Transmutation Doping

[Refs. *Neutron Transmutation Doping in Semiconductors* Jon Meese, ed.; Plenum Press (1979), *Neutron Transmutation Doping of Semiconductor Materials*, R.D. Larrabee, ed.; Plenum Press (1984).]

In 1973, a German high-power Si device manufacturer showed for the first time that neutron-transmutation-doped silicon is superior to standard bulk doped crystals for the fabrication of 4 kV, high-current thyristors and diodes [M.S. Schnöller, IEEE Trans. Electr. Dev. **ED-21**, 313, (1974).] The advantage of neutron transmutation doping (NTD) lies in the extreme doping homogeneity which can be achieved. The doping fluctuations observed in floating zone crystals lead to high-field spots in reverse-biased, high-voltage devices. These spots cause a severe reduction of the average maximum attainable reverse breakdown fields. Figure 92 shows the distribution of the breakdown voltage for NTD and conventional Si power devices.

YIELD IMPACT IN DEVICE MANUFACTURING



TOTAL YIELD $V_B > V_B^{MIN}$ ~ 90%

YIELD CONV. $V_B > V_B^1$ ~ 60%

YIELD NTD $V_B > V_B^1$ ~ 85%

Figure 92

The natural silicon isotopic composition is 92.23% ^{28}Si, 4.67% ^{29}Si and 3.10% ^{30}Si. These three stable isotopes undergo (n, γ) reactions in the neutron flux of a reactor:

Natural Abundance	Isotope
(92.23%)	$^{28}Si\ (n,\gamma)\ ^{29}Si$
(4.67%)	$^{29}Si\ (n,\gamma)\ ^{30}Si$
(3.10%)	$^{30}Si\ (n,\gamma)\ ^{31}Si \rightarrow {}^{31}P + \beta^- + \bar{\nu}_e;\ T_{1/2} = 2.62$ hr.

The thermal neutron capture cross section, σ_c, for the last reaction is 0.11 barn (1 barn = 10^{-24} cm^2). It is this reaction which is used to dope Si with P. For a total neutron dose n(cm^{-2}) the concentration of phosphorus [P] atoms is simply

$$[P] = n\,\sigma_c\,[^{30}Si]\,.$$

Unfortunately there are several radiation damage processes which occur during transmutation doping:

- fast neutron knock-on displacements
- fission gamma ray induced damage
- gamma ray recoil damage
- beta recoil damage
- charged particle knock-on from (n,p), (n,α), etc.

A summary of the radiation damage reactions in Fig. 93 shows that neutron knock-ons are the most detrimental reactions. But even in the case where thermal neutrons $\left(E_{neutron} \leq \frac{3}{2} k_B T\right)$ are dominant (a case which is approached by some modern reactors with special moderation), we are still faced with substantial gamma ray recoil damage. The damage consists mainly of displaced silicon atoms. Thermal annealing to ~ 900°C is necessary to remove most of the damage. The lifetime of minority carriers remains reduced after NTD and annealing, because of remaining deep level centers.

POSITION → / DAMAGE PARTICLE ↓	IN CORE	IN GRAPHITE MODERATOR
FAST NEUTRON	4.06×10^6	1.38×10^4
FISSION GAMMA	3.64×10^3	36.4
GAMMA RECOIL	1.29×10^3	1.29×10^3
BETA RECOIL	2.76	2.76
TOTAL DISP/(P)	4.06×10^6	1.51×10^4

Fig. 93. Number of displaced silicon atoms per phosphorus nucleus produced for various damage mechanisms shown for an in-core fission neutron spectrum and a graphite moderated neutron spectrum.

A further problem with Si NTD is the neutron capture by ^{31}P which was produced by neutron capture of ^{30}Si. The ^{32}P isotope obtained in the reaction

$$^{31}P\ (n,\gamma)\ ^{32}P \rightarrow\ ^{32}S + \beta^- + \bar{\nu}_e$$

decays into sulfur, a deep double donor in Si. The half life is about two weeks, much longer than the half life of the ^{31}Si decay, and it leads to prolonged radioactivity of the Si samples. This secondary reaction limits the NTD method to resistivity levels of $\rho \sim 5\ \Omega$ cm. ($\sim 10^{15}$ P cm^{-3}). Using very high neutron flux reactors ~ 10^{14} - 10^{15}n cm^{-1}s^{-1} reduces the relative ^{32}P production because the ^{31}P has not yet fully formed during the irradiation time.

Whereas Si is at present the only semiconductor which is neutron transmutation doped on a commercial scale, other semiconductors can be doped by NTD as well. Figure 94 shows a summary of NTD for a few common semiconductors.

			Resulting dopant concentration atoms (cm^{-3})	
Semiconductor	Nuclear reaction	Type of doping	N_A, N_D	$\lvert N_A - N_D \rvert$
Si	$^{30}Si\ (n,\gamma)\ ^{31}Si\ (\beta)\ ^{31}P$	n		$2.06 \times 10^{-4}\ \Phi t$
Ge	$^{70}Ge\ (n,\gamma)\ ^{71}Ge\ (EC)\ ^{71}Ga$	p	$3.0 \times 10^{-2}\ \Phi t$	
	$^{74}Ge\ (n,\gamma)\ ^{75}Ge\ (\beta)\ ^{75}As$	n	$1.1 \times 10^{-2}\ \Phi t$	$1.8 \times 10^{-2}\ \Phi t$
	$^{76}Ge\ (n,\gamma)\ ^{77}Ge(\beta)\ ^{77}As(\beta)\ ^{77}Se$	n	$1.1 \times 10^{-3}\ \Phi t$	
Se	$^{74}Se\ (n,\gamma)\ ^{75}Se\ (EC)\ ^{75}As$	p		$9 \times 10^{-3}\ \Phi t$
GaAs	$^{69}Ga\ (n,\gamma)\ ^{70}Ga\ (\beta)\ ^{70}Ge$	n	$2.5 \times 10^{-2}\ \Phi t$	
	$^{71}Ga\ (n,\gamma)\ ^{72}Ga\ (\beta)\ ^{72}Ge$	n	$4.4 \times 10^{-2}\ \Phi t$	$1.7 \times 10^{-1}\ \Phi t$
	$^{75}As\ (n,\gamma)\ ^{76}As\ (\beta)\ ^{76}Se$	n	$9.9 \times 10^{-2}\ \Phi t$	
GaP	$^{69}Ga\ (n,\gamma)\ ^{70}Ga\ (\beta)\ ^{70}Ge$	n	$2.8 \times 10^{-2}\ \Phi t$	
	$^{71}Ga\ (n,\gamma)\ ^{72}Ga\ (\beta)\ ^{72}Ge$	n	$4.9 \times 10^{-2}\ \Phi t$	$8.2 \times 10^{-2}\ \Phi t$
	$^{31}P\ \ (n,\gamma)\ ^{32}P\ \ (\beta)\ ^{32}S$	n	$4.7 \times 10^{-3}\ \Phi t$	

Fig. 94. Possibilities for semiconductor doping by irradiation with thermal neutrons [Φ = neutronfluence/cm^{2}s, t = time (s)].

Ge is the only semiconductor which becomes doped both with donors and acceptors by neutron transmutation. This leads to a fixed compensation ratio if one starts with high-purity crystals of natural isotopic composition. This fact has been used by H. Fritzsche and M. Cuevas [Phys. Rev. **119**, 1238 (1960)] to study conduction mechanisms at high homogenous doping levels and constant compensation.

Starting in the early 1980's we have doped natural, ultra-pure Ge via NTD to a wide range of doping levels. We choose the dopant concentrations high enough for hopping conduction to occur at low temperatures. In the hopping conduction regime carriers tunnel from occupied to empty impurity levels and do not require the thermal energy required for full ionization into the nearest band. In this conduction regime the resistivity, ρ, obeys the following dependence

$$\rho = \rho_o \exp\left(\Delta/k_B T\right)^{1/2}$$

where ρ_o and Δ are constants for a given neutron fluence. Figure 95 shows an Arrhenius like plot of log (ρ) as a function of $T^{-1/2}$ for several NTD Ge samples. The experimental points follow the law given in the above equation over up to ten order orders of magnitude. [For a derivation of $\rho(T)$, see B.I. Shklovskii and A.L. Efrös, *Electronic Properties of Doped Semiconductors*, Springer Series in Solid-State Sciences **45**, Springer Verlag (1984).]

In recent years we have been able to obtain isotopically separated Ge from countries of the former Soviet Union. The pure isotopes can be engineered into an artificial Ge crystal which can be doped n- or p-type with almost any compensation value by altering the starting isotopic ratios. Pure ^{70}Ge will become uncompensated p-type while pure ^{74}Ge will turn into uncompensated n-type material. This field of "Isotopically Engineered Semiconductors" is in its very early phase and whether it takes off on a larger scale will depend mostly on economical factors. The typical costs for stable isotopes are today still a few US$ per mg (or a few M$ per kg!) far too high to be of commercial viability. These costs can be reduced by orders of magnitude if large quantities are required. [See E.E. Haller, "Isotopically Engineered Semiconductors", Applied Physics Reviews, J. Appl. Phys. **77**(7), 2857-78 (1995).]

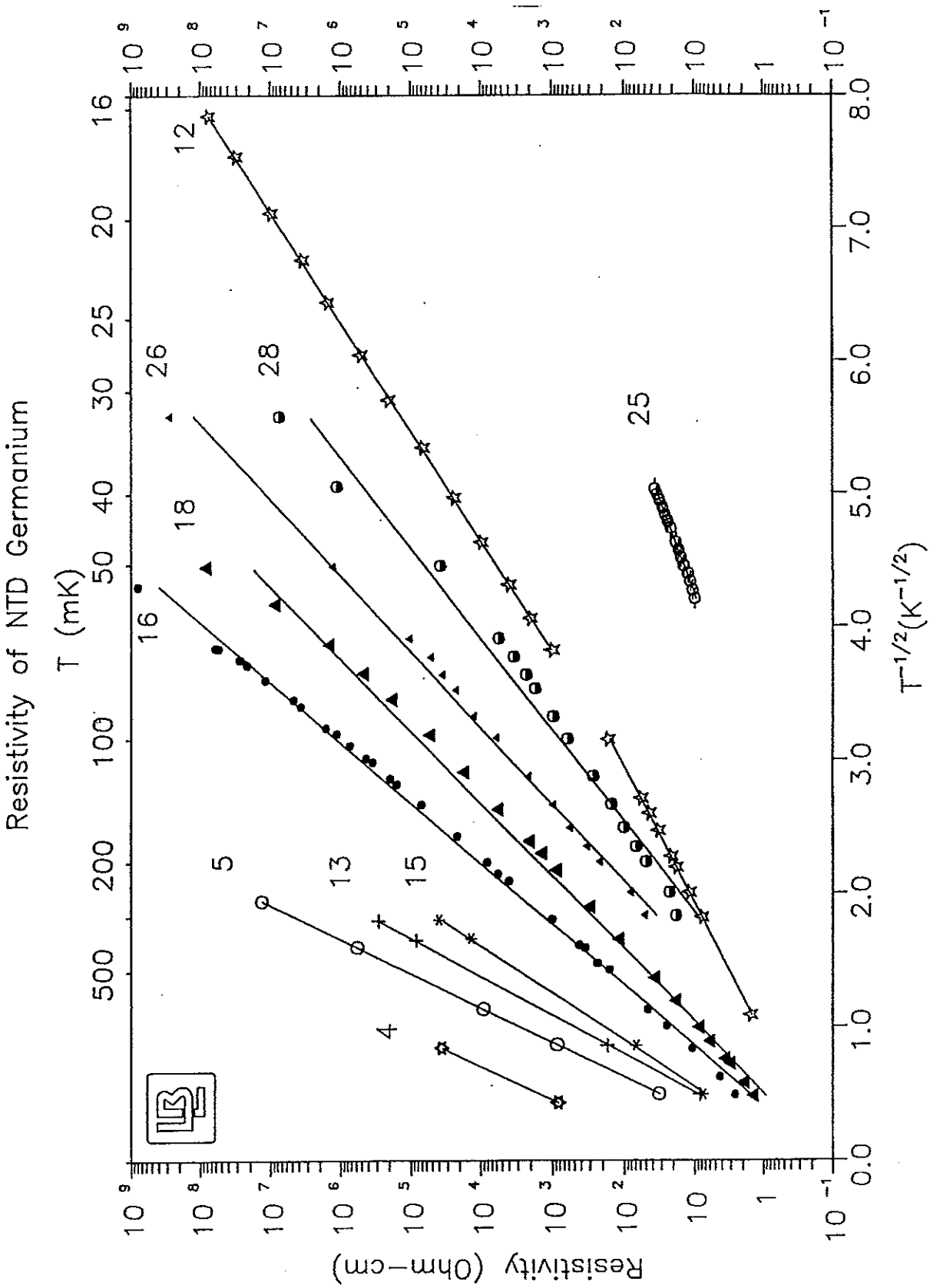

Resistivity of NTD Germanium
T (mK)
Resistivity (Ohm-cm)
$T^{-1/2}(K^{-1/2})$
16
12
26
28
18
25
5
13
15
4

Figure 95

5.4. Alloyed (or fused) Contacts

"Alloying" is a very old method of doping which has been widely used for the fabrication of "ohmic" contacts on semiconductor specimens or devices as well as for fabricating early junction transistors. Soldering a wire to a piece of semiconductor single crystal can lead to an alloyed contact (though many experimentalists do not realize this!). Using the Al-Si phase diagram as an example (Fig. 96), alloyed contact formation can be explained as follows: a thin (few μm) layer of Al is evaporated on a Si single crystal slice. Heating under an inert or reducing gas atmosphere leads to the formation of an aluminum-silicon eutectic at T = 577°C. The eutectic composition corresponds to 88 atomic % Al and 12 atomic % Si. Upon further heating more Si is dissolved because the amount of Al is limited. At 700°C the Si content has risen to 20%.

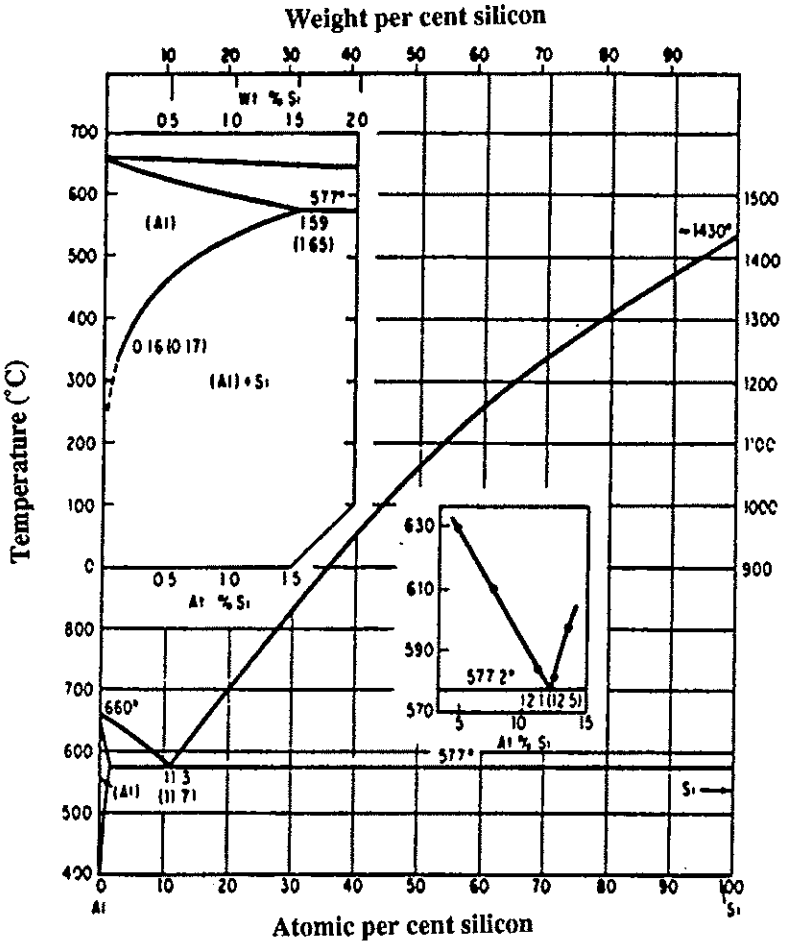


Fig. 96. Aluminum-silicon phase diagram (from *Constitution of Binary Alloy* by Max Hansen. Copyright 1958. McGraw-Hill Book Company. Used by permission.)

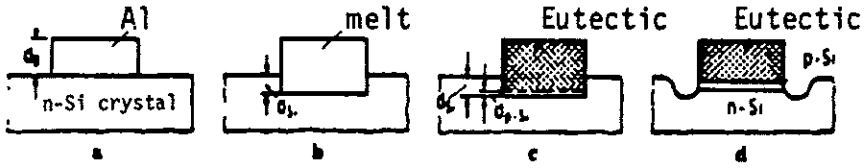


Figure 97

Consulting Fig. 85b we find that at 700°C the solubility of Al in Si is 9×10^{18}cm^{-3}. Cooling slowly from this temperature leads to epitaxial regrowth of Si on the silicon substrate and incorporation (doping) of Al acceptors in the epi layer. Passing through the eutectic temperature, the eutectic composition will separate into Al with a few % Si and pure Si, both solids at $T_{eutectic}$. The heavily-doped epi layer acts as a good ohmic contact on p-Si but "blocks" on n-Si. The described sequence is schematically shown in Fig. 97 for a n-Si wafer. The first widely used junction Si and Ge diodes and transistors were fabricated using In-alloyed emitter and collector regions (Fig. 98).

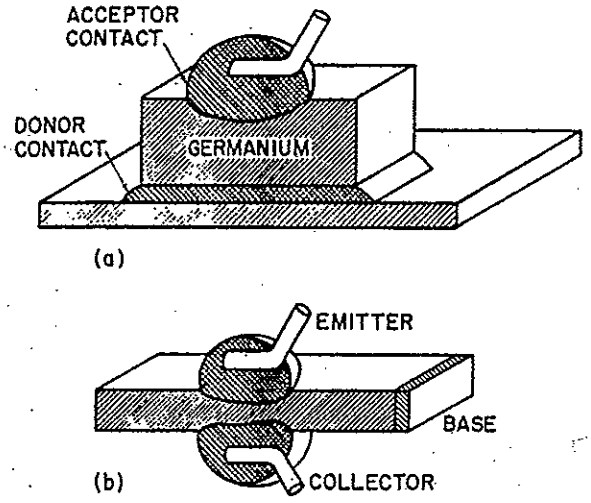


Fig. 98. Fused impurity contacts on (a) rectifiers and on (b) transistors. [Courtesy R.N. Hall, Proc. IRE **40** (11), 1512 (1952).]

In case one wants to separate the metallurgical aspects (eutectic formation, epi-growth) from the doping, one can choose an appropriate metal which does not become incorporated in the epi regrowth layer. By adding effective dopants to this metal, one can control both the metallurgical and doping aspects of the process. Such a case is gold with 1% Ga or 1% As which has been, and still is, used for alloyed contacts on Ge devices. In this case the gold-germanium eutectic forms at

356°C. Additional Ge dissolves upon further heating. During cooling, germanium crystallizes epitaxially on the bulk Ge. The solubility of Au in Ge at low temperatures is totally negligible. However the epi layer incorporates either Ga or As leading to p^+ or n^+ doping respectively.

The Au-Si eutectic temperature is also very low ($T_{eut.}$ = 370°C) and allows low temperature contact formation on Si.

N-type contacts on GaAs are often formed by evaporating a film of Au containing a few % Ge on a surface with a very thin (~ 200Å) Ni "wetting" layer. Upon heating for two minutes to ~ 450°C the various layers and the GaAs interact. It is generally believed that Ga moves out of the GaAs into the Au layer and Ge replaces the Ga atoms forming shallow donors .

5.5. Diffusion

5.5.1. Phenomenological, macroscopic aspects

[Refs. *Atomic Diffusion in Semiconductors*, D. Shaw (Plenum Press, 1973); *A Primer of Diffusion Problems*, R. Ghez (Wiley, 1988); *Silicon Semiconductor Technology*, W.R. Runyan; *Diffusion*, W. Jost; *Atomic Diffusion in III-V Semiconductors*, B. Tuck (Adam Hilger, 1988); *Diffusion in Crystalline Solids*, G.E. Murch and A.S. Nowick, eds. (Academic Press, 1984); see also original articles by A. Seeger and C.P. Chik, Phys. Stat. Sol. **29**, 455 (1968) and many other, recent authors.]

Impurity diffusion is still one of the most important technologies for the production of well-defined doped layers in semiconductors. Diffusion is a thermally activated process which tends to reduce concentration differences in a system through the random walk of individual dopant atoms and defects. Diffusion is not driven by concentration gradients! Fick's first and second law quantitatively describe matter transport by diffusion:

$$\vec{J} = - D \vec{\nabla} C \qquad \text{Fick's first law}$$

and

$$\frac{\partial C}{\partial t} = -\vec{\nabla}\vec{J} = \vec{\nabla}(D\vec{\nabla}C); \text{ and for D = const.: } \frac{\partial C}{\partial t} = \Delta C \qquad \text{Fick's second law}$$

Very often one deals with a one-dimensional case and can write

$$J = - D\frac{\partial C}{\partial x} \text{ and } \frac{\partial C}{\partial t} = \frac{\partial C}{\partial x}(D\frac{\partial C}{\partial x}); \text{ and for D = const.: } \frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} .$$

D = diffusion coefficient or diffusivity (cm^2s^{-1}),

C = concentration of diffusing species (cm^{-3}); and
J = flux of diffusing species ($cm^{-2}s^{-1}$).

In many cases one can describe the diffusion coefficient with an Arrhenius dependence:

$$D = D_o \exp(-H^M/k_BT) \ ,$$

with H^M = migration enthalpy for diffusion, k_B = Boltzmann constant = 8.62×10^{-5} eV/K.

The pre-exponential factor D_o can be expressed by

$$D_o = g\, a^2\, \nu_o \exp(S^M/k_B) \ ,$$

with g being a geometric crystal structure dependent factor close to unity, a = lattice constant and ν_o = highest vibrational mode frequency of the lattice (attempt frequency), S^M = migration entropy. Two boundary conditions are of general importance for many technical applications. Both deal with the case of an impurity which enters a "pure" single crystal from or near the surface. In one case, an inexhaustible source of impurities is present; in the other case one deals with a finite source (e.g., a "buried" implanted layer).

- Diffusion from an inexhaustible source into a semi-infinite space: For the one-dimensional problem with an inexhaustible source on the surface at x = 0 with no impurities in the crystal at t = 0 and under the condition that D = constant at a given temperature, we find the solution of Fick's second law in the form of a complementary error function:

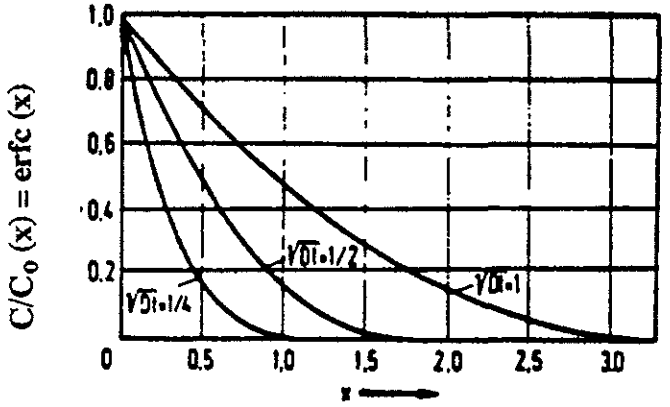
$$C(x,t) = C_o \mathrm{erfc}\left[\frac{x}{2\sqrt{Dt}}\right]$$

with C_o = surface concentration (constant). The error function is defined by an integral:

$$\mathrm{erfc}(y) = 1 - \mathrm{erf}(y) = 1 - \frac{2}{\sqrt{\pi}} \int_0^y e^{-z^2} dz \qquad \text{with } y = \frac{x}{2\sqrt{Dt}}$$

Pages 161-163 contain the values of erfc(y) for $0 < y \leq 4.5$. Figure 99 shows the values for erfc(x) graphically as a function of x for three values of $\sqrt{Dt}$. The value of N(x,t) is close to $^1/_2N_o$ for y =

$^1/_2$ or $x = \sqrt{Dt}$. $2\sqrt{Dt}$ is often used as a "normalized" diffusion depth when discussing or comparing diffusion processes. For $y = 1.82$ the value of erfc(y) has dropped to 1% and for $y = 4.0$ it is 1.5×10^{-8}, a very rapid drop-off indeed!



erfc-Profile

Figure 99

y	erfc(y)	y	erfc(y)	y	erfc(y)
0,00	+1,00000	0,22	+0,75570	0,44	+0,53377
0,01	+0,98871	0,23	+0,74497	0,45	+0,52451
0,02	+0,97743	0,24	+0,73429	0,46	+0,51534
0,03	+0,96615	0,25	+0,72367	0,47	+0,50625
0,04	+0,95488	0,26	+0,71310	0,48	+0,49725
0,05	+0,94362	0,27	+0,70258	0,49	+0,48833
0,06	+0,93237	0,28	+0,69211	0,50	+0,47950
0,07	+0,92114	0,29	+0,68171	0,51	+0,47075
0,08	+0,90992	0,30	+0,67137	0,52	+0,46210
0,09	+0,89871	0,31	+0,66109	0,53	+0,45353
0,10	+0,88753	0,32	+0,65087	0,54	+0,44506
0,11	+0,87637	0,33	+0,64077	0,55	+0,43667
0,12	+0,86524	0,34	+0,63063	0,56	+0,42838
0,13	+0,85413	0,35	+0,62061	0,57	+0,42018
0,14	+0,84305	0,36	+0,61067	0,58	+0,41207
0,15	+0,83200	0,37	+0,60079	0,59	+0,40406
0,16	+0,82098	0,38	+0,59099	0,60	+0,39614
0,17	+0,81000	0,39	+0,58126	0,61	+0,38831
0,18	+0,79906	0,40	+0,57160	0,62	+0,38058
0,19	+0,78816	0,41	+0,56203	0,63	+0,37295
0,20	+0,77729	0,42	+0,55253	0,64	+0,36541
0,21	+0,76647	0,43	+0,54311	0,65	+0,35797

y	erfc(y)	y	erfc(y)	y	erfc(y)	y	erfc(y)	y	erfc(y)	y	erfc(y)
0,66	+0,35062	1,01	+0,15318	1,36	+0,054438	1,71	+0,015592	2,06	+0,003576	2,41	+0,000653
0,67	+0,34337	1,02	+0,14916	1,37	+0,052687	1,72	+0,014997	2,07	+0,003417	2,42	+0,000620
0,68	+0,33621	1,03	+0,14521	1,38	+0,050983	1,73	+0,014421	2,08	+0,003265	2,43	+0,000589
0,69	+0,32915	1,04	+0,14135	1,39	+0,049326	1,74	+0,013865	2,09	+0,003119	2,44	+0,000559
0,70	+0,32219	1,05	+0,13756	1,40	+0,047714	1,75	+0,013328	2,10	+0,002979	2,45	+0,000530
0,71	+0,31532	1,06	+0,13385	1,41	+0,046147	1,76	+0,012809	2,11	+0,002845	2,46	+0,000503
0,72	+0,30856	1,07	+0,13022	1,42	+0,044623	1,77	+0,012309	2,12	+0,002718	2,47	+0,000477
0,73	+0,30189	1,08	+0,12667	1,43	+0,043142	1,78	+0,011825	2,13	+0,002592	2,48	+0,0004527
0,74	+0,29532	1,09	+0,12319	1,44	+0,041703	1,79	+0,011359	2,14	+0,002474	2,49	+0,0004292
0,75	+0,28884	1,10	+0,11979	1,45	+0,040304	1,80	+0,010909	2,15	+0,002361	2,50	+0,0004069
0,76	+0,28246	1,11	+0,11646	1,46	+0,038946	1,81	+0,010475	2,16	+0,002252	2,51	+0,0003857
0,77	+0,27617	1,12	+0,11321	1,47	+0,037627	1,82	+0,010056	2,17	+0,002148	2,52	+0,0003654
0,78	+0,26998	1,13	+0,11002	1,48	+0,036345	1,83	+0,009653	2,18	+0,002049	2,53	+0,0003462
0,79	+0,26389	1,14	+0,10691	1,49	+0,035102	1,84	+0,009264	2,19	+0,001954	2,54	+0,0003280
0,80	+0,25789	1,15	+0,10337	1,50	+0,033894	1,85	+0,008888	2,20	+0,001862	2,55	+0,0003106
0,81	+0,25199	1,16	+0,10090	1,51	+0,032723	1,86	+0,008527	2,21	+0,001775	2,56	+0,0002941
0,82	+0,24618	1,17	+0,09799	1,52	+0,031586	1,87	+0,008179	2,22	+0,001692	2,57	+0,0002784
0,83	+0,24047	1,18	+0,09516	1,53	+0,030483	1,88	+0,007843	2,23	+0,001612	2,58	+0,0002636
0,84	+0,23485	1,19	+0,09239	1,54	+0,029414	1,89	+0,007520	2,24	+0,001535	2,59	+0,0002494
0,85	+0,22935	1,20	+0,08968	1,55	+0,028377	1,90	+0,007209	2,25	+0,001462	2,60	+0,0002360
0,86	+0,22389	1,21	+0,08704	1,56	+0,027371	1,91	+0,006910	2,26	+0,001392	2,61	+0,0002232
0,87	+0,21856	1,22	+0,08446	1,57	+0,026397	1,92	+0,006621	2,27	+0,001326	2,62	+0,0002111
0,88	+0,21331	1,23	+0,08194	1,58	+0,025452	1,93	+0,006344	2,28	+0,001262	2,63	+0,0001997
0,89	+0,20815	1,24	+0,079494	1,59	+0,024537	1,94	+0,006077	2,29	+0,001201	2,64	+0,0001888
0,90	+0,20309	1,25	+0,077099	1,60	+0,023651	1,95	+0,005820	2,30	+0,001143	2,65	+0,0001784
0,91	+0,19811	1,26	+0,074764	1,61	+0,022793	1,96	+0,005573	2,31	+0,001087	2,66	+0,0001686
0,92	+0,19323	1,27	+0,072486	1,62	+0,021961	1,97	+0,005336	2,32	+0,001034	2,67	+0,0001593
0,93	+0,18843	1,28	+0,070265	1,63	+0,021157	1,98	+0,005107	2,33	+0,000983	2,68	+0,0001505
0,94	+0,18372	1,29	+0,068101	1,64	+0,020378	1,99	+0,004888	2,34	+0,000935	2,69	+0,0001422
0,95	+0,17910	1,30	+0,065992	1,65	+0,019624	2,00	+0,004677	2,35	+0,000889	2,70	+0,0001343
0,96	+0,17457	1,31	+0,063936	1,66	+0,018895	2,01	+0,004475	2,36	+0,000845	2,71	+0,0001268
0,97	+0,17012	1,32	+0,061934	1,67	+0,018189	2,02	+0,004280	2,37	+0,000803	2,72	+0,0001197
0,98	+0,16576	1,33	+0,059984	1,68	+0,017507	2,03	+0,004093	2,38	+0,000763	2,73	+0,0001130
0,99	+0,16149	1,34	+0,058086	1,69	+0,016847	2,04	+0,003914	2,39	+0,000724	2,74	+0,0001066
1,00	+0,15729	1,35	+0,056237	1,70	+0,016209	2,05	+0,003741	2,40	+0,000688	2,75	+0,0001006

y	erfc(y)	y	erfc(y)	y	erfc(y)	y	erfc(y)	y	erfc(y)
2,76	+0,0000949	3,11	+0,000010915	3,46	+0,000000992	3,81	+0,00000007118	4,16	+0,00000000403
2,77	+0,0000895	3,12	+0,000010225	3,47	+0,000000923	3,82	+0,00000006579	4,17	+0,00000000370
2,78	+0,0000844	3,13	+0,000009577	3,48	+0,000000859	3,83	+0,00000006080	4,18	+0,00000000339
2,79	+0,0000795	3,14	+0,000008969	3,49	+0,000000799	3,84	+0,00000005617	4,19	+0,00000000311
2,80	+0,0000750	3,15	+0,000008398	3,50	+0,000000743	3,85	+0,00000005188	4,20	+0,00000000286
2,81	+0,0000708	3,16	+0,000007861	3,51	+0,000000690	3,86	+0,00000004792	4,21	+0,00000000262
2,82	+0,0000666	3,17	+0,000007358	3,52	+0,000000642	3,87	+0,00000004425	4,22	+0,00000000240
2,83	+0,0000627	3,18	+0,000006885	3,53	+0,000000597	3,88	+0,00000004085	4,23	+0,00000000220
2,84	+0,0000591	3,19	+0,000006441	3,54	+0,000000554	3,89	+0,00000003770	4,24	+0,00000000202
2,85	+0,0000556	3,20	+0,000006025	3,55	+0,000000515	3,90	+0,00000003479	4,25	+0,00000000185
2,86	+0,0000524	3,21	+0,000005635	3,56	+0,000000478	3,91	+0,00000003210	4,26	+0,00000000170
2,87	+0,0000493	3,22	+0,000005269	3,57	+0,000000444	3,92	+0,00000002961	4,27	+0,00000000155
2,88	+0,0000464	3,23	+0,000004926	3,58	+0,000000412	3,93	+0,00000002731	4,28	+0,00000000142
2,89	+0,0000436	3,24	+0,000004604	3,59	+0,000000383	3,94	+0,00000002518	4,29	+0,00000000130
2,90	+0,0000410	3,25	+0,000004302	3,60	+0,000000355	3,95	+0,00000002322	4,30	+0,00000000119
2,91	+0,0000386	3,26	+0,000004020	3,61	+0,000000330	3,96	+0,00000002140	4,31	+0,00000000109
2,92	+0,0000363	3,27	+0,000003755	3,62	+0,000000306	3,97	+0,00000001972	4,32	+0,00000000100
2,93	+0,0000341	3,28	+0,000003507	3,63	+0,000000284	3,98	+0,00000001817	4,33	+0,00000000092
2,94	+0,0000321	3,29	+0,000003275	3,64	+0,000000263	3,99	+0,00000001674	4,34	+0,00000000084
2,95	+0,0000302	3,30	+0,000003057	3,65	+0,000000244	4,00	+0,00000001542	4,35	+0,00000000077
2,96	+0,0000282	3,31	+0,000002854	3,66	+0,000000226	4,01	+0,00000001420	4,36	+0,00000000070
2,97	+0,0000266	3,32	+0,000002663	3,67	+0,000000210	4,02	+0,00000001307	4,37	+0,00000000064
2,98	+0,0000250	3,33	+0,000002485	3,68	+0,000000194	4,03	+0,00000001203	4,38	+0,00000000059
2,99	+0,0000235	3,34	+0,000002318	3,69	+0,000000180	4,04	+0,00000001107	4,39	+0,00000000053
3,00	+0,0000220	3,35	+0,000002162	3,70	+0,000000167	4,05	+0,00000001019	4,40	+0,00000000049
3,01	+0,0000207	3,36	+0,000002016	3,71	+0,000000154	4,06	+0,00000000937	4,41	+0,00000000045
3,02	+0,0000194	3,37	+0,000001880	3,72	+0,00000014337	4,07	+0,00000000862	4,42	+0,00000000041
3,03	+0,0000182	3,38	+0,000001752	3,73	+0,00000013274	4,08	+0,00000000793	4,43	+0,00000000037
3,04	+0,0000171	3,39	+0,000001633	3,74	+0,00000012288	4,09	+0,00000000729	4,44	+0,00000000034
3,05	+0,0000160	3,40	+0,000001521	3,75	+0,00000011373	4,10	+0,00000000670	4,45	+0,00000000031
3,06	+0,0000150	3,41	+0,000001417	3,76	+0,00000010524	4,11	+0,00000000616	4,46	+0,00000000028
3,07	+0,0000141	3,42	+0,000001320	3,77	+0,00000009736	4,12	+0,00000000566	4,47	+0,00000000026
3,08	+0,0000132	3,43	+0,000001229	3,78	+0,00000009005	4,13	+0,00000000520	4,48	+0,00000000024
3,09	+0,0000124	3,44	+0,000001145	3,79	+0,00000008328	4,14	+0,00000000477	4,49	+0,00000000021
3,10	+0,000011648	3,45	+0,000001066	3,80	+0,00000007701	4,15	+0,00000000438	4,50	+0,00000000020

- Diffusion from a finite source into a semi-infinite space: The solution of Fick's second law with a fixed supply of S diffusing impurities per cm^2 located at x = 0 for t = 0:

$$C(x,t) = \frac{S}{\sqrt{\pi Dt}} \exp\left[-(x/2\sqrt{Dt})^2\right]$$

(for a perfectly "reflecting surface", i.e., a surface at which diffusing atoms do not get trapped (e.g. oxidized) but are reflected back into the crystal).

With increasing time the surface concentration, which is assumed to be infinite at t = 0, changes like

$$C(0,t) = \frac{S}{\sqrt{\pi Dt}}.$$

This expression is easily obtained from the equation for C(x,t) and the following integral

$$\int_{-\infty}^{+\infty} \exp\left[-(ax)^2\right]dx = \sqrt{\pi/a}.$$

- Outdiffusion from a semi-infinite space: During indiffusion of a dopant impurity from a semi-infinite or finite source, we observe outdiffusion of the bulk dopant. Again we have to solve Fick's second law with the appropriate boundary conditions

boundary conditions

$$\begin{cases} C(x > 0, t = 0) = C_{bulk} \\ C(x = 0, t > 0) = 0 \end{cases}$$

(surface acts as a sink opposite to our assumption made earlier for diffusion from a finite source!)

$$C(x,t) = C_{bulk}\,\mathrm{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

C(x,t) is plotted in Fig. 100 (page 165). Outdiffusion leads to a reduction of the bulk dopant concentration near the surface. If a p-n junction is produced by diffusion of a donor impurity into p-type Si, the depth of the junction is deeper with outdiffusion occurring than when it does not take place.

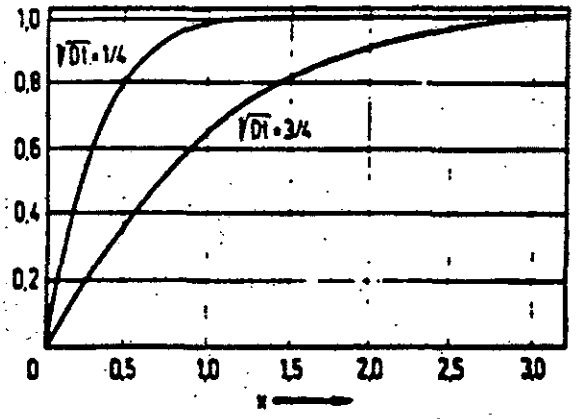


Figure 100. Outdiffusion profiles described by erf(x/2$\sqrt{Dt}$).

Figure 101 shows erf(z), erfc(z) as well as $(2/\sqrt{\pi})\exp(-z^2)$ all in one plot. The erf(z) is an odd function

$$\text{erf}(-z) = -\,\text{erf}(z)$$

with $\lim_{z\to\infty} \text{erf}(z) = 1$. The Gaussian function, with appropriate prefactor, is the derivative of erf(z)

$$(d/dz)\,\text{erf}(z) = (2/\sqrt{\pi})\exp(-z^2)\,.$$

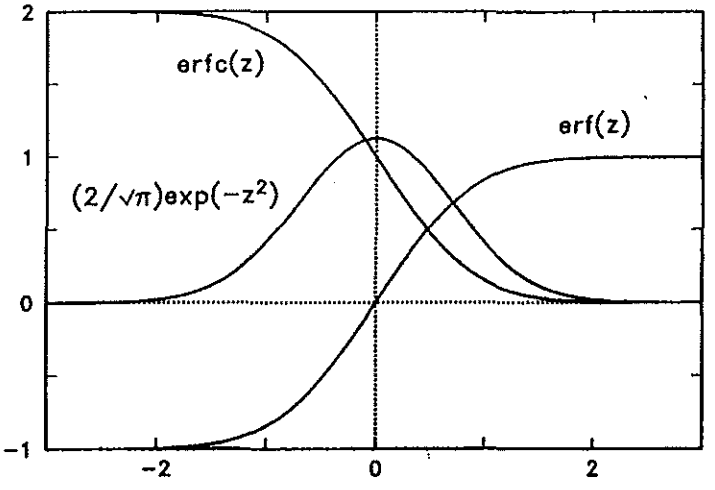


Fig. 101. Graph of the Gaussian and error functions [Courtesty R. Ghez, *A Primer of Diffusion Problems* (Wiley & Sons 1988); p.103.]

• Diffusion for D ≠ constant: Many cases have been studied in which the diffusivity D depends on the local dopant concentration. The physical reasons for this will be discussed in a later section (5.5.4 Dopant Diffusion). Figure 102 shows the logarithm of normalized concentration profiles ($C/C_{surface}$) as a function of the normalized length scale $x/(4Dt)^{1/2}$ for several dependencies of D on C:

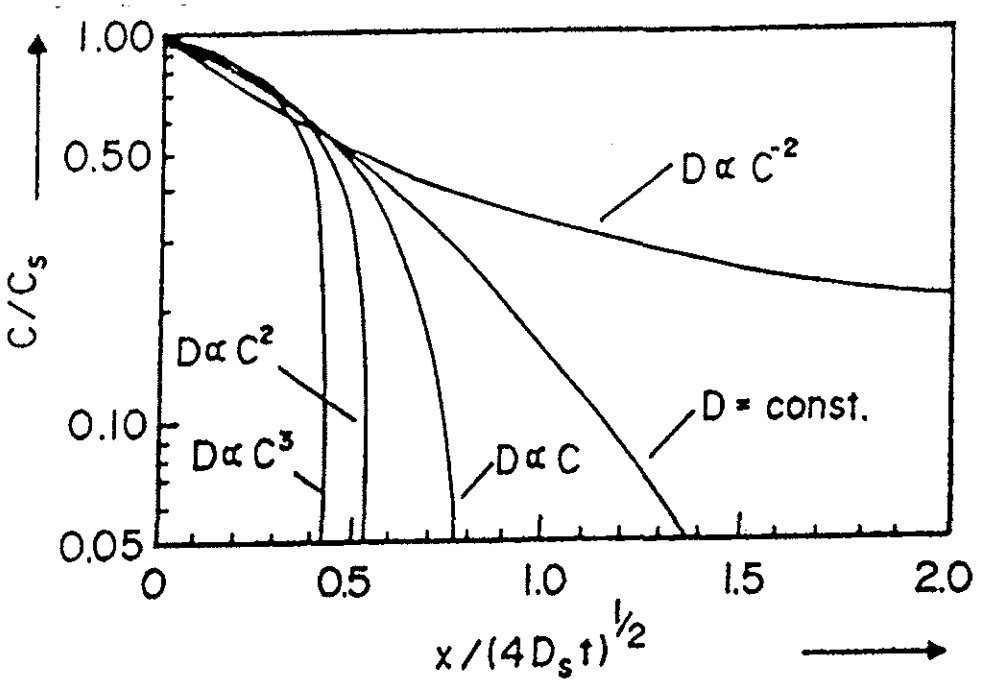


Fig. 102. Normalized diffusion profiles for several dependencies of D on C [Courtesy U. Goesele, in *Microelectronic Materials and Processes*, R.A. Levy, ed. (Kluwer Academic 1989), p. 588.]

5.5.2. Practical Aspects of Diffusion

In order to obtain predictable, well-defined diffusion profiles in semiconductor wafers, careful consideration must be given to the following points:

- doping homogeneity requires constant temperatures to within less than 0.5°C and perfect initial distribution of the dopant;
- cleanliness is required to avoid contamination with fast diffusers;
- control and minimization of crystal damage, i.e., the formation of point and extended defects (stacking faults, etc.) should be kept as small as possible; pitting through thermal etching has to be avoided;
- ease of control of the diffusion parameters; and
- reproducibility.

In general diffusion is performed inside a tube furnace. Appropriate design of the diffusion furnace and tube and choice of dopant introduction have been developed to the point where all or most of the above conditions can be fulfilled. Three zone furnaces, electrically heated and with electronically controllable temperatures/time profiles, are typically used for diffusion. Quartz or poly-Si tubes (for MOS work) contain the Si wafers. One end of the tube serves as the gas inlet, the other as the outlet and loading/unloading opening. The latter area is usually surrounded by a laminar flow box so that very low particulate-contamination conditions can be created.

The choice of dopants depends on the diffusion constant, the maximum solubility and the effectiveness of SiO_2 masks. Dopants are used in their elemental form but more often as chemical compounds, e.g. as oxides, halides or hydrogen compounds. Most oxides have sufficiently high vapor pressures to be used as vapor phase dopants. The dopant oxide is reduced directly on the silicon surface.

Typical overall reactions are:

$$2B_2O_3 + 3Si \rightarrow 4B + 3SiO_2$$
$$2P_2O_5 + 5Si \rightarrow 4P + 5SiO_2$$

If a predeposition/deglaze/drive-in cycle is used, the oxide (silicate glass) is removed in the deglaze step. It is believed that the liquid silicate glass which is formed during predeposition helps improve the uniformity of the distribution of dopant atoms on the surface.

Halides and hydrogen compounds are converted to oxides by the addition of oxygen gas or water vapor. The oxide is then reduced by the silicon as described above.

Typical overall reactions are:

$$4POCl_3 + 3O_2 \rightarrow 2P_2O_5 + 6Cl_2$$
$$2PH_3 + 4O_2 \rightarrow P_2O_5 + 3H_2O$$
$$4BBr_3 + 3O_2 \rightarrow 2B_2O_3 + 6Br_2$$
$$B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O$$

The following table lists some of the common dopant sources and the temperature which is required to obtain sufficiently high vapor pressures.

Solids	T(°C)
$B_2O_3(H_3BO_3)$	600 - 1200
P (elemental red phosphorus)	200 - 300
P_2O_5	200 - 300
Liquids	
$P(CH_3O)_3$	10 - 30
BBr_3	0 - 30
$POCl_3$	2 - 40
PCl_3	170
Gases	
BCl_3	15
B_2H_6	room temperature
PH_3	room temperature

Most diffusion processes are performed using the open-tube method. Figure 103 shows schematically the quartz or silicon diffusion tube in the furnace and appropriate gas manifolds for solid, liquid and gaseous dopant sources.

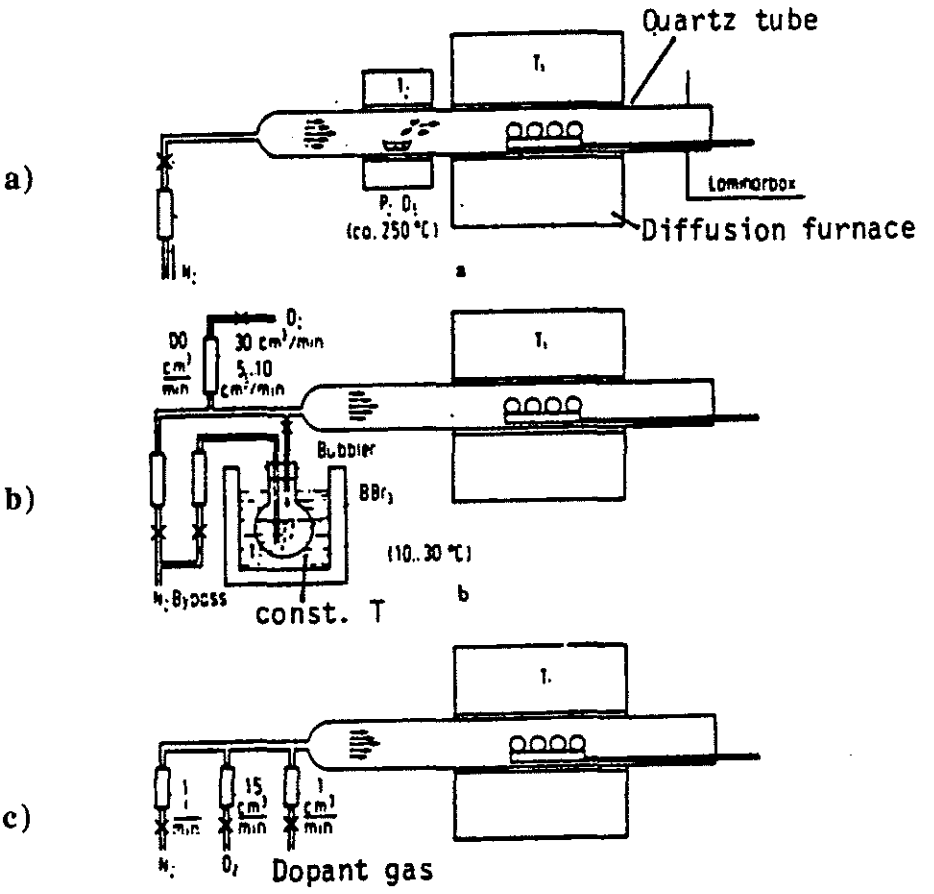


Fig. 103. Three diffusion furnace designs: a) the phosphorus source P_2O_5 is kept at its own temperature T_2, the source vapor is transported to the wafers with N_2 carrier gas, b) the boron source is liquid BBr_3 which is transported with N_2 streaming via a gas by-pass line through a bubbler; oxygen is provided separately for borosilicate glass formation; c) dopant gases (PH_3, B_2H_6, etc.), N_2 and O_2 flow through mass flow controllers into the diffusion tube.

• Predeposition and Drive-in Diffusion: In many practical cases a two-step procedure is used to introduce an impurity by diffusion. A predeposition step is followed by a drive-in step with a cleaning or deglazing step in between. The following advantages can be realized with two-step diffusion:

- choice of a particular surface impurity concentration which is typically smaller than the solid solubility at the drive-in temperature;
- choice of the doping gradient at a p-n-junction at a given depth;
- and use of the oxide which prevents outdiffusion as a mask in the following diffusion step.

The mathematics of two-step diffusions is more complicated than the simple cases discussed so far. Directly after predeposition the impurity concentration profile follows an erfc(y). After a prolonged drive-in, a Gaussian profile is approached. R.C.T. Smith [Australian J. Phys. **6**, 127 (1958)] has evaluated this case for a heat conduction problem:

$$C(x, t, t') = C_o \frac{2}{\sqrt{\beta}} \int_{\sqrt{\beta}}^{\infty} \exp(-m^2)\mathrm{erf}(\alpha m)dm$$

The primes indicate the drive-in diffusion step.

$$\alpha = D^+/D't' \begin{cases} \alpha = \infty \rightarrow \mathrm{erfc} \\ \alpha = 0 \rightarrow \mathrm{Gaussian} \end{cases}$$

$$\beta = \frac{x^2}{4(Dt + D't')}$$

C_o = surface concentration during predeposition. Values of the integral are listed for various values of α and β in the following table:

Smith Function

σ \ β	0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	1,0	1,1	1,2
0,1	0,09015	0,08155	0,07376	0,06672	0,06035	0,05459	0,04938	0,04467	0,04040	0,03655	0,03306	0,02990
0,2	0,17838	0,16119	0,14566	0,13162	0,11894	0,10748	0,09713	0,08777	0,07931	0,07167	0,06477	0,05853
0,3	0,26295	0,23723	0,21403	0,19310	0,17422	0,15719	0,14182	0,12795	0,11545	0,10416	0,09398	0,08479
0,4	0,34254	0,30837	0,27761	0,24993	0,22501	0,20259	0,18240	0,16422	0,14786	0,13314	0,11988	0,10794
0,5	0,41626	0,37374	0,33557	0,30132	0,27058	0,24299	0,21822	0,19599	0,17603	0,15812	0,14203	0,12759
0,6	0,48366	0,43290	0,38751	0,34692	0,31062	0,27814	0,24908	0,22308	0,19982	0,17900	0,16036	0,14368
0,7	0,54464	0,48580	0,43340	0,38673	0,34515	0,30809	0,27505	0,24562	0,21937	0,19596	0,17508	0,15645
0,8	0,59940	0,53264	0,47347	0,42100	0,37447	0,33317	0,29652	0,26398	0,23508	0,20940	0,18657	0,16628
0,9	0,64829	0,57380	0,50812	0,45017	0,39903	0,35385	0,31393	0,27864	0,24742	0,21979	0,19532	0,17365
1,0	0,69176	0,60975	0,53784	0,47475	0,41935	0,37066	0,32783	0,29013	0,25693	0,22765	0,20183	0,17903
1,1	0,73033	0,64100	0,56318	0,49529	0,43600	0,38415	0,33877	0,29900	0,26411	0,23348	0,20655	0,18286
1,2	0,76448	0,66808	0,58465	0,51232	0,44950	0,39486	0,34726	0,30574	0,26946	0,23772	0,20991	0,18553
1,3	0,79470	0,69148	0,60276	0,52634	0,46035	0,40327	0,35377	0,31078	0,27336	0,24074	0,21225	0,18734
1,4	0,82144	0,71164	0,61797	0,53781	0,46901	0,40979	0,35870	0,31449	0,27616	0,24286	0,21385	0,18855
1,5	0,84509	0,72899	0,63069	0,54714	0,47586	0,41482	0,36238	0,31720	0,27815	0,24431	0,21492	0,18933
1,6	0,86601	0,74388	0,64130	0,55469	0,48123	0,41865	0,36511	0,31914	0,27953	0,24530	0,21562	0,18983
1,7	0,88454	0,75666	0,65010	0,56076	0,48542	0,42153	0,36710	0,32051	0,28048	0,24595	0,21607	0,19014
1,8	0,90095	0,76759	0,65739	0,56562	0,48865	0,42369	0,36854	0,32147	0,28112	0,24638	0,21636	0,19033
1,9	0,91549	0,77693	0,66340	0,56948	0,49114	0,42529	0,36956	0,32213	0,28154	0,24665	0,21653	0,19045
2,0	0,92838	0,78491	0,66833	0,57254	0,49303	0,42646	0,37029	0,32258	0,28182	0,24682	0,21664	0,19051
2,5	0,97404	0,81009	0,68228	0,58029	0,49735	0,42887	0,37165	0,32335	0,28225	0,24707	0,21678	0,19059
3,0	0,99920	0,82094	0,68698	0,58234	0,49825	0,42928	0,37183	0,32343	0,28229	0,24708	0,21679	0,19059
∞	1,02843	0,82795	0,68892	0,58291	0,49843	0,42933	0,37184	0,32343	0,28229	0,24709	0,21679	0,19059

1,3	1,4	1,5	1,6	1,7	1,8	1,9	2,0	2,5	3,0	4,0	5,0	β / σ
0,02705	0,02446	0,02213	0,02002	0,01811	0,01638	0,01481	0,01340	0,00811	0,00491	0,00180	0,00066	0,1
0,05289	0,04779	0,04319	0,03903	0,03527	0,03187	0,02880	0,02603	0,01568	0,00945	0,00343	0,00125	0,2
0,07651	0,06903	0,06228	0,05620	0,05071	0,04575	0,04128	0,03725	0,02228	0,01333	0,00477	0,00171	0,3
0,09720	0,08752	0,07881	0,07097	0,06391	0,05756	0,05183	0,04668	0,02766	0,01640	0,00577	0,00204	0,4
0,11462	0,10297	0,09251	0,08312	0,07468	0,06711	0,06030	0,05419	0,03178	0,01866	0,00645	0,00224	0,5
0,12875	0,11538	0,10340	0,09268	0,08308	0,07448	0,06678	0,05988	0,03475	0,02021	0,00688	0,00236	0,6
0,13982	0,12499	0,11174	0,09992	0,08936	0,07993	0,07150	0,06398	0,03677	0,02120	0,00712	0,00242	0,7
0,14824	0,13219	0,11790	0,10519	0,09387	0,08379	0,07481	0,06680	0,03806	0,02180	0,00724	0,00244	0,8
0,15444	0,13741	0,12230	0,10889	0,09699	0,08642	0,07702	0,06867	0,03885	0,02213	0,00730	0,00245	0,9
0,15889	0,14109	0,12535	0,11141	0,09907	0,08814	0,07844	0,06985	0,03931	0,02231	0,00733	0,00246	1,0
0,16200	0,14361	0,12739	0,11307	0,10041	0,08923	0,07933	0,07056	0,03956	0,02240	0,00734	0,00246	1,1
0,16411	0,14529	0,12872	0,11412	0,10125	0,08989	0,07985	0,07098	0,03969	0,02244	0,00735	0,00246	1,2
0,16552	0,14638	0,12956	0,11478	0,10176	0,09028	0,08016	0,07122	0,03976	0,02246	0,00735	0,00246	1,3
0,16643	0,14706	0,13008	0,11517	0,10205	0,09051	0,08033	0,07134	0,03979	0,02247	0,00735	0,00246	1,4
0,16700	0,14749	0,13039	0,11540	0,10222	0,09063	0,08042	0,07141	0,03980	0,02247	0,00735	0,00246	1,5
0,16736	0,14774	0,13057	0,11552	0,10231	0,09070	0,08046	0,07144	0,03981	0,02247	0,00735	0,00246	1,6
0,16757	0,14789	0,13067	0,11559	0,10236	0,09073	0,08049	0,07146	0,03981	0,02247	0,00735	0,00246	1,7
0,16770	0,14797	0,13073	0,11563	0,10239	0,09075	0,08050	0,07147	0,03982	0,02247	0,00735	0,00246	1,8
0,16777	0,14802	0,13076	0,11565	0,10240	0,09075	0,08050	0,07147	0,03982	0,02247	0,00735	0,00246	1,9
0,16781	0,14804	0,13078	0,11566	0,10240	0,09076	0,08051	0,07147	0,03982	0,02247	0,00735	0,00246	2,0
0,16786	0,14807	0,13079	0,11567	0,10241	0,09076	0,08051	0,07147	.....	.....	.....	.....	2,5
0,16786	0,14807	0,13079	0,11567	0,10241	0,09076	0,08051	0,07147	.....	.....	.....	.....	3,0
0,16786	0,14807	0,13079	0,11567	0,10241	0,09076	0,08051	0,07147	0,03982	0,02247	0,00735	0,00246	∞

A very simple method for the application of dopants has found renewed interest. This is the spin-on oxide process. A drop of an organic silicon compound containing dopant elements dissolved in ethanol is placed in the center of a silicon wafer. The wafer is spun to ~ 1000 rpm which leads to a uniform film on the surface. After a drying process the wafers are heated to the desired diffusion temperature. Control of the various parameters is not as good as in the processes described above but the spin-on method is very fast and economical. There is no need for expensive high-purity gas manifolds.

- Influence of Oxide Layer Formation on Diffusion Profiles: When a doped semiconductor is oxidized, the impurities prefer to stay either in the semiconductor or in the oxide. A segregation coefficient k, analogous to the one used in crystal growing, can be defined:

$$k = \frac{\text{impurity concentration in semiconductor}}{\text{impurity concentration in oxide}} = k_o \exp(-E/k_B T)$$

$k > 1$ leads to a "pile-up" of impurities in the bulk; $k < 1$ to a "pile-down." Figure 104 shows the change of the doping concentration near the surface after the semiconductor has been oxidized.

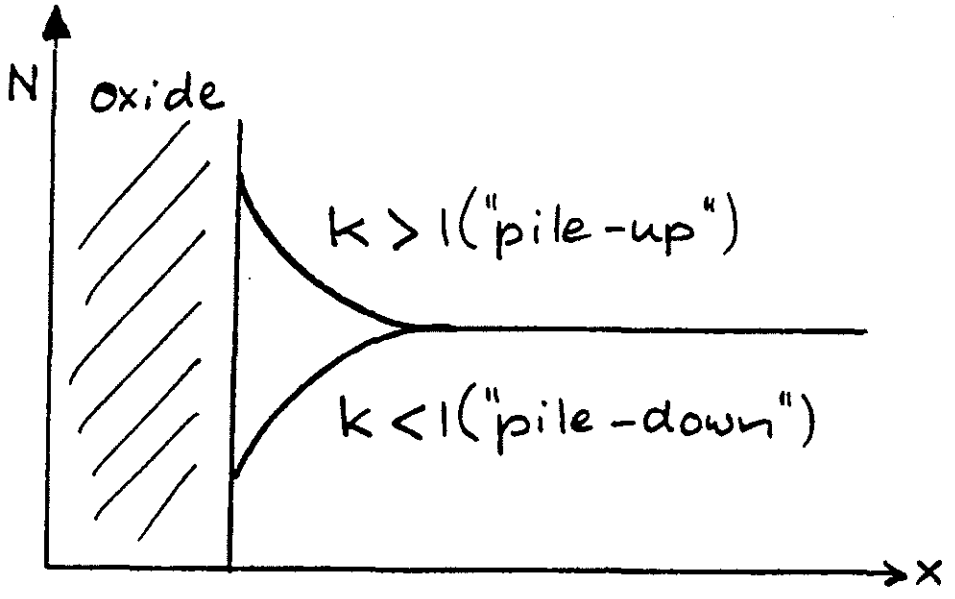


Fig. 104. Influence of oxidation on impurity distribution.

The following table shows the segregation coefficients for common dopants in the Si/SO_2 system:

dopant	k
B	~0.6 (1050°C); 10^{-1} (800°C)
P, Sb, As	~10
Ga	~20
In	~10^{-3} (1050°C); 10^{-1} (800°C)

[for a recent In study see: I. C. Kizilyalli et al., J. Appl. Phys. **80**, 4944 (1996).]

5.5.3. Microscopic Diffusion Mechanisms

We can distinguish between self diffusion, i.e., the random walk of semiconductor host atoms through a crystal, and dopant or impurity diffusion. Focusing on the latter type we can again form two groups: interstitial impurities (e.g. Li in Si and Ge) which diffuse rapidly through a crystalline solid on their own and substitutional impurities which must leave their host lattice site in order to move through the lattice. Native defects such as the vacancy and the self interstitial play a key role in the diffusion of substitutional impurities. Up to the 1970's the vacancy was thought to be the native defect playing the major role in diffusion. After Seeger's pioneering theoretical work and a very large number of experiments, we know that interstitials are as important as vacancies, at least in crystalline Si above 1050°C.

In vacancy assisted diffusion one assumes a substitutional impurity will move into a neighboring vacancy when possible. In a vacancy concentration gradient we would expect impurities which diffuse mostly via vacancies to move up the vacancy gradient.

Other impurities leave their host lattice site because a self interstitial arrives and "kicks-out" the impurity. In an interstitial position this impurity can move much more rapidly and eventually will kick-out a Si host lattice atom assuming a new substitutional position in the lattice.

A third kind of impurity occupies simultaneously both kinds of lattice sites, substitutional and interstitial, in a temperature and Fermi level dependent concentration ratio. Cu in Ge is the classical example for this kind of impurity.

A diffusion mechanism based on vacancies is termed the Frank-Turnbull or dissociative mechanism

$$A_S \Leftrightarrow A_I + V$$

$A_{S,\,I}$ = substitutional, interstitial impurity; V = vacancy.

Impurities which diffuse assisted by interstitials are described by the "kick-out" mechanism

$$A_S + I \Leftrightarrow A_I$$

with I = interstitial Si.

From this introductory discussion it becomes clear that native defects play a crucial role in the diffusion of the standard dopant impurities which predominantly occupy host lattice sites (e.g. B, P in Si and Ge; Be, Zn, Si, Sn, S, Se, Te in III-V semiconductors, etc.). Therefore an understanding of impurity diffusion requires a detailed knowledge of native defect formation and transport. Dopants and native defects are inseparable when we are interested in their random walks through the lattice.

It is then not surprising that many authors discuss first self diffusion, recognizing it as the most basic of all diffusion processes.

5.5.3.1. Self diffusion

In elemental semiconductors we have to deal with self interstitials and vacancies. In compound semiconductors and their alloys we must consider self interstitials and vacancies of each host latttice species as well as antisites, i.e., host atoms occupying a substitutional position not on their own sub-lattice (e.g. As_{Ga} in GaAs is an arsenic atom sitting on a Ga site forming a double donor!). We will focus on Si self diffusion in the following for two reasons. First, in silicon, one knows more than in other semiconductors about self interstitials and vacancies though this knowledge is still very incomplete, and second, silicon is and will remain the most important semiconductor from an applications point of view.

We know that both self interstitials as well as vacancies contribute to self diffusion. We write for the self diffusion coefficient, D^{SD}

$$D^{SD} = D_I\, C_I^{eq} + D_V\, C_V^{eq} \ .$$

$C_{I,\,V}^{eq}$ are the equilibrium concentrations of self interstitials and vacancies respectively, expressed as a fraction of the silicon host lattice atom concentration and $D_{I,\,V}$ are the corresponding diffusivities. Each of the four factors depends on Boltzmann factors of the form exp ($-H/k_BT$). The diffusivities

depend on migration enthalpies of the native defects (H_V^m, H_I^m) and the concentrations are described by formation enthalpies (H_V^f, H_I^f). When plotting log (D^{SD}) as a function of the inverse absolute temperature one finds two straight sections. The lower temperature part ($T \lesssim 1050°C$) is dominated by the vacancy assisted diffusion and the slope corresponds to $H_V^m + H_V^f$. At $T \gtrsim 1050°C$ the slope is steeper and is due to the sum $H_I^m + H_I^f$.

Some of the best experimental results for Si self diffusion have been obtained through the detailed study of Zn diffusion [H. Bracht, N.A. Stolwijk and H. Mehrer, *Semiconductor Silicon 1994*, H.R. Huff, W. Bergholz and K. Sumino, eds., The Electrochemical Society, Pennington, N.J., p. 593 (1994)]. These authors find:

$$C_I^{eq} D_I = \left(3.0\ {}^{+1.0}_{-0.8}\right) 10^3 \exp\left(-\frac{(4.95\pm0.03)\text{eV}}{k_B T}\right) \text{cm}^2\text{s}^{-1}$$

$$C_V^{eq} D_V = \left(8.6\ {}^{+9.9}_{-4.6}\right) 10^{-2} \exp\left(-\frac{(3.80\pm0.08)\text{eV}}{k_B T}\right) \text{cm}^2\text{s}^{-1}$$

Figure 105 shows the data with the straight lines corresponding to the above equations. For comparison the authors compiled radioactive tracer self diffusion data from earlier experiments and compared these with their newest data (Fig. 106). The agreement looks very good.

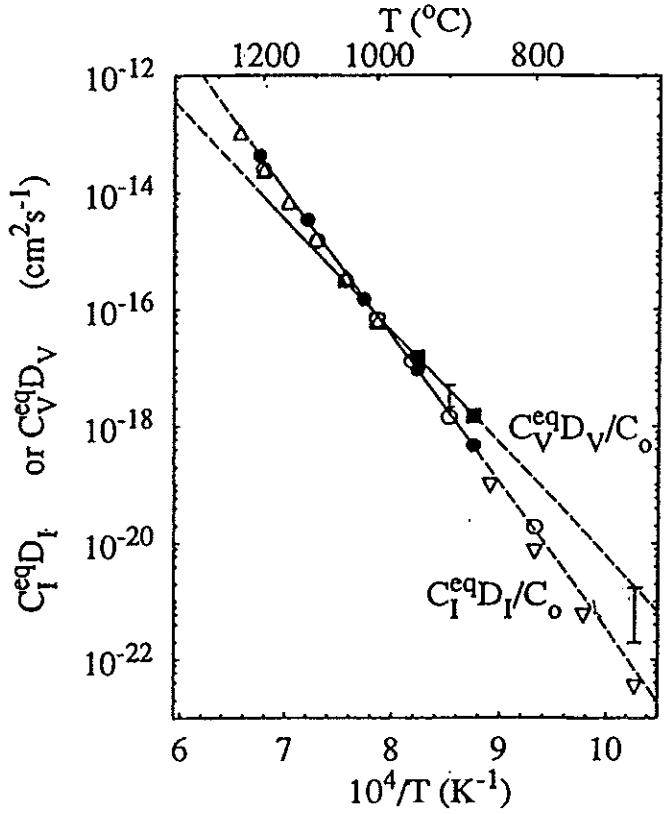


Fig. 105. Temperature dependence of $C_I^{eq}D_I$ and $C_V^{eq}D_V$ (Courtesy of H. Bracht et. al., see text for details.)

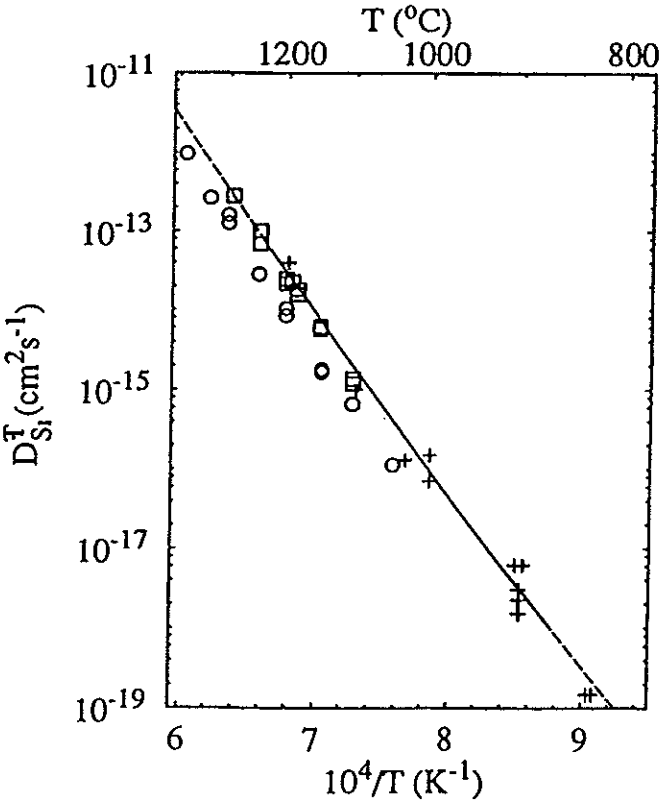


Fig. 106. Temperature dependence of the Si tracer self-diffusion (Courtesy of H. Bracht et. al., see text for details.)

The major task remaining is the decomposition of the concentration-diffusivity products into their individual factors. The same authors analyzed Au and Pt diffusion data which were obtained under a variety of conditions including oxidation, crystal growth and the effects of A-swirls in dislocation free Si crystals. (A-swirl defects have been shown to consist of extrinsic stacking faults, i.e., condensates of Si interstitiales forming during cooling of dislocation free crystals). Combining all this information, Bracht *et. al.* arrive at the following estimates

$$C_I^{eq} = \left(2.9\ {}^{+8.2}_{-2.1}\right) 10^{24} \exp\left(-\frac{(3.18\pm0.15)\text{eV}}{k_B T}\right) \text{cm}^{-3}$$

$$D_I = \left(5.1\ {}^{+9.4}_{-3.3}\right) 10 \exp\left(-\frac{(1.77\pm0.12)\text{eV}}{k_B T}\right) \text{cm}^2\text{s}^{-1}$$

$$C_V^{eq} \approx 1.4 \times 10^{23} \exp\left(-\frac{2.0\ \text{eV}}{k_B T}\right) \text{cm}^{-3}$$

$$D_V \approx 3.0 \times 10^{-2} \exp\left(-\frac{1.8\ \text{eV}}{k_B T}\right) \text{cm}^2\text{s}^{-1}$$

The equilibrium concentrations C_I^{eq} and C_V^{eq} must be divided by the Si atom concentration $C_O = 5 \times 10^{22}\text{cm}^{-3}$ to be comparable to the concentrations given earlier in the form of fractions of the Si atom concentration.

The large errors show the significant uncertainty which still adheres to all these data. The major cause for this is the inability to measure the individual diffusivities and concentrations directly and separately. In all standard diffusion experiments one always measures data which depend on the C × D products. New approaches for the separation of the products may become feasible with isotopically engineered Si multilayer structures.

A study of Ga self diffusion in GaAs has been conducted most recently using isotopically pure layers of GaAs on natural GaAs substrates. Secondary ion mass spectroscopy (SIMS) shows the concentrations of ^{69}Ga and ^{71}Ga in the substrate (concentration ratio 3:2) and in the two MBE layers grown from isotopically pure Ga effusion sources (Fig. 107). The substrate was semi-insulating and the two layers were not intentionally doped. Moving from the surface on the left side of the figure towards the substrate on the right side, one notices that the ^{71}Ga concentration profile rises more sharply than the ^{69}Ga concentration profile drops at the ^{69}Ga-^{71}Ga interface. The inverse situation occurs at the ^{71}Ga-natural Ga interface where the ^{69}Ga concentration increases more sharply than the ^{71}Ga concentration drops. The reason for this difference stems from the

knock-on of the SIMS primary ions (10-30keV Cs^+) pushing the Ga atoms deeper into the crystal. Going from a high concentration to a low concentration leads to a noticeable profile change while in the opposite direction the effect is practically undetectable.

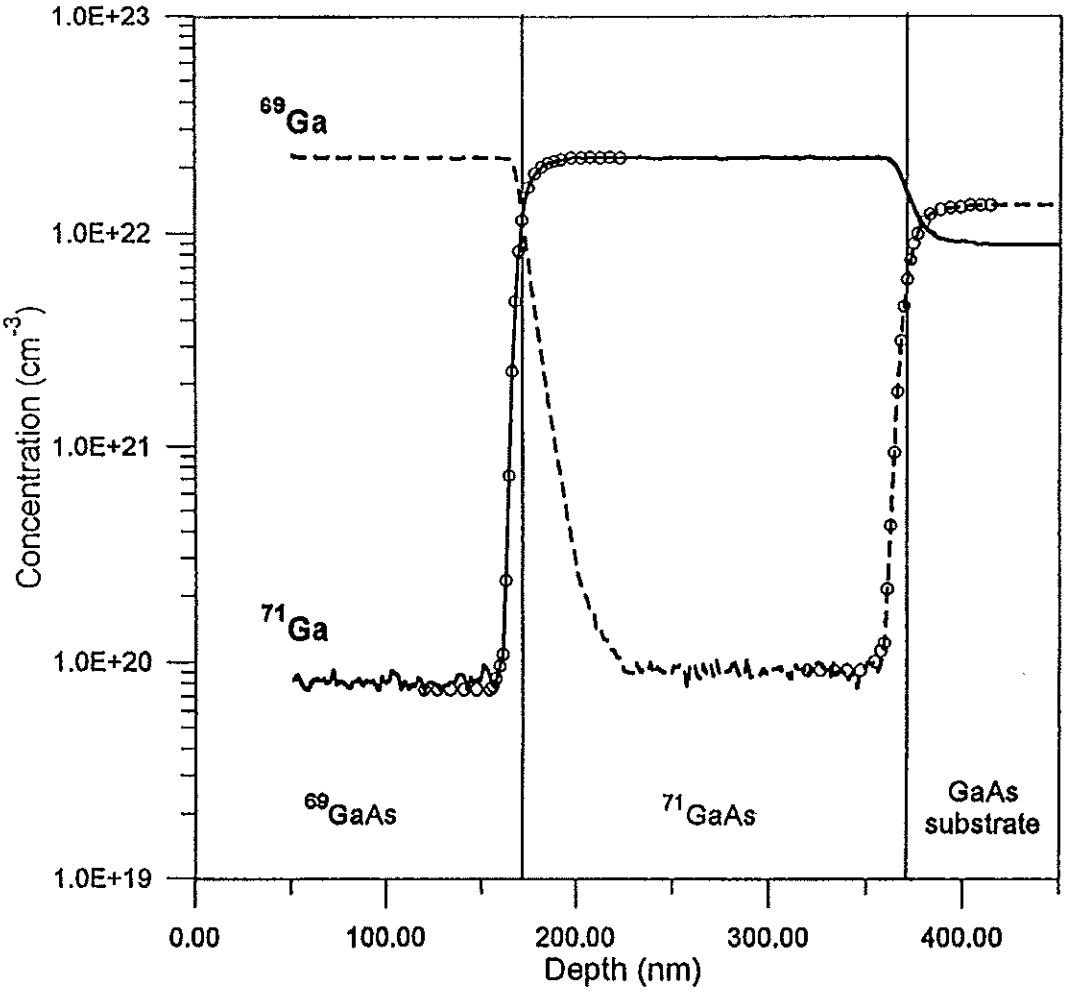


Fig. 107. SIMS Depth Profiles of ^{69}Ga and ^{71}Ga in as-grown GaAs isotopes epilayers on natural GaAs substrate. The vertical lines represent epilayer interfaces. The circles are calculated values considering SIMS broadening effects.

Exposing this structure to high temperatures induces random walk of the host lattice atoms. No chemical or impurity concentration gradients nor electric field gradients influence this random walk process and we expect to observe the purest possible case of Ga self diffusion. One issue which has to be addressed is the influence of the bare surface of the $^{69}GaAs$ layer on diffusion. At high temperatures As evaporation must be avoided by placing the diffusion samples into a sealed quartz ampoule with As overpressure.

In Fig. 108, we see the SIMS profile for the above sample after Ga self diffusion for 3321s at 974°C. Fitting error functions over two orders of magnitude leads to very precise values of the diffusivity. We have studied numerous samples and have compiled the data in Fig. 109. The straight line is accurately described by $H^M = 4.24$ eV and a pre-exponential factor of 43. Assuming that $g \sim 1$, using the lattice constant for GaAs $a = 5.4$ Å and a cut-off phonon frequency of 8×10^{12} Hz, we find $S^{SD} = 8k_B$, a value very comparable to the self diffusion entropies found for Ge and Si.

It would be most interesting to repeat this experiment for As. This is unfortunately not possible because there exists only one stable isotope of its element, ^{75}As.

Very similar experiments have been conducted with Ge and are planned for Si.

The foregoing discussion has dealt with equilibrium concentrations of vacancies and interstitials. In many instances C_I and C_V deviate by orders of magnitude from C_I^{eq} and C_V^{eq}: This in turn will change D^{SD}. We will discuss these effects in the following section which deals with substitutional diffusion.

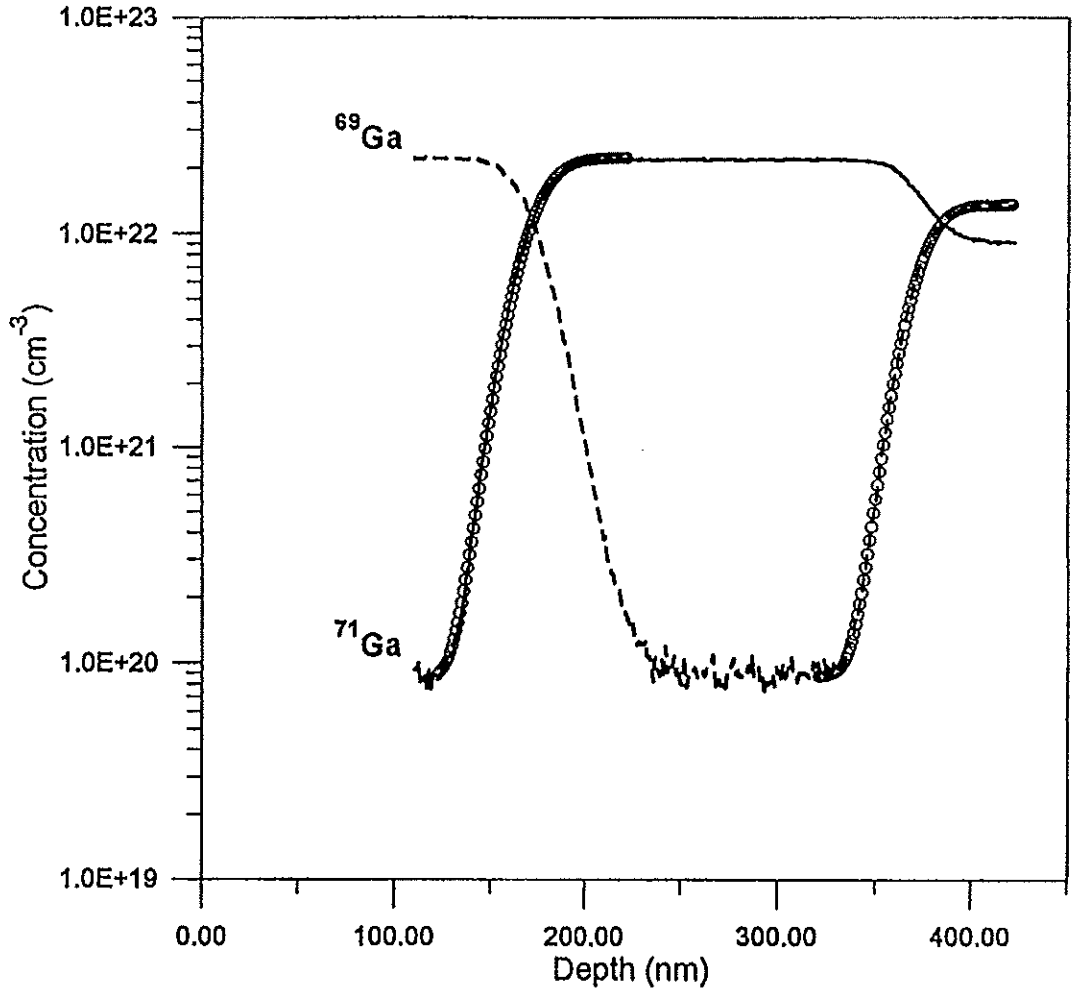


Fig. 108. SIMS Depth Profiles of ^{69}Ga and ^{71}Ga in GaAs isotope epilayers annealed at 974°C for 3321 s. The circles are theoretical fits.

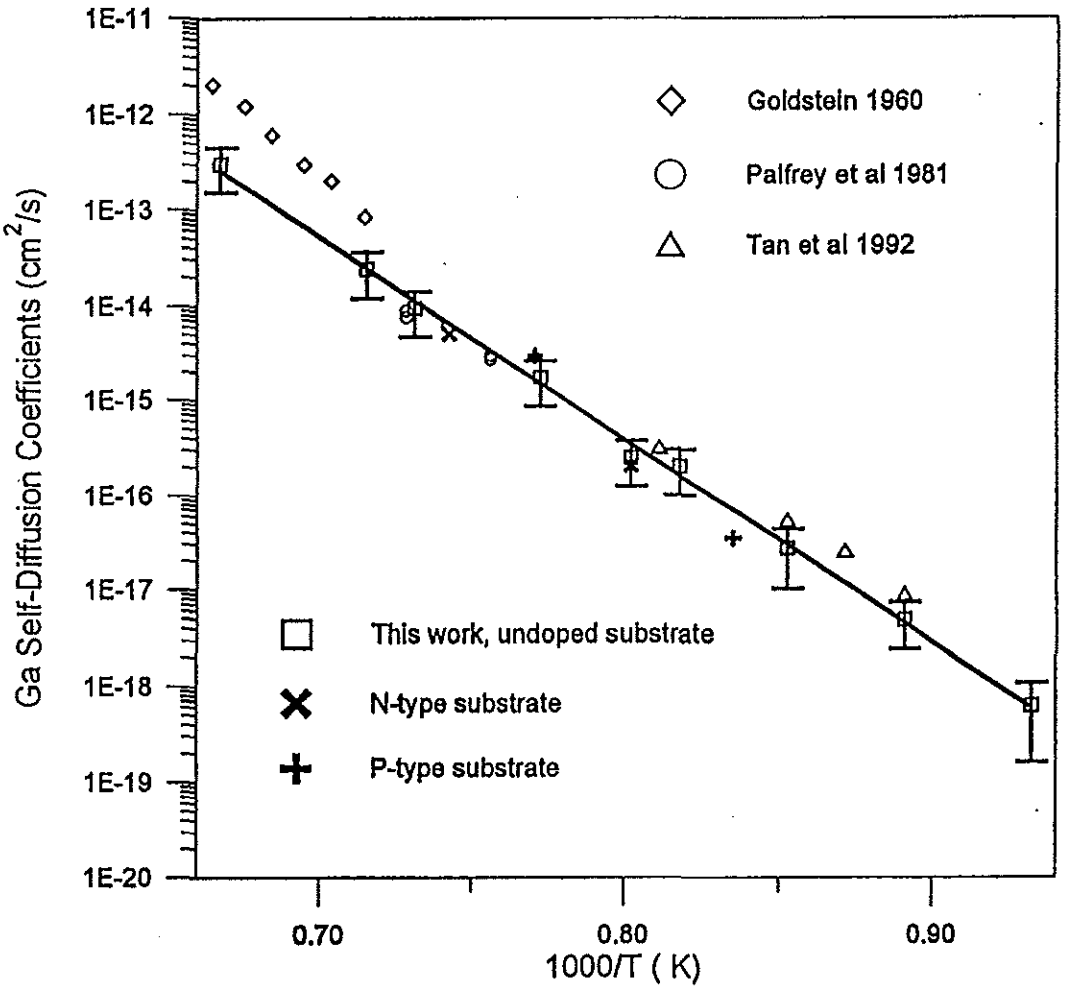


Fig. 109. Arrhenius plot of temperature dependence of Ga self-diffusion coefficients.

5.5.3.2. Dopant Diffusion

In Fig. 110, we find a compilation of the experimentally determined diffusivities of many dopants in Si presented in Arrhenius form.

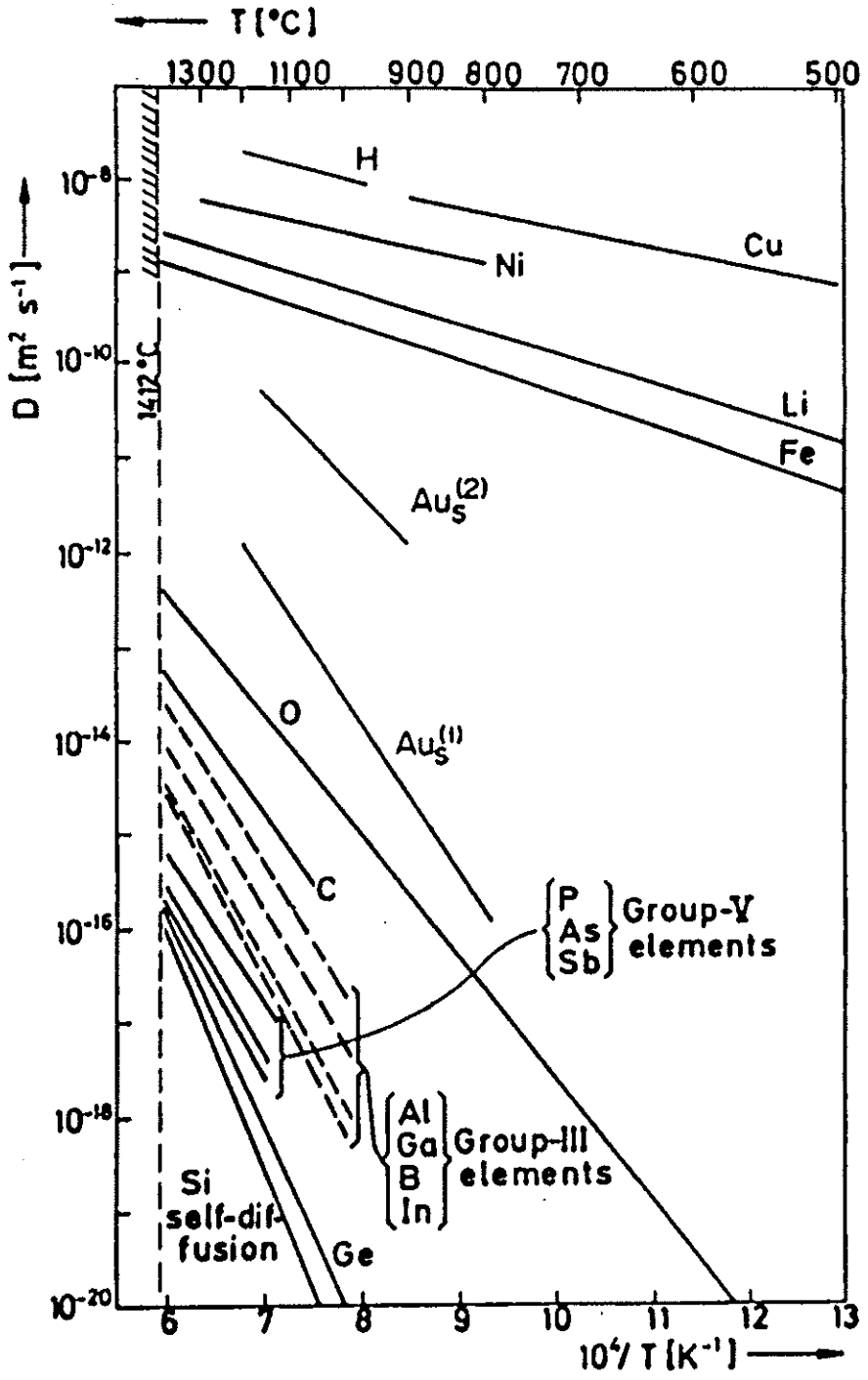


Fig. 110. Diffusivities of various elements in silicon as a function of inverse absolute temperature. $Au_S^{(1)}$ and $Au_S^{(2)}$ refer to effective diffusivities of substitutional gold in dislocation-free and highly dislocated silicon, respectively.

The data fall into three major groups: the fast diffusers such as Li, Cu, H, etc., the slow diffusers including Si, Ge and all shallow substitutional dopants including B, Al, Ga, In, P, As and Sb and a few impurities lying between these two groups. C, O and Au belong to the third group. All the rapid diffusers occupy interstitial sites where they are not strongly bound to the lattice.

In the case of lithium, it is assumed that all Li^+ ions reside on interstitial sites forming shallow donors. The Li^+ ions are very mobile down to temperatures as low as ~ 100°C and readily form complexes with negative acceptor ions. This kind of dopant passivation (not compensation in the usual sense!) is used to form p-i-n junction devices with very wide i-("intrinsic," a misnomer!) regions. The passivation can be extremely effective and net-dopant concentrations as low as $10^8 cm^{-3}$ have been achieved in these i-regions. Reverse biased p-i-n junctions are widely used as detectors in nuclear and x-ray spectrometers with large energy resolution.

Hydrogen shows a diffusion behavior similar to lithium with two major differences: first, no shallow donor state is formed by interstitial H, and second, hydrogen forms strongly bound H_2 molecules which appear to diffuse much more slowly than H or Li ions. A very large literature on H passivation has accumulated over the past ten years after the discoveries of the electronically active role of hydrogen in ultra pure Ge [see: E.E. Haller and other authors in *Hydrogen in Semiconductors*, editors J. Pankove and N. Johnson in *Semiconductors and Semimetals*, Vol. **34**, series editors Willardson and Beer (Wiley and Sons, 1991).]

The rapidly diffusing Fe and Au have been known for a long time because of their very strong effects on minority carrier lifetimes occurring already at very low concentrations. With the exception of Au doping used in very fast switching diodes, these two impurities are typically kept diligently away from all devices. Elaborate gettering mechanisms have been designed to remove Fe and Au from the active regions in Si wafers and chips. The interested reader is referred to the semiconductor processing literature for details.

Cu is one of the most intriguing impurities in Ge. This impurity occupies partially interstitial and partially substitutional sites. On the latter site Cu forms a triple acceptor with three well defined hole binding energies. The triple acceptor properties clearly indicate a host lattice site but the rapid diffusion behavior caused confusion in the early semiconductor days because it was not known that one kind of impurity would assume more than one kind of lattice site. Through high temperature, electric field drift studies of radioactive ^{64}Cu, it became clear that copper travels as a positive ion, i.e., it must be a donor in its interstitial form. The energy level of this donor has not yet been determined.

The substitiutional shallow dopant impurities diffuse assisted by vacancies and interstitials. The key problem is to determine the relative contributions of these two native defects as a function of temperature, Fermi level position, other dopants and external disturbances such as oxidation, nitridation and implantation to name a few. These dependencies lead to phenomena which puzzled semiconductor scientists and technologists alike for a long time. In the following, we will first state what the currently accepted views are on diffusion of group V donor and group III acceptor dopants. After this we will review what happens when external disturbances are turned on. This in turn will help explain the unusual experimental observations which posed serious problems for basic understanding.

Returning to Fig. 110, we see that above ~ 1000°C the group III acceptors diffuse faster than Si (self diffusion). One proposed reason is related to the Coulomb attraction between the negatively charged acceptor ions and the positive Si interstitials. On the other hand, the vacancy assisted diffusion component increases for the larger acceptors Ga and In but never becomes dominant because of the Coulomb repulsion between the negatively charged acceptors and vacancies. From a very extensive amount of diffusion data obtained under widely varying conditions by many experiementers, U. Gösele and T.Y. Tan [MRS Proc. Vol. **36**, 105 (1985)] arrived at the following approximate values for G_I^S , the fractional intersticialcy diffusion component ($G_I^S = D_I^S/D^S$ with D_I^S = self-interstitial diffusivity, D^S = total self-diffusivity):

G_I^S

T(°C) / Dopant	950	1000	1000	1100
B	0.16 - 0.38	0.17 - 0.45	0.17 - 0.6	0.19 - 1.0
Ga	—	—	—	0.6 - 0.7
Al	—	—	—	0.6 - 0.7
In	—	0.3	—	—
P	0.14 - 0.30	0.16 - 0.38	0.19 - 0.45	0.23 - 1.0
As	0.1 - 0.25	0.12 - 0.3	0.15 - 0.35	0.19 - 0.5
Sb	—	—	—	0.015 - 0.02

We should point out that the low value limits interestingly came mostly from staunch defenders of the vacancy-only diffusion camp. The true values lie more closely towards the upper limits.

Despite the large fluctuations in G_I^S two conclusions can be drawn. First one sees that G_I^S increases with temperature in support of the greater importance of interstitials at higher temperatures, and second, Sb diffuses predominantly via vacancies.

Assuming that the above described general picture is correct, we are now in a position to understand various effects related to disturbances of the vacancy and interstitial populations. One of the earliest and most detrimental deviations from normal diffusion has been the so-called emitter push effect. Figure 111 shows what happens to a boron diffusion profile in front of a second, high concentration phosphorus diffusion front. The high phosphorus concentration is assumed to increase the Si interstitial concentration producing an interstitial wind. The intersticialcy diffusion component of B increases in front of the diffusing P but not outside the range of the interstitial wind. This enhanced diffusion of B led to much broader base layers in n-p-n bipolar Si transistors

than originally designed. Such transistors exhibit much lower cut-off operating frequencies. Advanced modeling and processing overcame this problem.

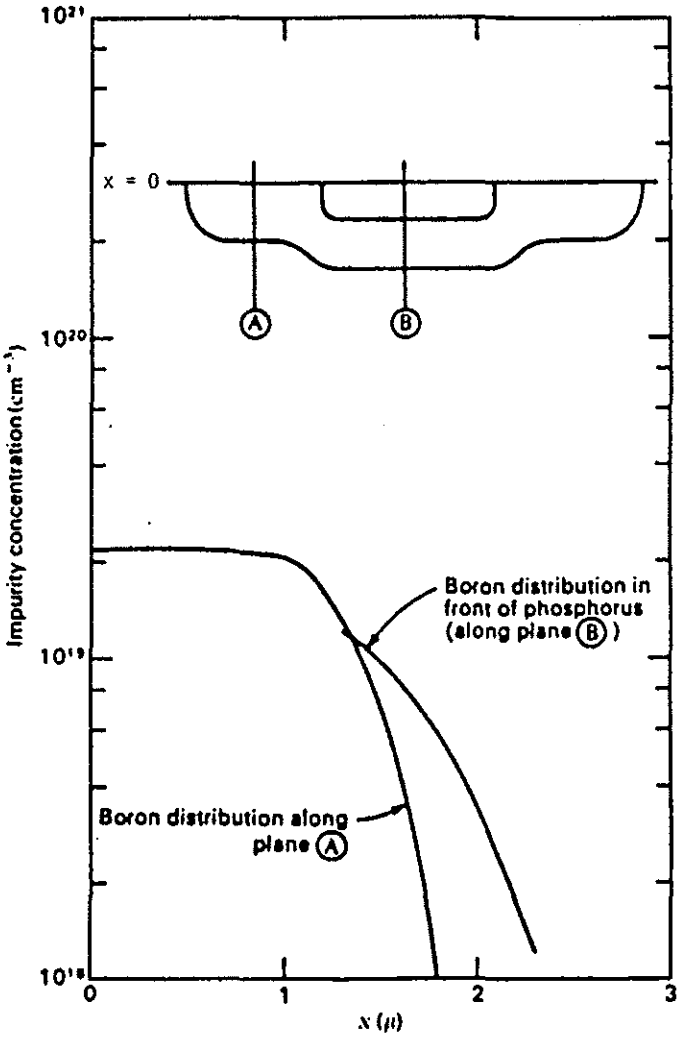


Fig. 111. The "emitter-push" effect: anomalous diffusion due to a second, high concentration, diffusion.

A second source for interstitial wind is believed to originate in Si oxidation. Upon oxidation of Si under a SiO_2 layer a volume expansion of 125% has to be accommodated by mostly visco-elastic flow. Vacancies from the bulk and injection of interstitials from the oxide-Si interface both help in reducing surface strains. Dopants which diffuse mostly via the intersticialcy mechanism (and these seem to be all except Sb) will exhibit increased diffusivities while Sb diffusion is retarded. The opposite observation has been made during nitridation, i.e., Sb diffuses at an accelerated rate while all the other group III and group V dopants diffuse at a slower pace.

Experimental results obtained by Mizuo and Higuchi [Japanese J. Appl. Phys. **21**, 281 (1982) and J. Electrochem. Soc. **130**, 1942 (1983)] are shown in Fig. 112. It should be pointed out that these transient diffusion effects vanish at very high temperatures and after very long times. A further surprising finding is the very long time it takes to establish a dynamic (steady state) equilibrium between non-equilibrium vacancy and interstitial concentrations. At 1100°C it takes on the order of one hour for this to occur in a typical Si wafer.

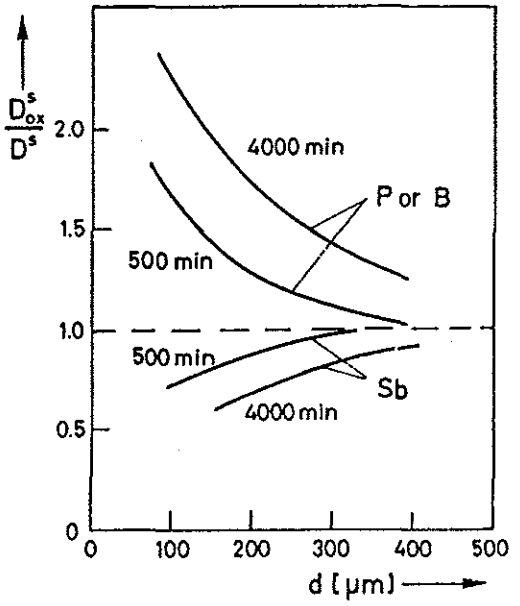


Fig. 112. Diffusion enhancement of P and B and diffusion retardation of Sb at the frontside of Si wafers oxidized at the backside [After Mizuo and Higuchi (1982b, 1983)]: D^S_{OX}/D^S is given as a function of the wafer thickness d for 500- and 4000-min oxidations.

A third source of excess interstitials has been studied quantitatively more recently. This source is related to ion implantation caused rodlike defects oriented along <110> directions on {131} planes. These defects are believed to emit interstitials during the early phase of radiation damage annealing, leading to the so-called Transient Enhanced Diffusion (TED). [See: D.J. Eaglesham, P.A. Stolk, H.-J. Gossmann and J.M. Poate, Appl. Phys. Lett. **65**, 2305 (1994).] Figure 113 clearly shows the enhanced diffusivity of the first three or four near surface boron delta layers grown by MBE into Si after a very shallow Si implantation which produces primary damage at a depth of less than 1000Å.

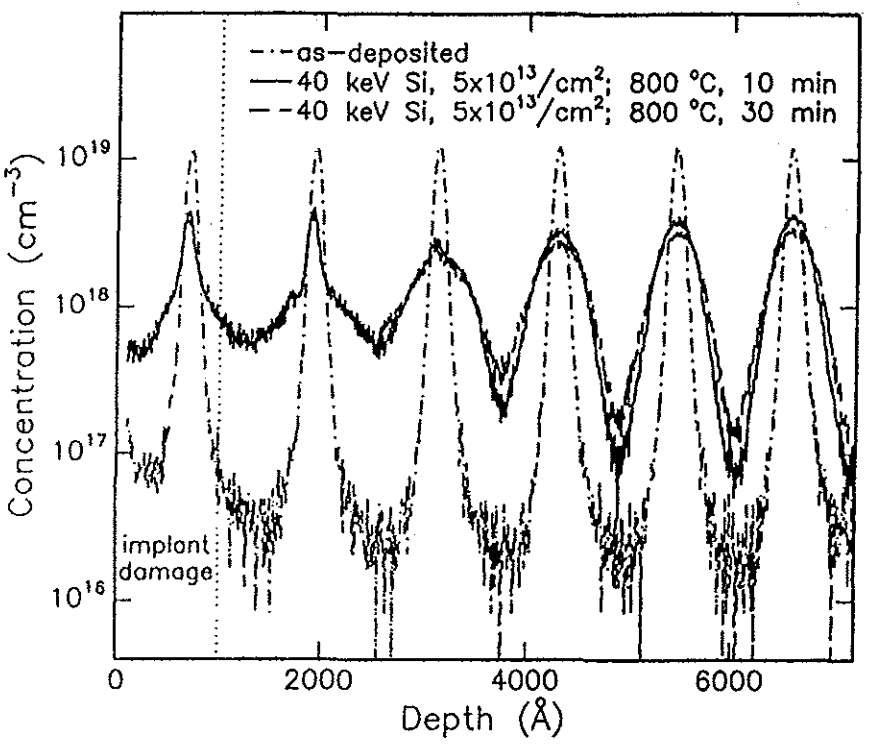


Fig. 113. TED in B delta-doped superlattice. 815°C, 10 and 30 min, showing enhanced broadening, particularly near the shallow, 5×10^{13}cm^{-2} 40 keV Si implant.

The same group of authors has demonstrated convincingly that carbon in Si is a very effective Si interstitial sink. What is believed to happen is a kick-out of substitutional C and subsequent recombination of C_I with C_S. At the C concentration typically encountered in Si, one has to assume that carbon plays an important role in all diffusion processes. This had not been realized until these recent experiments (and is still not accepted by many groups!). Figure 114 dramatically illustrates the effect of a carbon concentration of 2×10^{19}cm^{-3} (admittedly a very high concentration) on the diffusion of boron delta layers as compared to a layer with carbon concentration only 10^{18}cm^{-3}. Much lower carbon concentrations are expected to play a crucial role in dopant diffusion. The reason this effect is usually neglected appears to be related to the relatively smooth character of normal diffusion profiles which do not require many parameters to be fitted with good precision.

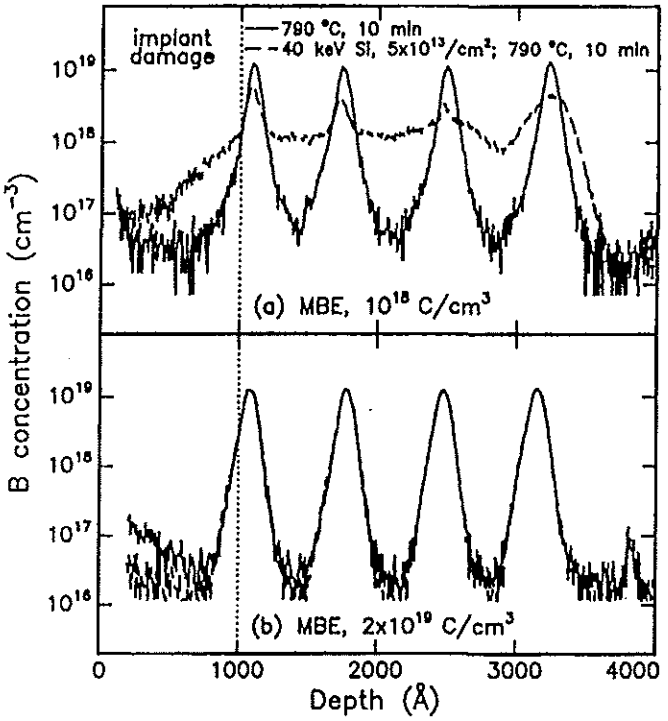


Fig. 114. Secondary ion mass spectroscopy measurements of B diffusion profiles in MBE-grown superlattices containing substitutional C levels of (a) 10^{18}cm-3 and (b) 2×10^{19}cm^{-3}. Profiles are shown for unimplanted and ion-damaged [5×10^{13}cm^{-2}, 40 keV Si) superlattices after diffusion in vacuum at 790°C for 10 min. (Courtesy P.A. Stolk, D.J. Eaglesham, H.-J. Gossman and J.M. Poate, Appl. Phys. Lett. **66**, 1371 (1995).]

As we have seen in Fig. 110, C and O have diffusivities which fall between those of the rapid interstitial diffusers and the typical substitutional shallow dopants. Oxygen is only bound to two Si neighbors resulting in a much reduced energy for oxygen interstitial formation. It is believed that when one Si-O bond breaks it can be reestablished with a different, neighboring Si host atom. This kind of transport will not require native defects.

Carbon on the other hand is believed to assume substitutional sites producing large strains because of its small size and the strong, short bonds. The microscopic diffusion mechanism for carbon is not well understood but it is assumed that a minor fraction of C diffuses interstitially. These arguments remind us of the rapid diffusion of Cu in Ge!

Closing this section, we show for completeness the experimentally established diffusion data for some impurities in Ge (Fig. 115).

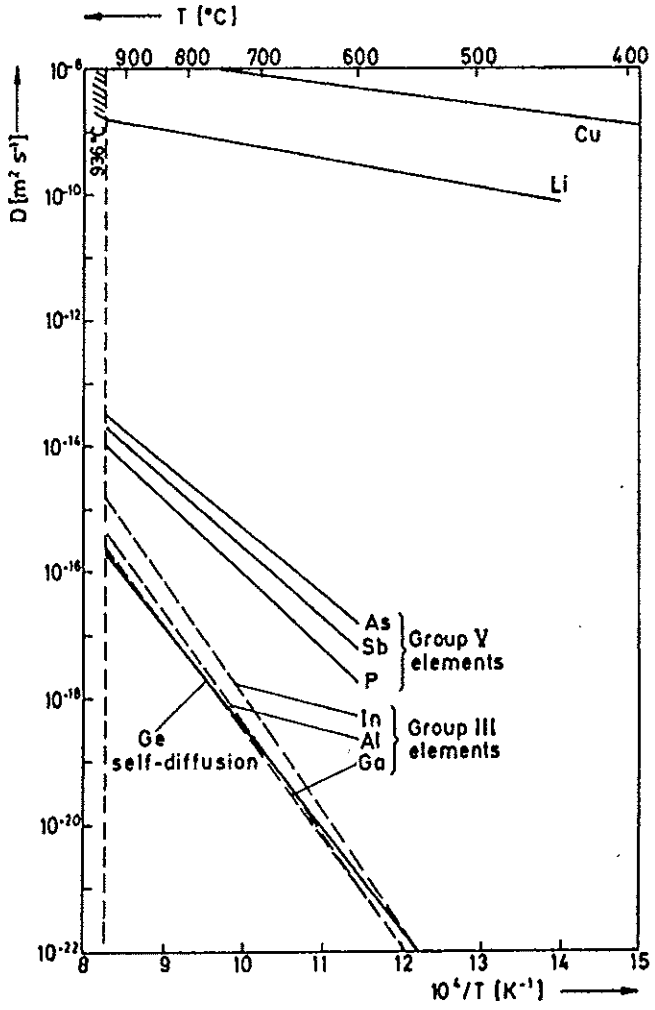


Fig. 115. Survey of the tracer diffusivities of foreign atoms in germanium [Cu (see Seeger and Chik, 1968), Li (Fuller and Ditzenberger, 1953), As, Sb and Bi (Dunlap, 1954), In, Al and Ga (Dorner, 1980).] For comparison, Ge tracer self-diffusion data (Vogel et. al., 1983) are included.

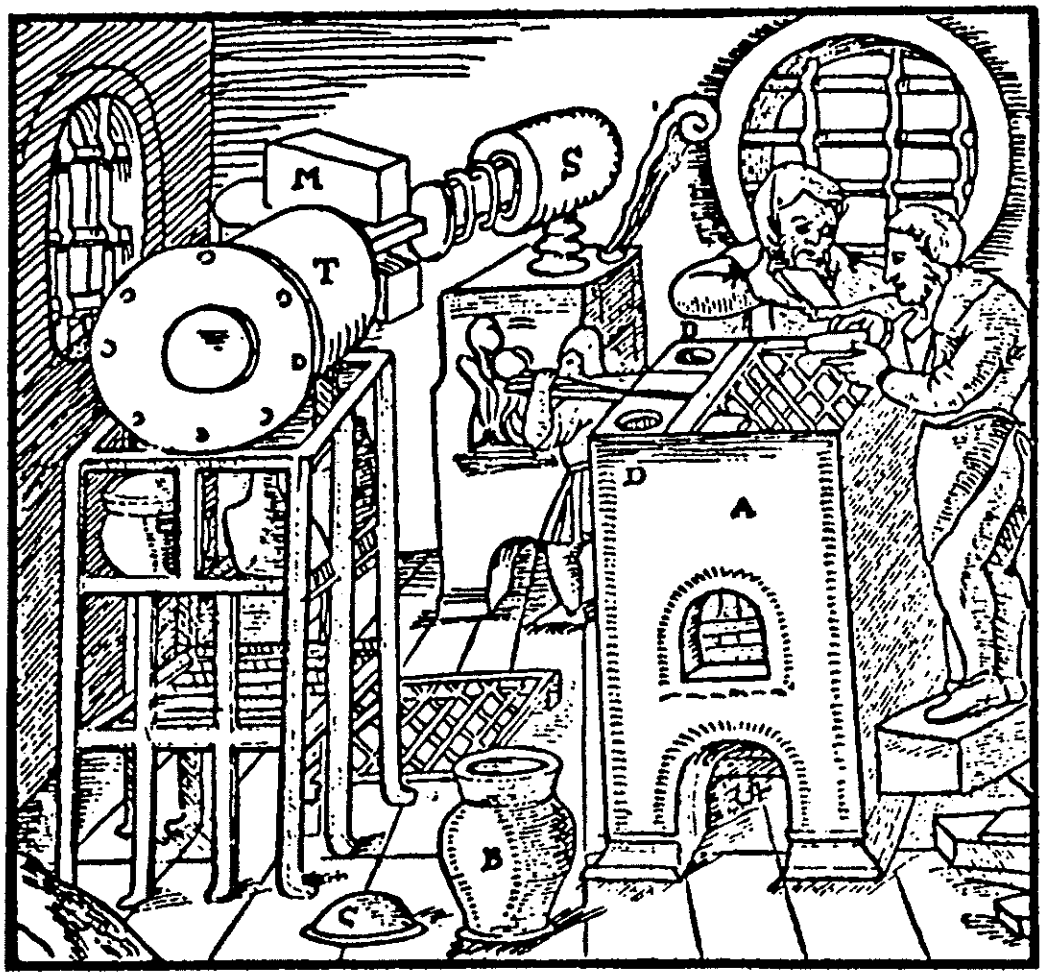
5.6. Ion Implantation

References:

- "Ion Implantation in Semiconductors" by J.W. Mayer, L. Eriksson and J.A. Davis (Academic Press, New York 1970).
- "Ion Implantation of Semiconductors" by G. Carter and W.A. Grant (John Wiley & Sons, New York 1976).
- "Ion Implantation" by H. Ryssel and I. Ruge (John Wiley & Sons, New York 1986).
- "Ion Implantation: Science and Technology," J.F. Ziegler, ed. (Academic Press, New York 1984).
- Proceedings of the International Conferences on Ion Implantation.
- Current literature: J. Appl. Phys., Appl. Phys. Lett., etc.

Ion implantation is a direct spin-off from nuclear science. It has become a preferred semiconductor doping technology because it gives the operator great freedom in the choice of the dopant species and depth distribution. When used in conjunction with rapid thermal annealing, it is truly a low temperature ($\leq 700^{\circ}C$) technique as compared to doping by diffusion. In contrast to diffusion, ion implantation is a process very far from equilibrium. Ions are accelerated to high energies and injected into the surface of a semiconductor. The ions are stopped by various interactions, some of which lead to the displacement of host material atoms. This is generally called radiation damage and it must be repaired for most applications by thermal or rapid thermal annealing in order to restore the properties of the crystalline semiconductor.

In order to reach the dopant distribution depth of interest, i.e., a few tens to hundreds of nm, one has to accelerate ions to energies of tens to hundreds of keV depending on the ion species and the target composition. The "basic concepts" below offer definitions for major expressions used in the field of ion implantation science and technology.



(G. Agricola, "De Re Metallica" 1556, amended by M. W. Thompson.)

Professor Thompson has drawn attention to the undue preoccupation with the diffusion furnace (front right) as opposed to the ion implantation equipment (rear left) apparent in this early woodcut, an attitude which persists in a few workshops even to this day.

Basic Concepts (after G. Carter and W.A. Grant, "Ion Implantation of Semiconductors," J. Wiley & Sons (1976), p. 223).

Binary collisions. Although ion implantation involves a beam of many ions interacting with a solid containing many atoms, the collision between one ion and one target atom is of basic importance. In many circumstances the problem of an ion entering a solid can be treated as a succession of binary collisions in which the ion interacts or collides with only one target atom at a time. Consequently, a study of the dynamics of binary collisions is essential in any study of ion implantation.

Interatomic potential. The collision between atoms depends intrinsically on the interatomic potential V(r). The forces acting on both particles and hence their trajectories are derived from V(r) so that a knowledge of the interatomic potential is of fundamental importance. We will need therefore to discuss the different approaches employed to obtain satisfactory expressions for V(r).

Energy loss. As an ion penetrates a solid it loses its energy in a series of collisions with the target atoms until it eventually comes to rest. The amount of energy lost in each collision will determine the total path length or range of the ion. Energy loss of the incoming particle occurs by either elastic or inelastic processes. In the former, kinetic energy is conserved and in the latter, is converted into another form by excitation of atomic electrons.

Ion range. The range of an ion is determined by the rate at which it loses energy. Having described and evaluated energy loss processes we can calculate ion ranges and predict the spatial distribution of implanted impurities within surface layers of targets. In non-crystalline or amorphous materials, energetic implanted ions have a Gaussian-shaped distribution but in single crystals the picture becomes complicated because the crystal structure assumes a role in determining the range profile.

Channeling. In a single crystal the lattice atoms are arranged periodically in space and in certain directions the structure has open channels bounded by densely packed walls. Ions entering in these directions often do not make a random sequence of collisions, as in an amorphous material, but rather are steered by a succession of correlated interactions with atoms in the channel walls. This has a profound effect on many aspects of ion implantation.

Damage. Energy is transferred from the projectiles to the target atoms as they slow down. An atom receiving sufficient kinetic energy will be displaced from its lattice site and the target will suffer radiation damage. Since the energy required to displace an atom is typically ≈ 25eV, heavy ions with energies of a few keV cause considerable damage. A recoiling target atom may itself be sufficiently energetic to act as a secondary projectile which in turn displaces further atoms and produces a cascade of displacement collisions. Radiation damage is often the inevitable outcome of ion implantation and may mask the desired doping effects. The mechanisms of radiation damage and annealing are of vital importance.

Annealing. Atoms (both host lattice and dopant atoms) displaced during implantation may at a later stage return to their usual positions. This annealing may occur over a short time scale (e.g. during the period that the collision cascades are moving through the lattice) or over much longer periods following completion of the implantation. The annealing is often assisted by providing thermal energy. Annealing is used to activate the dopants, i.e., to move them onto proper lattice sites.

Doping. Doping by ion implantation should be compared and contrasted to other methods, principally that of thermally-activated diffusion. In this latter technique, the dopant is deposited on to the substrate surface and driven in by high temperature annealing. The dopant concentration profile is then dependent on thermal equilibrium criteria for a particular dopant/target combination. Ion implantation is by contrast a non-equilibrium process, the dopant atoms being driven into the solid by violent use of their excess kinetic energy.

5.6.1. Ion-Solid Interactions and Ion Range

In order to gain an understanding of the ion implantation process, we will start with a brief review of the two major interactions of an energetic ion moving into a solid. It is generally assumed that the individual interactions are independent of each other which means that the slowing down of the ion is a statistical process.

5.6.1.1. Nuclear and Electronic Interactions

The interaction between an energetic ion and an atom in a solid (also called nuclear or elastic or large angle collision) is a simple process only for very large ion energies (> MeV). In this case the electrons can be neglected and only the repulsive Coulomb potentials of the two nuclei have to be considered. This extreme and simple case has been treated by Rutherford. Unfortunately this is not the typical situation we encounter in ion implantation of semiconductors. At energies of one to a few hundred keV and ion species ranging from simple

protons to Sb^+ or heavier, the effects of the electrons bound to the ion as well as to the target atom have to be taken into account. The degree to which the electrons screen (hide) the nuclear Coulomb potential is a very complicated matter which depends on many physical parameters. Bohr laid the foundation for these studies. In the 1960s, Lindhard, Sharff and Schiøtt (Kgl. Danske Videnskab Selskab. Mat. Fys. Medd. **33**, No. 14 (1963)) developed the first comprehensive theoretical framework (LSS theory) for ions moving in an amorphous solid. They derived a universal relationship for the nuclear stopping power $(d\varepsilon/d\rho)_n$ using dimensionless energy (ε) and length (ρ) parameters. These are given by:

$$\varepsilon = E\,(a/e^2)\left[M_2/Z_1Z_2\,(M_1 + M_2)\right]$$

and

$$\rho = R\pi a^2 N\left[M_1M_2/(M_1 + M_2)^2\right]$$

with

$$a = 0.8853\,a_0/\left(Z_1^{2/3} + Z_2^{2/3}\right)^{1/2}$$

where E is the ion energy, R is the ion range, a_0 is the Bohr radius (0.53 Å), e is the charge of the electron, N is the number of atoms per unit volume of the target. $Z_{1,2}$ and $M_{1,2}$ are the atomic numbers and the masses of the projectile (1) and the target atoms (2), respectively. The nuclear stopping power $(d\varepsilon/d\rho)_n$ is shown as a solid line in Fig. 116 as a function of $\sqrt{\varepsilon}$. The characteristic peak energy ε_1 is given in Table 14 for various elements in Si, Ge and Sn.

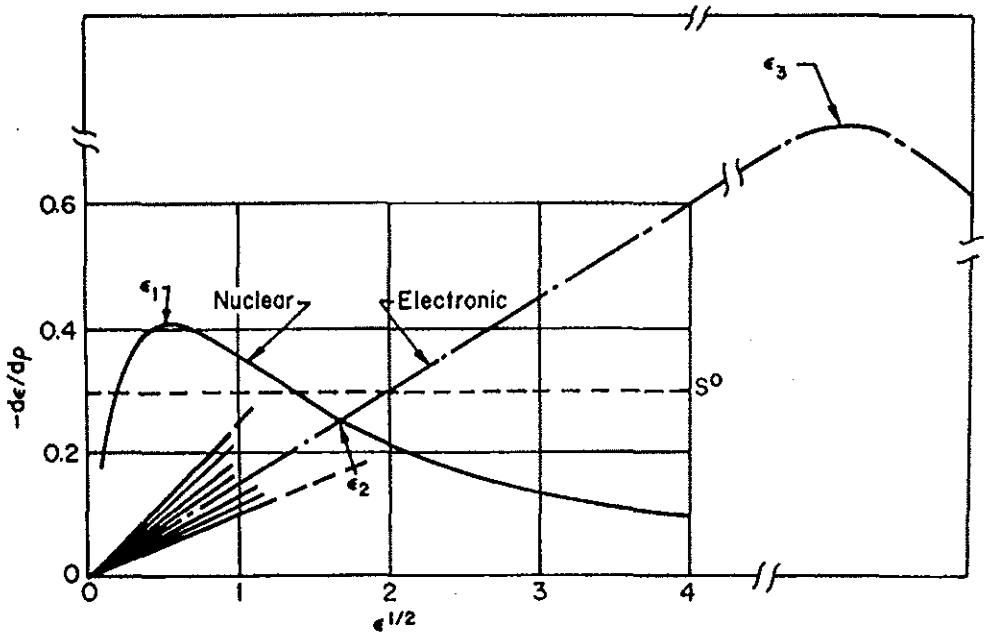


Fig. 116. Theoretical nuclear and electronic stopping-power curves expressed in terms of the reduced variables ρ and ε. For electronic stopping, a family of lines (one for each combination of projectile and target) is obtained; the majority of cases fall within the limits shown. The dot-dash line represents the electronic stopping for $k = 0.15$. The horizontal line labeled S^o represents the constant-stopping-power approximation suggested by Nielsen. Values of the characteristic energies ε_1, ε_2 and ε_3 are given in Table 12. [After J. Lindhard *et al.*, Kgl. Dan. Vid. Selsk. Mat. Fys. Medd. **33** (1963).]

Collisions between energetic ions and electrons are very different from nuclear collisions because of the very small mass of the electrons. Collisions with electrons lead to small angle scattering. The LSS theory shows that the electronic stopping power $(d\varepsilon/d\rho)_e$ can be described by:

$$(-d\varepsilon/d\rho)_e = k\,\varepsilon^{1/2}$$

with

$$k = \left[\frac{0.0793\,\xi_e\,Z_1^{1/2}\,Z_2^{1/2}\,(M_1+M_2)^{3/2}}{(Z_1^{2/3}+Z_2^{2/3})^{3/4}\,M_1^{3/2}\,M_2^{1/2}}\right]$$

and

$$\xi_e \approx Z_1^{1/6}$$

The electronic stopping power cannot be expressed by a universal curve as for nuclear stopping but by a family of dependencies for a range of k (Fig. 111). Again we find a peak value $(-d\varepsilon/d\rho)_e$ at ε_3 and a cross over between nuclear and electronic loss at ε_2.

Table 12 summarizes the LSS parameters ε, ρ and k for three target species and a wide range of ions and Table 13 shows ρ as a function of ε and of k. The values for k range between 0.1 and 1.0 for most ion/target combinations. Table 14 gives the values for ε_1, ε_2 and ε_3 for Si, Ge and Sn targets and ions from mass 10 to mass 209.

In summary, we have learned that for low energies nuclear interactions dominate while at large energies electronic energy loss is more important. It follows that the path of an energetic ion ($\varepsilon > \varepsilon_2$) into a solid is rather straight at first (electronic or low angle scattering) but when the ion has slowed down sufficiently ($\varepsilon > \varepsilon_2$) nuclear or large angle scattering will become dominant, leading to a random walk (Fig. 117).

TABLE 12

THE LSS PARAMETERS ε, ρ AND k FOR VARIOUS DOPANTS IN Si, Ge (≅ GaAs) AND Sn (≅ CdTe)

Ion	ε/E (keV)			ρ/R (microns)[a]			k		
	Si	Ge	Sn (CdTe)	Si	Ge	Sn (CdTe)	Si	Ge	Sn (CdTe)
Li	0.221	0.089	0.052	28.0	8.0	2.7	0.28	0.65	1.04
B	0.113	0.049	0.029	32.2	10.6	3.8	0.22	0.47	0.75
N	0.074	0.033	0.020	32.2	11.8	4.5	0.20	0.42	0.65
Al	0.028	0.015	0.0093	30.5	15.3	6.4	0.14	0.26	0.39
P	0.021	0.012	0.0078	29.0	15.7	6.8	0.14	0.24	0.36
Ga	0.0054	0.0037	0.0027	17.9	15.2	8.1	0.12	0.16	0.21
As	0.0048	0.0034	0.0025	17.0	14.8	8.1	0.12	0.16	0.20
In	0.0021	0.0017	0.00130	11.4	12.2	7.6	0.11	0.14	0.17
Sb	0.0019	0.0015	0.00121	10.7	11.9	7.5	0.11	0.14	0.16
Tl	0.00070	0.00062	0.00052	6.0	8.2	5.9	0.11	0.13	0.14
Bi	0.00066	0.00059	0.00050	5.8	8.0	5.8	0.11	0.13	0.14

[a] Using the following density values (in gm/cm³): 2.33 for Si; 5.32 for Ge; and 5.84 for Sn (i.e., the density for CdTe).

TABLE 13

ρ VALUES AS A FUNCTION OF ε AND k

ϵ	$k = 0.0$	$k = 0.10$	$k = 0.12$	$k = 0.14$	$k = 0.2$	$k = 0.3$	$k = 0.4$	$k = 1.0$
0.01	0.072	0.069	0.069	0.068	0.067	0.064	0.062	0.052
0.02	0.115	0.110	0.109	0.108	0.106	0.102	0.098	0.081
0.05	0.218	0.207	0.205	0.203	0.197	0.188	0.180	0.144
0.10	0.360	0.339	0.335	0.332	0.321	0.304	0.289	0.224
0.20	0.614	0.571	0.563	0.553	0.533	0.501	0.472	0.353
0.50	1.35	1.21	1.19	1.17	1.10	1.01	0.938	0.656
1.0	2.67	2.29	2.22	2.17	2.01	1.80	1.63	1.06
2.0	5.84	4.57	4.39	4.22	3.79	3.26	2.88	1.71
5.0	19.4	11.9	11.1	10.4	8.83	7.11	5.99	3.17
10.0	53.6	23.8	21.6	19.9	16.1	12.3	10.1	4.92

TABLE 14

CHARACTERISTIC ENERGIES (IN keV) CORRESPONDING TO ε_1, ε_2, AND ε_3 IN FIG. 116

Ion	ε_1			ε_2			ε_3
	in Si	in Ge	in Sn	in Si	in Ge	in Sn	
B	3	7	12	17	13	10	3×10^3
P	17	29	45	140	140	130	3×10^4
As	73	103	140	800	800	800	2×10^5
Sb	180	230	290	2000	2000	2000	6×10^5
Bi	530	600	700	6000	6000	6000	2×10^6

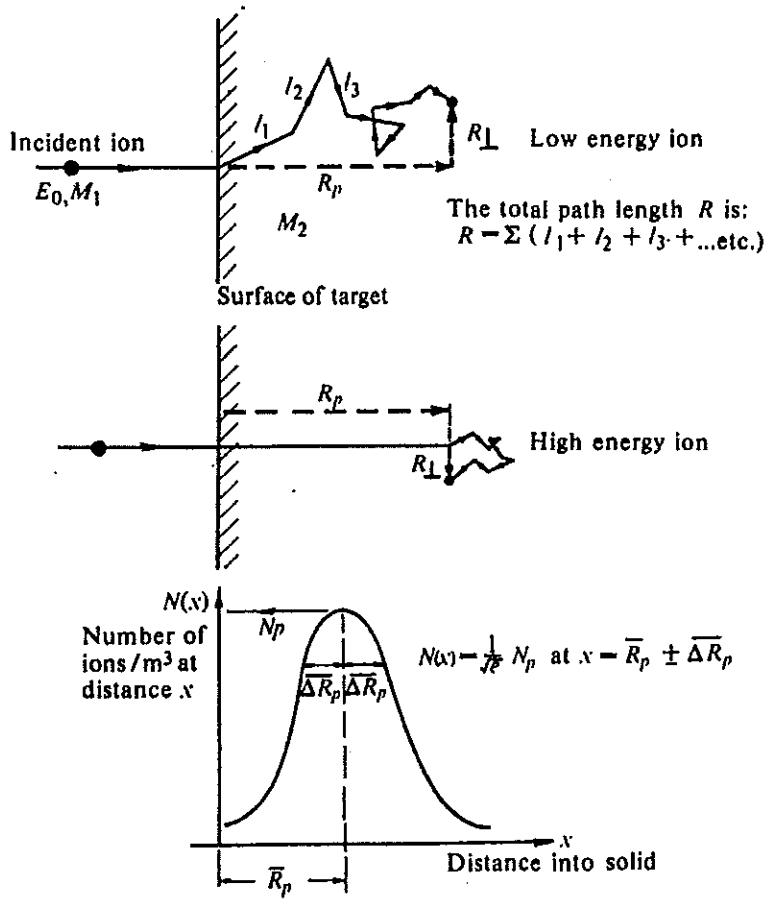


Fig. 117. Basic range concepts for low and high energy ions. [After G. Carter and W.A. Grant, "Ion Implantation of Semiconductors," John Wiley & Sons, New York, 1976.]

5.6.1.2. Ion Range and Range Straggling

In order to obtain the total ion range one has to integrate over the nuclear and electronic losses. Table 15 contains numerical values for the path lengths, R, for ion energies between 1 and 10^4 keV for three target materials and ion masses from 7 to 204. Inspection of a few relevant cases (e.g. B and P in Si) illustrates that very convenient depths of a few hundred to a few thousand Å can be reached with ion energies of tens to hundreds of keV. This energy range was of interest to nuclear physics some 30 to 40 years ago. At that time ion sources, acceleration, mass separation, beam guidance and diagnostics over large distances were developed to a very sophisticated level. With the current trend towards shallower junctions, ion implant energies are being reduced.

TABLE 15

TOTAL PATH LENGTH R (microns) CALCULATED FROM THE LSS THEORY[a]

	^{7}Li	^{11}B	^{14}N	^{27}Al	^{31}P	^{70}Ga	^{75}As	^{115}In	^{204}Tl
a. *In silicon*									
1 keV	0.020	0.011	0.008	0.004	0.004	0.002_5	0.002_5	0.002	0.002
3 keV	0.046	0.025	0.018	0.009	0.008	0.005	0.005	0.004	0.004
10 keV	0.131	0.069	0.048	0.023	0.020	0.012	0.012	0.010	0.009
30 keV	0.334	0.187	0.130	0.058	0.050	0.027	0.026	0.022	0.019
100 keV	0.826	0.527	0.384	0.184	0.155	0.071	0.067	0.052	0.043
300 keV	1.69	1.19	0.920	0.527	0.454	0.199	0.186	0.130	0.098
1 MeV	3.41	2.55	2.08	1.43	1.30	0.675	0.631	0.415	0.269
3 MeV	6.20	4.79	4.10	3.08	2.87	1.88	1.79	1.27	0.777
10 MeV	—	—	7.81	6.39	6.09	4.79	4.67	3.80	2.62
b. *In germanium*									
1 keV	0.030	0.016	0.011	0.005	0.005	0.002	0.002	0.002	0.001
3 keV	0.064	0.034	0.024	0.011	0.010	0.005	0.005	0.003_5	0.003
10 keV	0.153	0.083	0.057	0.026	0.023	0.011	0.010	0.008	0.006
30 keV	0.348	0.196	0.133	0.061	0.051	0.024	0.022	0.017	0.013
100 keV	0.81	0.501	0.353	0.169	0.141	0.059	0.056	0.040	0.029
300 keV	1.63	1.09	0.812	0.443	0.378	0.155	0.144	0.095	0.065
1 MeV	3.27	2.32	1.81	1.16	1.03	0.492	0.457	0.288	0.174
3 MeV	5.94	4.35	3.48	2.48	2.27	1.34	1.27	0.853	0.493
10 MeV	11.17	8.34	6.78	5.15	4.80	3.40	3.28	2.53	1.64
c. *In cadmium telluride*									
1 keV	0.053	0.029	0.020	0.009	0.008	0.003_5	0.003	0.002	0.001_5
3 keV	0.108	0.059	0.041	0.019	0.016	0.007	0.007	0.005	0.003
10 keV	0.241	0.134	0.093	0.043	0.036	0.016	0.015	0.011	0.007
30 keV	0.515	0.295	0.204	0.094	0.080	0.034	0.032	0.022	0.016
100 keV	1.15	0.709	0.501	0.242	0.203	0.083	0.077	0.052	0.036
300 keV	2.28	1.50	1.11	0.595	0.506	0.205	0.190	0.122	0.078
1 MeV	4.55	3.17	2.43	1.51	1.33	0.612	0.567	0.352	0.205
3 MeV	8.26	5.92	4.66	3.19	2.89	1.62	1.53	1.00	0.564
10 MeV	15.5	11.3	9.06	6.62	6.12	4.10	3.94	2.94	1.84

[a] Projected ranges (R_p) may be obtained by combining the above R values with the appropriate R_p/R ratios from Table X5.

There is one more refinement which is required before we can predict the average depth reached by an ion implanted into a target. The total ion path length R traveled by an ion in a solid is longer than the depth reached perpendicular to the surface, often called the "projected path length", R_p, because the ion changes direction many times before coming to rest. Especially during the nuclear collision phase the ion path deviates significantly from a straight line. In

Table 16 we see that a low energy Li ion may have a projected range as small as 22% of the total distance traveled (R) in Sn while a 500 keV Sb ions reaches a depth of 91% of R in Si.

TABLE 16

PROJECTED RANGE CORRECTIONS R_p/R

Ion	Substrate	R_p/R values[a]				Rule-of-thumb value $(1 + M_2/3M_1)^{-1}$
		20 keV	40 keV	100 keV	500 keV	
Li	Si	0.54	0.62	0.72	0.86	0.4
B		0.57	0.64	0.73	0.86	0.54
P		0.72	0.75	0.79	0.86	0.77
As		0.83	0.84	0.86	0.89	0.89
Sb		0.88	0.88	0.89	0.91	0.93
Li	Ge	0.33	0.40	0.53	0.74	0.2
B		0.34	0.40	0.50	0.71	0.3
P		0.50	0.52	0.58	0.71	0.56
As		0.67	0.69	0.72	0.77	0.76
Sb		0.76	0.76	0.78	0.81	0.83
Li	Sn	0.22	0.28	0.40	0.63	0.1_5
B		0.24	0.28	0.38	0.60	0.2
P		0.34	0.37	0.43	0.57	0.44
As		0.51	0.53	0.56	0.65	0.65
Sb		0.63	0.64	0.66	0.72	0.75
C	ZnO	0.49	0.54	0.62	0.78	—
Si		0.60	0.63	0.68	0.78	—
Ge		0.75	0.76	0.78	0.83	—
Sn		0.82	0.82	0.83	0.86	—
Pb		0.89	0.89	0.89	0.89	—
C	CdS	0.36	0.41	0.49	0.68	—
Si		0.48	0.51	0.56	0.68	—
Ge		0.65	0.66	0.69	0.75	—
Sn		0.73	0.74	0.75	0.80	—
Pb		0.82	0.82	0.82	0.84	—

[a] These R_p/R ratios have been supplied by Schiøtt.[52]

All the LSS theory results presented so far are mean values. The statistical nature of the ion scattering process leads to different values of R for every ion. The LSS theory also can be used to obtain the range distribution. To first order one finds a Gaussian distribution centered at R_p (Fig. 117, bottom). Schiøtt has calculated the standard deviation, ΔR_p, for several relevant cases (Fig. 118). These calculations again clearly show the effect of nuclear scattering (large

angle) at low energies and electronic scattering (small angle) at large energies. The higher the ion energy the smaller is the width of the ion range distribution. A simple, practical consequence of this is the need for several implantation runs with a number of ion energies and doses if one wants to produce a constant ion implant concentration profile over a large depth.

In the mid-1980s, sophisticated numerical computation programs were developed for ion implantation (see: J.F. Ziegler in "Ion Implantation Science and Technology," Academic Press (1984), pp. 51-108). These programs can now be purchased for use on personal computers and they are convenient for the prediction of multiple implant profiles over a wide range of energies, targets and ions.

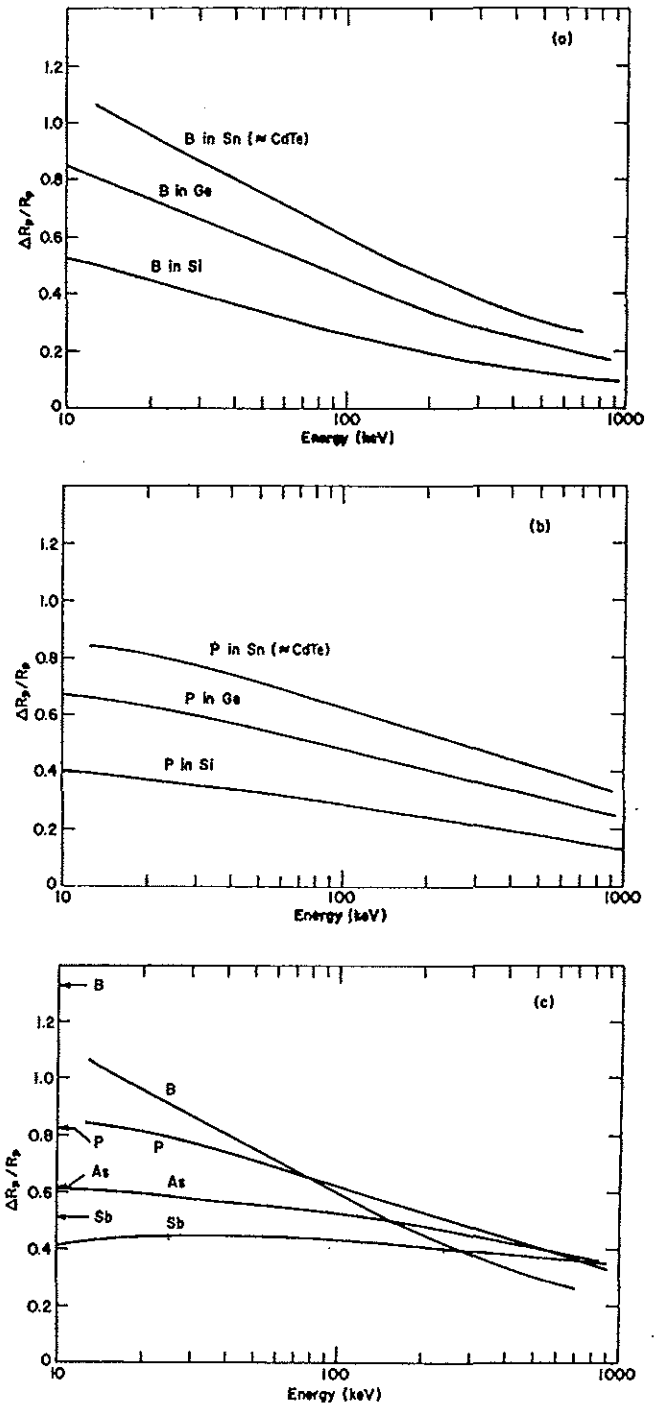


Fig. 118. Range straggling $\Delta R_p/R_p$ versus energy (in keV) for several common dopants, based on the calculations of Schiøtt: (a) boron in silicon, germanium, and cadmium telluride; (b) phosphorus in silicon, germanium, and cadmium telluride and (c) several dopants in cadmium telluride. In (c), the values calculated by Sigmund and Sanders are shown by arrows on the left-hand side; these values do not include electronic stopping. [After James W. Mayer, Lennart Eriksson and John A. Davies, "Ion Implantation in Semiconductors" Academic Press, New York, 1970.]

5.6.1.3. Channeling and Dose Rate Effects

The LSS theory assumes an amorphous target. In crystalline targets, i.e., most semiconductors (Si, Ge, GaAs), one observes large deviations from the LSS theory especially for small ion energies. This is not surprising since one finds wide open channels in crystalline solids (Fig. 119). The widest channels in the diamond structure are found in the <110> orientations. Ions entering such a channel are expected to reach a much larger R_p. The ion distribution is no longer Gaussian but instead has a long tail into the crystal. This effect, called channeling [see Gemmel D S, *Rev. Mod. Phys.* **46**, 129 (1974)] is easily observed and has been extensively studied and applied. An experimental result obtained with 40 keV phosphorus ions is shown in Fig. 120. It shows that "channeling" only occurs when the P ion direction is within a critical angle $\Psi_{C1/2}$ of a low index crystal orientation (<111>, <100> or <110>). Table 17 lists some critical angles for boron, phosphorus and antimony ions impinging on single crystal silicon:

TABLE 17

Ion	Energy (keV)	Critical angle (degrees) <100>	<110>	<111>
Boron	10	4.76	6.97	5.30
	100	2.67	3.47	2.98
	300	2.03	2.98	2.26
Phosphorus	10	5.79	7.51	6.45
	100	3.26	4.22	3.63
	300	2.47	3.21	2.76
Antimony	10	6.95	9.01	7.74
	100	3.91	5.07	4.35
	300	2.97	3.84	3.31

The critical angle can be approximated by:

$$\Psi_{C1/2} \cong \sqrt{\frac{2Z_1Z_2e^2}{4\pi\varepsilon_0 Ed}}$$

with:

Z_1, Z_2 = atomic number of ion/target atoms; E = ion energy; and d = interatomic distance of target atoms in Å.

This expression is only valid for energies $E > \frac{2Z_1Z_2e^2}{4\pi\varepsilon_0 a}$ with a = screening radius (~0.1 to 0.2 Å).

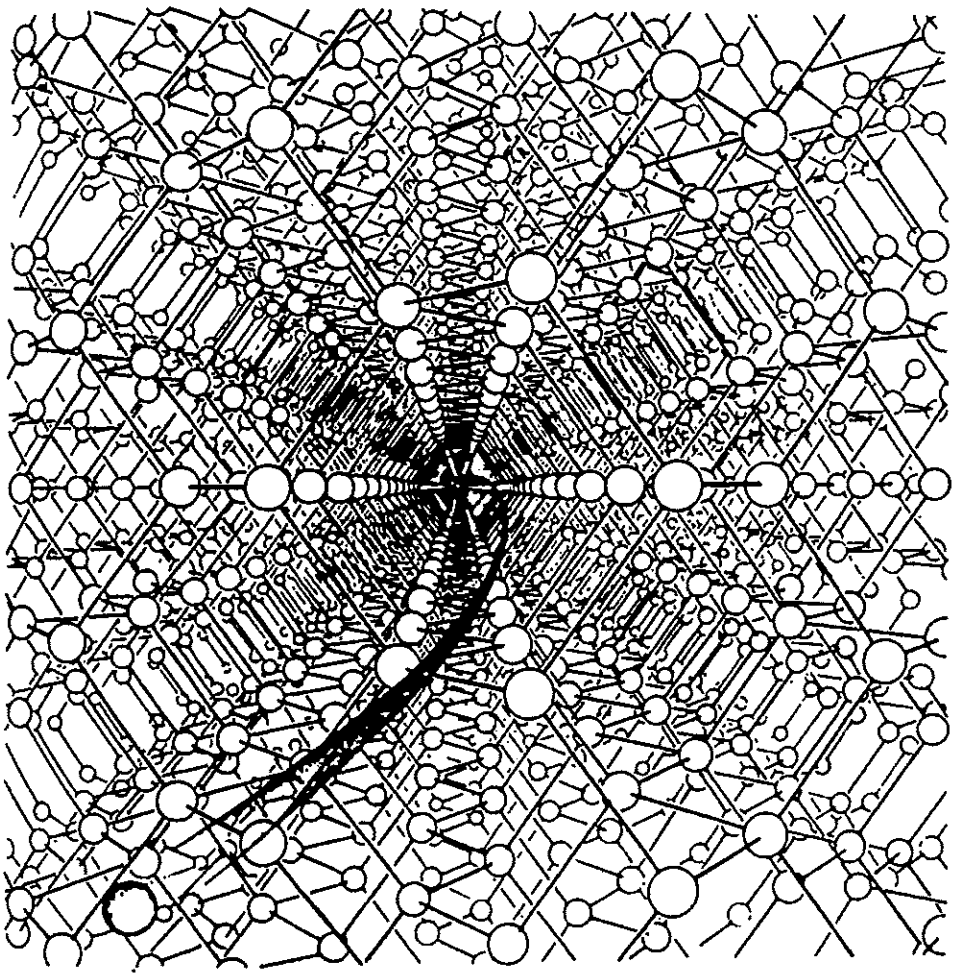


Fig. 119. Artists conception of the channeling process on a microscopic scale. [From "Channeling in Crystals" by W. Brandt. Copyright © 1968 by Scientific American, Inc. All rights reserved.]

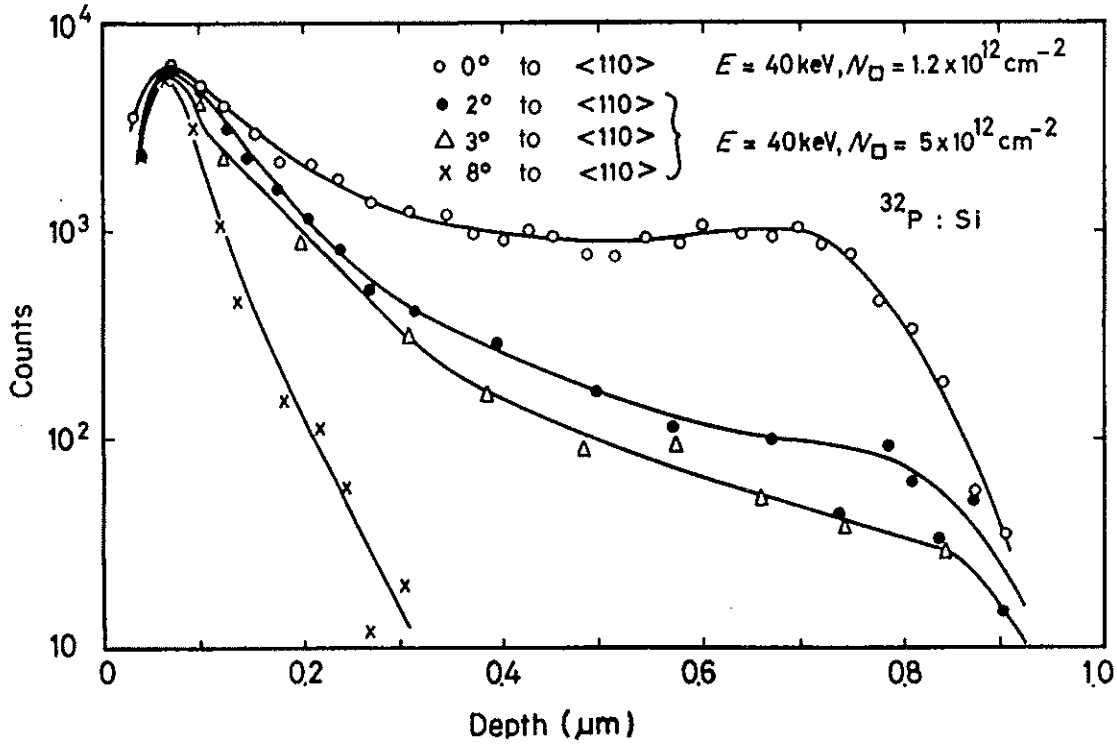


Fig. 120. The variation of ^{32}P concentration profiles with crystal orientation, for 40 keV ions.. [After G. Dearnaley et al., Can. J. Phys. **46**, 587 (1968).]

Because of the small values of $\Psi_{C1/2}$, it is impractical to use channeling for fabrication purposes. As mentioned earlier, the trend is towards shallower implants. Rather than utilizing channeling, one typically aligns a semiconductor deliberately a few degrees off any major crystal axis in relation to the ion trajectory during implantation. Channeling, though not utilized as a special implantation technique, is present as a secondary effect in all implantation profiles in single crystals regardless of the ions entry angle. Ions entering in a random orientation are scattered with a certain probability into one of the many crystal channels. This leads to strong deviations from Gaussian profiles. Figure 121 displays the In-impurity profile obtained in a random orientation in crystalline Si (filled circles) compared to amorphous Si (open circles).

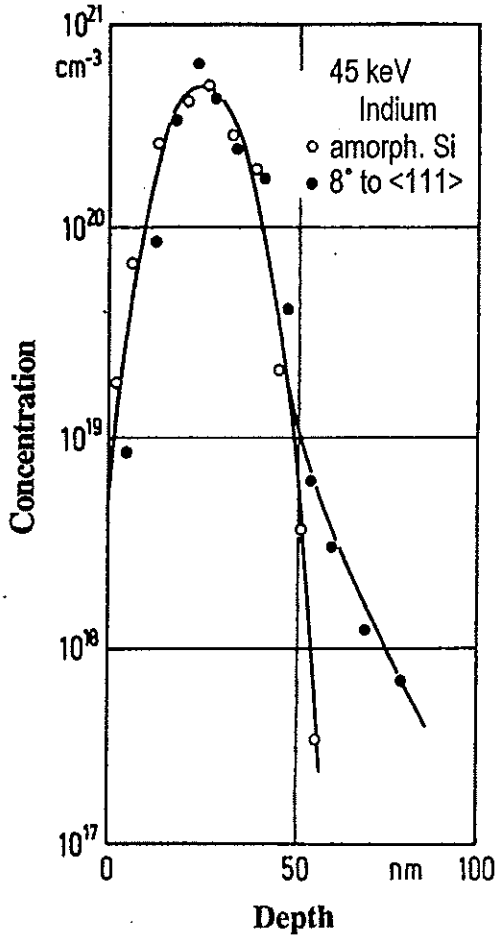


Fig. 121. Comparison of room temperature implants of In into amorphous (o) and crystalline (•) Si. [After J. Lindhard, "Influence of Crystal Lattice on Motion Energetic Charged Particles," Mat. Fys. Medd. Dan. Vid. Selsk. **34**, (1965).]

Beyond a depth of ~ 50 nm the profile strongly deviates from a Gaussian distribution. If the crystal is first rendered amorphous by nitrogen ions, the In profile does not show the far extending tail, a direct indication that channeling must have occurred in the crystalline case.

Several parameters affect the scattering of ions into channels:

- the orientation of the crystal with respect to the ion beam
- the total ion dose
- the temperature of the crystal
- the surface quality of the crystal

A majority of channeling events can be avoided by a misorientation of wafer and beam. In practice this means that the ion beam is not parallel to any major crystal axis. There are, however, still a few ions which, by random collisions, get scattered into a channel. After a high enough ion implant dose the crystal is rendered amorphous, and primary as well as secondary channeling disappears (see Fig. 122). Pre-amorphization of Si by implanting with Si has been used to obtain Gaussian dopant implant profiles.

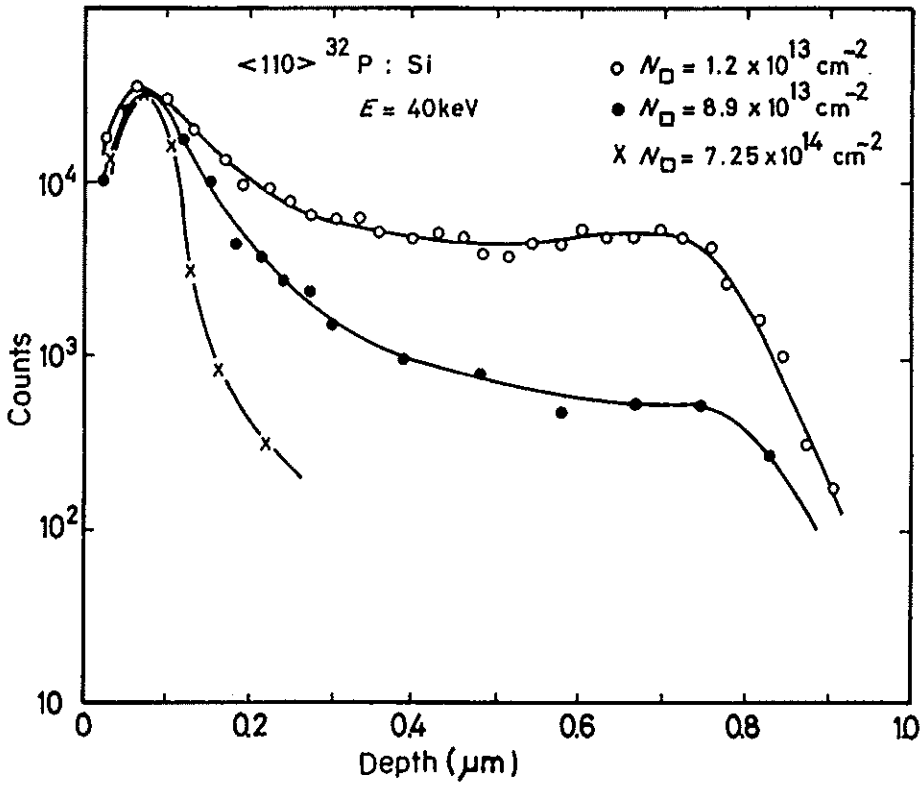


Fig. 122. The effect of increasing ion dose of 40-keV ^{32}P concentration profiles, at room temperature [After G. Dearnaley et al., Can. J. Phys. **46**, 587 (1968).]

Implantation at elevated temperatures leads to a decrease in the acceptance angle $\Psi_{C1/2}$ for channeling due to the increase in the amplitude of the target atom vibrations. On the other hand, it has been demonstrated experimentally that much higher ion doses are needed to render a crystal amorphous at elevated temperatures, where radiation damage annealing occurs during the implantation (dynamic or in-situ annealing). Because channeling is strongly dependent upon the degree of crystallinity, it can be suppressed by a thin oxide layer on the Si wafer. Experimental results using this technique are shown in Fig. 123. Implantation through a thin oxide is also

being used to move the maximum dopant concentration closer to the surface in order to minimize metal contact resistance.

There are further effects besides channeling which lead to deviations of the implantation profile from the LSS theory predicted distribution.

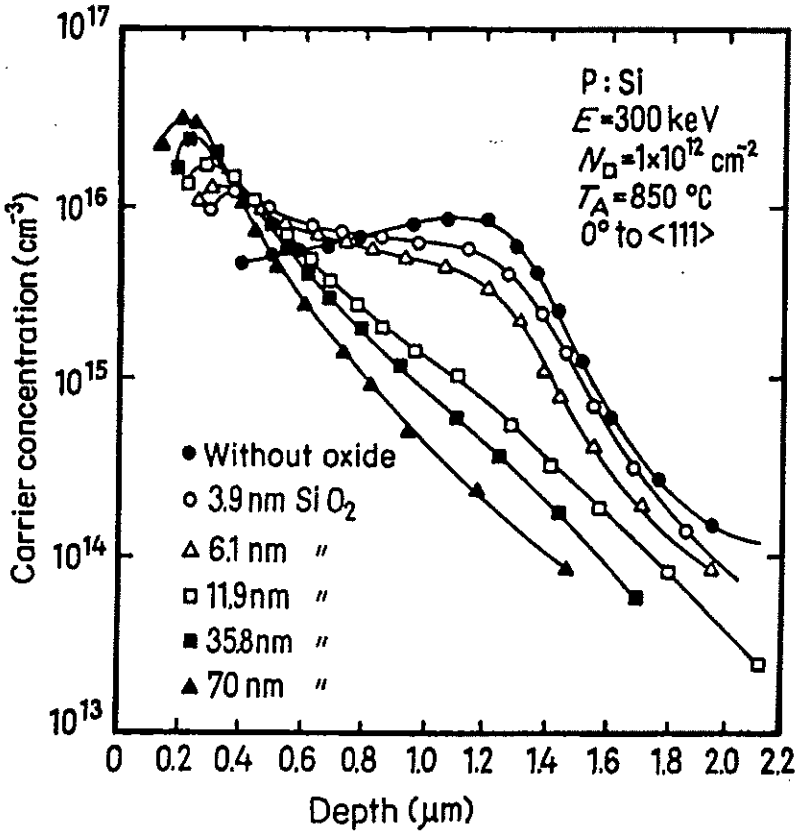


Fig. 123. Donor profile for various surface-oxide thicknesses for 300 keV ^{31}P ions [after R.A. Moline and G.W. Reutlinger, Proc. 2nd Int'l Conf. on Ion Implantation, eds. I. Ruge and J. Graul, Springer-Verlag (1971) p. 309.]

Radiation enhanced diffusion occurs at temperatures far below typical diffusion temperatures. This effect can be explained by the large number of vacancies and interstitials which are formed during implantation and by the large number of electrons and holes created during implantation. The recombination of an electron and a hole frees energy of the order of the semiconductor bandgap, e.g., ~ 1 eV in Si. This energy is released in a small volume leading to local "heating." The diffusion coefficient of many impurities has been found to increase with the implantation dose rate. An example is given in Fig. 124 where an antimony profile is plotted as a function of depth for two different dose rates and two substrate temperatures. The profiles are

clearly dominated by dose rate and not by temperature. Aluminum implantations in Si have yielded profiles which are a factor of thirty times wider than predicted by the LSS model, a case in which it was possible to explain the broadening by radiation defect-enhanced diffusion (vacancy and interstitial assisted).

Thermal diffusion during annealing is another effect which leads to broadening of the implantation profile (see next section).

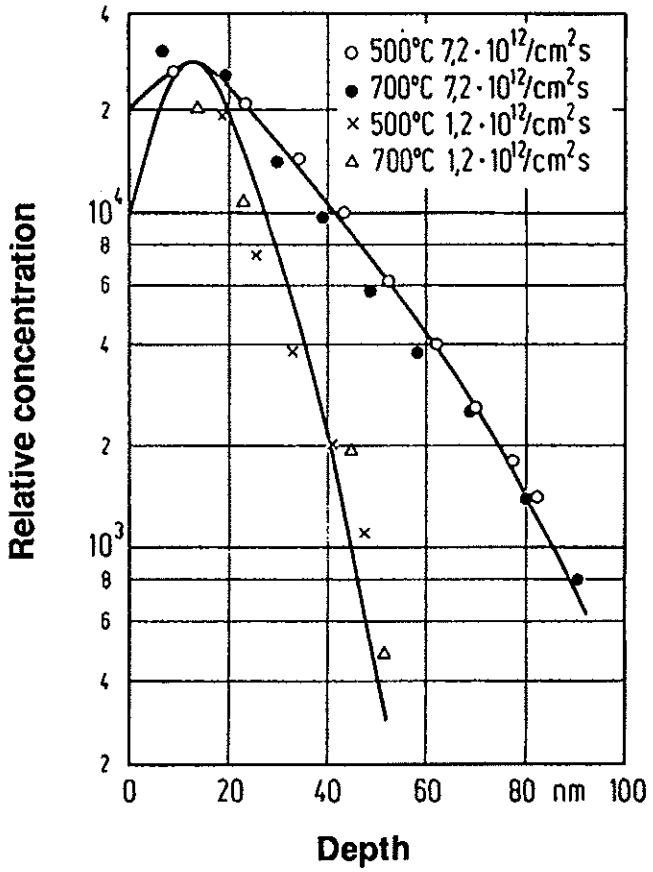


Fig. 124. Effect of dose rate on ion distribution [after K. Gamo et al., Appl. Phys. Lett. 17, 391 (1970).]

5.6.2. Annealing of the Radiation Damage and Impurity Activation

LSS theory and its refinements may tell us something about the impurity distribution with depth for any given ion, ion-energy, target material, crystal orientation, dose rate, etc. They do not tell us, however, if the ion has taken the appropriate position in the lattice in order to act as a donor or an acceptor. Most dopant impurities (except Li, which is not used in industry) must

occupy the correct tetrahedral substitutional lattice position to become an electrically active donor or acceptor.

Heating, or thermal activation, is the principal process used to move impurities into a substitutional position. An easy way to follow the degree of electrical activation of implanted ions with annealing temperature is to measure the room temperature sheet resistivity of the implanted surface between heating cycles. A four-point probe is typically used to obtain the resistivity (see section on "Electronic Materials Analysis"). Figure 125 shows the sheet resistivity (in Ω/□) for various ions implanted in Ge as a function of the annealing temperature. While P, Ga and As become active at ~300 and 500 °C, it is interesting to note that boron seems to find substitutional places in Ge during implantation and does not need thermal annealing to become electrically active. Thermal annealing does remove the implantation damage but the boron acceptors are already formed during implantation.

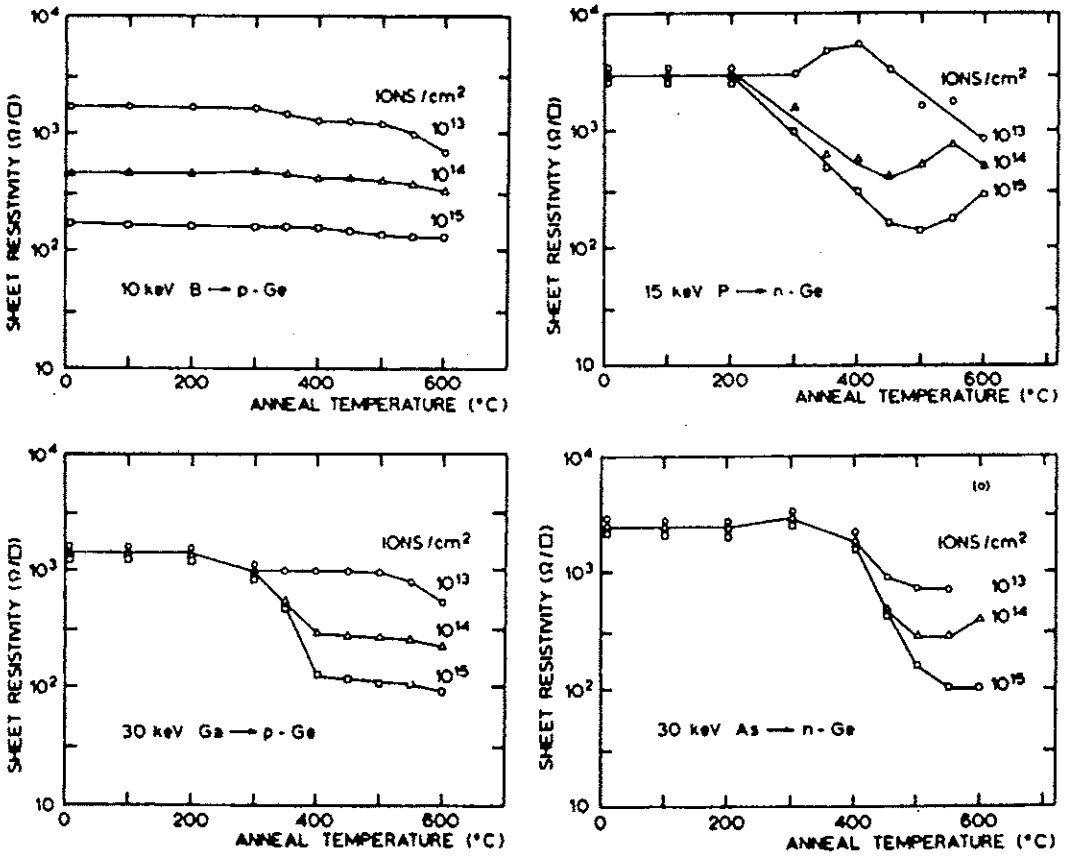


Fig. 125. Sheet resistivity versus anneal temperature for boron, phosphorus, gallium and arsenic implants, parameter being ion dose [after H. Herzer and S. Kalbitzer, Proc. 2nd Int'l Conf. on Ion Implantation, eds. I. Ruge and J. Graul, Springer-Verlag (1971), p. 309.]

A partial explanation for room temperature activation of boron in Ge may be found in the experimental observation that acceptors do not recombine with radiation damage [preferably vacancies in Ge; see I. S. Park and E. E. Haller, J. Appl. Phys. **64**, 6775 (1988).] However, this effect alone is not sufficient since other group III acceptors such as Al or Ga do require thermal activation after implantation. The one unusual aspect of boron in Ge is its segregation coefficient which is 15 at equilibrium. Does this mean boron finds vacant Ge sites and occupies these more preferentially than displaced Ge atoms?

Because of the commercial importance of Si, thermal annealing and dopant activation have been studied most extensively in this semiconductor and to a lesser degree in compound semiconductors. In Si one finds that for the most important dopant impurities (B, P, Sb) thermal dopant activation of close to 100% can be reached if the appropriate annealing cycle is chosen

In order to determine what annealing temperatures are required for high dopant activation, one typically performs a series of isochronal annealing steps and measures between each step the resistivity of the implanted layer. For a more complete picture, Hall effect and resistivity must be measured yielding the free carrier concentration and the mobility. The former tells us about the activation percentage, the latter about residual damage. In all discussions of dopant activation, it should be remembered that the solid solubility (see Fig. 85 a,b on pages 143 and 144) of any particular impurity cannot be exceeded for equilibrium conditions. If implantation leads to concentrations (atoms/cm^3) larger than the solubility limit, precipitation and cluster formation are observed during thermal annealing given sufficient time.

Figure 126 shows the activation of P implanted with 100 keV energy off-axis in Si at room temperature and at 450 °C. The room temperature doses range between 1 and 20 $\times$ 10^{15} cm^{-2} and activation begins near 400 °C and is complete near 600 °C except for the highest dose for which the sheet resistivity continues to drop beyond 900 °C. At first it appears counter-intuitive that the 450 °C implant leads to a low resistivity as-implanted but requires higher temperatures for full activation than its room temperature counterpart. This behavior has been convincingly explained by the formation of rather large defect-impurity aggregates during the high temperature implantation. These large defect clusters are rather stable and require high temperatures to be dissociated.

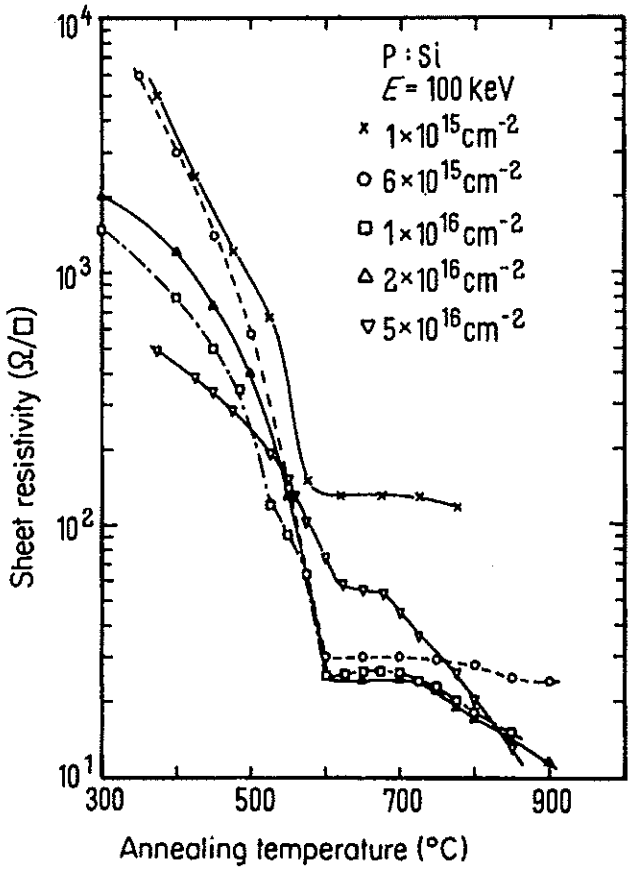


Fig. 126. Sheet resistance of phosphorus layers implanted at room temperature (full lines) and annealed between 300 and 900 °C. Dotted line at 450°C implantation. Measurements taken following each successive 30-minute isochronal anneal [after J.M. Shannon et al., Rad Effects **6**, 217 (1970).]

Figure 127 shows the activation process for hot implants of P in Si in more detail. As we have noticed above, the room temperature implantation is characterized by a single, large activation step near 600 °C. Hot implants show some activation at lower temperatures but require much higher temperatures to reach full activation. The probable cause for this behavior has been stated above.

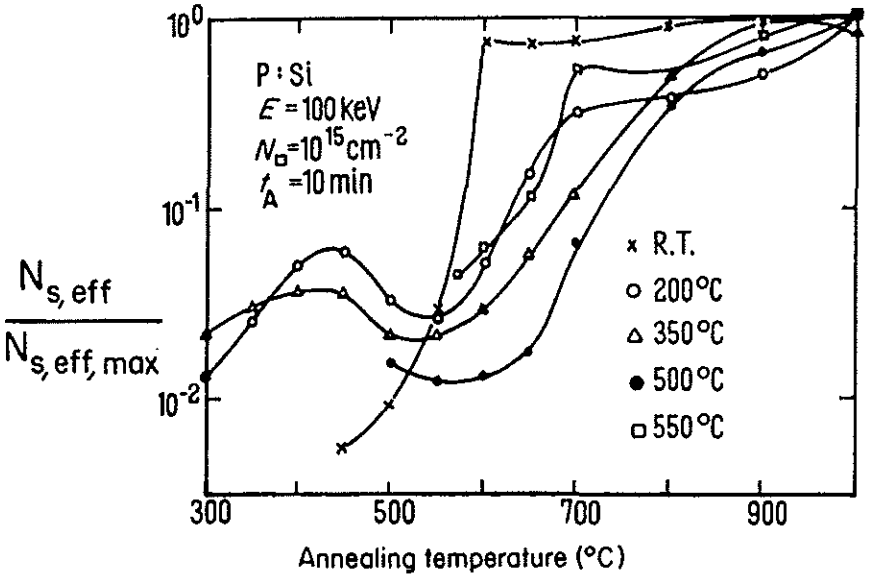


Fig. 127. Isochronal annealing of phosphorus-implanted silicon. E = 100 keV; $N_{\square} = 10^{16}$ cm^{-2}; implantation temperature between room temperature and 550 °C. [after N. Yoshihiro et al., Proc. 3rd Conf. on Solid State Devices, Tokyo (1971), Suppl. Oyo Buturi **41**, 225 (1972).]

During annealing complicated macroscopic defect structures appear and vanish. Dislocation loops and stacking faults are the major structural defects. Figure 128 impressively illustrates the evolution of the damage with annealing temperature and as a function of the implantation temperature. Again, as we have already seen from the electrical activation data, defects formed during hot implants are harder to remove than those created during room temperature implants. Micrographs of a P implanted Si wafer after 800°C annealing and three implant temperatures are displayed in Fig. 129. These structural defect pictures clearly show the enormous complexity of damage formation and removal associated with ion implantation.

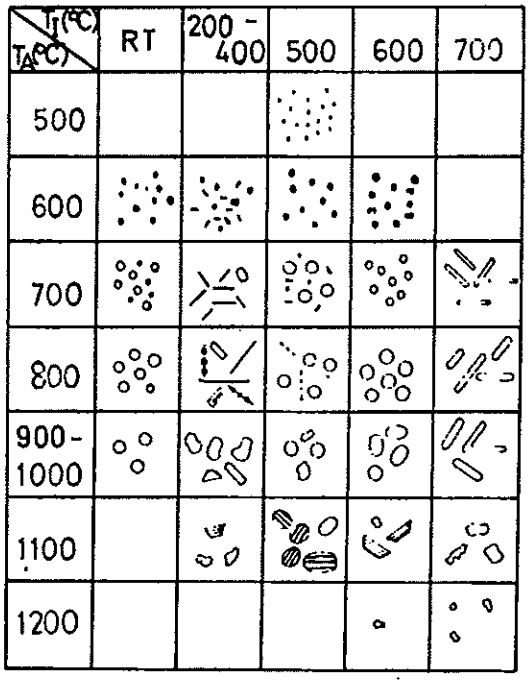


Fig. 128. Defects in phosphorus-implanted silicon depending on implantation and annealing temperature. Implantation energy: 100 keV; dose: 5×10^{14} cm^{-2}. [after M. Tamura, Appl. Phys. Lett. **23**, 651 (1973).]

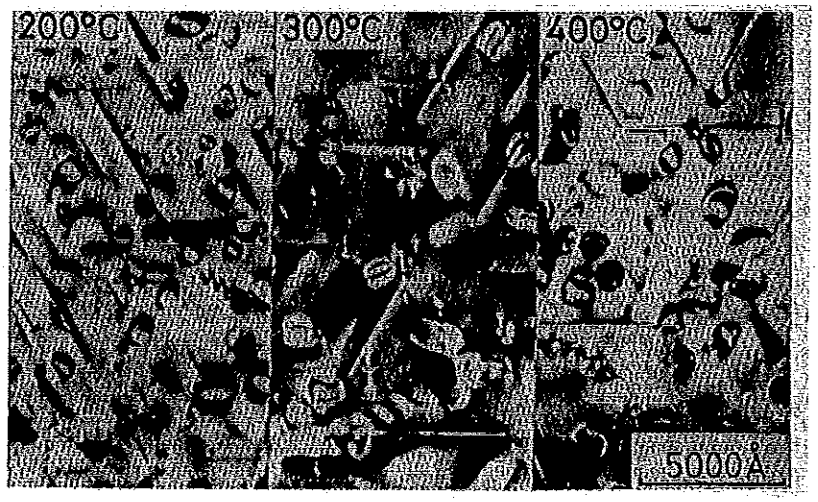


Fig. 129. Micrographs of defects after annealing at 800°C. These samples were implanted at 200 to 400 °C (cf. Fig. 128). [after M. Tamura, Appl. Phys. Lett. **23**, 651 (1973).]

Boron activation in Si proceeds similarly to P. With increasing dose one observes reverse annealing around 600°C. (Fig. 130). Contrary to B in Ge, full activation is difficult to achieve and is reached between 800°C and 1000°C. One explanation which has been proposed is that the small mass of boron results in relatively light damage to the crystal. Additional damage can be produced if instead of B^+ ions, one implants BF, BF_2 or B and F or similarly BCl, BCl_2 or B and Cl ions. All these ion species are produced in a radio frequency gas discharge source supplied with either BF_3 or BCl_3. The annealing results for high dose implants and co-implants (B and F, B and Cl) are shown in Figs. 131a and b. It is clear that all the B implants accompanied by F show excellent activation at temperatures as low as 600°C. While the reasons for this large reduction of the activation temperature are not fully understood, it is quite obvious that the reduced annealing temperature can be very advantageous for semiconductor wafer processing.

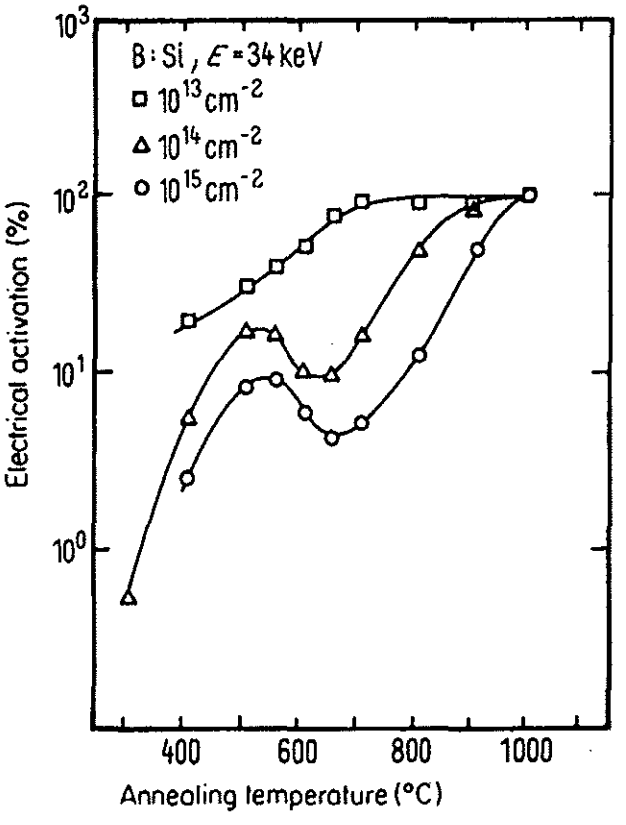
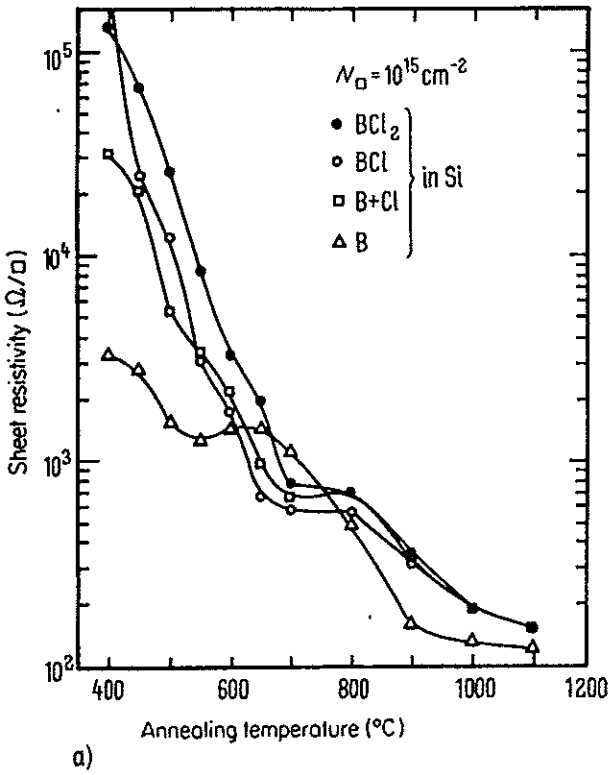


Fig. 130. Relative increase in electrical activation as a function of annealing temperature for 10^{13}, 10^{14} and 10^{15} cm^{-2} boron in silicon. [after I Ruge, H. Müller and H. Ryssel, "Festkörperprobleme XII," Braunschweig (1972).]



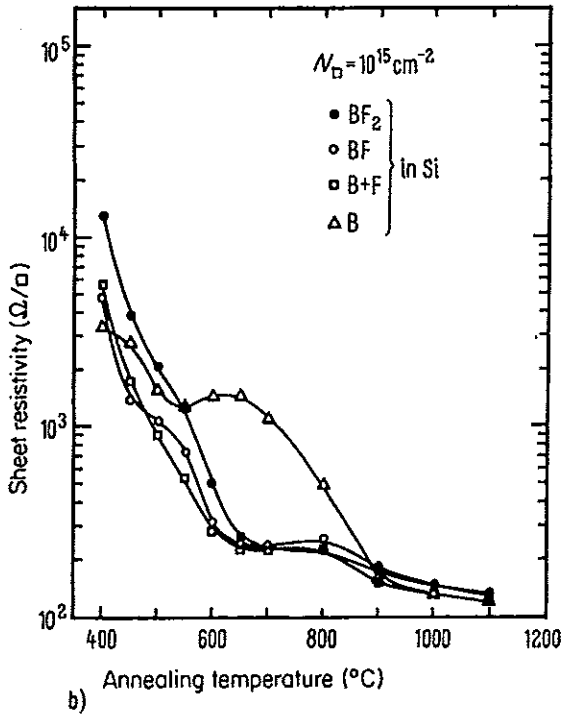


Fig. 131. Isochronal annealing behavior of sheet resistivity for implanted silicon (a: BCl_3 source, b: BF_3 source): B^+ 25 keV; Cl^+ 80 keV + B^+ 25 keV; BCl^+ 105 keV; BCl^+_2 184 keV; F^+ 43 keV + B^+ 25 keV; BF^+ 68 keV; BF_2^+ 111 keV. [after D.G. Beanland, "Ion Implantation in Semiconductors and other Materials," F. Chernow, J.A. Borders and D.W. Brice, eds., New York (1977).]

A class of non-equilibrium thermal annealing approaches has been developed in recent years which make use of powerful lasers or heating lamps. In these new annealing processes, rapid heating and cooling cycles are used to anneal the implanted layer. Two time regimes have to be discussed:

- ultra-short, high energy pulses (ns) which may involve melting and regrowth
- short energy pulses (up to s) which lead to rapid thermal annealing but do not involve melting.

Pulsed laser and electron beam annealing enjoyed great interest in the late 1970s to early 1980s. At that time powerful lasers had been developed and surface processing experiments using lasers were initiated in Russia. The underlying idea is simple. In order to restrict thermal annealing to the ion implanted near-surface region, one needs a very short pulse of photons which is absorbed near the surface. The large power density raises the temperature of this region in a time much shorter than the lattice phonons need to move away from the surface into the interior. By the time significant thermal transport into the bulk begins to occur, the laser pulse has been turned off. This clever, simple concept led to a large number of experiments which produced non-trivial results. One much debated question related to the definition of "melted."

Some scientists argued that optical excitation leads to many broken bonds but that the atoms would not realize for some time that they existed in a molten state instead of a solid state. Redistribution of impurities, mass transport leading to shallow craters and an abrupt change in the index of refraction upon melting eventually showed that the near-surface region does melt at the classical melting point of the material. [see D.H. Auston et al., Appl. Phys. Lett. **33**, 437 (1978).]

Very rapid surface layer heating has been achieved with pulsed lasers (ruby, Nd:glass, eximer). Mode locking and spatial filtering are used to obtain very short pulses with a uniform distribution of the power over the whole semiconductor wafer area. A Q-switch is needed to obtain the rapid pulse. Delivery of ~ 1 Joule cm^{-2} in ~ 100 ns leads to surface melting of silicon (pulse power ~ 10 MW). A simple equation gives a good estimate for the temperature rise T for a given pulse energy E and pulse duration Δt:

$$T = \frac{2(1 - r)E}{S\sqrt{\pi C \rho \chi \Delta t}}$$

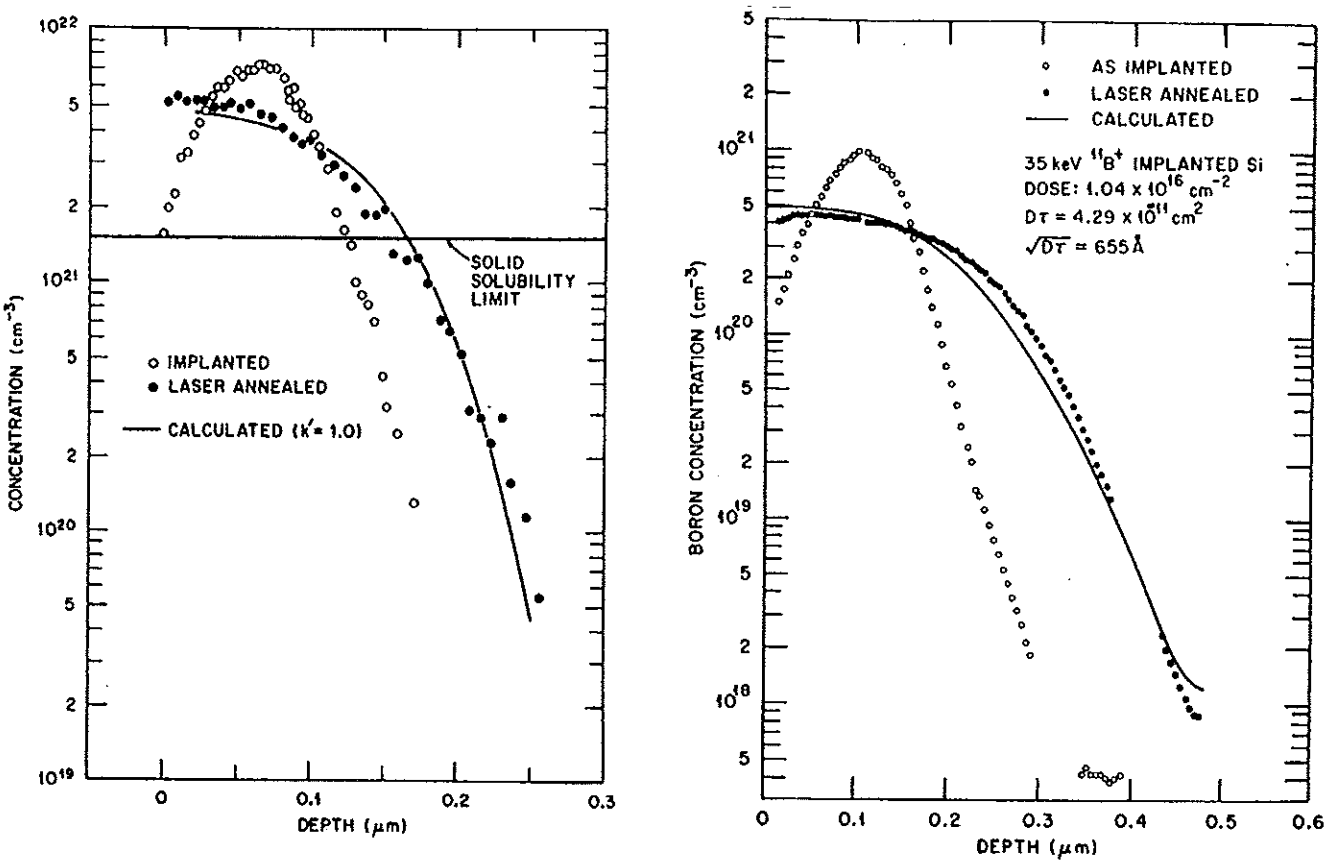
with:	Si	Ge
r = reflection coefficient	0.30	0.40
C = specific heat (J/g °C)	0.76	0.31
ρ = density (g cm^{-3})	2.30	5.30
χ = thermal conductivity (W/cm °C)	0.53	0.45

S = area (cm^2)

D = thermal diffusion coefficient = $\frac{\chi}{\rho C}$

Figure 132 shows implant profiles of B in Si before and after laser annealing. The implant energies and doses were chosen such that the resulting concentration profiles exceed the maximum solubility limits of the two impurities in Si. The post annealing profiles show that the substitutional fraction of the implanted dopant species remained above the maximum solubility limit, not too surprising a result considering that we are dealing with a process far from equilibrium. For a while it was thought that these high dopant concentrations could be exploited for higher performance devices.

In the past 10 to 15 years it has become clear that laser annealing in the ns time regime does not lead to improvements which justify the large investment in equipment and the reduction in throughput because of single wafer processing. As it turns out, rapid thermal annealing on the time scale of seconds is short enough to lead to a significant reduction in the in-diffusion of undesirable contaminants such as Fe, Au and to effective activation of the implanted dopants. Instead of lasers, two banks of high intensity quartz-halogen lamps are used as a photon source. Confocal mirrors guide the photons into a silica chamber which contains the semiconductor wafer and which can be purged with a gas during annealing. The total mass of all materials to be heated is kept to a minimum. This makes possible heating from room temperature to 1000 °C in a few short seconds. Rapid thermal annealers of this basic design are now commercially available and are widely used.



a) b)

Fig. 132. Depth profiles of a) As- and b) B-implanted Si derived from RBS and channeling spectra for as-implanted and laser-annealed cases. For the laser-annealed case, the substitutional fractions of the dopants are shown. Note the limit of the solid solubilities of As and B in Si [a) after C.W. White et al., J. Appl. Phys. **51**, 738 (1980) and b) C.W. White et al., Appl. Phys. Lett. **33**, 662 (1978).]

5.6.3. Solid Phase Epitaxy

An interesting observation regarding the activation of ion implanted dopants has been made for fully amorphized layers. Amorphization is achieved through implantation at low temperatures where in-situ annealing is effectively suppressed. Amorphization can be obtained through implantation of dopants (but this requires rather large doses), of noble gases or self-implantation (Si in Si, Ge in Ge, etc.). The amorphous layer can regrow, still in a solid state starting from the amorphous-crystalline interface and proceeding towards the free surface. This process has been named solid phase epitaxy (SPE), a kind of epitaxial crystal growth. The layer crystallized through SPE can be highly perfect and if it contains dopants, one finds very high activation as long as the maximum solubility is not exceeded. Careful SPE studies were

performed in the late 1970s for Si and Ge. Figures 133 and 134 show Arrhenius plots of the SPE growth rate as a function of inverse absolute temperature. In all cases multiple self-implantation was used to turn the near-surface region amorphous. The regrowth activation energies are constant for a given semiconductor but the prefactors depend strongly on the crystal orientation. For both Ge and Si, the SPE regrowth rate is highest in the <100> direction and slowest in the <111> direction. The crystalline perfection of the SPE layer is best for <100> regrowth and worst for the <111> direction. Pre-amorphization was achieved by self-implantation, followed by dopant implantation, effectively suppressing random channeling discussed above.

Figure 135 shows the boron activation results for pre-amorphization using Ne, Si, BF_2 and B (at 77 K) followed by SPE/annealing. In all four cases full activation is achieved near 600°C. We notice small differences in the hole mobility depending on the species used for the pre-amorphization.

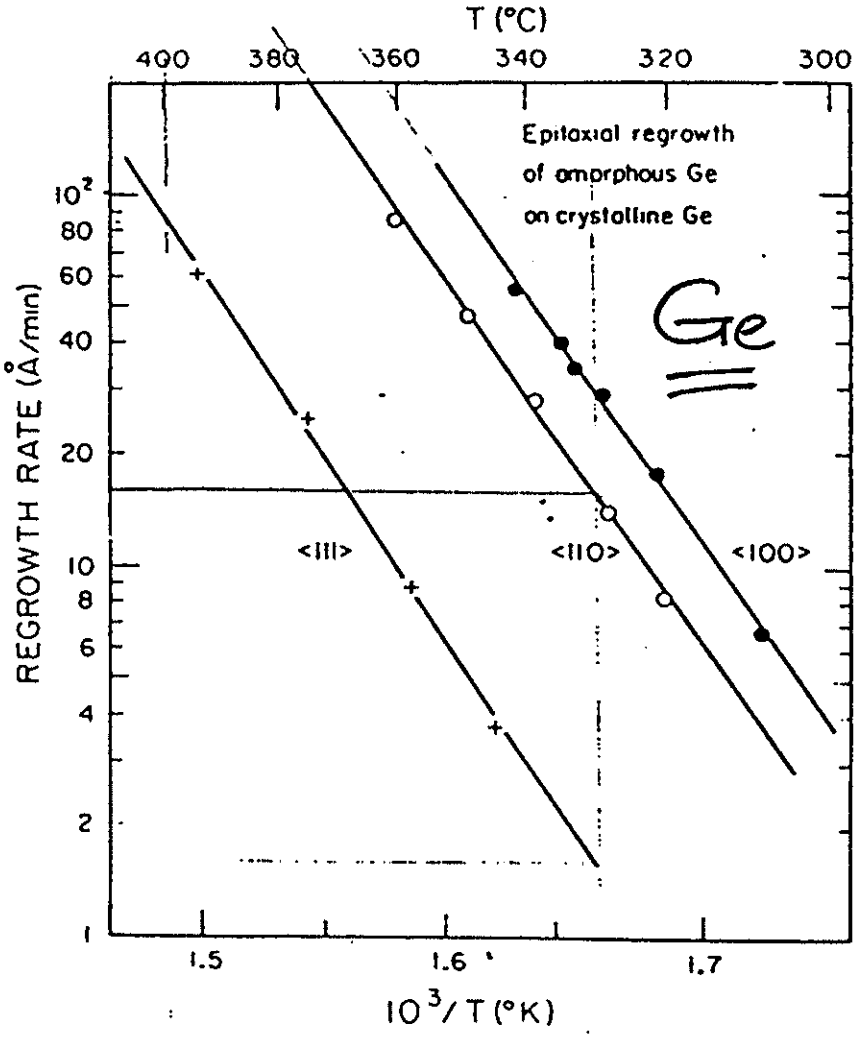


Fig. 133. Regrowth rate versus reciprocal temperature for amorphous layers on <100>, <110>, and <111> Ge. [Ref: L. Csepregi et al., Solid State Commun. **21**, 1019 (1977).]

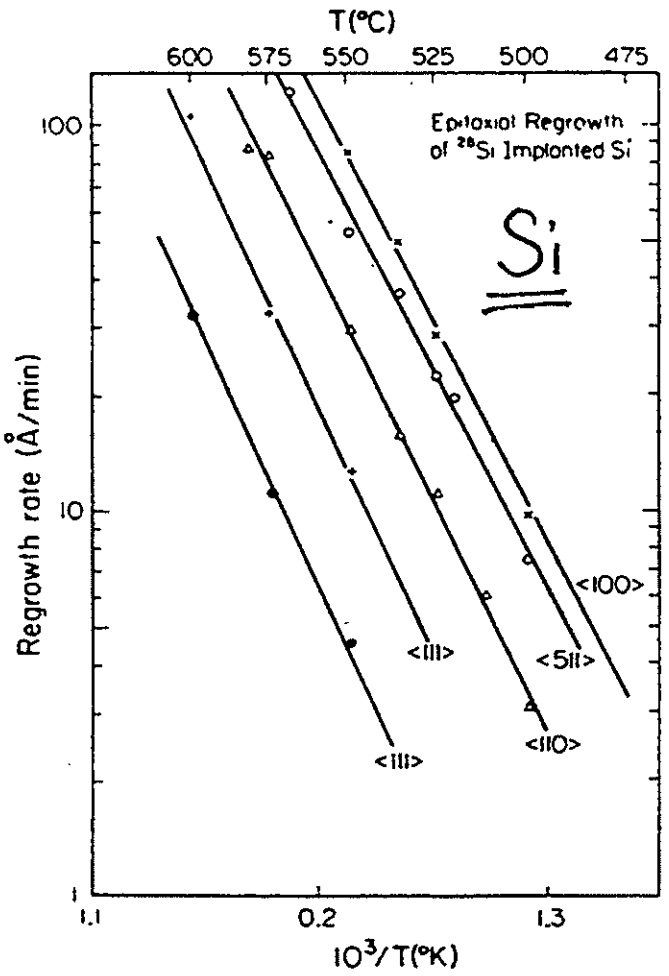


Fig. 134. Arrhenius plot of the regrowth rate for Si samples oriented along the <111>, <110>, <511>, and <100> directions. The closed circles and the crosses are for the slow and fast regrowth rates of the <111> samples, respectively. [Ref: L. Csepregi et al., J. Appl. Phys. **49**, 3707 (1978).]

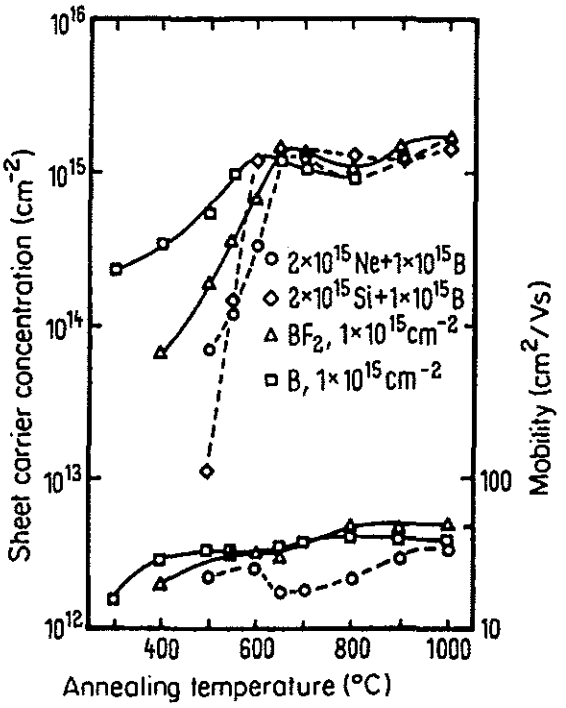


Fig. 135. Annealing behavior of sheet carrier concentration ($N_{S,eff}$) and sheet Hall mobility ($\mu_{S,eff}$) for boron-implanted silicon samples made amorphous through implantations of neon, silicon, BF_2 or boron (the latter at 77 K). [after I Ruge, H. Müller and H. Ryssel, *Festkörperprobleme XII*, Braunschweig (1972).]

5.6.4. Ion Implantation Technology

We have reviewed what happens in a semiconductor when it is implanted and annealed. But how does one create a beam of monoenergetic dopant ions and implant these homogeneously into a 6" diameter (or larger) Si wafer? How does one avoid contamination and how fast is the process? These and other aspects will be briefly reviewed in this section.

An ion implantation machine is a compact nuclear accelerator taking up the space of a medium size laboratory. Figure 136 a) gives a schematic top view of a late 1970s/early 1980s Varian-Extrion implanter. The ions are generated in a gas discharge (*ion source*) using the *ion source power supply.* The appropriate *gases* are stored in small lecture bottles in the ventilated *gas box.* The pre-accelerated *ion beam* (~ 25 keV) travels in a stainless steel high vacuum pipe into the *analyzer magnet.* This double focusing magnet generates an adjustable magnetic field. Depending on the ions which shall be implanted, the magnetic field is set to a value which bends

these ions through a total angle of 90°. Heavier ions are bent less and hit the vacuum chamber wall as do lighter ions which are bent more than 90°. The *resolving aperture* allows precise mass selection and the following *variable slit* controls the beam intensity. The ions which pass through the variable slit are accelerated while traveling through the *acceleration tube.* Indeed, the ions are generated at a high positive potential and are accelerated towards ground potential. This means that all the implanter parts before the acceleration tube are kept in a closed metal shield called *terminal enclosure* while the end station containing the semiconductor wafers is at ground potential. The voltage across the acceleration tube can be varied between 0 and 175 kV. After the electrostatic acceleration the ion beam is re-focused by a magnetic quadrupole lens. In order to obtain an even ion implantation density across the full wafer, the ion beam is deflected electrostatically by a pair of *x and y* electrodes (*plates*). Similar to a TV picture (magnetically deflected electron beam!), the ion beam raster covers the *semiconductor wafer.* The ion implant dose can be measured directly by integrating the current flowing into the wafer, as long as great care is taken to consider the secondary electrons created by the energetic ion beam hitting the semiconductor wafer. The secondary electron beam can be a multiple of the ion beam! The electrons are typically bent sideways by small permanent magnets and collected with a small positive potential on a ring electrode surrounding the wafer. The net-current is then integrated to yield the dose. Modern end stations contain Si wafer carousels which can rapidly move wafers into position for implantation while maintaining high vacuum at all times. Figure 136 b) provides a side view of this implanter. Its height approaches 10 feet and the footprint is of the order of 10×20 feet^2.

There are a number of critical parts and issues regarding ion implantation and we will review a few of them. The ion source is important as it should provide a high intensity, pure ion beam and should require minimal maintenance. A typical ion source for gases is shown in Fig. 137. A radio frequency coil is wrapped around a quartz tube containing a dopant gas (e.g., BF_3, B_2H_6, AsH_3, PH_3, etc.). The RF field ionizes and partially cracks the gas molecules (e.g., BF_3 —> B, BF, BF_2, F, F_2,...) forming singly and multiply charged ionic species. The density of the plasma can be increased by a longitudinal magnetic field which keeps the free electrons on small radius spirals increasing the chance for ionizing collisions. The ions are extracted by a negative DC bias applied to the electrode C. A bundle of fine quartz tubes at the exit is often used to give the ion beam a small divergence before it enters the analyzing magnet.

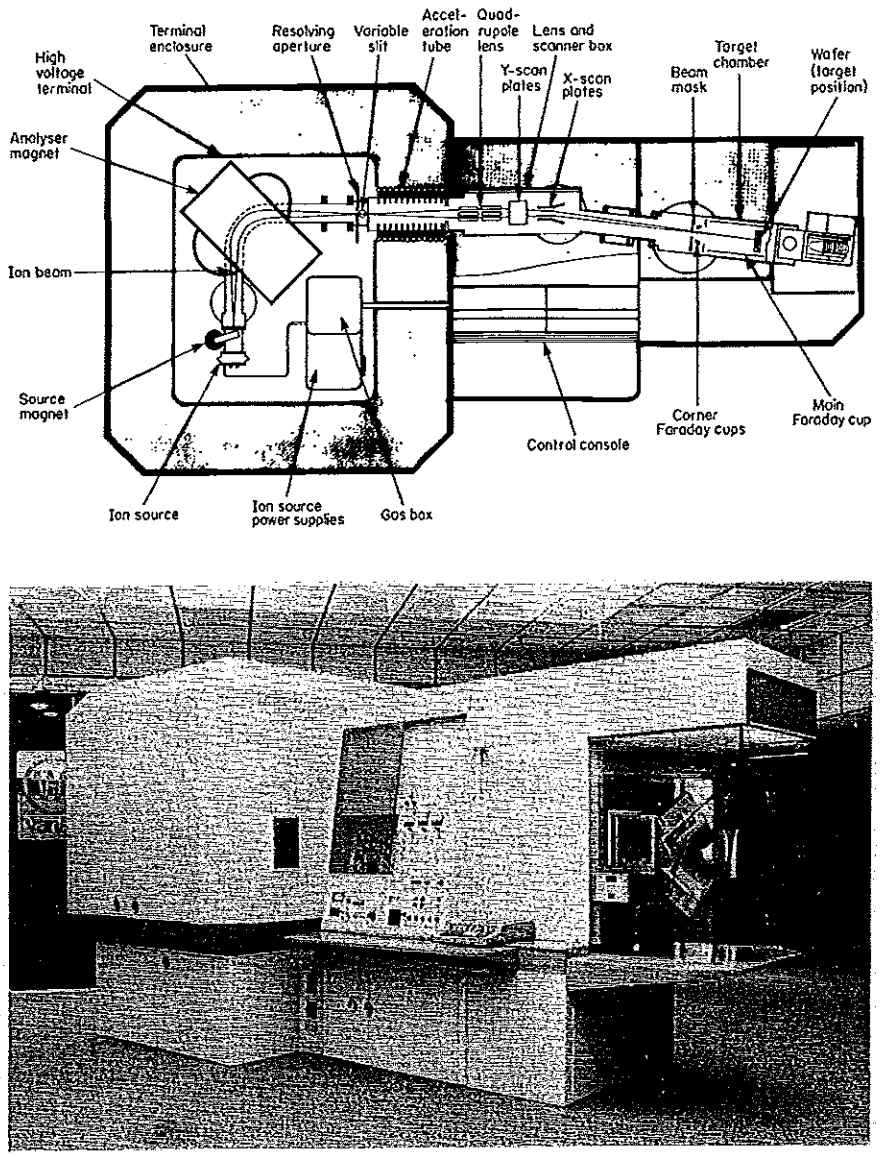


Fig. 136. a) Schematic illustration of an Extrion medium-current implanter; maximum energy 200 keV, maximum current 1.5 mA. (b) Photograph of this implanter. [after H. Ryssel and I. Ruge, "Ion Implantation," John Wiley & Sons, New York, 1986.]

In recent years a number of new sources have been developed which are both more efficient and safer. The low pressure arc discharge source cracks and ionizes the dopant gas. Again, a magnetic field forces the charged particles onto small radius spirals increasing the probability for ionizing collisions. In order to provide more source variety (not all ions of interest can be provided through a gaseous compound!) and to increase safety, a solid (in some cases elemental) source is used together with an argon ion arc (discharge).

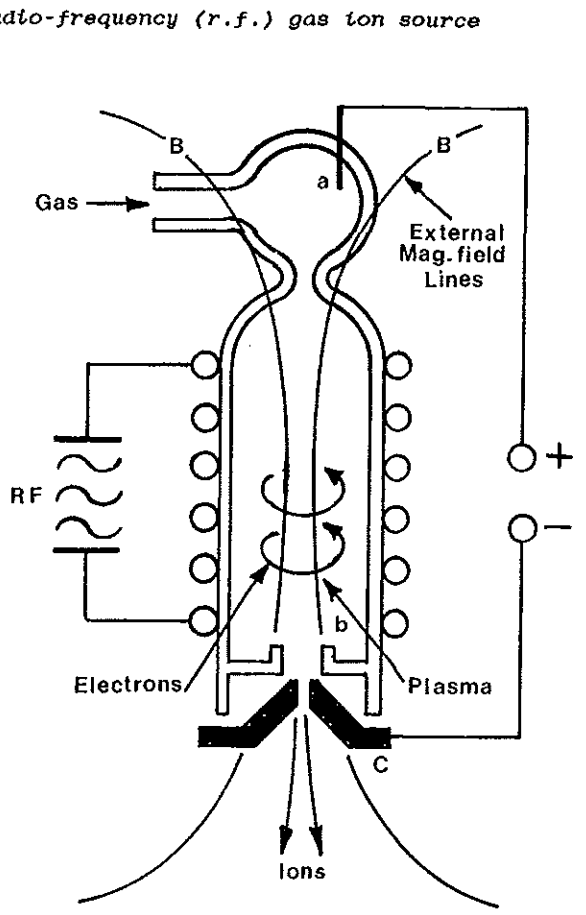
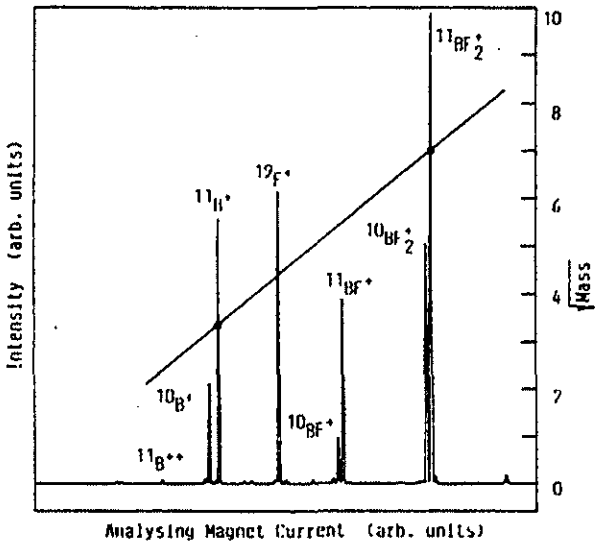
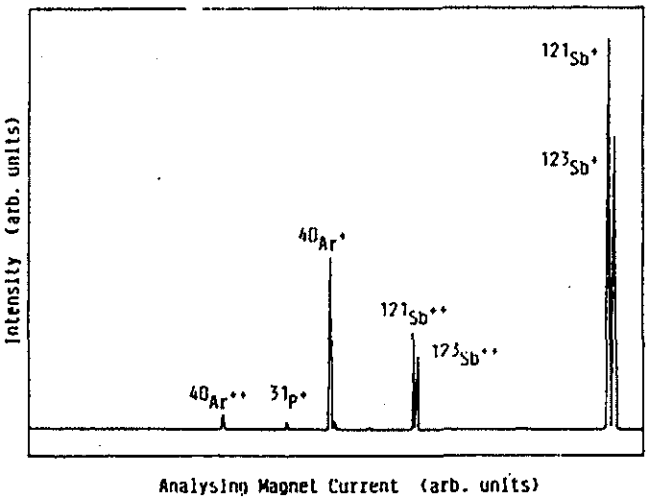


Fig. 137. This source is well established as one of the best sources for producing light ions from gases and is widely used, for example, in Van de Graaff accelerators and for low intensity requirements in implanters. It is mostly useful for natural gases since other materials might coat the quartz envelope and degrade its performance. [Courtesy K.G. Stephens in "Ion Implantation" Science and Technology, J.F. Ziegler, ed. (Academic Press, New York 1984), p. 414.]

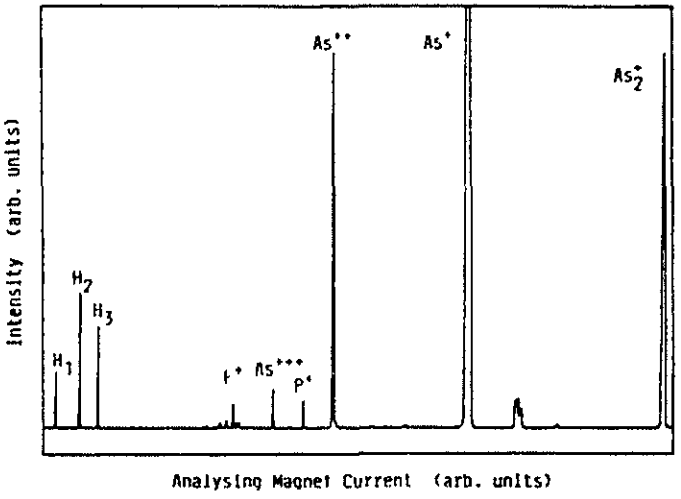
Typical mass spectra produced by BF_3 (Fig 138a), AsH_3 (Fig. 138b), a solid Sb source sputter by Ar gas (Fig. 138c) and SiF_4 (Fig. 138d) illustrate the rich variety of molecular species and charge states. It may appear fortuitous that the BF_2^+ species is the most abundant for BF_3 feed gas and is also desirable because of the relatively low temperature required for full B activation in Si!



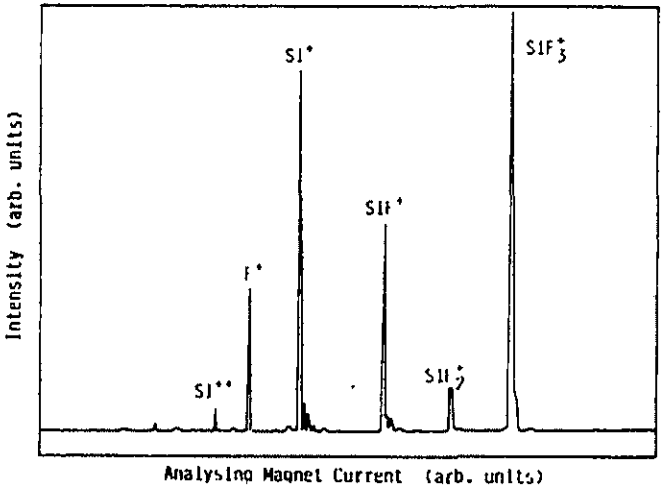
Mass spectrum of a medium-current implanter, feed gas BF_3.



Mass spectrum of a high current implanter, with solid Sb and argon as carrier gas.



Mass spectrum of a high-current implanter, feed gas AsH_3.



Mass spectrum of a high-current implanter, feed gas SiF_4.

Fig. 138.

[Courtesy H. Glawischnig and K. Noack in "Ion Implantation: Science and Technology," J.F. Ziegler, ed. Acadmeic Press, New York (1984), pp. 223-4.]

Ion beam apertures are critical components of an implanter, especially those downstream where the ions have their full energy. Sputtering of the aperture material can lead to serious contamination. A drastic case is shown in Fig. 139 where 60 keV Ge ions have been implanted into Ge. Sputtering of aperture material led to a massive Fe and Cr contamination of the surface. Obviously stainless steel is a very poor choice for apertures! Carbon or Si apertures suppress contamination.

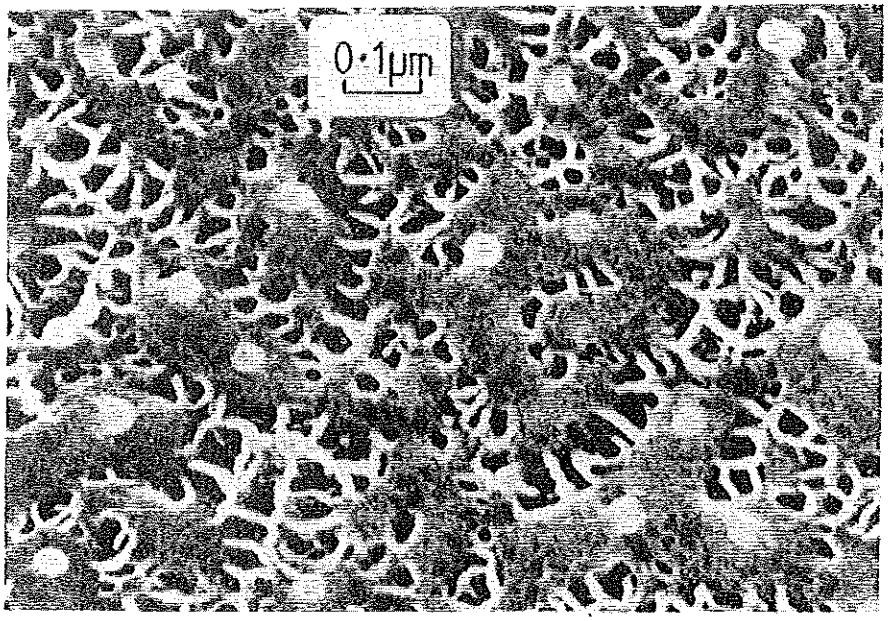


Fig. 139. Voids formed by 60 keV Ge^+ — Ge plus Fe/Cr from the beam defining aperture. [Courtesy I.H. Wilson and K.M. Barfoot in "Ion Implantation" Science and Technology, J.F. Ziegler, ed. (Academic Press, New York 1984), p. 569.]

A last point to be discussed concerns the implant homogeneity. We are all familiar with Lissajou figures formed on an oscilloscope screen when the X and Y deflection are run by sine wave voltages. The Lissajou figures become stationary when the ratio of the two sine wave frequencies is simple (e.g., 1:2). When this happens with the X and Y deflection generators in an implanter, we call it a scan lock-up. The result is a standing ion pattern which is shown in Fig. 140 as a grid of fluctuating ion dose. It is essential that the X- and Y-frequencies never form a simple ratio. Typically the two generators are quartz crystal controlled. A well-tuned implanter can provide at best an ion dose homogeneity of ± 0.5%. More typical are patterns shown in Fig. 141.

As a closing remark one should never forget that serious safety issues must be kept in mind when operating an ion implanter. Not only can toxic gases pose serious danger but also electrons which travel in the opposite direction of the ions. X-ray Bremstrahlung radiation of 200 keV electrons is highly penetrating and can harm the operator. This is the reason why all ion implanters must have an X-ray alarm system.

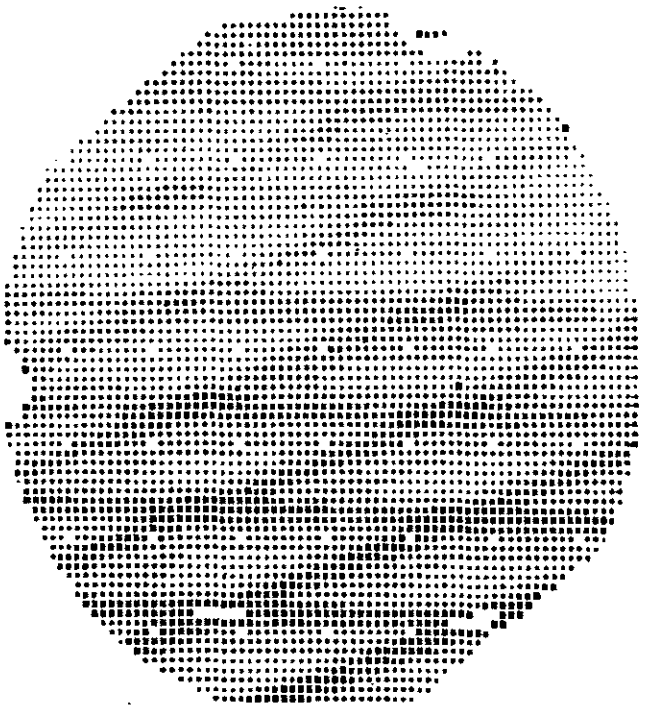


Fig. 140. Doping variations from C-V measurements at 4300 test sites on a 3-inch wafer. The standing wave pattern resulted from a scan lock-up condition with an electrostatic-scan system. The doping variation average was 13% over the entire wafer. [from M.I. Current and M.J. Markert, "Mapping of Ion Implanted Wafers," in "Ion Implantation" Science and Technology, J.F. Ziegler, ed. (Academic Press, New York 1984), p. 505.]

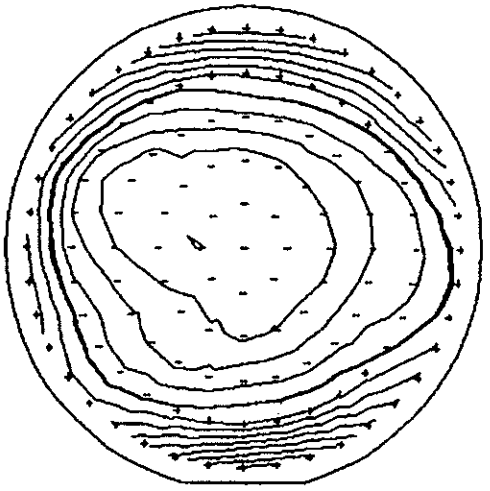


Fig. 141. Sheet resistance contour map of As implanted Si(100) annealed at 1200 °C for 10 seconds in a rapid thermal annealer. The prototype system used in this anneal produced strong thermal gradients along the radial direction. The lower sheet resistance values in the center of the wafer correspond to higher dopant activation levels and some additional diffusion in the hotter portion of the wafer. [from M.I. Current and M.J. Markert, "Mapping of Ion Implanted Wafers," in "Ion Implantation" Science and Technology, J.F. Ziegler, ed. (Academic Press, New York 1984), p. 530.]

5.6.5. Recent Developments

5.6.5.1. Ion Implantation in Compound Semiconductors

The preceding sections dealt mostly with doping of elemental semiconductors by ion implantation. Compound semiconductors offer additional challenges. Here are a few examples specific to compounds. How can one influence the implanted dopant atoms to assume a position on the appropriate sublattice? How is the local stoichiometry affected by implantation and the resulting damage, especially for compounds with very heavy and very light components (e.g., InP, InN, etc.)? Does thermal annealing restore stoichiometry? How is decomposition during thermal annealing suppressed? Does solid phase epitaxy work as well as in Si or Ge?

We will not address all of these points but will briefly review a topic which is interesting from a scientific standpoint and at the same time is very useful for the fabrication of devices. The topic is co-implantation. Co-implantation is the sequential implantation of two different types of ions into the same near-surface region. Specifically we will deal with co-implantation of carbon with other elements in GaAs.

It has been observed that implanted carbon cannot be effectively activated by thermal annealing. Only around one percent of the implanted C atoms become shallow acceptors. One immediately may think that C, a group IV element, is amphoteric and will occupy both sublattices, forming donors and acceptors which compensate each other effectively. This does not turn out to be the case as determined by local vibrational mode spectroscopy and mobility measurements.

Co-implantation with other elements strongly affects the level of activation. A. Moll et al. (*Appl. Phys. Lett.* **60**, 2383-5 (1992)) have conducted systematic studies of this co-implantation technique. Care is taken that the two ion species have identical depth distribution profiles in the host semiconductor. This requires that the two ions be implanted at different energies because they have different masses. The histogram in Fig. 142 summarizes the activation efficiency for several co-implanted elements after rapid thermal annealing at 800 °C and 950°C. Co-implantation with specific elements clearly improves the activation.

The large differences in co-implantation effectiveness can be interpreted based on two observations: i) the heavier the co-implanted ion species, the higher the carbon activation and ii) group III co-implants are much more effective than group V co-implants. The first trend suggests that heavier implant damage leads to higher activation. Carbon being only the sixth element with atomic mass 12 does relatively little damage when compared to gallium with an average atomic mass 70 implanted to the same depth as carbon! The second trend can be explained with stoichiometry arguments. Implantation of group III elements leads to a deviation from stoichiometry with either group III elements assuming group V sites (antisites) or with excess group V vacancies. Carbon in such an environment will preferably assume sites which restore stoichiometry. These are the group V sublattice sites on which carbon forms shallow acceptors. Figure 143 displays the sheet carrier concentration of the carbon co-implanted and annealed layers. Implantation of N or P indeed reduces the carbon activation below carbon-only implants. The stoichiometry effect leads to a difference in sheet carrier concentration of over one full order of magnitude.

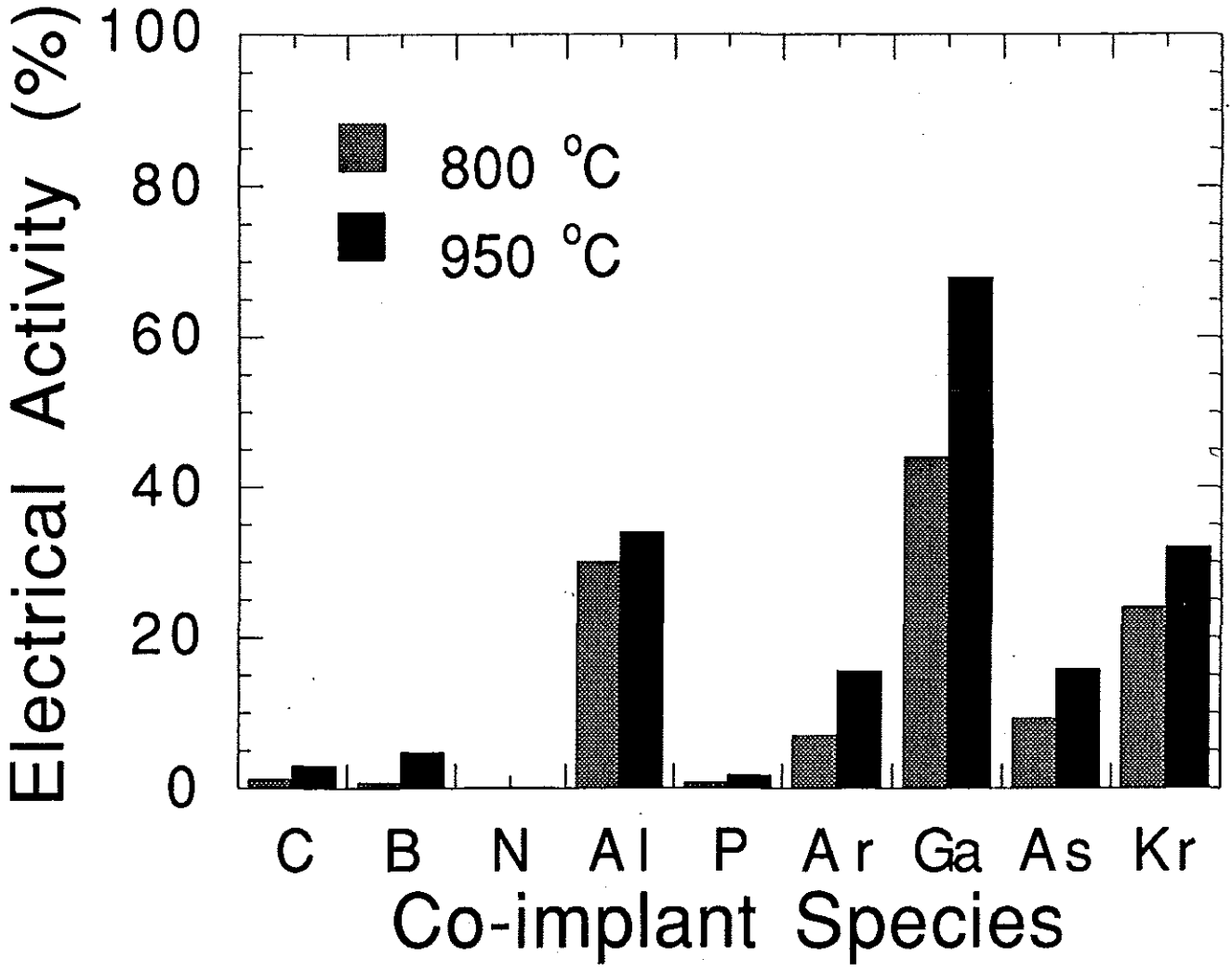


Fig. 142. Electrical activity of implanted carbon as a function of co-implant species for samples annealed at 800 °C and 950°C. Electrical activity is the ratio of sheet carrier concentration to implant dose.

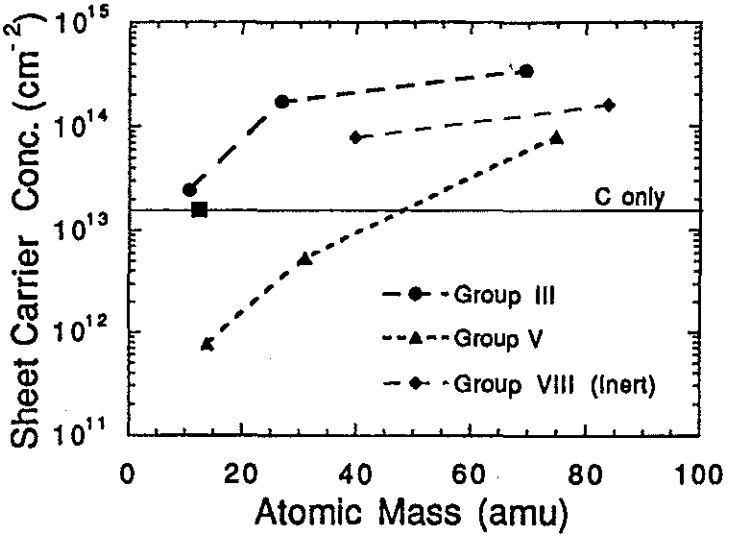


Fig. 143. Sheet carrier concentration as a function of atomic mass for samples annealed at 950°C for 10 seconds.

Co-implantation has been used to improve activation of other impurities in GaAs and other semiconductors. The large effects observed for C in GaAs are not reproduced for all of these cases. This specific combination appears to be especially amenable to co-implantation effects.

We are in the process of studying solid phase epitaxy of GaAs amorphized by either Ga or As or co-implantation of the two. Again, stoichiometry and damage are expected to play important roles. Depending on the outcome of these studies, we may investigate dopant activation through SPE in compound semiconductors.

5.6.5.2. Plasma Immersion Ion Implantation (PIII)

(see for example: P.K. Chu et al. in *Semiconductor International,* June 1996, pp. 165-172 and articles in MRS Bulletin Vol. 21, No. 8, August 1996).

The dimensions of individual transistors in an integrated circuit are continuously shrinking. Submicron and deep submicron devices require ever shallower p-n junctions and lower implant energy. At the same time Si wafers are increasing in size. Today's standard is 200

mm diameter but 300 mm diameter CZ Si crystals are being developed by all the leading manufacturers. These two trends have led to the exploration of a new ion implantation technique in which a Si wafer (or many wafers) are immersed in a plasma containing dopant ions. The wafers are periodically pulsed to negative voltages (up to 50 kV). During these pulses the positive ions are accelerated towards the wafers and are implanted. The modest voltages lead to the shallow implant depths required for modern circuits. Depending on the gas pressure the mean free path of the ions may be rather short. This in turn allows implantation from all angles into deep grooves or holes or other geometric features. No scanning of an ion beam is required and the implant ion flux can be of the order of mA/cm^2. This makes the implant time independent of the wafer size, in sharp contrast to standard ion implantation for which the total ion beam current cannot be easily increased beyond a few tens of mA. In this case the implant time increases with the square of the wafer diameter.

These advantages of PIII must be contrasted with several shortcomings. Since there is no ion mass selection, contamination with a wide variety of elements could occur. Particularly detrimental contaminants are iron, a deep level impurity in Si and Na, a highly detrimental ion in the gate oxide. Lack of ion energy control leads to implant profiles which can no longer be tailored precisely, one of the basic advantages of ion implantation. It is too early to conclude how widespread the use of PIII will become. It certainly looks like a promising technique for very high dose implants which are required for the formation of buried oxides or for non-semiconductor applications such as surface modifications of metal parts used in internal combustion engines (bearings, pistons, piston rings, etc.).

6. Metal-Semiconductor and Metal-Oxide-Semiconductor (MOS) Junctions

References:

- "Physics of Semiconductor Devices, S.M. Sze (John Wiley & Sons, New York 1981), second edition, chapter 5.
- "Metal-Semiconductor Schottky Barrier Junctions and their Applications," B.L. Shane, ed. (Plenum Press, New York 1984).
- "Semiconductor Contacts," H.K. Henisch (Clarendon Press, Oxford 1984).
- "Semiconductor Material and Device Characterization," D.K. Schroder, (John Wiley & Sons, New York 1990), chapter 3.
- "Electronic Thin Film Science," K.N. Tu, J.W. Mayer and L.C. Feldman, (MacMillan Publishing Co., New York 1992), chapter 9.
- "Electrons in Solids: An Introductory Survey," R.H. Bube (Academic Press, New York 1981).
- "Semiconductor Surfaces and Interfaces," W. Mönch (Springer, Berlin 2001), 3rd edition, chapters 6 and 19.1.

6.1. Introduction

A semiconductor device requires a number of electrical contacts which may provide electrical power for operation and/or transfer signals to and from the device. These contacts are made with a variety of metals. A great amount of effort has been spent on the understanding of the physics of metal-semiconductor and metal-oxide semiconductor junctions and on the fabrication of reliable, stable contacts which have close to ideal properties.

Based on the specific function, we distinguish between the kinds of metal-semiconductor junctions. Metal-semiconductor junctions may be formed to obtain an excellent rectifier which passes current with little hindrance for one polarity but blocks current flow efficiently for the opposite polarity. High power and very high frequency rectifiers use this metal-semiconductor junction configuration. When Ohmic contacts are required, the rectifying properties must be avoided or circumvented. This is achieved in one of two ways: either a metal is found which forms a junction with the particular semiconductor offering little resistance to current flow in both directions or the near surface region of the semiconductor is very heavily doped, leading to

efficient transfer of charges via tunneling processes. In the metal-oxide-semiconductor structure, the metal electrode is used to impart an electric field controlling lateral current flow in the near-surface region of the semiconductor. A major issue for this MOS structure is the reduction of electrically active states at the oxide-semiconductor interface.

This chapter summarizes the major aspects of such junctions. For an in-depth study of the topic, the reader is referred to the references given above and to the original literature.

6.2. The Ideal Metal-Semiconductor Junction

In the early 1870s, Ferdinand Braun, high school teacher in Leipzig, connected metal wires to natural semiconducting minerals such as galena (PbS).[6.1] To his surprise, he found the current flow to depend on the polarity of the applied bias. Ohm's law did not hold for these metal-mineral contacts! Thus was born the solid state rectifier. Wireless telegraphy and radio engineers soon used "crystal detectors" for the demodulation of amplitude modulated radiofrequency signals though the underlying physical principals were not understood. It took roughly 50 years until Walter Schottky developed a theory explaining the effect.[6.2]

The ideal metal-semiconductor junction, commonly referred to as *Schottky barrier*, can be understood using simple band theory. One may start the discussion by asking what the difference is in the minimum energy required to remove an electron from the metal and from the semiconductor. In the metal, we remove the electron from the Fermi level to the vacuum level (=> free electron). The energy required for this process divided by the electron charge is the metal work function Φ_M. In the semiconductor, one defines two physical entities: in analogy to the metal we define i.) the semiconductor work function Φ_S which is the energy between the Fermi level and the vacuum level divided by the electron charge and ii.) the electron affinity χ_S which is the energy difference between the conduction band minimum and the vacuum level. When a metal and a semiconductor are brought into contact, their Fermi levels will equilibrate. The consequence for $\Phi_M > \Phi_S$ (typical!) and an n-type semiconductor is the transfer of electrons

[6.1] F. Braun, Annal. Phys. Chem. **153**, 556-63 (1874); translation in S. Sze, "Semicond. Pioneering Papers," World Scientific, 1991. As an aside we also credit Braun for the invention of the TV picture tube, in German "Braunsche Röhre."
[6.2] W. Schottky, Naturwissenschaften **26**, 843 (1938); Z. Physik **118**, 539-92 (1942).

from the semiconductor to the metal. The semiconductor is no longer neutral because a depletion layer forms in which the positive donor ions constitute a positive space charge. Using Poisson's equation, one can calculate the electric field and the potential created by the space charge.

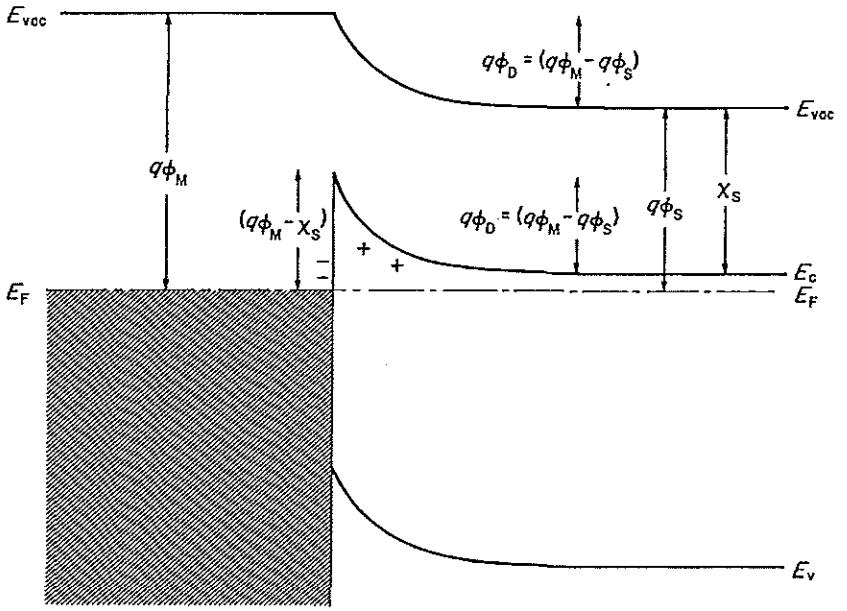


Fig. 144. (a) A Schottky barrier between a metal and an n-type semiconductor, formed when the work function of the metal is larger than that of the semiconductor. Note: q ≡ e = charge of the electron. [After R.H. Bube, "Electrons in Solids: An Introductory Survey," Academic Press, New York, 1981, p. 164.]

For constant donor doping, the semiconductor bands will bend parabolically upwards. Figure 144a illustrates this situation. An energy barrier $e\Phi_M$-χ_S exists for electrons attempting to travel from the metal to the semiconductor. Note that this barrier, the Schottky barrier, cannot be changed by the application of a bias across the junction! The barrier against electron travel from the semiconductor to the metal is $e(\Phi_M-\Phi_S)$ and it can be changed with an applied bias. The Schottky barrier "blocks" current flow when the metal is biased negatively relative to the semiconductor (reverse bias) but electric current flows toward the metal for a positive bias applied to the metal (forward bias). One finds for the current density:

$$J = J_o\left[\exp\left(\frac{e\Phi_{applied}}{k_B T}\right) - 1\right]$$

$\Phi_{applied}$ = externally applied bias.

The constant J_o depends on the particular theory which applies to the case in question. Schottky developed the diffusion theory which is based on the assumption that the Schottky barriers are much larger than k_BT. For moderately doped, high mobility semiconductors (Ge, Si, GaAs) and electric fields near the junction of 10^4 to 10^5 Vcm^{-1}, thermionic emission over the barrier more appropriately describes the constant J_o (see the discussion in S.M. Sze's book). The current-voltage characteristics and the band diagrams for forward and reverse bias are shown in Figs. 144b-d.

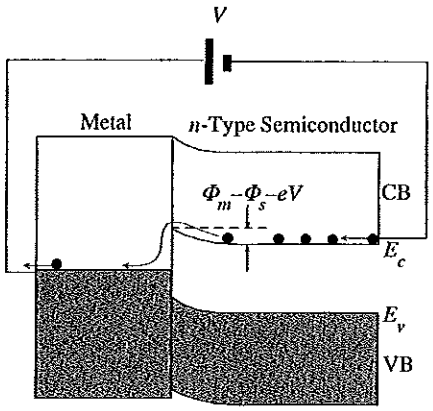
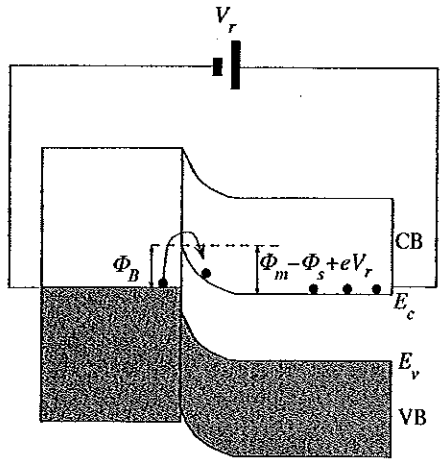
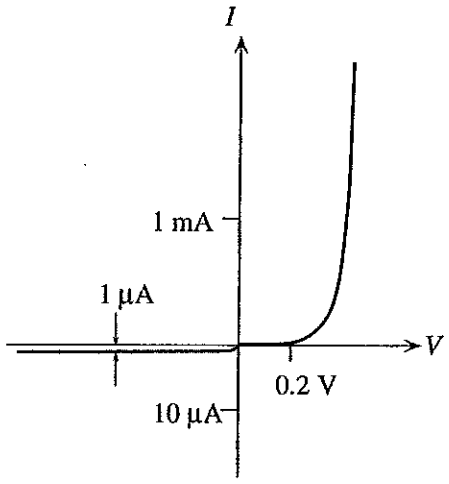


Fig. 144. (b) Forward biased Schottky junction. Electrons in the CB of the semiconductor can readily overcome the small PE (Potential Energy) barrier to enter the metal.



(c) Reverse-biased Schottky junction. Electrons in the metal cannot easily overcome the PE barrier Φ_B to enter the semiconductor.



(d) I-V characteristics of a Schottky junction exhibit rectifying properties (negative current axis is in microamps). [After S.O. Kasap, "Principles of Electrical Engineering Materials and Devices," McGraw-Hill, New York, 1997, p. 370.]

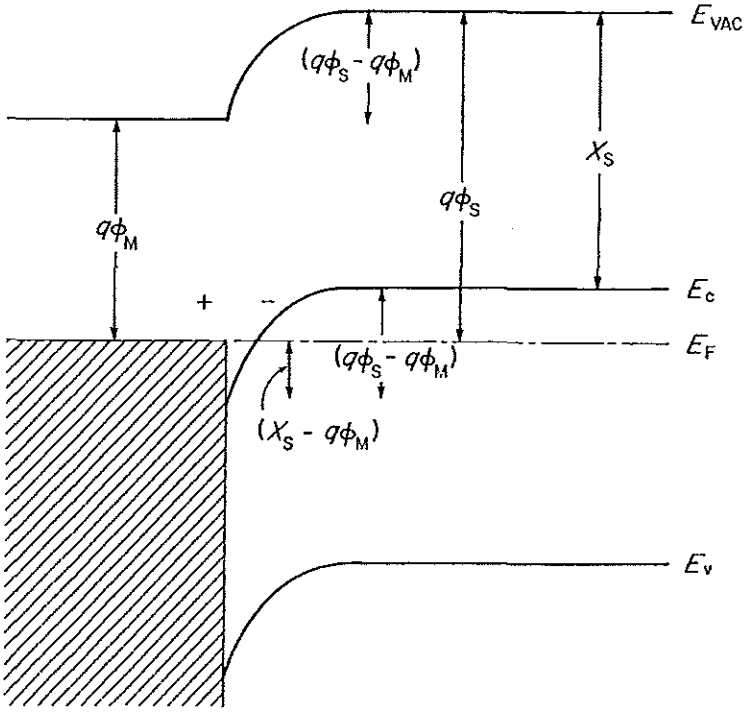
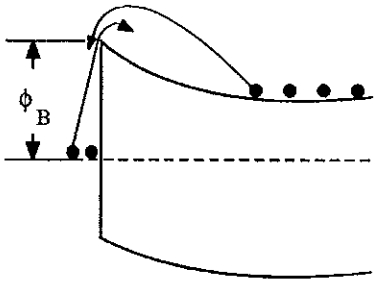
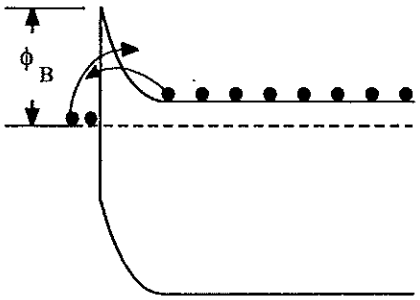


Fig. 145. An ohmic contact between a metal and an n-type semiconductor, formed when the work function of the metal is smaller than that of the semiconductor. Note: q ≡ e = charge of the electron. [After R.H. Bube, "Electrons in Solids: An Introductory Survey," Academic Press, New York, 1981, p. 166.]

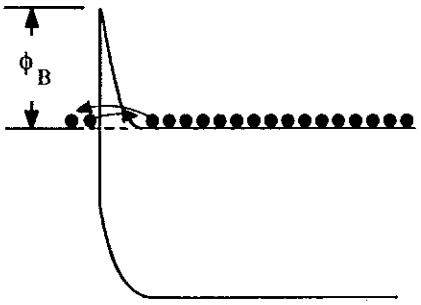
Do all metal-semiconductor junctions act as rectifiers? Not if $\Phi_M < \Phi_S$ for an n-type semiconductor can be realized. In this case (Fig. 145), a reservoir of free electrons accumulates in the semiconductor. This, in turn, permits the unimpeded exchange across the junction for either bias condition. The contact is Ohmic. But what can be done if we are not in a position to realize the condition for an Ohmic contact but still need one? This condition is encountered very often. In this case, we resort to quantum mechanical tunneling between the metal and the semiconductor. Making the depletion layer thinner than few tens of Angstroms by very high doping permits electrons to tunnel both ways unhindered through the Schottky barrier (Fig. 146).



(a) Low N_D - Thermionic Emission



(b) Intermediate N_D - Thermionic/Field Emission



(c) High N_D - Field Emission

Fig. 146. Depletion-type contacts to n-type substrates with increasing doping concentrations. The electron flow is schematically indicated by the electrons and their arrows. [After D.K. Schroder, "Semiconductor Material and Device Characterization," John Wiley & Sons, New York, 1990, p. 103.]

The reader is encouraged to evaluate the situation for a p-type semiconductor and both cases: $\Phi_M > \Phi_S$ and $\Phi_M < \Phi_S$.

How realistic is the model of the Schottky barrier shown in Figs. 144a and 145? For a moment, recall the conditions we considered when developing very simple band theory. We assumed to have an infinitely extended semiconductor with lateral translational symmetry. Surfaces with broken bonds, oxidized surfaces or reconstructed surfaces were never part of the picture. But the simple Schottky barrier model assumes that both the metal and the semiconductor energy bands can be described by bulk properties right to the very junction between the two. While this simple picture applies rather well to wide bandgap and ionic semiconductors, it does not describe the situation accurately for metals on Si, Ge or GaAs. The next section describes real metal-semiconductor junctions.

6.3. Real Metal-Semiconductor Junctions (including Silicides)

A large number of early Schottky barrier investigations showed that the measured barrier was not $\Phi_M - \Phi_S$. Indeed, for silicon, the metal work function did not seem to have any effect on the barrier height at all. Barrier heights of various metals on Si, GaSe and SiO_2 are plotted in dependence of the metal work function in Fig. 147a. Covalent silicon barely shows any dependence, with a slope of $S = d\Phi_{B_n}/d\Phi_M \cong 0.05$. On the other hand, the more ionic semiconductor GaSe yields S = 0.6 and for SiO_2 the metal work function is fully effective ($S \cong 1.0$). Figure 147b shows a compilation of the slopes S for numerous semiconductors and wide bandgap oxides as a function of the electronegativity difference of the elements forming the semiconductor (Pauling electronegativity). The strongly ionic materials (large electronegativity difference) experience the full metal work function in the barrier height ($S \cong 1$).

Various modifications of the simple Schottky barrier model have been proposed to explain this behavior. Most of these extensions of the simple model are based on electronic bandgap states residing at the metal-semiconductor interface. These states are either derived from surface states[6.3] or induced in the semiconductor by the nearby metal [metal induced gap states (MIGS)[6.4]]. A reduction of the bandgap and metallurgical reactions between the metal and silicon (silicide formation) have been considered. We will first discuss the surface/interface state picture and then review silicide formation.

[6.3] J. Bardeen, Phys. Rev. **71**, 717 (1947).
[6.4] S.G. Louie, J.R. Chelikowsky and M.L. Cohen, Phys. Rev. B **15**, 2104 (1977).

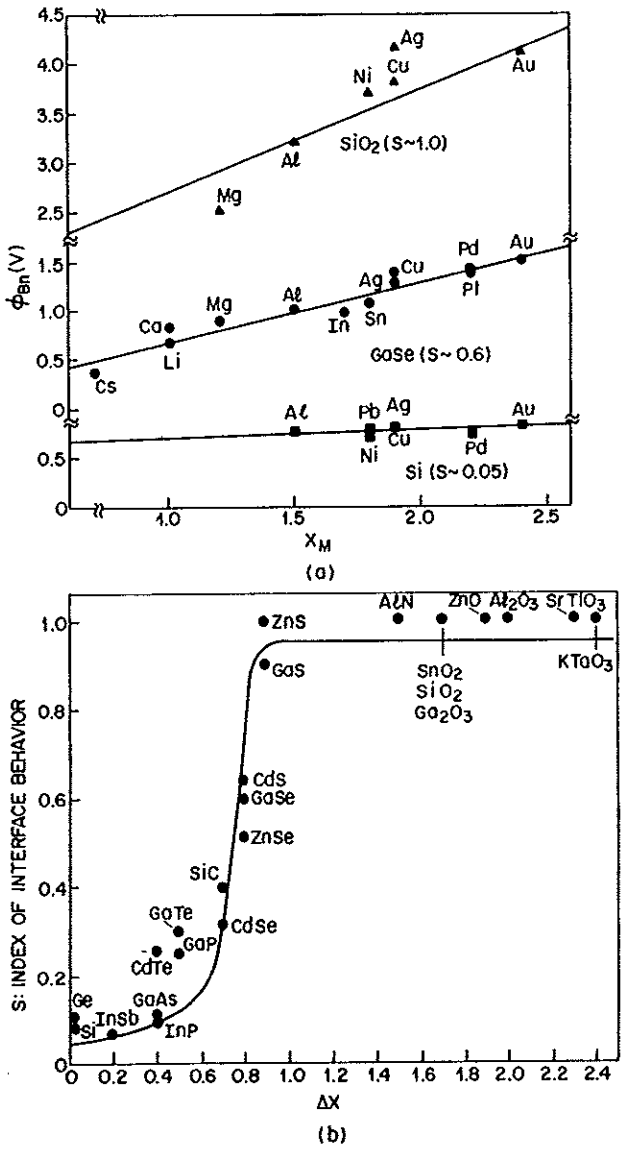


Fig. 147. (a) Barrier height versus electronegativity of metals deposited on Si, GaSe, and SiO_2. (b) Index of interface behavior S as a function of the electronegativity difference of the semiconductors. [After S.M. Sze, "Physics of Semiconductor Devices," John Wiley & Sons, New York, 1981, p. 278; see S. Kurtin, T.C. McGill and C.A. Mead, Phys. Rev. Lett. **22**, 1433 (1969).]

Terminating a silicon surface abruptly leads to a variety of possible structures which depend in part on the ambient. In an ultra-high vacuum, a cleaved silicon surface will reconstruct in regular atomic patterns, depending on the crystalline orientation of the surface. This new atomic configuration cannot be expected to extend the bulk bandstructure out to the very surface. The reconstruction and its imperfections form energy states which lie within the bandgap. In an

imperfect vacuum, a very thin oxide film forms readily, resulting in surface states with energy levels within the bandgap. At a sufficiently high concentration, these surface states "pin" the Fermi level at the very surface at an energy position independent of the underlying bulk material. In the case of a "real" silicon surface, the states density peaks roughly at E_{VB} + 1/3 E_{GAP}, pinning the Fermi level. The Fermi levels in the silicon bulk and at the surface have to equilibrate which in turn leads to bending of the bands. For n-type silicon, the surface energies are pushed up by close to 2/3 E_{GAP} with the surface Fermi level pinned at E_{VB} + 1/3 E_{GAP} (Fig. 148). For a well-pinned surface, the Fermi level no longer moves when a metal is deposited. This is the basic explanation for the insensitivity of the barrier height of Schottky contacts formed on real surfaces. Not only do "real" silicon surfaces become pinned, but a similar phenomenon is also observed for many other semiconductors.

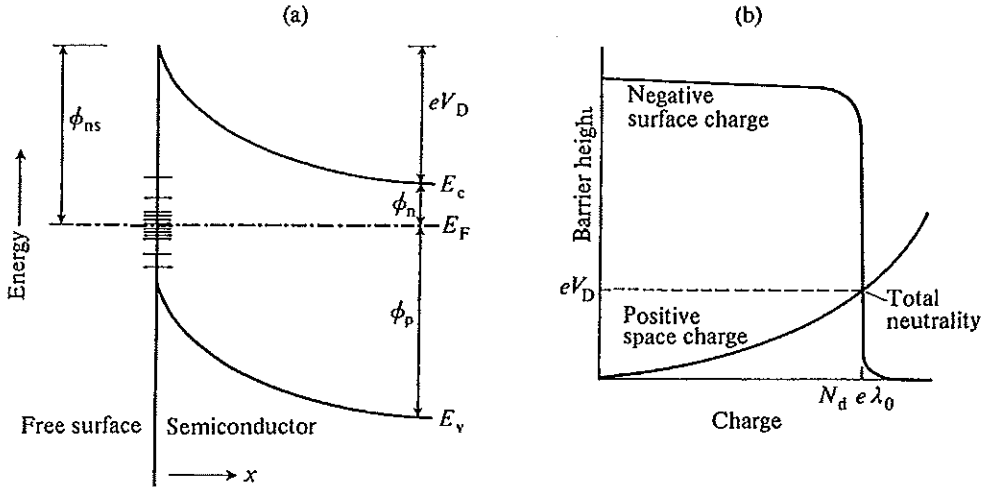


Fig. 148. Barrier on a free surface due to bunched surface states; zero current. (a) Band structure. (b) Conditions governing total neutrality and hence equilibrium. Barrier height, as affecting the electrons in the semiconductor: eV_D. [After H.K. Henisch, "Semiconductor Contacts," Clarendon Press, Oxford, 1984, p. 31.]

Figure 149 displays results of Fermi level pinning studies for GaAs, GaSb and InP.[6.5] While in GaAs pinning occurs between 1/3 to 1/2 of the bandgap energy above E_{VB}, it is close to the valence band edge in GaSb and near the conduction band edge in InP. This characteristic behavior of specific semiconductors is caused most likely by native donor and acceptor defect

[6.5] W.E. Spicer *et al.*, Phys. Rev. Lett. **44**, 420 (1980).

centers (no impurities are required!) which compensate each other electrically, minimizing the energy of the system. Heterojunction bandoffsets, doping concentration limits, heterojunction intermixing and further phenomena have been shown to depend on the same native defects.[6.6]

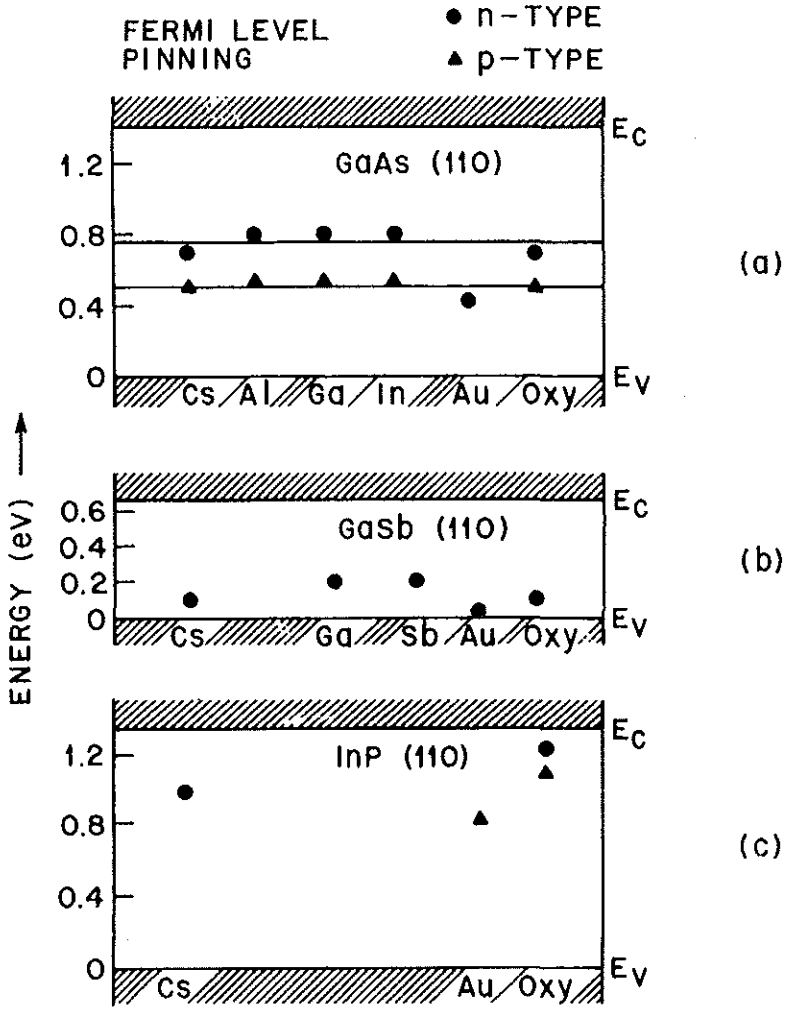


Fig. 149. Surface Fermi-level position for a number of metals and oxygen on GaAs, GaSb, and InP. Note that there is little dependence on the chemical nature of the metals and oxygen. [After S.M. Sze, "Physics of Semiconductor Devices," John Wiley & Sons, New York, 1981, p. 276; see W.E. Spicer, I. Lindau, P. Skeath, C.Y. Su and P. Chye, Phys. Rev. Lett. **44**, 420 (1980).]

While there is general agreement that some kind of surface defect or a family of defects (native and/or with impurities) dominate the electrical behavior and the Fermi level at the semiconductor surface, the identification of these defects (structure, composition, charge states)

[6.6] W. Walukiewicz, "Amphoteric Native Defects in Semiconductors," Appl. Phys. Lett. **54**, 2094 (1989).

has been exceedingly difficult and remains an important subfield of modern semiconductor research.

From a practical point of view, Schottky barrier contacts formed by metal evaporation are more an art than a science. Especially highly blocking barriers may be difficult to form and recipes for success abound. This author recalls the long days and nights spent on forming pure Au barriers on large area planar ultra-pure Ge p-i-n diodes for gamma ray detection. When things worked well, diodes with tens of cm^3 depletion volume would exhibit reverse leakage currents in the pA range at 77K. At other times, weeks would pass without one working device!

Semi-conductor	Type	E_g (eV)	Ag	Al	Au	Cr	Cu	Hf	In	Mg	Mo	Ni	Pb	Pd	Pt	Ta	Ti	W
Diamond	*p*	5.47			1.71													
Ge	*n*	0.66	0.54	0.48	0.59		0.52		0.64			0.49	0.38					0.48
Ge	*p*		0.50		0.30				0.55									
Si	*n*	1.12	0.78	0.72	0.80	0.61	0.58	0.58		0.40	0.68	0.61		0.81	0.90		0.50	0.67
Si	*p*		0.54	0.58	0.34	0.50	0.46				0.42	0.51	0.55				0.61	0.45
SiC	*n*	3.00		2.00	1.95													
AlAs	*n*	2.16			1.20										1.00			
AlSb	*p*	1.63			0.55													
BN	*p*	7.50			3.10													
BP	*p*	6.00			0.87													
GaSb	*n*	0.67			0.60													
GaAs	*n*	1.42	0.88	0.80	0.90		0.82	0.72							0.84	0.85		0.80
GaAs	*p*		0.63		0.42			0.68										
GaP	*n*	2.24	1.20	1.07	1.30	1.06	1.20	1.84		1.04	1.13	1.27			1.45		1.12	
GaP	*p*				0.72													
InSb	*n*	0.16	0.18[a]		0.17[a]													
InAs	*p*	0.33			0.47[a]													
InP	*n*	1.29	0.54		0.52													
InP	*p*				0.76													
CdS	*n*	2.43	0.56	Ohmic	0.78		0.50					0.45	0.59	0.62	1.10		0.84	
CdSe	*n*	1.70	0.43		0.49		0.33								0.37			
CdTe	*n*		0.81	0.76	0.71										0.76			
ZnO	*n*	3.20		0.68	0.65		0.45		0.30					0.68	0.75	0.30		
ZnS	*n*	3.60	1.65	0.80	2.00		1.75		1.50	0.82				1.87	1.84	1.10		
ZnSe	*n*		1.21	0.76	1.36		1.10		0.91				1.16		1.40			
PbO	*n*		0.95						0.93			0.96	0.95					

[a] 77 K.

Table 18. Measured Schottky Barrier Heights (Volts at 300K). [After S.M. Sze, "Physics of Semiconductor Devices," John Wiley & Sons, New York, 1981, p. 291.]

An extensive summary of measured Schottky barrier heights for many semiconductor/metal combinations is presented in Table 18. Note that in many cases for which the barrier of a certain metal is known for the n- as well as p-type form of a semiconductor, the two barrier heights add up, as expected, to approximately the bandgap energy. Also, the metals Pt, Pd and Au consistently form the highest barriers on n-type semiconductors.

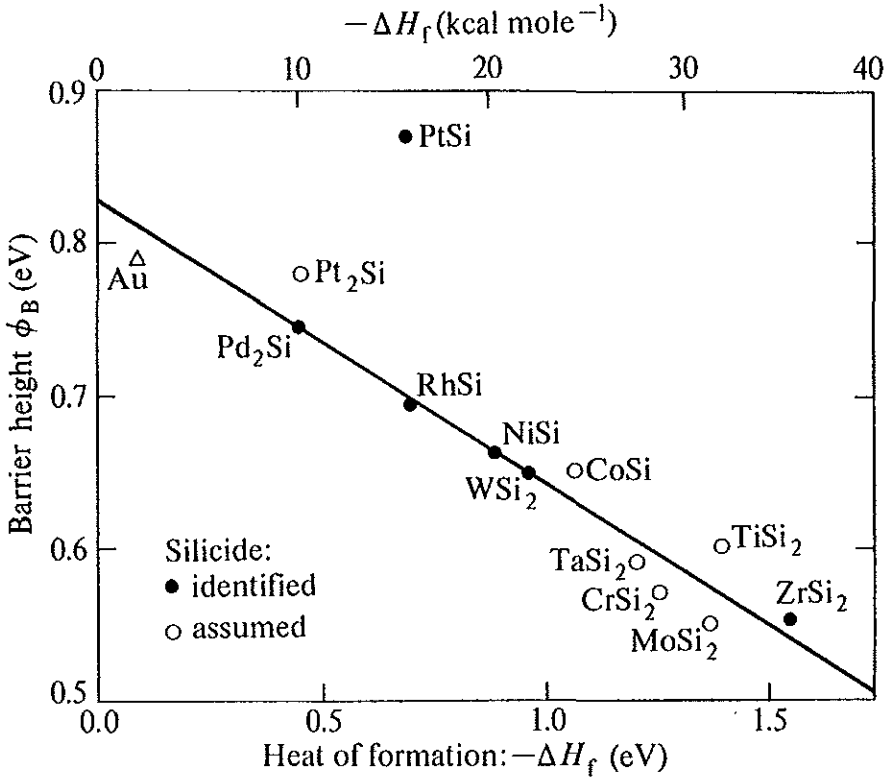


Fig. 150. Relationship between the barrier height at silicide-silicon interfaces and the heat of the formation of the silicide involved. [After H.K. Henisch, "Semiconductor Contacts," Clarendon Press, Oxford, 1984, p. 31; see J.M. Andrews and J.C. Phillips, Phys. Rev. Lett. **35**, 56 (1975).]

Because of reproducibility and reliability problems, Schottky barriers in the past have found limited applications in electronic devices. This limitation has been largely overcome for silicon by the introduction of silicides, compounds or eutectoids formed between a large number of metals (foremost transition metals) and silicon. To form a silicide, a thin metal film is deposited on clean, oxygen-free silicon. A diffusion driven solid phase reaction can be initiated at the silicon-metal interface at very moderate temperatures. In general, the formation of the silicide proceeds into the silicon and after a very short time, the original interface is completely consumed. The new silicide-silicon interface appears to be free from the surface state pinning effects described earlier. The ultimate perfection can be achieved with cobalt silicide which grows epitaxially on the silicon substrate. An interesting linear relationship between the silicide-silicon barrier height and the heat of formation of the silicide Δ H_f (eV) was first reported by Andrews and Phillips[6.7] (Fig. 150). This finding showed that chemical bond formation at the

[6.7] J.M. Andrews and J.C. Phillips, Phys. Rev. Lett. **35**, 56 (1975).

interface does play a decisive role in the resulting Schottky barrier height. An improved picture was offered by Ottaviani *et al.*[6.8] who showed that the silicide barrier height depends linearly on the eutectic temperature over a broad range of silicides (Fig. 151). Tu *et al.*[6.9] extended this plot to include rare earth silicide which form the lowest barriers on n-Si (Fig. 152). Table 19 summarizes the barrier heights of a number of silicides on silicon together with the minimum silicide formation and the melting temperatures.

Metal Silicide	ϕ_B (V)	Structure	Forming Temperature (°C)	Melting Temperature (°C)
CoSi	0.68	Cubic	400	1460
$CoSi_2$	0.64	Cubic	450	1326
$CrSi_2$	0.57	Hexagonal	450	1475
HfSi	0.53	Orthorhombic	550	2200
IrSi	0.93	—	300	—
MnSi	0.76	Cubic	400	1275
$Mn_{11}Si_{19}$	0.72	Tetragonal	800[a]	1145
$MoSi_2$	0.55	Tetragonal	1000[a]	1980
Ni_2Si	0.7–0.75	Orthorhombic	200	1318
NiSi	0.66–0.75	Orthorhombic	400	992
$NiSi_2$	0.7	Cubic	800[a]	993
Pd_2Si	0.72–0.75	Hexagonal	200	1330
PtSi	0.84	Orthorhombic	300	1229
RhSi	0.69	Cubic	300	—
$TaSi_2$	0.59	Hexagonal	750[a]	2200
$TiSi_2$	0.60	Orthorhombic	650	1540
WSi_2	0.65	Tetragonal	650	2150
$ZrSi_2$	0.55	Orthorhombic	600	1520

[a]Can be $\leq$ 700°C under clean interface condition.

Table 19. Barrier Height of Metal Silicide on n-Type Si. [After S.M. Sze, "Physics of Semiconductor Devices," John Wiley & Sons, New York, 1981, p. 292.]

The silicide-silicon technology has rapidly matured in the past few years and has led to applications in various kinds of electronic devices. Silicides are preferred over more simple metal-silicon contacts because they are highly stable, in many cases to over 800 °C, especially the disilicides (MSi_2). The interfaces exhibit excellent planarity and reproducible electrical barrier properties. The fundamental understanding and theory of the barrier formation have, however, not progressed to the point where first principles calculations can predict barrier heights or formation temperatures based on fundamental properties of silicon and the metal in question can be made.

[6.8] G. Ottaviani, K.N. Tu and J.W. Mayer, Phys. Rev. Lett. **44**, 284 (1980).
[6.9] K.N. Tu, R.D. Thompson and B.Y. Tsaur, Appl. Phys. Lett. **38**, 626 (1981).

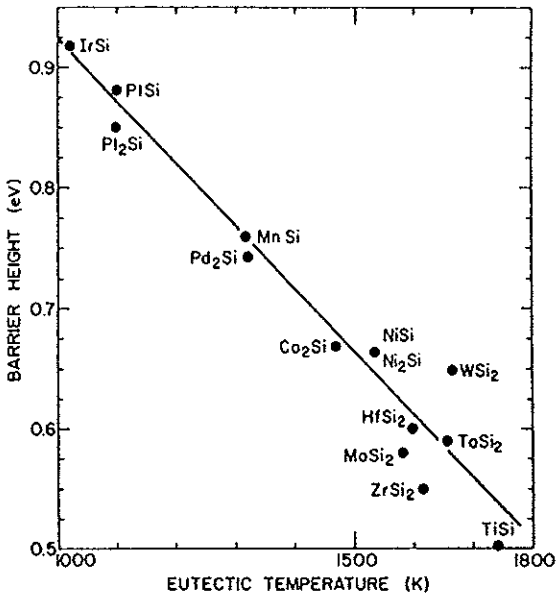


Fig. 151. A plot of Schottky-barrier height against eutectic temperature for transition-metal-silicide-Si systems. The eutectic temperature selected follows the rule described in the text. [G. Ottaviani, K.N. Tu and J.W. Mayer, Phys. Rev. Lett. **44**, 284 (1980).]

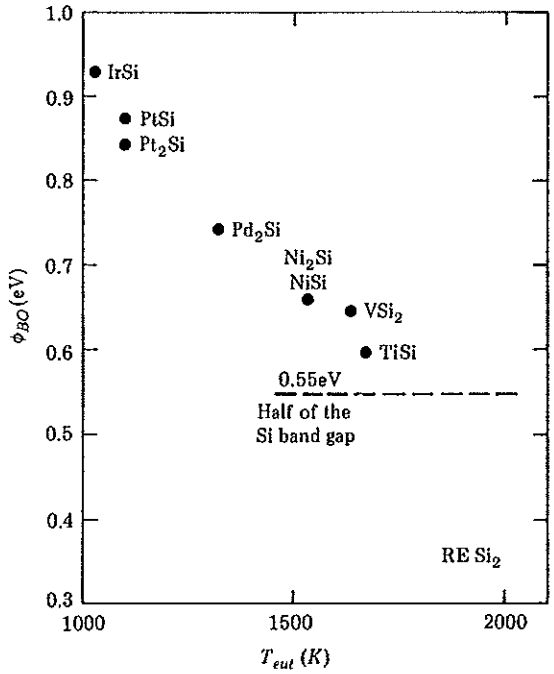


Fig. 152. Plot of Schottky barrier height against eutectic temperature of near-noble, refractory and rare-earth metal silicides. The barrier height values (except that of TiSi) were measured by *I-V* technique at room temperature using 1/16 mil diameter diodes on n-Si of ~ 10 Ω-cm. [After K.N. Tu, J.W. Mayer and L.C. Feldman, "Electronic Thin Film Science for Electrical Engineers and Materials Scientists," Macmillan, New York, 1992, p. 234; taken from Tu, Thompson, and Tsaur, 1981.]

6.4. The Metal-Oxide-Semiconductor (MOS) Junction[6.10]

From a practical and applications point of view, the MOS structure is by far the most important materials configuration for modern silicon electronics. The MOS field effect transistor (FET) is most likely the electronic device function produced not only at the highest rate but also the most abundant in absolute numbers. Each modern Pentium 4 processor holds over 45 million MOSFETs on a single silicon chip! What are the reasons for this phenomenal popularity and success? Basically, it is the exceptionally low interface state density which can be obtained at a properly formed silicon-silica (SiO_2) interface. This special property of silicon and its oxide is unique among semiconductors and at least as important for the success of silicon technology as are the convenient bandgap of silicon, its excellent free carrier mobility, high doping capability, abundance and non-toxicity.

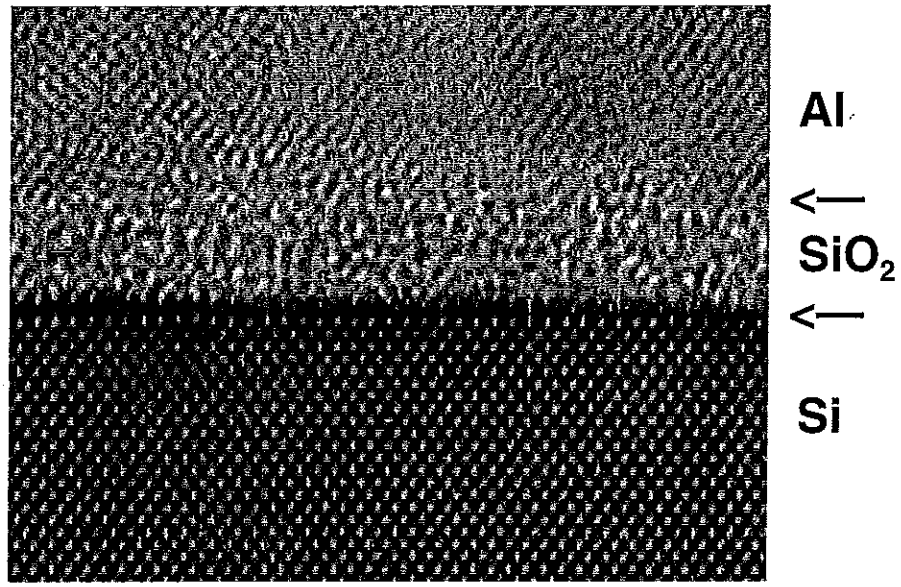


Fig. 153. Modern CMOS gate oxide area. The crystalline Si abruptly ends with a (100) plane. The extremely thin oxide displays a mottled contrast because the atoms are randomly distributed in the SiO_2 glass. The Al metallization on top of the thin oxide shows lattice fringes, indicating its polycrystalline nature. [Courtesy of R. Jamison and R. Gronsky, UC Berkeley.]

Figure 153 shows a micrograph of a small section of a Si-SiO_2 interface imaged at atomic resolution with high resolution transmission electron microscopy. The crystalline silicon lattice

[6.10] M.L. Green, E.P. Gusev, R. Degraeve and E.L. Garfunkel, Appl. Phys. Lett. **90**, 2057 (2001).

terminates sharply within one plane. No defects are visible in this terminating plane. This structural perfection results in surface state densities as low as 10^8 $cm^{-2}V^{-1}$ and lower. With such a low surface density, it is rather easy to have an electric field originating at a metal film on the oxide layer (commonly called "gate") penetrate through the Si-SiO_2 interface into the underlying silicon. The field inside the crystalline silicon will control the space charge density and the free carrier concentration in this near surface region. The basic MOSFET structure is shown in Fig. 154. The gate electrode, isolated by an oxide of thickness d (d < 20Å in a Pentium 4!), controls the current flow in the channel. The channel, in turn, is connecting the n-type source and drain regions. This transistor is formed in the near surface region of a p-type silicon wafer. An inversion of the free carrier population from holes to electrons has to be induced by the gate voltage to turn the transistor "on."

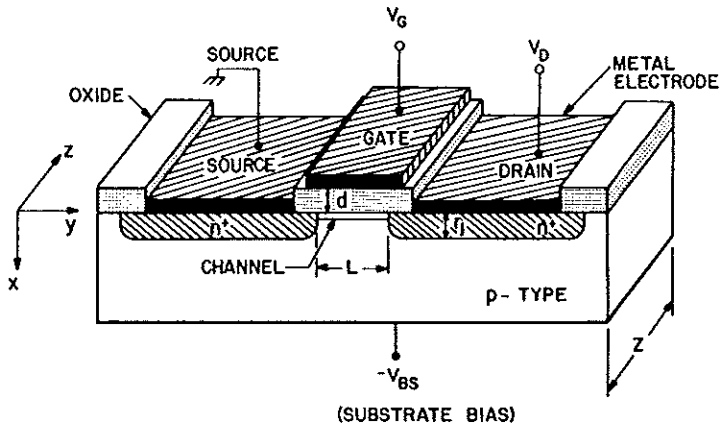


Fig. 154. Schematic diagram of a MOSFET. [After S.M. Sze, "Physics of Semiconductor Devices," John Wiley & Sons, New York, 1981, p. 434; see D. Kahng and M.M. Atalla, IEEE Trans. Electron Devices, **ED-23**, 655 (1976).]

Figure 155 displays the metal-oxide-semiconductor band configurations for various voltages applied to the gate metal on both n- and p-type silicon. In the accumulation mode (a), excess electrons (holes) are pulled to the n-Si (p-Si) surface, increasing the lateral (parallel to the surface) electrical conductivity. In the depletion condition (b), electrons (holes) are pushed away from the surface by the gate field, resulting in a reduction of the lateral, near-surface conductivity. The inversion condition, used to operate the MOSFET displayed in Fig. 154, is shown at the bottom of the figure (c). The bands are pulled up (down) sufficiently strongly to bring minority holes (electrons) to the surface of the n-type (p-type) semiconductor. The hole

concentration exceeds the electron concentration greatly, turning the near-surface region effectively into p-type silicon. While the potential on the gate electrode leads to all of these free carrier concentration changes in the semiconductor, no appreciable exchange of free carriers between the gate electrode and the semiconductor takes place.

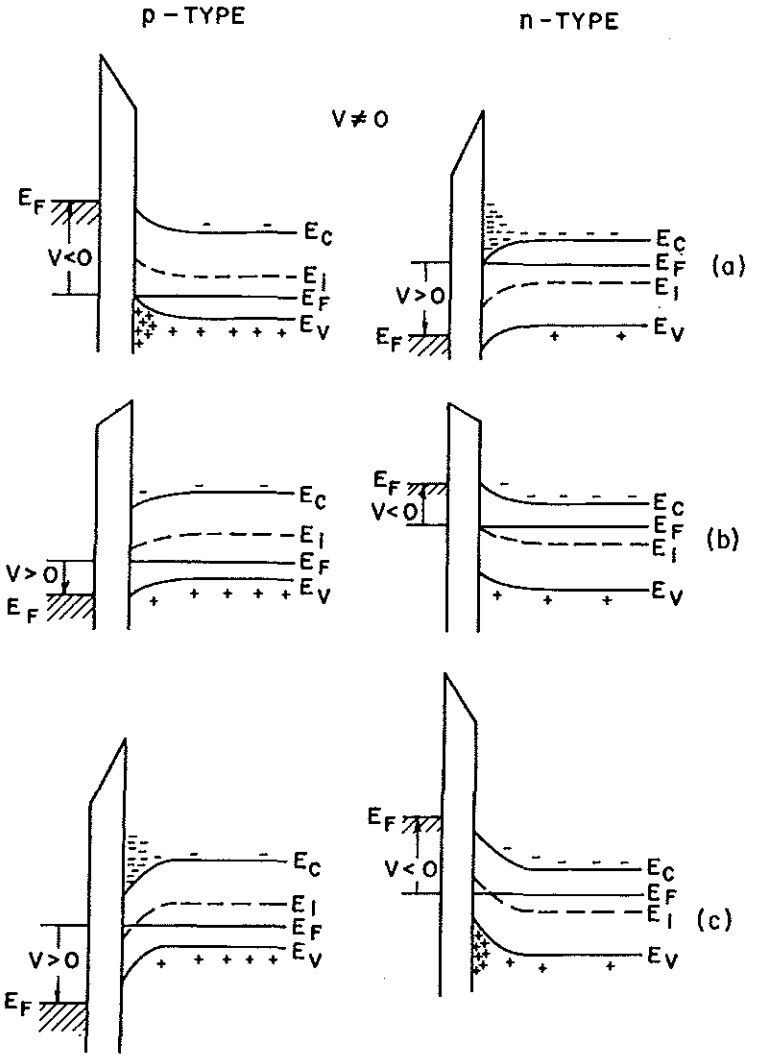


Fig. 155. Energy-band diagrams for ideal MIS diodes when $V \neq 0$, for the following cases: (a) accumulation; (b) depletion; (c) inversion. [After S.M. Sze, "Physics of Semiconductor Devices," John Wiley & Sons, New York, 1981, p. 365.]

The gate oxide is a good insulator! Today, the gate oxide thickness has shrunk to dimensions (< 20Å) which allow effective electron tunneling between the gate metal and the semiconductor. The search is on for a replacement of SiO_2 with insulators of higher dielectric constant and a low silicon-insulator interface state density.

7. Electronic Materials Analysis

References:

- "Semiconductor Material and Device Characterization," D.K. Schroder, (John Wiley & Sons, New York 1990).
- "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., (Plenum Press, New York 1979).
- "Semiconductor Characterization Techniques," P.A. Barnes and G.A. Rozgonyi, eds., Electrochemical Society Proc. Vol. **78-3** (1979).
- "Characterization of Semiconductor Materials," P.F. Kane and G.B. Larrabee, (McGraw-Hill, New York 1970).

7.1. Introduction

The ever increasing demands on better control of practically all semiconductor properties (i.e., doping and deep level type and concentration, compensation, point-and-line defects, lifetime of minority carriers, dimensions, etc.) can only be met if sufficiently sensitive and reliable analytical methods and tools are available. A good method should allow the direct and independent measurement of a certain property. In addition, one often wants to know exactly what is causing a semiconductor crystal to exhibit a certain property (e.g., which impurity – P, As or Sb – is making a crystal n-type). The number of characterization techniques and their level of sophistication which have been developed are truly impressive. This sophistication has been possible in large part because of the development of more complex instrumentation using, among other components, improved integrated circuits. In turn, this shows how synthesis of novel materials and novel devices and analysis go hand in hand, supporting each other. It is the aim of this chapter to discuss a few important analytical tools which are useful for semiconductor materials fabrication and processing. This discussion is restricted to characterization techniques, specifically developed or mostly used for semiconductors. More generally applicable techniques, such as x-ray diffraction, the various kinds of electron microscopy, Auger electron spectroscopy (AES), secondary ion mass spectrometry (SIMS), to a name a few, have been discussed in many outstanding texts.

We will also discuss deliberately some very simple techniques which should demonstrate that not all equipment needs to be expensive and may therefore often not be available. The reader is encouraged to set up some of the simple circuits and experiment with semiconductor reject samples.

7.2. Electrical Measurements

7.2.1. Type of Conduction

The thermoprobe (Fig. 156) and the rectifying properties of a metal tip on a semiconductor (Fig. 157) are very simple tools used to determine the type of conduction. In the first case, one measures a current in the external circuit generated by charge carrier flow in the semiconductor from the area under the hot tip to the area under the cold tip. External currents of opposite sign are generated for n- and p-type crystals. In the case of germanium, the method is useful at room temperature with specimens up to a resistivity close to ~ 30-40 Ω cm (or $\geq 4 \times 10^{13}$ carriers cm^{-3}). For higher purity crystals, the intrinsic carrier concentration dominates, leading to an indication of n-type. The reason for this is the carrier mobility which is higher for electrons than for holes. The type of conduction of silicon crystals can be determined over a resistivity range from 1 m Ω cm to ~ 100 Ω cm.

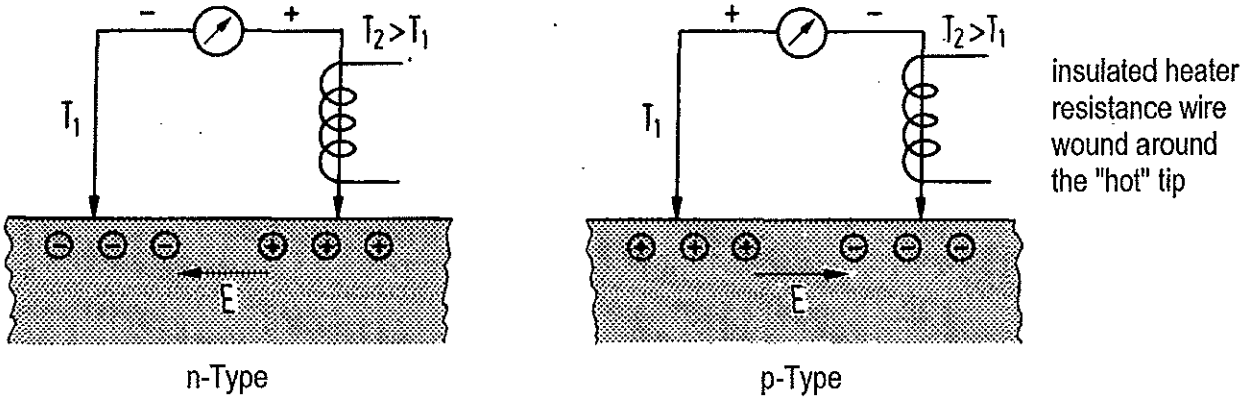


Fig. 156. Flow of charges from the hot to the cold tip. [After I. Ruge, "Halbleiter Technologie,", Springer-Verlag, Berlin, 1975, p. 187.]

A second simple method relies on the fact that most metals form a high Schottky barrier of some finite height on n-type Ge and Si and a lower barrier on p-type crystals. This leads to a

large difference in the I/V characteristic between a metal tip touching an n- or a p-type crystal. A large area contact serves as the "ohmic" contact to the semiconductor. This is essentially the experiment which led Ferdinand Braun to discover the metal-semiconductor rectifier in 1874. It is convenient to display the I/V characteristic on an oscilloscope by supplying the x-axis input with a 60 Hz AC-voltage and the y-axis input with the current signal (via the voltage drop across a resistor of appropriate value!). The thermoprobe and the metal tip rectifier are qualitative techniques. They are easy to set up and serve as a simple and quick test for the type of conduction.

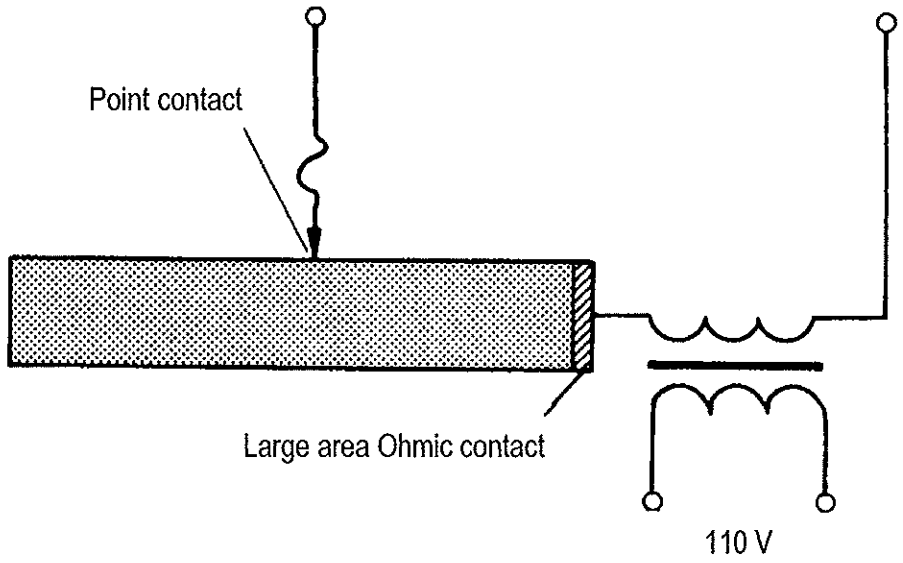


Fig. 157. Determination of the crystal type from I/V characteristic of a metal point contact on the semiconductor (e.g., Au on n-type Si: high barrier -> good diode; Au on p-type Si: low barrier -> poor diode). [After I. Ruge, "Halbleiter Technologie," Springer-Verlag, Berlin, 1975, p.188.]

The Hall effect, which actually led to the discovery of the existence of two types of carriers in solids – electrons and holes – can obviously be used for the determination of the type of conduction. It will be discussed shortly.

7.2.2. Electrical Conductivity and Resistivity

Conductivity, σ, or resistivity, ρ, measurements are routinely used to determine the net-doping concentration $|N_A - N_D|$. Commercial silicon is specified by type of conduction, resistivity and minority carrier lifetime. The conductivity is:

$$\sigma = \frac{1}{\rho} = e\ (n\mu_n + p\mu_p)$$

with n, p = electron/hole concentration

and

μ_n, μ_p = electron/hole mobility.

For extrinsic semiconductors, [$|\ N_A - N_D\ | > n_i$], one can neglect the minority carrier term in good approximation. The free carrier concentration is equal to net-doping concentration only in the case where all the impurity levels are ionized. We write:

$$n = \frac{\sigma}{e\mu_n} = N_D - N_A\ \ (n-\text{type})$$

and

$$p = \frac{\sigma}{e\mu_p} = N_A - N_D\ \ (p-\text{type})$$

The precise knowledge of the mobilities $\mu_{n,p}$ is necessary to make the above equations useful. Mobility values can be found in tables and diagrams for crystals with a low degree of compensation, i.e.,

$$\frac{N_D}{N_A} \ll 1 \text{ for } p-\text{type and } \frac{N_A}{N_D} \ll 1 \text{ for } n-\text{type crystals.}$$

Figures 158 and 159 display the resistivity ρ and the mobility for electrons and holes in Ge, Si and GaAs, respectively.

The four-point probe in its collinear arrangement (Fig. 160) is most commonly used to determine the resistivity of semiconductor samples exceeding 1 cm in at least two dimensions. The outer two probes are connected to a current source. The two probes in between are high impedance voltage sensors.

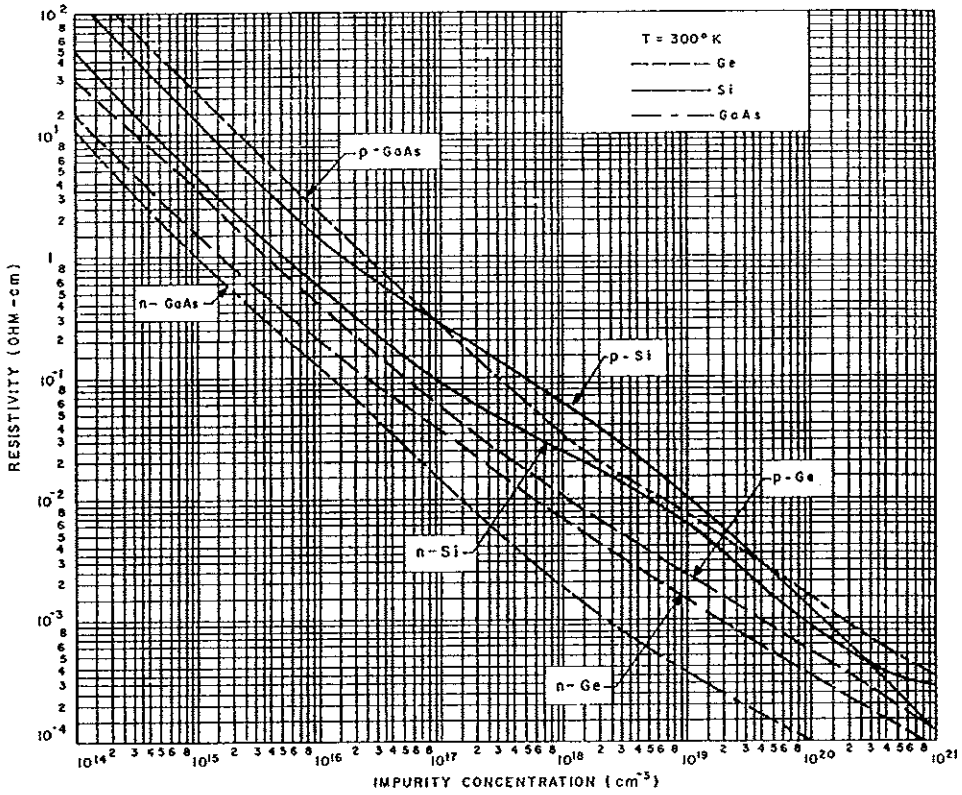


Fig. 158. Resistivity vs. impurity concentration for Ge, Si and GaAs at 300 K. [After S.M. Sze and J.C. Irvin, Solid State Elec. **11**, 599 (1968).]

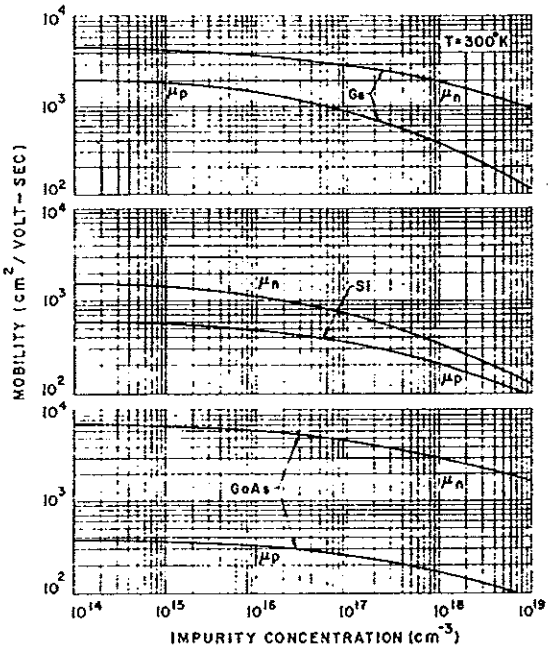


Fig. 159. Drift mobility of Ge and Si and Hall mobility of GaAs at 300 K vs. impurity concentration. [After S.M. Sze and J.C. Irvin, Solid State Elec. **11**, 599 (1968).]

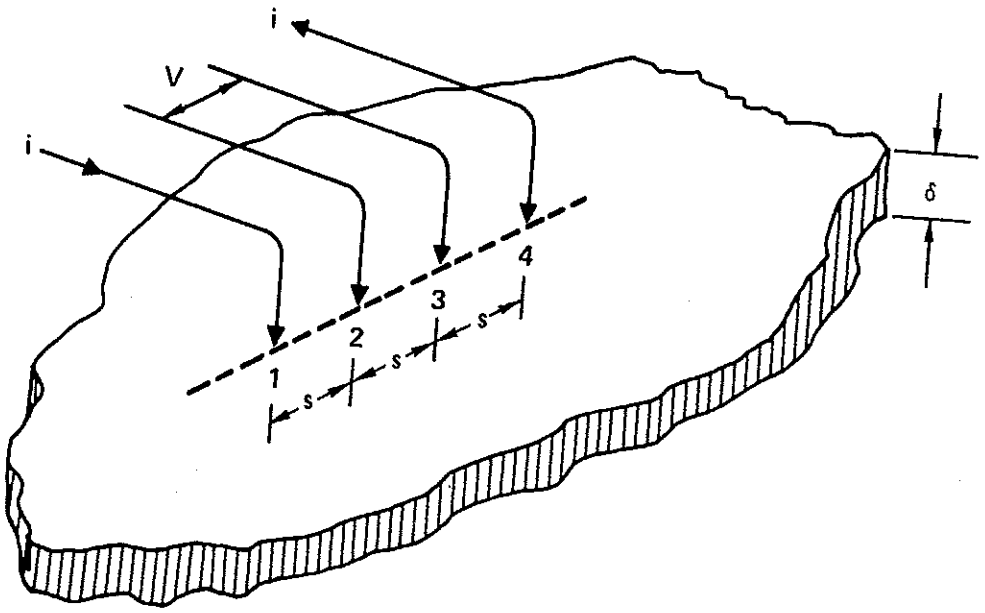


Fig. 160. Collinear four-probe array on a semi-infinite sheet of thickness, δ; outer probes 1 and 4 are current input and output probes, inner probes 2 and 3 measure the potential difference, V, between them. The distances between adjacent probes are all equal to s. [After H.H. Wieder, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 68.]

In order to determine the voltage V_{23} developed between the inner two electrodes, we first have to decide if the distance between the probes s is much smaller or much larger than the thickness of the semiconductor sample to be measured. Let us first analyze the case for which the sample thickness $\delta >> s$. The equipotential surfaces for a current entering at a point on the surface of a semi-infinite homogeneous material (resistivity ρ) are hemispheres (Fig. 161). The voltage drop dV across a hemispherical shell of radius r and thickness dr is

$$dV = IdR = I\rho\frac{dr}{2\pi r^2}$$

The total voltage drop on the surface between points 2 and 3 is

$$\Delta V_{23} = \int_s^{2s} dV = I\rho\ \frac{1}{2\pi}\ \frac{1}{r}\Big|_s^{2s} = I\rho\ \frac{1}{2\pi}\left(\frac{1}{s} - \frac{1}{2s}\right) = I\rho\ \frac{1}{2\pi}\cdot\frac{1}{2s}$$

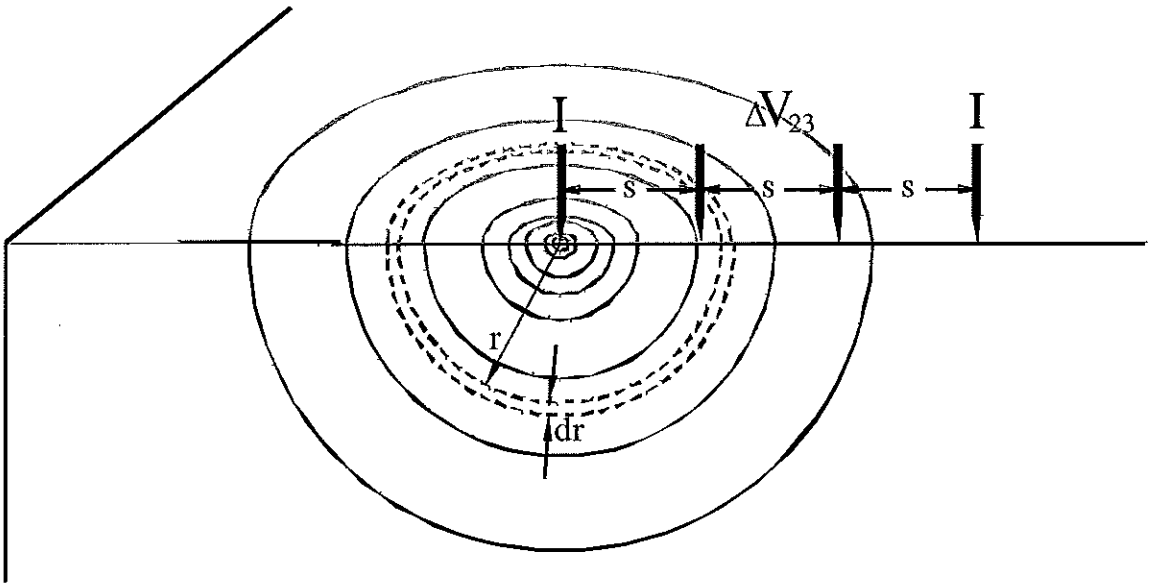


Fig. 161. Current flow from a point (1) into a semi-infinite solid with resistivity ρ.

This is the voltage drop between 2 and 3 produced by current I flowing into a semi-infinite space. An opposite current of the same magnitude flowing out of the semi-infinite space into point 3 generates the same voltage drop across 2 and 3 with the same polarity, i.e., ΔV doubles:

$$\Delta V_{23} = I\rho \frac{1}{2\pi s}$$

and

$$\rho = \frac{1}{\sigma} = 2\pi s \frac{\Delta V_{23}}{I}$$

Some commercial four point probes choose $s = \frac{1}{2\pi}$ (cm) so that $\rho = \Delta V_{23}/I$.

The second case we consider applies to $s >> \delta$, the thin sample case. The current I entering at point (1) produces circular equipotential lines. We find

$$dV = I\rho \frac{dr}{2\pi r\delta}$$

The voltage drop between points 2 and 3 is

$$\Delta V = \int_s^{2s} I\rho \frac{dr}{2\pi rs} = \frac{I\rho}{2\pi\delta} \ln(r)\Big|_s^{2s}$$
$$= \frac{I\rho}{2\pi\delta}\left[\ln(2s) - \ln(s)\right]$$
$$= \frac{I\rho}{2\pi\delta} \ln\left(\frac{2s}{s}\right) = \frac{I\rho}{2\pi\delta} \ln(2)$$

As before, we superpose an outflowing current at point (3) of the same magnitude. This doubles the voltage drop:

$$\Delta V_{23} = \frac{I\rho}{\pi\delta} \ln 2$$

or

$$\rho = \frac{\Delta V_{23}}{I} \cdot \frac{\pi}{\ln 2} \qquad \left(\frac{\pi}{\ln(2)} = 4.5324\right)$$

Often one does not know the precise value of δ but one is still interested in the sheet resistivity $\frac{\rho}{\delta} = \frac{\Delta V_{23}}{I} \frac{\pi}{\ln(2)}$ (in units of Ω per square). We have successfully discussed the two extremes $\delta >> s$ and $\delta << s$ but what about $\delta \sim s$? This situation is encountered quite often, especially in a research laboratory where semiconductor samples of a wide variety of sizes and shapes are studied. F.M. Smits[7.1] studied this case. He gives the following equation $\rho = \frac{\Delta V_{23}}{I} \frac{\delta\pi}{\ln(2)} F(\delta/s)$ and a table with a correction factor F which depends on the ratio δ/s:

[7.1] F.M. Smits, Bell System Technol. J. **37**, 711-18 (1958).

δ/s	F(δ/s)
0.4	0.9995
0.5	0.9974
0.5555	0.9948
0.6250	0.9898
0.7143	0.9798
0.8333	0.9600
1.0	0.9214
1.1111	0.8907
1.25	0.8490
1.4286	0.7938
1.666	0.7225
2.0	0.6336

Collinear Factor Point Probe Correction Factor F

For $\delta/s = 0.4$, we are in the "thin" film regime as $F(0.4) = 0.9995$, i.e., the value of ρ deviates by .05% from the one obtained with the thin film equation. For $\delta/s = 2.0$, Smits gives $F = 0.6336$ and we are close to the "thick" sample regime:

$$\frac{\delta\pi}{\ln(2)} F(2.0) \cong 2\pi s$$

or

$$\frac{2\pi s \ln(2)}{\delta\pi} = \ln(2) = 0.693.$$

The spreading resistance probe allows the measurements of spacially resolved resistivity profiles (Fig. 162). The crowding of the current near a very small diameter contact results in about 80% of the total potential drop to occur within a distance of about five times the probe contact radius. If the second contact is a large area ohmic contact placed somewhere on the crystal sample (e.g., back of a wafer), we find for a non-indenting contact and in the absence of any contact barrier:

$$R = \frac{\rho}{4a} = \frac{V}{I}$$

More commonly one uses a second identical fine probe, in close proximity to the first one. If one wishes to measure, for example, the depth dependence of the resistivity in a diffusion or ion-implantation doped wafer, the wafer is beveled under a very shallow angle (1° to 5°) to geometrically magnify the layer to be studied (Fig. 163). The two probes are used to measure the voltage drop created by a complicated current flow pattern in the sample. With a detailed knowledge of the mobility as a function of dopant concentration, calibration measurements of known resistivity profiles and significant modeling efforts, it has become possible to determine profiles as narrow as a few tens of nanometers.

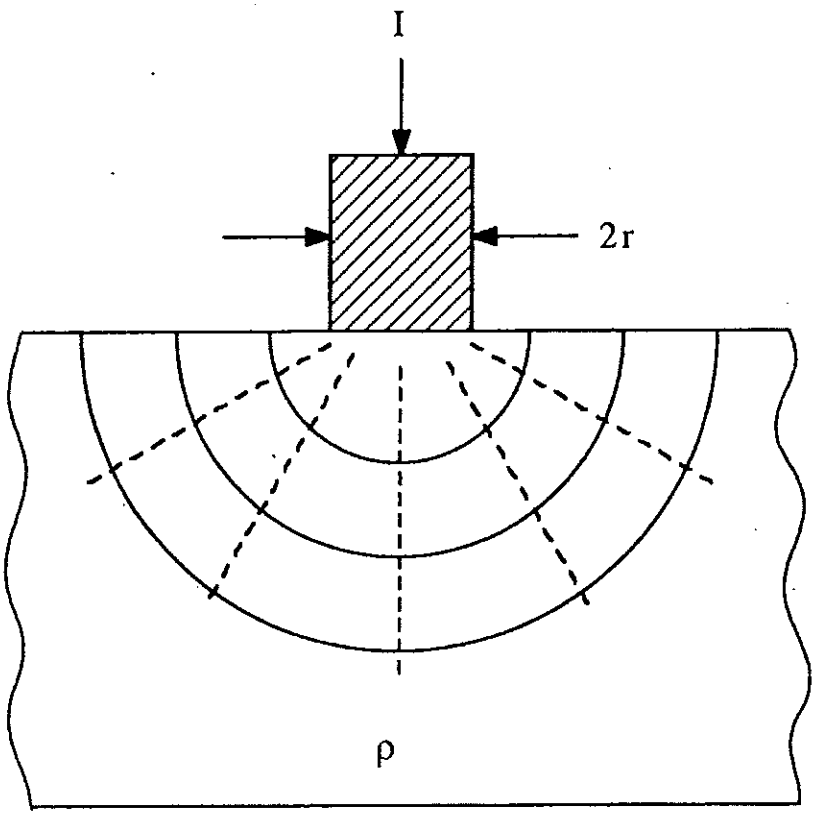


Fig. 162. A cylindrical contact of diameter 2r to a semiconductor. The equipotential lines are indicated by the solid lines and the electric field lines by the dashed lines. [After D.K. Schroder, "Semiconductor Material and Device Characterization," John Wiley & Sons, New York, 1990, p. 24.]

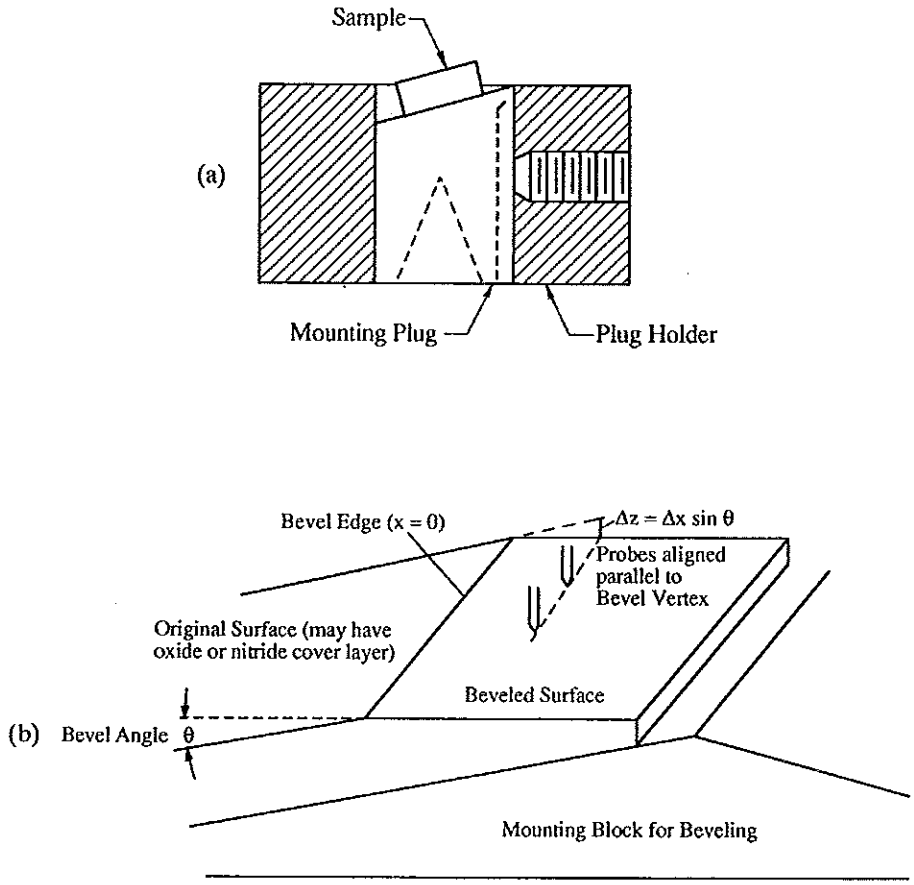


Fig. 163. Spreading resistance bevel block in (a), and the bevelled sample with the probes in (b). [After D.K. Schroder, "Semiconductor Material and Device Characterization," John Wiley & Sons, New York, 1990, p. 23.]

The accuracy and repeatability of all characterization methods involving fine metal point contacts depend to a large degree on the quality of the area of the probe which is in direct contact with the semiconductor. In order to obtain satisfactory results, modern four point and two point spreading resistance probes make use of an automatic probe lowering mechanism which places the probe tips without any lateral motion and with a well-calibrated force on the surface. The tips consist of very hard tungsten-osmium alloys. With an applied contact pressure exceeding 1000 kgf/mm^2, the probe tips deform elastically while the semiconductor gets slightly damaged. Such measurements are clearly invasive and may leave metal contamination on the surface. In order to avoid premature tip destruction, the lowering and raising of the contacts must proceed without any bias applied. An electrical bias may lead to spark erosion.

While there is significant engineering value in measuring the resistivity of a bulk semiconductor or, using a spreading method, in determining a resistivity-depth profile with high spatial resolution, we must recall that the resistivity is proportional to the product of the concentration of the free carriers and their mobility. To decouple the two, we resort to Hall effect measurements and combine them with resistivity measurements.

7.2.3. Hall Effect Measurements

The Hall effect is used to determine the free carrier type and concentration. It is the only method which yields both quantities directly and independently of other materials parameters. Figure 164 shows the standard Hall effect configuration (Hall bar).

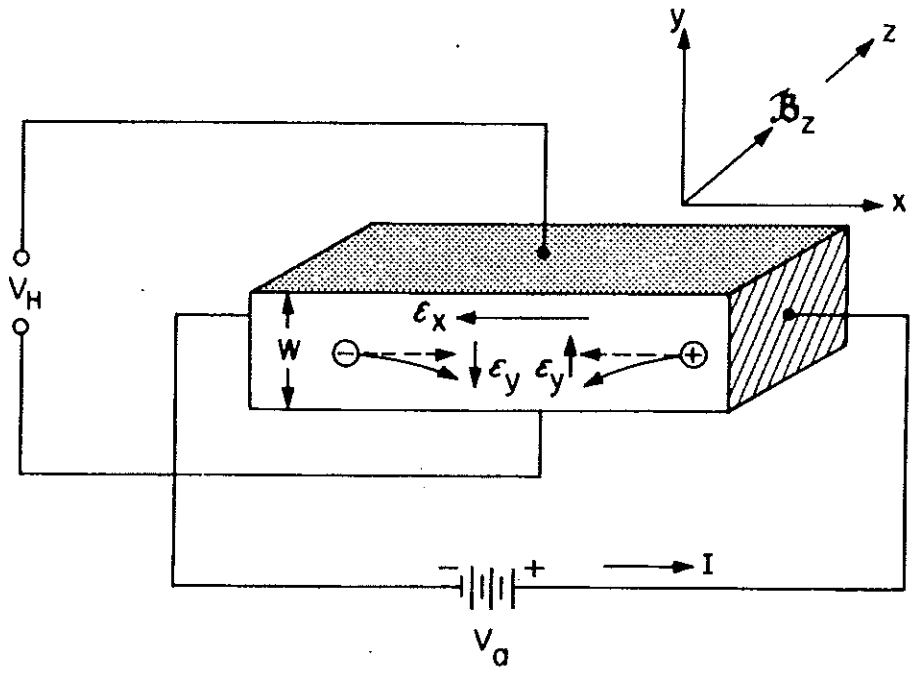


Fig. 164. Basic setup to measure carrier concentration using the Hall effect. [After S.M. Sze, "Physics of Semiconductor Devices," John Wiley & Sons, New York, 1981, p. 33.]

A current I is passed through a sample in the negative x-direction. The Lorentz force acting on the electric charges moving in a magnetic field B (in the direction z) displaces some carriers in the y-direction. This leads to an internal electric field E_H which cancels precisely the Lorentz force when equilibrium is reached:

$$e\left(\vec{v} \times \vec{B}\right) = e\vec{E}_H$$

or for the simple geometrical configuration in Fig. 164, we can rewrite the above equation without vectors:

$$\frac{V_H}{w} = E_{H_y} = B_z v_x = B_z R_H j_x$$

j_x = current density = I/area, and
R_H = Hall coefficient

In the case where only one type of carrier contributes to conduction, we find:

$$j_x = v_x en \quad \text{and} \quad R_H = \frac{1}{ne}$$

and

$$n = \frac{B_z j_x}{E_{H_y}}$$

with n = carrier concentration and e = charge of the electron.

When both electrons and holes contribute to the conduction, one finds:

$$R_H = \frac{1}{e} \frac{p - b^2 n}{(p + bn)^2}$$

with $b = \mu_n/\mu_p$ (= ratio of the electron and the hole mobility).

In this simple derivation of the Hall constant R_H, no consideration has been given to the fact that conduction is not a continuous process but is of a statistical nature. The distribution of charge carrier velocities can be very complicated because it depends on temperature, electric field, the shape of the iso-energy surfaces and on several scattering mechanisms (e.g., phonon

scattering, ionized impurity scattering). For nondegenerate semiconductors with spherical energy surfaces (e.g., InAs, GaAs), one finds for a small magnetic induction ($B << 1/\mu$):

$$R = \frac{3\pi}{8}\,\frac{1}{ne}$$

The factor $\frac{3\pi}{8}$ is called Hall factor and it is due to phonon scattering. For $\mu B >> 1$, this factor approaches unity. For an in-depth discussion of the Hall effect in semiconductors, the reader is encouraged to study the article by Beer.[7.2]

In many practical situations, a Hall effect sample configuration suggested by L.J. van der Pauw[7.3] is used. Van der Pauw applied conformal transformation of an arbitrarily shaped sample of constant thickness δ (lamellae) onto a semi-infinite half-plane (Fig. 165). The four contacts on the periphery of the lamellae transform to four contacts on the boundary of the semi-infinite plane (1 -> I, 2 -> II, 3 -> III and 4 -> IV). The resistivity ρ can be expressed as:

$$\rho = \frac{\pi\delta}{\ln(2)} \cdot \frac{R_{12,34} + R_{23,41}}{2} \cdot f(R_{12,34}/R_{23,41})$$

δ = thickness of lamellae;

$R_{12,34}$ = resistance value obtained from ratio of voltage across contacts 3 and 4 and current j through contacts 1 and 2; and

$R_{23,41}$ = analogous to $R_{12,34}$

The factor f shown in Fig. 166 depends on the ratio of the resistances.

In most common cases, the samples are circular or quadratic and f =1.

[7.2] A.C. Beer, "The Hall Effect and its Applications," C.L. Chien and C.R. Westgate, eds., Plenum Press, New York, 1990, pp. 299-338.

[7.3] L.J. Van der Pauw, Phillips Res. Rept. 13, 1 (1958).

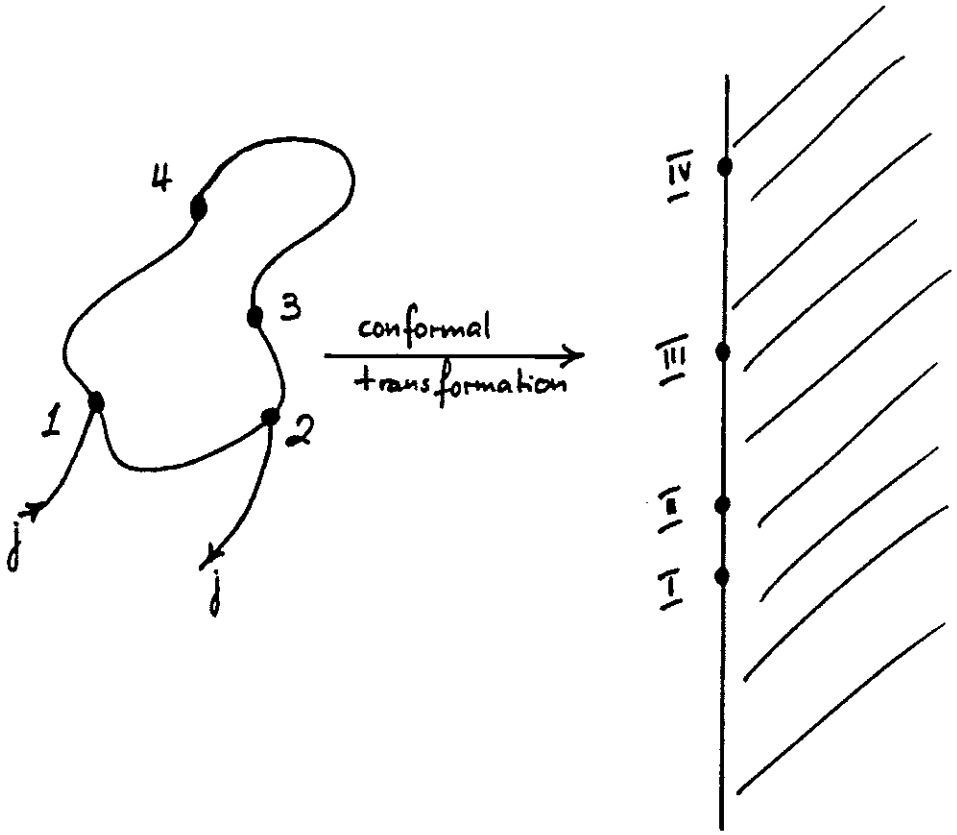


Fig. 165. Van der Pauw's approach to determining the resistivity of a homogeneous arbitrarily shaped lamellae of thickness δ.

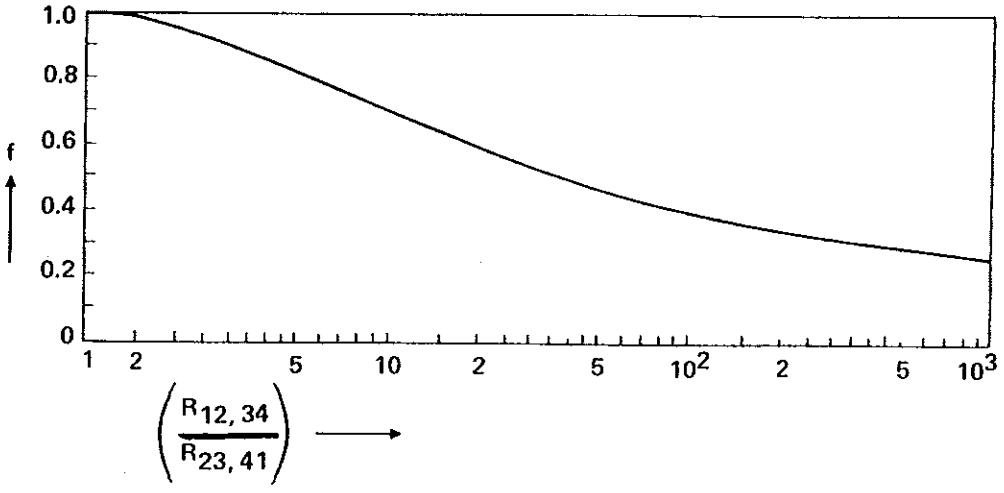


Fig. 166. The function f used to determine the specific resistivity of a specimen with an arbitrary contour as a function of the resistance ratio $R_{12,34}/R_{23,42}$. [After L.J. van der Pauw, Phillips Res. Repts., **13**, 1, (1958).]

Hall effect measurements can be performed with the same sample geometry by passing the current from 1 to 3 and measuring the voltage between 2 and 4 or sending the current from 2 to 4 and determining the voltage between 3 and 1. The magnetic induction direction has to be perpendicular to the lamellae. The Hall coefficient R_H is given by:

$$R_H = \frac{\delta \Delta R_{24,13}}{B}$$

$\Delta R_{24,13}$ = change in resistance due to the B field. The free carrier concentration n (one type case) is found to be:

$$n = \frac{B}{\delta \ \Delta R_{24,13} e}$$

With the knowledge of the resistivity and the carrier concentration, one can determine the mobility:

$$\mu = \frac{1}{en\rho} = \frac{2(\ln(2))}{\pi B} \cdot \frac{\Delta R_{24,13}}{R_{12,34} + R_{23,41}} \cdot \frac{1}{f}$$

Van der Pauw and later other authors as well have evaluated the influence of contact size and contact position on the measurement results. These studies show that the Van der Pauw geometry is very insensitive to deviations from the ideal case. Figure 167 shows that the error in sheet resistance depends only to second order on the ratio of the contact length to the length of one side of a square-shaped sample. The cloverleaf-shaped sample (Fig. 168) is a most common geometry.

Contactless conductivity and Hall effect techniques have been developed more recently. They are based on the free carrier power absorption in an RF-field. A feedback circuit keeps the RF-voltage across the inductor coil which couples to the semiconductor sample constant. The feedback correction signal is a direct measure of the conductivity. Because conductivity is an additive property, one can measure the linear average conductivity in wafers with multiple doping layers:

$$\overline{\sigma t} = \sum_{i=1}^{n} \sigma_i t_i$$

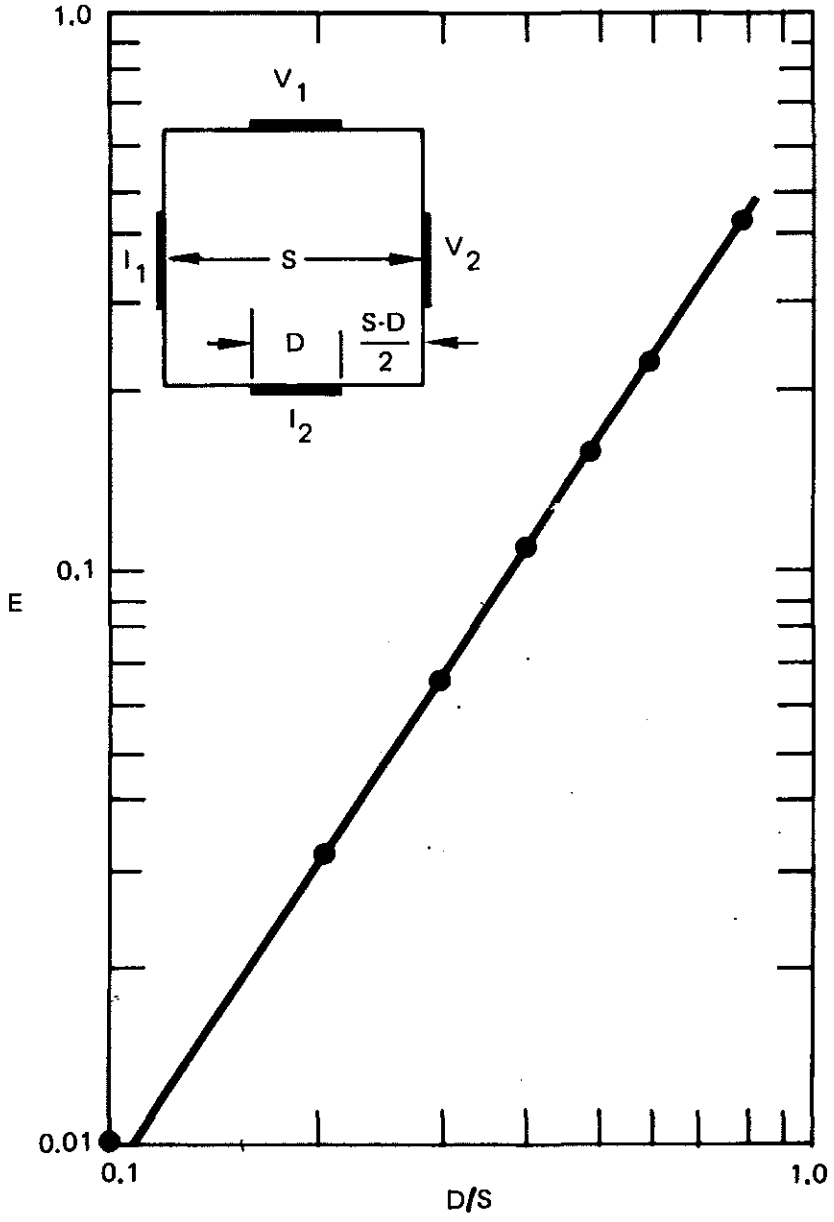


Fig. 167. Normalized sheet resistance error, E, with contacts centered on each side of a square specimen with dimensions as shown in the insert as a function of the ratio of contact length to side of the square. [After M.G. Buehler and J.M. David, Natl. Bureau of Stds., Special Publ., 400-29 (1967) 64.]

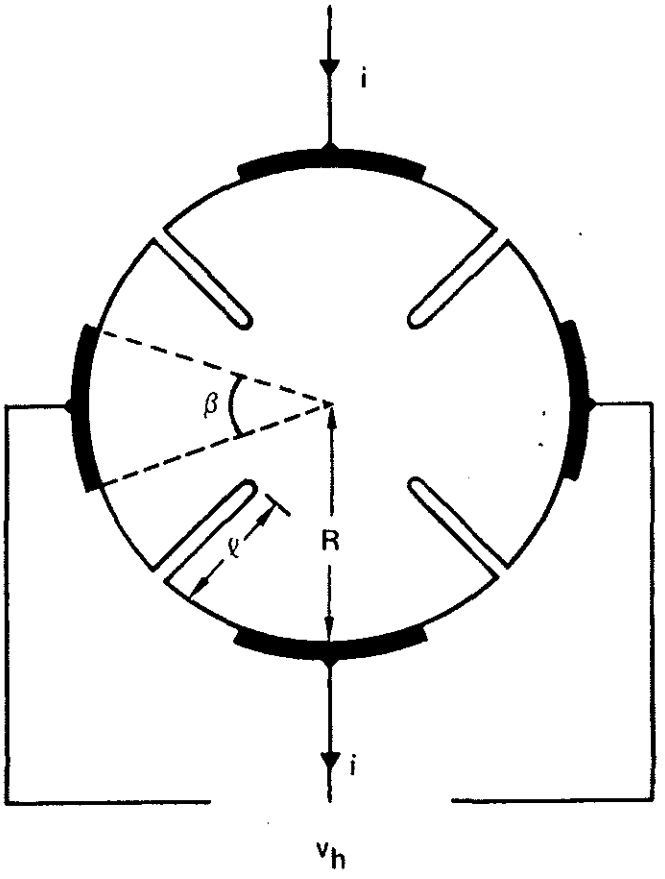


Fig. 168. Cloverleaf-shaped structure for Hall effect measurements shows slots of length ℓ cut into disc of radius R. Contacts subtend angle β. [After E. Jacobs and G. de Mey, Electron. Lett., **12**, 4 (1967).]

7.2.4. Space Charge and Diode Capacitance

At this point, we have exhausted the kind of characterization measurements that are based on charge transport. The methods discussed so far are based on electrons or holes moving in a semiconductor. In an extrinsic semiconductor, the concentration of these free carriers depends intimately on the dopant concentrations. Specifically, in the saturation region where all the shallow donors and acceptors are fully ionized, the free electron (hole) concentration in equilibrium is equal to the net-donor (net-acceptor) concentration:

$$n = N_D^+ - N_A^-$$

$$p = N_A^- - N_D^+$$

With this in mind, we can consider measuring the net-ionized dopant concentration instead of the free carrier concentration. This can be best achieved by measuring the capacitance of a reverse biased p-n junction or a Schottky barrier junction. We choose a metal Schottky barrier on an n-type silicon or germanium crystal. Work function differences and surface state pinning lead to a potential barrier which forms a depletion region in the semiconductor. The net-ionized donor concentration in this depletion region constitutes a positive space charge. The Poisson equation quantitatively relates the potential Φ and space charge concentration (Fig. 169).

$$\frac{d^2\Phi}{dx^2} = -\frac{e(N_D - N_A)}{\varepsilon\ \varepsilon_0}$$

with ε = relative dielectric constant ($\varepsilon_{Si} = 11.8$, $\varepsilon_{Ge} = 16$) and $\varepsilon_0 = 8.85 \times 10^{-12}$ Fm^{-1}.

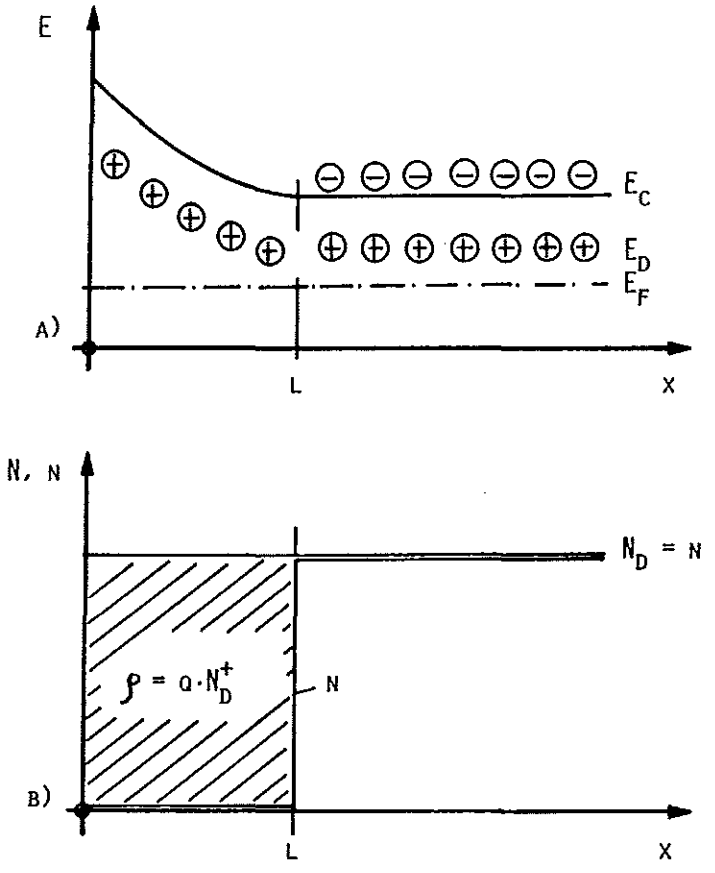


Fig. 169. (A) Band diagram and charge distribution in a Schottky contact. (B) Space charge in the depletion region. [After K. Heime, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York 1979, Chapter 5, p. 206.]

Two integration steps lead to

$$\Phi(x) = (x-L)^2 \ \frac{e(N_D^+ - N_A^-)}{2\varepsilon\ \varepsilon_0} \text{ for } 0 < x < L$$

and

$$\Phi(x) = V_{bi} + V_{ex}$$

The total potential difference across the junction corresponds to the sum of the externally applied voltage V_{ex} and the built-in potential V_{bi} (diffusion potential for a p-n junction, Schottky barrier height in a metal-semiconductor junction) and yields:

$$V_{bi} + V = L^2 \ \frac{e(N_D^+ - N_A^-)}{2\varepsilon\ \varepsilon_0} = L^2 \frac{eN}{2\varepsilon\ \varepsilon_0}$$

with $N = N_D^+ - N_A^-$

The space charge region, depleted of free carriers, is sandwiched between a conducting metal and the undepleted, conducting part of the semiconductor. This three-layer structure is a planar capacitor. Its capacitance per unit area is given by:

$$C = \frac{dQ}{dV} = \frac{\varepsilon\ \varepsilon_0}{L}$$

$$dQ = -eN(L)dL$$

$$C = -eN(L)\ \frac{dL}{dV}$$

Geometrically we find:

$$C = \varepsilon\ \varepsilon_0\ \frac{A}{L} = \frac{\varepsilon\ \varepsilon_0}{L} \quad \text{(for A = 1).}$$

Differentiation with respect to V leads to

$$\frac{dC}{dV} = \varepsilon\ \ \varepsilon_0\ \ L^{-2} \frac{dL}{dV}$$

Substituting L^{-2} with $\left(\frac{C}{\varepsilon\ \varepsilon_0}\right)^2$, we find:

$$\frac{dC}{dV} = \varepsilon\ \varepsilon_0\ \frac{C^2}{\varepsilon^2\varepsilon_0^2}\ \frac{dL}{dV}$$

or

$$\frac{dL}{dV} = \frac{\varepsilon\ \varepsilon_0}{C^2}\ \frac{dC}{dV}$$

Substituting $\frac{dL}{dV}$ with $-\frac{C}{eN(L)}$ leads to

$$C = -eN(L)\cdot\frac{\varepsilon\ \varepsilon_0}{C^2}\ \frac{dC}{dV}$$

And finally, one obtains:

$$N(L) = -\frac{C^3}{e\ \varepsilon\ \varepsilon_0\ (dC/dV)} = -\frac{1}{e\varepsilon\ \varepsilon_0}\ C^3\ \frac{dV}{dC}$$

Using the identity $\frac{d(1/C^2)}{dV} = -\frac{(2/C^3)dC}{dV}$, we can write:

$$\frac{d(1/C^2)}{dV} = \frac{2}{\varepsilon\ \varepsilon_0\ eN}$$

The last equation shows that the inverse slope of a $1/C^2$ versus V plot is proportional to the net-dopant concentration (only ionized centers!). For a single-sided junction with constant N, one obtains a straight line. The intercept of the line with the abscissa gives the built-in potential (see Fig. 170). In the case where N varies with L, we use the more general equation.

The determination of N(x) by C-V measurements is challenging because it requires extremely accurate data. This is due to the fact that differentiation has to be used to obtain the slope of a curve. Various schemes have been proposed to reduce the accuracy problem. A very clever method has been published by Miller.[7.4] He modulates a reverse biased diode with a small voltage ΔV which is proportional to the depletion layer width x. The resulting depletion layer

[7.4] G.L. Miller, IEEE Trans. Elec. Dev. **ED-19**, 1103 (1972).

modulation Δx is inversely proportional to the net-dopant concentration. The block diagram of this "Miller-profiling technique" is shown in Fig. 171.

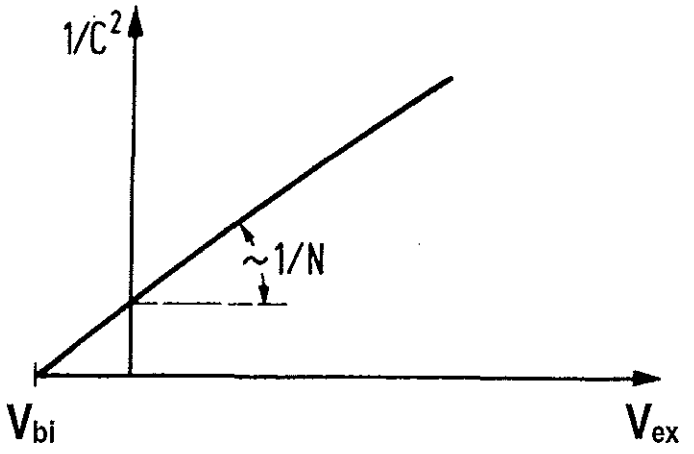


Fig. 170. $1/C^2$ versus V_{ex} plot for an abrupt, single-sided junction (e.g., Schottky diode or p^{++}-n diode). The slope B is inversely proportional to the ionized net-dopant concentration. [After I. Ruge, "Halbleiter Technologie," Springer-Verlag, Berlin, 1975, p. 202.]

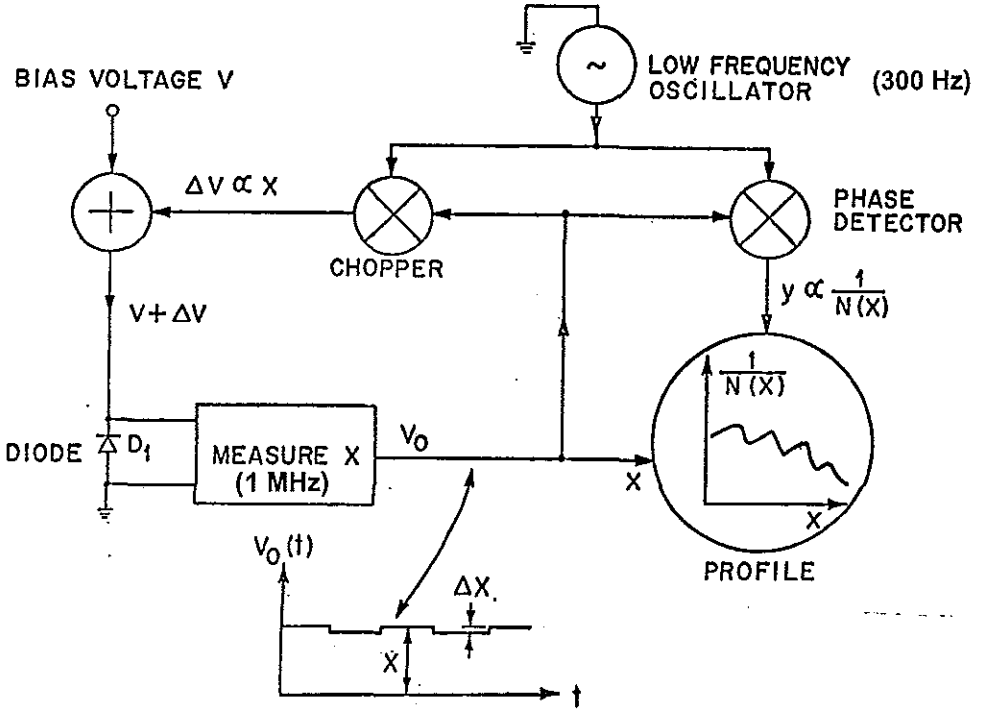


Fig. 171. Simplified block diagram of the feedback method of semiconductor junction profiling. [After G.L. Miller, D.A.H. Robinson and S.D. Ferris, in "Semiconductor Characterization Techniques," P.A. Barnes and G.A. Rozgonyi, Electrochemical Society Proc., Vol. **78-3**, 22 (1978).]

Summarizing this section on capacitive measurements of the space charge region in a Schottky or p-n junction, we recall both advantages as well as disadvantages. First, most active devices of a complicated electronic circuit make use of junctions, making in turn measurements on junctions very convenient. Second, it is easy with a modern capacitance bridge to measure accurately very low capacitances down to the femto Farad (10^{-15} F) range. Third, the net-dopant concentration is measured directly (similar to the Hall effect which measures the free carrier concentration directly). Among the difficulties encountered are the effects of stray capacitances, temporal drifts in the capacitance because of surface states and/or deep levels in the bulk of the semiconductor and the condition that measurements should be taken in a temperature range where all the dopant atoms are ionized and the semiconductor is extrinsic ($|N_D-N_A)| > n_i$). The latter condition is not fulfilled for p-type GaN because the shallow acceptors are quite "deep" with their energy level located at E_V + 160 meV. At room temperature, only ~1% of all the acceptors are ionized.

7.2.5. Deep Level Transient Spectroscopy (DLTS)

The early difficulties encountered in space charge measurements have been used advantageously in a technique called Deep Level Transient Spectroscopy (DLTS). This technique makes use of the direct observation of the charging and discharging of deep levels in the semiconductor through the observation of capacitance transients. Three properties of deep levels can be determined: concentration, position in the bandgap and capture cross section for free carriers. With some additional sophistication, the thermal activation energy of the capture cross section can be measured as well.

Let us discuss an n^+-p junction to explore the principles of DLTS. We start with the junction that is reverse biased during the time interval (1) (Fig. 172). At time (2), the bias is rapidly reduced (pulsed) to a lower value and the depletion layer width shrinks significantly. Beginning at time (3), the bias reduction pulse is removed and the original bias is reapplied. During time interval (4), we observe the recovery of the p-n junction. What happens to a "deep" acceptor level (acceptor "trap") during the bias reduction pulse and what happens afterwards, and how may the changes be reflected in the junction capacitance (Fig. 173)?

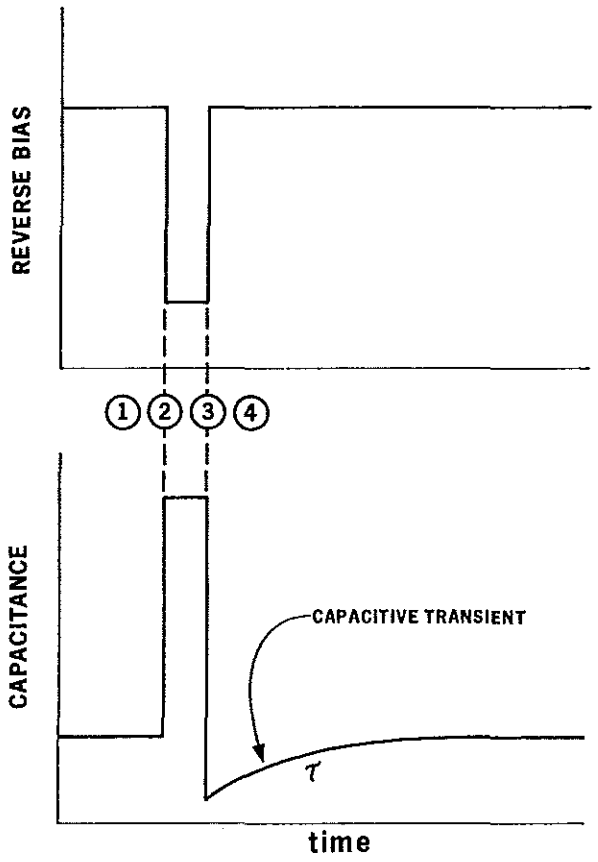


Fig. 172. The dependence of bias and of capacitance of a partially depleted p-n or p–i-n semiconductor junction in a Deep Level Transient Spectrometer.

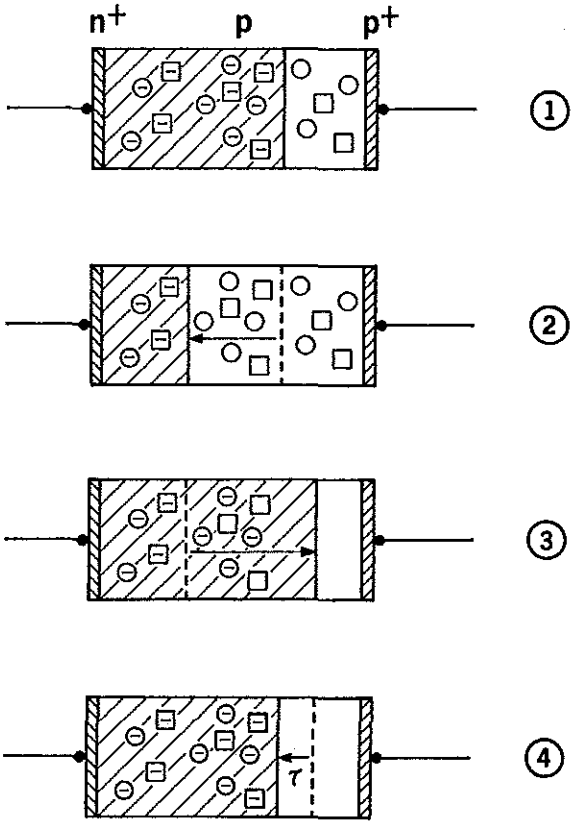


Fig. 173. Change of the depletion layer and the charge state of shallow (circles) and deep (squares) acceptors in a p-i-n device. Shallow acceptors follow the bias changes given instantaneously while the deep traps get refilled much more slowly with a time constant t.

Deep acceptors which are ionized and reside during time interval (1) in the depletion layer and during the time (2) -> (3) in the volume which is flooded with free holes may capture a hole with a capture rate C:

$$C = \sigma \langle v \rangle p$$

with

σ = capture cross section (cm^2)

$<v>$ = thermal velocity of holes (cms^{-1}) = $\left(\frac{3k_BT}{m^*}\right)^{1/2}$

p = free hole concentration (cm^{-3})

Let us assume that the temperature is sufficiently high for all shallow dopants to remain ionized during bias reduction. Upon reapplication of the original bias at time (3), the deep neutral acceptors begin to release their holes exponentially with a thermal emission rate:

$$e = \sigma < v > N_{band} \exp(-E_T / k_B T)$$

with

N_{band} = effective density of states in the band

E_T = energy of acceptor trap above the valence band.

The pre-exponential factor is derived from a "detailed balance" argument for the steady state for which the capture and the emission rate must be equal.

The change of the charge state of the deep levels can directly be observed by measuring the capacitance. Figure 173 schematically displays what happens to the space charge during and after the bias reduction pulse. The deep traps which become neutral during the bias reduction pulse no longer contribute any space charge. Because of the lower total space charge, the depletion layer widens beyond the steady state width immediately after the bias pulse is removed at time (3). Every deep acceptor trap which thermally releases its hole becomes again negatively charged, leading to a progressing reduction of the depletion layer width until all deep acceptors have released their holes and we have returned to the pre-pulse condition.

The three parameters which characterize a deep level can be determined from a measurement of the capacitance as a function of time as displayed in Fig. 172. The amplitude of the capacitive transient ΔC starting at time (3) leads to the concentration of the acceptor traps N_T:

$$N_T \cong 2\ (N_A - N_D)\ \frac{\Delta C}{C}$$

with

$(N_A\text{-}N_D)$ = shallow net-acceptor concentration in the p-region

ΔC = amplitude of capacitance transient

C = steady state capacitance

The above expression is valid for $N_T << N_A\text{-}N_D$. This is the case for most semiconductors which have not been deliberately doped with deep levels.

The position of the acceptor trap level above the valence band E_T can be determined from the emission rate time constant as a function of temperature. By repetitively pulsing the diode and recording and averaging many capacitance transient time constants over a wide temperature range, one can obtain sufficient data for an Arrhenius plot of the logarithm of the capacitance transient time constants (1/emission rate) as a function of the inverse absolute temperature. The slope of the straight line is $-E_T/k_B$. In the early 1970s, D. Lang at Bell Labs proposed an electronic scheme to select capacitive transients with a given time constant. Lang used a double box car integrator, an instrument which instantaneously measures repetitive signals, S_1 and S_2, at two times, t_1 and t_2, and forms the difference of these two signals. The difference signal is integrated and averaged over many signal periods, leading to an excellent signal to noise ratio and high sensitivity. Figure 174 graphically conveys the formation of a peak at a specific temperature where the $S_1(t_1)$ - $S_2(t_2)$ is a maximum for a single type of traps emitting their charges exponentially with time (τ = time constant). At the bottom of the figure, the temperature is very low and emission proceeds slowly, leading to long time constants; $S_1\text{-}S_2$ is small. At the upper edge of the figure, the temperature is high enough for the emission event to be completed before S_1 is recorded; again $S_1\text{-}S_2$ is close to zero. Somewhere between these two extremes, there will be a maximum in the difference $S_1\text{-}S_2$. This maximum occurs for

$$\tau = \frac{t_2 - t_1}{\ln\,(t_2/t_1)}$$

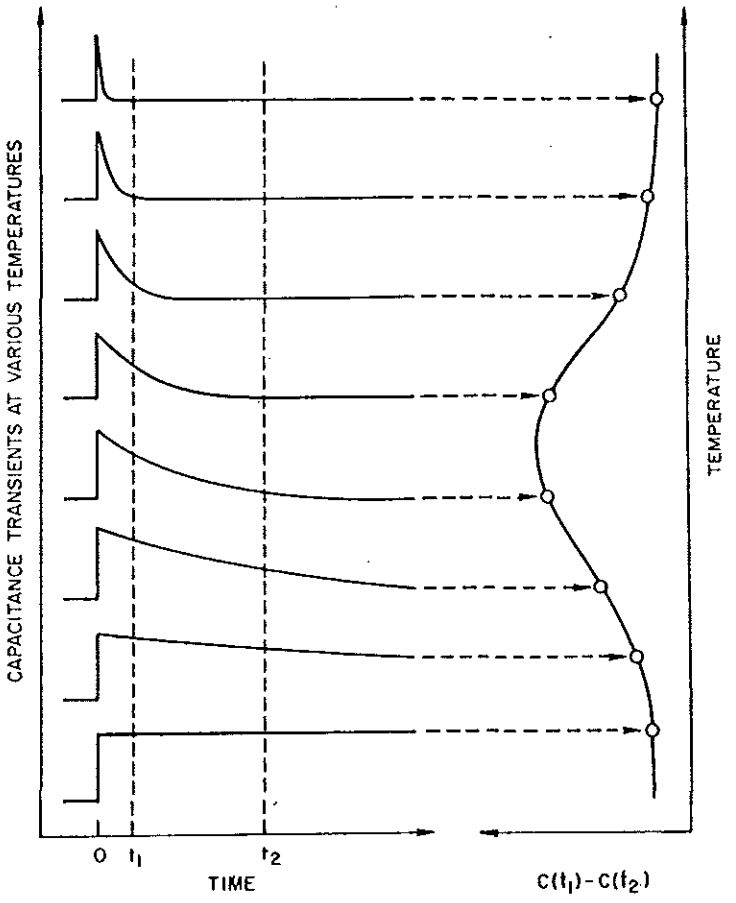


Fig. 174. Implementation of a rate window by means of a double-boxcar integrator. The output corresponds to the (average) difference of the amplitudes at the sample times t_1 and t_2. [After G.L. Miller, D.V. Lang and L.C. Kimerling, Ann. Rev. Mater. Sci., **7**, 417 (1977).]

Other time constant "selection" or "filtering" methods have been developed with various attributes. Based on the fact that the ionization of a group of identical deep traps is governed by an exponential decay, leading to an exponentially decaying capacitance transient, Miller *et al.*[7.5] showed that a correlation method using an exponential weighting function leads to the most effective selection process or the best signal to noise. Despite this fundamental advantage, this method has not been used broadly, mostly because the electronic equipment has never been produced commercially. A third approach for electing a particular time constant uses the well known lock-in amplifier. Figure 175 shows a DLTS signal with noise and the three weighting functions used for selecting a specific time constant.

[7.5] G.L. Miller, J.V. Ramirez and D.A.H. Robinson, J. Appl. Phys. **46**, 2638 (1975).

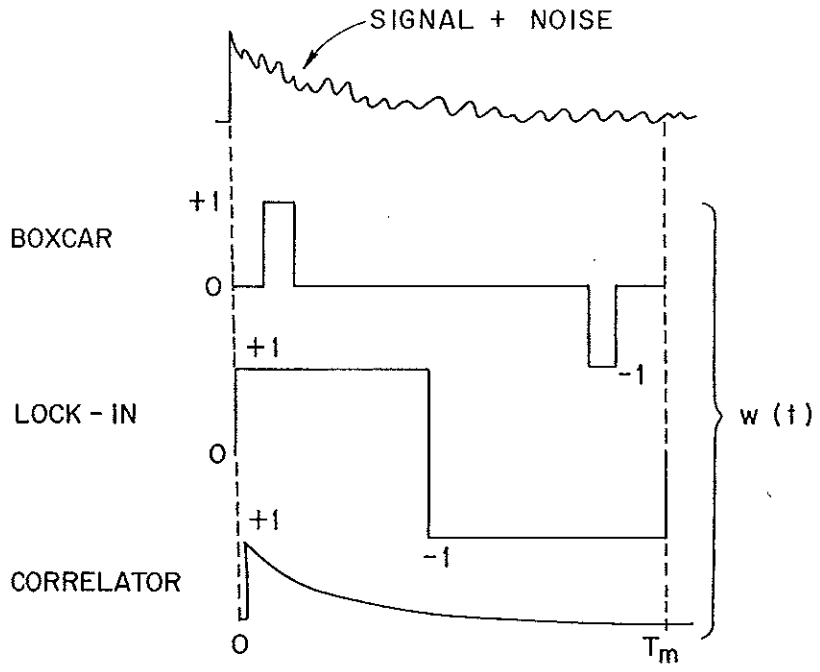


Fig. 175. Diagram comparing a representative input signal from a capacitance transient experiment with three different weighting operations that have been used to process such signals. [After G.L. Miller, J.V. Ramirez and D.A.H. Robinson, J. Appl. Phys. **46**, 2638 (1975).]

In order to determine the exponential decay time constant with sufficient accuracy for a useful Arrhenius plot, one wants to cover at least two to three orders of time constants (typically 0.3 μs to 300 μs), each time constant setting requiring a full cooling run! Today's DLTS equipment uses fast analog-to-digital converters and powerful computers which record all the transients over several orders of time constants in one cooling cycle. Sophisticated software has been developed to obtain the best sensitivity and selectivity at the same time.

For the mathematically inclined reader, we mention that an exponential decay is transformed into a delta function by an inverse Laplace transformation. Wouldn't such a transformation lead to the ultimate resolution? Unfortunately, exponentials are not orthogonal functions like trigonometric functions, and this inverse transformation does not cope well with noise. Nolte *et al.*[7.6] treated the basic problem.

[7.6] D.D. Nolte and E.E. Haller, J. Appl. Phys. **62**, 900 (1987).

When a diode with several kinds of deep traps is cooled, one obtains signals only at the temperatures for which at least one kind of trap has the emission rate corresponding to the pre-set time constant of the DLTS apparatus. Each kind of trap leads to a peak at its specific temperature in a DLTS spectrum. Several examples are shown in the attached paper on DLTS with ultra-pure Ge diodes.

The third parameter, the capture cross section σ, can be determined from the dependence of the capacitive transient amplitude ΔC on the length of the bias reduction pulse t[(2)-(3)]. One easily recognizes that it takes several capture rate time constants to fill most of the traps during the bias reduction pulse. By going to ever shorter reduction pulse lengths, fewer deep traps get filled during the bias reduction period and the capacitive transient ΔC becomes smaller:

$$\frac{d(N_T^o)}{dt} = C_h \ (N_T - N_T^0)$$

with

N_T^o = concentration of neutral acceptor traps

N_T = total concentration of acceptor traps

C_h = hole capture rate.

Integration of the above expression from t = 0 to t = τ_c (= pulse length) leads to

$$N_T^o(\tau_c) = N_T \ [1 - \exp(-C_h \tau_c)]$$

and

$$\frac{N_T - N_T^0(\tau_c)}{N_T} = \frac{\Delta \ N_T(\tau)}{N_T} = \exp \ (-C_h \tau_c)$$

It follows

$$\ell n\left(\frac{\Delta N_T(\tau_c)}{N_T}\right) = -C_n \tau_c$$

A plot of the left side as a function of the bias reduction pulse length τ_c yields the capture cross section in the form of the slope of the straight line.

The discussion so far has focused on deep levels or traps which capture and release majority carriers, i.e., holes in a p-type semiconductor or electrons in an n-type material. Can DLTS be used to characterize traps which are of minority type, e.g., a deep donor in a p-type semiconductor? The answer is affirmative if a way is found to provide a supply of minority carriers. This can be achieved in at least two ways. Either the bias reduction pulse is made larger than the steady state bias, leading to forward current injecting minority carriers or light with $h\nu > E_{GAP}$ is used to illuminate the junction. Such photons ionize electrons across the gap, forming free holes in the valence band and free electrons in the conduction band. Minority type traps can capture the appropriate free carrier, turning neutral. The situation at time (3) when the forward bias pulse is removed and the steady state bias is reapplied is now quite different from the majority trap case.

Whereas the net space charge is *reduced* when deep majority traps become neutral, the net charge *increases* when minority traps are capturing a charge and turn neutral. This is because ionized minority deep centers are charged with the opposite polarity as compared to ionized majority traps. With their opposite charge, they decrease the net-majority space charge. This compensating space charge disappears when the minority traps become neutral, leading to an increased majority space charge. The consequence of this space charge increase is a decrease in depletion layer width and an increase in capacitance. We see that the capacitance transient ΔC of minority traps is of the opposite polarity than the transient of the majority traps. Peak #11 in Fig. 6 of the attached paper on the very first DLTS study of deep levels in ultra-pure Germanium is due to minority traps. In this case, light was shone on the diode, generating both free holes and free electrons.

The importance of DLTS can be fully appreciated when one recalls that the minority carrier lifetime is controlled by deep traps. The minority carrier lifetime, in turn, is of great importance, affecting the gain of transistors, the leakage current in p-n junctions, the efficiency of solar cells and the time a metal oxide semiconductor (MOS) or charged coupled device (CCD) can store charge. Great precaution is taken in every step of semiconductor device fabrication to avoid contamination of the semiconductors with deep level impurities or defects. DLTS is one of

the tools of choice to characterize the traps with excellent sensitivity. It is a parametric technique and its sensitivity tracks the net shallow level space charge concentration. A well designed and constructed DLTS system can detect 1 deep level impurity in 10^5 shallow levels, the best systems are ~ 10 times more sensitive (1 deep level in 10^6 shallow centers). Such a system should be able to detect 10^9 cm^{-3} deep levels in a semiconductor doped to 10^{15} cm^{-3}.

On the following pages, the very first DLTS paper dealing with ultra-pure germanium is reproduced. A large number of traps were discovered, and artificial peaks developed with changing surface conditions. Majority as well as minority deep level centers were found. In order to obtain a maximum signal-to-noise ratio, a Miller correlator and integrator were built. In the meantime, DLTS systems with excellent performance have become commercially available, spoiling the fun and satisfaction derived from building unique electronic equipment!

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DEEP LEVEL TRANSIENT SPECTROSCOPY OF HIGH-PURITY GERMANIUM DIODES/DETECTORS*

Eugene E. Haller, Pearl P. Li, G. Scott Hubbard and William L. Hansen

Abstract

Deep Level Transient Spectroscopy (DLTS)[5] has been applied for the first time to high-purity germanium p-i-n diodes. Using the correlator technique,[6] a large number of peaks due to acceptor levels in the forbidden band have been observed. The levels due to substitutional copper, to copper-hydrogen complexes and to divacancy-hydrogen defects have been positively identified. Several unknown levels have been discovered. The results obtained with DLTS are in excellent agreement with results from Hall-effect measurements. DLTS is the perfect tool to follow the creation and annealing of radiation defects.

Introduction

Impurities, point and line defects can introduce energy levels in the forbidden band of a semiconductor. If the levels are close to one of the band-edges they are called "shallow" (i.e. ~ 10 meV in Ge; ~ 40 meV in Si). Their concentration determines the width of the depletion layer of a reverse biased diode at a given voltage. If the energy levels do not lie close to either band-edge they are called "deep". Such levels can retain free charge carriers (electrons/holes) for prolonged times. Deep levels are often called "traps". The presence of a small concentration is detrimental to the performance of semiconductor p-n or p-i-n radiation or particle detectors since retention of 10^{-4} of the charge produced by an ionizing event can lead to an unsymmetric line-shape in a spectrum. Therefore, much effort has been spent in the characterization and in the removal of trapping centers from semiconductor crystals.

From the analysis of the shape of gamma-ray photopeaks, one can draw certain conclusions about traps in a p-i-n detector.[1,2] However, this approach is very indirect and a large number of additional parameters besides the traps influence the photopeak line shapes. More quantitative answers can be obtained from Hall Effect measurements on single crystals.[3,4] This method works well for trap depths up to about one-third of the band gap. Levels close to or in the middle of the band gap are hidden by the generation of intrinsic carriers. Furthermore, in the presence of several trap levels with similar energies, it becomes very difficult to obtain accurate values for trap concentration and position. A further disadvantage of the Hall Effect method is that it does not work for the actual detector diode but only for unprocessed single crystal samples. Since the detector processing may introduce impurities, this is a significant disadvantage of the Hall Effect method.

Deep-Level Transient Spectroscopy, DLTS

The Deep-Level Transient Spectroscopy (DLTS) method was introduced by Lang[5] in 1974. This kind of spectroscopy uses the strong dependence of the emission rate of carriers from deep traps on temperature and on the energy of the trap. Deep traps are periodically filled by a pulsed bias reduction across a reverse biased junction (p-n or p-i-n). Figure 1 shows the change of the bias and capacitance with time. In the presence of traps, a slow recovery of the capacitive value occurs after the pulse. Figure 2 displays the condition of the depletion layer inside a n^+-p-p^+ device at the different times shown in Fig. 1. The depletion layer is shaded in each case. Capacity change

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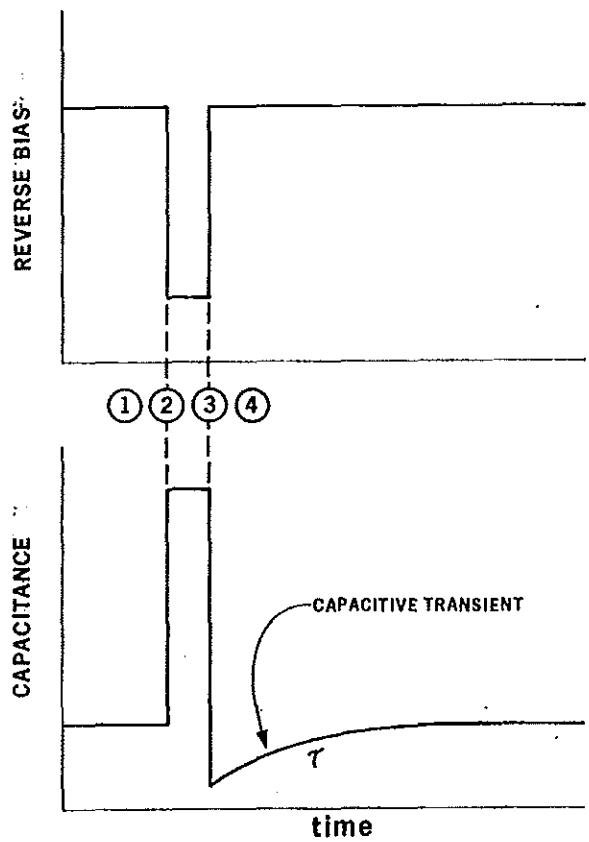


Fig. 1 Time dependence of bias and of capacitance of a partially depleted p-n or p-i-n semiconductor junction in a Deep Level Transient Spectrometer.

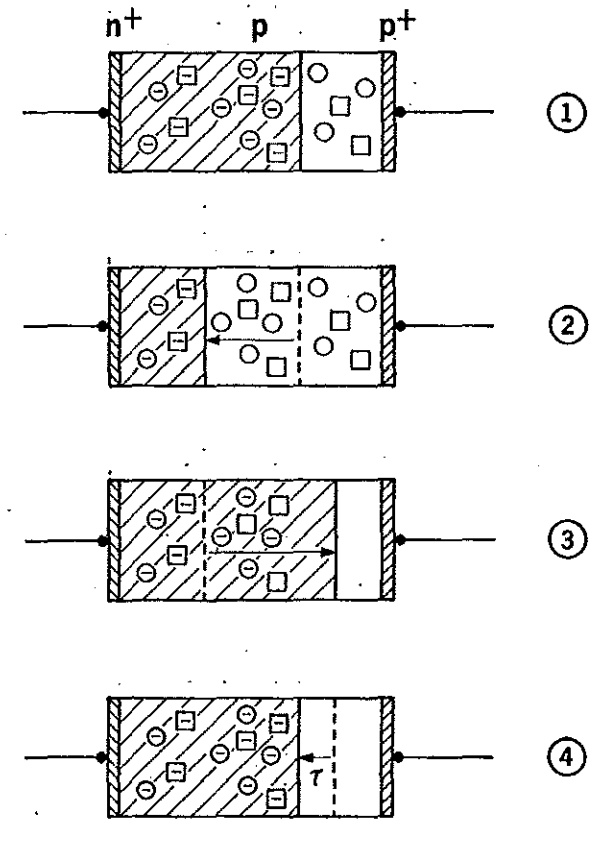


Fig. 2 Change of the depletion layer and the charge state of shallow (circles) and deep (squares) acceptors in a p-i-n device. Shallow acceptors follow the bias changes given instantaneously while the deep traps get refilled much more slowly with a time constant τ.

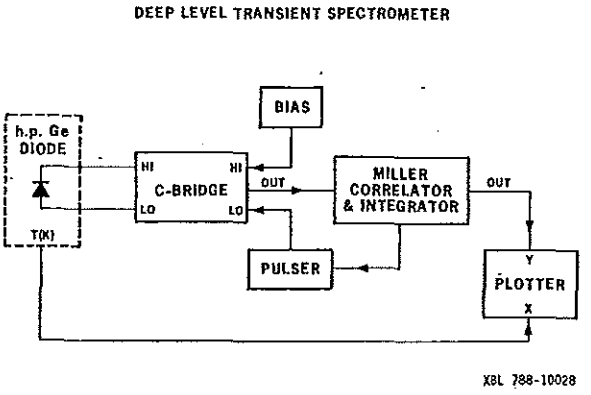


Fig. 3 Block diagram of a simple Deep Level Transient Spectrometer.

in period # 3 to 4 occurs because the deep levels (in our example deep acceptors, i.e. majority traps) do not get filled with electrons instantaneously when the full bias is reapplied. The filling proceeds exponentially with time. The time constant τ depends on the trap energy E (E = energy of the trap level measured from the corresponding bandedge) and the temperature T:

$$\tau = C_0 T^{-2} \exp(\frac{E}{kT}) \qquad (1)$$

where C_0 is a constant and k is the Boltzman constant (= .08633 meV K^{-1}). The emission rate e is the inverse of τ:

$$e = C_1 T^2 \exp(-\frac{E}{kT}) \qquad (2)$$

Various schemes have been used to measure the capacitve transient and its recovery time constant. The correlator technique employed by Miller[6] uses the fact that the best signal to noise ratio can be obtained if the filtering function has the same time dependence as the signal. We have used the correlator technique for our measurements.

Experimental

A block diagram of the electronics of our Deep Level Transient Spectrometer is shown in Fig. 3. The diode under test is mounted inside a variable temperature cryostat. A Stirling engine cryocooler which reaches a minimum temperature of 28 K with room temperature water cooling and about 22 K with precooled methanol cooling is used. Great care is taken to avoid all light leaks to the detector. The temperature is measured by a forward biased silicon diode.[7] The capacitance of the test diode is continuously measured with a 1 MHz bridge.[8] A variable DC-bias is applied through the HI-lead of the bridge and the bias reduction pulse is fed through the LO-lead to the diode. In a few cases, a IR-light emitting diode next to the diode under test is used to change the charge state of present traps. The output signal of the capacitance bridge is fed to the Miller correlator where it is weighed by an exponentially decaying waveform with a time constant τ_r that can be set to values between 1 and 100 ms. The product is integrated and drives the Y-axis of a plotter. The X-axis is directly connected to the temperature measurement diode. The correlator output is a maximum when the time constants of signal and weighting function τ_r are equal. This happens at a well defined temperature for each trap (Eq. 1). We have used five time constants τ_r (1.3, 3, 10, 30 and 100 ms) to determine E and C_1 of each trap.

Results

A large number of high-purity germanium diodes have been studied. Most of the devices were standard diodes of the kind used for gamma and X-ray or particle detection (area ~ 10 cm^2, thickness ~ .5 cm). The internal time-constant due to the resistance of the undepleted material and the capacitance of the depletion layer is always made smaller than the inverse of the C-bridge measurement frequency. In some cases (very high-purity), the diodes had to be made thinner than standard (<.5 cm) to satisfy this requirement.

The first diodes measured contained well known traps. Three crystals grown in vacuum, a nitrogen atmosphere and a hydrogen atmosphere were copper doped by diffusion from a surface source. Figure 4 shows the DLT spectra of the three diodes made from these crystals. Peak 4 is due to the doubly ionized copper acceptor (Cu^{--}). Peaks 1 and 3 are due to copper-hydrogen complex acceptors. The origin of peak 2 is unknown. It shows up in all crystals containing oxygen concentrations $\geq 5 \times 10^{13}$ cm^{-3}. The emission rate parameters are compiled in Table 1. Next we investigated two diodes which exhibited trapping (Figs. 5 and 7). In diode S24-7.7 (Fig. 5) all copper and oxygen related peaks (1-4) are clearly visible. Two further peaks 5 and 6 have not been identifed. The same diode produces the DLT spectrum shown in Fig. 6 when a light pulse instead of a bias reduction is used to change the charge state of the traps. A new negative peak 11 is present. We assume that peak 11 is due to the triply ionized copper acceptor (Cu^{---}).

Table 1

Peak No.	C_1	E(meV)	Impurity Correlation
1	5.2×10^7	60.4	copper, hydrogen
2	1.6×10^6	68.7	oxygen
3	2.3×10^7	179.5	copper, hydrogen
4	2.3×10^8	300	copper, doubly ionized acceptor (Cu^{--})
5	not identified		
6			
7	3.6×10^7	67.4	divancy-hydrogen, singly ionized acceptor
8	not identified		
9			
10	6.7×10^7	151.8	copper, lithium, hydrogen
11	--	--	minority carrier trap; copper, triply ionized acceptor (Cu^{---}).

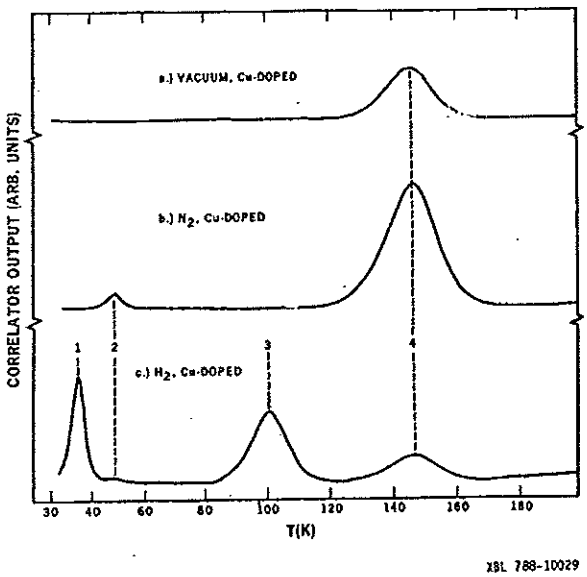


Fig. 4 DLT spectra of three high-purity germanium diodes which were intentionally doped with copper.

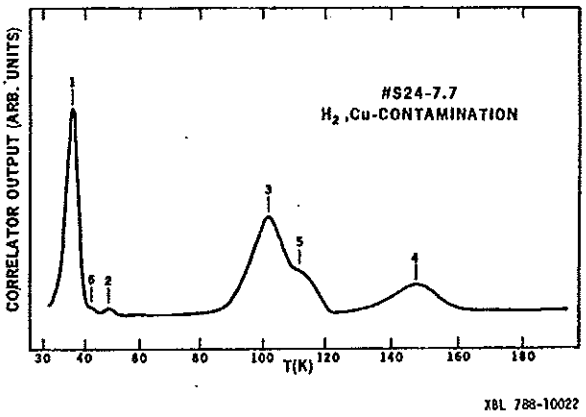


Fig. 5 DLT spectrum of diode S24-7.7. Peaks 1, 3 and 5 clearly indicate the presence of copper.

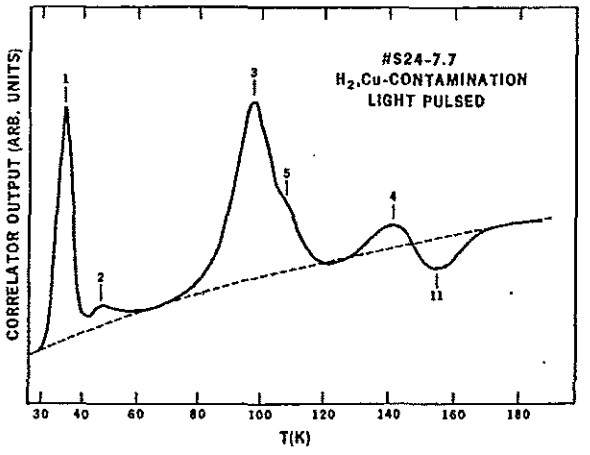


Fig. 6 Same diode as in Fig. 5 but light instead of bias pulses are changing the charge state of deep traps. A minority trap produces a negative peak 11.

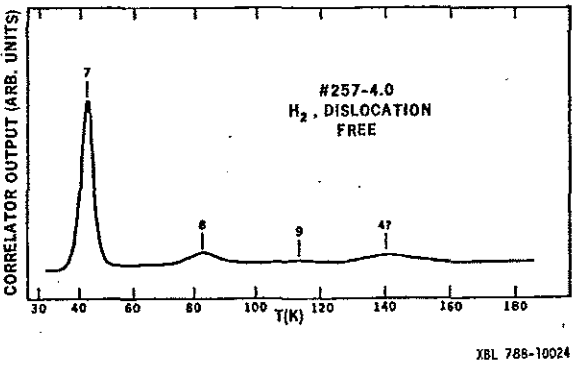


Fig. 8 Diode 257-4.0 is made from dislocation free high-purity germanium. Peak 7 is due to the divancy-hydrogen complex. Spurious peaks 8 and 9 have not been identified.

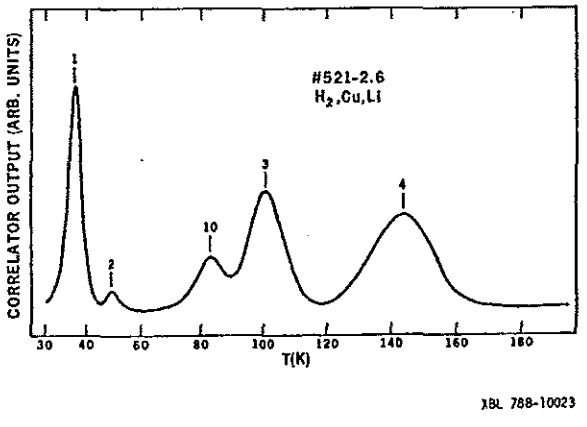


Fig. 7 DLT spectrum of diode 521-2.6. The Li-n^+-contact was diffused for prolonged times. Peak 10 is most probably due to a lithium-copper complex. All peaks characteristic for copper are clearly present (peaks 1, 3 and 4).

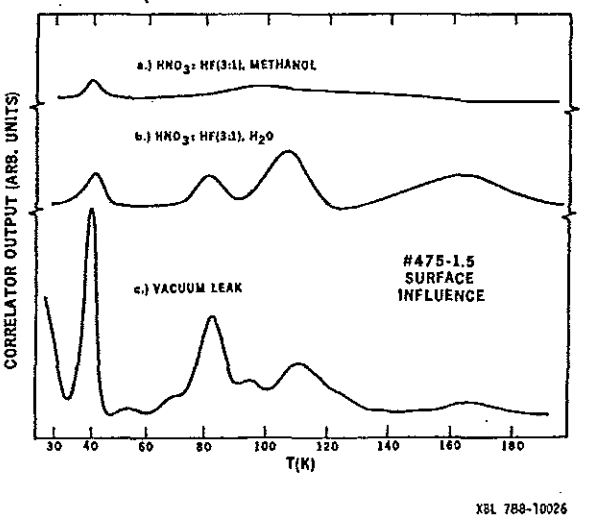


Fig. 9 DLT spectra of the same n-type high-purity germanium diode 475-1.5. Different surface conditions lead to a number of peaks.

Diode 521-2.6 exhibits all copper and oxygen related peaks (Fig. 7). A new peak 10 is present. This diode has a very deeply diffused lithium n^+ -contact. We correlate peak 10 with a lithium-copper acceptor. There is no doubt that detectors S24-7.7 and 521-2.6 contain copper, and it is most probable that the various copper related acceptors are the source of trapping.

Dislocation free germanium contains the well known divacancy-hydrogen complex.[9] A diode made from dislocation free material exhibits a very pronounced peak around 43 K (Fig. 8). The energy determined from Hall effect measurements and from DLTS are in good agreement. To caution future DLT spectroscopists, we include DLT spectra of the very pure n-type diode 475-1.5. Different surface treatment or contamination produces a series of peaks (Fig. 9). These peaks are due to surface states. This result suggests the use of grounded guard-ring structures for these measurements. Such devices would allow the separate study of bulk and surface levels.

Discussion and Conclusions

It is not the goal of this paper to present a complete study of a large number of deep levels in high-purity germanium. Our intention is to demonstrate how powerful DLT spectroscopy is and what kind of studies can be undertaken. From our experiments, which deal exclusively with majority traps in n^+-p-p^+ diodes, it becomes clear that the method is very sensitive and selective. Both levels in the germanium bulk and at the surface can produce DLTS peaks. We foresee the possibility of using this new spectroscopy for the identification of surface contaminants and pollution studies. Experiments to evaluate the use of very pure Ge detectors as fast neutron integrating flux monitors are in progress. It is known that the defects produced by the fast neutrons create deep levels in the germanium. If these levels are not too broadly spread in the forbidden band, it should be possible to obtain a rather sensitive fast neutron monitor. With the new possibility to determine the energy and concentration of deep traps in high-purity Ge, it will be possible to predict the performance of large volume gamma-ray detectors.

Acknowledgments

We are indebted to F. S. Goulding for his steady interest and encouragement in our work. T. Sun has performed many of our DLTS measurements. We would like to thank G. L. Miller, Bell Telephone Laboratories, for his assistance in the construction of the DLT spectrometer.

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7.2.6. Minority Carrier Lifetime

In chapter 1.3, we discussed free carrier generation and recombination processes. The dominant physical parameter describing the return of a semiconductor to its equilibrium condition is the minority carrier lifetime τ. Using deep level defects and impurities, this lifetime can be changed and controlled over many orders of magnitude. Short lifetimes are important for fast switches while long lifetimes lead to efficient p-n junction rectifiers and solar cells as well as high gain junction n-p-n and p-n-p transistors.

Most of the techniques developed to measure the minority carrier lifetime make use of optical excitation and an electrical determination of the change in conductivity with time. Optical excitation combines several advantages over other methods. It covers a very wide range of time scales. It is non-invasive and the intensity can be controlled over a very wide range. We will briefly discuss the Haynes-Shockley experiment[7.7] which makes use of minority carrier injection from a metal point contact and allows the measurement of several fundamental properties: free carrier drift mobility, diffusivity and lifetime. Figure 176 shows the experimental setup. A relatively small bias is applied across a long bar of an n-type semiconductor generating a constant electric field along the bar. At position x = 0 resides a point contact (either a metal tip pressed onto the semiconductor surface or better, a small p-type region formed by acceptor ion implantation, acceptor diffusion or appropriate alloying) which can inject a short pulse of holes, minority carriers. The holes drift in the electric field to the right and are collected at a point x = d. For this purpose, a reverse biased p-type region is formed. Upon closer inspection, the experimental setup is nothing but a p-n-p transistor with a very wide base layer! The drift mobility is readily obtained from the time t it takes the holes to move from point x = 0 to x = d.

$$v = \frac{d}{t} = \mu_h E$$

and

$$\mu_h = \frac{d}{Et}$$

[7.7] J.R. Haynes and W. Shockley, Phys. Rev. **81**, 835 (1951).

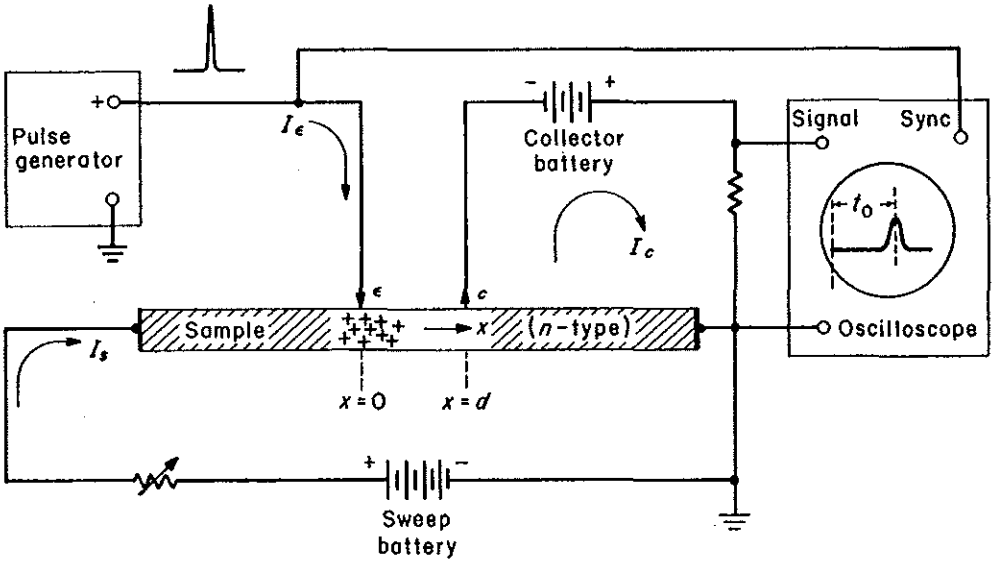


Fig. 176. A schematic diagram of the Haynes-Shockley experiment. [After J.P. McKelvey, "Solid State and Semiconductor Physics," Krieger Publishing Co, Malabar, Florida, 1982, p. 343.]

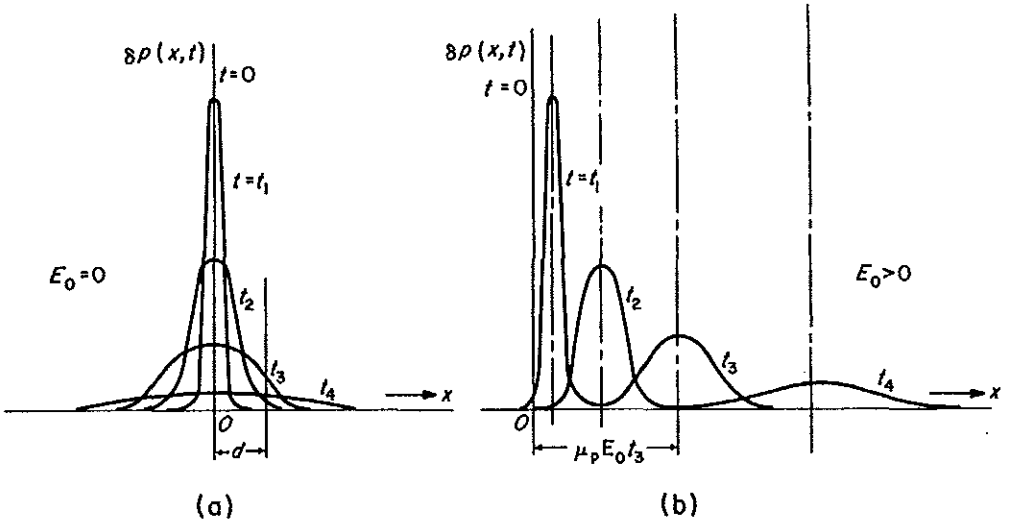


Fig. 177. (a) Diffusive spread of excess carriers injected initially as a δ-function distribution at x=0, with no applied electric field. (b) Diffusive spread and drift of the distribution in (a) with a constant electric field in the x-direction. In both cases, $0 < t_1 < t_2 < t_3 < t_4$. [After J.P. McKelvey, "Solid State and Semiconductor Physics," Krieger Publishing Co, Malabar, Florida, 1982, p. 341.]

Figure 177 illustrates the hole pulse assumed to be a δ-function at injection. The hole packet drifts and diffuses apart with time into a Gaussian shape described by

$$\Delta n_h(x,t) = \frac{\Delta n_o e^{-t/\tau_h}}{\sqrt{4\pi Dt}} \exp[-(x-\mu_h Et)^2/4Dt]$$

with Δn_h = excess hole concentration as a function of time and space, Δn_o = total number of holes injected per pulse at t = 0, D = hole diffusion coefficient, τ_h = hole lifetime.

The holes arriving at the depletion layer under the extracting contact will be collected rapidly, producing a voltage drop across the load resistor. This voltage drop is displayed on the y-axis of an oscilloscope. The peak signal voltage V_{max} is related to the drift time t by

$$V_{max} = \frac{1}{\sqrt{t}} \exp(-t/\tau)$$

Plotting the logarithm of $V_{max}\sqrt{t}$ as a function of t leads to a straight line with a slope $-1/\tau$.

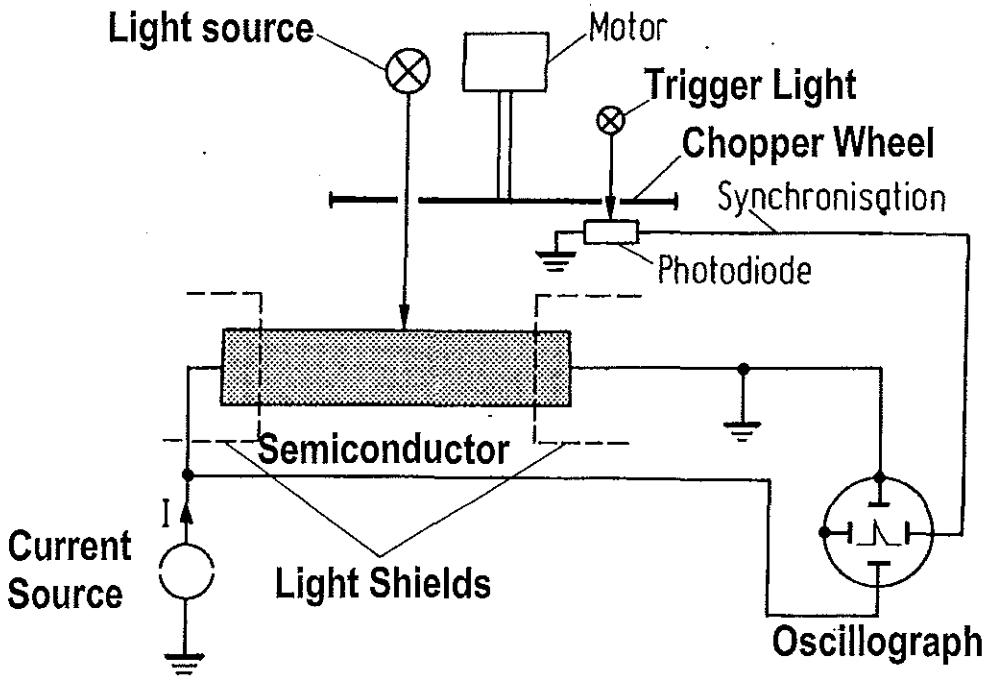


Fig. 178. Measurement of lifetimes by conductance change due to injection of carriers by illumination. [After I. Ruge, "Halbleiter Technologie,", Springer-Verlag, Berlin, 1975, p. 205.]

Photoconductivity decay measurements requiring optical excitation are often used to determine the minority lifetime (Fig. 178). A semiconductor bar is illuminated by brief optical pulses ($h\nu > E_{GAP}$) produced by the light source and the motor-driven chopper wheel. The increase in conductivity is measured by applying a constant current source across the semiconductor bar. The change in the bias voltage ΔV across the bar is at time $t = 0$, ΔV_0 and decreases exponentially as the excess carriers disappear through recombination to the point where the steady state bias value is reached.

$$\Delta V = \Delta V_0 \exp(-t/\tau).$$

These two examples shall suffice to illustrate how the decay of excess carrier concentrations can be measured. A word of caution should, however, be added. The lifetime or times one measures typically depend on many important details of the excitation and recombination processes. After all, many different physical processes, each with its characteristic time constant may proceed over some time interval in parallel.

In summary, it should be stated that the minority carrier lifetime of a semiconductor is often a good measure of its quality. For silicon, one finds that the lower the deep level concentration caused by impurities and defects[7.8] is, the longer is the minority carrier lifetime. The best floating zone Si single crystals exhibit lifetimes exceeding 10 ms.

7.3. Optical Characterization

References:

- "Optical Characterization of Semiconductors," S. Perkowitz, (Academic Press, New York 1993)
- "Fundamentals of Semiconductors," P.Y. Yu and M. Cardona, (Springer, Berlin 1996).

[7.8] H.-J. Queisser and E.E. Haller, Science **281**, 945 (1998).

7.3.1. Introduction

Optical studies belong to one of the richest areas of solid state and semiconductor physics. The number of possible excitation paths and recombination processes in semiconductors is very large indeed, covering an energy range from below meV to tens of eV. Energy absorption, as well as emission, can be provided by photons alone but in many cases, electrons, atoms or ions can partake. Photoelectron spectroscopy, for example, uses photons to emit electrons out of a semiconductor surface while cathodoluminescence is the process of light emitted from a semiconductor surface upon bombardment with energetic electrons. A discussion of all possible techniques involving photons would fill many volumes. Here we will concentrate on near and far infrared linear absorption spectroscopy, most useful for measuring bandgaps and identifying deep and shallow level defects and impurities. The text of S. Perkovitz provides a focused and most useful description of photoluminescence and Raman spectroscopy, the other two widely used optical characterization techniques.

There are many reasons why photons are exceptionally well suited for semiconductor characterization. Photons essentially provide "pure" energy and can be produced monochromatically at very high intensities (e.g., lasers, electron synchrotrons) over an extremely wide energy range. Photons can be manipulated easily and sent over large distances. At energies exceeding ~ 1 eV, single photons can be counted. Photons travel fast and dominate the highest frequency processes in semiconductors. They also offer the highest time resolution (femtoseconds, 10^{-15} s).

7.3.2. Electronic Transitions

7.3.2.1. Bandgap (or Fundamental) Absorption Spectroscopy

Perhaps the simplest use of photons for the characterization of a semiconductor is linear absorption spectroscopy. One measures the intensity of a monochromatic photon beam with intensity I_0, arriving at a semiconductor sample and after passing through the sample, I_t (Fig. 179).

For many materials, one finds experimentally the transmission T to be

$$T = \frac{I_1}{I_0} = \exp\ (-\alpha x)$$

with x = sample thickness and α = linear absorption coefficient. This simple equation neglects reflection from the sample surfaces.

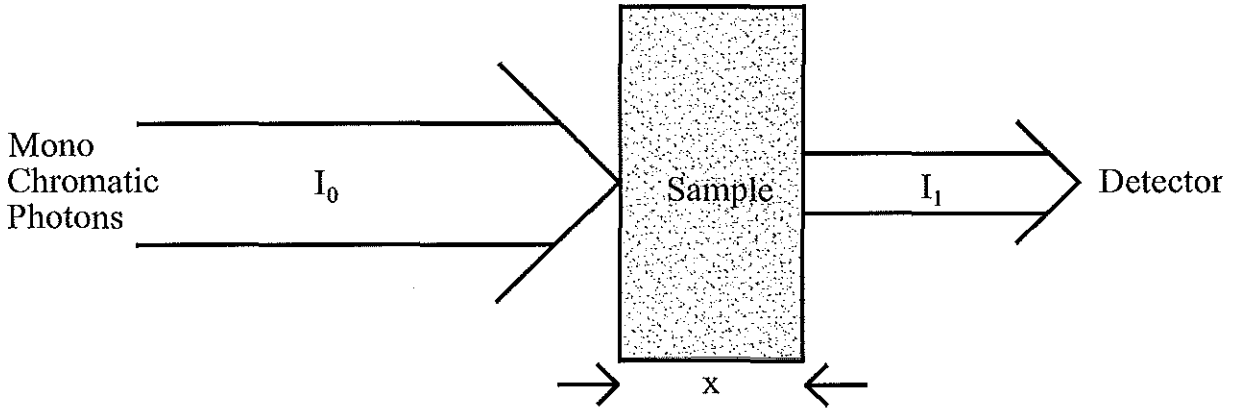


Figure 179.

When taking into account multiple reflections from the same surfaces in a vacuum, one finds:

$$T = \frac{(1-R)^2\ \exp(-\alpha x)}{1-R^2\ \exp(-2\alpha x)}$$

with the reflectance R given by

$$R = \frac{(1-n)^2 + k^2}{(1+n)^2 + k^2}$$

with n = refractive index and k = extinction coefficient of the semiconductor at the photon energy under consideration.

The extinction coefficient k and the linear absorption coefficient α are related by

$$\alpha = \frac{4\pi k}{\lambda}$$

with λ = wavelength.

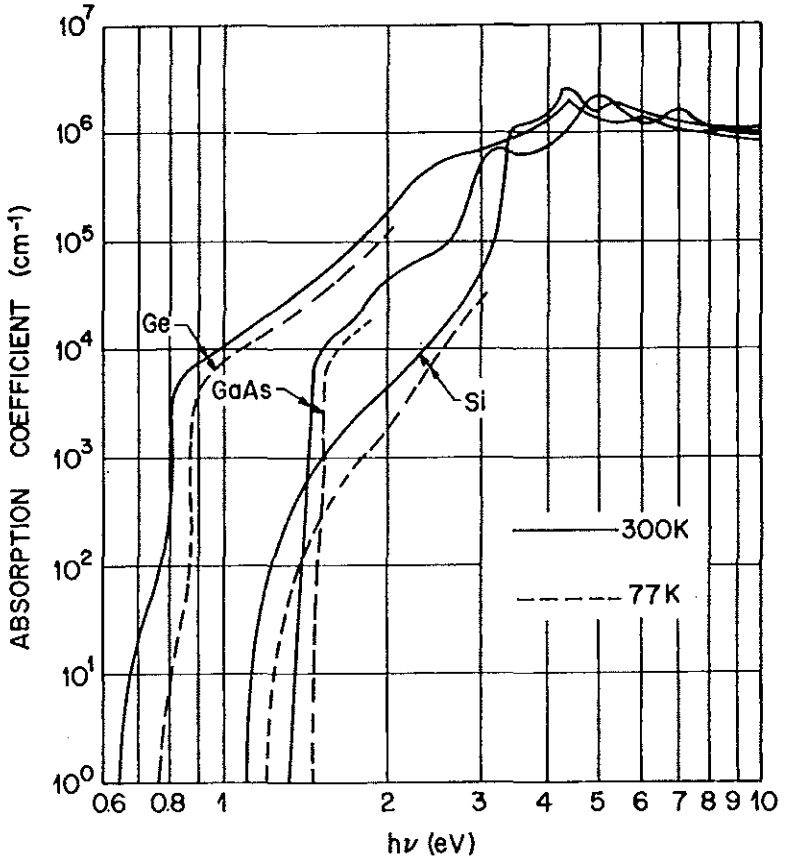


Fig. 180. Measured absorption coefficients near and above the fundamental absorption edge for pure Ge, Si and GaAs. [After S.M. Sze, "Physics of Semiconductor Devices," John Wiley & Sons, New York, 1981, p. 42; see W.C. Dash and R. Newman, Phys. Rev. **99**, 1151 (1955); H.R. Philipp and E.A. Taft, Phys. Rev. **113**, 1002 (1959) and Phys. Rev. Lett. **8**, 13 (1962); D.E. Hill, Phys. Rev. **133**, A866 (1964).; H.C. Casey, Jr., D.D. Sell and K.W. Wecht, J. Appl. Phys. **46**, 250 (1975).]

Figure 180 shows the extremely strong dependence of α on photon energy for GaAs, Si and Ge at room temperature and at 77K. The linear absorption coefficient α changes by over six orders of magnitude near the fundamental absorption edge caused by the emission of electrons across the direct bandgap. The change in α is especially abrupt for the direct bangap semiconductor GaAs. The rapid ascent of α to values exceeding ~ 10^4 cm^{-1} for Ge and Si is preceded by an energy region of lower α values. This is the indirect bandgap range where in addition to the photon, phonons are required for the conservation of k, the crystal momentum (Fig. 181). With the introduction of several new binary, ternary and quaternary semiconductors in recent years (e.g., GaN, InGaN, AlGaN, GaAsN, GaAsSb, ZnSeS, etc.), the optical

determination of the bandgap has experienced a kind of comeback. It is fortuitous that α reaches values as high as 10^6 cm^{-1} because it allows absorption (and reflection!) measurements of very thin semiconductor films. This is especially advantageous for some of the new semiconductor alloys that can only be grown in thin film form (e.g., $GaAs_{1-x}N_x$; $x \leq 0.05$).

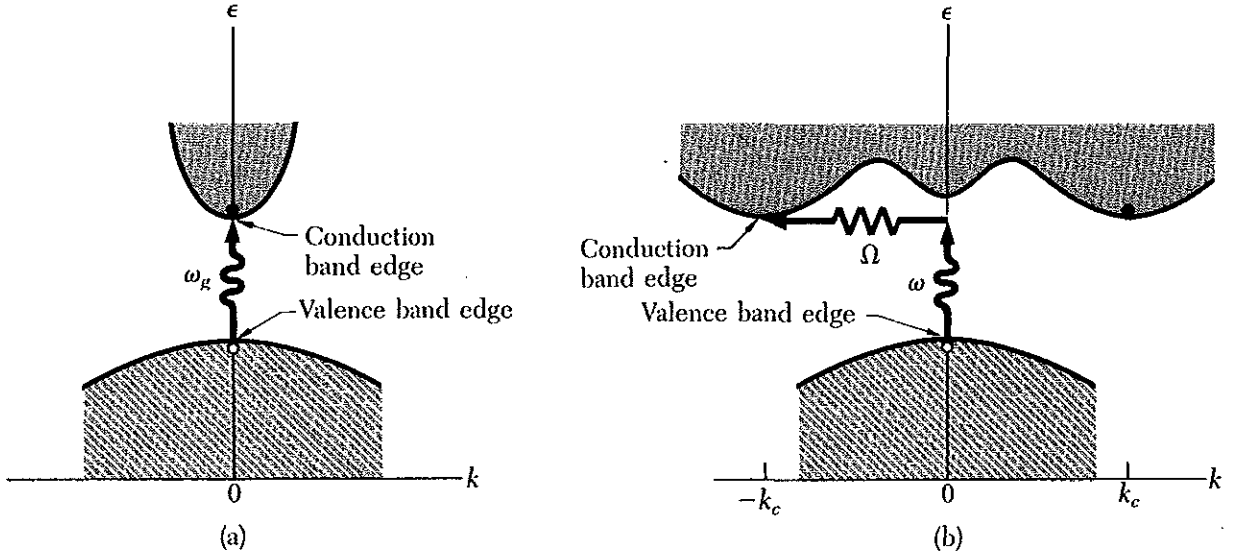


Fig. 181. In (a) the lowest point of the conduction band occurs at the same value of k as the highest point of the valence band. A direct optical transition is drawn vertically with no significant change of k, because the absorbed photon has a very small wavevector. The threshold frequency ω_g for absorption by the direct transition determines the energy gap $E_g = \hbar\omega_g$. The indirect transition in (b) involves both a photon and a phonon because the band edges of the conduction and valence bands are widely separated in k space. The threshold energy for the indirect process in (b) is greater than the true band gap. The absorption threshold for the indirect transition between the band edges is at $\hbar\omega = E_g + \hbar\Omega$, where Ω is the frequency of an emitted phonon of wavevector $K \cong -k_c$. At higher temperatures phonons are already present; if a phonon is absorbed along with a photon, the threshold energy is $\hbar\omega = E_g - \hbar\Omega$. Note: The figure shows only the threshold transitions. Transitions occur generally between almost all points of the two bands for which the wavevectors and energy can be considered. [From C. Kittel, "Introduction to Solid State Physics, 6th Ed., John Wiley & Sons, New York, 1986, p. 186.]

7.3.2.2. Far Infrared Spectroscopy of Shallow Acceptors and Donors

Shallow donors and acceptors are quite well described by the hydrogenic model (see Chapter 1). As hydrogen atom analogs, we expect them to have a ground state and bound excited states for the bound carrier (electron for donors, hole for acceptors). Photons can induce transitions between the s-like ground state and all non-s-like bound excited states. Several

examples of such far infrared absorption spectra have been presented in the first chapter. For the typical semiconductors Si, Ge, GaAs, the ground state binding energy for electrons or holes lies in the range from a few meV to a few tens of meV. While this range also covers donors in the currently very popular group III-nitrides and their alloys, acceptors in these materials are rather deep, of the order of 150 to 280 meV. Far infrared Fourier transform spectroscopy is ideally suited for the study and characterization of shallow level impurities.

In order to be able to induce far IR transitions in donors or acceptors, these dopants must have a carrier occupy the ground state. This means the semiconductor has to be deep in the freeze-out regime. Temperatures in the liquid Helium range are suitable. Compensating donors or acceptors remain ionized at all temperatures and are not accessible with far IR spectroscopy. This limitation can be overcome by shining low intensity bandedge light (i.e., light with $h\nu \gtrsim E_{GAP}$) on the sample. Many of the generated free electrons and holes can get trapped by acceptors and donors, rendering them neutral.

Why are Fourier transform spectrometers used in far IR spectroscopy? In order to appreciate the advantages which can be realized with a Michaelson interferometer over a grating spectrometer, we briefly review the operation of an interferometer.[7.9] Figure 182 shows a two-arm Michaelson interferometer. Broadband light from a source A (e.g., an electronically heated SiC rod called a glowbar or a Hg arc lamp) is directed through a small aperture and a chopper B onto an off-axis parabolic mirror C which forms a parallel beam. This beam is split by a semi-transparent mirror D (beam splitter, BS) into two beams at an angle of 90°. The two beams are reflected from plane mirrors, one stationary G, the other mobile E. The motion of mirror E, precisely parallel to the optical beam axis, can be computer controlled via a stepper motor. The beams reflected from mirrors G and E return to the beam splitter and recombine, forming a fourth beam which is focused by a mirror H through some spectral filters I onto the detector. Imagine for a moment that the stationary and the mobile mirror are at precisely the same distance from the beam splitter. The two components of photons split at the beam splitter will recombine precisely in phase and, neglecting all losses, the exit beam looks like the entrance beam. If, however, one

[7.9] R.J. Bell, "Introductory Fourier Transform Spectroscopy," (Academic Press, New York 1972).

optical paths is longer or shorter than the other, there will be an interference pattern in the intensity as a function of photon frequency in the exit beam.

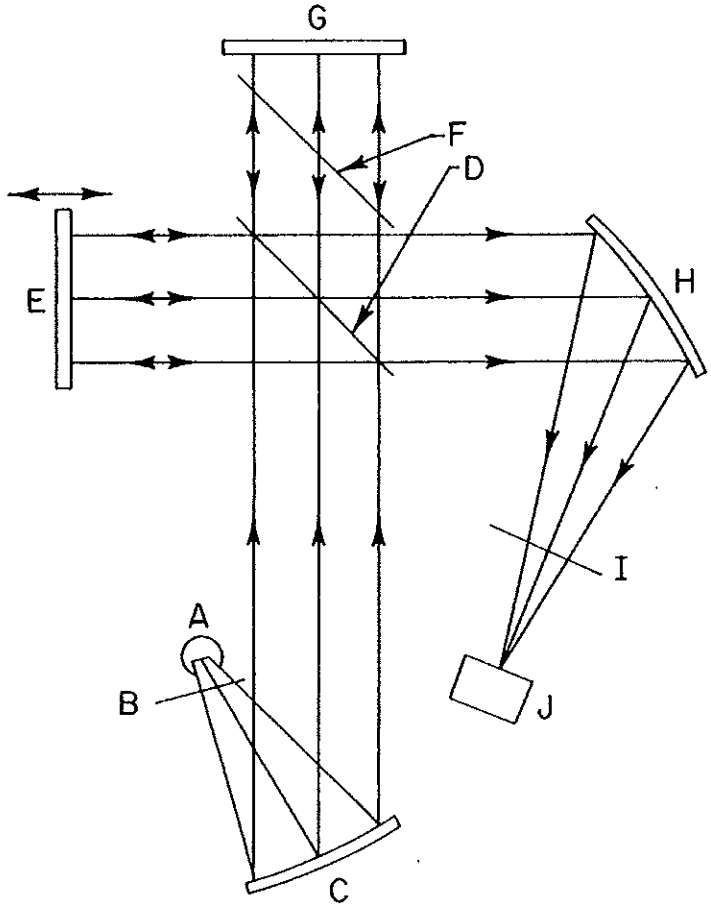


Fig. 182. Basic Michelson interferometer. A: Source. B: Chopper. C: Collimator. D: Beamsplitter. E: Movable mirror. F: Compensator. G: Fixed mirror. H. Focusing mirror. I: Spectral filters. J: Detector. [R.J. Bell, "Introductory Fourier Transform Spectroscopy," Academic Press, New York, 1972, p. 18.]

If the optical path difference Δ (= $2\Delta x$, Δx = mechanical path difference) is equal to $n\lambda$, the two beams recombine constructively but they annihilate each other if $\Delta = (2n+1)\frac{\lambda}{2}$ (n=0, 1, 2...). Except for the unique condition $\Delta = 0$, the exit beam will show an intensity modulation pattern called interferogram. Interferograms can be transformed mathematically back into spectra by Fourier transformation. Let us consider a few simple cases. We assume that the

source A is a monochromatic laser. The interferogram will be a cosine pattern with constant amplitude:

$$I = I_0 \cos\left(\frac{2\pi\Delta}{\lambda}\right) = I_0 \cos\left(\frac{4\pi\Delta x}{\lambda}\right)$$

In case the laser has a finite bandwidth, the cosine amplitude will decay slowly.

In case two lasers with closely spaced frequencies are located at A, two cosine patterns which will "beat" with each other will be generated. The interferogram will be again a cosine but with an amplitude varying with the difference frequency of the two laser frequencies. The example shown in the next figure approximates this case quite closely. Figure 183a shows an interferogram with strong "beats" slowly diminishing in amplitude as the moving mirror travels away from the zero path position. The Fourier transformed interferogram is shown in Fig. 183b. It is the sum of the ground state to bound excited state spectra of B, Al and Ga acceptors in a sample of ultra-pure germanium. This is the same spectrum as shown in Fig. 18 on page 37.

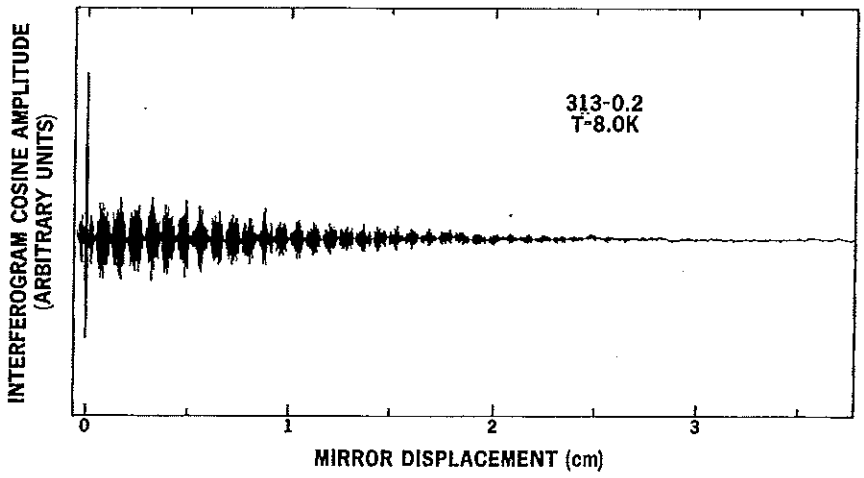


Fig. 183a. Cosine part of the photoconductivity interferogram of an ultra-pure germanium sample containing mainly aluminum as the residual acceptor impurity. The beat character is caused by two strong lines (C and D). [After E. E. Haller, W. L. Hansen and F. S. Goulding, Advances in Physics **30**, 93 (1981).]

Next, let us place a sample of a material with very narrow optical band path at the filter position I and a broadband source at A. The sample lets only pass photons with a specific wavelength and the detector will only "see" what happens to photons of this wavelength. This is

the same case as discussed above for the single laser and the interferogram will be a cosine pattern with a slowly decaying amplitude. These examples suffice to show how specific interferograms are transformed back into the original spectrum with a Fourier transformation.

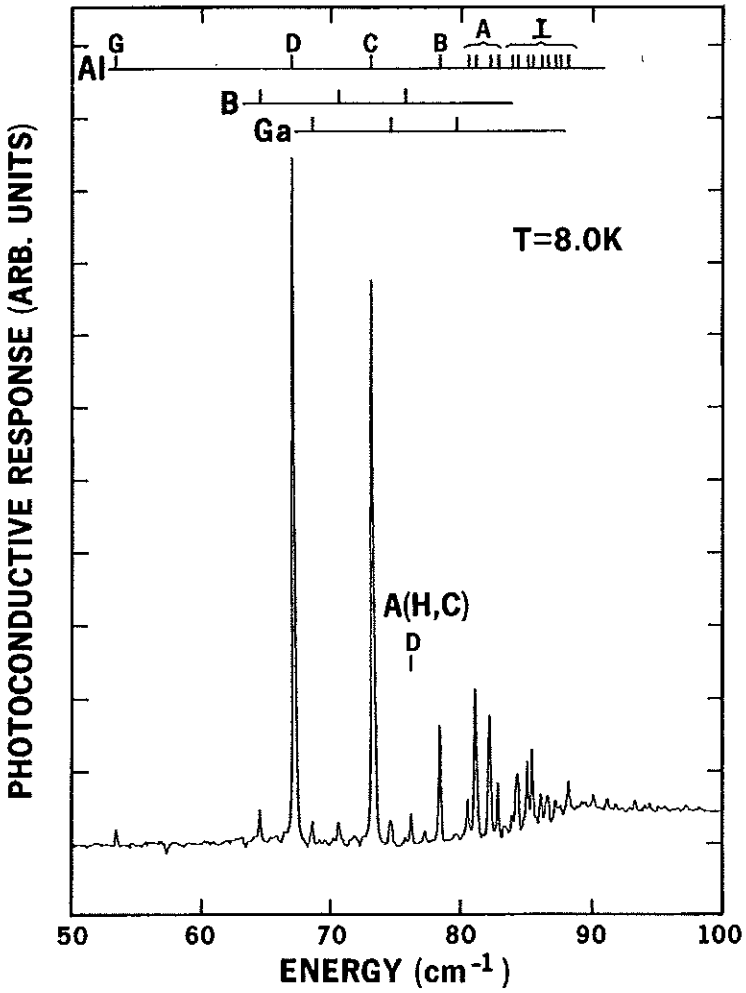


Fig. 183b. Fourier transform spectrum of the interferogram shown in Fig. 183a. The assignment of the lines is according to [*For donors:* J.H. Resuzer and P. Fisher, Phys. A **135**, 1125 (1964); *For acceptors:* R.L. Jones and P. Fisher, J. Phys. Chem. Solids **26**, 1125 (1965).]. Aluminum (Al) is the dominant acceptor. The hydrogenic sets of lines due to boron (B) and gallium (Ga) are also present. [After E. E. Haller, W. L. Hansen and F. S. Goulding, Advances in Physics **30**, 93 (1981).]

After this brief introduction to Fourier transform spectroscopy based on the Michaelson interferometer, we give reasons why this technique is preferred over normal grating spectroscopy, especially in the far IR region of the spectrum. There are two major advantages called

Jacquinot[7.10] or throughput advantage and Fellgett[7.11] or multiplex advantage. The Jacquinot advantage can be understood easily when one recognizes that the optical throughput of an interferometer is to first order independent of the resolution one wishes to achieve. The resolution is simply proportional to the number of times a certain frequency is "probed" by the moving mirror. In contrast, the grating spectrometer has a slit which has to be closed down as the required resolution is increased. It becomes evident that very high resolution far IR spectroscopy profits greatly from using interferometry.

The Fellgett advantage is clearly related to the signal-to-noise ratio at the detector. For constant white noise, one finds that the signal-to-noise ratio is better by $\sqrt{M}$ with M = optical bandwidth/optical resolution. M is the number of frequency intervals into which we divide the full spectral range used. In simple terms: in an interferometer the same optical frequency is seen by the detector every time the two reflected beams recombine constructively while in a grating spectrometer, each optical interval is detected once.

Why did Michelson not use his interferometer to build Fourier transform spectrometers? The reason is very simple: Fourier transformation was a laborious mathematical process which took a lot of time for every spectral element. Indeed, Michelson built a mechanical analog computer which allowed him to compute 80 Fourier components. He called his machine "Harmonic Analyzer." To fully resolve donor or acceptor spectra in pure, high quality semiconductors, we require thousands of spectral points. While this can be accomplished in seconds with modern computers, the author remembers the time when special vector processors were required to obtain acceptably short computing times. In the 1960s and 1970s, researchers in the Soviet Union could not get access to sufficiently powerful computers and they had to resort to huge gratings (1m × 1m) ruled on polished granite slabs!

A great leap forward in Fourier transform spectroscopy was made when the Cooley-Tuckey algorithm[7.12] for rapid Fourier analysis was invented. This algorithm depends only

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[7.10] A. Girard and P. Jacquinot in "Advanced Optical Techniques," A.C.S. Van Hell, ed. (North Holland Publ. Co., Amsterdam 1967), p. 71, 109.
[7.11] P. Fellgett, J. Phys. Radium **19**, 187 (1958).
[7.12] J.W. Cooley and J.W. Tuckey, Math. Comp. **19**, 297 (1965).

linearly on the number of spectral points N while traditional Fourier analysis goes with N^2. Fourier analysis also has become a crucial component in the processing and transmission of electrical signals.

But let us return to the spectroscopy of donors and acceptors in semiconductors. The examples of shallow dopants shown in the first chapter were all generated with Fourier Transform Infrared (FTIR) spectroscopy . The improvement in semiconductor crystals and in FTIR have gone hand in hand and linewidth of $\leq$ 10 μeV can be fully resolved with today's instruments.

As the semiconductor to be investigated with FTIR becomes purer, the absorption lines become weaker and eventually disappear in the noise. In this case, the combination of two measurement techniques has led to spectacular results.

What happens when we attach ohmic contacts to the sample under infrared absorption study and measure its conductivity? Instead of measuring the reduction of photon flux, we detect directly what is going on inside the sample, namely the optical generation of free carriers from neutral donors or acceptors. In a cooled semiconductor which contains very few carriers in the absence of a photon flux, we can create free carrier *changes* of many orders of magnitude. Shallow levels which have very rich "hydrogenic" bound excited states spectra are especially well suited for photoconductive studies because their concentration may be small in pure materials (sensitivity) and because there exist no intense broadband photon sources in the far infrared range. Figure 184 shows the change in conductivity of a small germanium single crystal sample as a function of the photon energy (or wavenumber $\nu = \lambda^{-1}$ (cm^{-1})). At low sample temperatures (T = 4.2 K), one only observes the photoconductive continuum (1) which is due to the excitation of a bound electron from antimony donors into the conduction band. As the temperature is raised to 10 K, we begin to detect lines below the onset of the photoconductive continuum. These lines (or peaks) are due to a *two-step process* (Fig. 185): (1) excitation from the ground state of an impurity into one of the bound excited states followed by (2) the absorption of a phonon which transports the bound carrier into the band, thereby creating a free charge carrier. This two-step process, called photothermal ionization, has been developed into a very sensitive and very high

resolution technique. Figure 186 shows such a spectrum of boron and aluminum acceptors in an ultra-pure germanium crystal sample. Two additional sets of hydrogenically spaced lines are visible [$A(D,C)_1$ and $A(D,C)_2$]. These lines originate from a shallow acceptor complex of carbon and hydrogen (in this particular case, deuterium was used instead of hydrogen). The "activation" of neutral dopants such as Si or C on Ge lattice places was one of the interesting discoveries that were made possible through the development of ultra-pure germanium. The lines are sufficiently sharp to allow a clear separation between A1, B and Ga acceptor lines. The PTIS technique has been extensively used in recent years to investigate high purity semiconductors, especially germanium and silicon. It has led to the discovery and understanding of numerous novel acceptor and donor levels.[7.13]

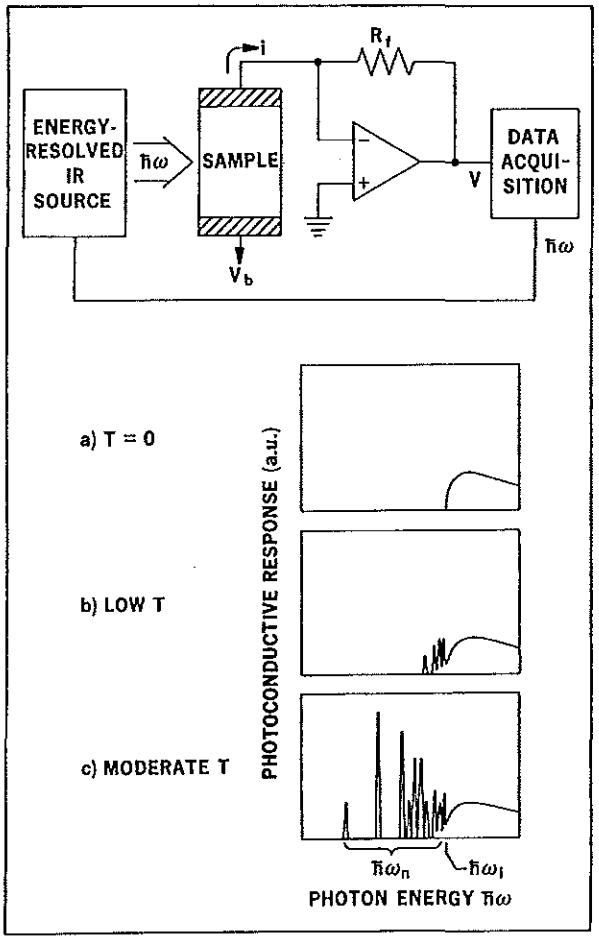


Fig. 184. Photothermal ionization spectroscopy (schematic). Typical spectra recorded at: (a) T = 0; (b) low T; (c) moderate T. [After J.M. Kahn, "Hydrogen-Related Acceptor Complexes in Germanium," PhD Thesis, UC Berkeley, 1986, p. 26.]

[7.13] E.E. Haller, W.L. Hansen and F.S. Goulding, Advances in Physics **30**, 93-138 (1981).

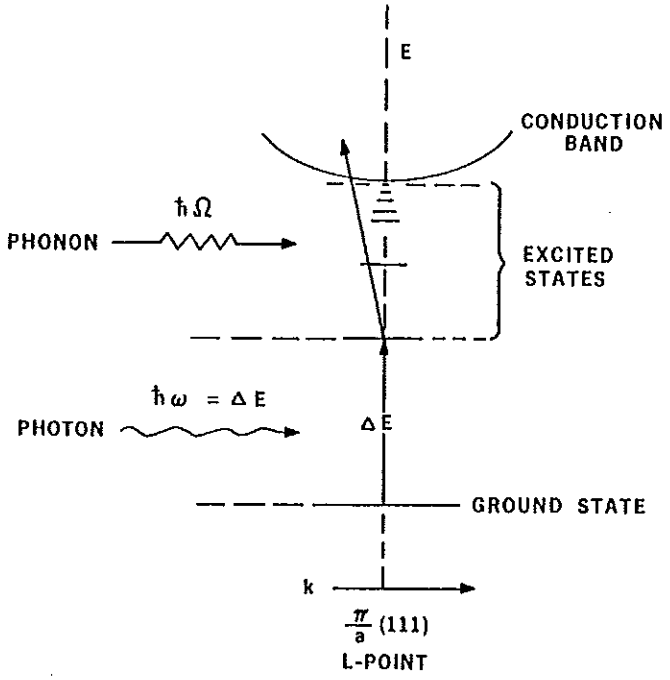


Fig. 185. Absorption of a photon followed by a phonon leads to a free carrier. [After E. E. Haller, W. L. Hansen and F. S. Goulding, Advances in Physics **30**, 93 (1981).]

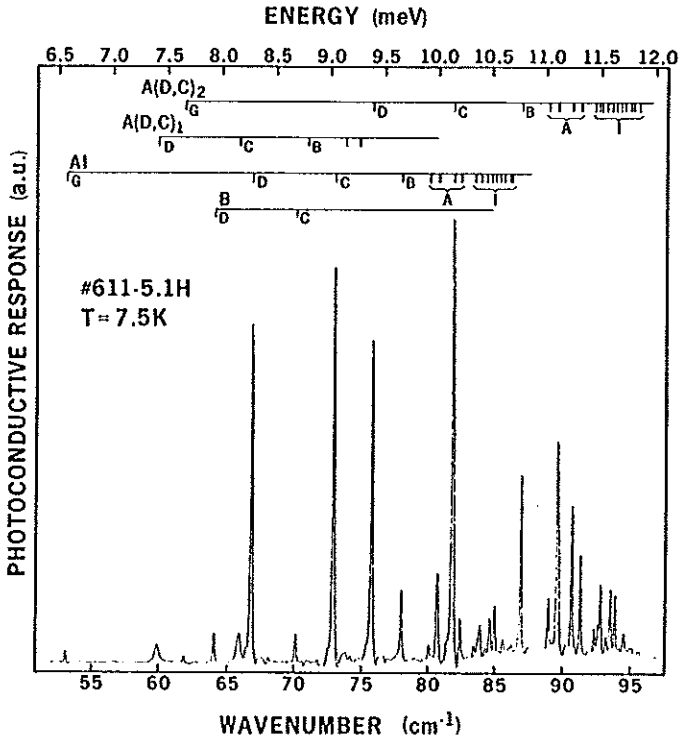


Fig. 186. PTI spectrum of an ultra-pure, p-type Ge sample containing the acceptors B, Al and A (D, C). The narrowest lines are ≈ 17 μeV wide. [After E. E. Haller, Advances in Solid State Physics **26**, 203 (1986).]

It is worthwhile to point out that the phenomenon of photoconductivity has also led to the development of very sensitive photon detectors, transducers which convert a photon signal into an electrical signal. Semiconductors doped with specific impurities can be optimized for photon energy ranges over a wide spectral region. Most spectacular scientific results have been published in recent years (1983-2000) with far-infrared photoconductors which were mounted and cooled to 2 K inside the Infrared Astronomical Satellite (IRAS). Various stages of star formation, intergalactic dust and over 200,000 new galaxies were discovered. The next liquid He cooled far infrared telescope satellite, the Space Infrared Telescope Facility (SIRTF) with many of our photoconductive detectors (which have been manufactured in my research group) on board will be launched in the year 2003.

7.3.3. Vibronic Transitions: Local Vibrational Mode Spectroscopy

The intrinsic vibrational modes of a semiconductor lattice have been discussed in the first chapter. We recognized that such vibrations cannot be excited in a purely covalently bonded semiconductor such as silicon or germanium. Only combination modes are optically active. Partially ionically bonded semiconductors on the other hand show strong "Reststrahlen" bands originating in excitation of optical phonons through the absorption of photons.

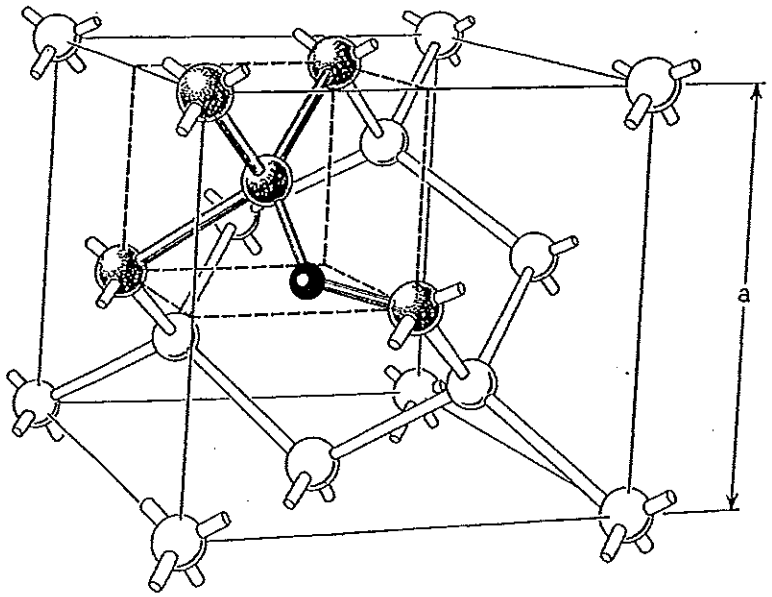


Fig. 187. The diamond structure, showing how each atom forms four bonds with its nearest neighbors. [After W. Shockley, "Electrons and Holes in Semicoductors," D. Van Nostrand Co., New York, 1950, p. 6.] The small black sphere near the center represents an oxygen impurity in a bond centered position.

A special and important situation arises when impurities with atomic masses smaller than the masses of the host lattice atoms are present in a semiconductor. Depending on the details of the local bonding structure, the symmetry and the geometry of the impurity (or impurity-defect complex), one or more characteristic vibrational modes characterize the center. For example, oxygen in silicon and germanium occupies a bond centered interstitial position (Fig. 187). Modeling the oxygen and its nearest neighbors with a Si-O-Si molecule leads to three vibrational modes, one of them of high energy (high frequency). This mode couples well to photons and leads to a strong absorption feature in a transmission spectrum at $\nu = 1107$ wavenumbers* for a room temperature silicon sample (Fig. 188a). At low temperature, the absorption line becomes significantly sharper, showing isotope mass features (Fig. 188b). The intensity of this local vibrational mode peak is proportional to the oxygen concentration in the semiconductor sample. The generally agreed upon conversion factor between oxygen concentration [O] and the peak linear absorption coefficient α is

$$[O] \cong 3.1\times10^{17}(cm^{-2})\cdot\alpha(cm^{-1})^{7.14}$$

This dependence is widely used in industry and research laboratories to measure the concentration of atomically dispersed oxygen impurities in silicon. Upon cooling of the semiconductor sample, local vibrational mode spectra become significantly sharper because the lattice phonon density decreases.

Focusing for a moment on germanium, we find the low temperature oxygen vibrational mode spectrum to consist of 11 individual peaks bunching near 862.5 cm^{-1}. Germanium has five stable isotopes, each present at significant concentrations. Each combination of germanium atoms (^{x}Ge-^{16}O-^{y}Ge) neighboring the oxygen impurity leads to a unique frequency (Fig. 189a). Combinations of germanium neighbor masses (^{x}Ge + ^{y}Ge) which are identical (e.g., ^{72}Ge + ^{74}Ge = ^{73}Ge + ^{73}Ge or ^{72}Ge + ^{76}Ge = ^{74}Ge + ^{74}Ge) lead to identical frequencies, reducing the multiplicity.

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* wavenumber $\nu = \dfrac{1}{\lambda\ (cm)} = \dfrac{f\ (Hz)}{c\ (cms^{-1})}$

7.14 K. Krishnan, P.J. Stout and M. Watanabe, in "Practical Transform Infrared Spectroscopy," J.R. Ferrero and K. Krishnan, eds., (Academic Press, San Diego 1990), pp. 285-349.

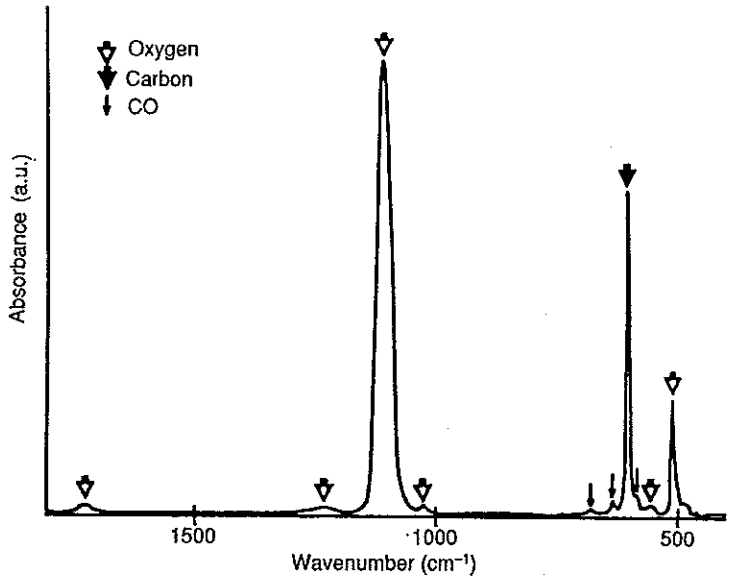


Fig. 188a. Difference between measured absorbances for float zone silicon and Czochralski silicon. The phonon peaks are eliminated, showing peaks due to interstitial oxygen, carbon and oxygen-carbon. [After S. Perkowitz, "Optical Characterization of Semiconductors," Academic Press, San Diego, 1993, p. 174; see K. Krishnan, P.J. Stout and M. Watanabe, in "Practical Fourier Transform Infrared Spectroscopy," J.R. Ferraro and K. Krishnan, eds., Academic Press, San Diego, 1990, pp. 285-349.]

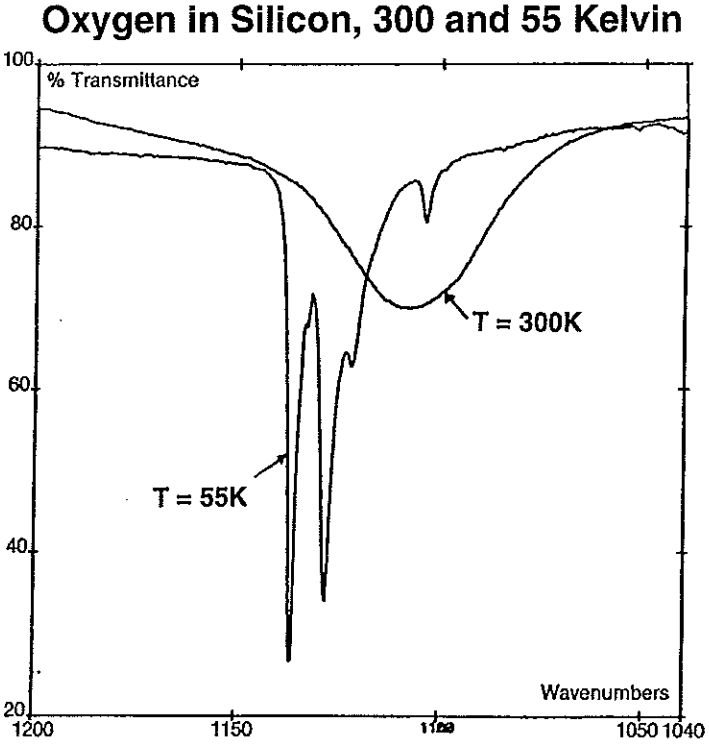


Fig. 188b. Far IR transmittance of silicon containing oxygen at 300K and 55K. The low temperature spectrum shows fine structure due to the isotopic composition of natural Si (^{28}Si, ^{29}Si, ^{30}Si).

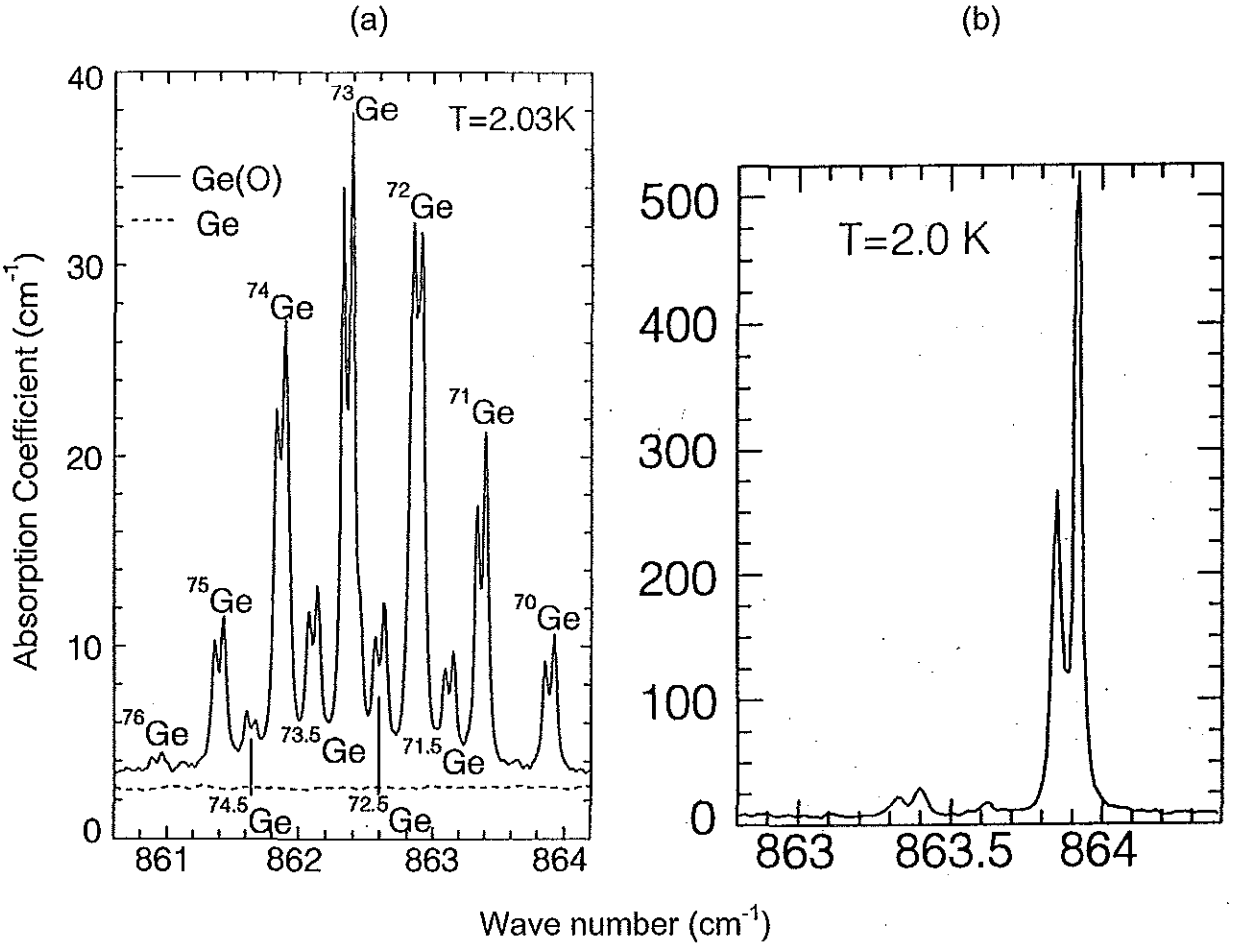


Fig. 189. Low temperature absorption spectrum of ^{nat}Ge doped with O. The eleven peaks arise from the eleven unique mass combinations of the Ge atoms neighboring the O atoms. The splitting of each line is due to non-linear super position of other vibrational modes of O. (b) is same spectrum as in (a) but with an isotopically pure ^{70}Ge crystal. [After A.J. Mayur, M. Dean Sciacca, M.K. Udo, A.K. Ramdas, K. Itoh, J. Wolk, and E.E. Haller, Phys. Rev. B **49**, 16293 (1994).]

That this absorption line multiplicity is indeed due to the various stable isotopes present in natural germanium has been proven directly in a recent experiment in which isotopically enriched germanium single crystals doped with oxygen were studied.[7.15] The spectrum of ^{70}Ge:O is shown in Fig. 189b. The eleven line absorption spectrum of ^{nat}Ge:O is reduced to one major peak. Residues of other Ge isotopes lead to small additional peaks.

[7.15] G. Sirmain, O.D. Dubon, W.L. Hansen, C.S. Olsen, and E.E. Haller, J. Appl. Phys. **79**, 209 (1996).

Oxygen in silicon (and germanium) is just one example (though a very important one!) demonstrating the power of LVM spectroscopy. Carbon in silicon is perhaps the second most important impurity measured by LVM spectroscopy. Barker and Sievers have written a comprehensive review on LVM spectroscopy in semiconductors.[7.16] The attached tables from this article, listing numerous impurities in many semiconductors which have been investigated with LVM spectroscopy, shall convey an impression of the possibilities offered by this characterization tool (Table 20). The sharpness of vibrational modes absorption lines, especially at low temperatures, and the sensitivity of the vibrational mode frequencies to the atomic mass of the impurity and its immediate neighbors make this tool highly selective and suitable for the identification of the vibrating atoms and their geometric configuration. M.D. McCluskey[7.17] has summarized numerous LVM studies of hydrogen containing impurity complexes in III-V and II-VI compound semiconductors

[7.16] A.S. Barker, Jr. and A.J. Sievers, Rev. Mod. Phys. **47**, Suppl. 2, S1 (1975).
[7.17] M.D. McCluskey, Reviews J. Appl. Phys., **87**, 3593 (2000).

Table 20. Localized modes in semiconductors and metals
[After A.S. Barker, Jr. and A.J. Sievers, Rev. Mod. Phys. 47, Suppl. 2, S1 (1975).]

Host and Impurity	Mode frequency (temperature) °K	Defect symmetry, Method of observation,[a] References,[b] Comments
Diamond		
N?	1340(300)	T_d?, A, 6
Silicon		
^{10}B	644(300), 646(80)	T_d, A, 1, 5, 7, 9, 14, 15, 16; R, 67
^{11}B	620(300), 622(80), ~230(300)	T_d, A, 1, 5, 7, 9, 14, 15, 16; R, 67; T, 79
^{10}B–^{7}Li	681, 584, 522(300); 683, 586, 523(80)	C_{3v}, A, 1, 5, 14, 17
^{10}B–^{6}Li	683, 584, 534(300)	C_{3v}, A, 1, 14, 17
^{11}B–^{7}Li	655, 564, 522(300); 656, 566, 523(80)	C_{3v}, A, 1, 5, 14, 17
^{11}B–^{6}Li	657, 564, 534(300); 659, 566, 536(80)	C_{3v}, A, 1, 5, 14, 17
^{11}B–^{11}B	615, 552.3(80)	D_{3d}, A, 2, 3
^{10}B–^{11}B	570(80)	D_{3d}, A, 2, 3
^{10}B–^{11}B	560(80)	C_{3v}, A, 2, 3
^{10}B–P	622, 653(80)	C_{3v}, A, 3, 5
^{11}B–P	600, 628(80)	C_{3v}, A, 3, 4, 5
^{10}B–As	625, 662(80)	C_{3v}, A, 3
^{11}B–As	604, 637(80)	C_{3v}, A, 3, 4
^{10}B–Sb	635, 668(80)	C_{3v}, A, 3
^{11}B–Sb	612, 643(80)	C_{3v}, A, 3, 4
As	366(80) Reson.	T_d, A, 6
P	441 Reson., 491(80) Reson.	T_d, A, 6
Ga–^{6}Li	521(300) Reson.?	C_{3v}, A, 6
Ga–^{7}Li	515(300) Reson.	C_{3v}, A, 6
Al–^{6}Li	525(300) Reson.?	C_{3v}, A, 6
Al–^{7}Li	520(300) Reson.	C_{3v}, A, 6
^{12}C	570(300), 573(80)	T_d, A, 3
^{13}C	586(300), 589(80)	T_d, A, 3
^{14}C	605(300), 650(80)	T_d, A, 3
O	Bands near 30, 500, 1100, 1200	10, 11 Interstitial oxygen in bent Si–O–Si molecule.
^{12}C–^{16}O	1103.9(4.2), 586(300), 637(300), 664(300), 589(80), 640(80), 690(80)	A, 12, 13 Some transition may be rotational as well as vibrational.
^{12}C + irrad.	922, 932(88)	Axial, A, 16, carbon probably interstitial

Localized modes in semiconductors and metals. *(Continued)*

Host and Impurity	Mode frequency (temperature) °K	Defect symmetry, Method of observation,[a] References,[b] Comments
Germanium		
Si	389(300)	T_d, A, 19; R, 20; neutron, 87
Si–Si	476, 448(300)	D_{3d}, R, 20
^{16}O	855(300), 862(4)	C_{2v}, A, 21
^{18}O	818(?)	C_{2v}, A, 22
^{10}B	571(80)	T_d, A, 23
^{11}B	547(80)	T_d, A, 23
^{10}B–Li	518.5, 610(80)	C_{1v}, A, 23
^{11}B–Li	497, 582.5(80)	C_{1v}, A, 23
Ga–^{6}Li	379, 405(80)	C_{1v}, A, 24
Ga–^{7}Li	356, 380(80)	C_{1v}, A, 24
Ga–P	~343, 350(80)	C_{1v}, A, 24
^{31}P	343(80)	T_d, A, 24
SiC		
N	635(4)	C_{1v}, R, T, 25
H_{Si}	2977(1.3)	Axial, 73
D_{Si}	2210(1.3)	Axial, 73
Impurity–vacancy	536(45)	Resonant mode, 74
GaAs		
Al	362(80) ~371(4.2)	T_d, A, Refl., 27, 42 T_d, T, 85
P	355(80), 353(300) ~363(4.2)	T_d, A, R, Refl., 27, 40, 41, 58 T_d, T, 85
^{6}Li	350, 389, 401, 409.5, 426, 452(80)	?, A, 29, 30 — The Li defect is now known to be complex involving two or more Li atoms and a native defect.
^{7}Li	364, 389, 394, 379, 383, 421(80)	?, A, 29, 30 — The Li defect is now known to be complex involving two or more Li atoms and a native defect.
Te–^{6}Li	419, 510(80)	C_{3v}, A, 29, 31
Te–^{7}Li	391, 475(80)	C_{3v}, A, 29, 31
Mn–^{6}Li	391, 413, 419(80)	C_{3v}, A, 32
Mn–^{7}Li	365, 386, 391(80)	C_{3v}, A, 32
Cd–^{6}Li	377, 401, 423(80)	C_{3v}, A, 29, 32
Cd–^{7}Li	354, 375, 395(80)	C_{3v}, A, 29, 32
Zn–^{6}Li	361, 385, 404, 433(80)	?, A, 29, 32
Zn–^{7}Li	340, 361, 379, 405(80)	?, A, 29, 32
Si_{Ga}	384(80)	T_d, A, 33, 34, 35, 36
Si_{As}	399(80)	T_d, A, 33, 34, 35, 36
Si_{Ga}–Si_{As}	367, 393, 464(80)	C_{3v}, A, 33, 34, 35, 36
Si_{Ga}–$^6Li_{Ga}$	374, 379, 405, 470, 480, 487(80)	C_s, A, 33, 34, 35, 36
Si_{Ga}–$^7Li_{Ga}$	374, 379, 405, 438, 448, 455(80)	C_s, A, 33, 34, 35, 36
Si_{Ga}–Cu_{Ga}	374, 376, 399(80)	C_s, A, 33, 34, 35, 36
Si_{Ga}–Zn_{Ga}	378, ~382, 395(80)	C_s, A, 37
Mg–^{7}Li	318, 338, 348, 366, 377, 392(80)	?, A, 38, ^{6}Li also studied
Mg_{Ga}	326(80)	T_d, A, 38
$^{11}B_{Ga}$; $^{10}B_{Ga}$	617(77); 640(77)	T_d, A, 68
^{13}C; ^{12}C	682(77); 661(77)	T_d, A, 68

Localized modes in semiconductors and metals. *(Continued)*

Host and Impurity	Mode frequency (temperature) °K	Defect symmetry, Method of observation,[a] References,[b] Comments
GaP		
Al	444.7(77)	T_d, A, R, 50, 51, 52
Si_{Ga}	465(80)	T_d, A, 39, 50
Si_{Ga}–Si_P	455.7(80)	T_d, A, 39, 50
^{11}B	569.7(77)	T_d, A, R, 50, 51, 52
^{10}B	593.8(77)	T_d, A, R, 50, 51, 52
^{14}N	493(4.2)	T_d, L, A, 53
O_i	1005(80)	T_d, A, 69
O_P	199, 229(1.6)	S, L, 84; Isotope shifts done
O_P?	464(77)	T_d, A, 50
C	606(20)	T_d, A, 57
As	272(300)	T_d, Refl., 58
S_P	ω_{LO}–6(20)	Bound phonon, L, R, Refl., 63, 64
Te_P	193(1.6) ω_{LO}–10(20)	S, L, 80, 82 Bound phonon, L, R, Refl., 63, 64
Si_{Ga}	ω_{LO}–10(20)	Bound phonon, L, R, Refl., 63, 64
Sn_{Ga}	72(5) ω_{LO}–19(20)	S, L, 83 Bound phonon, L, R, Refl., 63, 64
Se_P	ω_{LO}–7(20)	Bound phonon, Refl., 64
Cd	56(20.4)	S, L, 81, Complexed with a neighboring oxygen
Zn	48(1.6)	S, L, 81, Complexed with a neighboring oxygen
C_P	ω_{LO}–10(20)	Bound phonon, assignment uncertain, Refl., 67
Be_{Ga}	629(20)	A, 70
GaSb		
Al_{Ga}	316.7(77)	T_d, A, 47, 48
InSb		
Al	296(77)	T_d, A, 43, 44
P	293(77)	T_d, A, 45
AlAs		
Ga	252(300)	T_d, Refl., 46
InP		
$^{10}B_i$;^{11}B	543.5(77); 522.8(77)	T_d, A, 77
As_P	223(300)	Refl., 75
InAs		
Ga_{In}	~240(300)	T_d, Refl., 50
P_{As}	303(300)	Refl., 75
CdS		
^6Li	474(100)	?, A, 59
^7Li	457(100)	?, A, 59
I_1 transition unknown impurity		
I_1	ω_{LO}–19, –26(1.2)	Anisotropic bound phonon, L, 65
Be_{Cd}	446(∥), 453(⊥)(100)	C_{3v}, A, R, 66
Se_S	182(∥), 186(⊥)(300)	C_{3v}, Refl., 60, 71

[a] Rev. Mod. Phys., Vol. 47, Suppl. No. 2, Fall 1975

Localized modes in semiconductors and metals. *(Continued)*

Host and Impurity	Mode frequency (temperature) °K	Defect symmetry, Method of observation,[a] References,[b] Comments
CdSe		
Be_{Cd}	411(∥), 420(⊥)(100)	C_{3v}, A, 66
Te_{Se}	153(293)	S, Refl., 86, this mode may lie in a gap
S_{Se}	269(∥), 270(⊥)(18)	Refl., 60, 71
CdTe		
Zn_{Cd}	167(293)	S, Refl., 86
Be	391(4), 61(4) Reson.	T_d, A, 55, 56
Se_{Te}	170(300)	T_d, A, 54, Refl., 86
Li(?)	270(?)	?, A, 54
N. ZnS		
Be	490(100), 486(300)	R, A, 76
Se	220	Refl., R, 64
ZnSe		
^{7}Li	353(77), 336(77), 328(77), 318(77)	?, A, 60
Al	359(77)	T_d, A, 61
Al-O_{Zn} ?		?, A, 61
Be	450(100), 447(300)	S, R, A, 76, 85, 89
S	297(100)	S, A, Refl., R, 60, 62, 72
ZnTe		
Cd_{Zn}	153(293)	S, Refl., R, 86, this mode may be in the gap
S	270	S, Refl., 72
Be	415(100), 411(300)	S, A, 76
Metals		
In in Pb	77(0.83)	S, T, 78, Localized mode band (host crystal has $\omega_m = 73\ cm^{-1}$)
Au in Cu	80(300)	S, Neutron scattering, 90, well-defined resonant mode in acoustic band for 3% Au
W in Cr	Broad resonance	S, Neutron scattering, 91, resonant mode

[a] Key: A = absorption, R = Raman, Refl. = reflectivity, T = electron tunneling, L = luminescence, S = substitutional.

[b] The references for Table IV.1 are the following:

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7.4. Rutherford Backscattering (RBS) and Ion Channeling Spectrometry

References:

- "Backscattering Spectrometry," W.-K. Chu, J.W. Mayer and M.-A. Nicolet, (Academic Press, New York 1978).
- "Materials Analysis by Ion Channeling," L.C. Feldman, J.W. Mayer and S.J. Picreaux, (Academic Press, New York 1982).

Ernest Rutherford discovered the atomic nucleus when he observed that most energetic helium ions would pass through a very thin gold foil unaffected except for a few which suffered large angle scattering (backscattering). This experimental finding meant that the scattering cross section had to be much smaller than the size (projected area) of an atom in a lattice. Rutherford came to the conclusion that most of the atomic matter is concentrated in the nucleus. Rutherford Backscattering Spectrometry (RBS), therefore, was born around the turn of the 20th Century. It is surprising that it took such a long time until RBS was introduced to electronic materials characterization, especially thin films analysis. One of the pioneers in this field was J.W. Mayer. He saw in RBS a great opportunity for using the large number of early 1960s 2-4 MV accelerators which had been developed for nuclear physics experiments. Rutherford Backscattering Spectrometry has yielded a lot of crucial information on the composition of thin films (up to ~ 1 μm), interface reactions and crystallinity. RBS is ideally suited for many thin film or near-surface type phenomena.

The billiard ball collision principle is the basis of RBS. The kinematic factor K_M is used to determine the energy of elastically scattered particles:

$$K_M = E_n / E = \left(\frac{\sqrt{M^2 - m^2 \sin^2(\theta)} + m\cos\theta}{M+m} \right)^2$$

Figure 190 illustrates the collision kinematics. For $\theta = \pi$ (perfect backscattering), one observes the largest energy transfer:

$$K_M = \left[\frac{M-m}{M+m}\right]^2$$

For RBS, θ is usually chosen to be very close to π. In this case, one can approximate:

$$K_M \cong 1 - 4\frac{M}{m} + \delta^2 \frac{m}{M}. \quad (\delta = |\pi\text{-}\theta|)$$

For $\delta << \pi$ and $m << M$. Table 21 lists of K_M for helium projectiles backscattered from the elements Z = 5 to 83 at ten different backscatter angles θ.

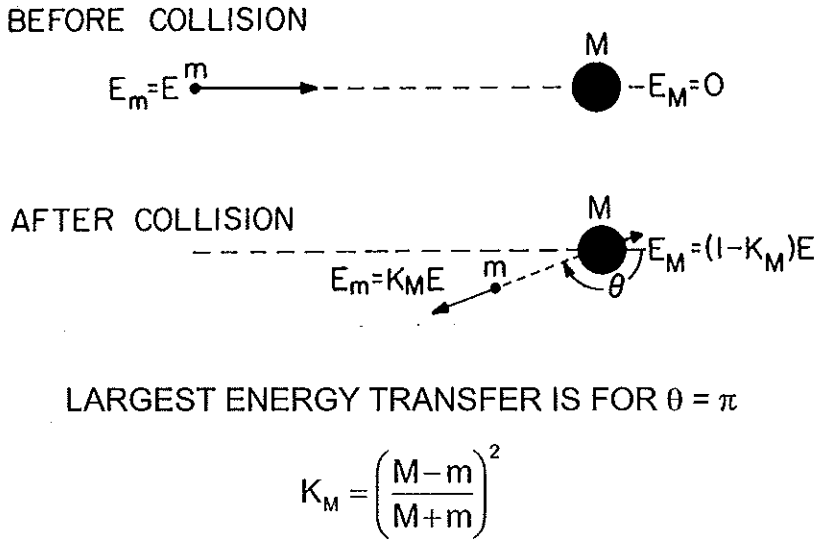


Fig. 190. Schematic representation of an elastic large-angle collision of a light particle of mass m with a stationary heavy mass M. The kinematic factor K_M gives the fraction of the initial energy that resides in the light particle after the collision. [After M.-A. Nicolet, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 583.]

From this very simple derivation, we know the energy of backscattered particles. However, we do not know the backscatter rate. To obtain rate information, we consider the forces acting between the colliding atoms in more detail. At helium energies of a few MeV, we can neglect all the electrons in the system. Figure 191 explains the differential scattering cross section for large

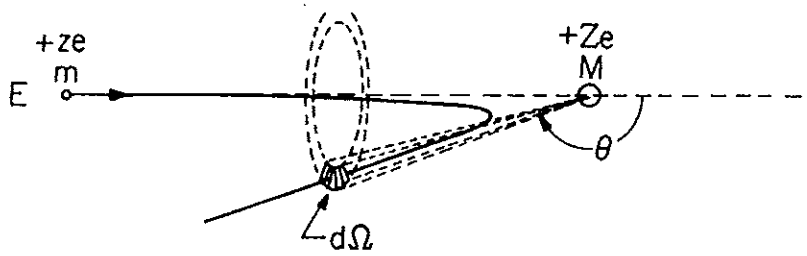
angle Coulomb scattering. One sees that the differential scattering cross-section $d\sigma/d\Omega$ is proportional to the square of the product of nuclear charges divided by the energy:

$$\frac{d\sigma}{d\Omega} \alpha \frac{(zZ)^2}{E^2}.$$

For small projectile mass ($m << M$) and for $\theta = \pi$, one can approximate:

$$\frac{d\sigma}{d\Omega} = \left(\frac{zZe^2}{4E}\right)^2 \left[\frac{1}{\sin^4 \frac{\theta}{2}} - 2\left(\frac{m}{M}\right)^2\right],$$

($e = 4.803 \times 10^{-10}$ stat C; when E is given in MeV, use $e^2 = 1.44 \times 10^{-13}$ MeVcm).



$$\frac{d\sigma_z}{d\Omega} = \left(\frac{zZe^2}{2E \sin^2\theta}\right)^2 \frac{\left[\cos\theta + \sqrt{1-\left(\frac{m}{M}\sin\theta\right)^2}\right]^2}{\sqrt{1-\left(\frac{m}{M}\sin\theta\right)^2}}$$

for $\theta = \pi$

$$\frac{d\sigma_z}{d\Omega} = \left(\frac{zZe^2}{2E}\right)^2 \left[\frac{1-(m/M)^2}{2}\right]^2 \alpha \left(\frac{zZ}{E}\right)^2$$

Fig. 191. Schematic representation of an elastic large-angle collision of a swift light particle of mass m and charge ze with a stationary heavy mass M and charge Ze. The differential scattering cross section $dx\sigma_z/d\Omega$ gives the likelihood that the scattering results in the detection of light particle in the differential solid angle $d\Omega$. [After M.-A. Nicolet, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 587.]

Both the dependence of the energy of the backscattered projectile on the target mass M, as well as the backscatter rate on the atomic number Z have been experimentally verified using thin films of Au, Ag and Cu (see Fig. 192).

Table 21. $\overline{K}$ for ^{4}He as Projectile

ELEMENT	AT. NO. (Z_2)	SCATTERING ANGLE θ 180°	170°	160°	150°	140°	130°	120°	110°	100°	90°
B	5	0.2112	0.2136	0.2209	0.2334	0.2517	0.2767	0.3091	0.3500	0.4000	0.4593
C	6	0.2501	0.2526	0.2604	0.2736	0.2929	0.3187	0.3518	0.3929	0.4423	0.5001
N	7	0.3086	0.3113	0.3194	0.3333	0.3531	0.3795	0.4127	0.4532	0.5009	0.5555
O	8	0.3597	0.3625	0.3708	0.3848	0.4047	0.4309	0.4635	0.5027	0.5483	0.5998
F	9	0.4251	0.4278	0.4360	0.4498	0.4693	0.4946	0.5258	0.5627	0.6050	0.6520
NE	10	0.4472	0.4499	0.4580	0.4717	0.4909	0.5158	0.5463	0.5822	0.6233	0.6687
NA	11	0.4948	0.4974	0.5053	0.5185	0.5370	0.5607	0.5896	0.6233	0.6615	0.7034
MG	12	0.5143	0.5169	0.5246	0.5375	0.5556	0.5788	0.6069	0.6397	0.6767	0.7171
AL	13	0.5500	0.5525	0.5600	0.5724	0.5897	0.6117	0.6384	0.6693	0.7040	0.7416
SI	14	0.5632	0.5657	0.5730	0.5852	0.6022	0.6238	0.6499	0.6801	0.7139	0.7505
P	15	0.5946	0.5970	0.6040	0.6156	0.6318	0.6523	0.6770	0.7054	0.7370	0.7711
S	16	0.6053	0.6076	0.6144	0.6259	0.6417	0.6619	0.6860	0.7138	0.7447	0.7779
CL	17	0.6352	0.6374	0.6440	0.6548	0.6698	0.6887	0.7114	0.7374	0.7662	0.7970
AR	18	0.6689	0.6709	0.6770	0.6871	0.7010	0.7186	0.7395	0.7634	0.7897	0.8179
K	19	0.6630	0.6650	0.6712	0.6814	0.6955	0.7133	0.7346	0.7588	0.7856	0.8142
CA	20	0.6698	0.6718	0.6779	0.6880	0.7019	0.7194	0.7403	0.7641	0.7904	0.8184
SC	21	0.6997	0.7016	0.7073	0.7166	0.7295	0.7457	0.7650	0.7869	0.8109	0.8365
TI	22	0.7152	0.7170	0.7224	0.7314	0.7437	0.7592	0.7776	0.7985	0.8214	0.8457
V	23	0.7298	0.7316	0.7368	0.7454	0.7572	0.7720	0.7896	0.8095	0.8312	0.8543
CR	24	0.7345	0.7362	0.7414	0.7498	0.7615	0.7761	0.7934	0.8129	0.8344	0.8570
MN	25	0.7468	0.7485	0.7534	0.7615	0.7727	0.7867	0.8033	0.8221	0.8425	0.8642
FE	26	0.7504	0.7520	0.7569	0.7649	0.7760	0.7898	0.8062	0.8247	0.8449	0.8662
CO	27	0.7618	0.7634	0.7681	0.7758	0.7864	0.7997	0.8154	0.8331	0.8524	0.8728
NI	28	0.7612	0.7628	0.7675	0.7752	0.7859	0.7992	0.8150	0.8328	0.8522	0.8726
CU	29	0.7770	0.7785	0.7829	0.7902	0.8002	0.8128	0.8276	0.8442	0.8624	0.8815
ZN	30	0.7825	0.7839	0.7883	0.7954	0.8052	0.8175	0.8319	0.8482	0.8659	0.8846
GA	31	0.7946	0.7960	0.8001	0.8069	0.8162	0.8279	0.8416	0.8570	0.8738	0.8914
GE	32	0.8020	0.8033	0.8073	0.8139	0.8229	0.8342	0.8475	0.8624	0.8786	0.8956
AS	33	0.8074	0.8087	0.8126	0.8191	0.8279	0.8389	0.8518	0.8663	0.8820	0.8986
SE	34	0.8163	0.8176	0.8213	0.8275	0.8359	0.8464	0.8588	0.8727	0.8877	0.9035
BR	35	0.8183	0.8195	0.8232	0.8293	0.8377	0.8481	0.8603	0.8741	0.8890	0.9046
KR	36	0.8259	0.8271	0.8307	0.8366	0.8446	0.8546	0.8664	0.8796	0.8938	0.9088
RB	37	0.8290	0.8302	0.8337	0.8395	0.8474	0.8573	0.8688	0.8818	0.8958	0.9105
SR	38	0.8329	0.8340	0.8375	0.8431	0.8509	0.8605	0.8718	0.8845	0.8982	0.9126
Y	39	0.8351	0.8362	0.8396	0.8452	0.8529	0.8624	0.8736	0.8861	0.8996	0.9138
ZR	40	0.8389	0.8400	0.8433	0.8488	0.8563	0.8656	0.8765	0.8888	0.9020	0.9159
NB	41	0.8416	0.8427	0.8460	0.8514	0.8588	0.8679	0.8787	0.8907	0.9038	0.9174
MO	42	0.8461	0.8471	0.8503	0.8556	0.8628	0.8717	0.8822	0.8939	0.9066	0.9198
TC	43	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RU	44	0.8532	0.8542	0.8573	0.8623	0.8692	0.8777	0.8877	0.8989	0.9110	0.9236
RH	45	0.8558	0.8569	0.8599	0.8648	0.8716	0.8800	0.8898	0.9008	0.9127	0.9251
PD	46	0.8603	0.8613	0.8642	0.8690	0.8756	0.8838	0.8933	0.9040	0.9155	0.9276
AG	47	0.8620	0.8630	0.8659	0.8706	0.8771	0.8852	0.8946	0.9052	0.9165	0.9284
CD	48	0.8673	0.8682	0.8710	0.8756	0.8818	0.8896	0.8987	0.9089	0.9199	0.9313
IN	49	0.8698	0.8707	0.8735	0.8780	0.8841	0.8917	0.9007	0.9106	0.9214	0.9326
SN	50	0.8738	0.8747	0.8773	0.8817	0.8877	0.8951	0.9037	0.9134	0.9239	0.9348
SB	51	0.8767	0.8776	0.8802	0.8845	0.8903	0.8976	0.9060	0.9155	0.9257	0.9363
TE	52	0.8820	0.8829	0.8854	0.8895	0.8951	0.9020	0.9102	0.9192	0.9290	0.9392
I	53	0.8814	0.8823	0.8848	0.8889	0.8945	0.9015	0.9097	0.9188	0.9286	0.9388
XE	54	0.8852	0.8860	0.8885	0.8925	0.8979	0.9047	0.9126	0.9215	0.9310	0.9409
CS	55	0.8865	0.8873	0.8897	0.8937	0.8991	0.9058	0.9136	0.9223	0.9317	0.9415
BA	56	0.8899	0.8907	0.8931	0.8969	0.9021	0.9086	0.9162	0.9247	0.9338	0.9434
LA	57	0.8911	0.8919	0.8942	0.8980	0.9032	0.9096	0.9172	0.9256	0.9346	0.9440
CE	58	0.8919	0.8927	0.8950	0.8988	0.9039	0.9103	0.9178	0.9261	0.9350	0.9444
PR	59	0.8926	0.8933	0.8956	0.8994	0.9045	0.9109	0.9183	0.9266	0.9355	0.9448
ND	60	0.8949	0.8956	0.8979	0.9016	0.9066	0.9128	0.9201	0.9282	0.9369	0.9460
PM	61	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
SM	62	0.8989	0.8997	0.9018	0.9054	0.9102	0.9162	0.9232	0.9310	0.9394	0.9481
EU	63	0.9000	0.9007	0.9028	0.9064	0.9111	0.9171	0.9240	0.9317	0.9400	0.9487
GD	64	0.9032	0.9039	0.9059	0.9094	0.9140	0.9197	0.9265	0.9339	0.9420	0.9504
TB	65	0.9041	0.9048	0.9069	0.9103	0.9149	0.9206	0.9272	0.9346	0.9426	0.9509
DY	66	0.9061	0.9067	0.9088	0.9121	0.9166	0.9221	0.9287	0.9359	0.9437	0.9518
HO	67	0.9075	0.9081	0.9101	0.9134	0.9178	0.9233	0.9298	0.9369	0.9446	0.9526
ER	68	0.9087	0.9094	0.9113	0.9145	0.9189	0.9244	0.9307	0.9378	0.9454	0.9533
TM	69	0.9096	0.9102	0.9122	0.9154	0.9197	0.9251	0.9314	0.9384	0.9459	0.9537
YB	70	0.9116	0.9122	0.9142	0.9173	0.9215	0.9268	0.9329	0.9398	0.9471	0.9548
LU	71	0.9125	0.9132	0.9151	0.9182	0.9224	0.9276	0.9337	0.9404	0.9477	0.9553
HF	72	0.9143	0.9149	0.9168	0.9198	0.9239	0.9290	0.9350	0.9417	0.9488	0.9562
TA	73	0.9153	0.9159	0.9178	0.9207	0.9248	0.9299	0.9358	0.9423	0.9494	0.9567
W	74	0.9166	0.9172	0.9190	0.9219	0.9260	0.9309	0.9368	0.9432	0.9502	0.9574
RE	75	0.9176	0.9182	0.9200	0.9229	0.9269	0.9318	0.9375	0.9439	0.9508	0.9579
OS	76	0.9197	0.9203	0.9221	0.9249	0.9288	0.9337	0.9393	0.9456	0.9523	0.9593
IR	77	0.9201	0.9207	0.9224	0.9252	0.9291	0.9339	0.9394	0.9456	0.9523	0.9592
PT	78	0.9212	0.9218	0.9235	0.9263	0.9301	0.9348	0.9403	0.9464	0.9530	0.9598
AU	79	0.9219	0.9225	0.9242	0.9270	0.9307	0.9354	0.9408	0.9469	0.9534	0.9602
HG	80	0.9234	0.9239	0.9256	0.9283	0.9320	0.9366	0.9420	0.9479	0.9543	0.9610
TL	81	0.9246	0.9252	0.9268	0.9295	0.9332	0.9377	0.9429	0.9488	0.9551	0.9616
PB	82	0.9255	0.9260	0.9276	0.9303	0.9339	0.9383	0.9435	0.9493	0.9555	0.9619
BI	83	0.9262	0.9268	0.9284	0.9310	0.9346	0.9390	0.9442	0.9499	0.9560	0.9624

θ	180°	178°	176°	174°	172°	170°	168°	166°	164°	162°	160°
δ^2	0	1.218 x10^{-3}	4.874 x10^{-3}	1.097 x10^{-2}	1.950 x10^{-2}	3.046 x10^{-2}	4.386 x10^{-2}	5.971 x10^{-2}	7.798 x10^{-2}	9.870 x10^{-2}	1.218 x10^{-1}

Table 21. $\overline{K}$ for ^{4}He as Projectile. This table gives the mean kinematic factors KM_2 of the istopes of an element. The projectile is a ^{4}He atom (M_1 = 4.002603 amu). The kinematic factor KM_2 of an individual isotope is calculated by Eq. (2.6), for the isotopic masses listed in Table I for M_2. For the average, the kinematic factor of each isotope listed in Table I is weighted by the relative abundance given for that isotope in the same table. The parameter θ is the scattering angle measured in the laboratory frame of reference, as shown in Fig. 2.1.

The factor $\overline{K}$ is approximately described by $1-4(\overline{M_1/M_2})+\delta^2(\overline{M_1/M_2})$, where δ measures the deviation of θ from 180° in arc units and where $\overline{M_1/M_2}$ is the average of the ratio M_1/M_2 weighted by the relative abundance of each isotope given in Table I for a particular element. A table of δ_2 versus $\pi-\theta$ is given to facilitate the evaluation of $\overline{K}$ by this expression.

$\overline{K}$ given here is numerically very close to $K_{\overline{M}}$. Usually no distinction is made between the two, and the symbol K is used for both (see Section 3.8). [After W.-K. Chu, J.W. Mayer and M.-A. Nicolet, "Backscattering Spectrometry," Academic Press, New York, 1978, p. 360.]

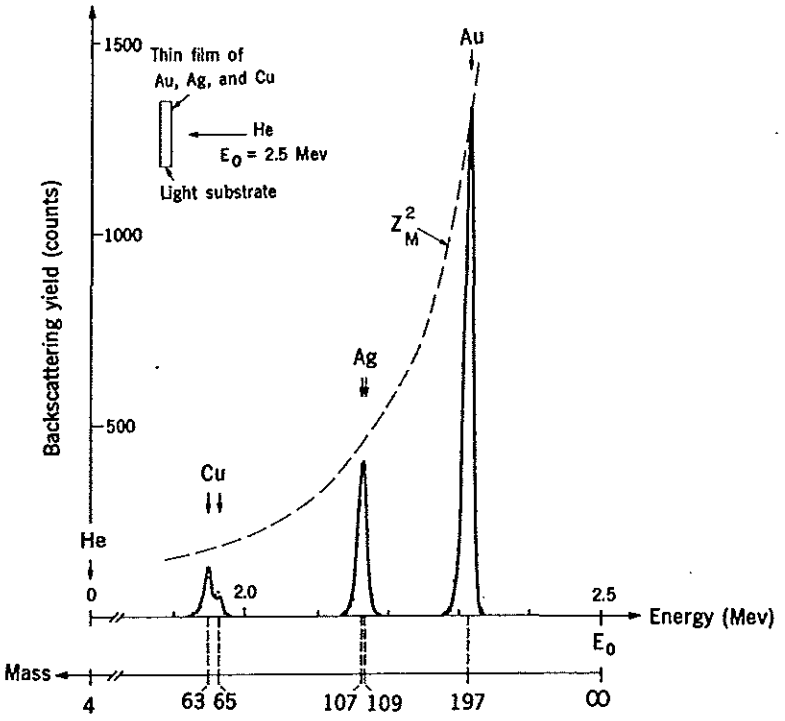


Fig. 192. Energy spectrum of 2.5 MeV $^4He^+$ ions backscattered from Cu, Ag and Au atoms deposited on a light substrate (signal not shown) in approximately equal amounts of about 10^{15} atoms/cm^2 (~ one monolayer). The width of the signal of monoisotopic Au is determined by the system resolution. The change in the scattering cross section follows the Z_M^2 dependence shown as a dashed line. The second abscissa gives the positions of $K_M E_0$ for the five isotopes ($K_M = \left(\frac{\sqrt{M^2 - m^2 \sin^2\theta} + m\cos\theta}{M+m} \right)^2$ with θ = 170°). [After M.-A. Nicolet, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 602.]

The third effect which has to be considered in order to understand backscatter spectra is the energy loss dE/dX of the projectile during passage through a medium. Large energy losses as they occur during large angle scattering (the basis of RBS) are extremely rare compared to the numerous small energy losses which do not influence the trajectory appreciably. Figure 193 defines the stopping cross section ε. Typical values for ε lie between 20-150 eV/(10^{15} atoms/cm^{-2}). Stopping cross sections are additive because they are related to a fixed number of corresponding target atoms:

$$\varepsilon^{A_mB_n} = m\varepsilon^A + n\varepsilon^B.$$

A_mB_n = compound molecule consisting of m A-atoms and n B-atoms.

$$\frac{dE}{dX}\left(A_mB_n\right) = N^{A_mB_n} \cdot \varepsilon^{A_mB_n},$$

$N^{A_mB_n}$ = volume density of the molecule in the compound.

This is known as Bragg's Rule[7.18].

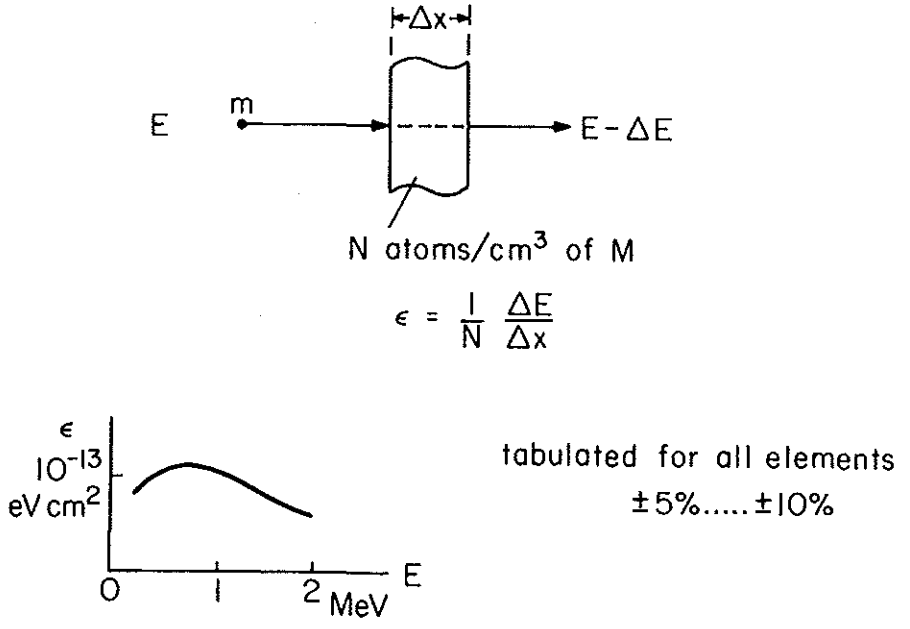


Fig. 193. Schematic representation of a transmission experiment to determine the dE/dx loss of a swift particle moving through a medium (top). The thickness Δx is assumed to be quite small, and in the limit, one defines

$$\varepsilon = (1/N) \lim_{\Delta x \to 0} \frac{\Delta E}{\Delta x}$$

as the stopping cross section. The general dependence of ε is sketched also (bottom). Theoretical calculations of stopping cross sections in the energy range of interest to BS are difficult. ε is best considered as an experimentally determined quantity. Typical accuracies are 5%, and sometimes worse. [After M.-A. Nicolet, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 589.]

[7.18] W. H. Bragg and R. Kleeman, Phil. Mag. **10**, S, 318 (1905).

Because the stopping of energetic helium nuclei is composed of a random sequence of independent small energy losses (called electronic stopping), we expect that the integral energy loss over a length Δx shows fluctuations from one particle to the next. This energy straggling can be estimated with the following equation:

$$\overline{\delta\Delta E^2} = \Delta x \; 4\pi \; NZ \left(ze^2\right)^2,$$

$\overline{\delta\Delta E^2}$ = variance of ΔE;
z = atomic number of projectile;
Z = atomic number of target; and
N = number of target atoms per unit volume.

With the knowledge of the kinematic factor K_M, the scattering cross section $d\sigma/d\Omega$ and the stopping cross section (or dE/dx), one is prepared to analyze Rutherford Backscattering spectra. Multichannel pulse height analyzers (PHA) in connection with a thin window silicon surface barrier detector and appropriate electronic pulse amplifiers are used to record the number of backscattered particles, measure their energies and store the counts in narrow energy bins. In a typical PHA, the energy axis is displayed as the X-axis (horizontal) and the number of counts per channel (energy slice ΔE) is displayed vertically (Y-axis). The number of backscattered particles ΔQ is:

$$\Delta Q = Q \; \frac{d\sigma}{d\Omega} \; (\theta) \; N\Delta x,$$

Q = total number of incident particles;
Ω = solid angle of detector;
N = volume density of atoms (cm^{-3}) and
Δx = thickness of a layer at a certain depth.

Figure 194 illustrates the above equation. The energy spread ΔE of the particles which are backscattered from Δx defines the energy loss factor $[S_o]$ (Fig. 195). Table 22 contains average values of $[S_o]$ for a backscatter angle of 10° ($\pi - \theta$) and various He energies. Figure 196 contains

superimposed spectra of 2 MeV He backscattered from Pt films on silicon. Reading the maximum He energy off the upper edge (taken at half height) of the spectrum ($\cong$ 1.85 MeV) and dividing by 2 MeV, we find the kinematic factor $K_M \cong .925$. From Table 21, we find that θ lies between 150° and 160°. The actual experiment was run at θ = 170° but we cannot read the maximum energy off the plot with good accuracy. The energy loss for the 2000 Å thick film is ΔE = (1.85-1.53) MeV $\cong$ .32 MeV. In the energy-loss factor table (Table 23) we find for Pt and 2 MeV He:[S_o] - 148.5 eV/Å, which yields x = .32 MeV/148.5 eV/Å, a value very close to the one stated in the figure.

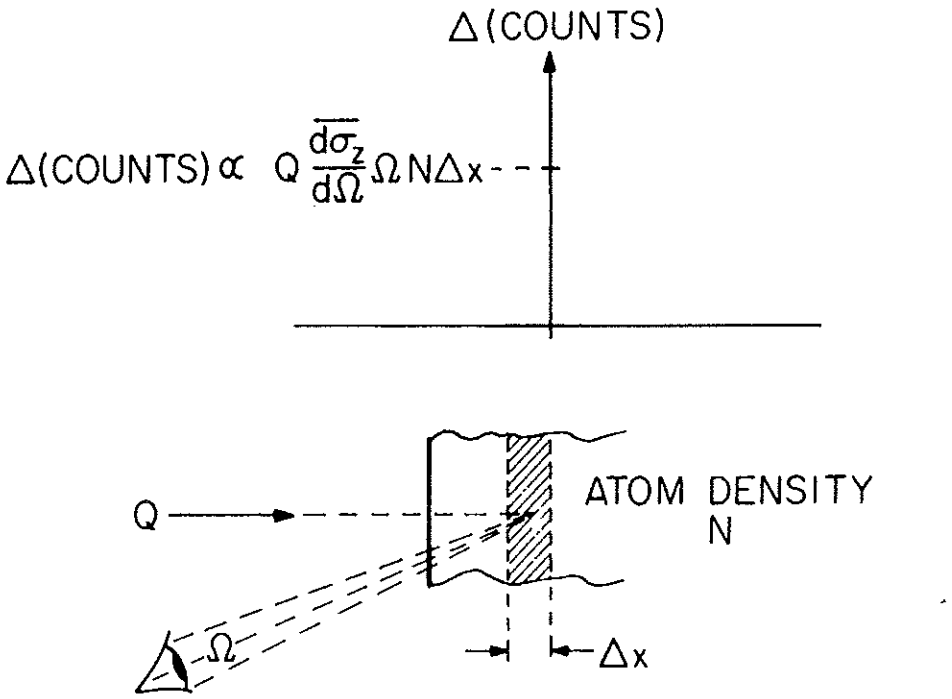


Fig. 194. Schematic representation of the connection that exists between backscattering events in a slice of thickness Δx at some depth in a sample, and the number of ΔQ in a channel of the backscattering spectrum. [After M.-A. Nicolet, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 593.]

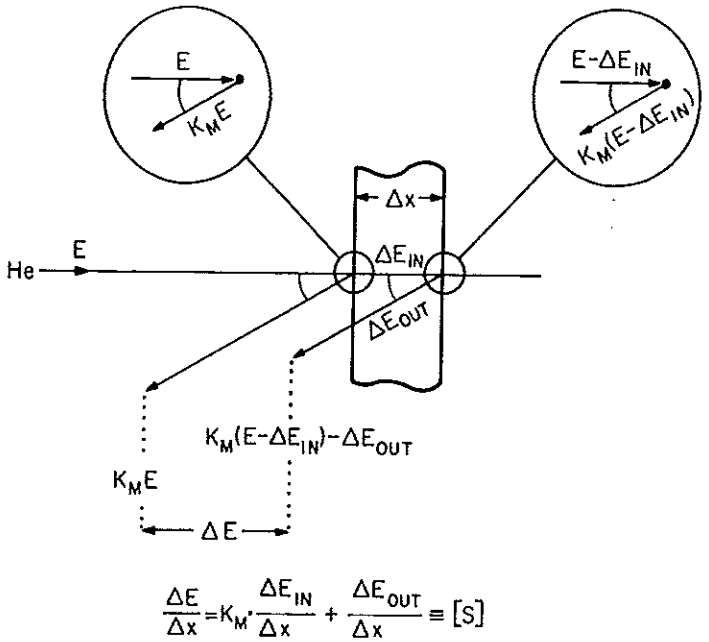


Fig. 195. Schematic representation of the energy spread ΔE of the particles backscattered at the front and back of a slice of thickness Δx, and the resulting definition of the energy loss factor [S]. [After M.-A. Nicolet, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 593.]

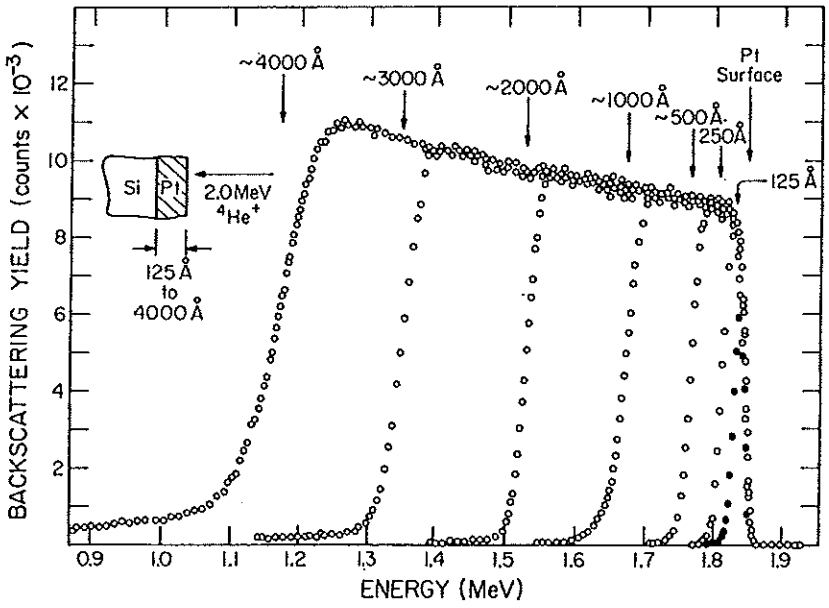


Fig. 196. Superimposed energy spectra of 2.0 MeV 4He+ ions backscattered from Pt films of various thicknesses to show the closely linear relationship between the width of the Pt signal and its thickness. The signals of the Si substrates are not shown. [After M.-A. Nicolet, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 605.]

Table 22. ^{4}He Stopping Cross Section Factor [ε_0]

ELEMENT	AT. NO. (Z_2)	ENERGY (in MeV) OF INCIDENT ^{4}He										
		1.0	1.2	1.4	1.6	1.8	2.0	2.4	2.8	3.2	3.6	4.0
BE	4	31.0	30.9	30.8	30.7	30.5	30.4	30.1	29.8	29.5	29.2	28.9
B	5	37.9	38.2	38.3	38.3	38.3	38.2	38.0	37.7	37.2	36.5	35.9
C	6	38.7	40.0	40.9	41.6	42.1	42.4	42.8	42.9	42.5	41.7	40.8
N	7	59.5	60.1	60.3	60.2	59.8	59.3	58.1	56.6	54.6	52.4	50.2
O	8	61.3	62.3	62.8	62.8	62.5	62.0	60.5	58.7	56.4	53.9	51.5
F	9	61.0	62.4	63.1	63.3	63.1	62.6	61.0	59.0	56.7	54.2	51.8
NE	10	61.9	63.6	64.4	64.6	64.4	63.9	62.4	60.4	58.2	55.7	53.3
NA	11	66.2	66.7	66.8	66.5	65.9	65.1	63.2	61.0	58.6	56.3	54.1
MG	12	86.4	85.6	84.3	82.7	80.9	79.0	75.2	71.5	67.9	64.5	61.7
AL	13	84.8	83.4	81.9	80.3	78.6	76.8	73.1	69.5	66.1	62.9	59.9
SI	14	109.6	107.0	103.7	100.1	96.3	92.6	85.8	79.7	74.3	69.6	65.9
P	15	107.6	106.0	103.5	100.3	96.9	93.5	86.9	81.0	75.6	70.8	67.3
S	16	110.3	109.8	107.8	104.8	101.4	97.8	90.9	84.5	78.7	73.6	70.0
CL	17	138.9	134.5	128.8	122.6	116.3	110.3	99.8	91.4	84.4	78.5	24.6
AR	18	145.8	140.9	134.2	126.9	119.5	112.6	101.0	92.1	85.1	79.2	75.6
K	19	150.6	148.7	144.8	139.6	133.9	128.0	116.8	107.2	99.2	92.3	87.0
CA	20	162.6	159.3	154.5	148.7	142.5	136.2	124.5	114.4	105.8	98.5	92.6
SC	21	164.0	160.7	155.9	150.2	144.1	138.0	126.5	116.6	108.2	101.0	95.2
TI	22	164.5	161.2	156.3	150.6	144.6	138.7	128.1	119.2	111.6	104.8	99.3
V	23	156.9	154.0	149.7	144.7	139.4	134.3	124.9	117.0	110.1	103.8	98.7
CR	24	151.1	149.6	146.3	142.1	137.4	132.7	124.1	116.8	110.6	105.0	100.2
MN	25	148.0	146.4	143.3	139.2	134.7	130.3	122.2	115.5	109.7	104.3	99.8
FE	26	157.2	155.4	151.4	146.3	140.8	135.4	125.7	118.2	112.2	107.0	102.8
CO	27	145.8	145.4	143.0	139.5	135.3	131.1	123.1	116.5	111.1	106.2	102.1
NI	28	138.5	139.2	138.2	136.1	133.2	130.0	123.2	116.3	109.4	102.8	97.4
CU	29	130.0	131.9	132.2	131.2	129.4	127.0	121.6	116.2	111.4	107.2	103.3
ZN	30	131.5	131.8	130.8	128.7	126.1	123.1	117.1	111.4	106.5	102.1	98.2
GA	31	143.6	142.5	140.2	137.0	133.4	129.6	122.2	115.7	110.2	105.3	101.1
GE	32	150.0	148.2	145.0	141.0	136.6	132.2	124.0	117.0	111.1	105.9	101.6
AS	33	159.8	157.6	153.7	149.0	143.9	138.9	129.7	122.1	115.7	110.0	105.3
SE	34	162.9	159.6	154.6	149.0	143.3	137.8	128.4	120.9	114.6	108.8	104.1
BR	35	182.4	176.2	168.2	159.7	151.5	144.1	132.4	124.2	117.8	111.7	106.8
KR	36	195.1	187.8	178.8	169.4	160.4	152.4	140.0	131.4	124.6	117.9	112.6
RB	37	200.5	195.2	187.2	178.3	169.3	161.1	147.6	137.8	129.8	122.2	116.3
SR	38	214.0	207.4	198.6	188.8	179.2	170.3	155.5	144.6	135.7	127.6	121.1
Y	39	220.7	213.2	203.2	192.5	182.4	173.5	159.8	150.3	141.9	132.9	126.0
ZR	40	231.9	225.2	215.7	205.3	194.9	185.3	169.6	157.9	148.3	139.1	131.8
NB	41	235.8	228.9	219.6	209.3	199.1	189.7	174.3	162.8	153.2	143.8	136.3
MO	42	225.2	219.7	211.5	202.2	192.9	184.2	170.0	159.2	149.9	140.8	133.4
TC	43	238.2	231.7	222.2	211.5	200.9	190.9	174.7	162.7	153.0	143.7	136.1
RU	44	224.8	220.7	213.3	204.4	195.1	186.4	171.9	161.1	152.1	143.2	135.7
RH	45	220.3	216.7	210.0	201.7	193.2	185.1	171.8	161.8	153.2	144.4	137.0
PD	46	211.8	209.2	202.7	194.2	185.5	177.5	165.1	157.2	150.8	143.7	136.9
AG	47	206.5	205.4	200.5	193.7	186.2	178.9	166.9	158.3	151.7	144.7	136.5
CD	48	212.7	211.0	206.1	199.4	192.2	185.1	172.9	163.5	156.0	148.9	141.9
IN	49	217.0	216.4	213.3	208.5	202.6	196.2	183.7	173.1	165.1	159.0	152.5
SN	50	224.1	220.8	215.0	208.0	200.7	193.7	181.5	172.2	164.7	157.7	151.3
SB	51	228.9	224.5	218.5	212.0	205.5	199.3	188.6	179.8	171.7	163.7	156.9
TE	52	236.7	230.7	223.2	215.4	207.7	200.7	188.9	179.7	171.5	163.3	156.5
I	53	261.0	249.4	235.8	222.2	209.7	199.1	183.7	174.3	167.0	159.0	152.1
XE	54	274.9	263.9	251.3	238.8	227.2	217.1	201.9	191.5	182.8	173.8	166.3
CS	55	268.5	261.2	251.1	240.1	229.4	219.6	204.1	192.7	182.8	172.7	164.7
BA	56	283.6	274.8	263.5	251.5	239.8	229.1	212.0	199.3	188.6	177.9	169.4
LA	57	294.0	284.0	271.2	257.6	244.5	232.8	214.4	201.4	190.6	179.8	171.2
CE	58	277.9	270.3	260.0	248.7	237.7	227.5	210.7	198.1	187.5	177.3	168.9
PR	59	274.6	267.5	257.8	246.9	236.2	226.2	209.7	197.2	186.7	176.7	168.4
ND	60	271.3	264.7	255.4	245.0	234.6	224.9	208.6	196.3	185.8	175.9	167.8
PM	61	268.1	261.9	253.0	242.9	232.8	223.4	207.4	195.2	184.9	175.1	167.1
SM	62	264.8	258.9	250.3	240.6	230.8	221.7	206.1	194.1	183.9	174.2	166.3
EU	63	261.4	255.9	247.7	238.3	228.8	219.9	204.7	192.8	182.7	173.2	165.5
GD	64	270.0	264.2	255.6	245.9	236.0	226.7	210.8	198.3	187.7	177.7	169.7
TB	65	255.2	250.2	242.6	233.8	224.9	216.4	201.8	190.3	180.4	171.2	163.8
DY	66	246.7	243.6	237.4	229.6	221.5	213.6	200.1	189.4	180.0	170.9	163.5
HO	67	235.9	233.3	227.9	221.0	213.5	206.1	193.1	182.4	173.0	164.2	157.2
ER	68	233.0	230.6	225.4	218.7	211.4	204.2	191.4	180.9	171.8	163.1	156.1
TM	69	230.0	227.7	222.7	216.2	209.1	202.2	189.7	179.5	170.4	161.8	155.0
YB	70	227.6	225.4	220.5	214.2	207.3	200.5	188.3	178.3	169.3	160.8	154.1
LU	71	232.6	230.0	224.6	217.9	210.7	203.7	191.1	180.7	171.5	162.7	155.7
HF	72	240.8	238.1	232.6	225.6	218.0	210.7	197.6	186.7	177.0	167.8	160.5
TA	73	235.9	233.5	228.1	221.2	213.8	206.6	193.9	183.6	174.5	165.6	158.3
W	74	229.0	227.2	222.6	216.6	210.1	203.6	191.9	181.9	172.7	163.7	156.5
RE	75	251.5	248.7	242.9	235.6	227.8	220.2	206.6	195.3	185.1	175.2	167.4
OS	76	250.9	248.6	243.3	236.4	228.8	221.4	208.1	196.9	186.7	176.6	168.7
IR	77	250.1	248.5	243.7	237.2	230.0	222.8	209.8	198.8	188.6	178.5	170.5
PT	78	244.4	244.9	241.8	236.7	230.7	224.4	212.6	202.3	192.3	182.3	174.4
AU	79	250.5	249.3	244.9	239.0	232.5	226.2	215.2	206.1	197.4	187.9	179.8
HG	80	241.1	241.2	237.9	232.9	227.1	221.2	210.1	200.3	190.8	181.0	173.1
TL	81	252.7	251.1	246.6	240.6	234.0	227.4	215.6	205.4	195.5	185.5	177.3
PB	82	275.7	272.7	266.8	259.5	251.8	244.4	231.3	220.3	209.8	199.0	190.1
BI	83	269.7	265.9	259.4	251.6	243.7	236.1	223.2	212.6	202.5	191.9	183.4
PO	84	276.8	272.2	264.9	256.3	247.7	239.7	226.3	215.6	205.4	194.5	185.9
AT	85	283.0	278.1	270.1	260.8	251.5	243.0	229.2	218.4	208.1	196.9	188.2
RN	86	288.7	283.7	275.1	265.2	255.3	246.4	232.2	221.4	210.9	199.2	190.4
FR	87	310.1	302.7	292.0	280.3	269.0	258.7	242.4	230.1	218.6	206.3	196.9
RA	88	325.6	316.6	304.7	291.9	279.5	268.4	250.5	237.0	224.6	211.8	202.0
AC	89	333.5	324.2	311.8	298.4	285.4	273.6	254.8	240.7	228.0	214.8	204.9
TH	90	339.6	330.3	317.7	303.9	290.5	278.22	258.8	244.2	231.1	217.7	207.5
PA	91	328.9	321.2	310.1	297.7	285.4	274.1	255.6	241.5	228.9	216.2	206.5
U	92	326.2	319.2	308.7	296.8	284.9	273.9	255.6	241.5	229.1	216.6	207.1

θ	180°	178°	176°	174°	172°	170°	168°	166°	164°	162°	160°
δ^2	0	1.218 x10^{-3}	4.874 x10^{-3}	1.097 x10^{-2}	1.950 x10^{-2}	3.046 x10^{-2}	4.386 x10^{-2}	5.971 x10^{-2}	7.798 x10^{-2}	9.870 x10^{-2}	1.218 x10^{-1}

Table 22. ^{4}He Stopping Cross Section Factor [ε_0]. This table gives the stopping cross section factor in 10-15 *electron voltage centimeters per atom* in the surface energy approximation, as defined by Eq. (3.12), for ^{4}He of normal incidence ($\theta_1 = 0$) and a scattering angle θ of 170° ($\theta_2 = 10°$) for incident energies E0 from 1.0 to 4.0 MeV. The table is based on the semiempirical stopping cross sections given in Table VI and on values of $K_{\overline{M}}$ from J.F. Ziegler, Thin Solid Films **19**, 289 (1973). The values of KM are practically equal to those given in Table 21 for $\overline{K}$. [After W.-K. Chu, J.W. Mayer and M.-A. Nicolet, "Backscattering Spectrometry," Academic Press, New York, 1978, p. 366.]

Table 23. ^{4}He Energy Loss Factor [S_0]

ELEMENT	AT. NO. (Z_2)	ENERGY (in MeV) OF INCIDENT ^{4}He 1.0	1.2	1.4	1.6	1.8	2.0	2.4	2.8	3.2	3.6	4.0	
BE	4	12.046	37.3	37.2	37.1	36.9	36.8	36.6	36.2	35.9	35.5	35.1	34.8
B	5	13.093	49.7	50.0	50.1	50.2	50.1	50.0	49.7	49.3	48.7	47.8	46.9
C	6	11.364	44.0	45.4	46.5	47.3	47.8	48.2	48.7	48.7	48.3	47.4	46.4
N	7	3.482	20.7	20.9	21.0	20.9	20.8	20.7	20.2	19.7	19.0	18.2	17.5
O	8	4.302	26.4	26.8	27.0	27.0	26.9	26.7	26.0	25.2	24.3	23.2	22.1
F	9	3.522	21.5	22.0	22.2	22.3	22.2	22.0	21.5	20.8	20.0	19.1	18.2
NE	10	3.585	22.2	22.8	23.1	23.2	23.1	22.9	22.4	21.7	20.8	20.0	19.1
NA	11	2.541	16.8	17.0	17.0	16.9	16.7	16.5	16.1	15.5	14.9	14.3	13.7
MG	12	4.302	37.2	36.8	36.3	35.6	34.8	34.0	32.3	30.7	29.2	27.7	26.5
AL	13	6.023	51.1	50.3	49.4	48.4	47.3	46.2	44.0	41.8	39.8	37.9	36.1
SI	14	4.978	54.5	53.3	51.6	49.8	47.9	46.1	42.7	39.7	37.0	34.6	32.8
P	15	3.543	38.1	37.6	36.7	35.5	34.3	33.1	30.8	28.7	26.8	25.1	23.8
S	16	3.886	42.9	42.7	41.9	40.7	39.4	38.0	35.3	32.8	30.6	28.6	27.2
CL	17	3.221	44.7	43.3	41.5	39.5	37.5	35.5	32.2	29.4	27.2	25.3	24.0
AR	18	2.489	36.3	35.1	33.4	31.6	29.7	28.0	25.1	22.9	21.2	19.7	18.8
K	19	1.330	20.0	19.8	19.3	18.6	17.8	17.0	15.5	14.3	13.2	12.3	11.6
CA	20	2.014	32.8	32.1	31.1	29.9	28.7	27.4	25.1	23.0	21.3	19.8	18.7
SC	21	4.015	65.9	64.5	62.6	60.3	57.9	55.4	50.8	46.8	43.4	40.5	38.2
TI	22	5.682	93.4	91.6	88.8	85.6	82.2	78.8	72.8	67.7	63.4	59.5	56.4
V	23	7.213	113.2	111.1	108.0	104.4	100.6	96.8	90.1	84.4	79.4	74.9	71.2
CR	24	8.331	125.8	124.6	121.9	118.4	114.5	110.6	103.3	97.3	92.1	87.5	83.5
MN	25	8.150	120.6	119.4	116.8	113.4	109.8	106.2	99.6	94.1	89.4	85.0	81.4
FE	26	8.483	133.3	131.8	128.5	124.1	119.5	114.9	106.7	100.3	95.2	90.8	87.2
CO	27	8.990	131.1	130.7	128.6	125.4	121.7	117.8	110.7	104.8	99.8	95.5	91.8
NI	28	9.126	126.3	127.0	126.1	124.2	121.6	118.6	112.4	106.1	99.8	93.8	88.9
CU	29	8.483	110.3	111.9	112.1	111.3	109.8	107.8	103.1	98.6	94.5	90.9	87.6
ZN	30	6.547	86.1	86.3	85.6	84.3	82.5	80.6	76.6	73.0	69.7	66.8	64.3
GA	31	5.104	73.3	72.8	71.5	69.9	68.1	66.1	62.4	59.1	56.2	53.8	51.6
GE	32	4.429	66.4	65.6	64.2	62.4	60.5	58.6	54.9	51.8	49.2	46.9	45.0
AS	33	4.598	73.5	72.5	70.7	68.5	66.2	63.9	59.6	56.1	53.2	50.6	48.4
SE	34	3.650	59.5	58.2	56.4	54.4	52.3	50.3	46.9	44.1	41.8	39.7	38.0
BR	35	2.563	46.8	45.1	43.1	40.9	38.8	36.9	33.9	31.8	30.2	28.6	27.4
KR	36	1.870	36.5	35:1	33.4	31.7	30.0	28.5	26.2	24.6	23.3	22.1	21.1
RB	37	1.077	21.6	21.0	20.2	19.2	18.2	17.4	15.9	14.8	14.0	13.2	12.5
SR	38	1.787	38.2	37.1	35.5	33.8	32.0	30.4	27.8	25.8	24.3	22.8	21.6
Y	39	3.042	67.1	64.8	61.8	58.6	55.5	52.8	48.6	45.7	43.2	40.4	38.3
ZR	40	4.272	99.1	96.2	92.2	87.7	83.3	79.2	72.4	67.5	63.3	59.4	56.3
NB	41	5.577	131.5	127.7	122.4	116.7	111.0	105.8	97.2	90.8	85.4	80.2	76.0
MO	42	5.407	144.3	140.8	135.5	129.6	123.6	118.1	108.9	102.0	96.0	90.2	85.5
TC	43	0.000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RU	44	7.257	163.2	160.2	154.8	148.3	141.6	135.3	124.8	116.9	110.3	103.9	98.5
RH	45	7.257	159.8	157.3	152.4	146.4	140.2	134.3	124.7	117.4	111.2	104.8	99.4
PD	46	6.767	143.3	141.6	137.1	131.4	125.6	120.1	111.8	106.4	102.1	97.3	92.7
AG	47	5.848	120.7	120.1	117.2	113.3	108.9	104.6	97.6	92.6	88.7	84.6	79.8
CD	48	4.598	97.8	97.0	94.7	91.7	88.4	85.1	79.5	75.2	71.7	68.4	65.2
IN	49	3.836	83.2	83.0	81.8	80.0	77.7	75.3	70.5	66.4	63.4	61.0	58.5
SN	50	3.695	82.8	81.6	79.5	76.9	74.2	71.6	67.1	63.6	60.8	58.3	55.9
SB	51	3.273	74.9	73.5	71.5	69.4	67.3	65.2	61.7	58.8	56.2	53.6	51.4
TE	52	2.938	69.5	67.8	65.6	63.3	61.0	59.0	55.5	52.8	50.4	48.0	46.0
I	53	2.344	61.2	58.4	55.3	52.1	49.2	46.7	43.1	40.9	39.1	37.3	35.7
XE	54	1.404	38.6	37.0	35.3	33.5	31.9	30.5	28.3	26.9	25.7	24.4	23.3
CS	55	0.860	23.1	22.5	21.6	20.7	19.7	18.9	17.6	16.6	15.7	14.9	14.2
BA	56	1.544	43.8	42.4	40.7	38.8	37.0	35.4	32.7	30.8	29.1	27.5	26.2
LA	57	2.677	78.7	76.0	72.6	69.0	65.5	62.3	57.4	53.9	51.0	48.1	45.8
CE	58	2.858	79.7	77.5	74.6	71.3	68.2	65.2	60.4	56.8	53.8	50.8	48.5
PR	59	2.896	79.5	77.5	74.6	71.5	68.4	65.5	60.7	57.1	54.1	51.2	48.8
ND	60	2.924	79.3	77.4	74.7	71.6	68.6	65.7	61.0	57.4	54.3	51.4	49.1
PM	61	0.000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
SM	62	3.027	80.1	78.4	75.8	72.8	69.9	67.1	62.4	58.8	55.7	52.7	50.3
EU	63	2.084	54.5	53.3	51.6	49.7	47.7	45.8	42.7	40.2	38.1	36.1	34.5
GD	64	3.027	81.7	80.0	77.4	74.4	71.4	68.6	63.8	60.0	56.8	53.8	51.4
TB	65	3.137	80.1	78.5	76.1	73.3	70.5	67.9	63.3	59.7	56.6	53.7	51.4
DY	66	3.170	78.2	77.2	75.3	72.8	70.2	67.7	63.4	60.0	57.1	54.2	51.8
HO	67	3.221	76.0	75.2	73.4	71.2	68.8	66.4	62.2	58.7	55.7	52.9	50.6
ER	68	3.273	76.3	75.5	73.8	71.6	69.2	66.9	62.7	59.2	56.2	53.4	51.1
TM	69	3.328	76.5	75.8	74.1	71.9	69.6	67.3	63.1	59.7	56.7	53.8	51.6
YB	70	2.429	55.3	54.7	53.6	52.0	50.3	48.7	45.7	43.3	41.1	39.1	37.4
LU	71	3.384	78.7	77.8	76.0	73.7	71.3	68.9	64.7	61.2	58.0	55.0	52.7
HF	72	4.429	106.6	105.5	103.0	99.9	96.6	93.3	87.5	82.7	78.4	74.3	71.1
TA	73	5.526	130.4	129.0	126.0	122.2	118.1	114.1	107.1	101.5	96.4	91.5	87.5
W	74	6.320	144.7	143.6	140.7	136.9	132.8	128.7	121.3	115.0	109.1	103.4	98.9
RE	75	6.805	171.2	169.2	165.3	160.3	155.0	149.9	140.6	132.9	126.0	119.2	113.9
OS	76	7.145	179.2	177.6	173.8	168.9	163.5	158.2	148.7	140.7	133.4	126.2	120.6
IR	77	7.053	176.4	175.3	171.9	167.3	162.2	157.1	148.0	140.2	133.0	125.9	120.3
PT	78	6.619	161.8	162.1	160.1	156.7	152.7	148.5	140.7	133.9	127.3	120.7	115.4
AU	79	5.905	147.9	147.2	144.6	141.1	137.3	133.6	127.1	121.7	116.5	111.0	106.2
HG	80	4.070	98.1	98.1	96.8	94.8	92.4	90.0	85.5	81.5	77.6	73.7	70.5
TL	81	3.502	88.5	87.9	86.4	84.2	81.9	79.6	75.5	71.9	68.5	65.0	62.1
PB	82	3.291	90.7	89.7	87.8	85.4	82.9	80.4	76.1	72.5	69.1	65.5	62.6
BI	83	2.828	76.3	75.2	73.3	71.2	68.9	66.8	63.1	60.1	57.3	54.3	51.9
PO	84	2.653	73.4	72.2	70.3	68.0	65.7	63.6	60.1	57.2	54.5	51.6	49.3
AT	85	0.000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RN	86	0.000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
FR	87	0.000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RA	88	1.338	43.6	42.4	40.8	39.1	37.4	35.9	33.5	31.7	30.1	28.3	27.0
AC	89	0.000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
TH	90	3.027	102.8	100.0	96.2	92.0	87.9	84.2	78.3	73.9	69.9	65.9	62.8
PA	91	4.015	132.1	129.0	124.5	119.5	114.6	110.1	102.6	97.0	91.9	86.8	82.9
U	92	4.818	157.2	153.8	148.8	143.0	137.3	132.0	123.1	116.4	110.4	104.4	99.8

θ	180°	178°	176°	174°	172°	170°	168°	166°	164°	162°	160°
δ^2	0	1.218 x10^{-3}	4.874 x10^{-3}	1.097 x10^{-2}	1.950 x10^{-2}	3.046 x10^{-2}	4.386 x10^{-2}	5.971 x10^{-2}	7.798 x10^{-2}	9.870 x10^{-2}	1.218 x10^{-1}

Table 23. ^{4}He Energy-Loss Factor [S_0]. This table gives the energy-loss factor in *electron volts per angstrom* in the surface energy approximation, as defined by Eq. (3.11), for ^{4}He of normal incidence ($\theta_1 = 0$) and a scattering value of θ - 170°C ($\theta_2 = 10°$) and for incident energies from 1.0 to 4.0 MeV. The table is based on the values given in Table 22 and the atomic densities listed in the third column. The value of [S_0] depends linearly on the value of the atomic density N through the relation between dE/dx and ε: $\varepsilon \equiv (1/N)dE/dx$ [Eq. (2.36)]. Values of $K_{\overline{M}}$ were taken from J.F. Ziegler, Thin Solid Films **19**, 289 (1973). The values of K_M are practically equal to those given in Table 21 for $\overline{K}$. [After W.-K. Chu, J.W. Mayer and M.-A. Nicolet, "Backscattering Spectrometry," Academic Press, New York, 1978, p. 368.]

Figure 197 shows a different example. Spectra of three metal films on Si are displayed. Determine which backscatter angle was used in this experiment! The height of each spectrum is proportional to the backscattering cross section $d\sigma/d\Omega$ (αZ^2) and the width ΔE of the energy axis channels. It is inversely proportional to the stopping cross section factor because we know from Fig. 193:

$$N\Delta x = \frac{\Delta E}{\varepsilon}.$$

The ratio of the height of the spectra should be:

$$Z^2_{Pt}/\varepsilon_{Pt} : Z^2_{Pd}/\varepsilon_{Pd} : Z^2_{Ti}/\varepsilon_{Ti} = 78^2/224.4 : 4.46^2/224.4 : 46^2/177.5 : 22^2/138.7 = 27.11 : 11.92 : 3.49$$

These ratios correspond very closely to the ratios which can be extracted from the figure.

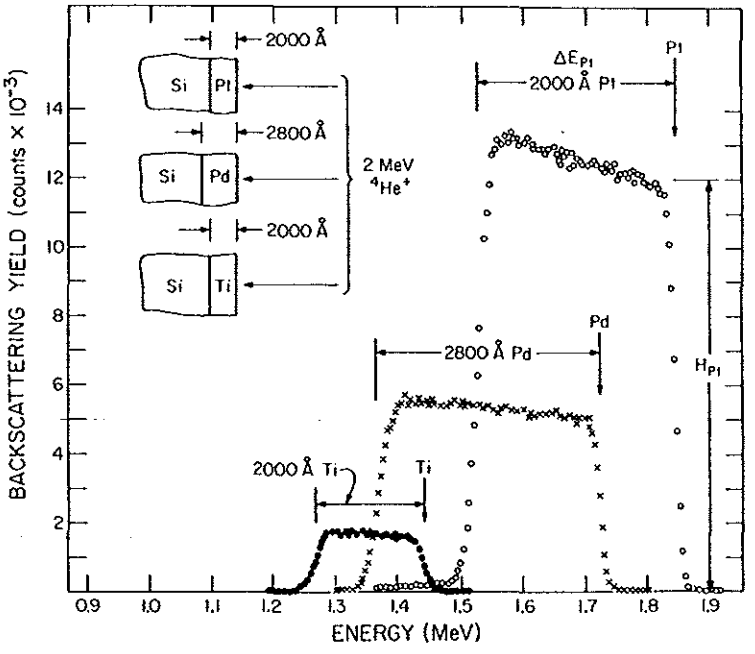


Fig. 197. Superimposed energy spectra of 2.0 MeV $^4He^+$ ions backscattered from three different thin films. The spectra were recorded under unchanged systems conditions, so that the differences in the three spectra reflect those in the scattering cross sections (yields), in the same mass (positions of energy edge given by the downward vertical arrows), and in the energy loss factors [S] (signal widths). The substrate signals are not shown. [After M.-A. Nicolet, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 606.]

The ratio of concentrations of elements in a thin compound film can be obtained with RBS. For a binary compound, one finds:

$$\frac{H_M/\sigma_M}{H_m/\sigma_m} = \frac{N_M^{comp}}{N_m^{comp}} \cdot \frac{[S]_m^{comp}}{[S]_M^{comp}}$$

The ratio of energy loss factors ([S]) is generally close to unity. In reasonable approximation, one obtains the composition ratio from the heights (H) of the normalized spectra. A slightly better value is obtained when the total number of counts (A) under each peak is used instead of the height:

$$\frac{A_M/\sigma_M}{A_m/\sigma_m} = \frac{N_M^{comp}}{N_m^{comp}}$$

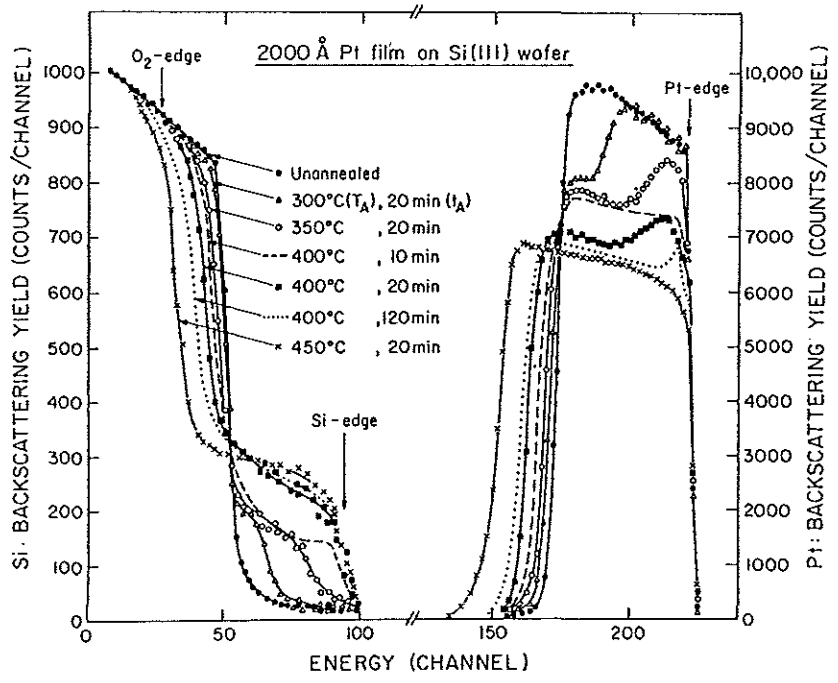


Fig. 198. A thin layer of platinum bonds strongly to silicon after a short anneal at temperatures well below the eutectic point (980°C). This fact has long been known and exploited in the semiconductor devices. This series of backscattering spectra reveals what happens. The Pt_2Si compound is formed, first at the interface and then throughout the whole film (300°, 350° and 400° curves). Upon further anneal, the layer finally turns into the stable from PtSi (450°C). The composition follows directly from the ratio of the two backscattering yield Y_{Si} and Y_{Pt} and the (known) values of scattering cross sections and stopping power for silicon and platinum [After A. Hiraki, *et al.*, Appl. Phys. Lett. **18**, 178 (1971).]

Figure 198 shows the various stages of platinum-silicide formation. After the 10 min/400°C annealing step, the first compound has formed. Using the first of the two approximations to determine the composition, we find:

$$\frac{H_{Pt}/\sigma_{Pt}}{H_{Si}/\sigma_{Si}} = \frac{7300/78^2}{130/14^2} = \frac{1.20}{.66} \approx 2 \text{ (i.e., } Pt_2Si)$$

After the last annealing step (20 min/450°C), we obtain:

$$\frac{H_{Pt}/\sigma_{Pt}}{H_{Si}/\sigma_{Si}} = \frac{6300/78^2}{230/14^2} = \frac{1.04}{1.17} \approx 1 \text{ (i.e., PtSi)}$$

Practically all silicide formation studies have been conducted using RBS.

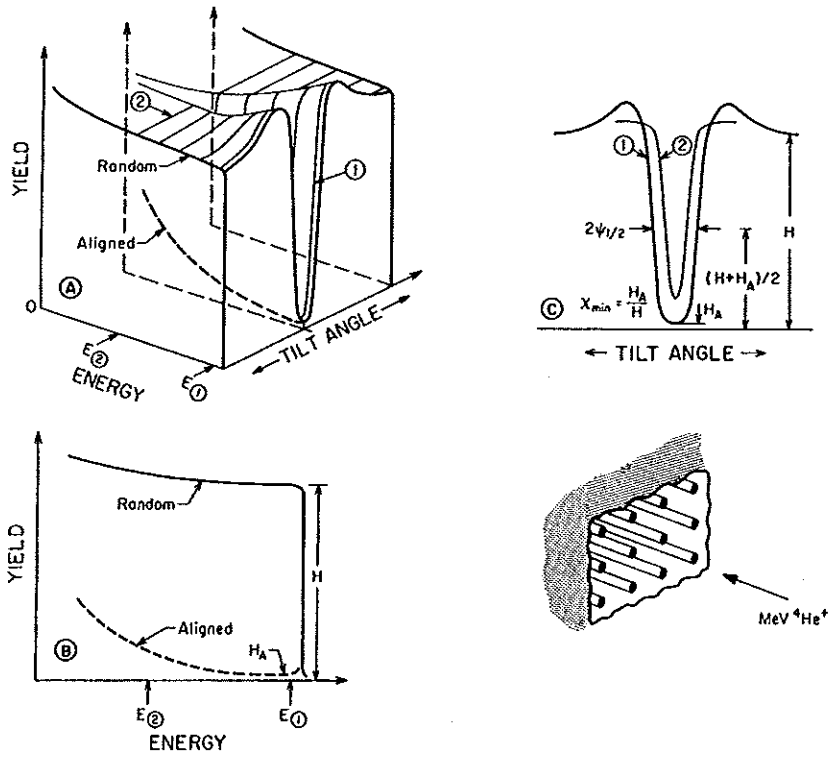


Fig. 199. Schematic representation of a channeling experiment in an axial direction. If the single-crystal sample is tilted away from the direction of exact axial alignment with the incident beam, the backscattering yield increases to its "random" value as shown in (A). (B) gives the energy spectrum of backscattered particles under exact axial alignment and for random direction of incidence. (C) gives the change in backscattering yield as a function of tilt angle for two energies. [After M.-A. Nicolet, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 611.]

For all the preceding discussion, we have assumed non-crystalline materials. In the majority of semiconductor applications, however, one uses single crystal bulk and thin films. As we have already seen in the section on ion implantation, one expects very marked effects when a collimated parallel ion beam hits a crystal along one of the major axes. The ion range for a given energy increases and for RBS, the backscattering yield decreases. The number of backscattered helium ions can be one hundred times less for the <110> directions than for a random orientation. Figure 199 shows RBS yield versus tilt angle and energy. This so-called "channeling" is effectively used to study the regrowth of amorphous layers (rendered amorphous by ion implantation), to determine the radiation damage in thin layers and to find the position of impurities by triangulation (channeling in several channels) (Fig. 200).

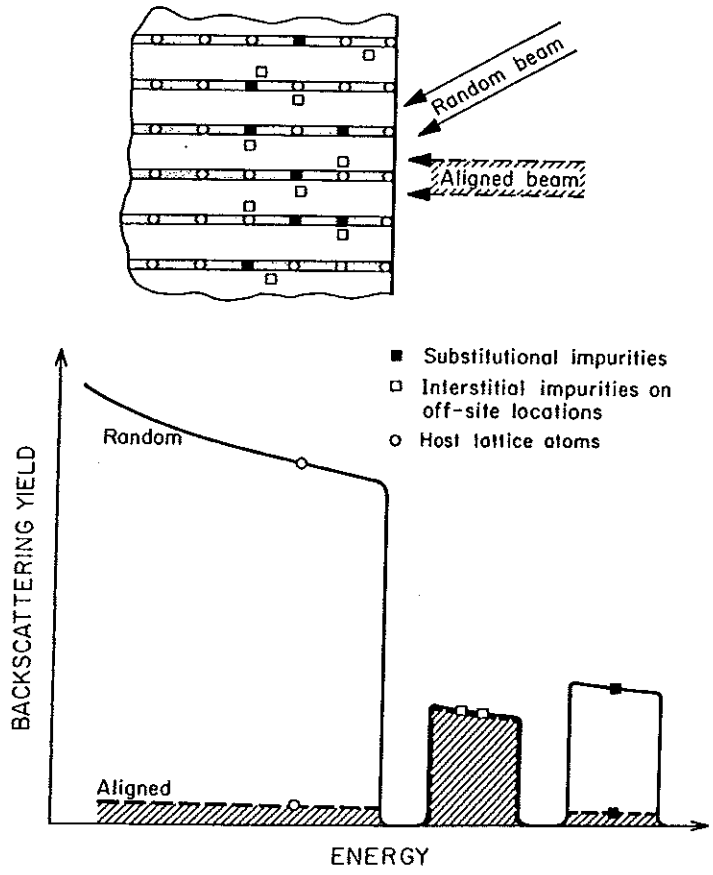


Fig. 200. The comparison of backscattering spectra taken with a collimated beam in a random and an axially aligned direction of incidence reveals if a heavy impurity atom is located in the row of atoms of the host lattice aligned with the beam (full squares) or not (open squares). [After M.-A. Nicolet, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 613.]

Tables 24-26 summarize the most important aspects of RBS.

Table 24
Summary of Backscattering Capabilities

Mass Perception	1 amu up to 40 amu 10 amu for heavy elements	
Depth Perception	Range	$\lesssim 1\ \mu m$
	Resolution	$\sim 200\ \mathring{A}$
	Accuracy	$\sim 5\%$
	No external standards required	
Composition Analysis	Atomic ratios or atoms per unit area	
	Accuracy	$\sim 5\%$ or better
	No external standards required	
Sensitivity	Heavy elements in light matrix	$\gtrsim 10^{18}/cm^2$
	Light elements in heavy matrix	$\gtrsim 10^{-1}$
	Similar masses - poor	
Crystal Structure	Lattice location	
	Epitaxy	
	Point defects sensitivity	$\gtrsim 5 \times 10^{20}/cm^3$
	Line defects sensitivity	$\gtrsim 10\ cm/cm^2$

Table 25
Relevant Parameters of a Backscattering Experiment and Typical Values

Particle	$^1H^+$, $^4H^+$, $^4He^{++}$, $^7Li^+$, $^{14}N^+$, . . .	
Beam Energy E	1 - 3	MeV
Accessible Depth	~ 5000	$\mathring{A}$
Beam Cross Section	~ 1	mm^2
Beam Current	$\lesssim 50$	nA
Power in Target	$\lesssim 100$	mW
Total Dose	~ 10	μC
Exposure Time	~ 10	min
Vacuum	$\sim 10^{-6}$	Torr

Table 26

Deficiencies of BS and Remedies

Deficiency	Remedy
Only atomic composition, no chemical information	x-rays AUGER ESCA
No microscopic lateral resolution	SEM TEM μ probe
Poor sensitivity to light elements	MS SIMS μ probe PIX nuclear reactions
Radiation damage	

[Tables 24-26: After M.-A. Nicolet, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 615-6.]

7.5. Electron Beam Induced Current Microscopy (EBIC)

References:

- C.J. Varker, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J. N. Zemel, ed. (Plenum Press, New York 1978), chapter 10, pp. 515-80.

This is the second charged-particle beam technique which we will discuss. RBS provides information on the elemental composition of the top few thousand Å of a semiconductor. EBIC, on the other hand, depends mainly on carrier lifetime and charge collection properties in a semiconductor diode. RBS has very poor lateral resolution but is quite a good "depth" microscope. EBIC, though, has the resolution of a scanning electron microscope (SEM) but the large variance in electron range and the large number of possible interactions lead to rather poor depth resolution.

Figure 201 illustrates the EBIC technique. An energetic electron beam (tens of keV) of a scanning electron microscope is focused on a semiconductor structure. The electrons and holes produced during the stopping of the incoming electrons diffuse through the crystal. Some of the

minority carriers reach the depletion region of the reverse biased p-n junction and are collected (Fig. 202). The collected charge corresponds to a current which is used to modulate the Z-axis (intensity) of the scanning electron microscope display. In a simplified approximation, one can write for the diode reverse current:

$$I = e\ G_D(x)(L_n + L_p) + eG_W(x)WA$$

with e = charge of the electron (= 1.6×10^{-19} C);

G_D, G_W = carrier generation rates outside (D) and inside (W) the depletion layer;

W = width of the depletion layer;

A = area of the depletion layer;

Ln, Lp = minority carrier diffusion length; and

$= \sqrt{D_{n,p} \cdot \tau_{n,p}}$ (τ = lifetime).

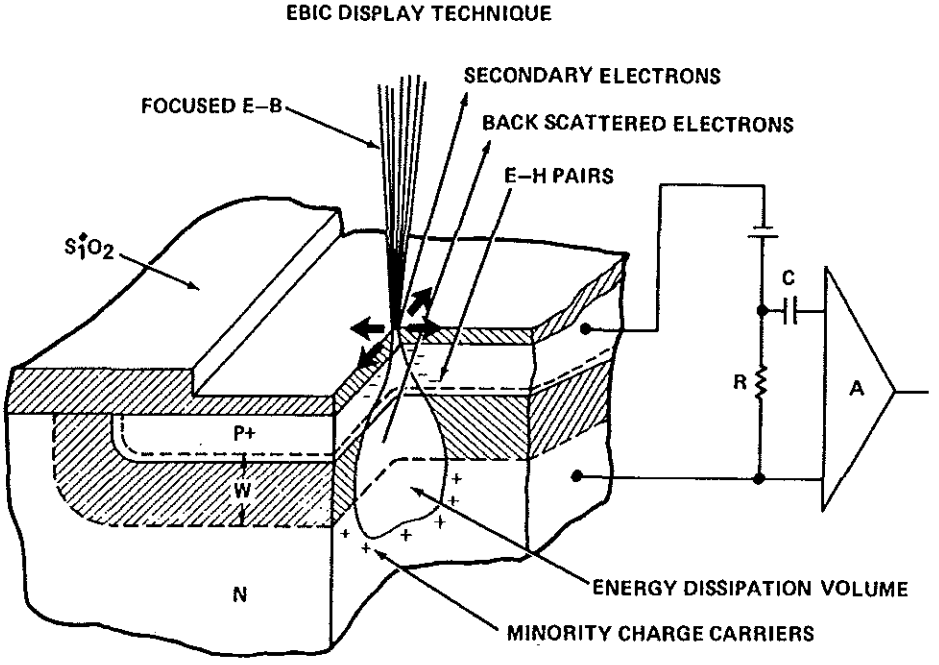


Fig. 201. Schematic illustrating EBIC microscopy. [After C.J. Varker, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 543.]

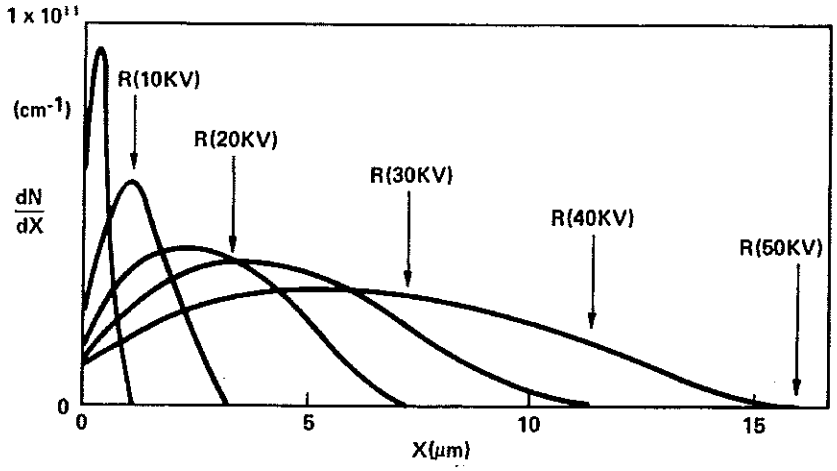


Fig. 202. Hypothetical steady state minority carrier distribution in silicon, resulting from electron penetration and energy dissipation. [After C.J. Varker, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 543.]

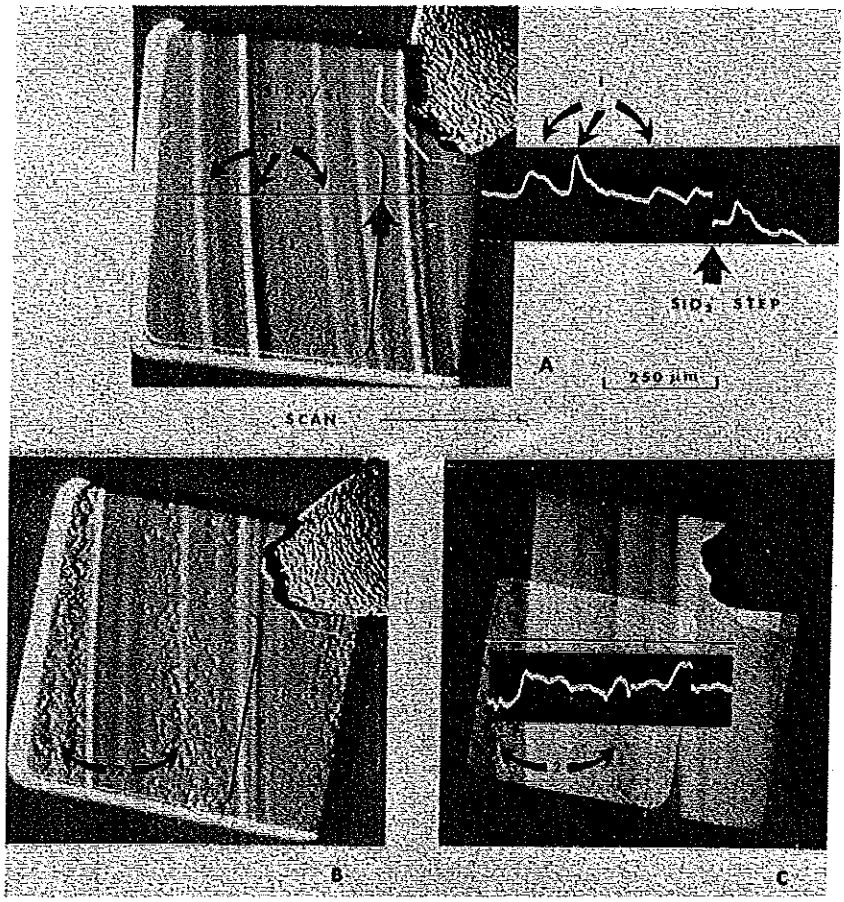


Fig. 203. Crystal inhomogeneities revealed by EBIC microscopy (a) dopant striations due to microsegregation, (b,c) microdefects resulting from impurity point defect complexes (swirl). [After C.J. Varker, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 563.]

The lifetime is the parameter which can change over orders of magnitude within small distances. The lifetime can be strongly correlated by all kinds of defects such as A and B swirls in dislocation-free silicon, oxygen precipitates, deep impurities (Au, Fe in Si; Cu in Ge), stacking faults and a host of impurity-defect complexes. By choosing the appropriate geometry of a test device, one can place the EBIC sensitive p-n junction at the location of interest. Figure 203 shows EBIC signals created by dopant striations and swirls. Figure 204 contains EBIC photographs of emitter "pipes" through the base layer to the collector of a transistor. The pipes consist in this particular case of copper precipitates along partially dissociated stacking faults.

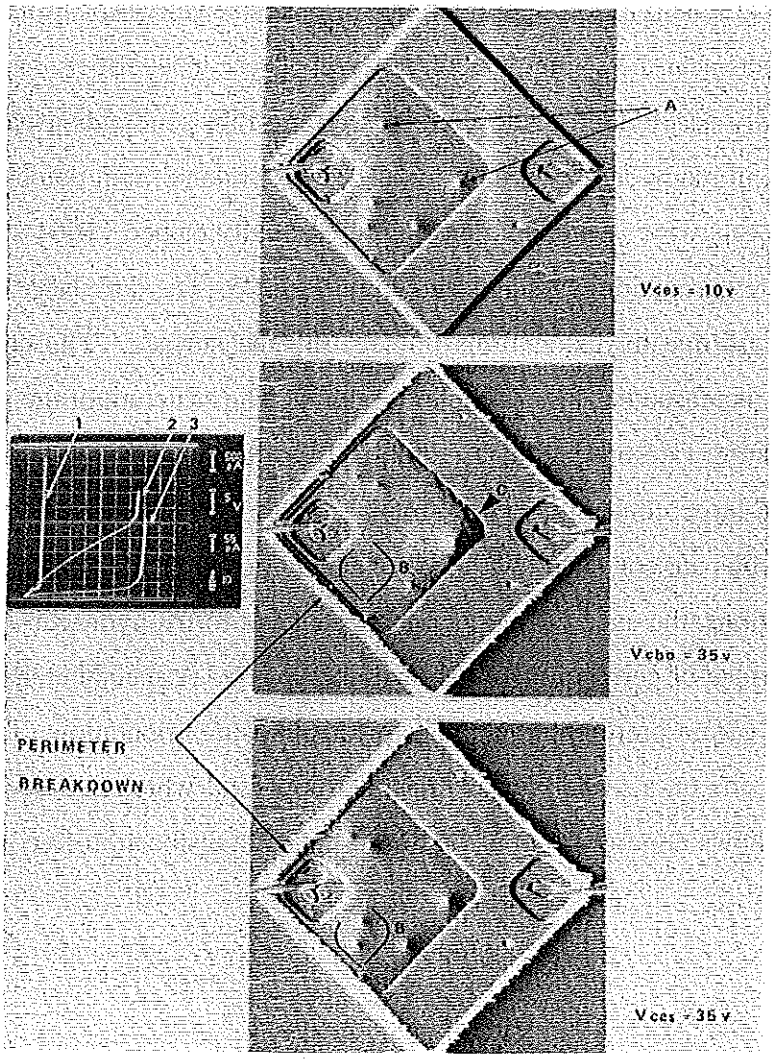


Fig. 204. EBIC micrograph showing the locations of typical emitter pipes. [After C.J. Varker, in "Nondestructive Evaluation of Semiconductor Materials and Devices," J.N. Zemel, ed., Plenum Press, New York, 1979, p. 573.]

7.6 Chemical Methods

Chemical etching of semiconductors is not only important in many device technology process steps but also in several analytical preparation techniques. After the desired crystal geometry (e.g., Si wafer) has been approximated with mechanical means (diamond saw, lapping, etc.), chemical etching (chemomechanical polishing (CMP)) is used to obtain damage-free surfaces. A large number of etching formulae have been developed for many different purposes. The following tables (Table 27) contain formulations for several semiconductors.

The chemical reactivity of etching compositions is sensitive to strains, impurities (Fermi level) or other defects near the semiconductor surface. These local effects can lead to preferential etching. The locally accelerated or decelerated etching rate generates a pattern on the crystal surface which can, in many cases (though not all!), be correlated with defects. Dislocations can be made visible in this way. An example of a preferentially etched {100} surface of a partially dislocated ultra-pure germanium crystal is shown in Fig. 205. The dark large etch pits in the right half of the photo are due to dislocations intersecting the surface. The etching rate near the dislocation core is increased because of the local strain. The structure of the pits is caused by the dependence of etching rate on crystal orientation. On the left-hand side, the dislocation-free section, one sees a large concentration of small, smooth pits without internal structures. It is believed that these pits are correlated with hydrogen and/or vacancy clusters in the crystal. In the absence of dislocations, the excess hydrogen nucleates and forms hydrogen-filled bubbles. Etch pits on a {113} surface of germanium are shown in Fig. 206. A special etchant composition had to be developed to obtain preferential etching for this unusual orientation. The next figure (Fig. 207) displays etch pits on a {100} surface of a GaAs crystal which had been etched in a KOH melt. Figure 208 shows doping striations on the surface of a dislocation-free p-type FZ Si crystal.

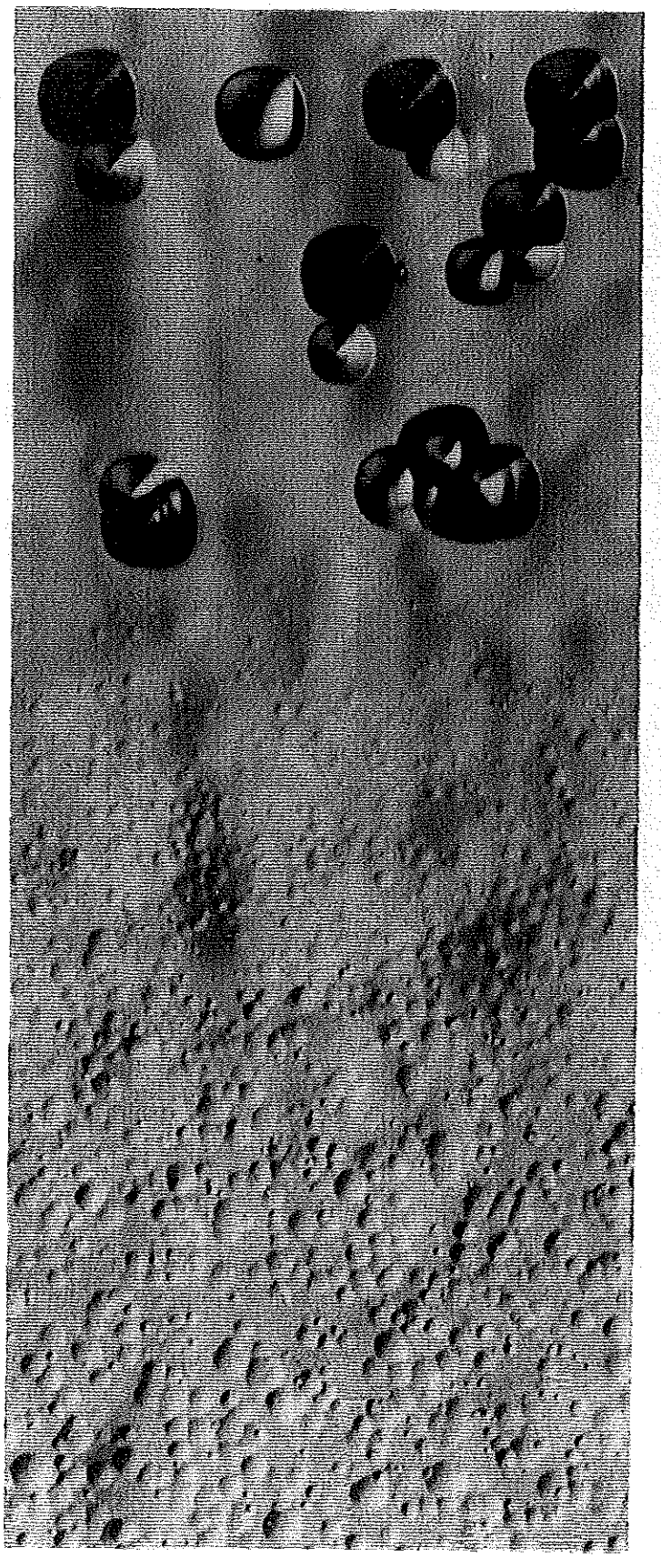


Figure 205.

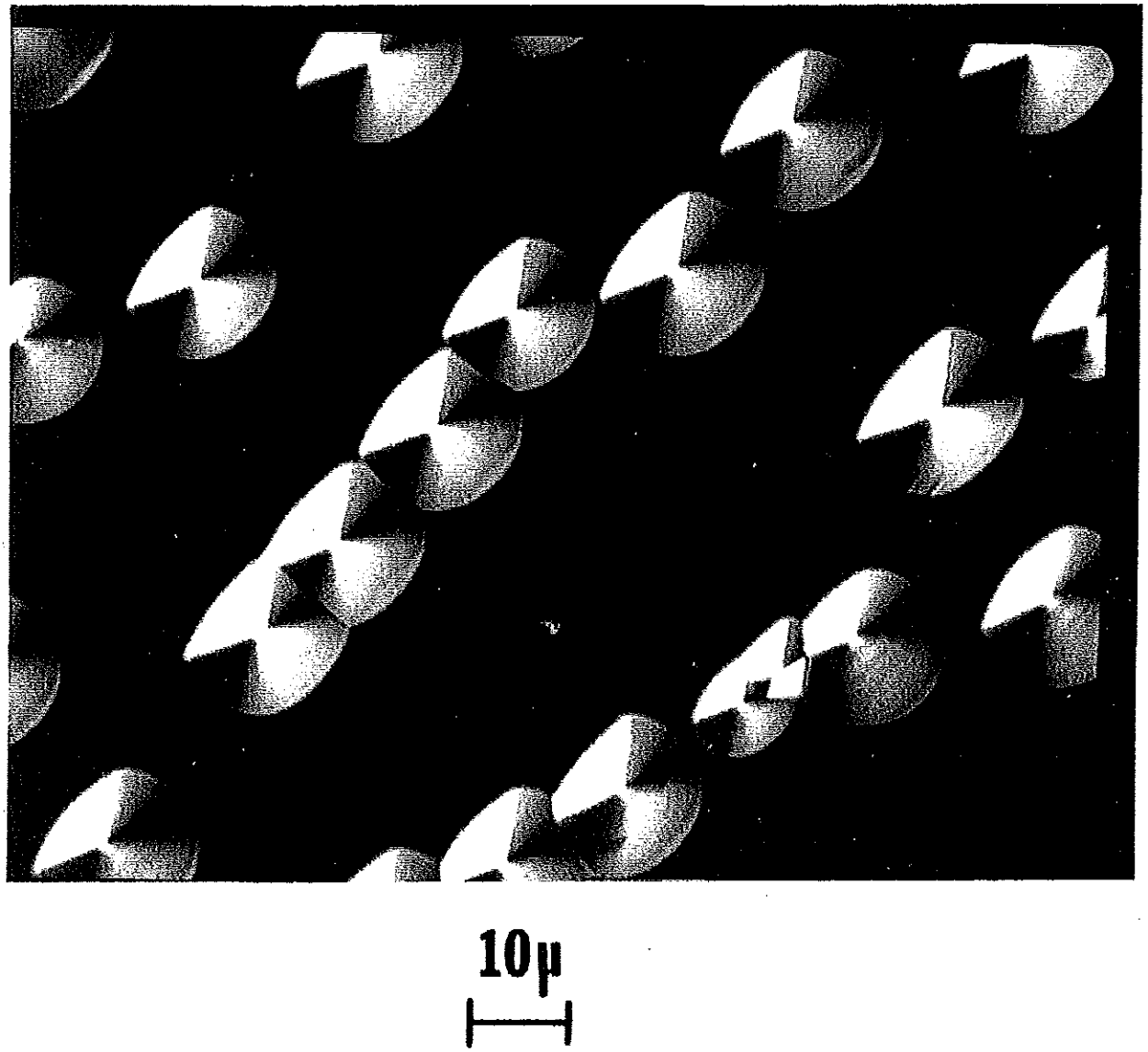


Fig. 206. Dislocation etch pits on a {113} surface of a Ge crystal. The pits are formed by four {111} planes. Note the well-developed planes pointing to the upper left corner.

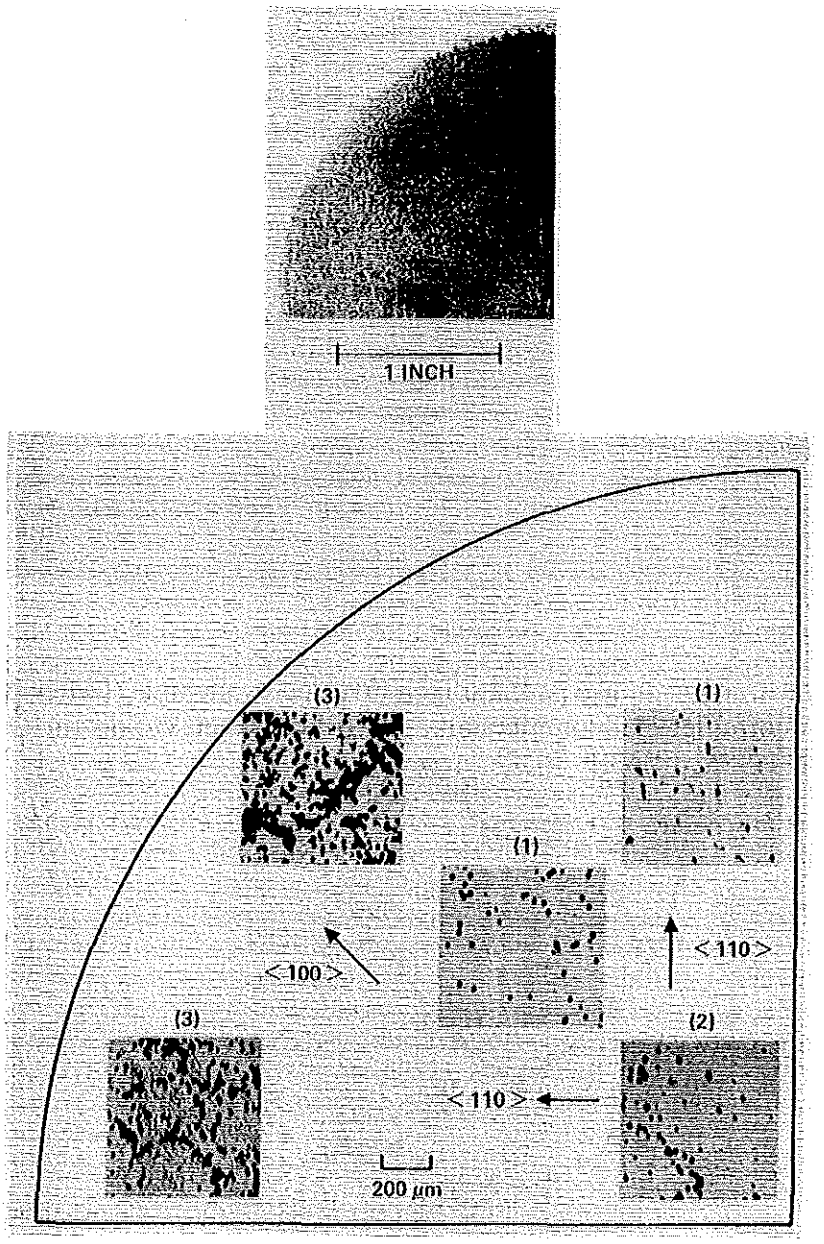


Fig. 207. Photomicrographs of selected regions on (100) LEC GaAs wafer after KOH etch, showing radial dislocation distribution. [After J. Angilello, R.M. Potenski and G.R. Woolhouse, J. Appl. Phys. **46**, 2315 (1975).]

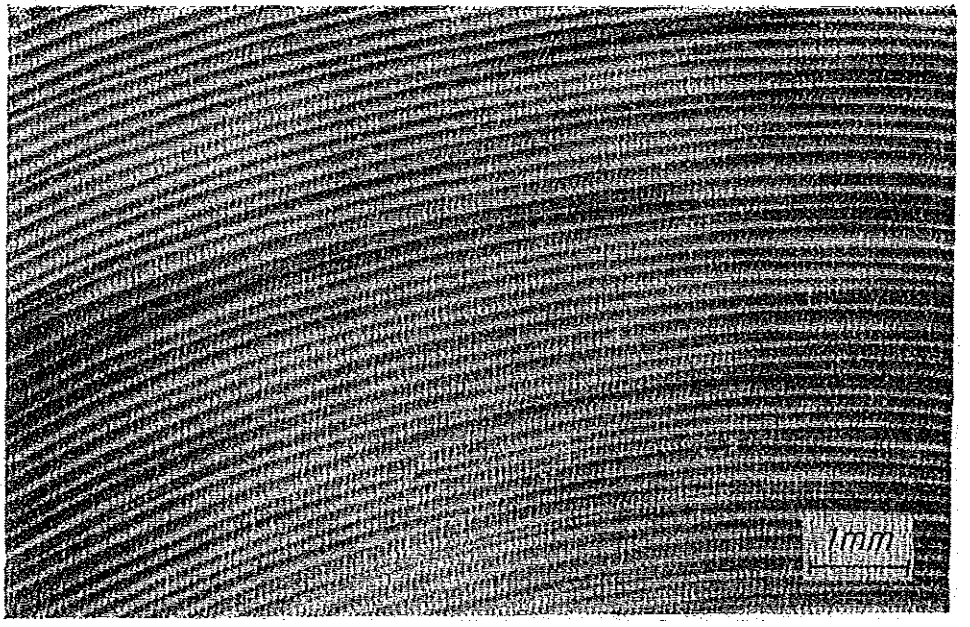


Fig. 208. Dopant striations in p-type FZ silicon revealed by etching. [After H.-F. Hadamovsky, "Werkstoffe der Halbleitertechnik," VEB Deutscher Verlag für Grundstoffindustrie, Leipzig, 1985, p. 353.]

Table 27. Etchants for the More Important Semiconductors
[After P.J. Holmes, "The Electrochemistry of Semiconductors," Academic Press, New York, 1962, pp. 368-77.]

A. GERMANIUM

Name of Etchant	*Recipe*	*Etching Time*	*Principal Uses, and Comments*	*Refs.*
Iodine etch A	5 HF 10 HNO_3 11 acetic acid with 30 mg I_2 dissolved	4 min	Polishing and etching (100) and (111) surfaces. Better than CP.–4 for etching (100)	29
CP.–4	15 HF 25 HNO_3 15 acetic acid with 0·3 Br_2 dissolved	1½ min for etching. 2 min or more for polishing	Polishing, etching (111) and (100), revealing sharp p–n junctions, and grain and twin boundaries	76 77
CP.–4A†	3 HF 5 HNO_3 3 acetic acid	2 to 3 min	Chemical polish. Much slower than CP.–4 at room temperature, so can be used warm (70°C). Absence of bromine means better prospects of a chemically clean surface	
Dash germanium etch	2 HF 4 HNO_3 15 acetic acid	20 sec to 1 min	Etching out of dislocations after precipitation of lithium (see Section IV C 2)	42

† Some confusion over the correct designation of this mixture has led to its being described both as CP.–6 and CP.–8. Current usage is tending to agree on the name CP.–4A.

White etch	1 HF 3 HNO_3	1 to 2 min warm	Chemical cleaning, and revealing p–n junctions	
WAg (silver nitrate etch)	2 HF 1 HNO_3 2 5% $AgNO_3$ solution	1 min	Etching of (111) planes, revealing grain boundaries. Liable to deposit silver which must be removed chemically, preferably by a cyanide wash. Also gives dull background pitting	78
No. 1	1 HNO_3 2 HF 1 10% $Cu(NO_3)_2$ solution	1 to 2 min	Etching of (111) planes. Also used in certain etches for silicon	12

Name of Etchant	*Recipe*	*Etching Time*	*Principal Uses, and Comments*	*Refs.*
Hydrogen peroxide	H_2O_2 (100 vols)	Hot, or 1 hour cold	Gives a clean matt surface, revealing grain boundaries and lineage. P–n junctions shown quite sharply. Good for preparing surfaces for electrical probing (the slight roughening prevents a probe from slipping)	79
No. 2 (Superoxol)	1 H_2O_2 1 HF 4 H_2O	1 to 3 min	Etching of (100) and (111) planes. Slow to attack polished surfaces, but otherwise a good etch for (100). Etch rate has been studied by Camp and Batterman	32 80
Dilute No. 2	1 No. 2 etch 50 H_2O	2 to 16 hours	Produces etch pits, with fine structure, on (111), (100) and (110), and is one of the best etches for (110). Intermediate dilutions between this and No. 2 are also useful for other orientations	81
Alkaline peroxide	8 g NaOH 100 ml 10-vol H_2O_2	70°C	Controlled removal of material (etch rate falls off with age to an approximately constant value when the etch is old: 0·2 mil/min new, 0·05 mil/min at 1 hour)	
Sodium hypochlorite	1 NaClO (10%) 10 H_2O (i.e. 0·1 M)	40 min (40°C), or as required for thinning	Etching on (100) and (111), and for thinning slices for electron microscopy. Interesting as a single-component etchant	75
Ferricyanide	6 g KOH 4 g $K_3Fe(CN)_6$ 50 ml H_2O	1 min boiling	Etching of (111) planes, showing up lineage structure and grain boundaries. This is a very good etch for producing clear triangular pits on (111)	37

B. SILICON

Name of Etchant	*Recipe*	*Etching Time*	*Principal Uses, and Comments*	*Refs.*
White etch	1 HF 3 HNO_3	15 sec	Chemical polish	
Dash etch	1 HF 3 HNO_3 8 to 12 acetic acid	1 to 16 hours	Etching of all planes. Deep pits, following the dislocation lines into the crystal. Smaller proportions of acetic acid give a faster etch, which is often useful	41
CP.–4A	3 HF 5 HNO_3 3 acetic acid	2 to 3 min	Slow chemical polishing revealing twins, twin lamellae, p–n junctions, etc., and sometimes dislocations	28
SD1	25 HF 18 HNO_3 5 acetic acid containing 0·1 Br_2 10 H_2O 1 g $Cu(NO_3)_2$ (i.e. 5 of No. 1 + 3 of CP.–4)	2 to 4 min	Reveals edge and mixed dislocations on all planes. A good etch for rapid estimates of pit density, but the copper deposition is undesirable if the material is to be used again	83
	100 HF 0·1 to 0·5 HNO_3	Applied in drops under a strong light on lightly ground surfaces	P-type material darkened more than n-type	57
	50 ml dilute $Cu(NO_3)_2$ 1 to 2 drops HF	Applied under white light, to a polished surface	Cu deposited on n-type (more deposit formed on material of lower resistivities)	58 60
Hot NaOH/KOH	1 to 30% solution	1 to 5 min 50 to 100°C	Rapid revealing of structural details: especially large pits associated with twin lamellae on (111) surfaces. Simple to apply, but deposits traces of iron on surface, which must be removed chemically by washing in hydrochloric acid	84 3
	4% NaOH Add 40% NaOCl until no hydrogen evolution on Si	As required about 80°C	Used for thinning-down slices for electron microscopy. The specimen should float on the surface of the etch. (Too much NaOCl, or covering the dish, causes specimen to sink.)	85

C. INDIUM ANTIMONIDE

Name of Etchant	*Recipe*	*Etching Time*	*Principal Uses, and Comments*	*Refs.*
	1 HF 1 HNO_3	2 to 5 sec (polishing)	Polish-etch for (111) and ($\bar{1}\bar{1}0$). No etching on (111) or (100)	54
	5 HF 5 HNO_3 2 H_2O	20 sec	Etching (100) and (110)	86
CP.–4A	3 HF 5 HNO_3 3 acetic acid	5 to 30 sec	Chemical polish. May be further diluted with water and HNO_3. Ordinary CP.–4 has also been used for etching ($\bar{1}\bar{1}\bar{1}$).	87 88
111 etch	1 CP.–4A 1 H_2O 1 acetic acid	1 min	Etching ($\bar{1}\bar{1}\bar{1}$) and planes from there out to ($\bar{1}\bar{1}2$)	87
No. 2 etch	1 HF 1 H_2O_2 4 H_2O	5 to 10 sec	Etching ($\bar{1}\bar{1}\bar{1}$)	52

D. GALLIUM ANTIMONIDE

	1 HF 1 HNO_3 1 H_2O	1 min	Etching ($\bar{1}\bar{1}\bar{1}$)	52
	1 HF 9 HNO_3	1 to 5 min	Chemical polish	89
	1 H_2O_2 1 HCl 2 H_2O		Etching ($\bar{1}\bar{1}\bar{1}$), and (100), with asymmetry of pits corresponding to different rates of attack on (111) and ($\bar{1}\bar{1}\bar{1}$)	52

E. GALLIUM ARSENIDE

	5 5% NaOH solution 1 30% H_2O_2	5 min	Removing material at a rate of 10 to 15 μ/min	90
	2 HCl 1 HNO_3 2 H_2O	10 min	Etching ($\bar{1}\bar{1}\bar{1}$)	50 91
	1 HF 5 HNO_3 8 to 12 1%$AgNO_3$ solution		Etching (111) and ($\bar{1}\bar{1}\bar{1}$). (The Ag inhibits attack on the normally fast-etching (111).)	92
	1 HF 3 HNO_3 2 H_2O		Chemical polish	92

F. INDIUM ARSENIDE

Name of Etchant	*Recipe*	*Etching Time*	*Principal Uses, and Comments*	*Refs.*
	15 HF 75 HNO_3 15 acetic acid 0·06 Br_2	5 sec	Etching ($\bar{1}\bar{1}\bar{1}$)	51

G. INDIUM PHOSPHIDE

	0·4 M solution of ferric ion in 6 N HCl	1·5 min 25°C	Etching ($\bar{1}\bar{1}\bar{1}$). This solution also etches InSb, InAs, and GaAs, at various temperatures (see reference)	49

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